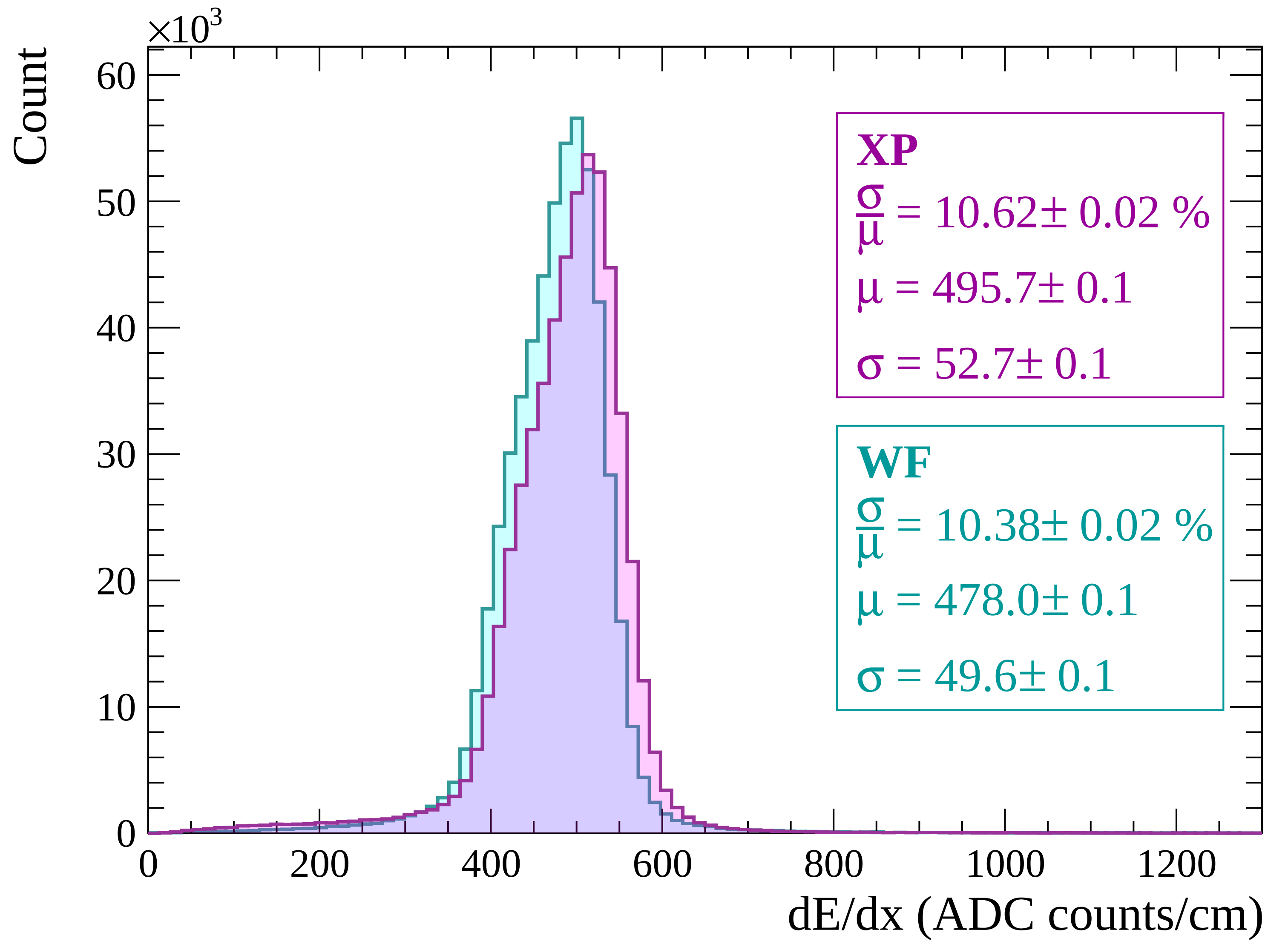
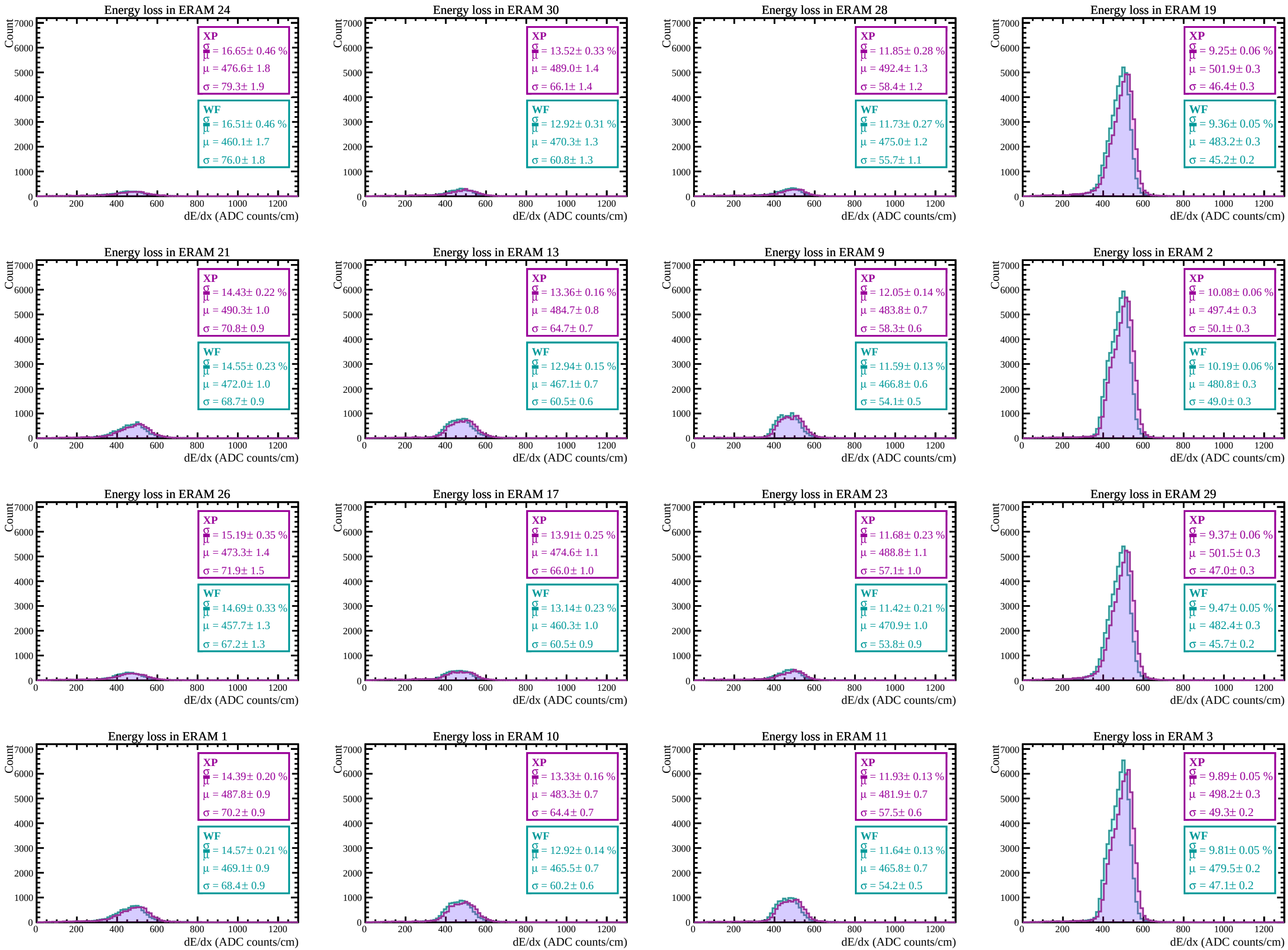
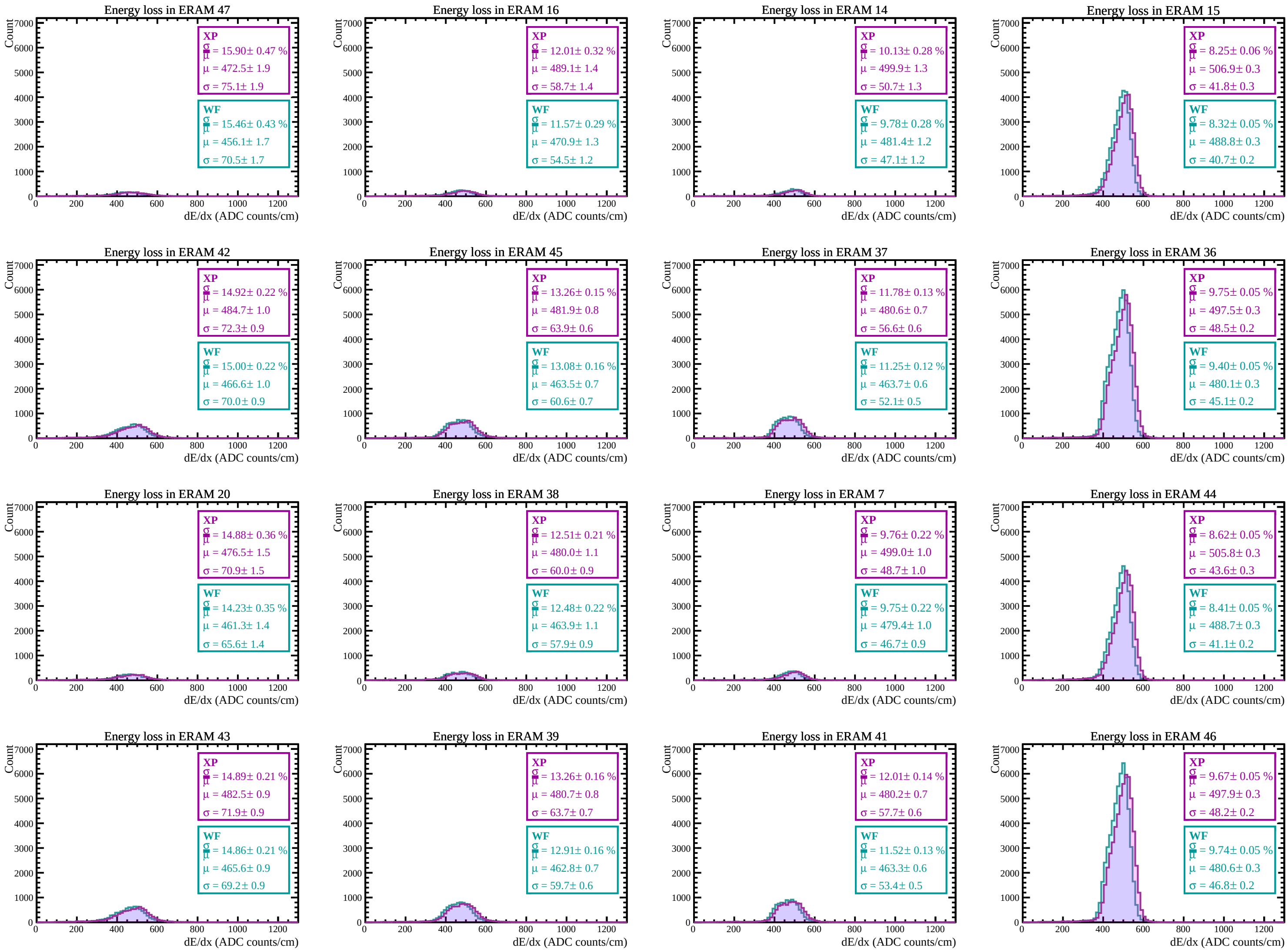


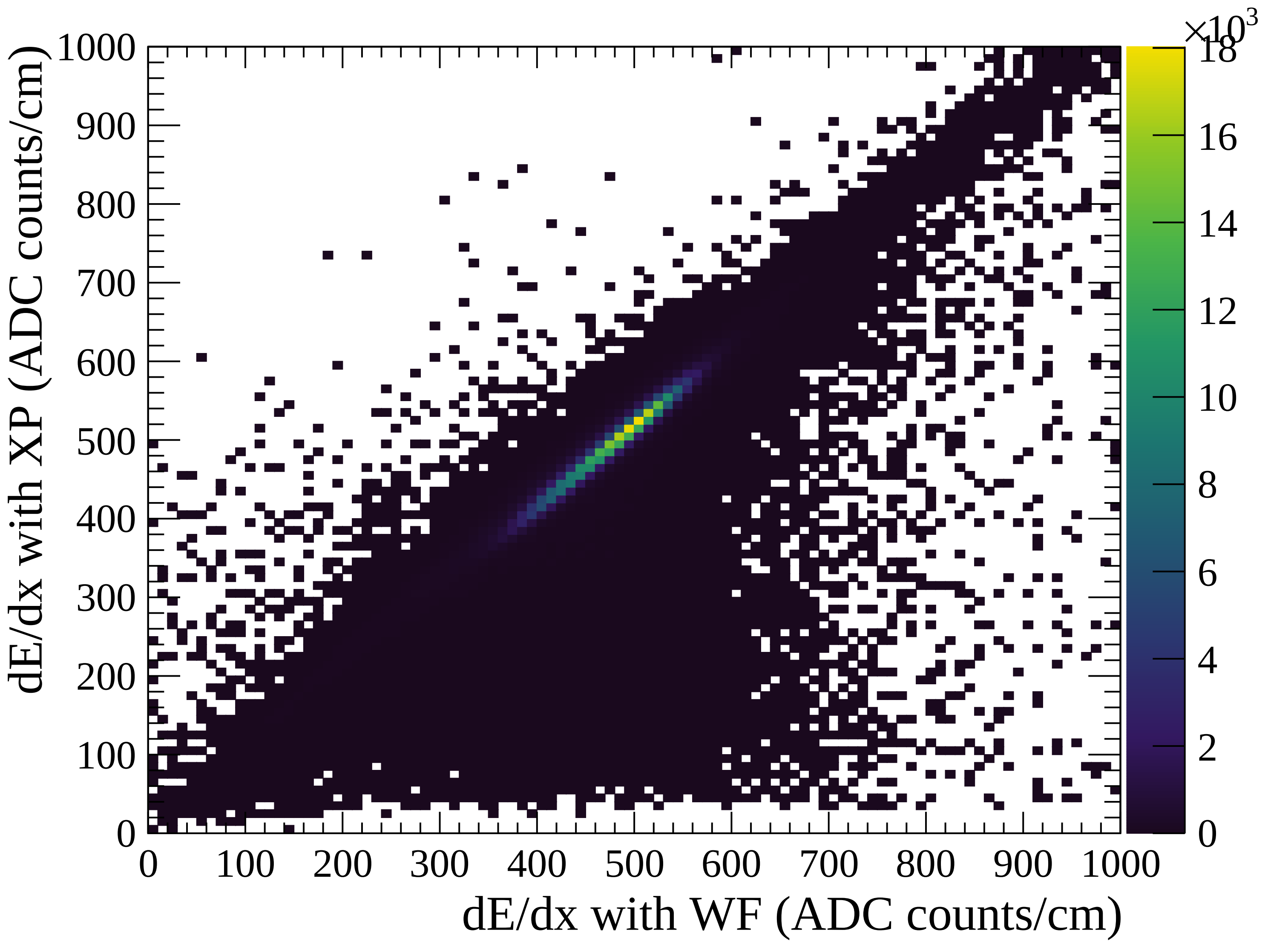
×10^3
Count
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.62± 0.02 %
μ = 495.7± 0.1
σ = 52.7± 0.1
WF
σ/μ = 10.38± 0.02 %
μ = 478.0± 0.1
σ = 49.6± 0.1



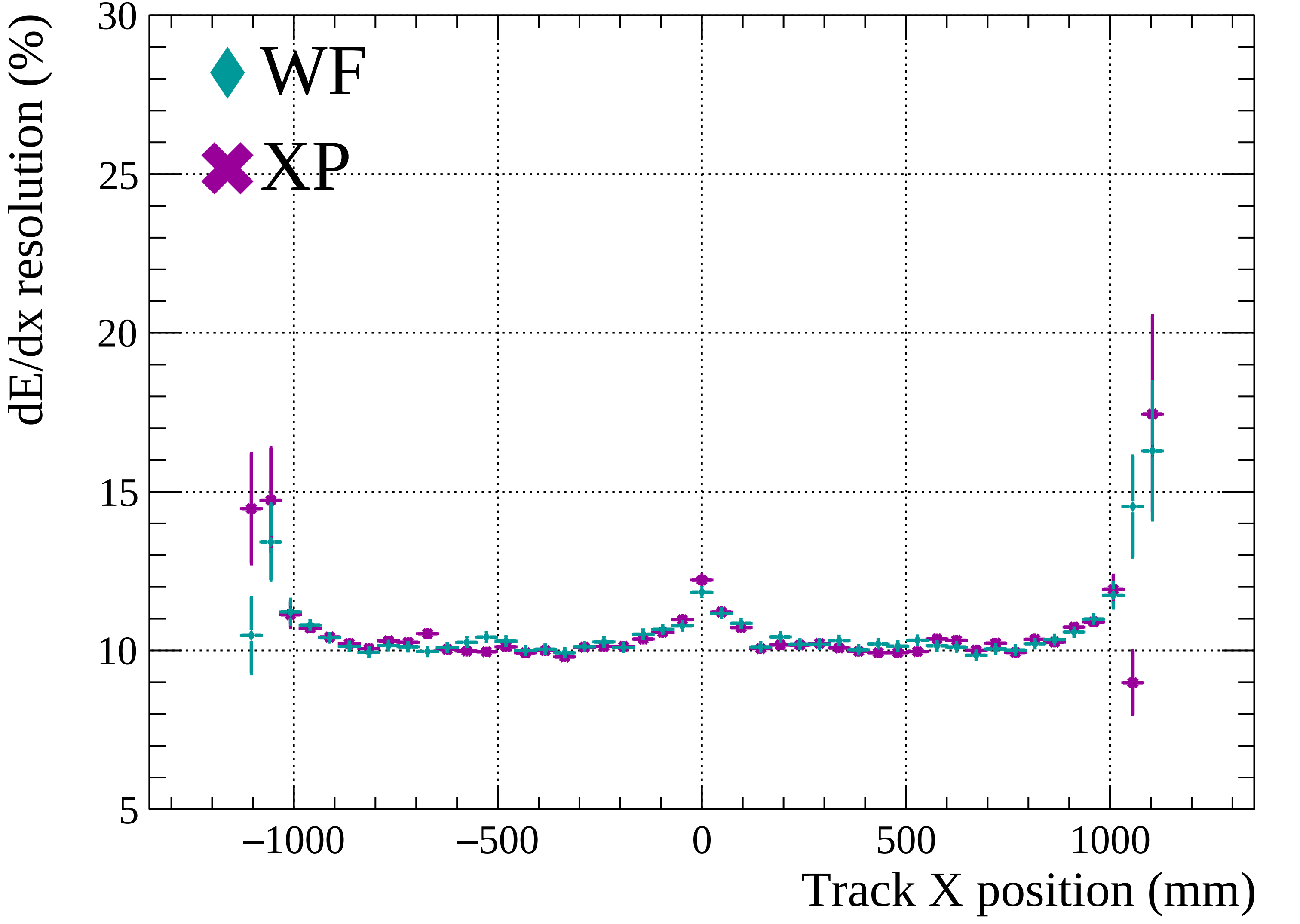
Energy loss in ERAM 24
XP
σ/μ = 16.65± 0.46 %
μ = 476.6± 1.8
σ = 79.3± 1.9
WF
σ/μ = 16.51± 0.46 %
μ = 460.1± 1.7
σ = 76.0± 1.8
Energy loss in ERAM 30
XP
σ/μ = 13.52± 0.33 %
μ = 489.0± 1.4
σ = 66.1± 1.4
WF
σ/μ = 12.92± 0.31 %
μ = 470.3± 1.3
σ = 60.8± 1.3
Energy loss in ERAM 28
XP
σ/μ = 11.85± 0.28 %
μ = 492.4± 1.3
σ = 58.4± 1.2
WF
σ/μ = 11.73± 0.27 %
μ = 475.0± 1.2
σ = 55.7± 1.1
Energy loss in ERAM 19
XP
σ/μ = 9.25± 0.06 %
μ = 501.9± 0.3
σ = 46.4± 0.3
WF
σ/μ = 9.36± 0.05 %
μ = 483.2± 0.3
σ = 45.2± 0.2
Energy loss in ERAM 21
XP
σ/μ = 14.43± 0.22 %
μ = 490.3± 1.0
σ = 70.8± 0.9
WF
σ/μ = 14.55± 0.23 %
μ = 472.0± 1.0
σ = 68.7± 0.9
Energy loss in ERAM 13
XP
σ/μ = 13.36± 0.16 %
μ = 484.7± 0.8
σ = 64.7± 0.7
WF
σ/μ = 12.94± 0.15 %
μ = 467.1± 0.7
σ = 60.5± 0.6
Energy loss in ERAM 9
XP
σ/μ = 12.05± 0.14 %
μ = 483.8± 0.7
σ = 58.3± 0.6
WF
σ/μ = 11.59± 0.13 %
μ = 466.8± 0.6
σ = 54.1± 0.5
Energy loss in ERAM 2
XP
σ/μ = 10.08± 0.06 %
μ = 497.4± 0.3
σ = 50.1± 0.3
WF
σ/μ = 10.19± 0.06 %
μ = 480.8± 0.3
σ = 49.0± 0.3
Energy loss in ERAM 26
XP
σ/μ = 15.19± 0.35 %
μ = 473.3± 1.4
σ = 71.9± 1.5
WF
σ/μ = 14.69± 0.33 %
μ = 457.7± 1.3
σ = 67.2± 1.3
Energy loss in ERAM 17
XP
σ/μ = 13.91± 0.25 %
μ = 474.6± 1.1
σ = 66.0± 1.0
WF
σ/μ = 13.14± 0.23 %
μ = 460.3± 1.0
σ = 60.5± 0.9
Energy loss in ERAM 23
XP
σ/μ = 11.68± 0.23 %
μ = 488.8± 1.1
σ = 57.1± 1.0
WF
σ/μ = 11.42± 0.21 %
μ = 470.9± 1.0
σ = 53.8± 0.9
Energy loss in ERAM 29
XP
σ/μ = 9.37± 0.06 %
μ = 501.5± 0.3
σ = 47.0± 0.3
WF
σ/μ = 9.47± 0.05 %
μ = 482.4± 0.3
σ = 45.7± 0.2
Energy loss in ERAM 1
XP
σ/μ = 14.39± 0.20 %
μ = 487.8± 0.9
σ = 70.2± 0.9
WF
σ/μ = 14.57± 0.21 %
μ = 469.1± 0.9
σ = 68.4± 0.9
Energy loss in ERAM 10
XP
σ/μ = 13.33± 0.16 %
μ = 483.3± 0.7
σ = 64.4± 0.7
WF
σ/μ = 12.92± 0.14 %
μ = 465.5± 0.7
σ = 60.2± 0.6
Energy loss in ERAM 11
XP
σ/μ = 11.93± 0.13 %
μ = 481.9± 0.7
σ = 57.5± 0.6
WF
σ/μ = 11.64± 0.13 %
μ = 465.8± 0.7
σ = 54.2± 0.5
Energy loss in ERAM 3
XP
σ/μ = 9.89± 0.05 %
μ = 498.2± 0.3
σ = 49.3± 0.2
WF
σ/μ = 9.81± 0.05 %
μ = 479.5± 0.2
σ = 47.1± 0.2
Count
dE/dx (ADC counts/cm)



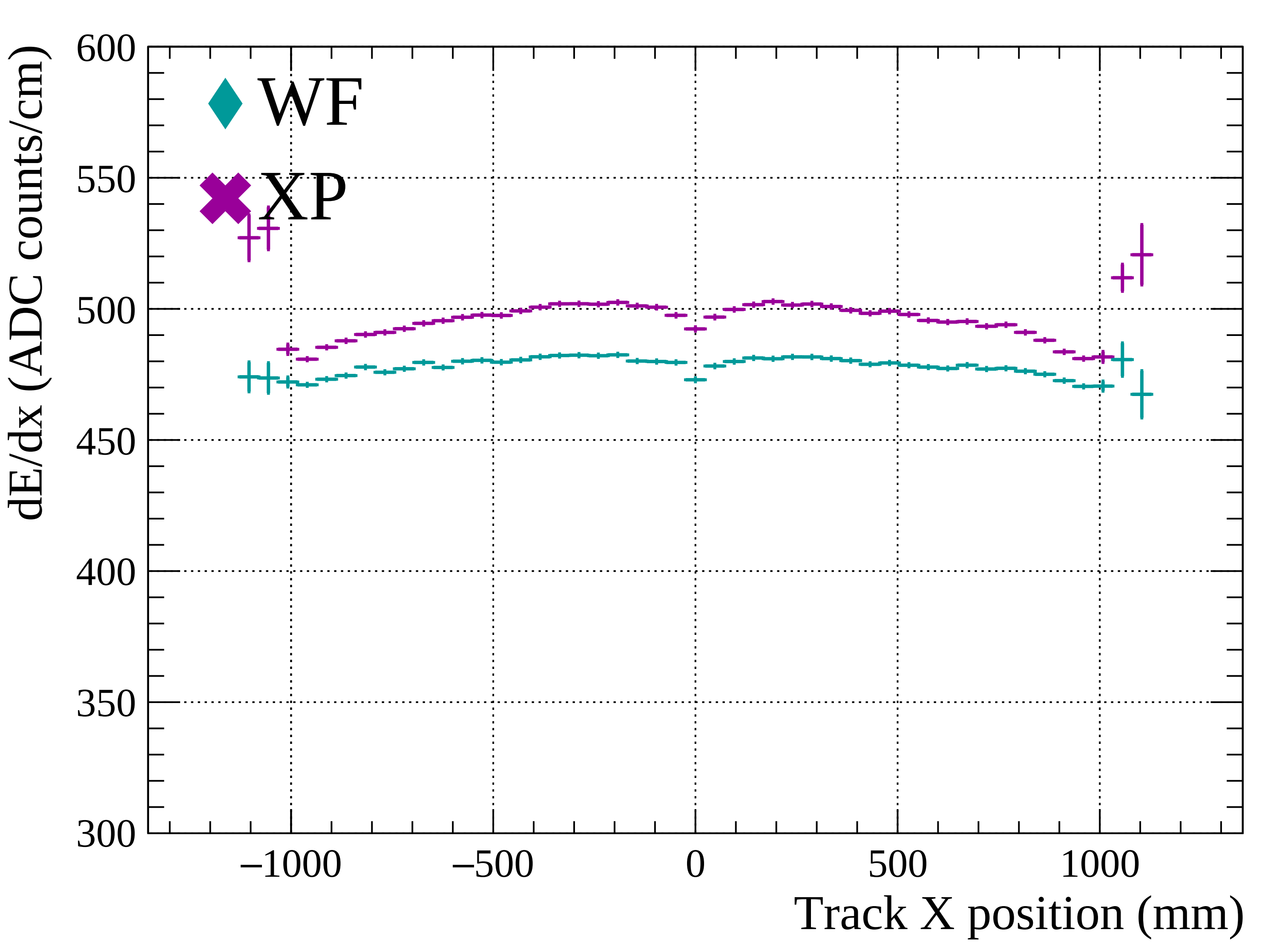
Energy loss in ERAM 47
XP
σ/μ = 15.90± 0.47 %
μ = 472.5± 1.9
σ = 75.1± 1.9
WF
σ/μ = 15.46± 0.43 %
μ = 456.1± 1.7
σ = 70.5± 1.7
Energy loss in ERAM 16
XP
σ/μ = 12.01± 0.32 %
μ = 489.1± 1.4
σ = 58.7± 1.4
WF
σ/μ = 11.57± 0.29 %
μ = 470.9± 1.3
σ = 54.5± 1.2
Energy loss in ERAM 14
XP
σ/μ = 10.13± 0.28 %
μ = 499.9± 1.3
σ = 50.7± 1.3
WF
σ/μ = 9.78± 0.28 %
μ = 481.4± 1.2
σ = 47.1± 1.2
Energy loss in ERAM 15
XP
σ/μ = 8.25± 0.06 %
μ = 506.9± 0.3
σ = 41.8± 0.3
WF
σ/μ = 8.32± 0.05 %
μ = 488.8± 0.3
σ = 40.7± 0.2
Energy loss in ERAM 42
XP
σ/μ = 14.92± 0.22 %
μ = 484.7± 1.0
σ = 72.3± 0.9
WF
σ/μ = 15.00± 0.22 %
μ = 466.6± 1.0
σ = 70.0± 0.9
Energy loss in ERAM 45
XP
σ/μ = 13.26± 0.15 %
μ = 481.9± 0.8
σ = 63.9± 0.6
WF
σ/μ = 13.08± 0.16 %
μ = 463.5± 0.7
σ = 60.6± 0.7
Energy loss in ERAM 37
XP
σ/μ = 11.78± 0.13 %
μ = 480.6± 0.7
σ = 56.6± 0.6
WF
σ/μ = 11.25± 0.12 %
μ = 463.7± 0.6
σ = 52.1± 0.5
Energy loss in ERAM 36
XP
σ/μ = 9.75± 0.05 %
μ = 497.5± 0.3
σ = 48.5± 0.2
WF
σ/μ = 9.40± 0.05 %
μ = 480.1± 0.3
σ = 45.1± 0.2
Energy loss in ERAM 20
XP
σ/μ = 14.88± 0.36 %
μ = 476.5± 1.5
σ = 70.9± 1.5
WF
σ/μ = 14.23± 0.35 %
μ = 461.3± 1.4
σ = 65.6± 1.4
Energy loss in ERAM 38
XP
σ/μ = 12.51± 0.21 %
μ = 480.0± 1.1
σ = 60.0± 0.9
WF
σ/μ = 12.48± 0.22 %
μ = 463.9± 1.1
σ = 57.9± 0.9
Energy loss in ERAM 7
XP
σ/μ = 9.76± 0.22 %
μ = 499.0± 1.0
σ = 48.7± 1.0
WF
σ/μ = 9.75± 0.22 %
μ = 479.4± 1.0
σ = 46.7± 0.9
Energy loss in ERAM 44
XP
σ/μ = 8.62± 0.05 %
μ = 505.8± 0.3
σ = 43.6± 0.3
WF
σ/μ = 8.41± 0.05 %
μ = 488.7± 0.3
σ = 41.1± 0.2
Energy loss in ERAM 43
XP
σ/μ = 14.89± 0.21 %
μ = 482.5± 0.9
σ = 71.9± 0.9
WF
σ/μ = 14.86± 0.21 %
μ = 465.6± 0.9
σ = 69.2± 0.9
Energy loss in ERAM 39
XP
σ/μ = 13.26± 0.16 %
μ = 480.7± 0.8
σ = 63.7± 0.7
WF
σ/μ = 12.91± 0.16 %
μ = 462.8± 0.7
σ = 59.7± 0.6
Energy loss in ERAM 41
XP
σ/μ = 12.01± 0.14 %
μ = 480.2± 0.7
σ = 57.7± 0.6
WF
σ/μ = 11.52± 0.13 %
μ = 463.3± 0.6
σ = 53.4± 0.5
Energy loss in ERAM 46
XP
σ/μ = 9.67± 0.05 %
μ = 497.9± 0.3
σ = 48.2± 0.2
WF
σ/μ = 9.74± 0.05 %
μ = 480.6± 0.3
σ = 46.8± 0.2
Count
dE/dx (ADC counts/cm)



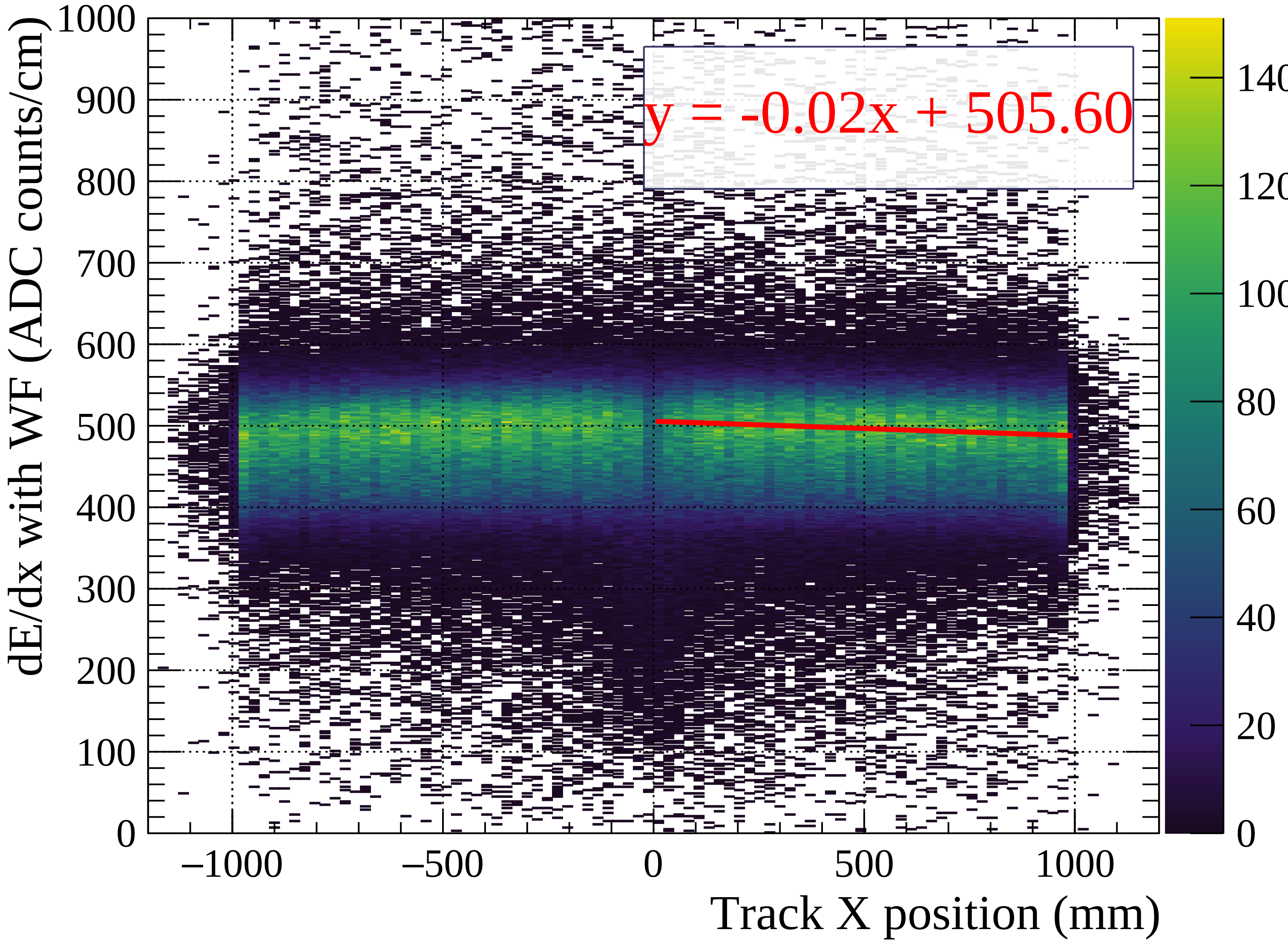
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
×10^3
0
2
4
6
8
10
12
14
16
18



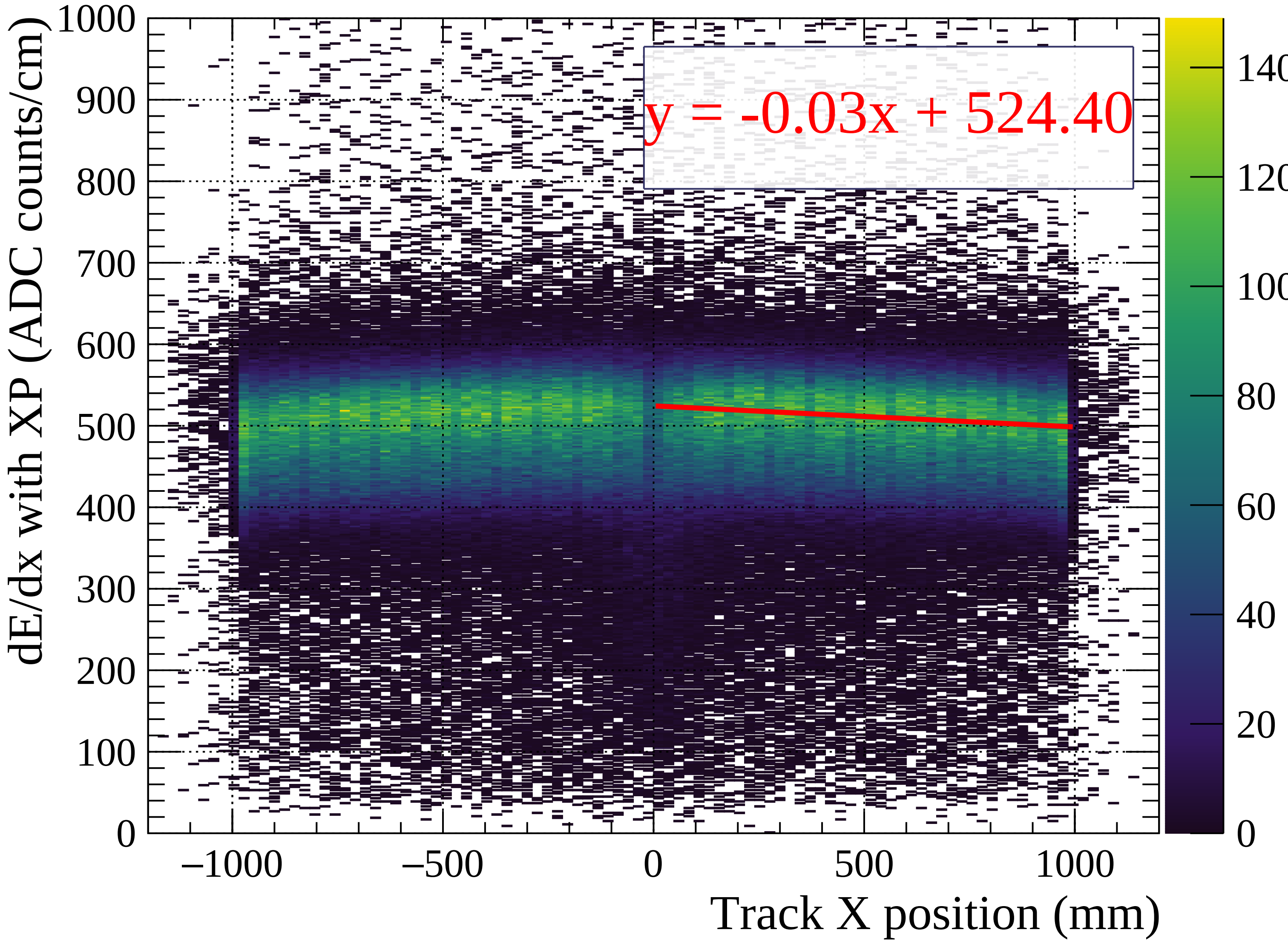
WF
XP
dE/dx resolution (%)
Track X position (mm)
30
25
20
15
10
5
−1000
−500
0
500
1000



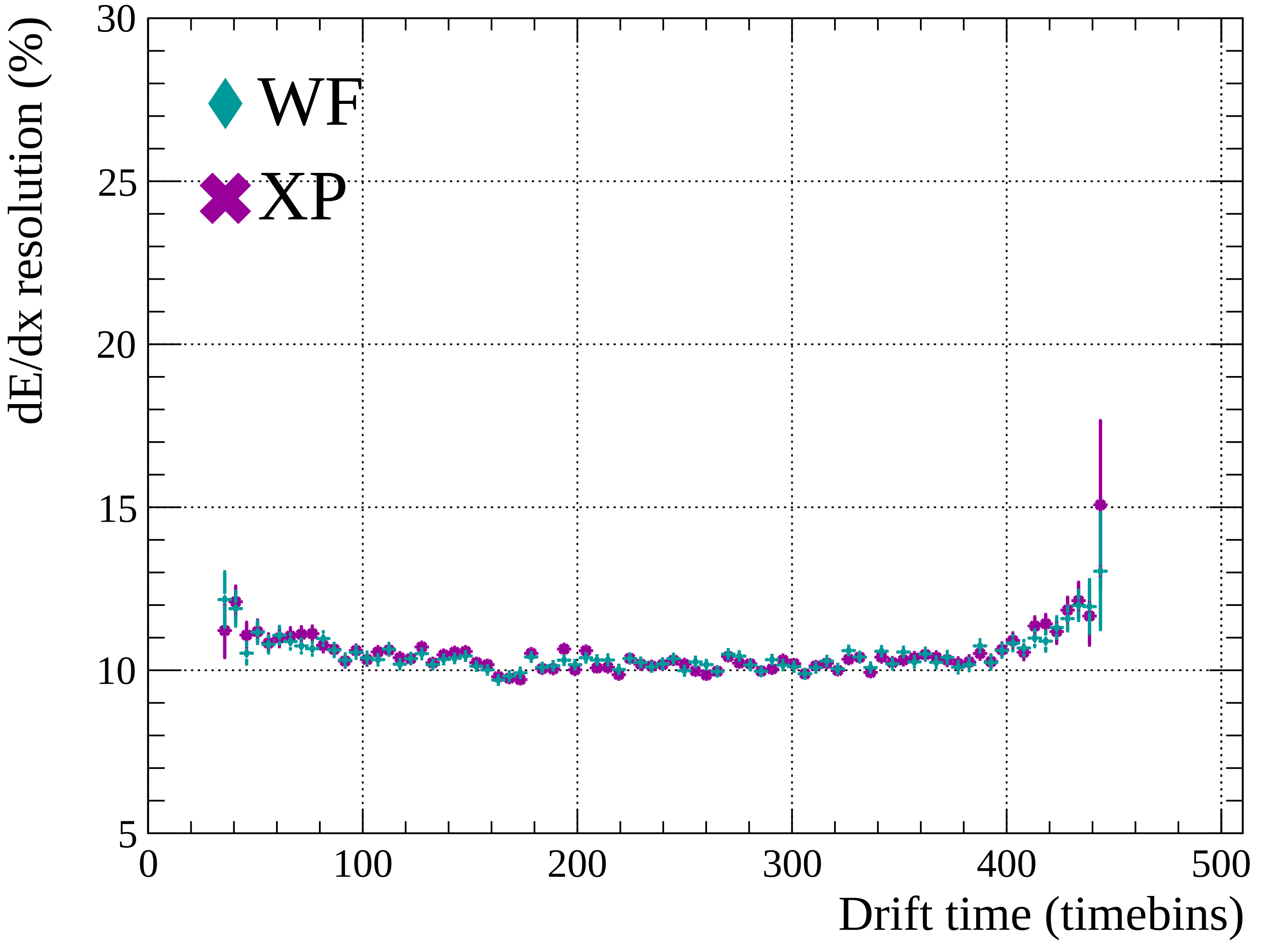
WF
XP
600
550
500
450
400
350
300
−1000
−500
0
500
1000
dE/dx (ADC counts/cm)
Track X position (mm)



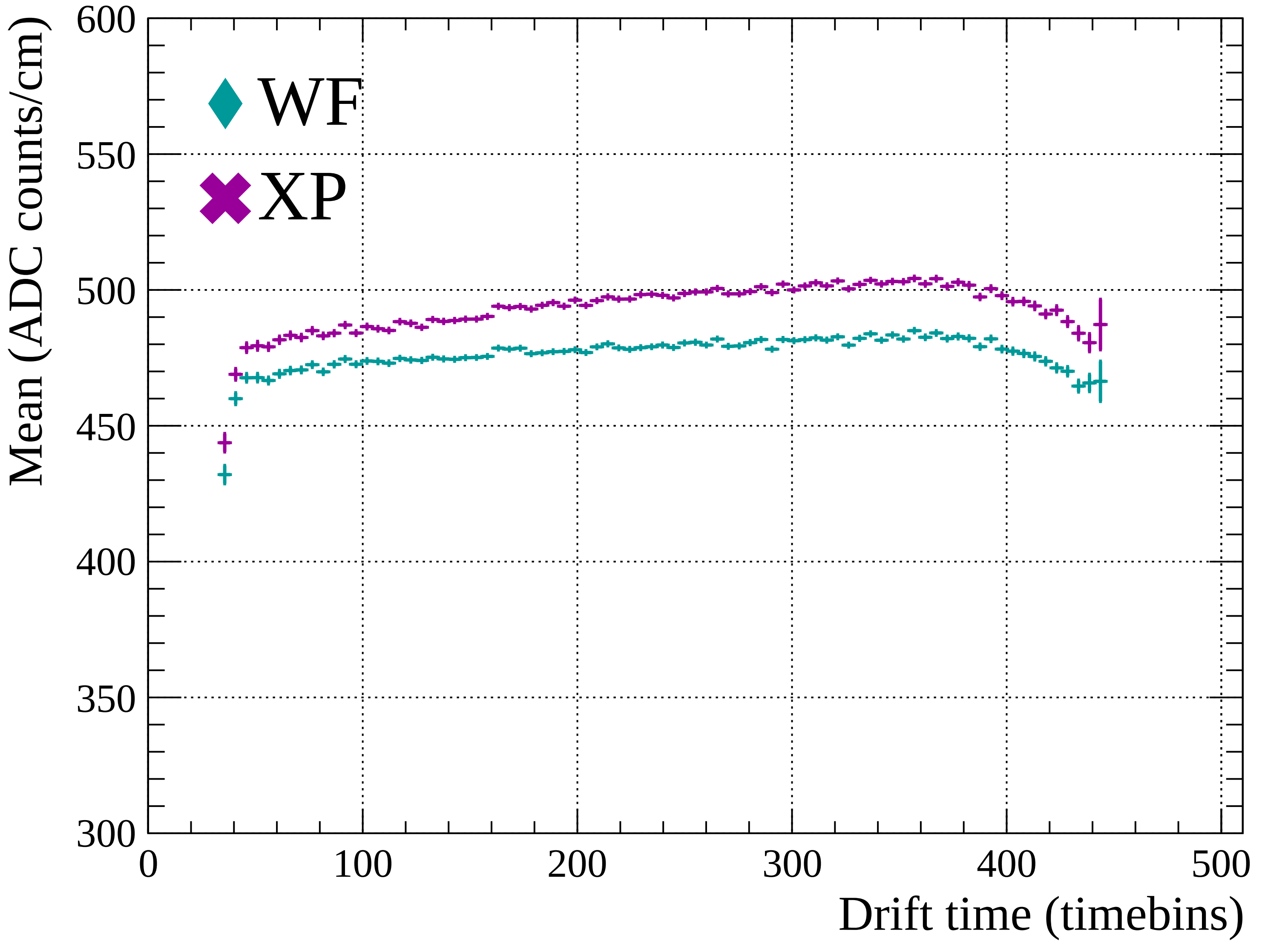
y = -0.02x + 505.60
dE/dx with WF (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
20
40
60
80
100
120
140



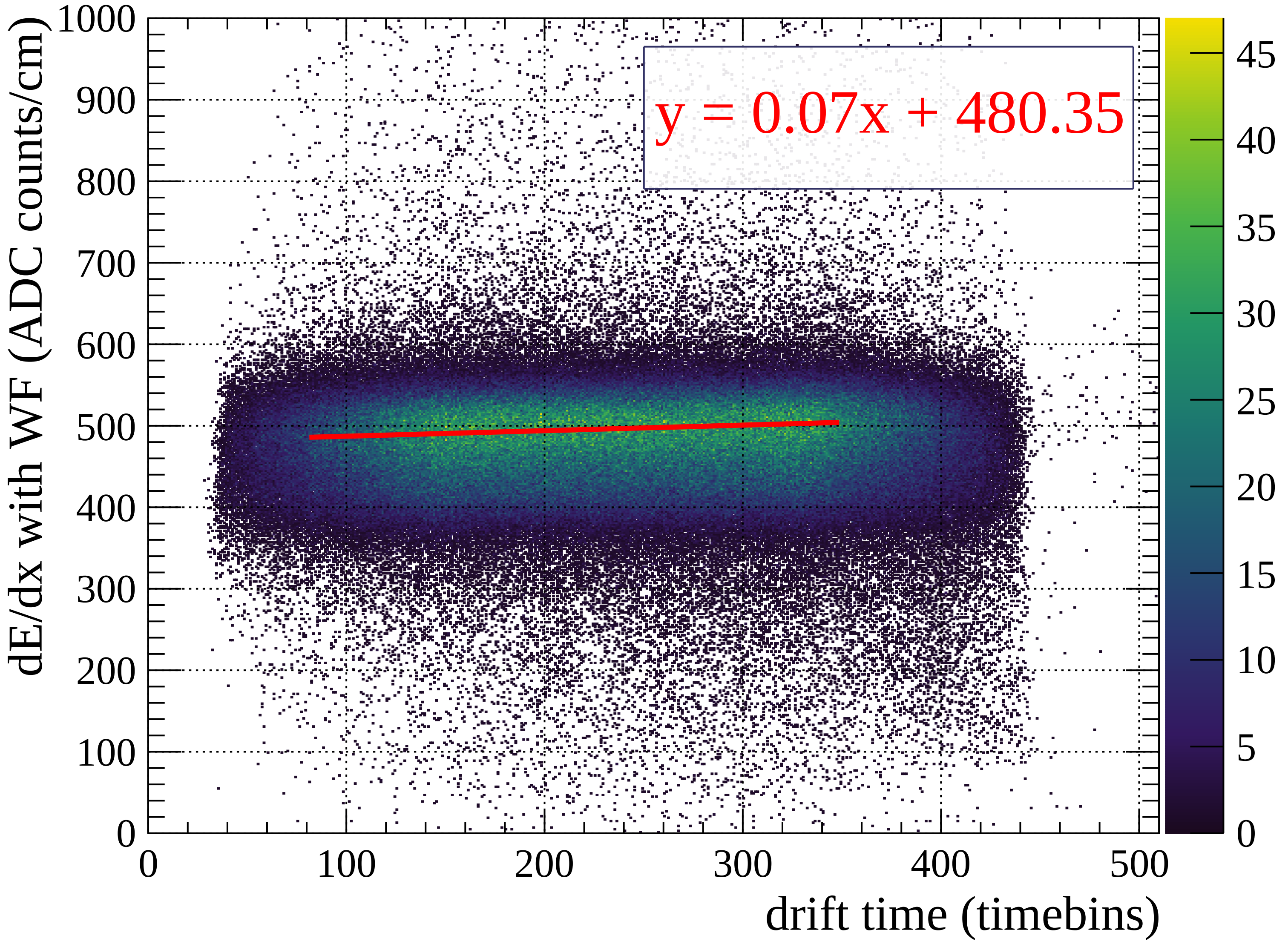
y = -0.03x + 524.40
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
20
40
60
80



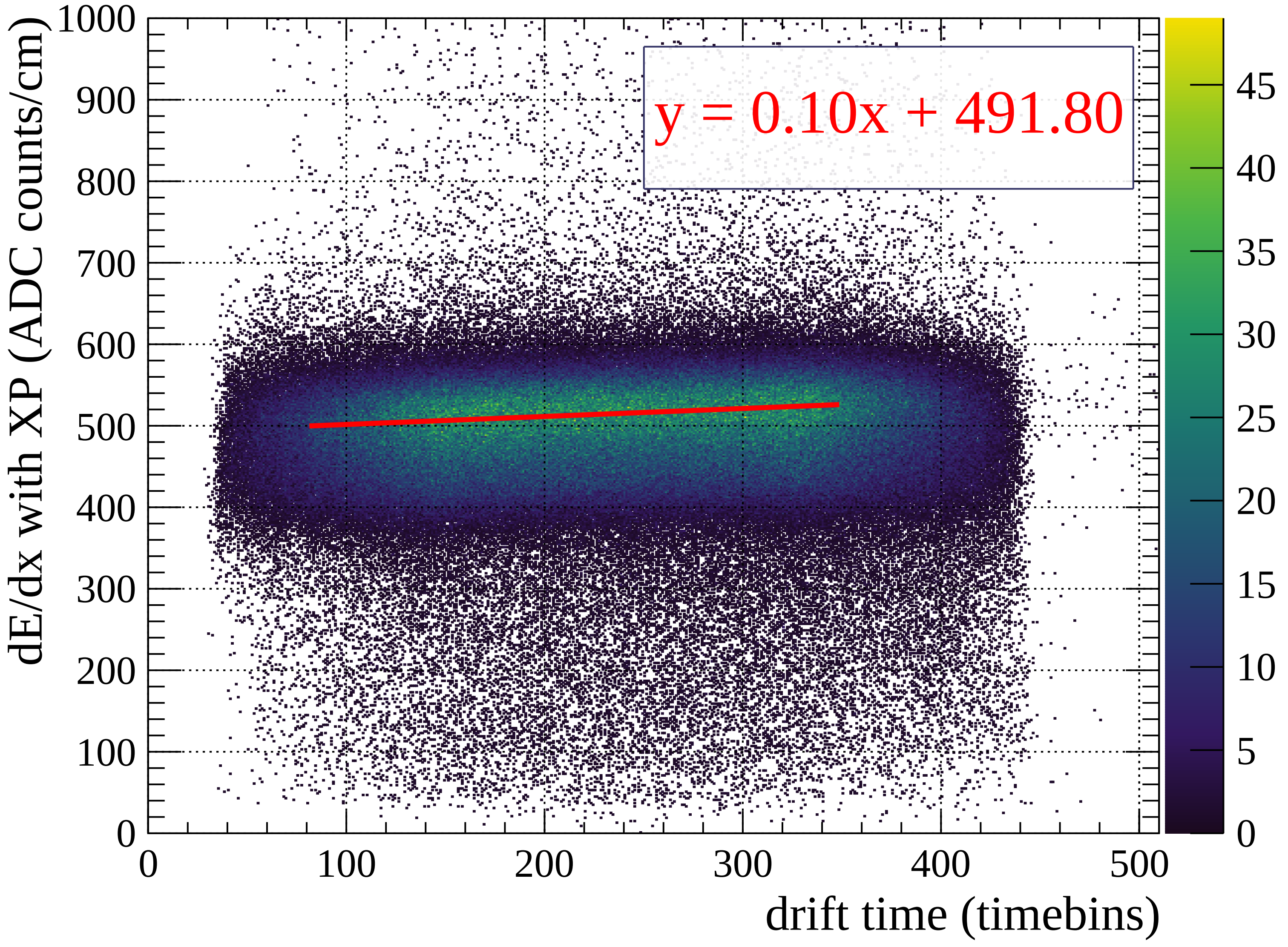
WF
XP
dE/dx resolution (%)
30
25
20
15
10
5
0
100
200
300
400
500
Drift time (timebins)



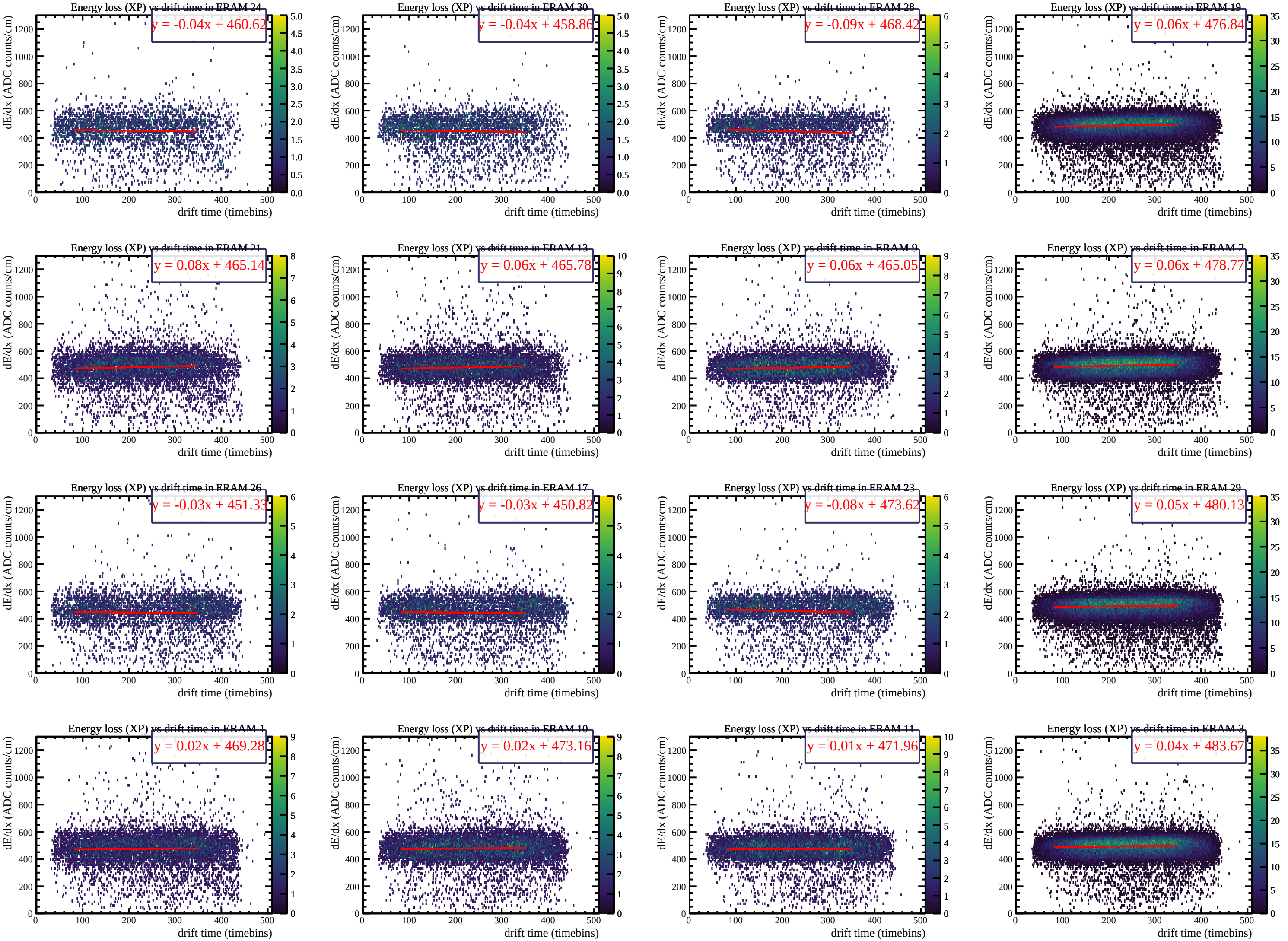
WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
100
200
300
400
500
Drift time (timebins)



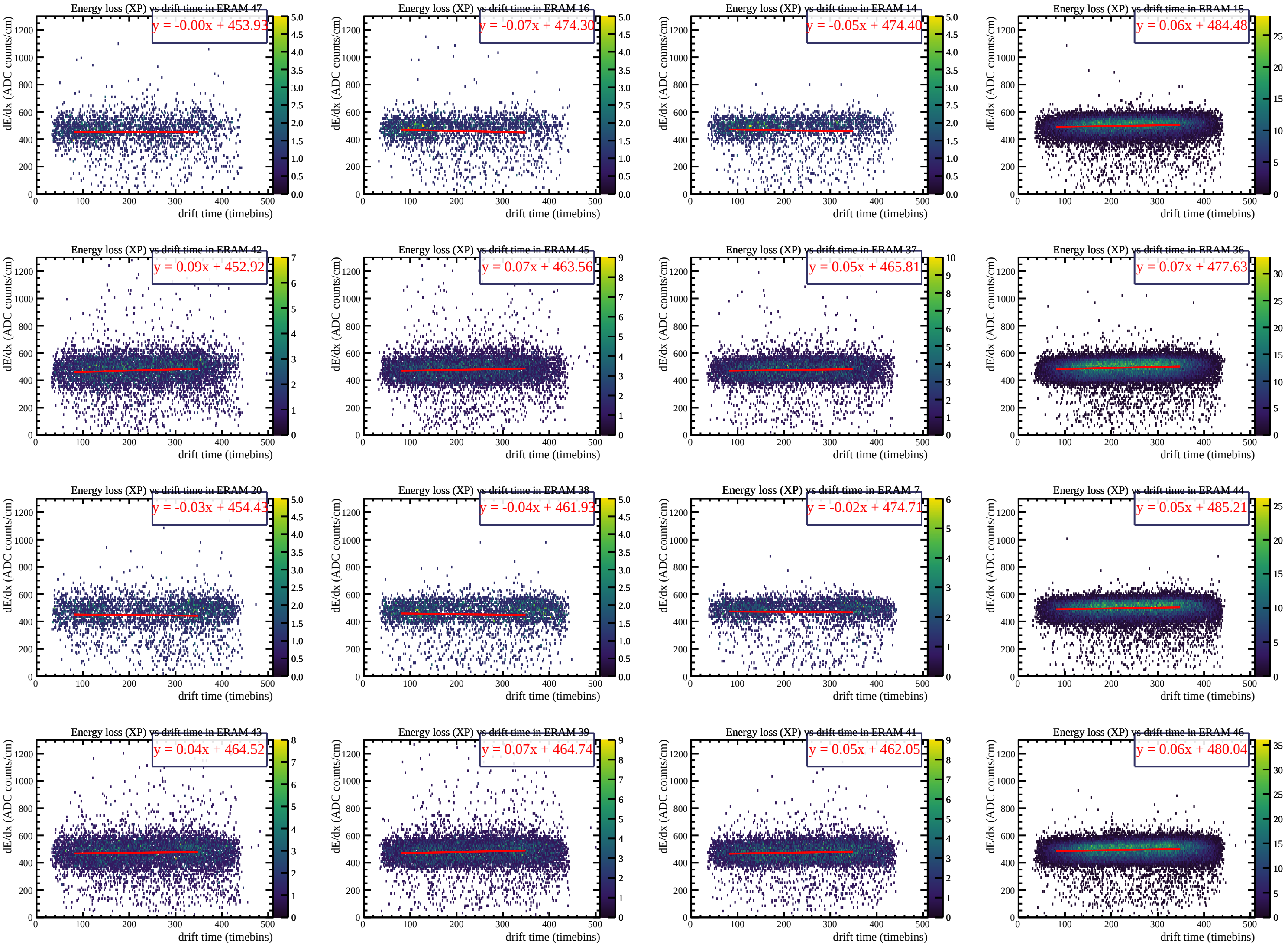
y = 0.07x + 480.35
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
5
10
15
20
25
30
35
40
45



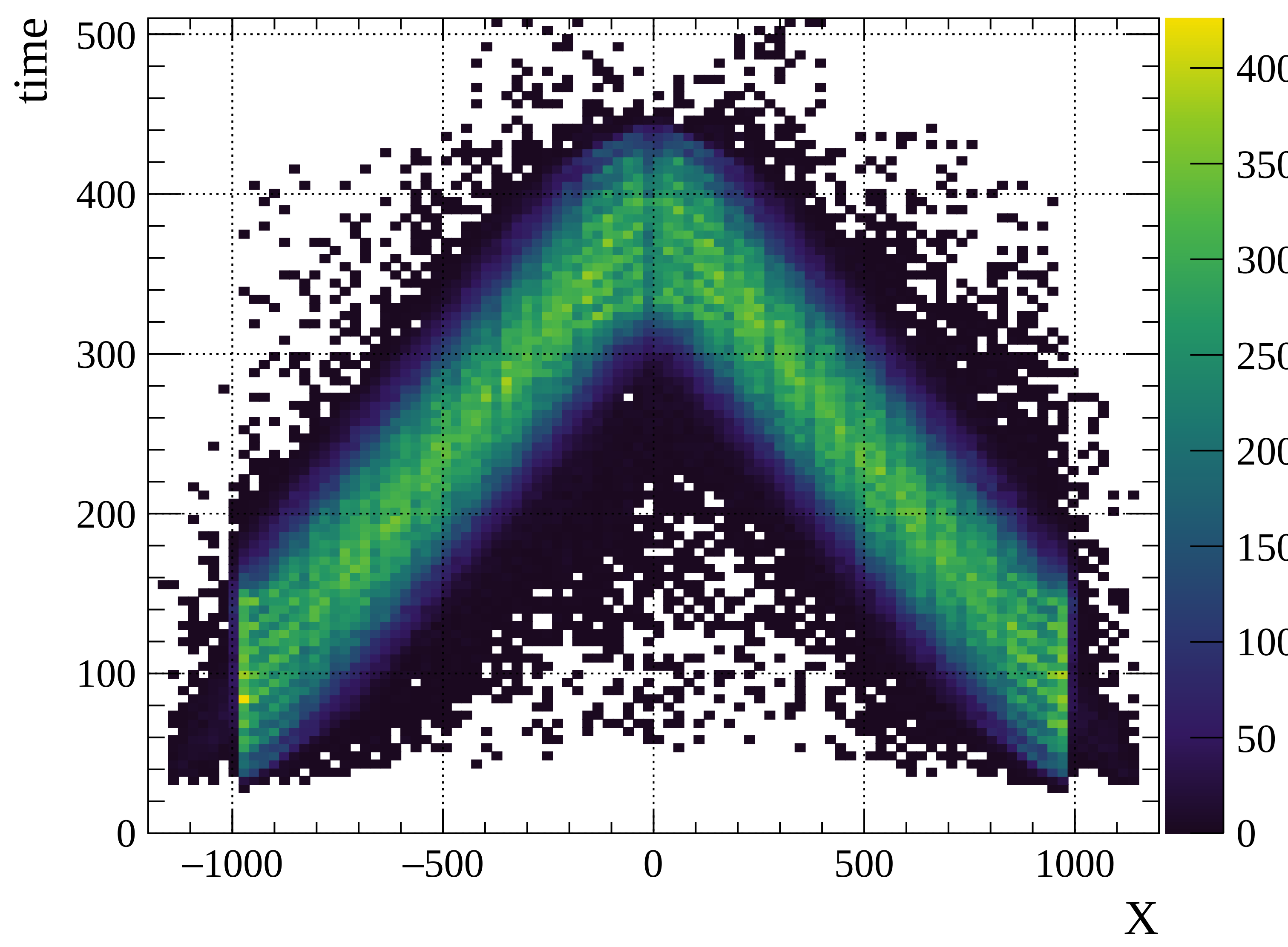
y = 0.10x + 491.80
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
5
10
15
20
25
30
35
40
45



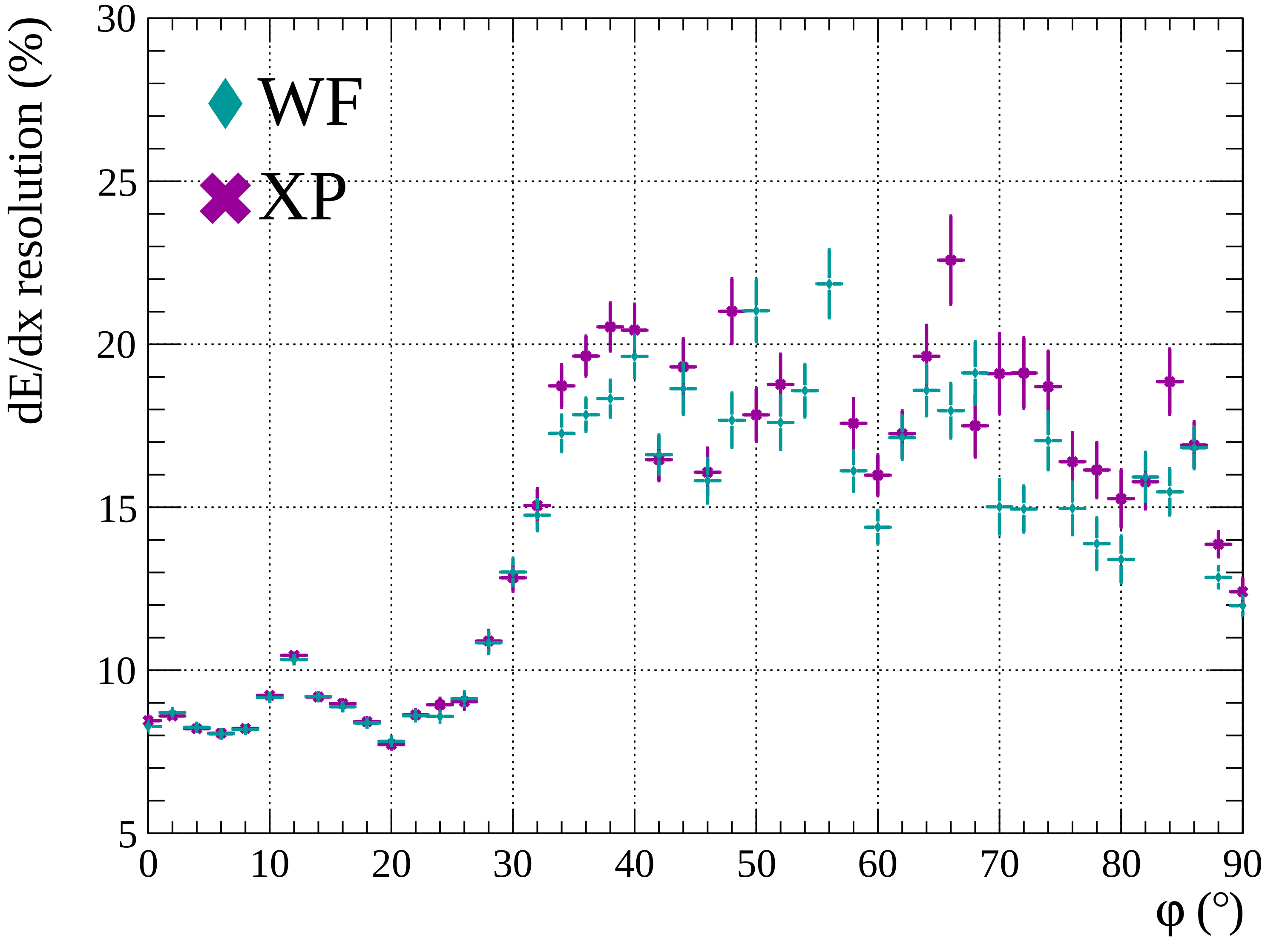
Energy loss (XP) vs drift time in ERAM 24
y = -0.04x + 460.62
Energy loss (XP) vs drift time in ERAM 30
y = -0.04x + 458.86
Energy loss (XP) vs drift time in ERAM 28
y = -0.09x + 468.42
Energy loss (XP) vs drift time in ERAM 19
y = 0.06x + 476.84
Energy loss (XP) vs drift time in ERAM 21
y = 0.08x + 465.14
Energy loss (XP) vs drift time in ERAM 13
y = 0.06x + 465.78
Energy loss (XP) vs drift time in ERAM 9
y = 0.06x + 465.05
Energy loss (XP) vs drift time in ERAM 2
y = 0.06x + 478.77
Energy loss (XP) vs drift time in ERAM 26
y = -0.03x + 451.33
Energy loss (XP) vs drift time in ERAM 17
y = -0.03x + 450.82
Energy loss (XP) vs drift time in ERAM 23
y = -0.08x + 473.62
Energy loss (XP) vs drift time in ERAM 29
y = 0.05x + 480.13
Energy loss (XP) vs drift time in ERAM 1
y = 0.02x + 469.28
Energy loss (XP) vs drift time in ERAM 10
y = 0.02x + 473.16
Energy loss (XP) vs drift time in ERAM 11
y = 0.01x + 471.96
Energy loss (XP) vs drift time in ERAM 3
y = 0.04x + 483.67
dE/dx (ADC counts/cm)
drift time (timebins)



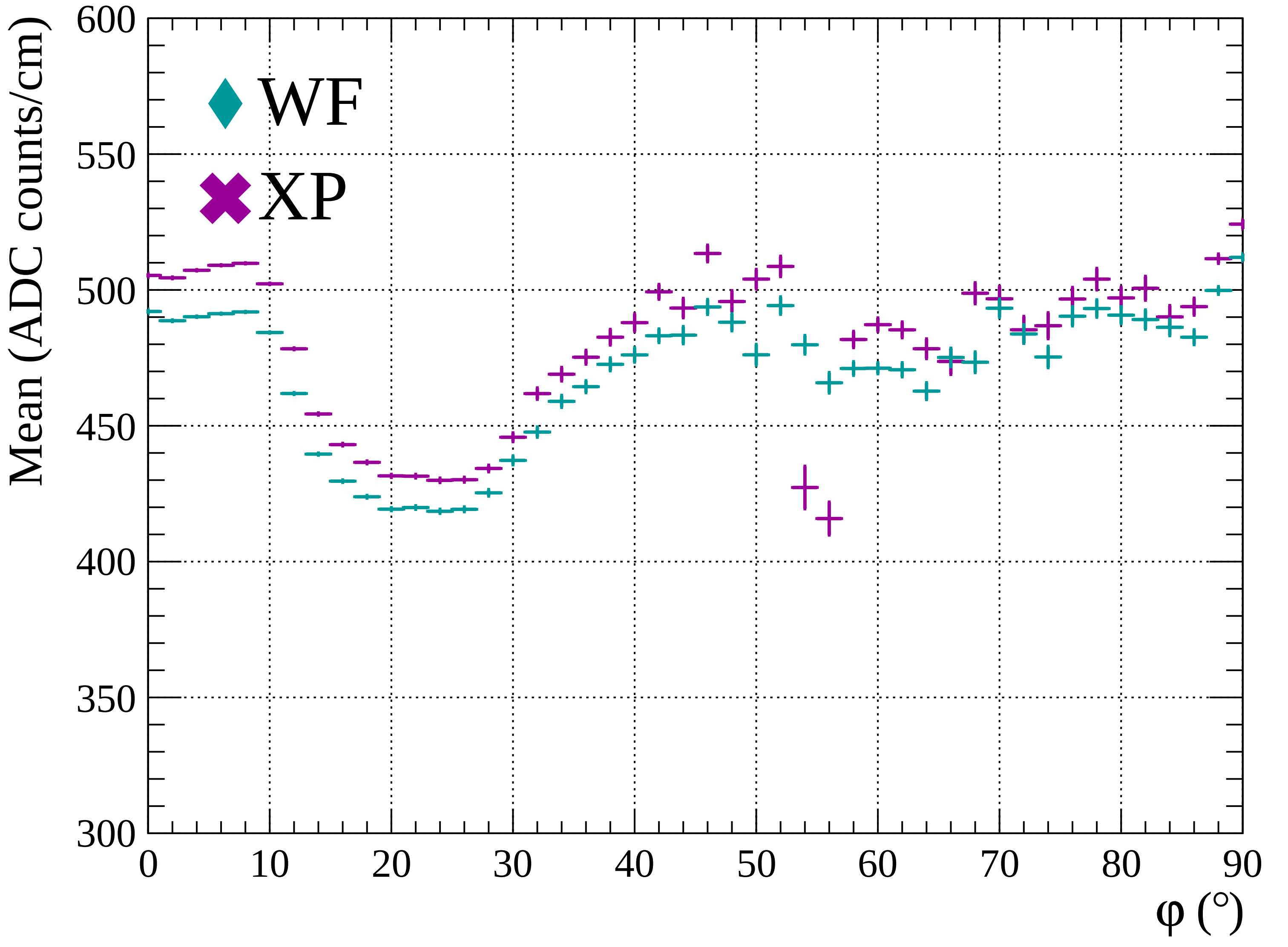
Energy loss (XP) vs drift time in ERAM 47
y = -0.00x + 453.93
Energy loss (XP) vs drift time in ERAM 16
y = -0.07x + 474.30
Energy loss (XP) vs drift time in ERAM 14
y = -0.05x + 474.40
Energy loss (XP) vs drift time in ERAM 15
y = 0.06x + 484.48
Energy loss (XP) vs drift time in ERAM 42
y = 0.09x + 452.92
Energy loss (XP) vs drift time in ERAM 45
y = 0.07x + 463.56
Energy loss (XP) vs drift time in ERAM 37
y = 0.05x + 465.81
Energy loss (XP) vs drift time in ERAM 36
y = 0.07x + 477.63
Energy loss (XP) vs drift time in ERAM 20
y = -0.03x + 454.43
Energy loss (XP) vs drift time in ERAM 38
y = -0.04x + 461.93
Energy loss (XP) vs drift time in ERAM 7
y = -0.02x + 474.71
Energy loss (XP) vs drift time in ERAM 44
y = 0.05x + 485.21
Energy loss (XP) vs drift time in ERAM 43
y = 0.04x + 464.52
Energy loss (XP) vs drift time in ERAM 39
y = 0.07x + 464.74
Energy loss (XP) vs drift time in ERAM 41
y = 0.05x + 462.05
Energy loss (XP) vs drift time in ERAM 46
y = 0.06x + 480.04
dE/dx (ADC counts/cm)
drift time (timebins)



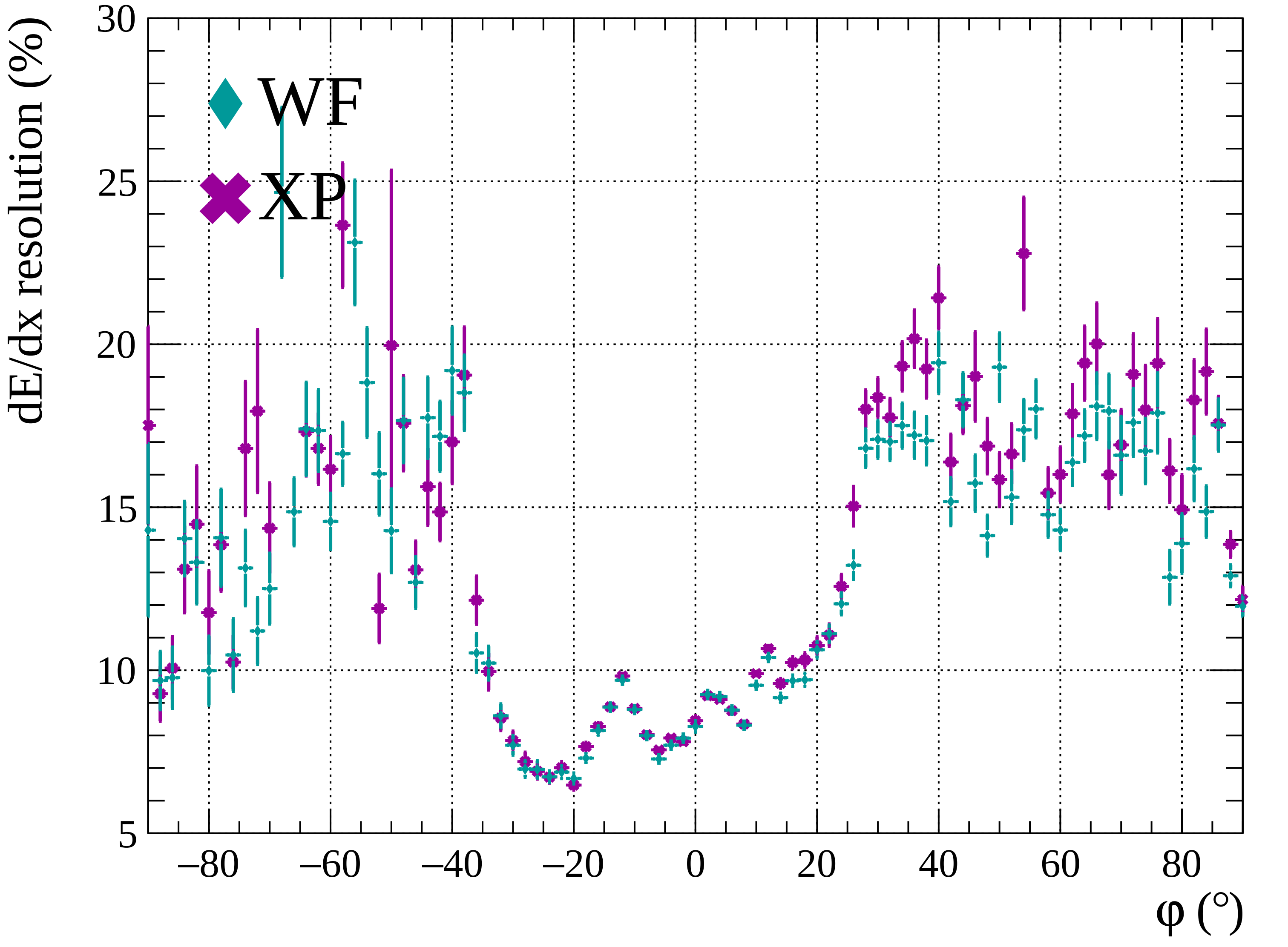
time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
400
350
300
250
200
150
100
50
0



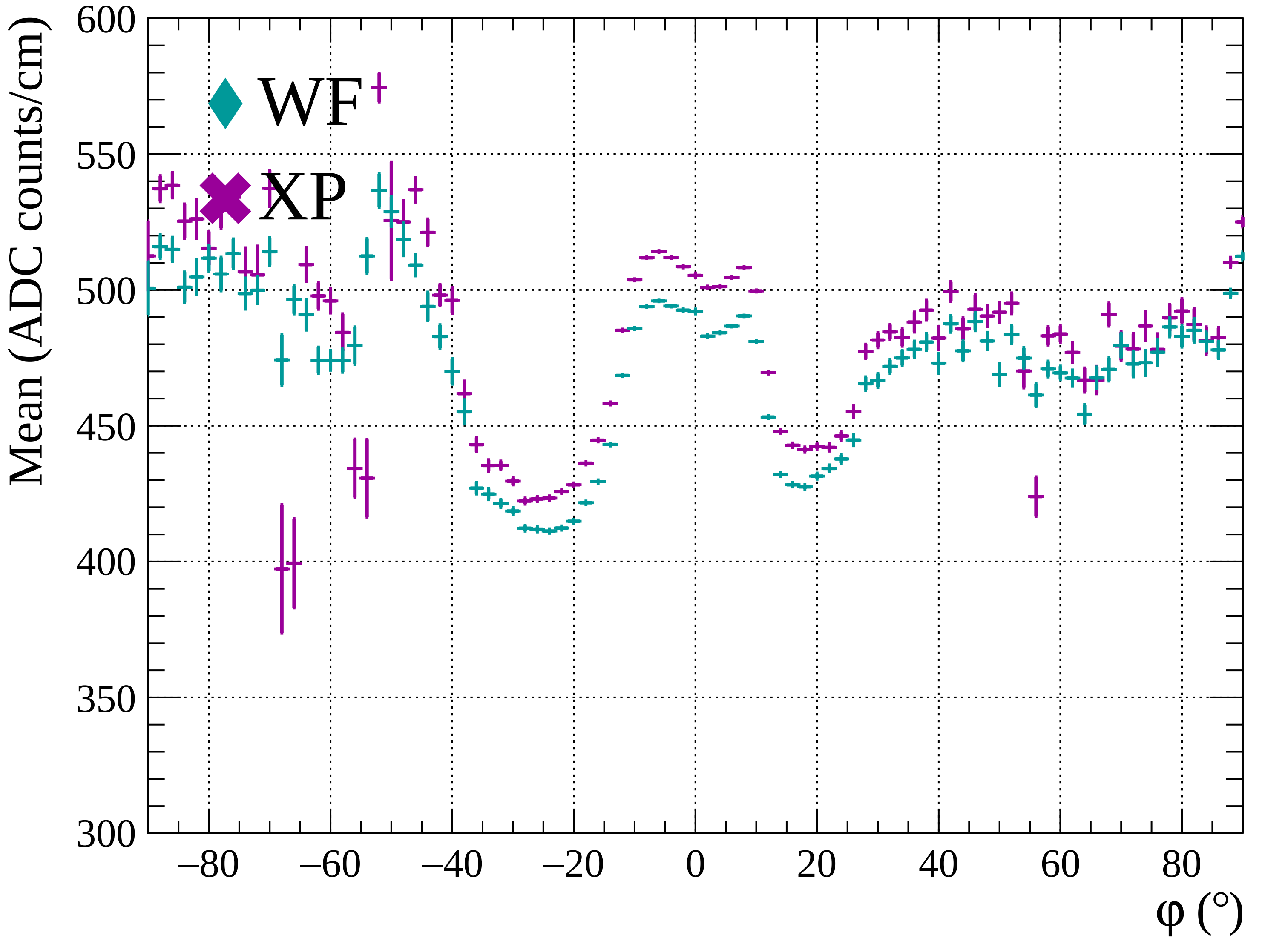
WF
XP
dE/dx resolution (%)
φ (°)
5
10
15
20
25
30
0
10
20
30
40
50
60
70
80
90



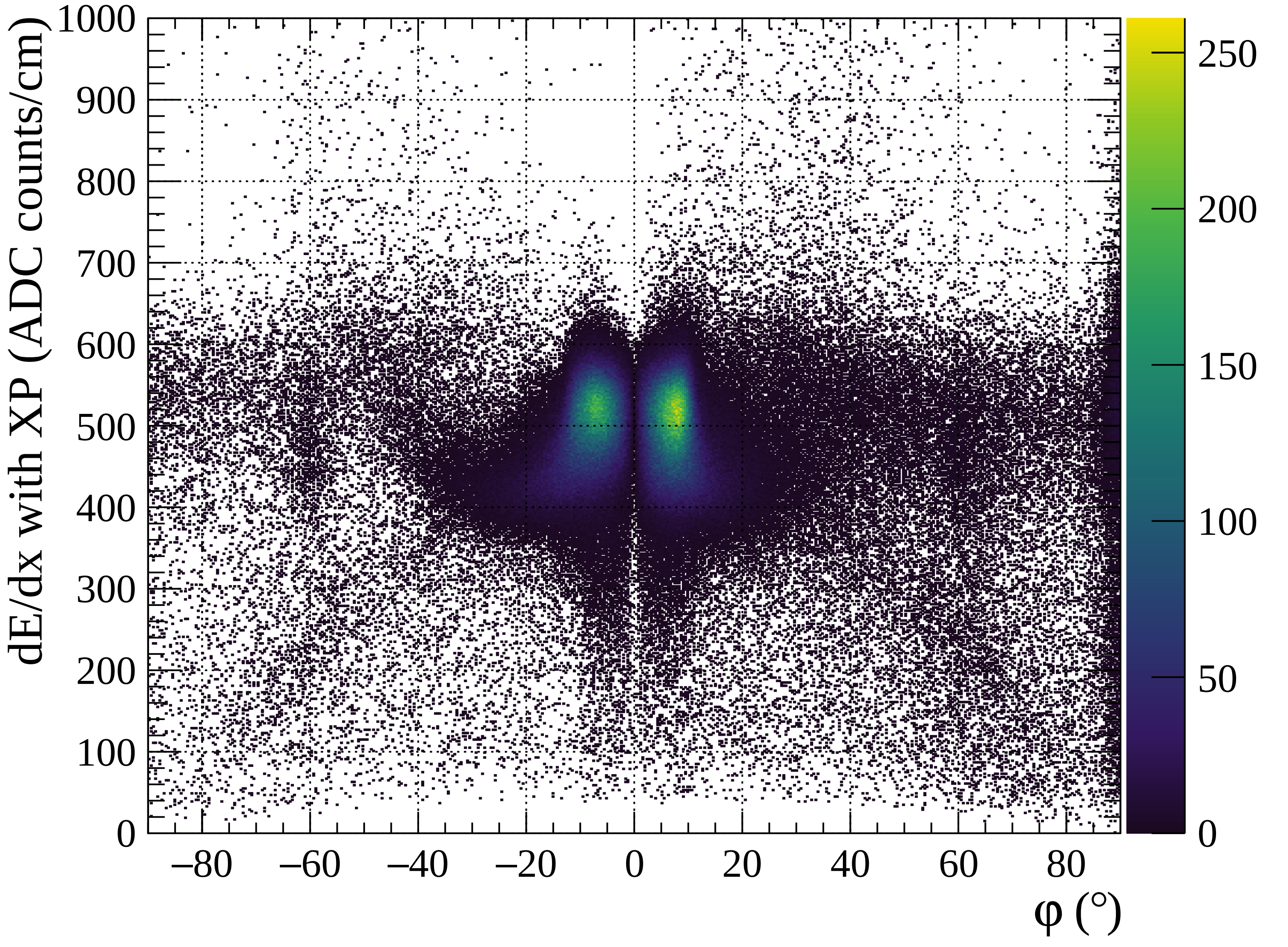
WF
XP
Mean (ADC counts/cm)
φ (°)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90

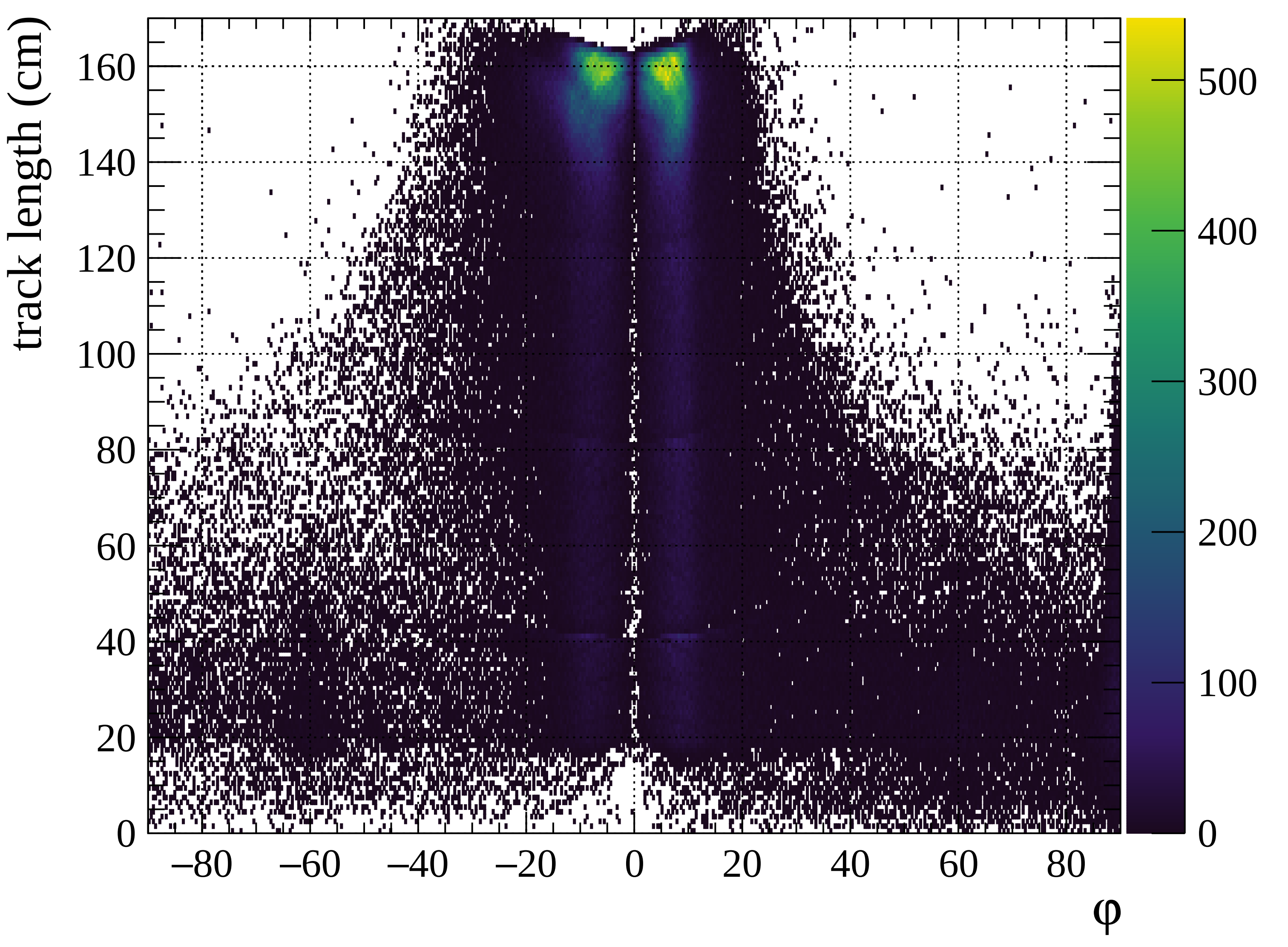


dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−80
−60
−40
−20
0
20
40
60
80
φ (°)

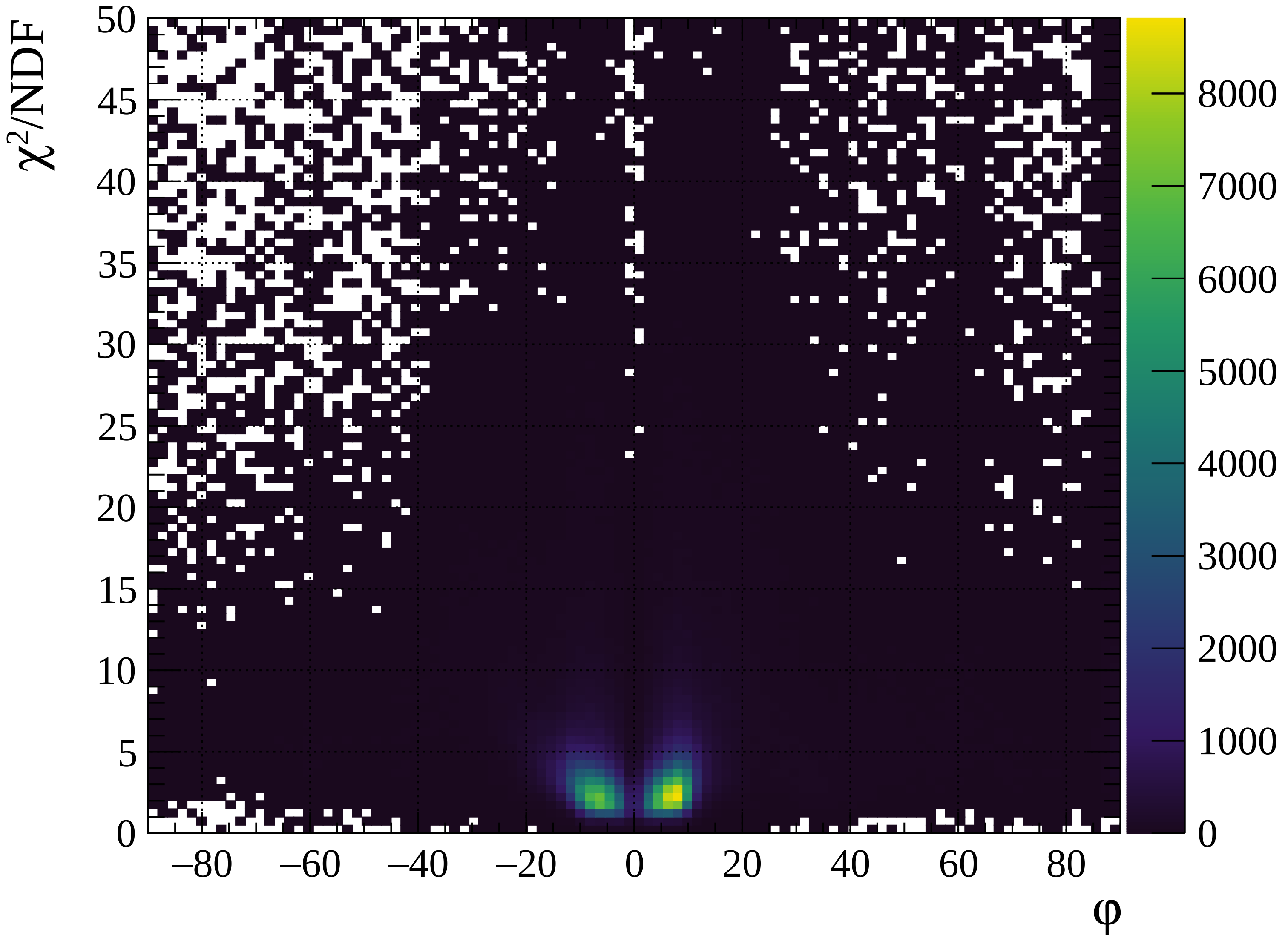


WF
XP
Mean (ADC counts/cm)
φ (°)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80

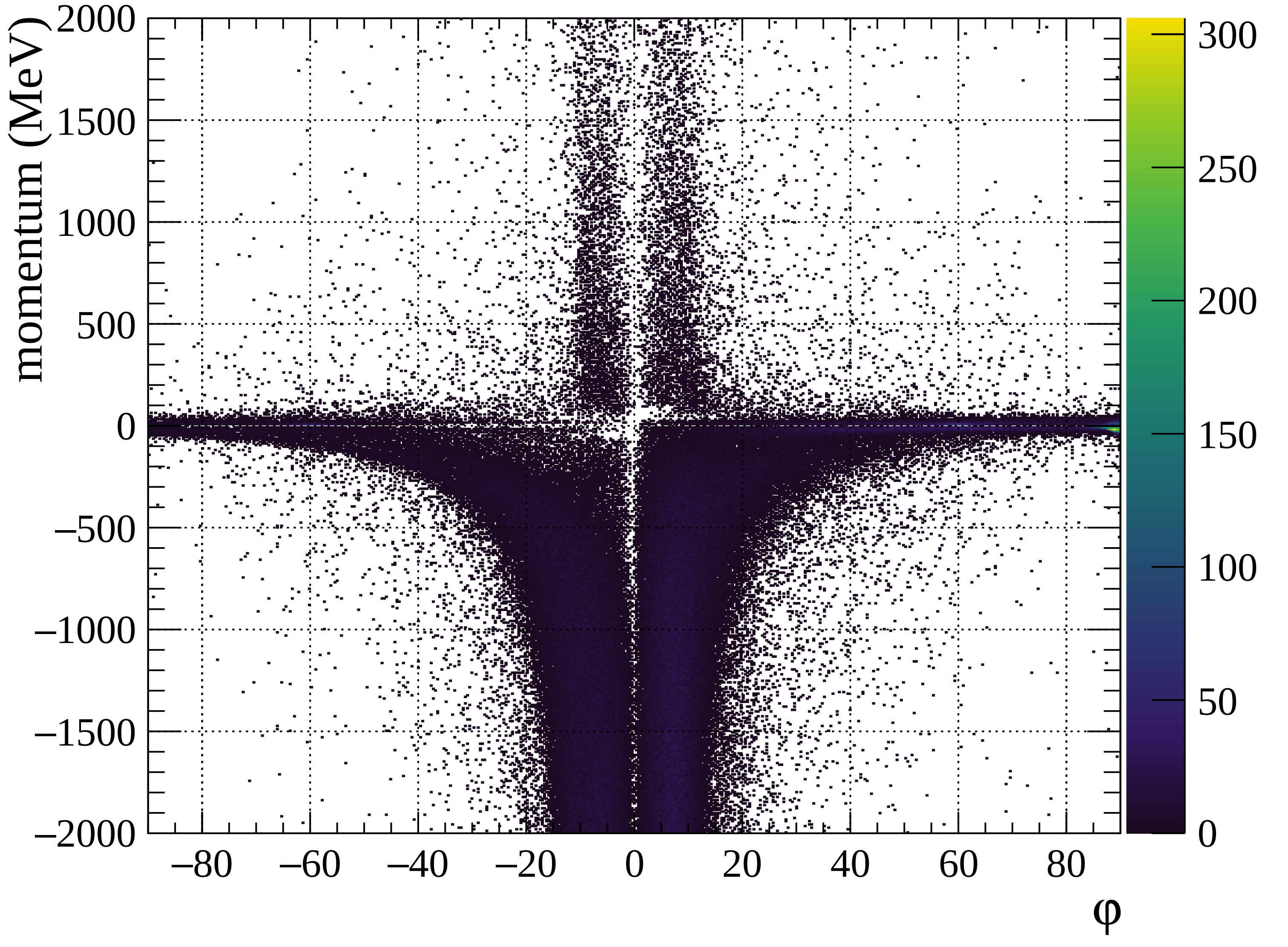

dE/dx with XP (ADC counts/cm)
φ (°)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
0
50
100
150
200
250

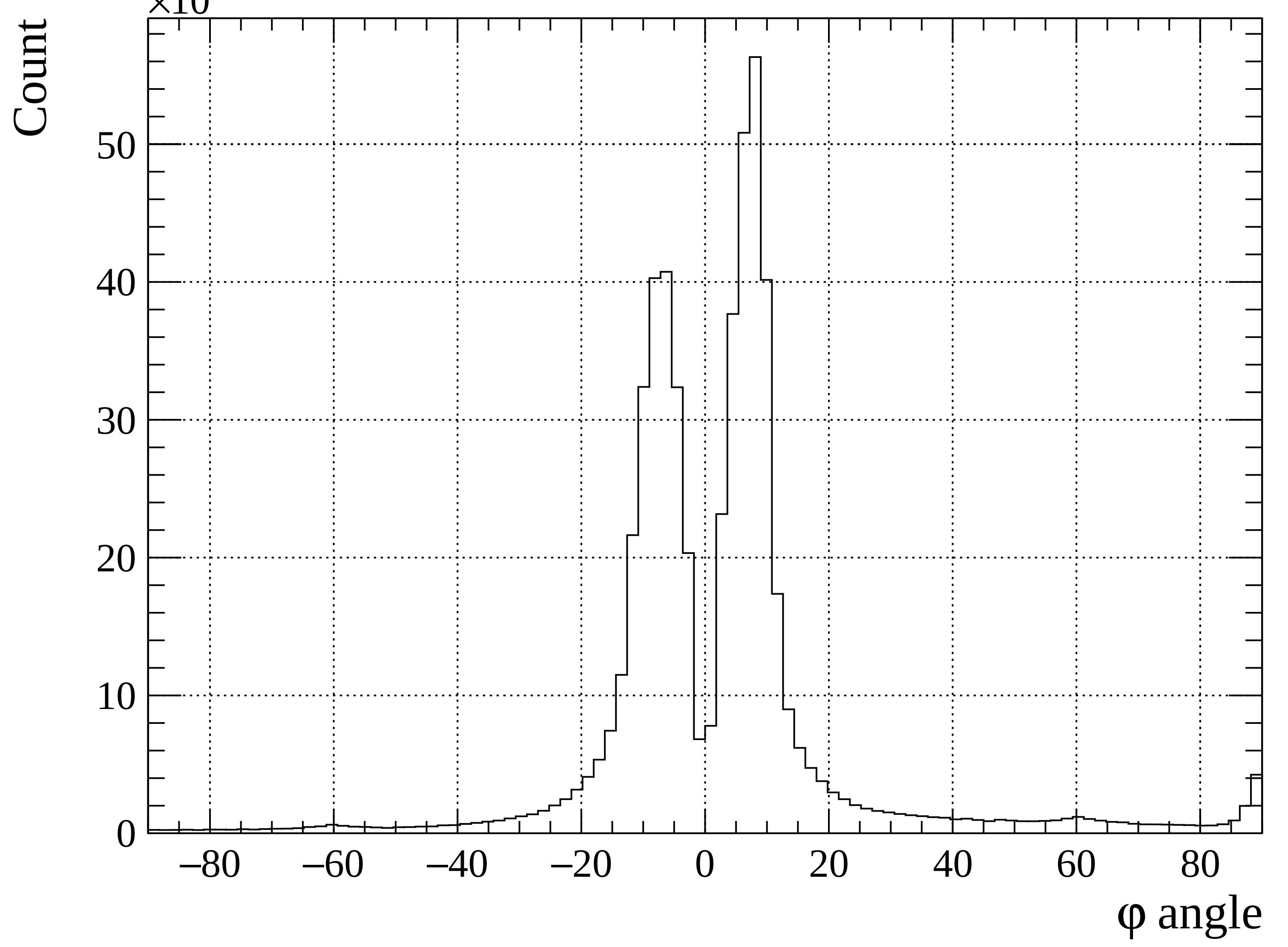


track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
φ
500
400
300
200
100
0

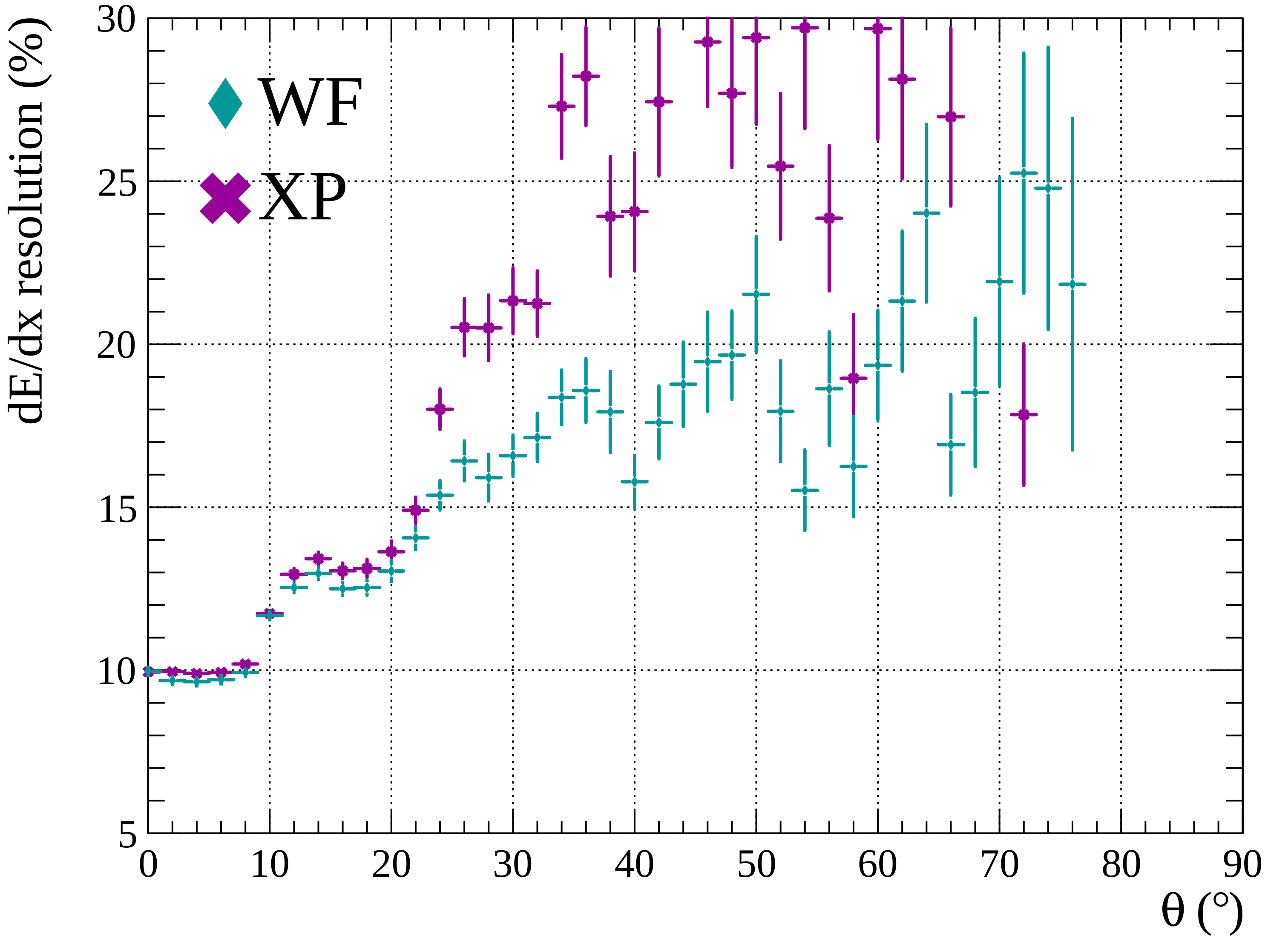


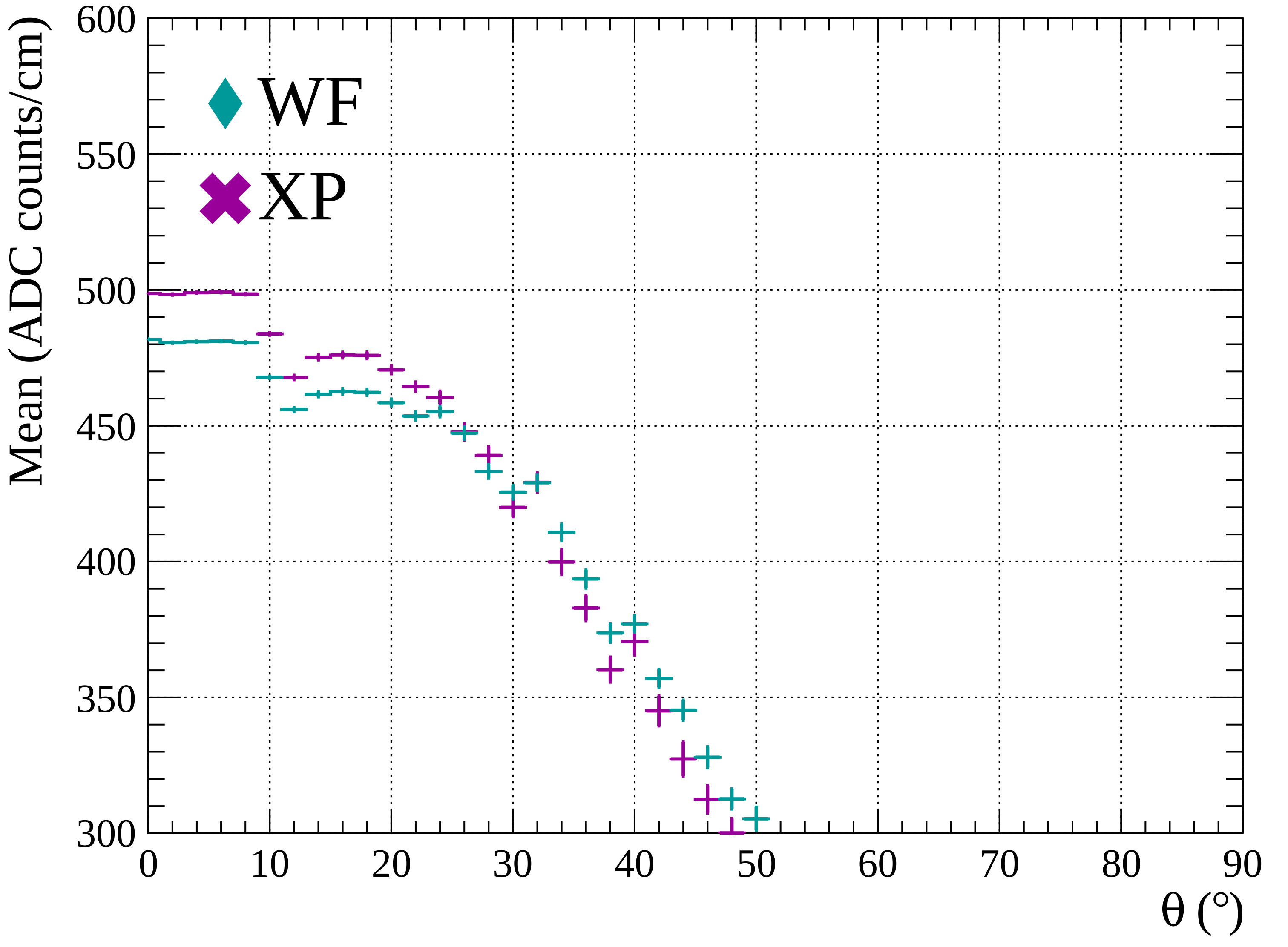
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
8000
7000
6000
5000
4000
3000
2000
1000
0


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
300
250
200
150
100
50
0

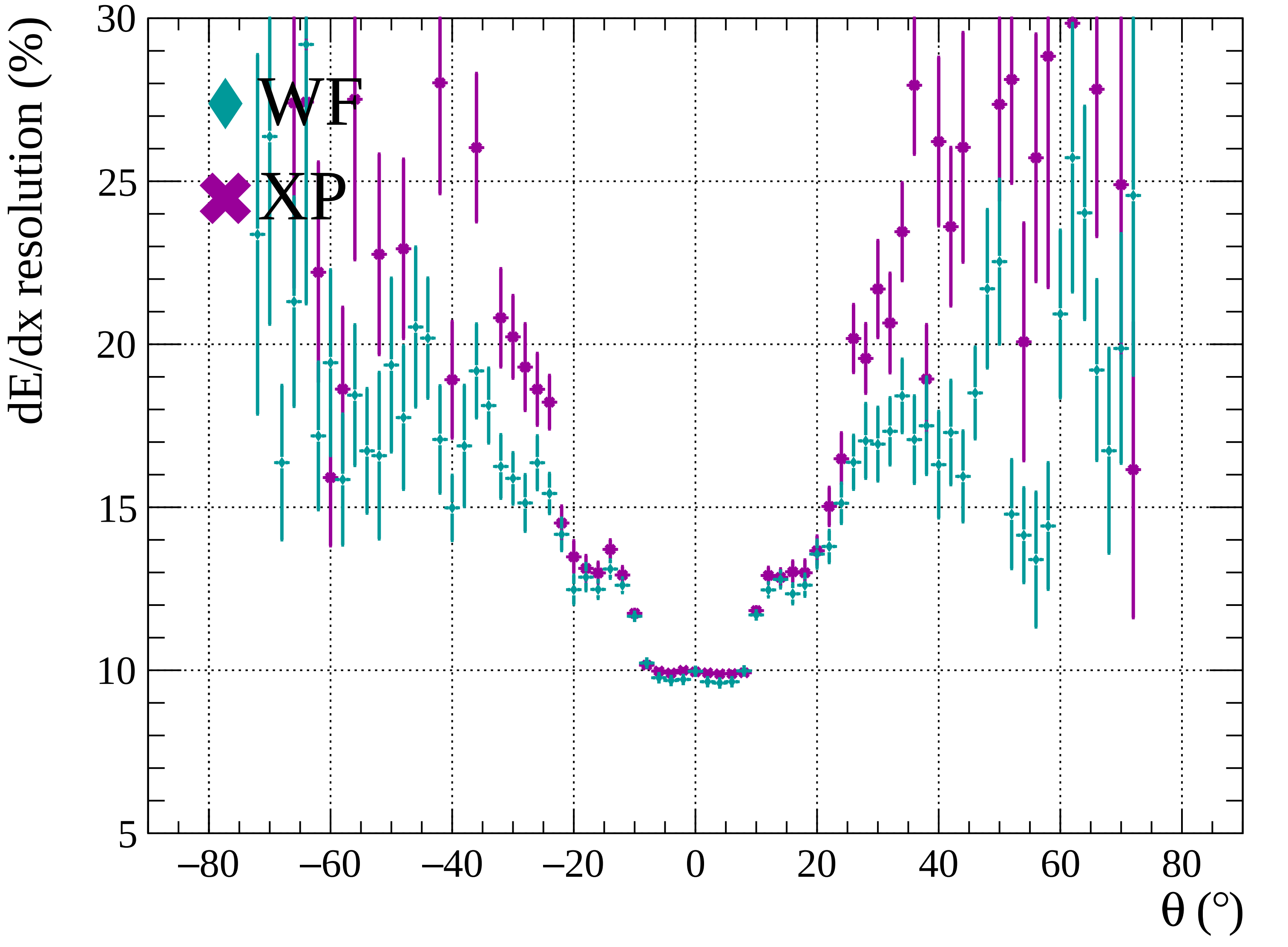


×10
Count
0
10
20
30
40
50
−80
−60
−40
−20
0
20
40
60
80
φ angle

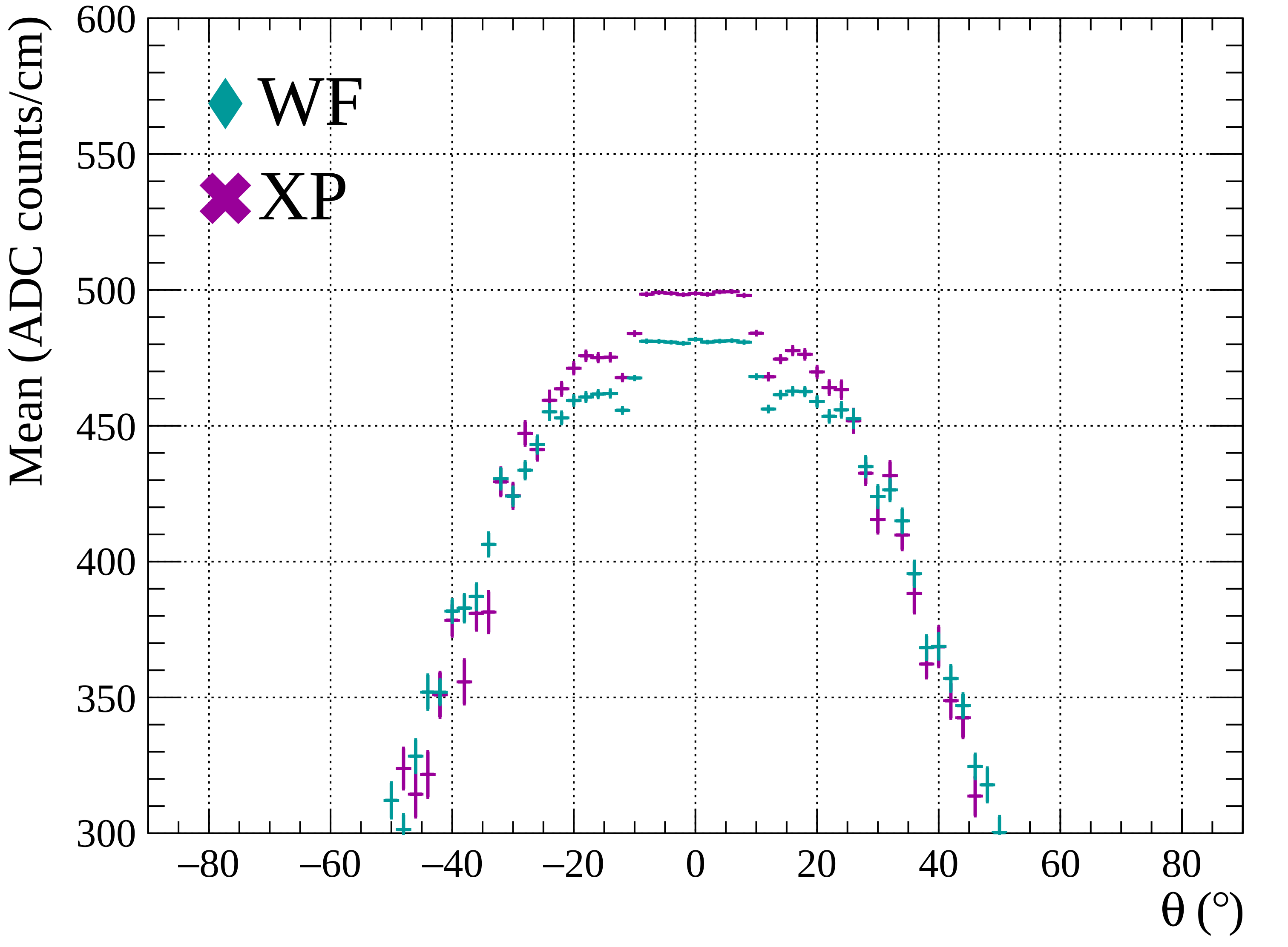

WF
XP
dE/dx resolution (%)
θ (°)
5
10
15
20
25
30
0
10
20
30
40
50
60
70
80
90

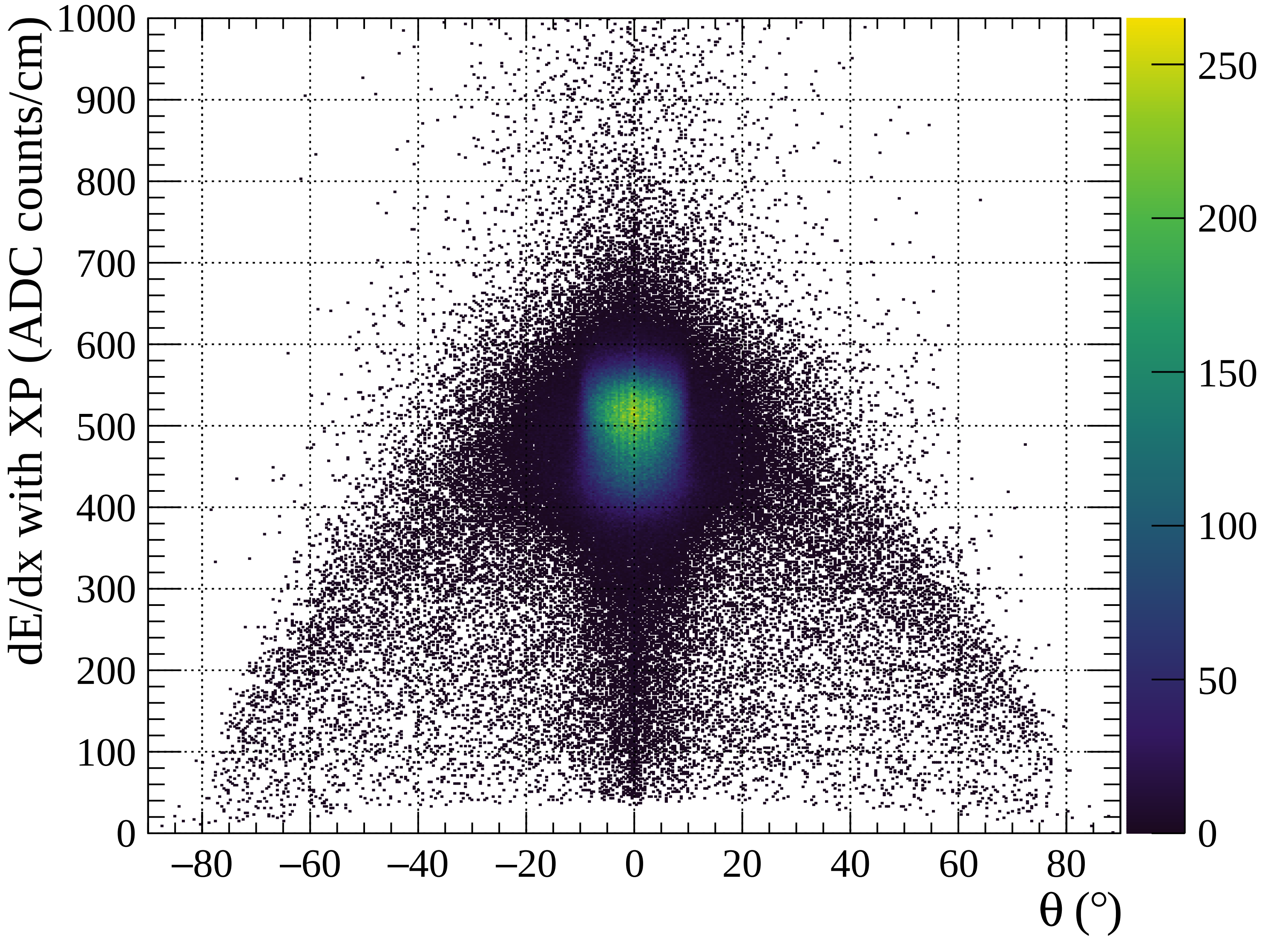


WF
XP
Mean (ADC counts/cm)
θ (°)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90

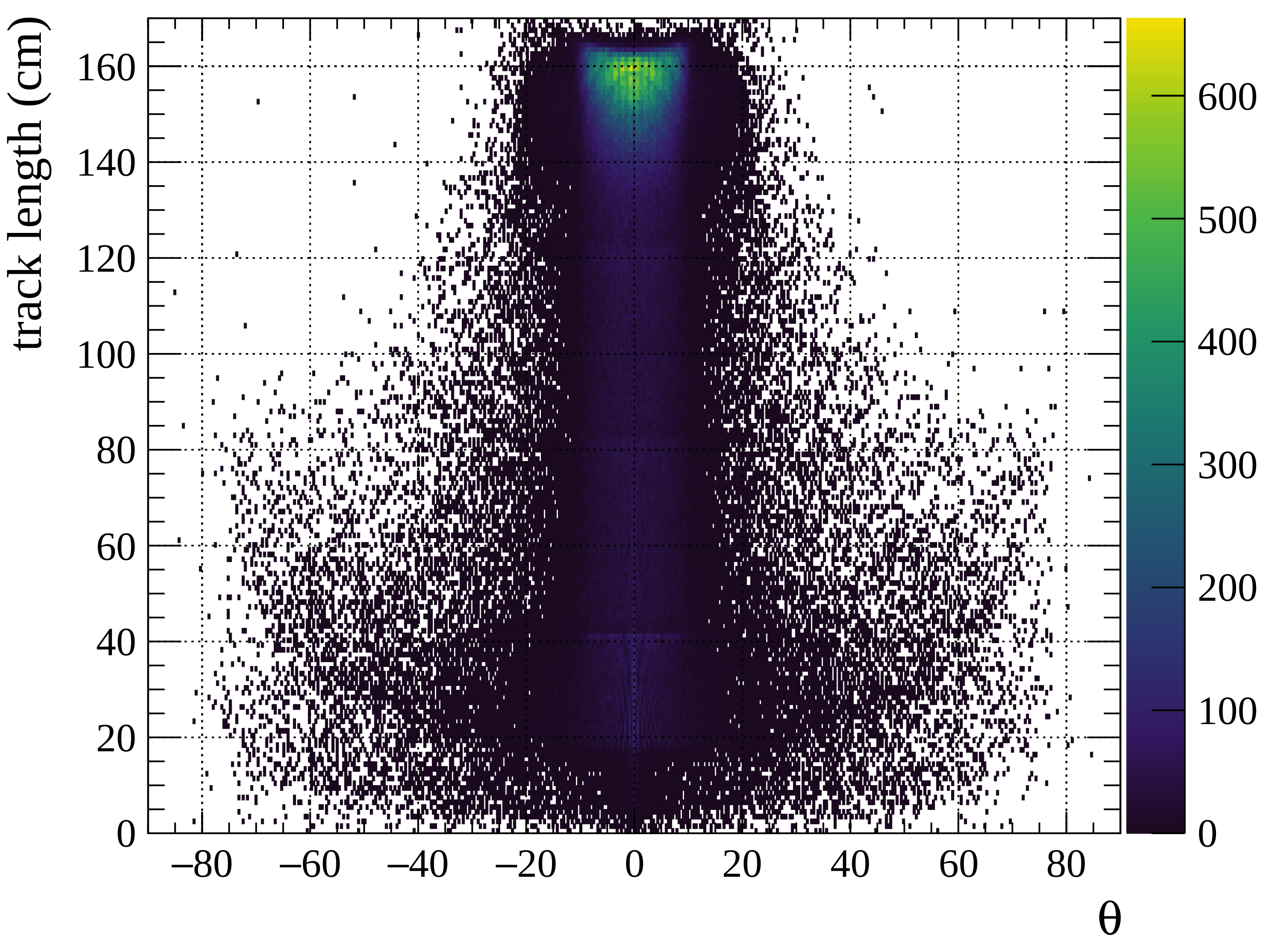


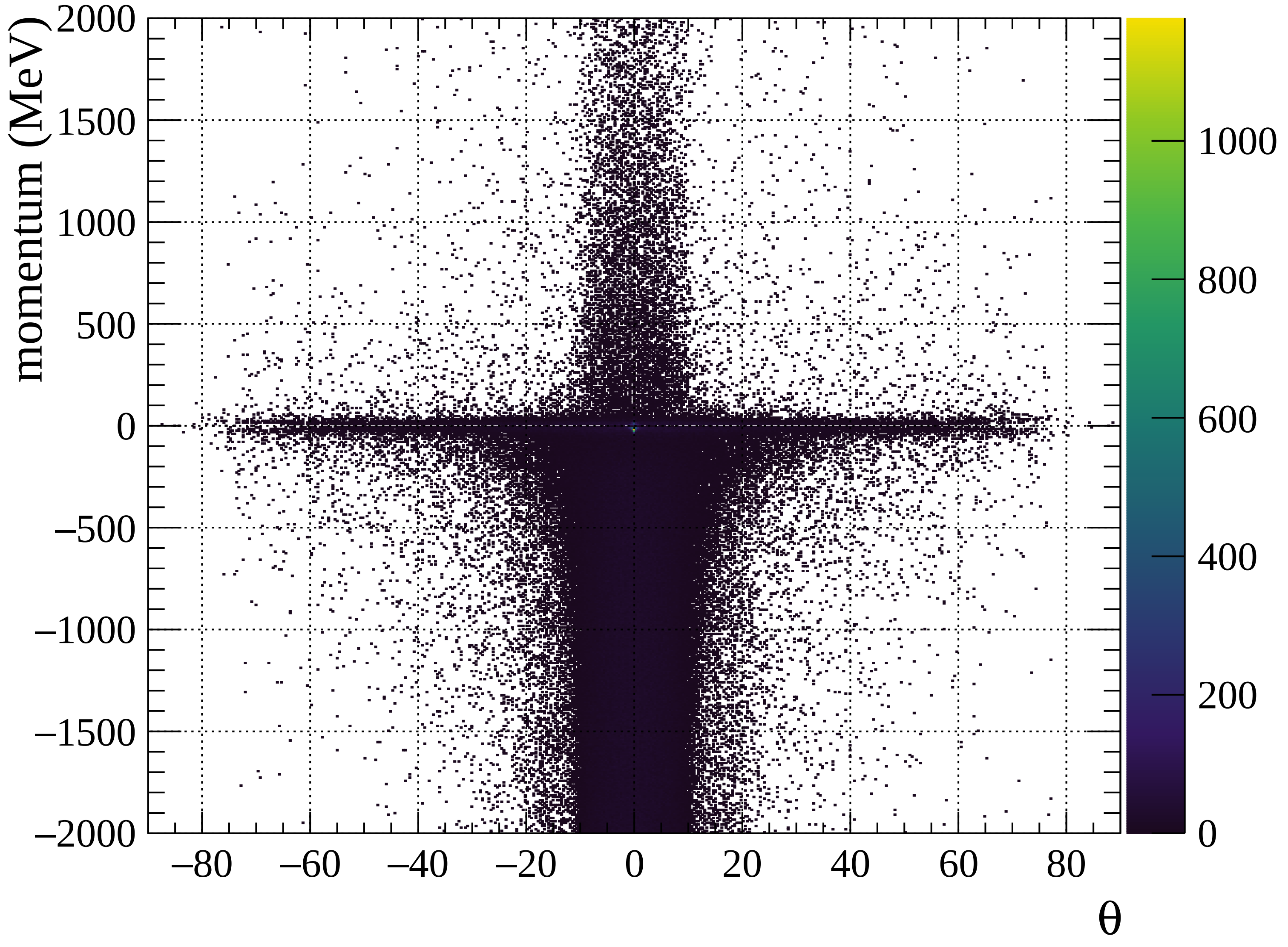
WF
XP
dE/dx resolution (%)
θ (°)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80


WF
XP
Mean (ADC counts/cm)
θ (°)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80

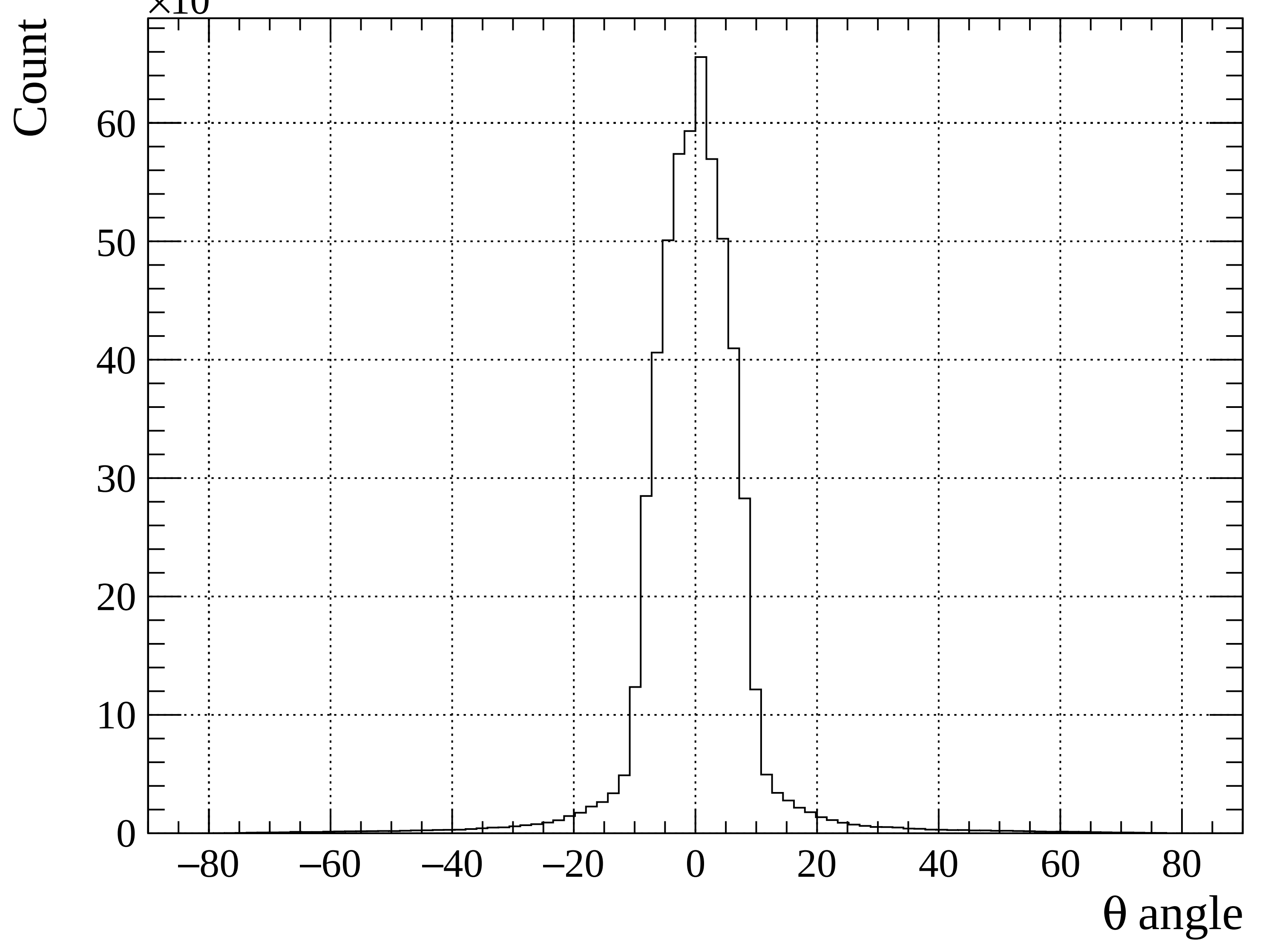


dE/dx with XP (ADC counts/cm)
θ (°)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
0
50
100
150
200
250

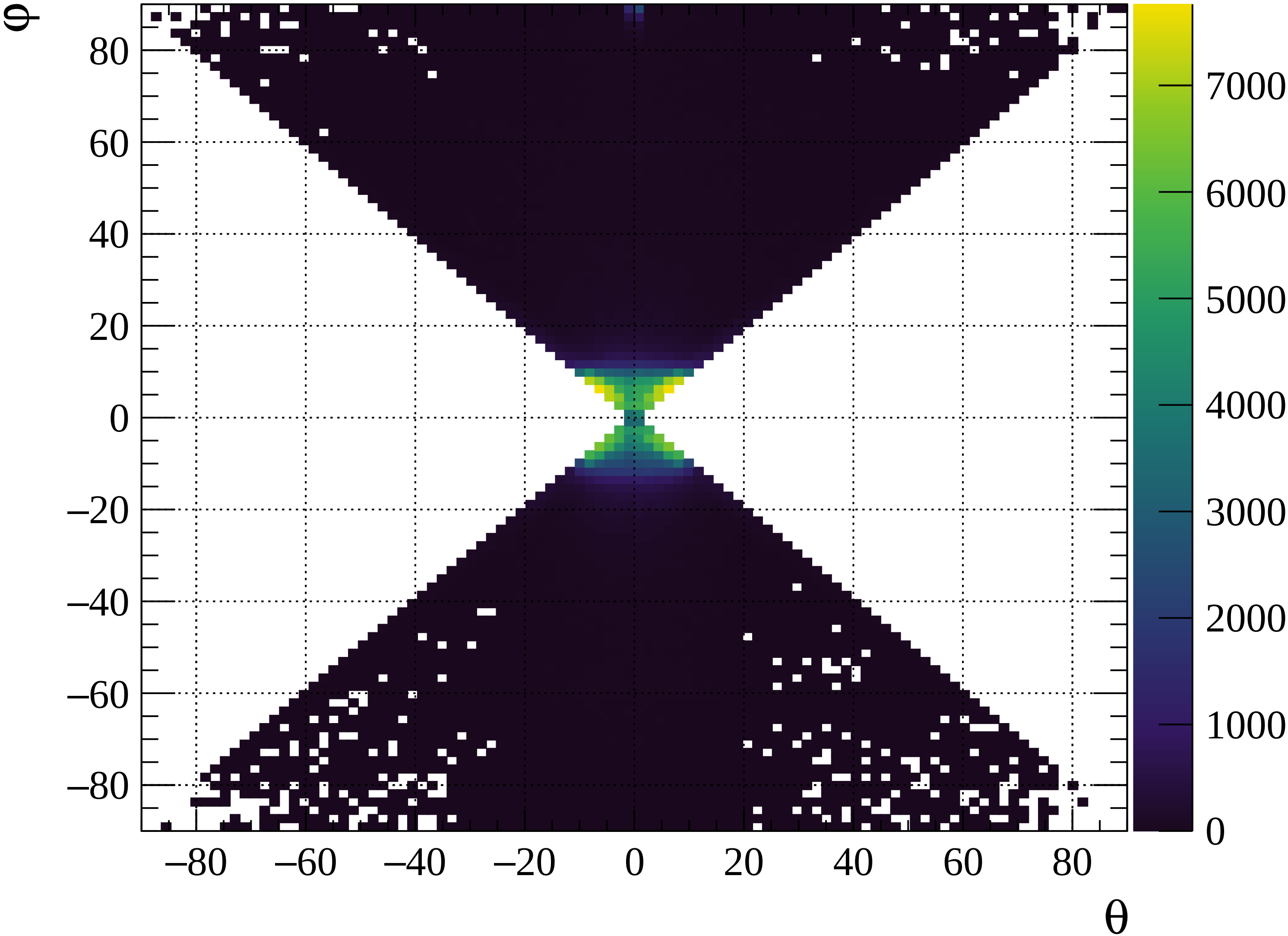

track length (cm)
θ
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
600
500
400
300
200
100
0



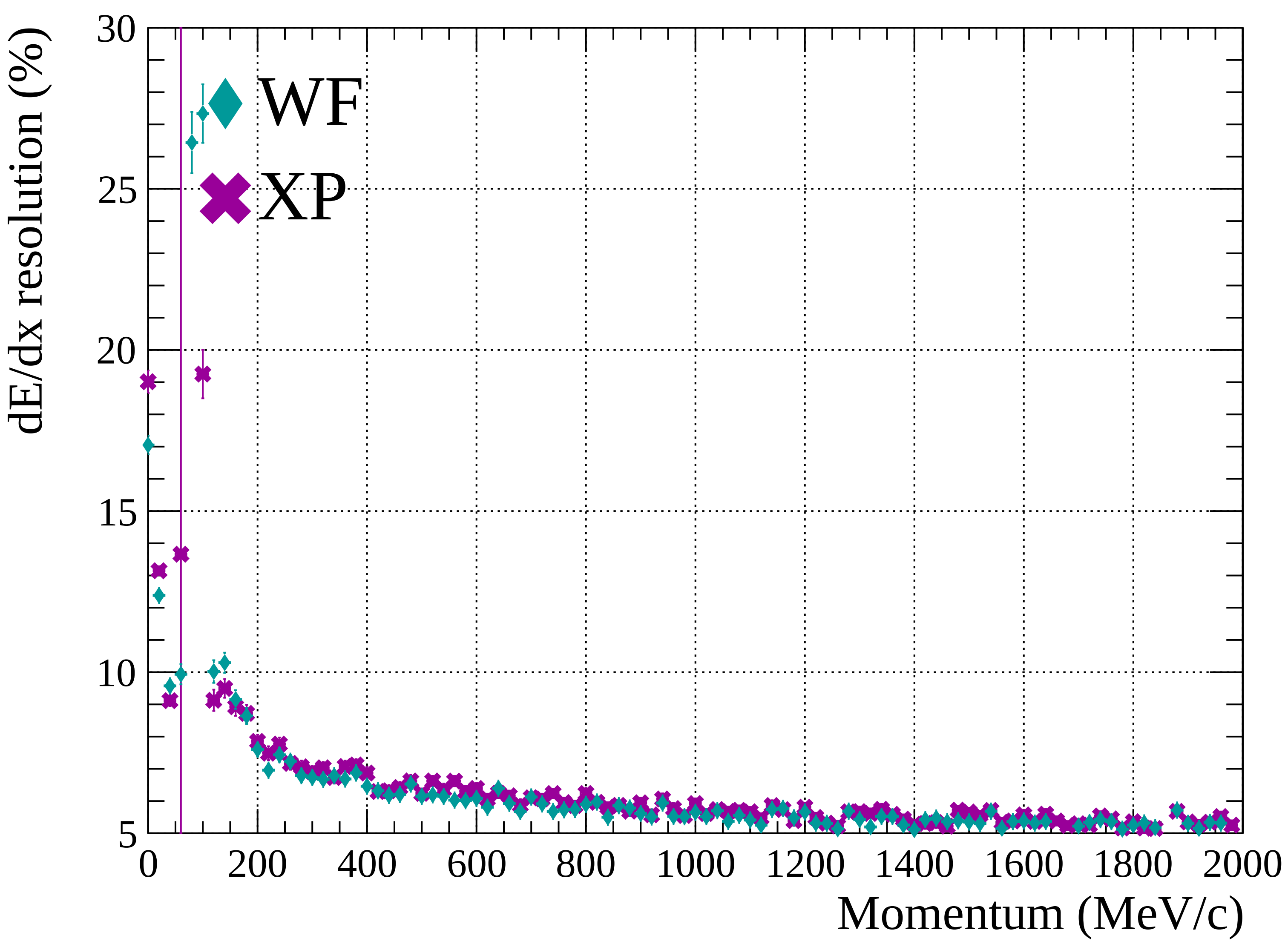
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
1000
800
600
400
200
0



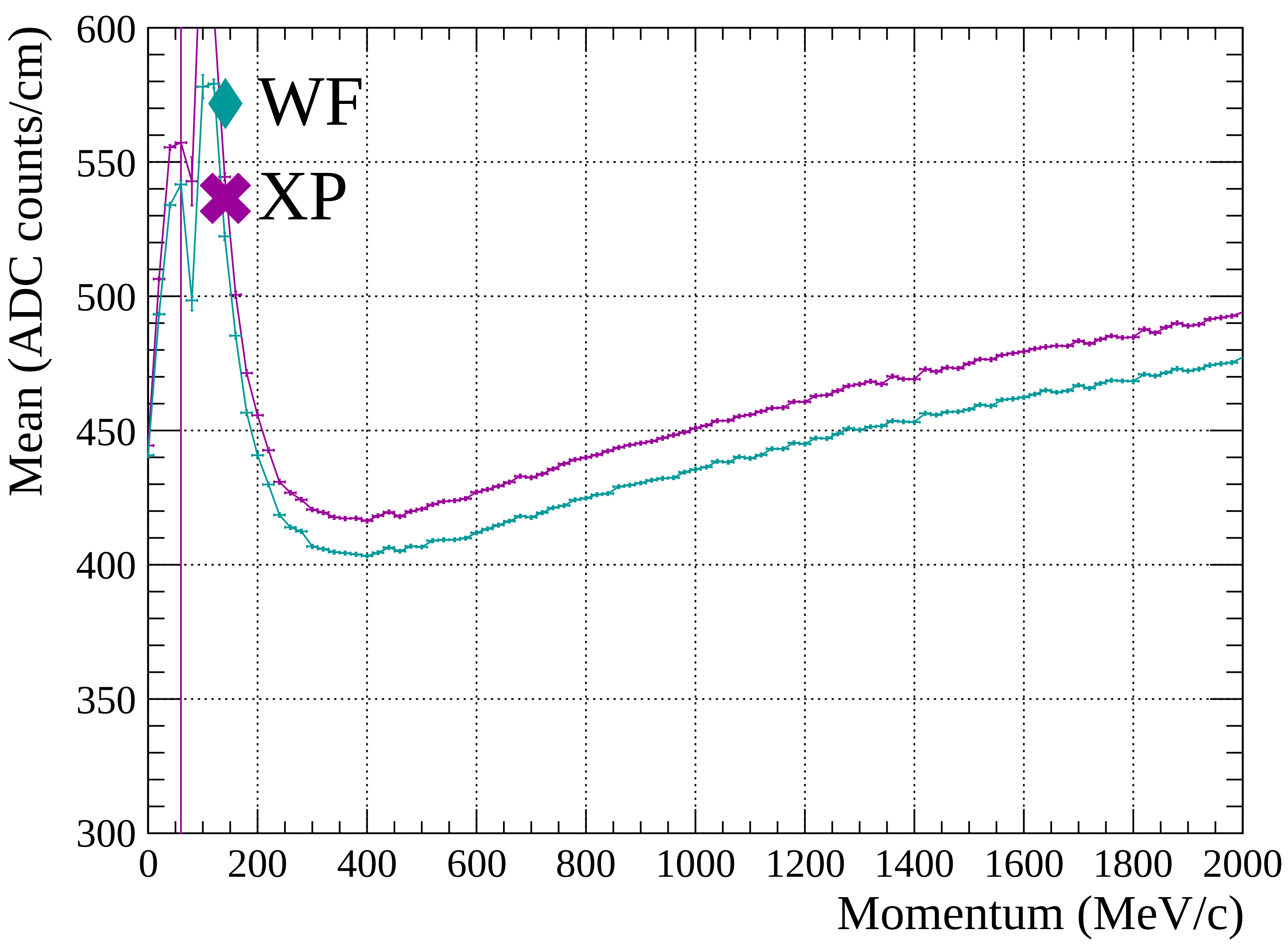
×10
Count
60
50
40
30
20
10
0
−80
−60
−40
−20
0
20
40
60
80
θ angle



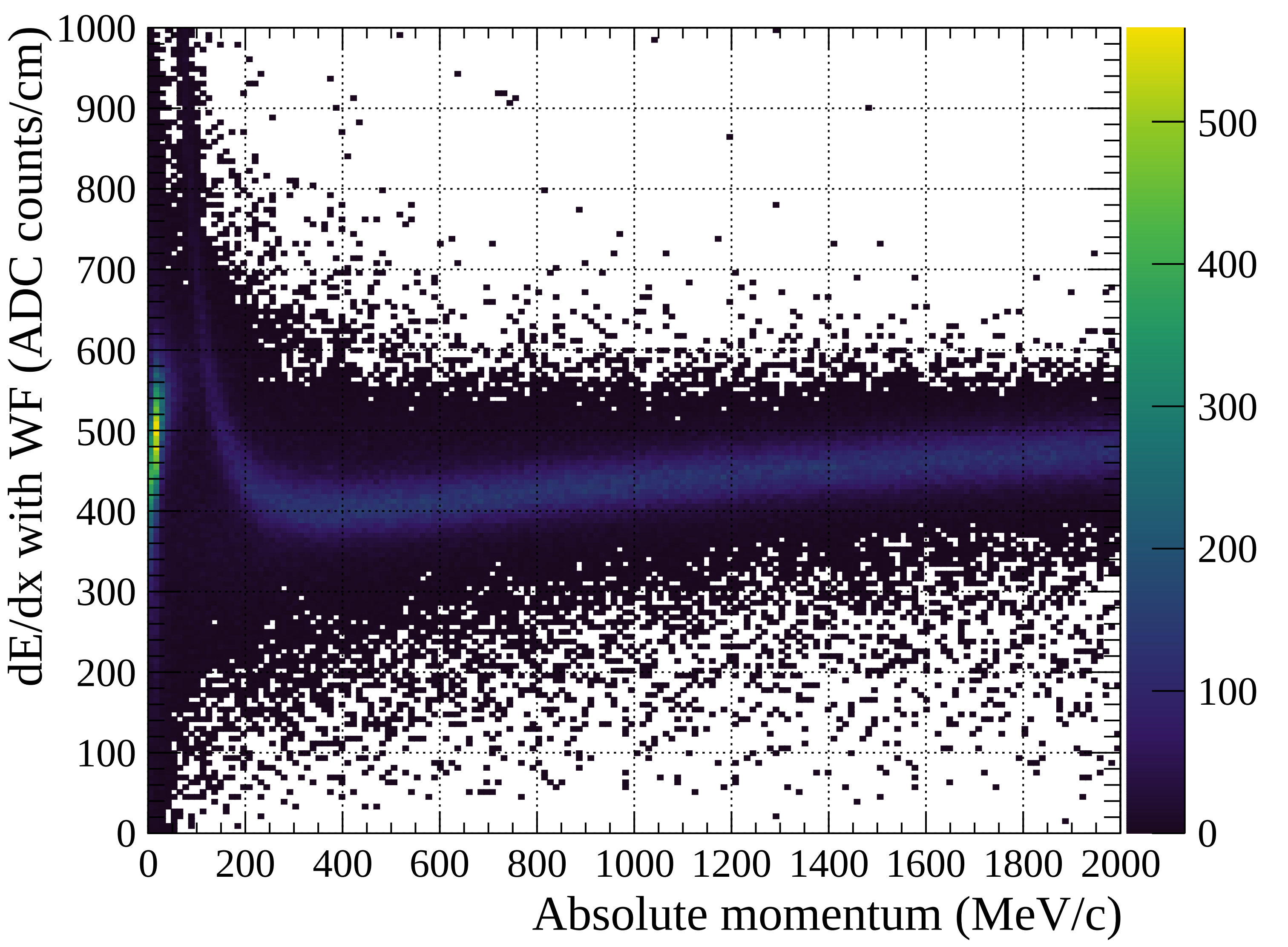
φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
7000
6000
5000
4000
3000
2000
1000
0

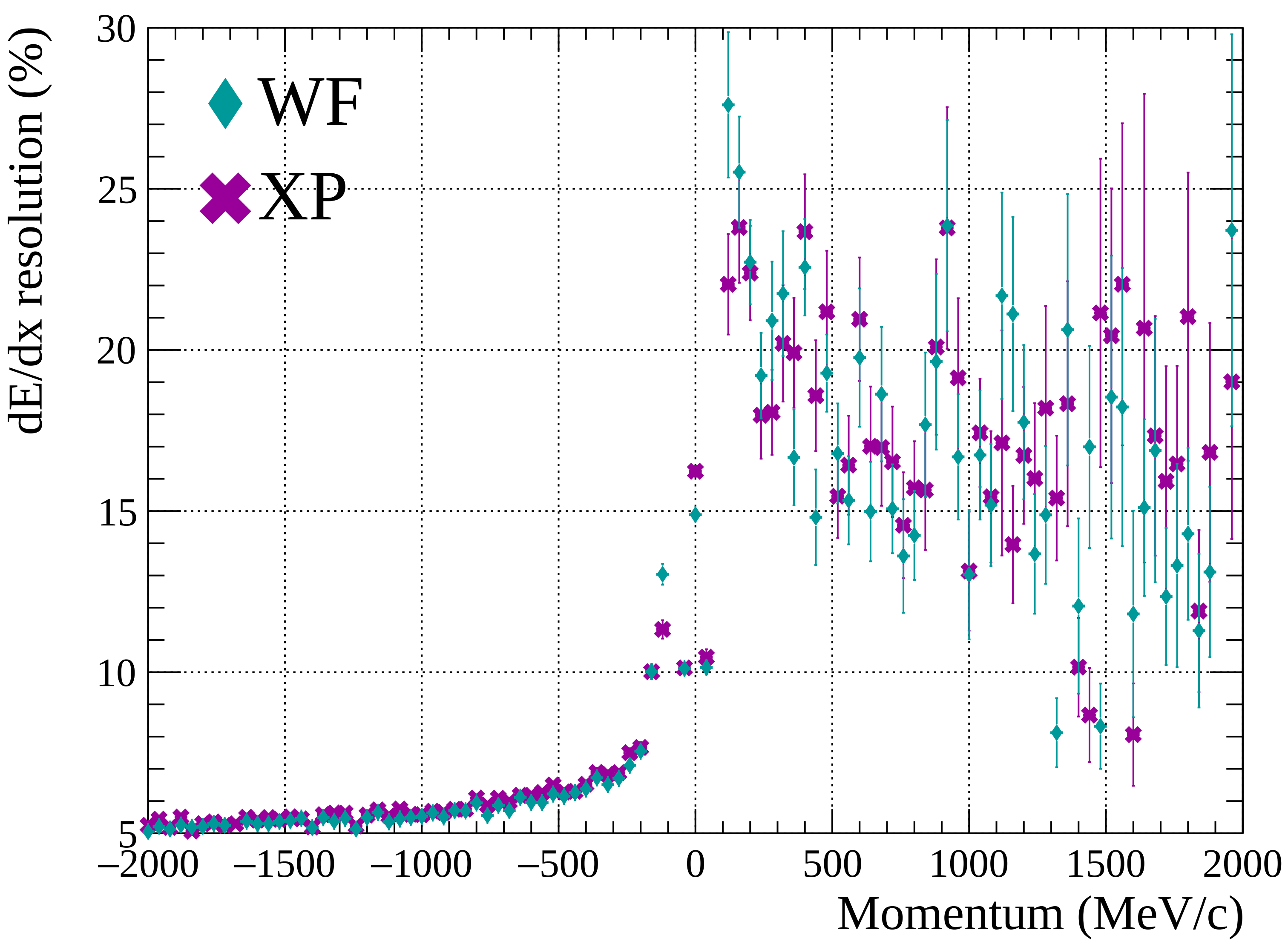


dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Momentum (MeV/c)

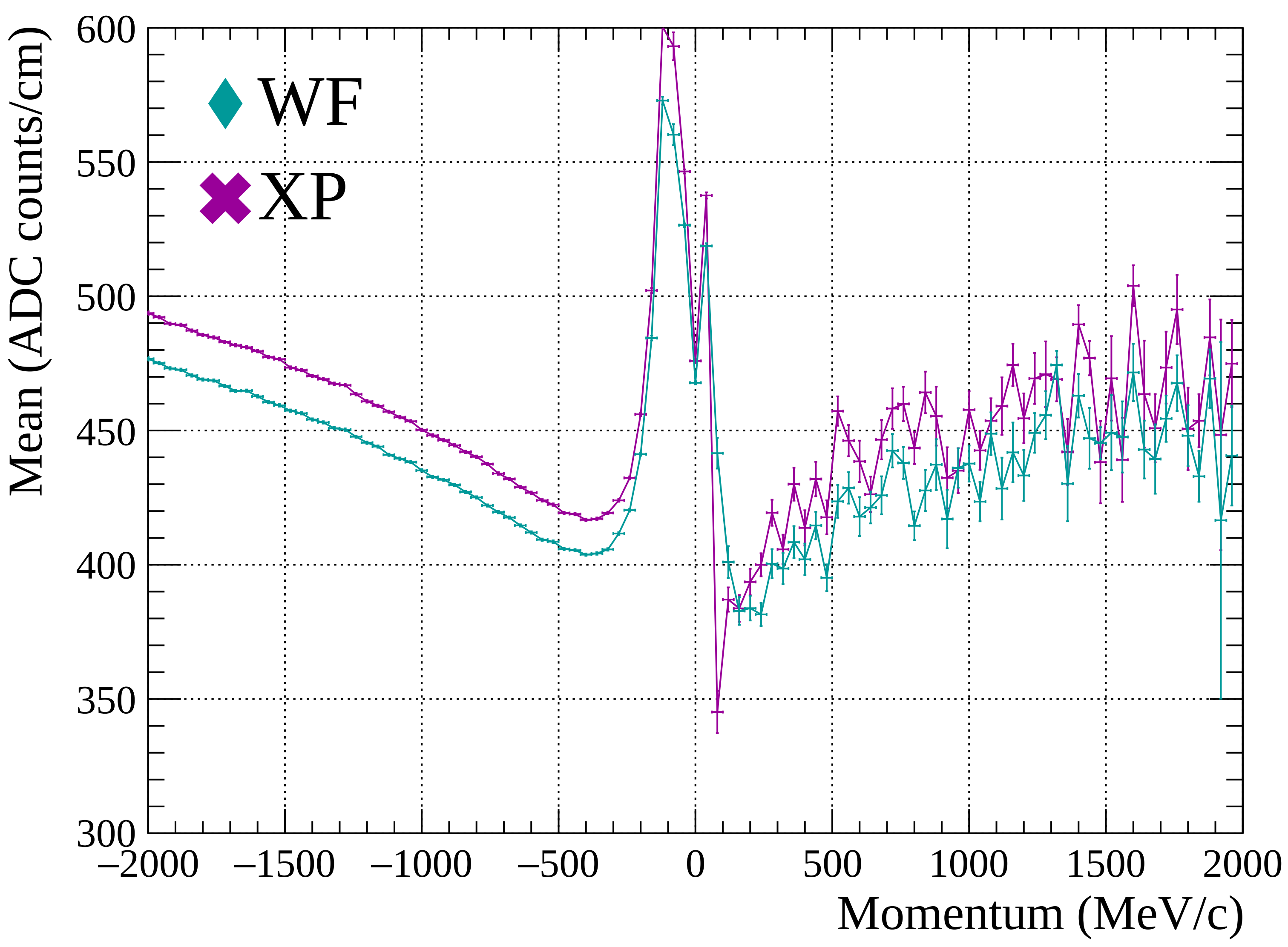


Mean (ADC counts/cm)
Momentum (MeV/c)
WF
XP
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000

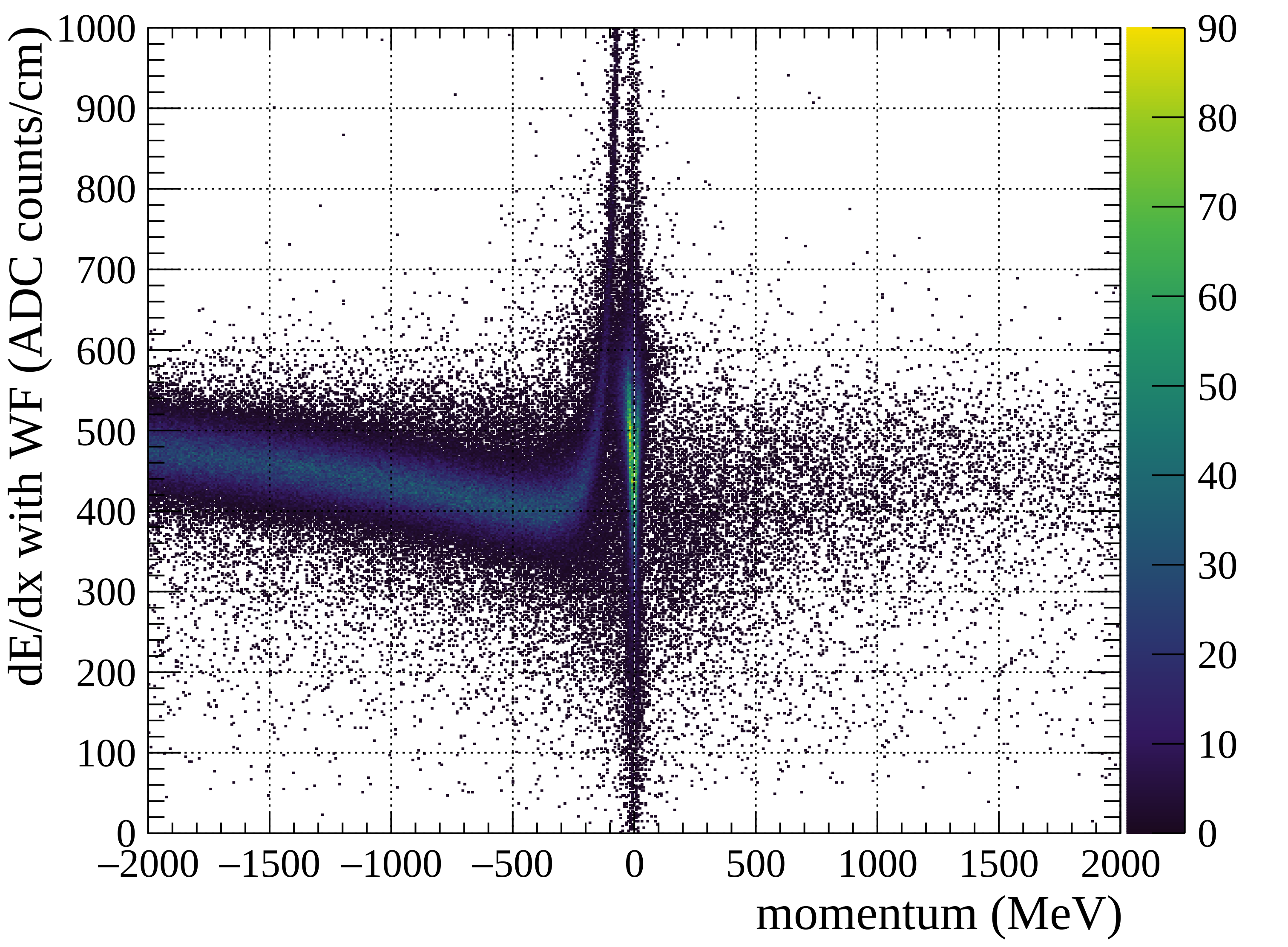

dE/dx with WF (ADC counts/cm)
Absolute momentum (MeV/c)
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
100
200
300
400
500



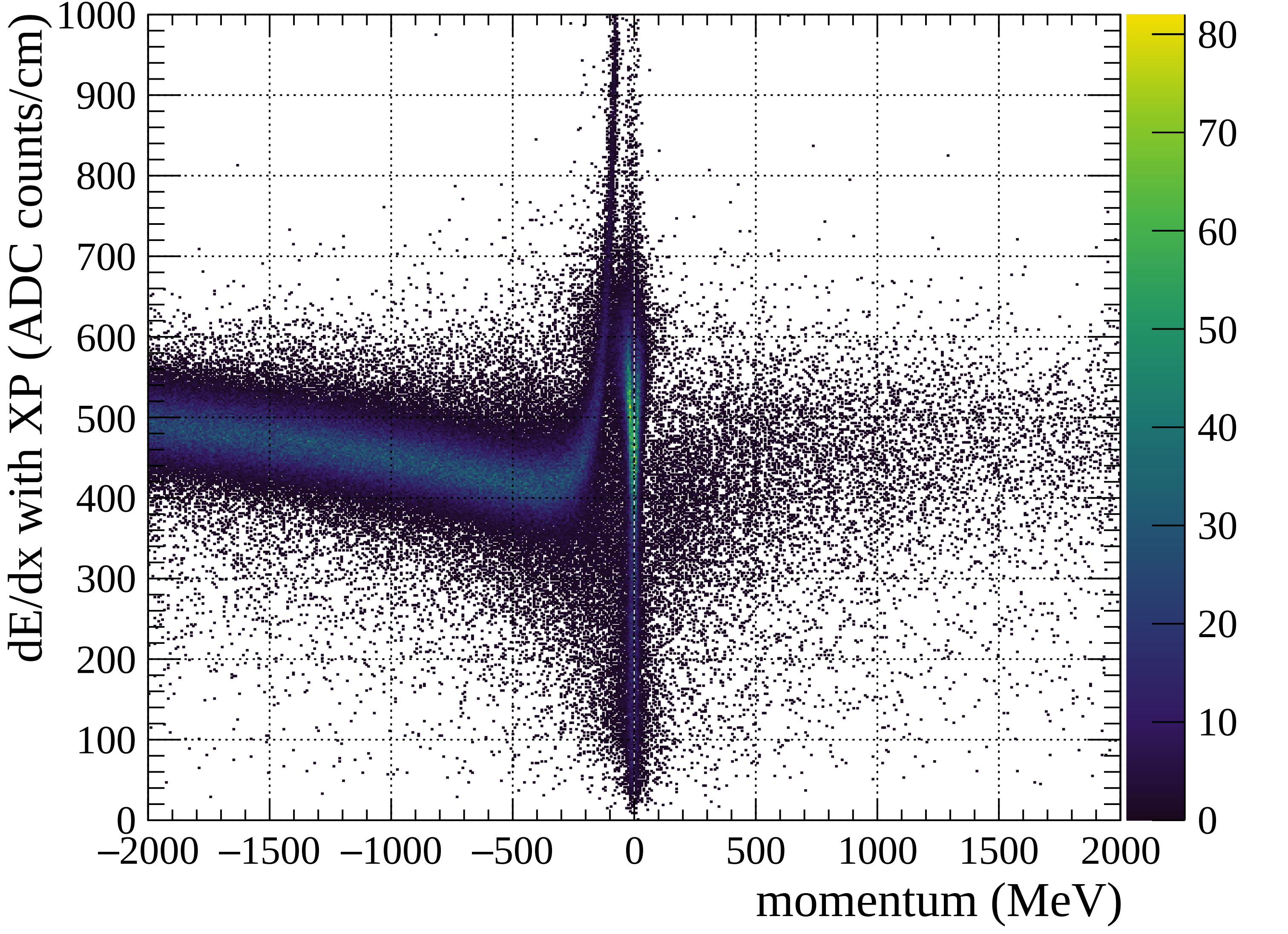
dE/dx resolution (%)
WF
XP
Momentum (MeV/c)
30
25
20
15
10
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000



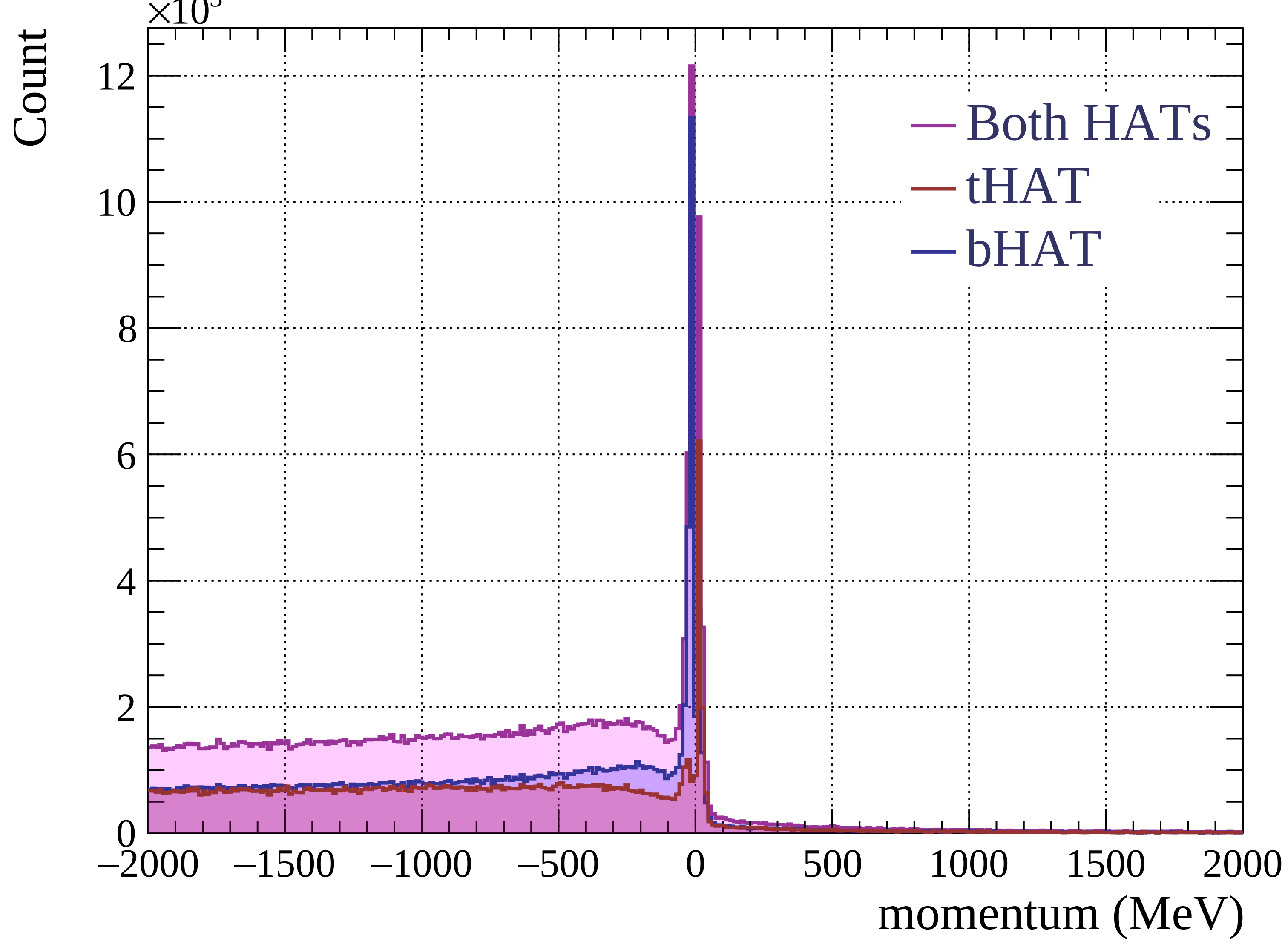
WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000



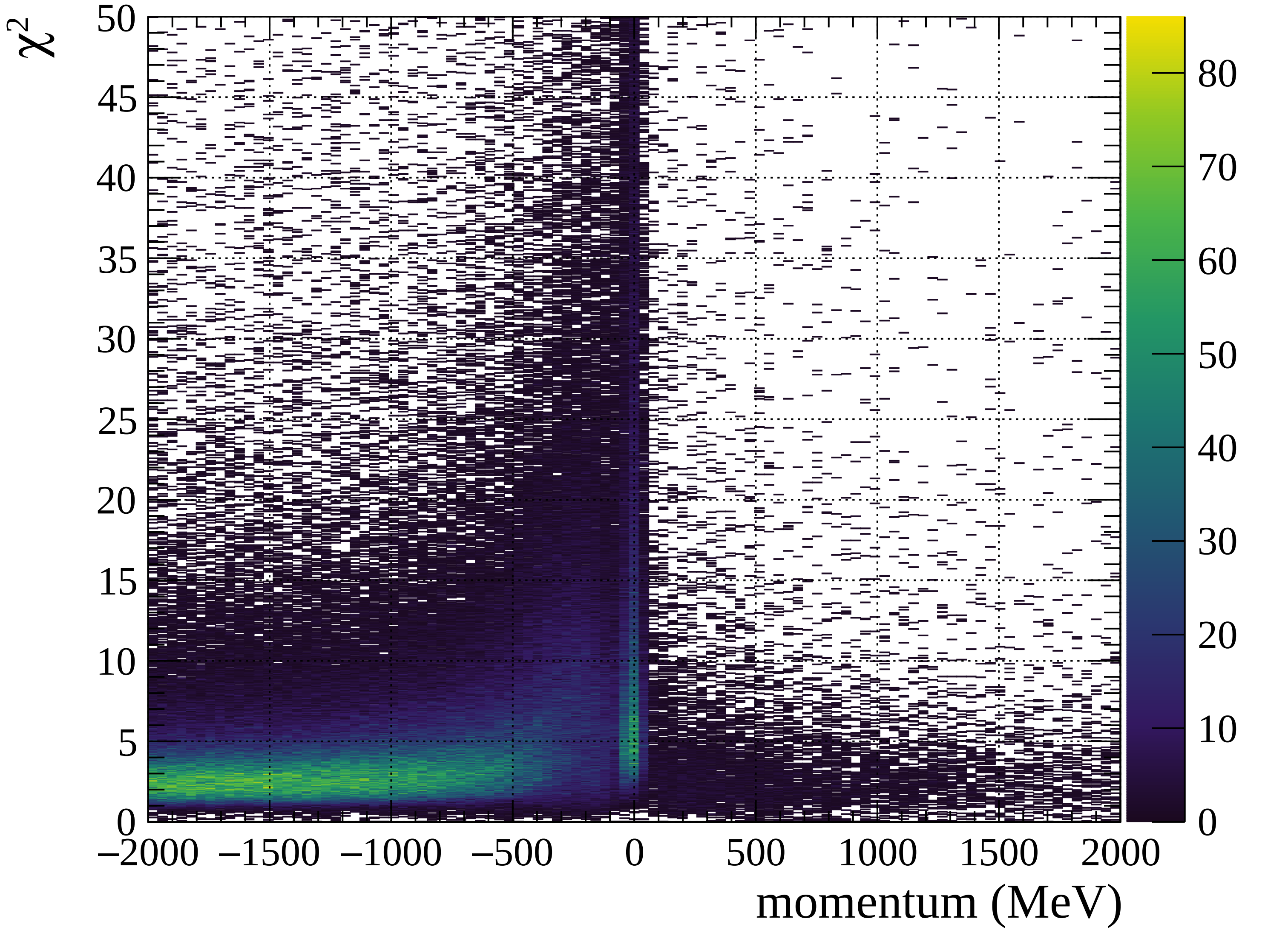
dE/dx with WF (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
10
20
30
40
50
60
70
80
90



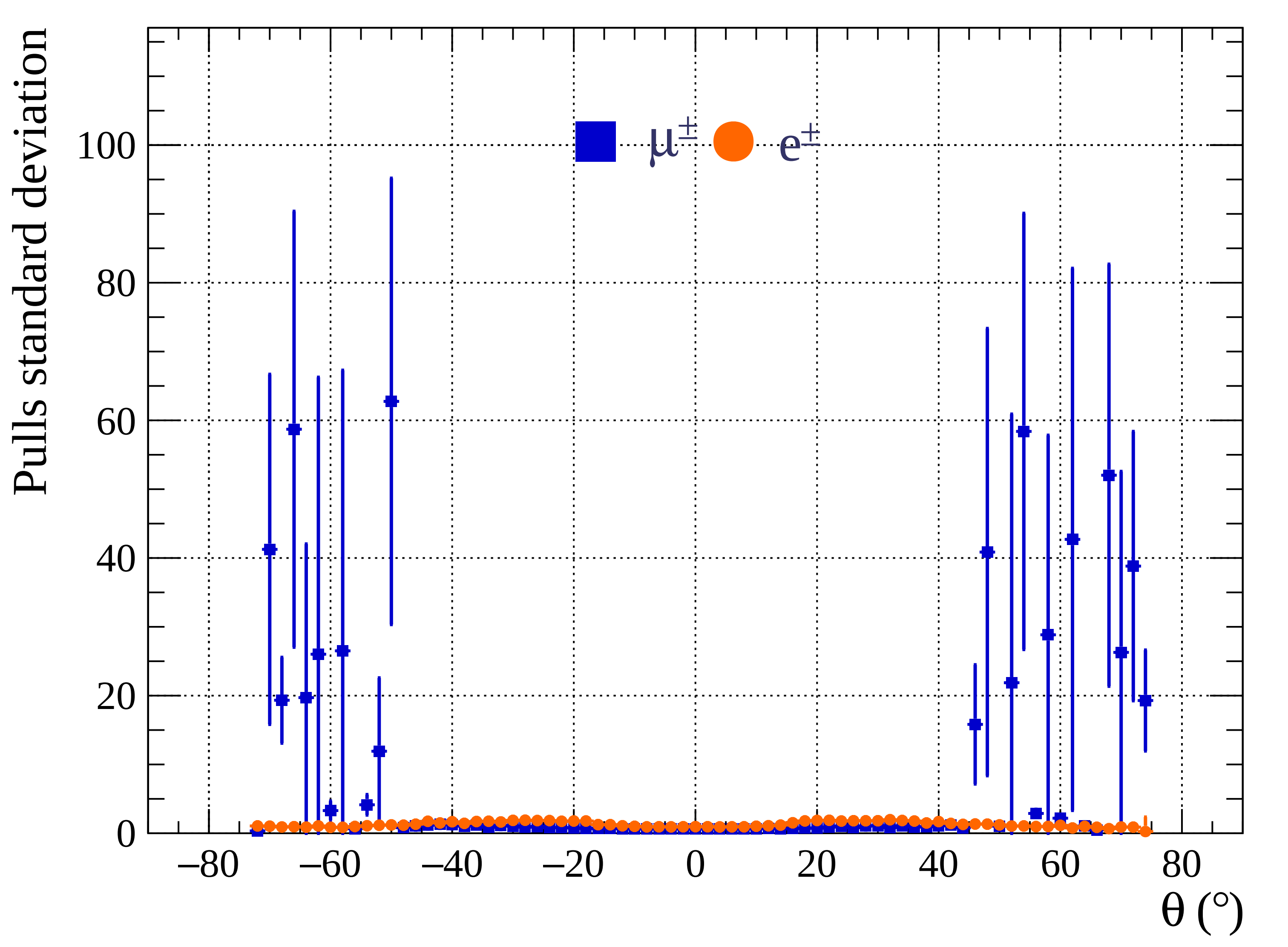
dE/dx with XP (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
10
20
30
40
50
60
70
80



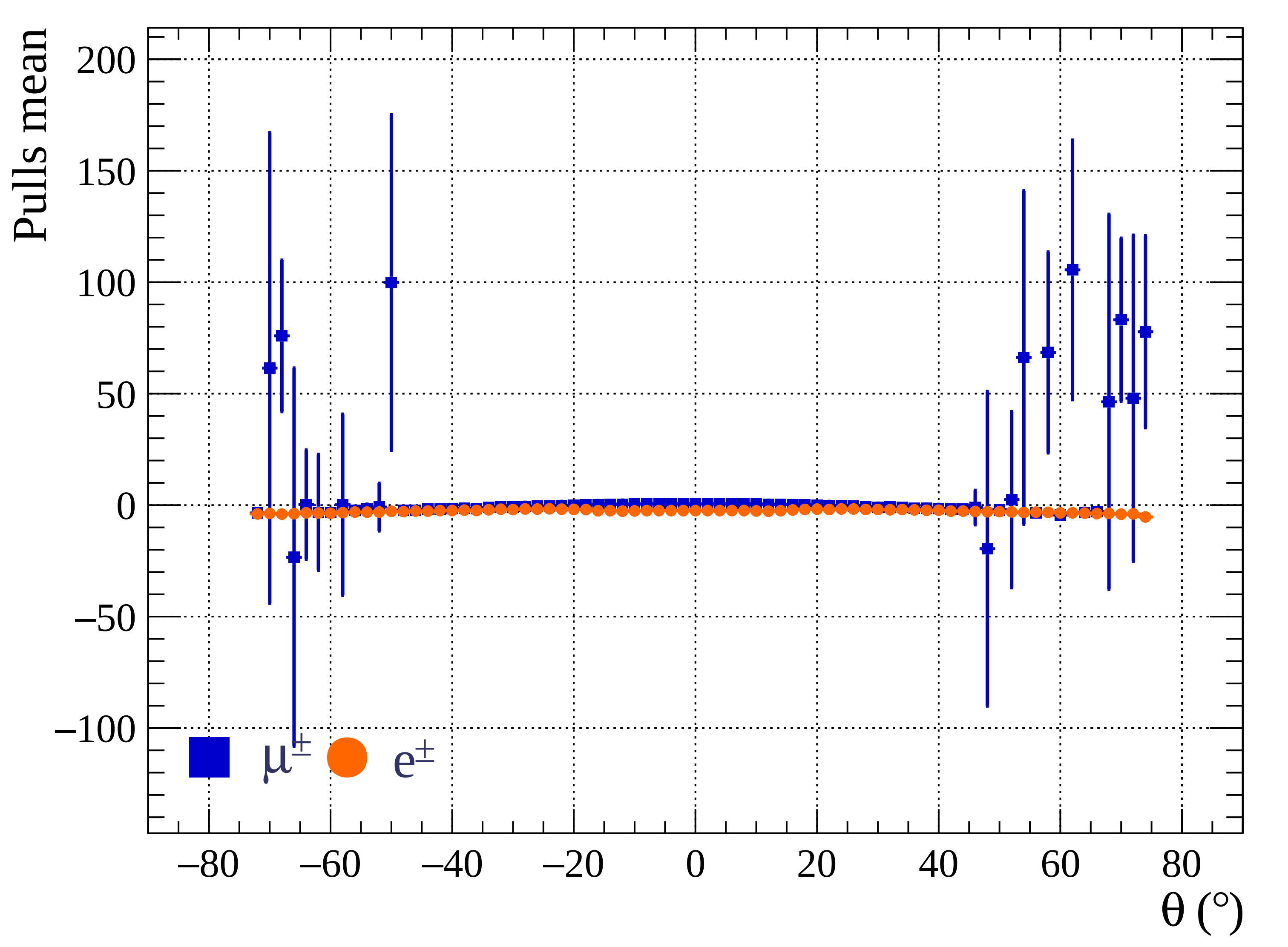
×10^3
Count
12
10
8
6
4
2
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT

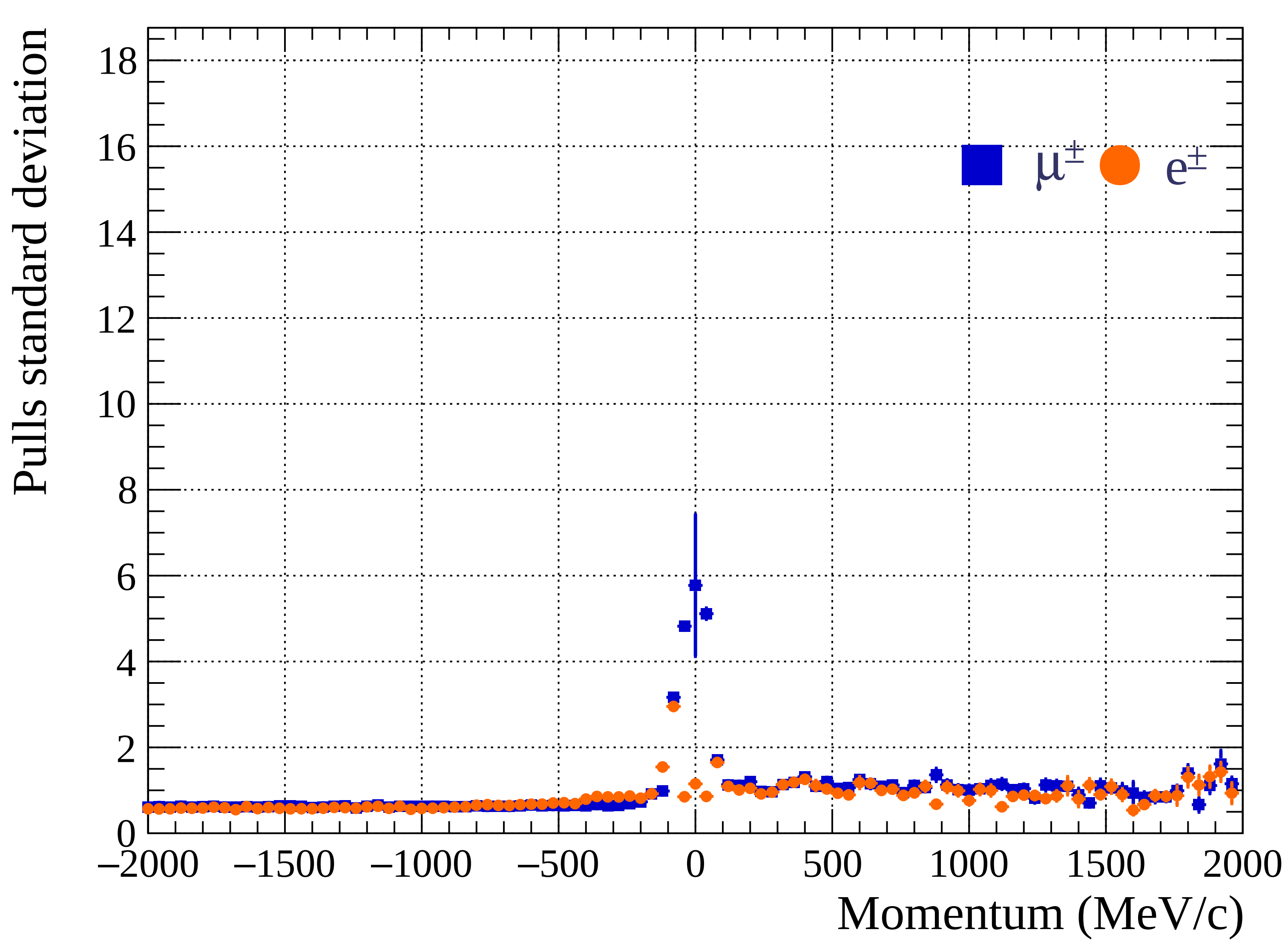


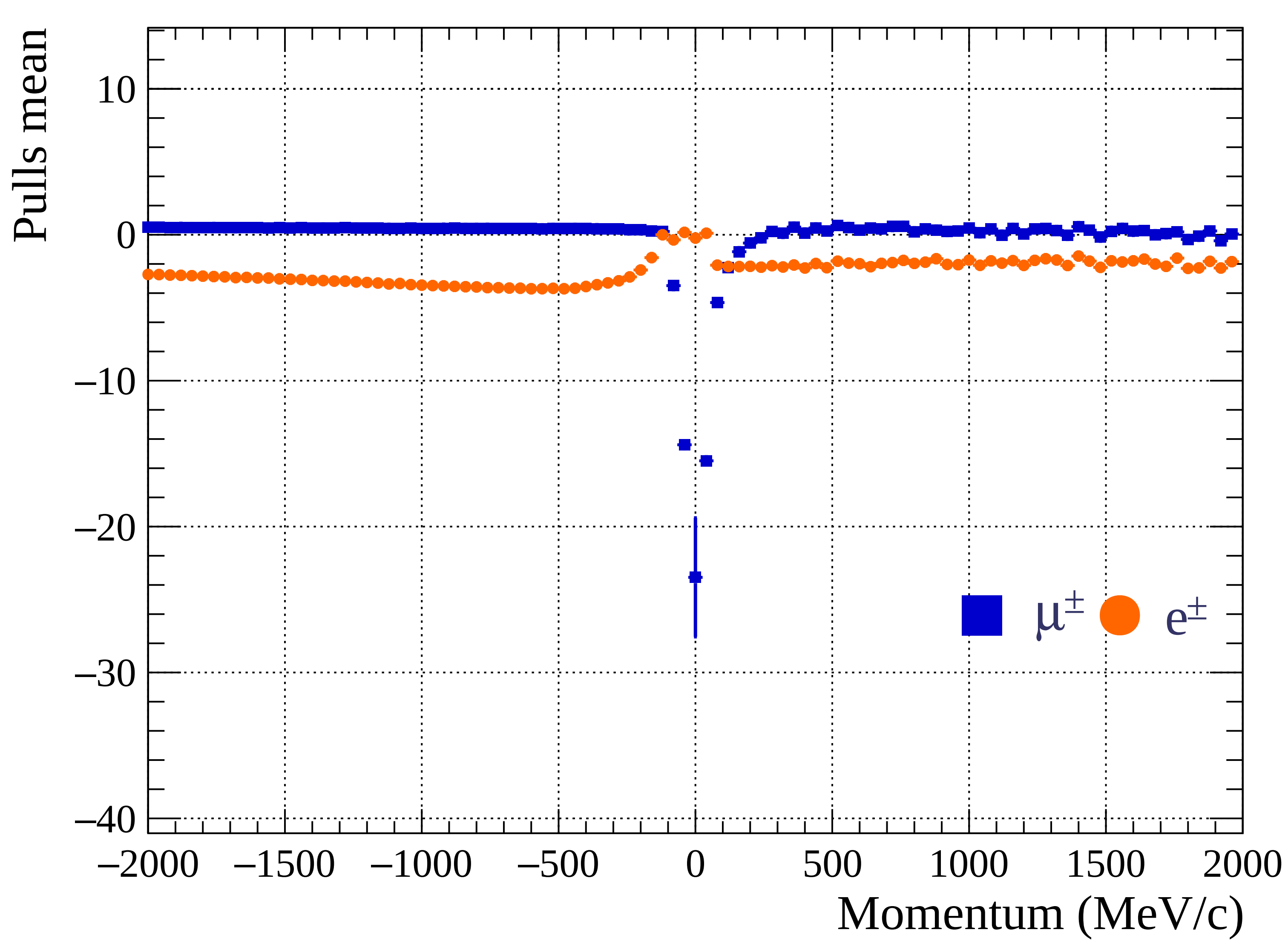
χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
80
70
60
50
40
30
20
10
0



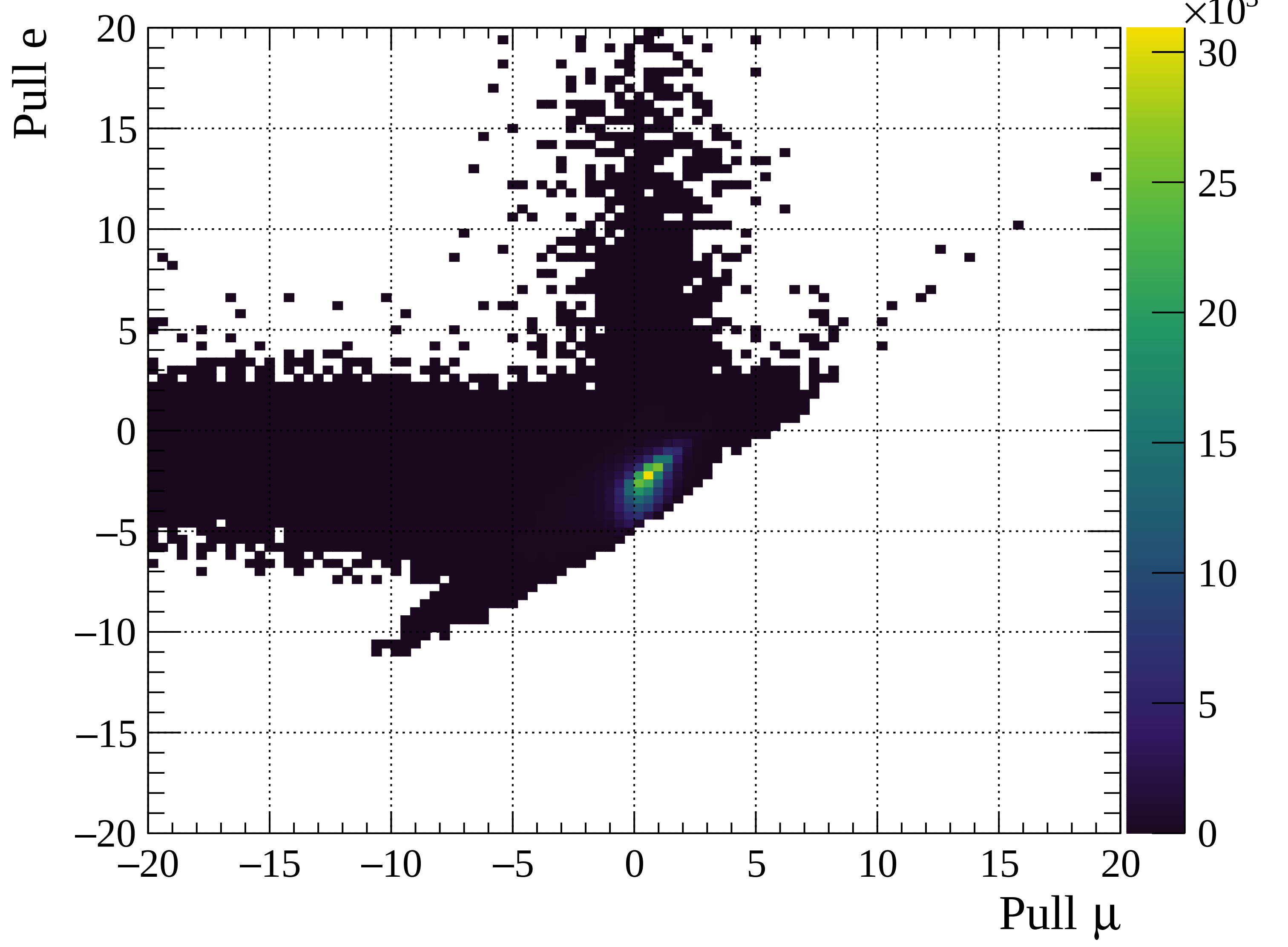
Pulls standard deviation
θ (°)
μ±
e±
0
20
40
60
80
100
−80
−60
−40
−20
0
20
40
60
80

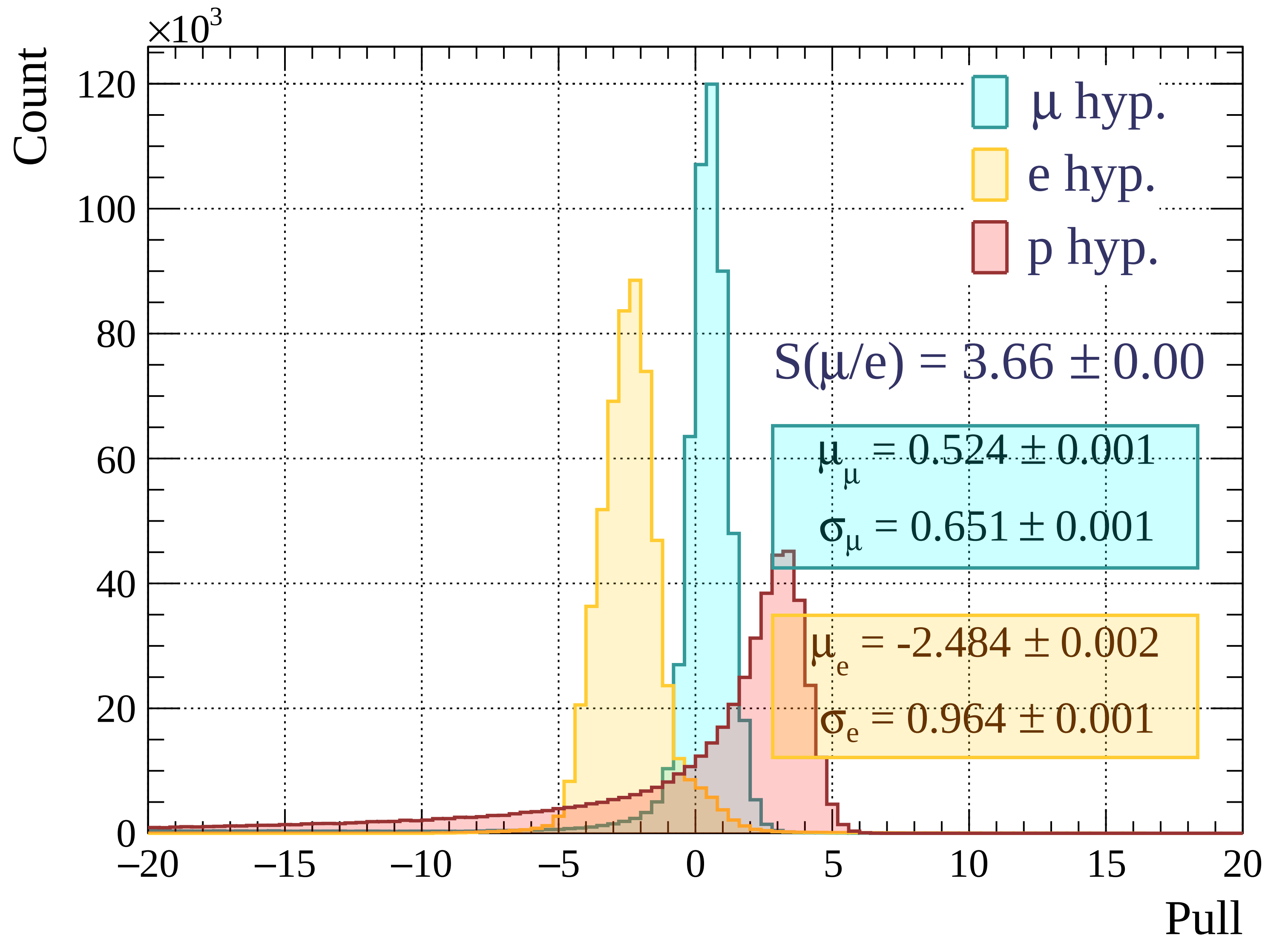

Pulls mean
200
150
100
50
0
−50
−100
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±


Pulls standard deviation
18
16
14
12
10
8
6
4
2
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
μ±
e±

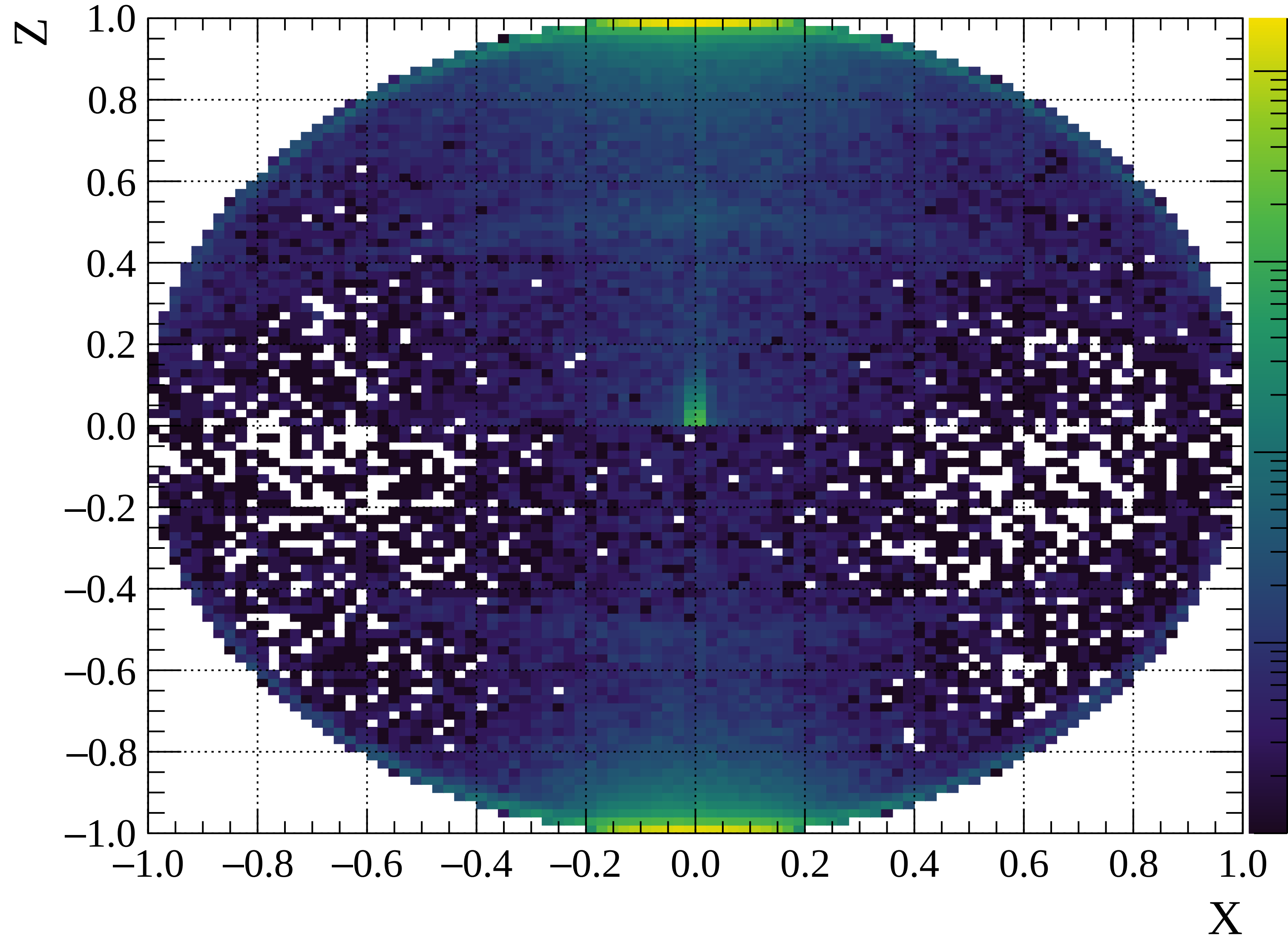


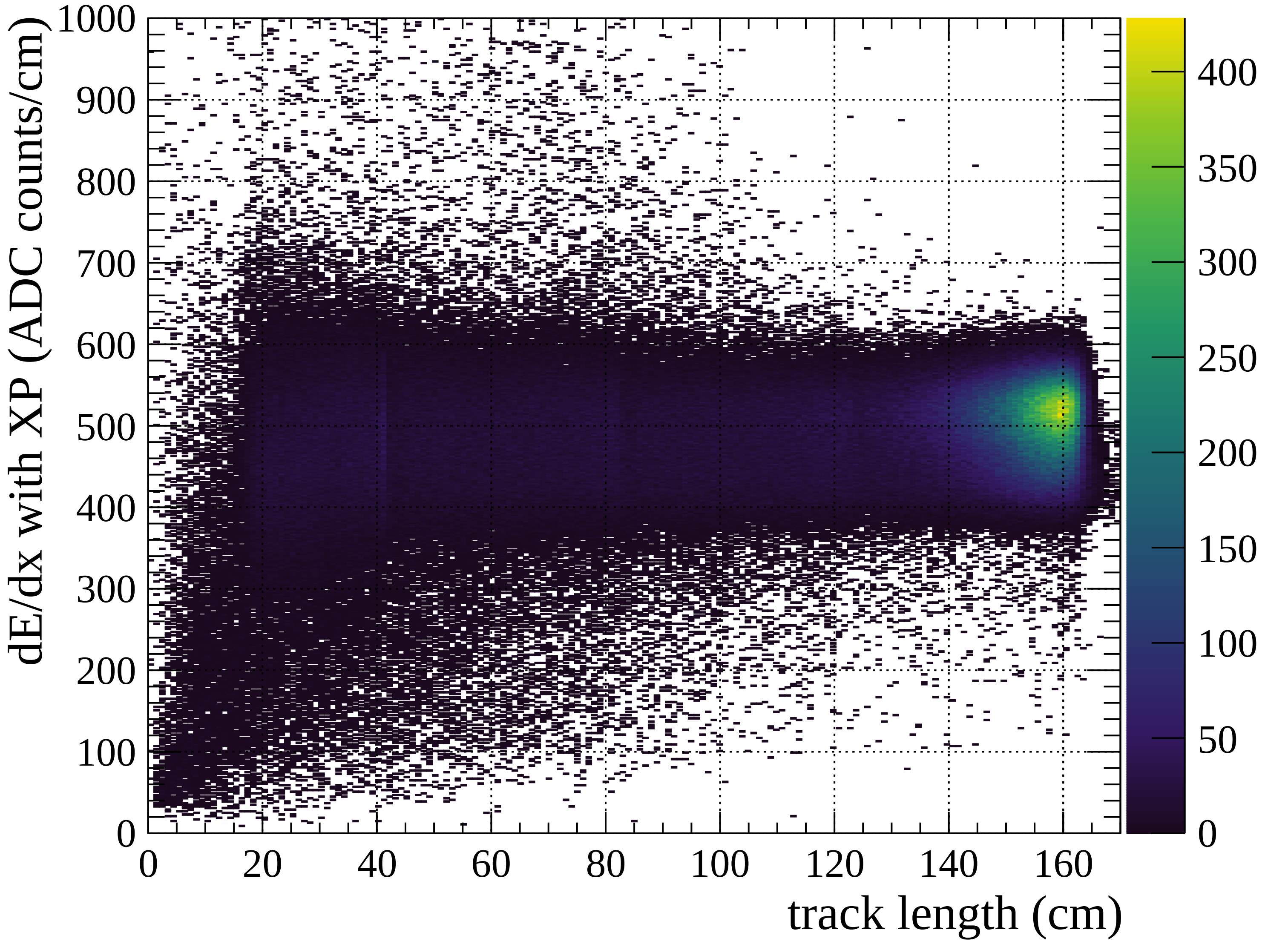
Pulls mean
10
0
−10
−20
−30
−40
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$


Pull e
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
Pull μ
×10³
30
25
20
15
10
5
0

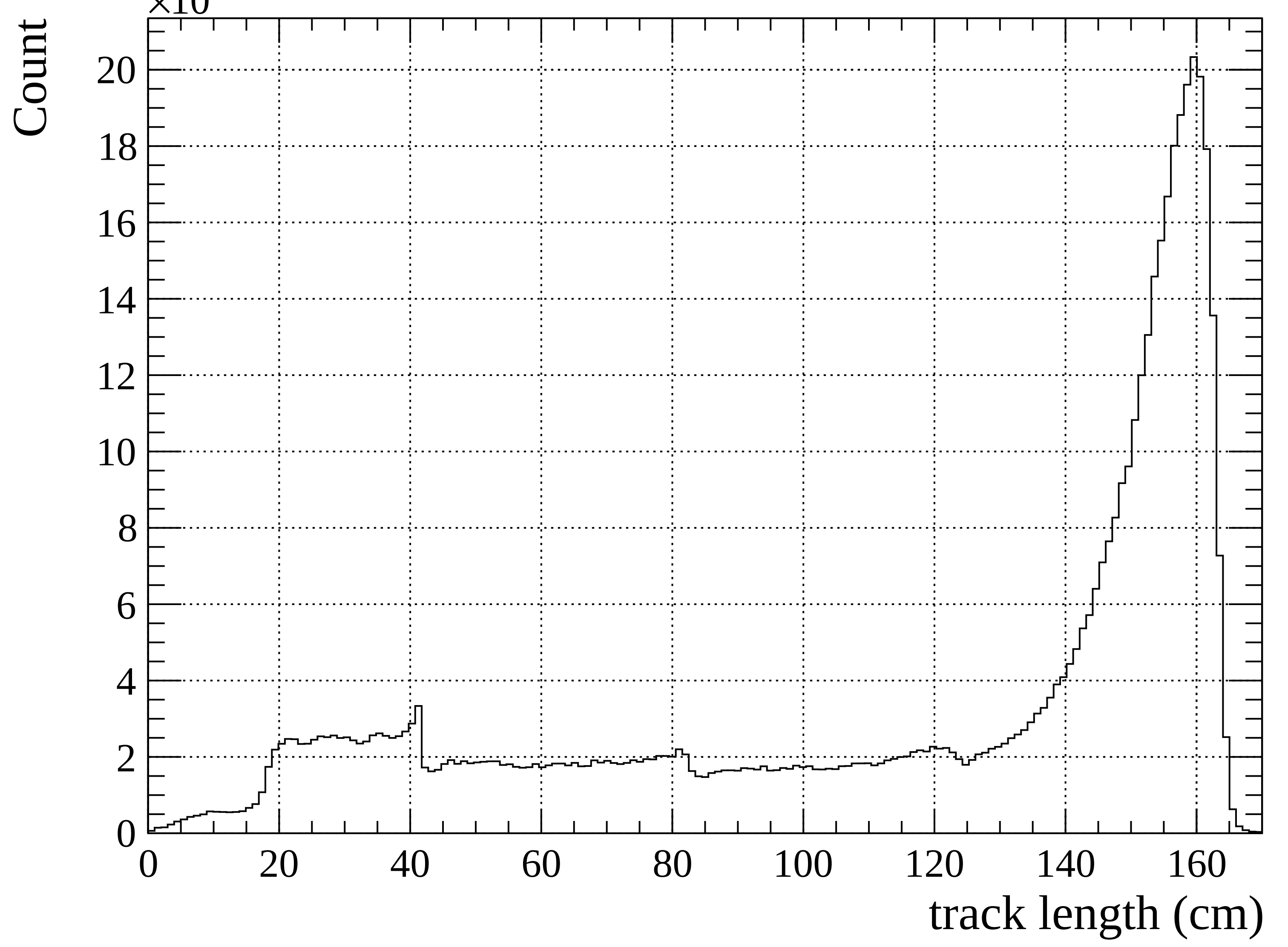


Count
×10^3
120
100
80
60
40
20
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 3.66 ± 0.00
μ_μ = 0.524 ± 0.001
σ_μ = 0.651 ± 0.001
μ_e = -2.484 ± 0.002
σ_e = 0.964 ± 0.001

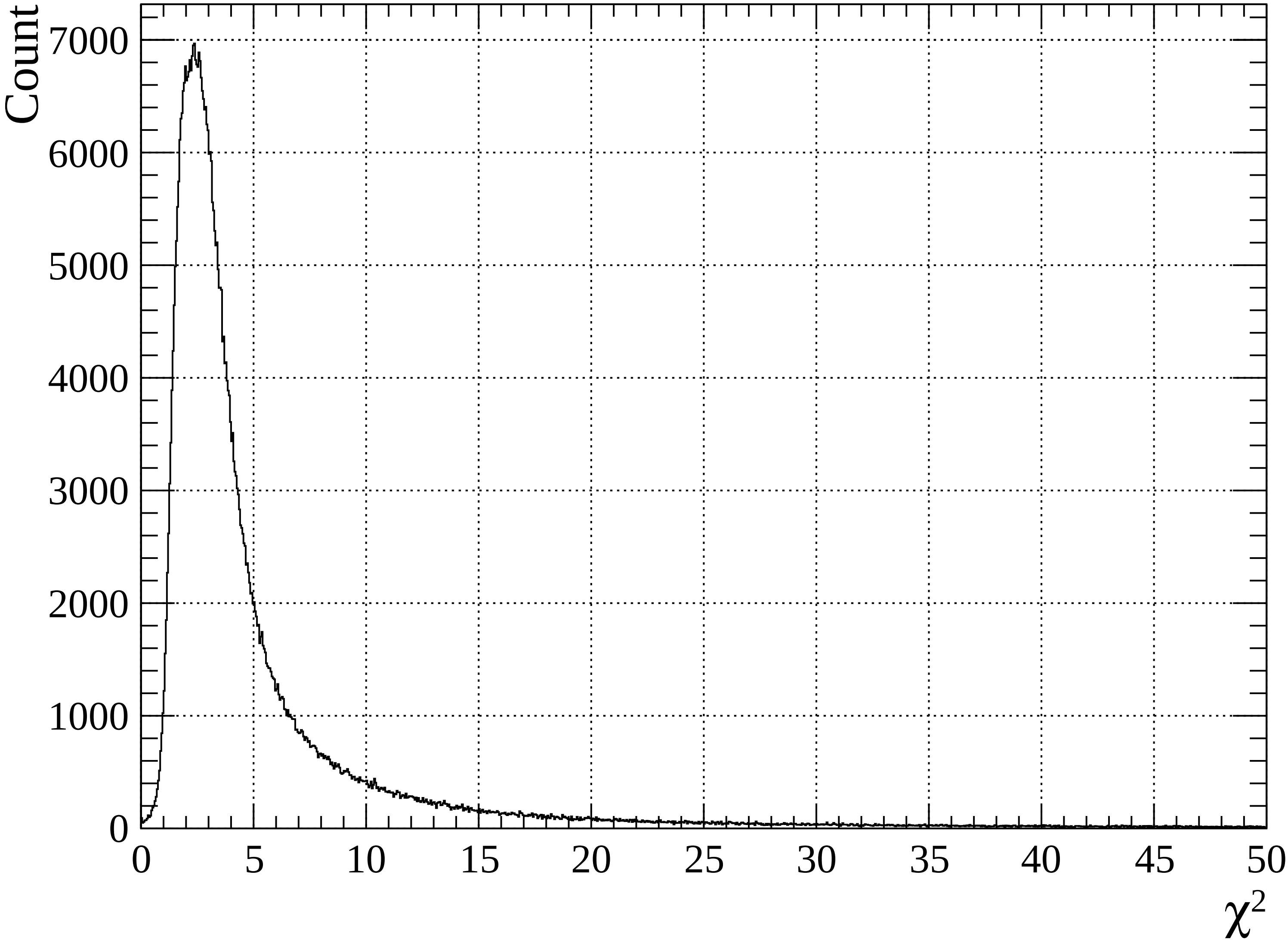

Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X

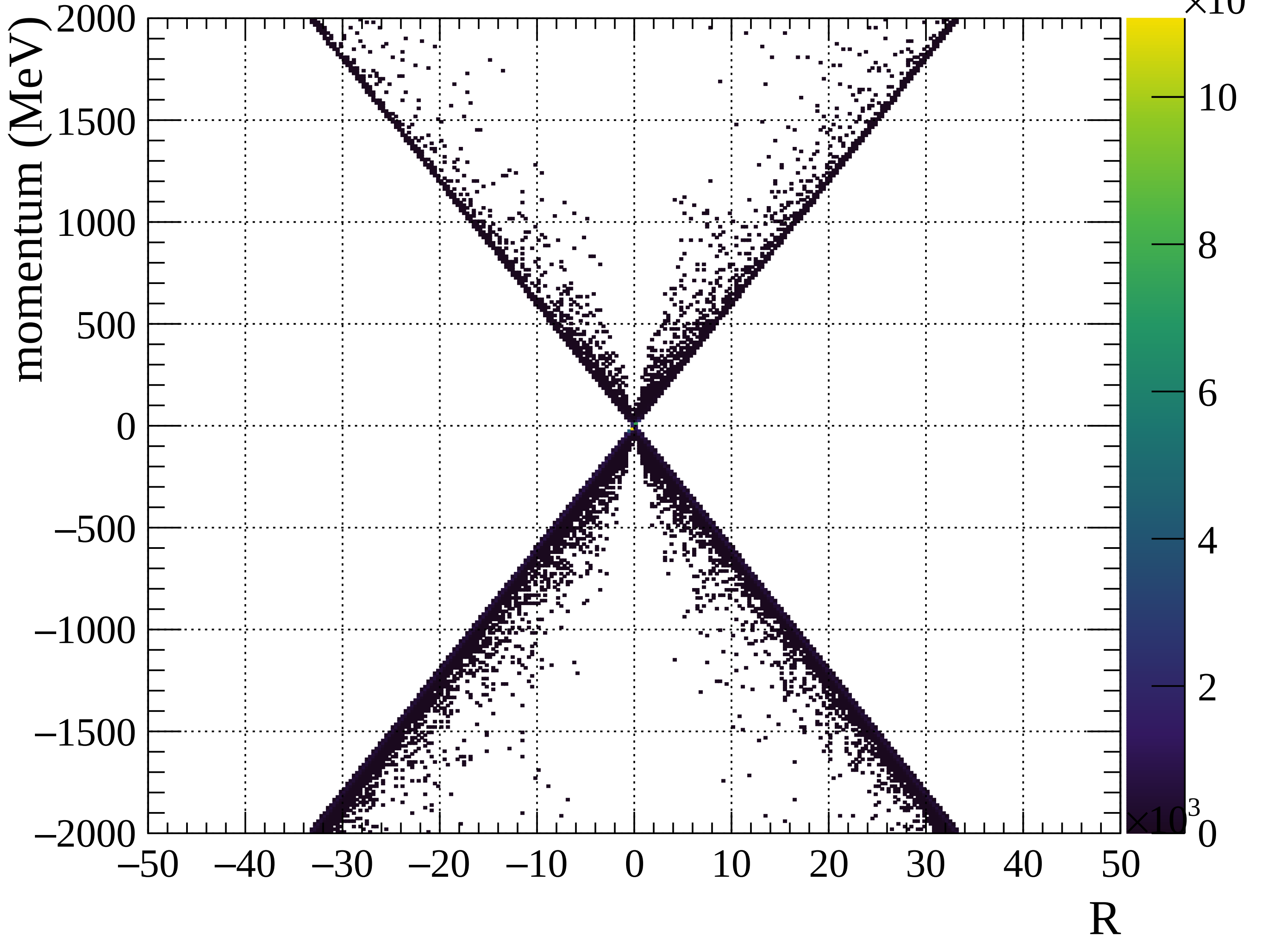


dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
50
100
150
200
250
300
350
400

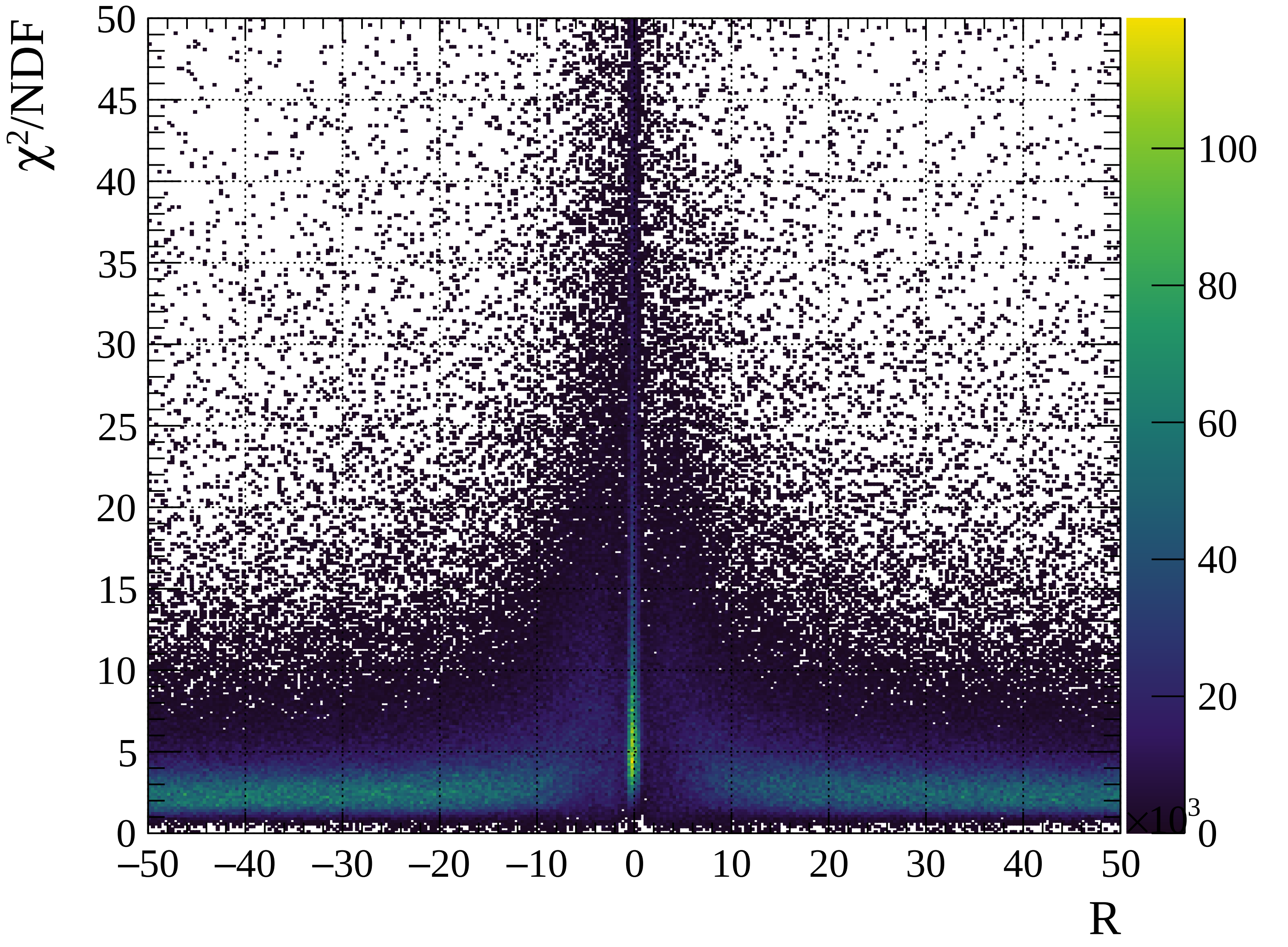


×10
Count
0
2
4
6
8
10
12
14
16
18
20
0
20
40
60
80
100
120
140
160
track length (cm)

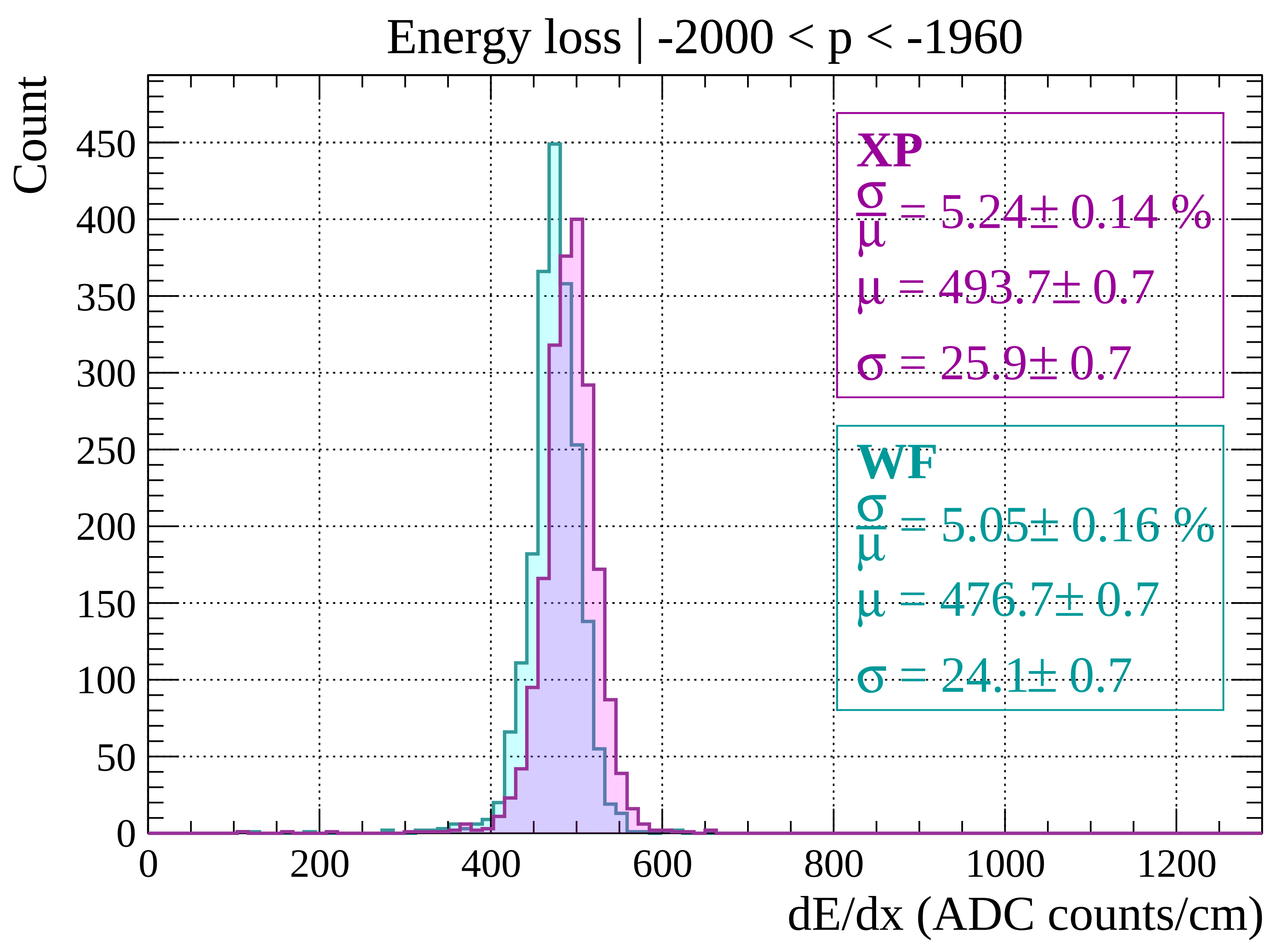

Count
7000
6000
5000
4000
3000
2000
1000
0
0
5
10
15
20
25
30
35
40
45
50
χ^2



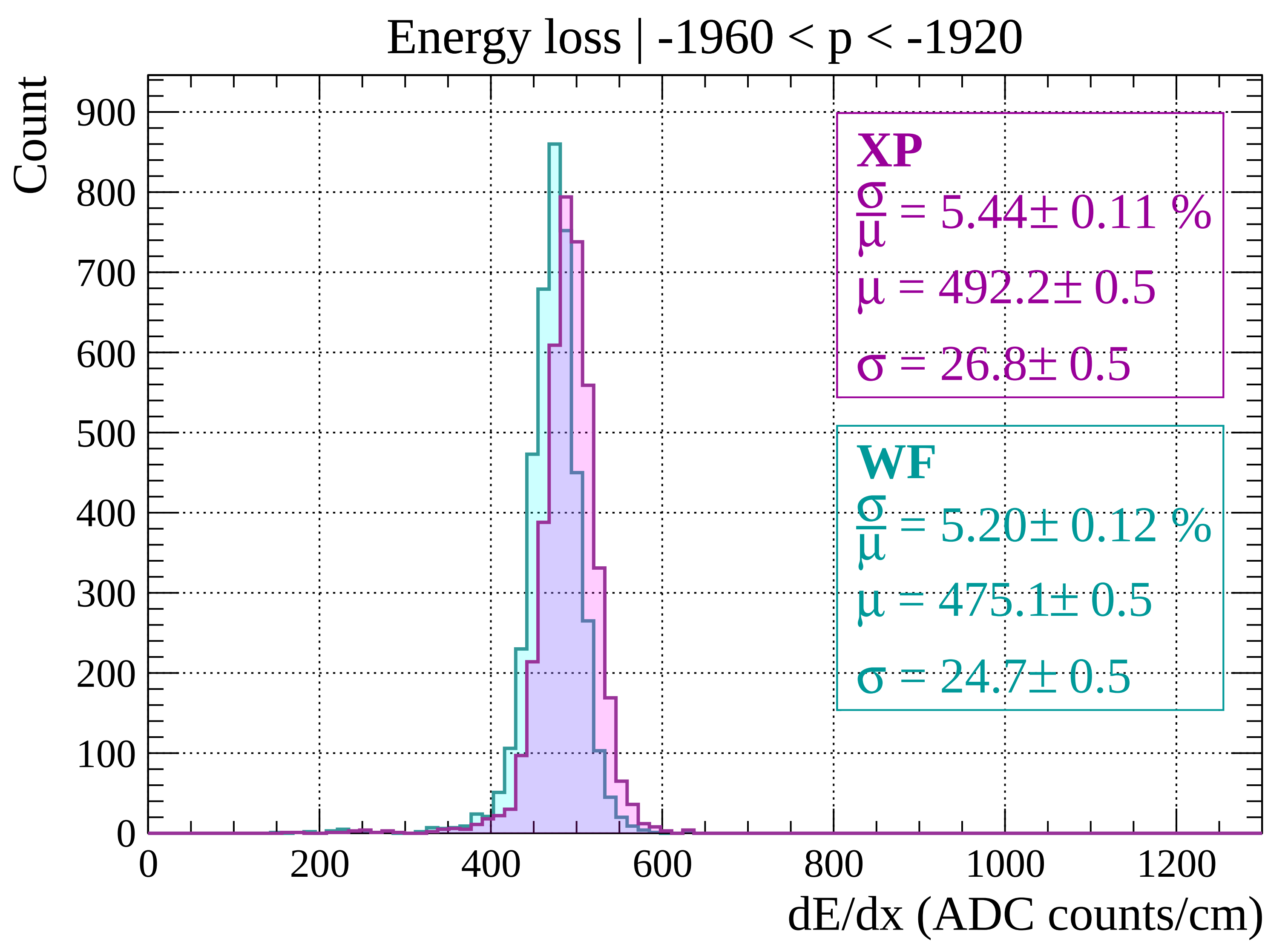
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
×10³
10
8
6
4
2
0



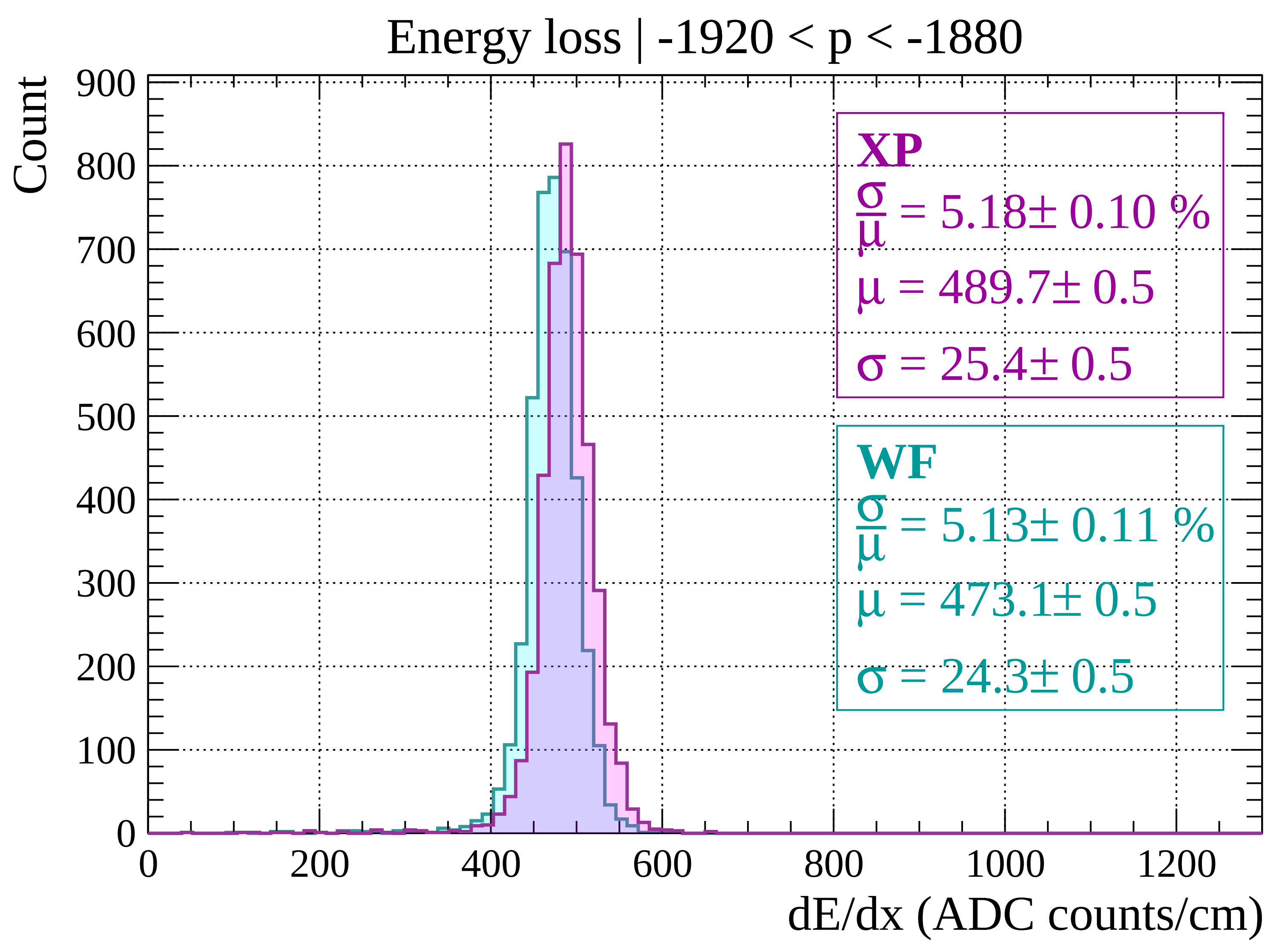
χ^2/NDF
R
$\times 10^3$
0
5
10
15
20
25
30
35
40
45
50
−50
−40
−30
−20
−10
0
10
20
30
40
50
20
40
60
80
100



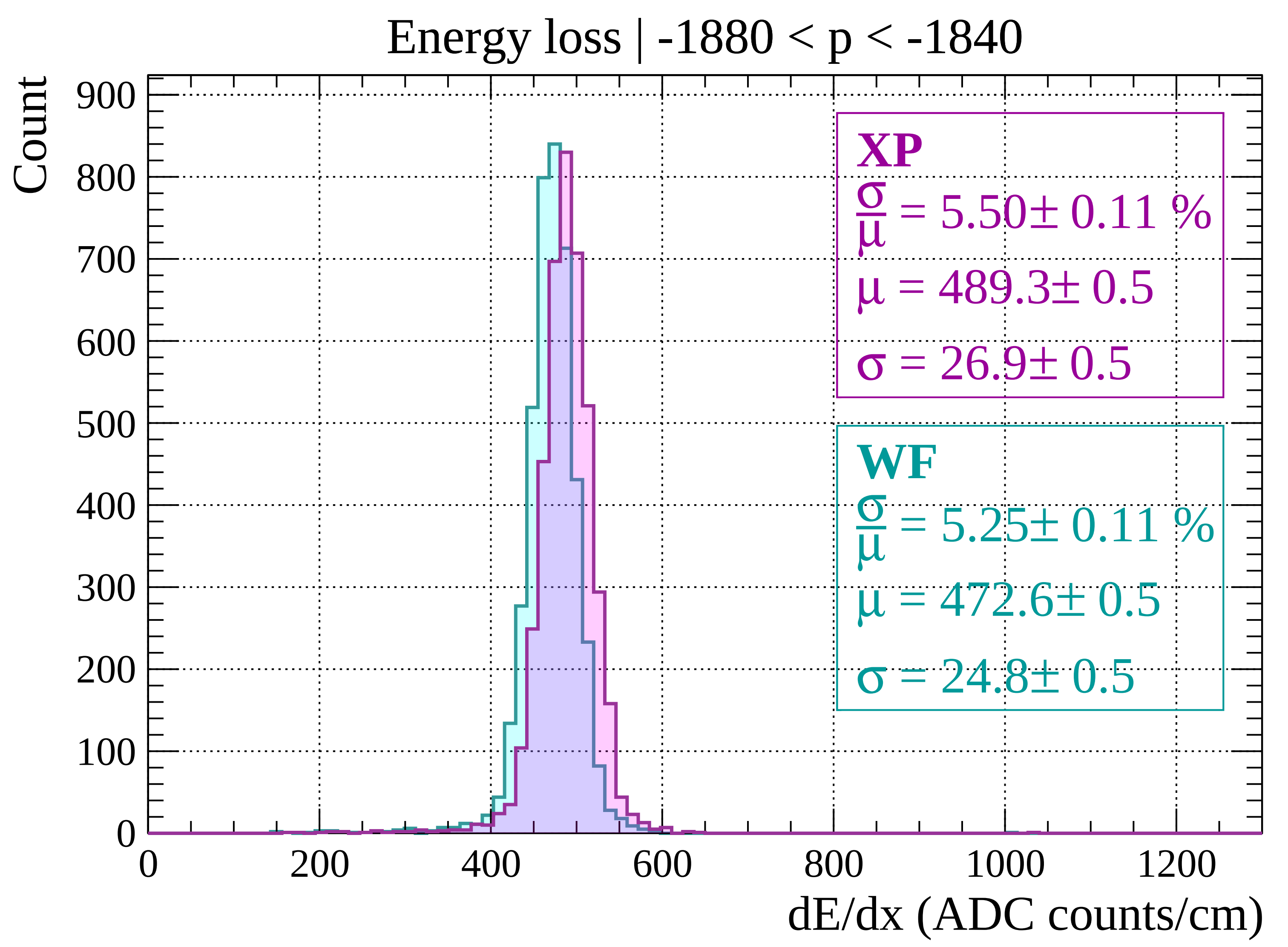
Energy loss | -2000 < p < -1960
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
σ/μ = 5.24± 0.14 %
μ = 493.7± 0.7
σ = 25.9± 0.7
WF
σ/μ = 5.05± 0.16 %
μ = 476.7± 0.7
σ = 24.1± 0.7

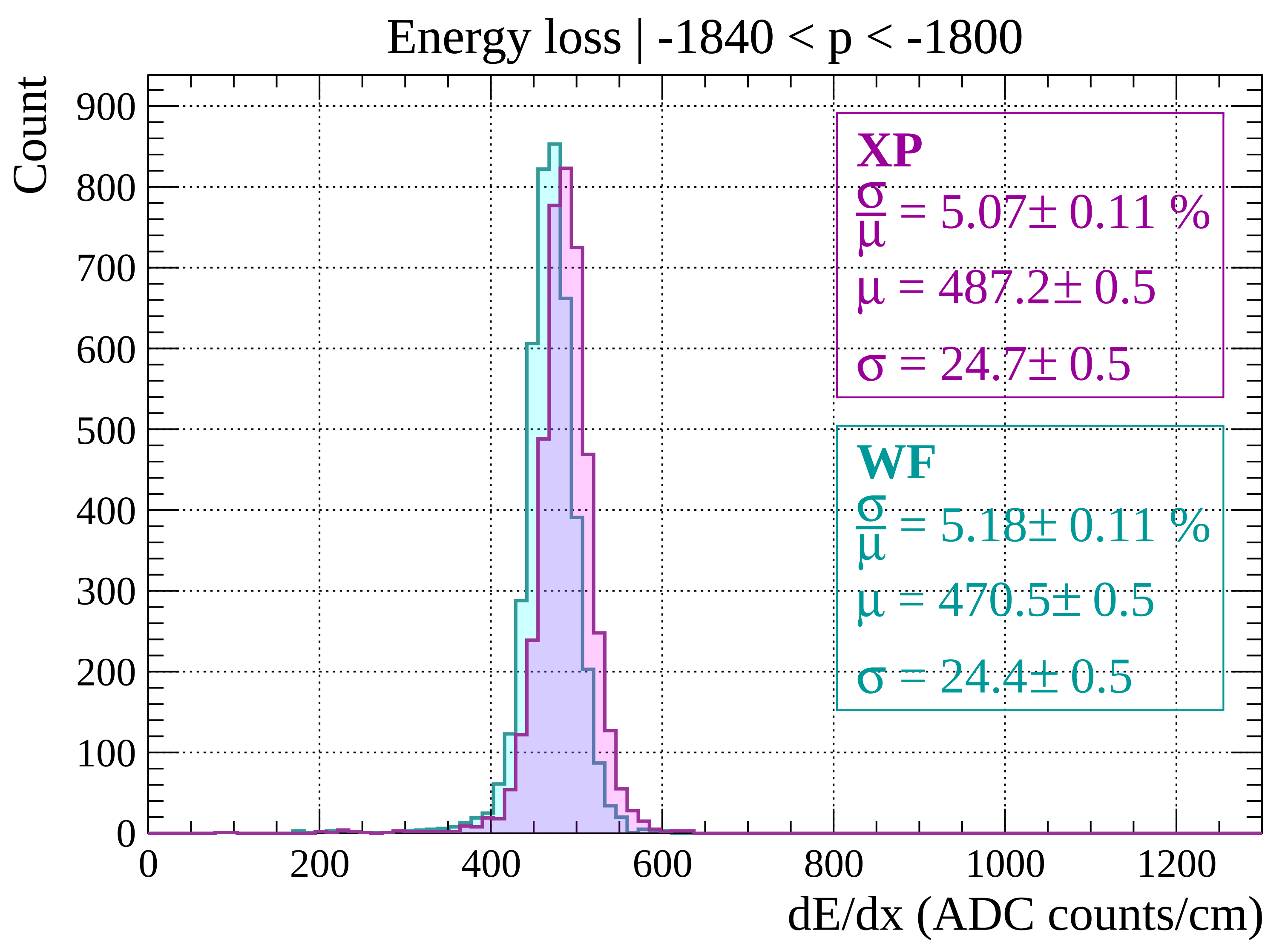


Energy loss | -1960 < p < -1920
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 5.44 \pm 0.11$ %
$\mu = 492.2 \pm 0.5$
$\sigma = 26.8 \pm 0.5$
WF
$\frac{\sigma}{\mu} = 5.20 \pm 0.12$ %
$\mu = 475.1 \pm 0.5$
$\sigma = 24.7 \pm 0.5$

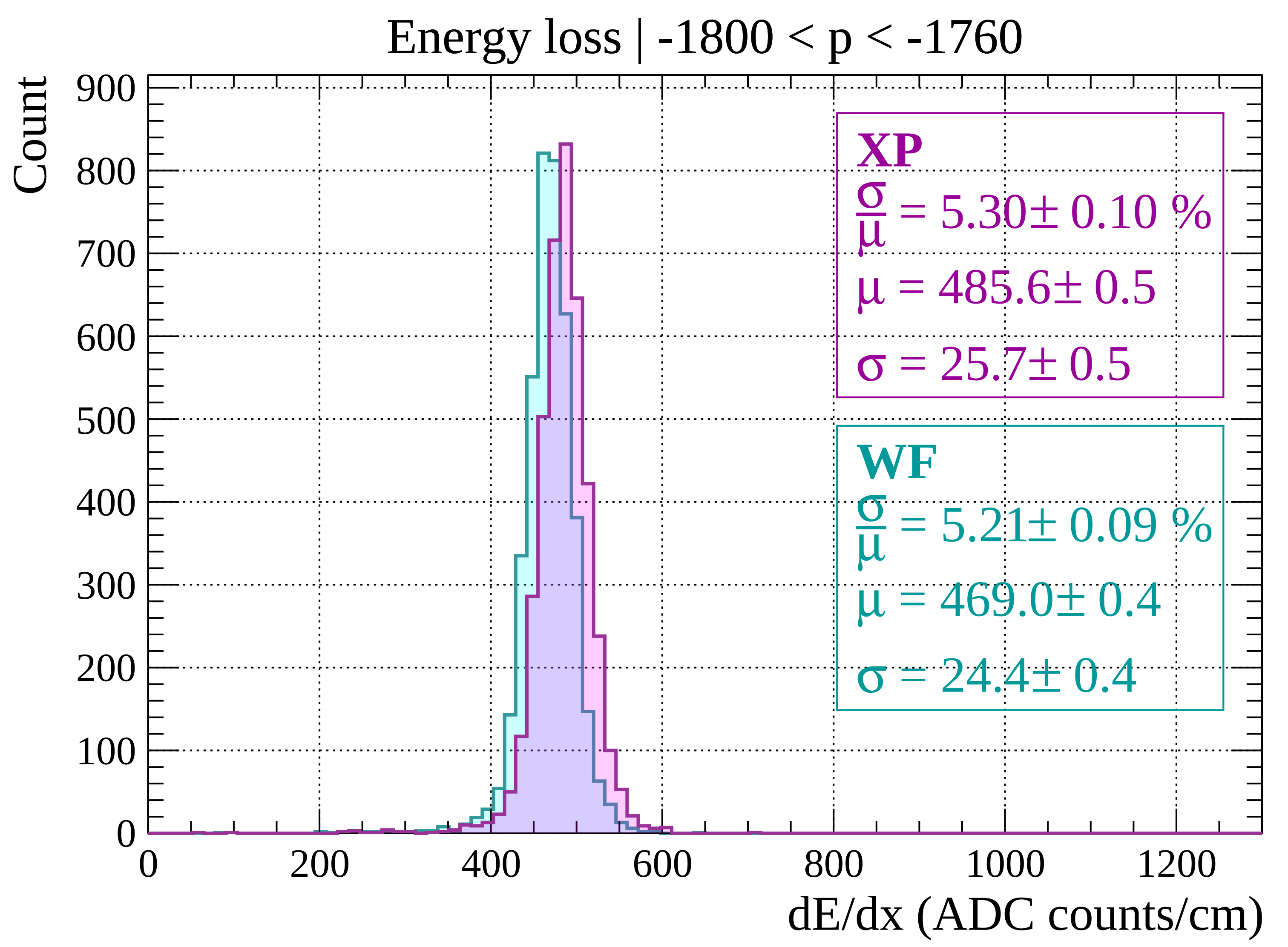


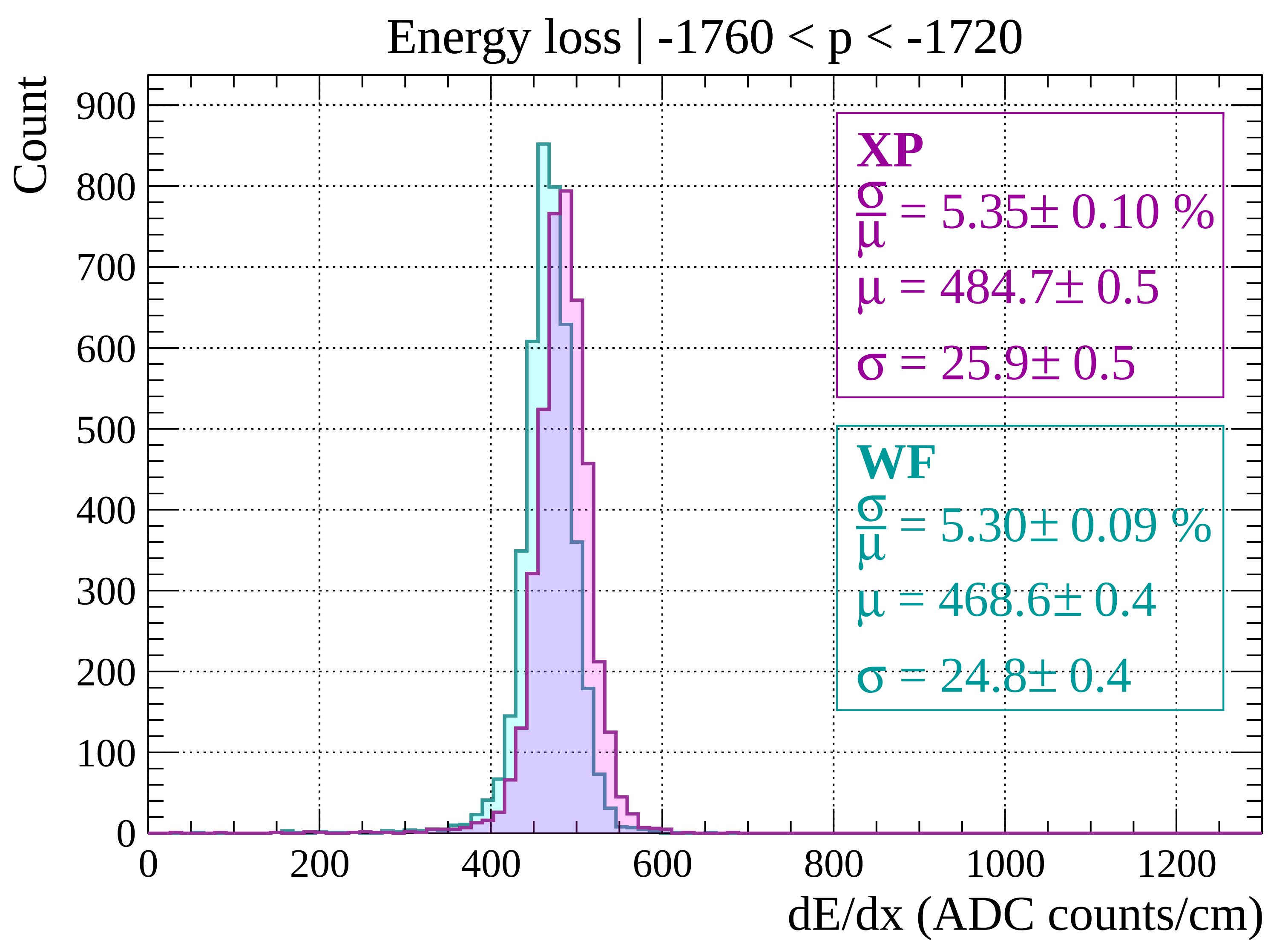
Energy loss | -1920 < p < -1880
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.18± 0.10 %
μ = 489.7± 0.5
σ = 25.4± 0.5
WF
σ/μ = 5.13± 0.11 %
μ = 473.1± 0.5
σ = 24.3± 0.5
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200


Energy loss | -1880 < p < -1840
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 5.50± 0.11 %
μ = 489.3± 0.5
σ = 26.9± 0.5
WF
σ/μ = 5.25± 0.11 %
μ = 472.6± 0.5
σ = 24.8± 0.5

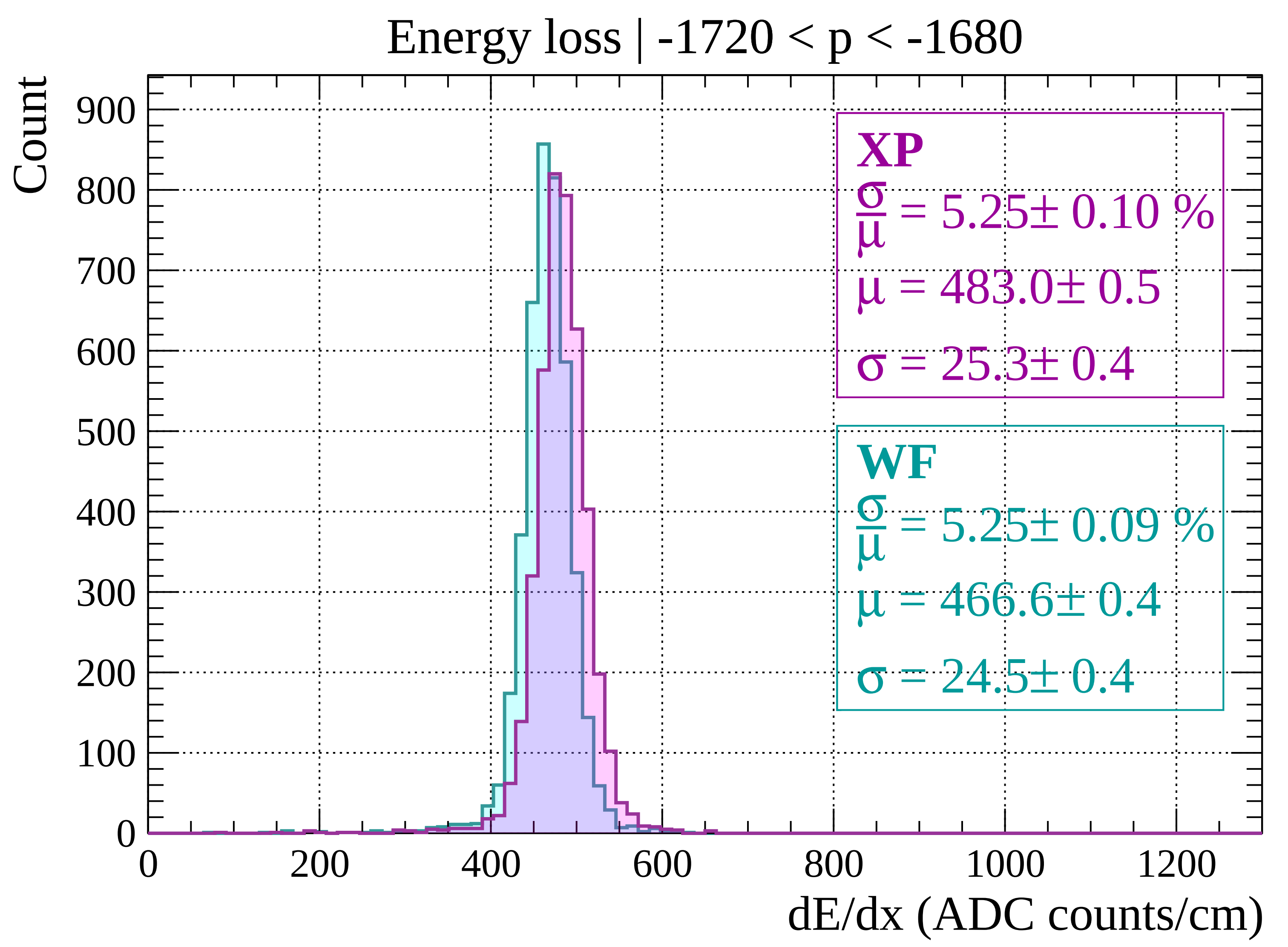


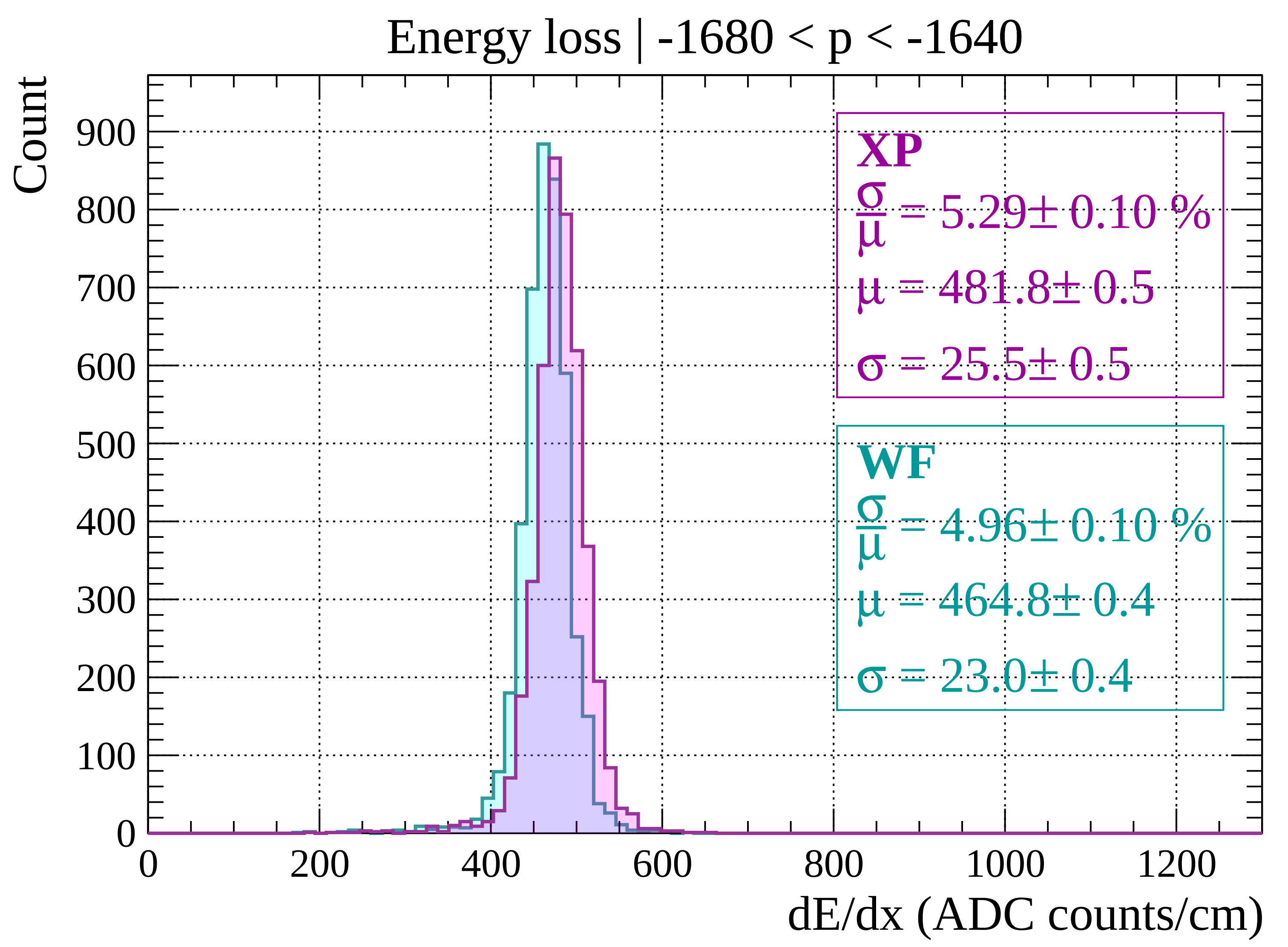
Energy loss | -1840 < p < -1800
Count
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.07 ± 0.11 %
μ = 487.2 ± 0.5
σ = 24.7 ± 0.5
WF
$\frac{\sigma}{\mu}$ = 5.18 ± 0.11 %
μ = 470.5 ± 0.5
σ = 24.4 ± 0.5


Energy loss | -1800 < p < -1760
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 5.30± 0.10 %
μ = 485.6± 0.5
σ = 25.7± 0.5
WF
σ/μ = 5.21± 0.09 %
μ = 469.0± 0.4
σ = 24.4± 0.4

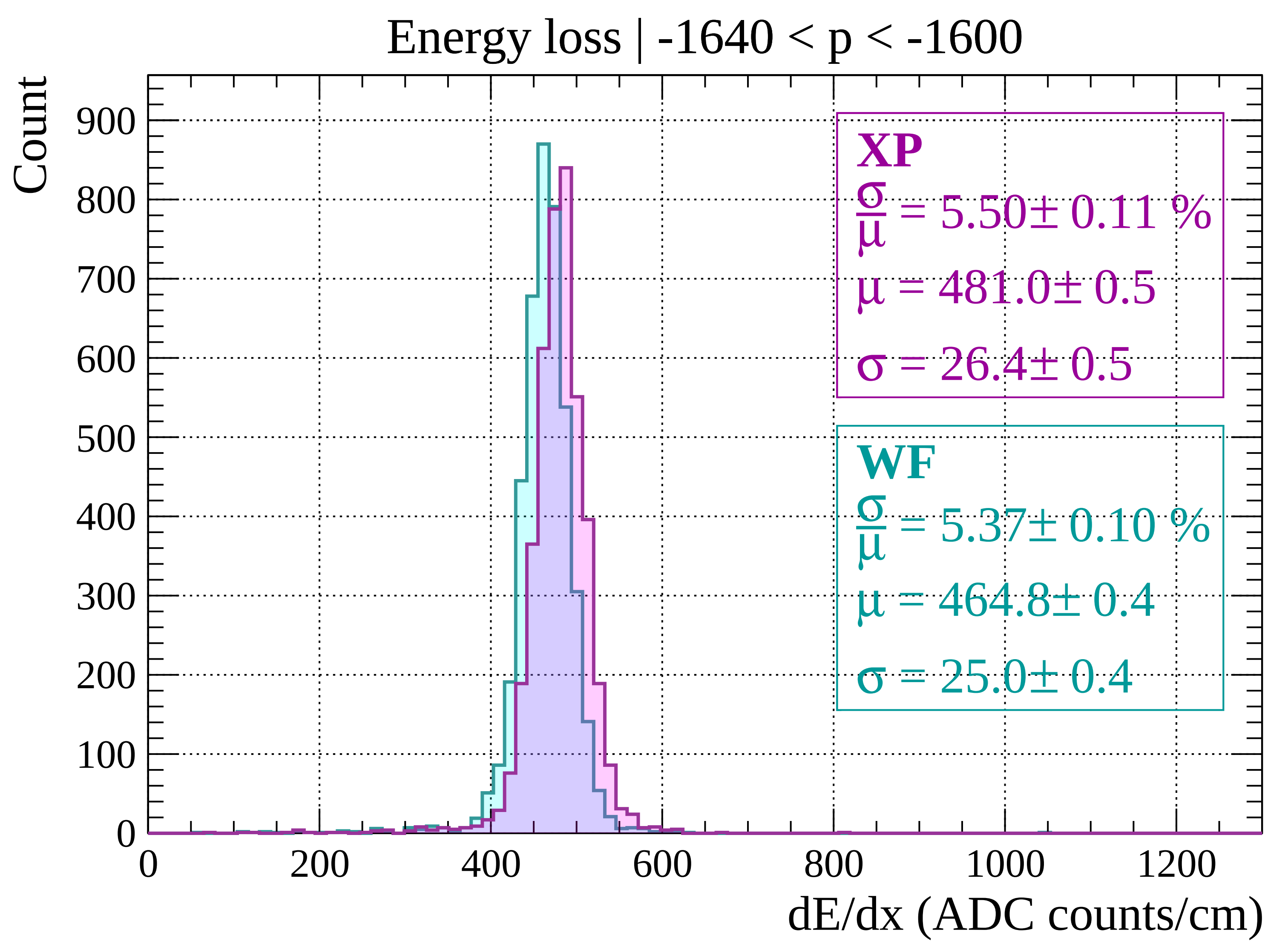


Energy loss | -1760 < p < -1720
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.35± 0.10 %
μ = 484.7± 0.5
σ = 25.9± 0.5
WF
$\frac{\sigma}{\mu}$ = 5.30± 0.09 %
μ = 468.6± 0.4
σ = 24.8± 0.4

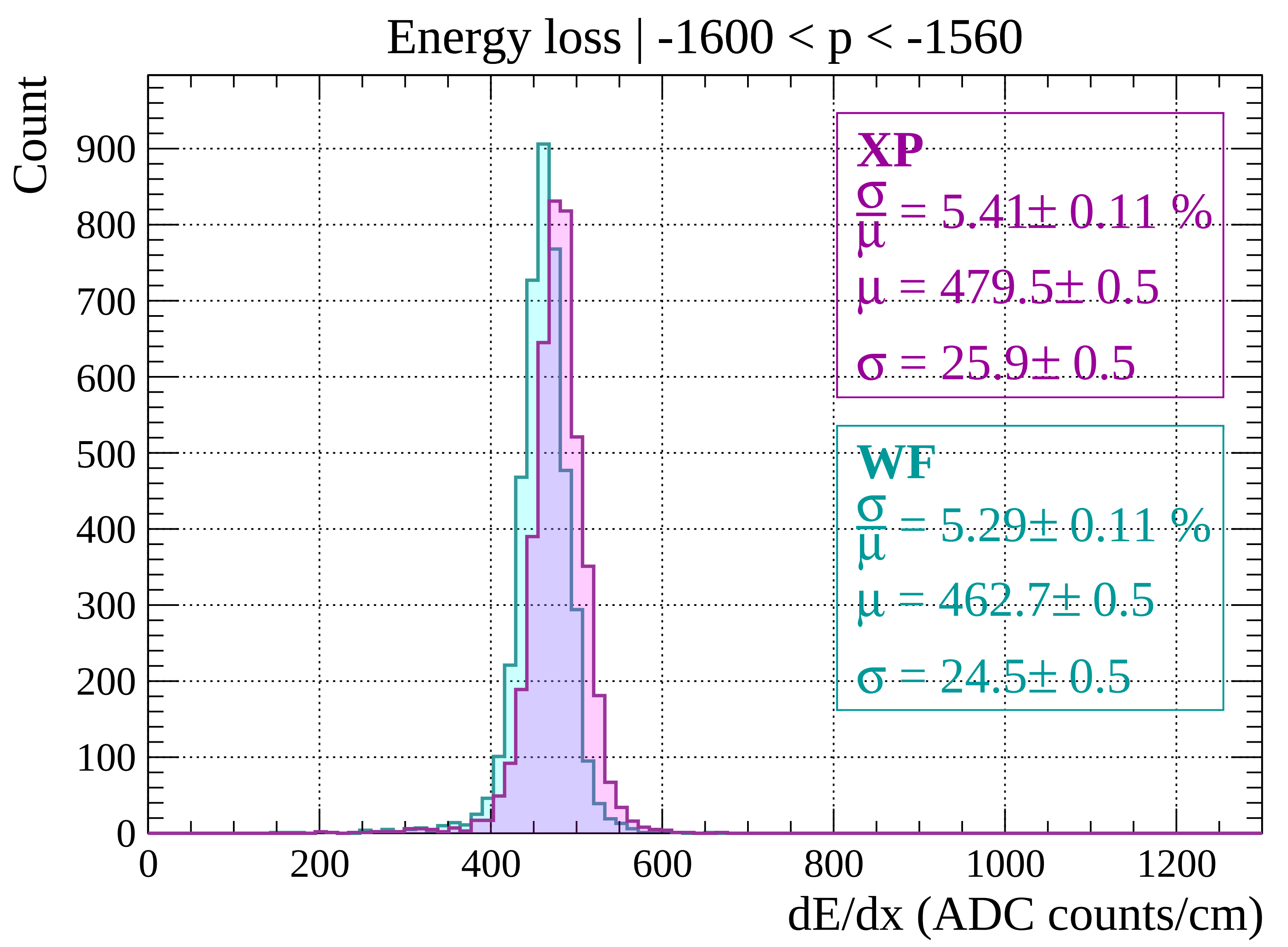

Energy loss | -1720 < p < -1680
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.25± 0.10 %
μ = 483.0± 0.5
σ = 25.3± 0.4
WF
$\frac{\sigma}{\mu}$ = 5.25± 0.09 %
μ = 466.6± 0.4
σ = 24.5± 0.4

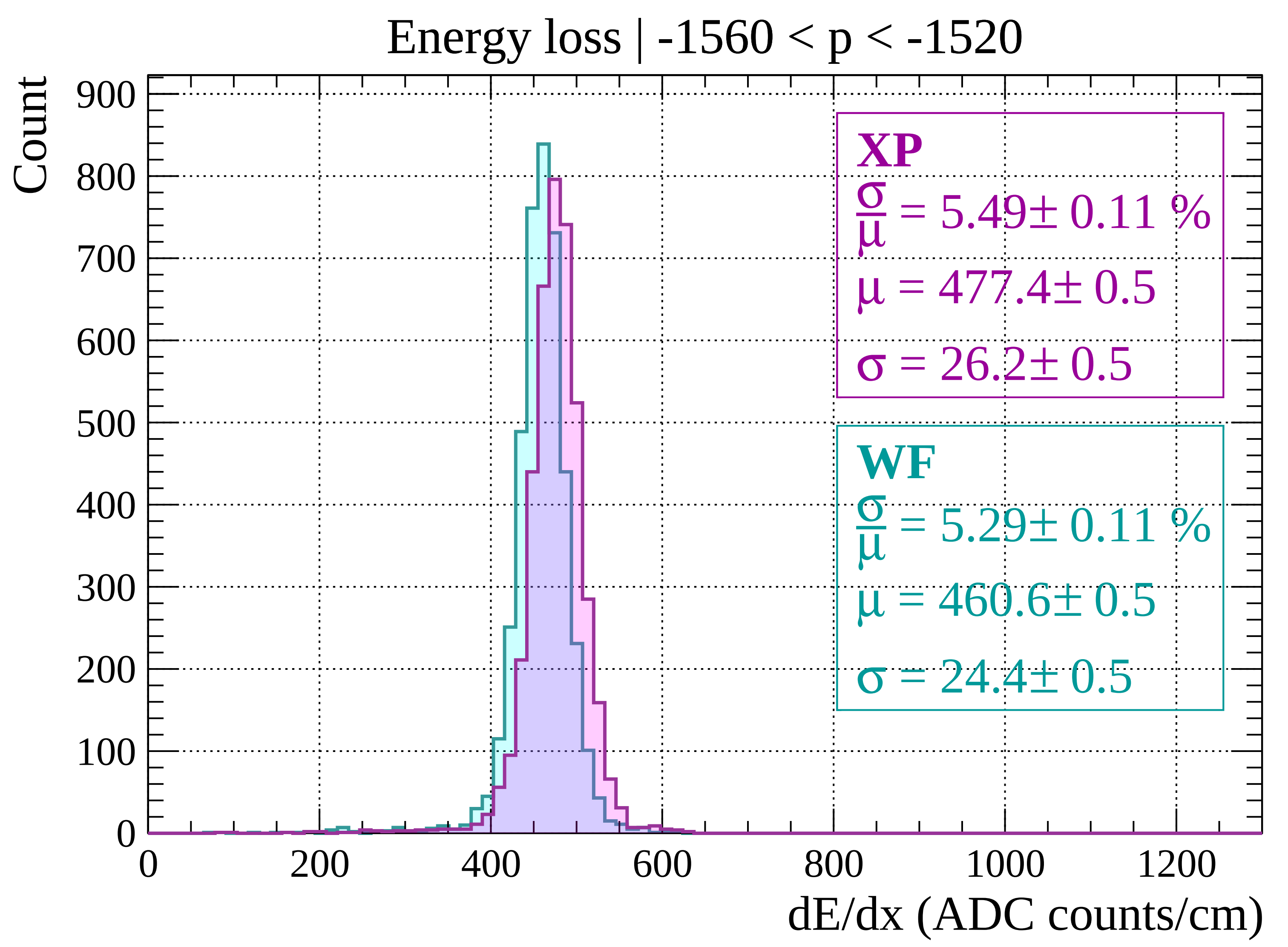


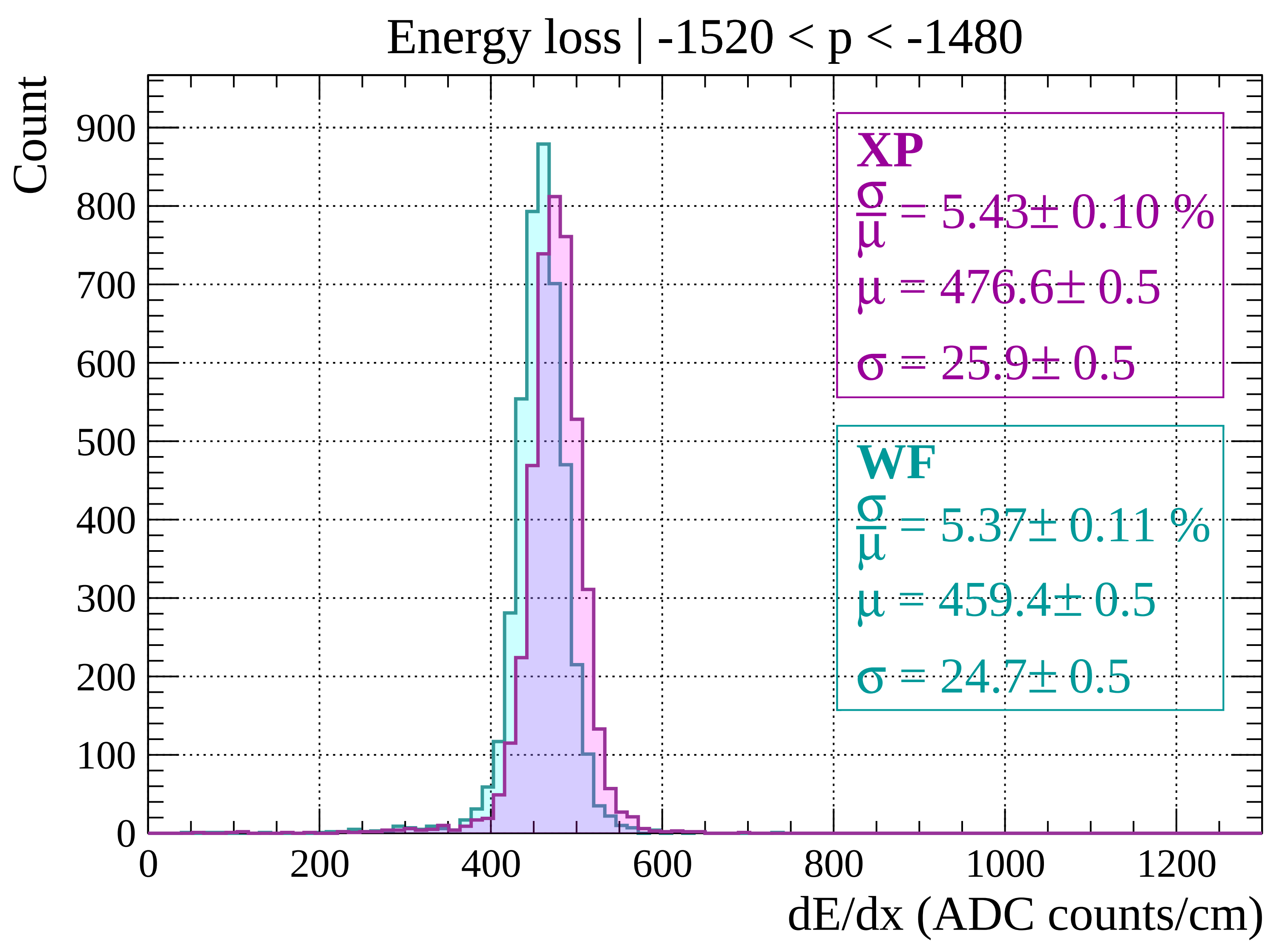
Energy loss | -1680 < p < -1640
Count
XP
$\frac{\sigma}{\mu}$ = 5.29± 0.10 %
μ = 481.8± 0.5
σ = 25.5± 0.5
WF
$\frac{\sigma}{\mu}$ = 4.96± 0.10 %
μ = 464.8± 0.4
σ = 23.0± 0.4
dE/dx (ADC counts/cm)



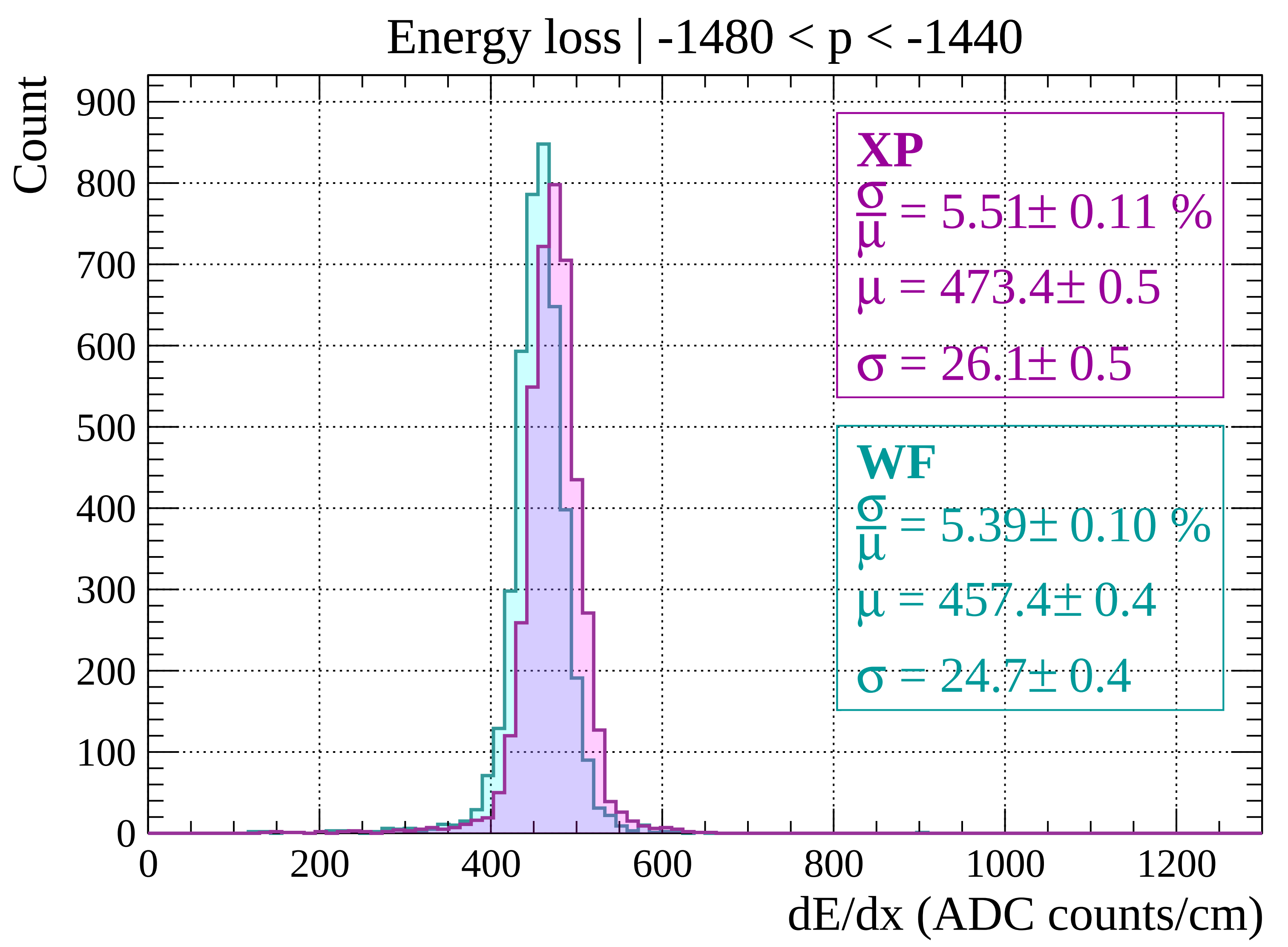
Energy loss | -1640 < p < -1600
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 5.50± 0.11 %
μ = 481.0± 0.5
σ = 26.4± 0.5
WF
σ/μ = 5.37± 0.10 %
μ = 464.8± 0.4
σ = 25.0± 0.4


Energy loss | -1600 < p < -1560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 5.41± 0.11 %
μ = 479.5± 0.5
σ = 25.9± 0.5
WF
σ/μ = 5.29± 0.11 %
μ = 462.7± 0.5
σ = 24.5± 0.5

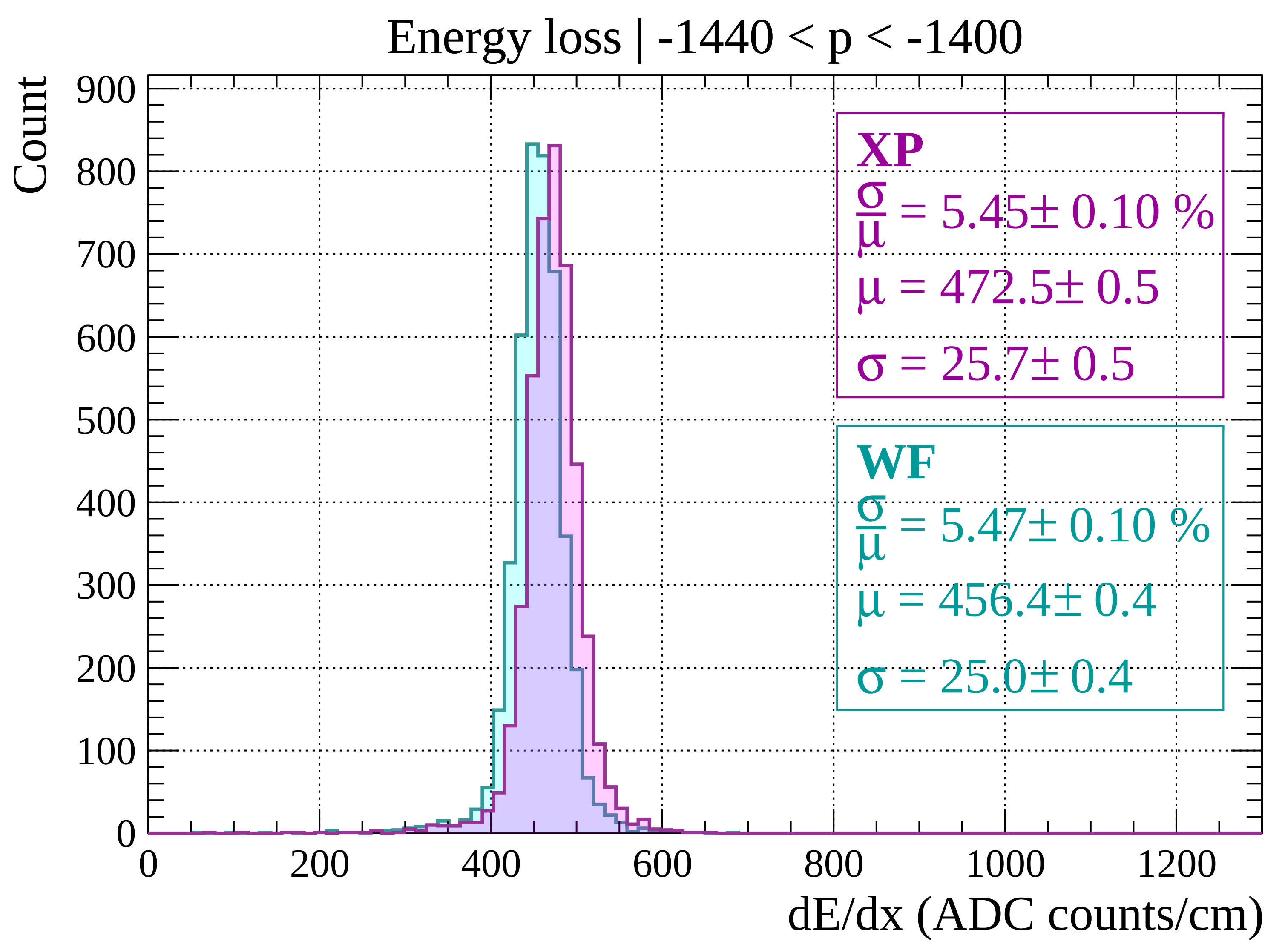

Energy loss | -1560 < p < -1520
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.49± 0.11 %
μ = 477.4± 0.5
σ = 26.2± 0.5
WF
$\frac{\sigma}{\mu}$ = 5.29± 0.11 %
μ = 460.6± 0.5
σ = 24.4± 0.5



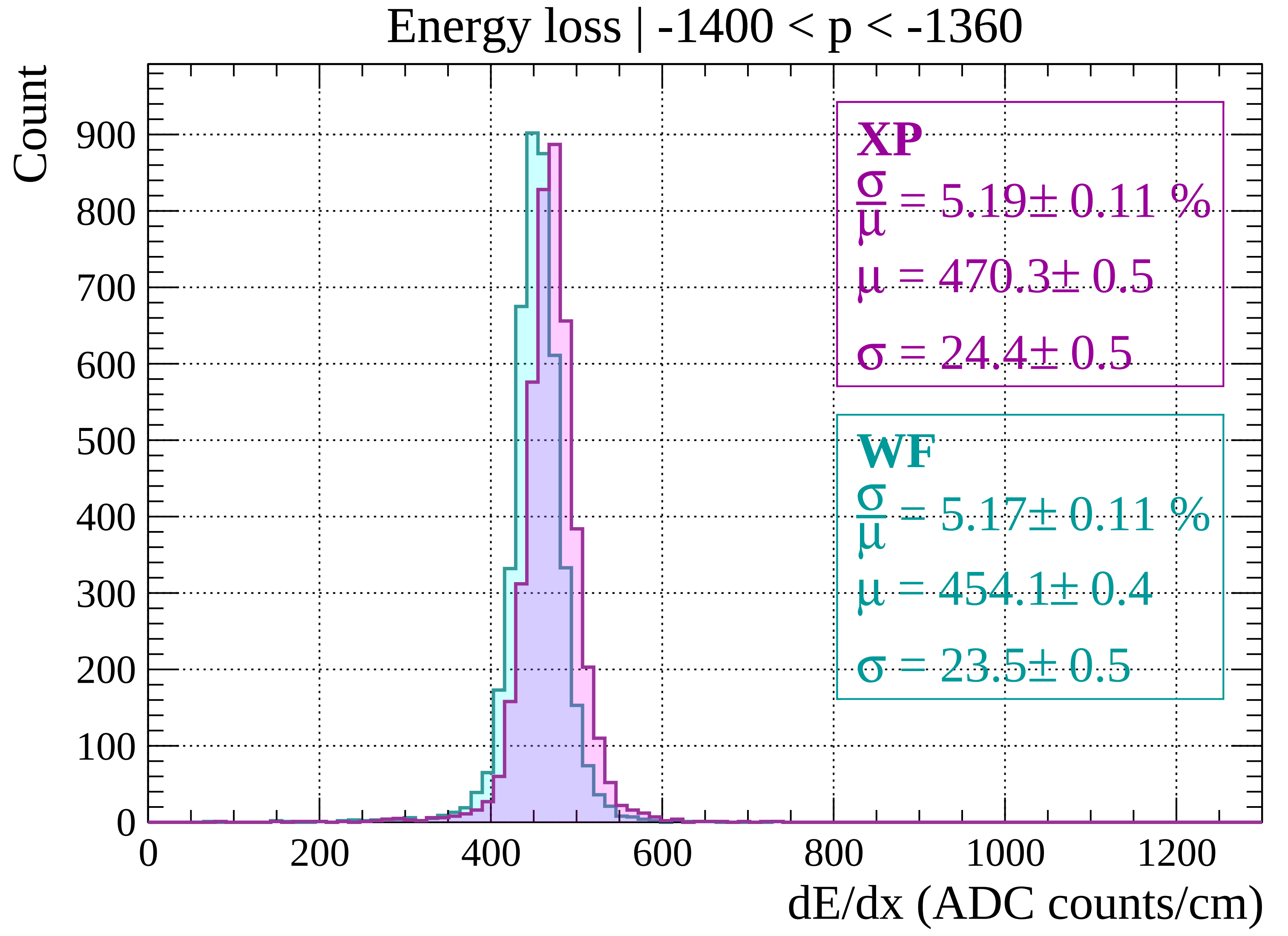
Energy loss | -1520 < p < -1480
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 5.43± 0.10 %
μ = 476.6± 0.5
σ = 25.9± 0.5
WF
σ/μ = 5.37± 0.11 %
μ = 459.4± 0.5
σ = 24.7± 0.5

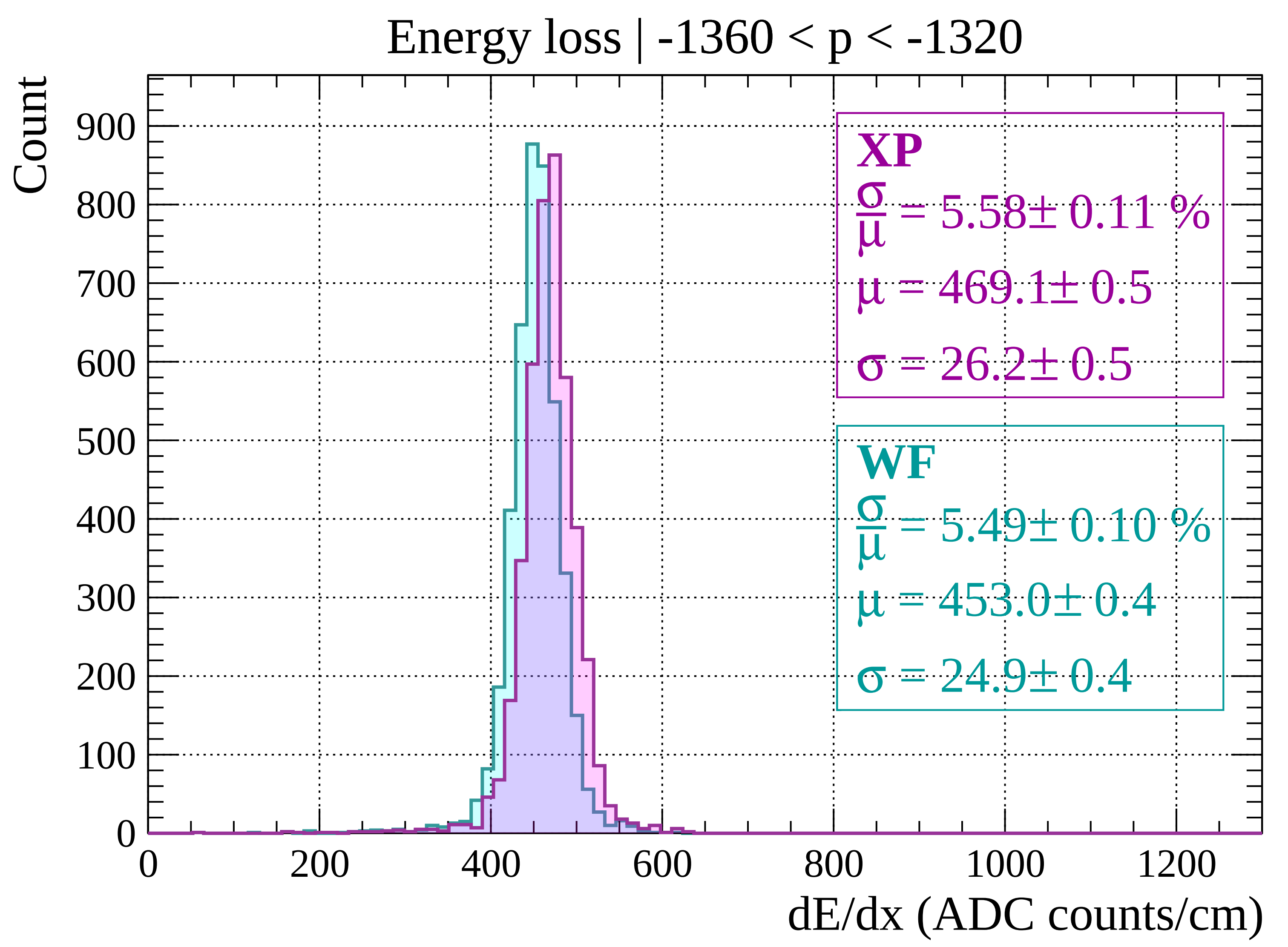


Energy loss | -1480 < p < -1440
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.51± 0.11 %
μ = 473.4± 0.5
σ = 26.1± 0.5
WF
σ/μ = 5.39± 0.10 %
μ = 457.4± 0.4
σ = 24.7± 0.4

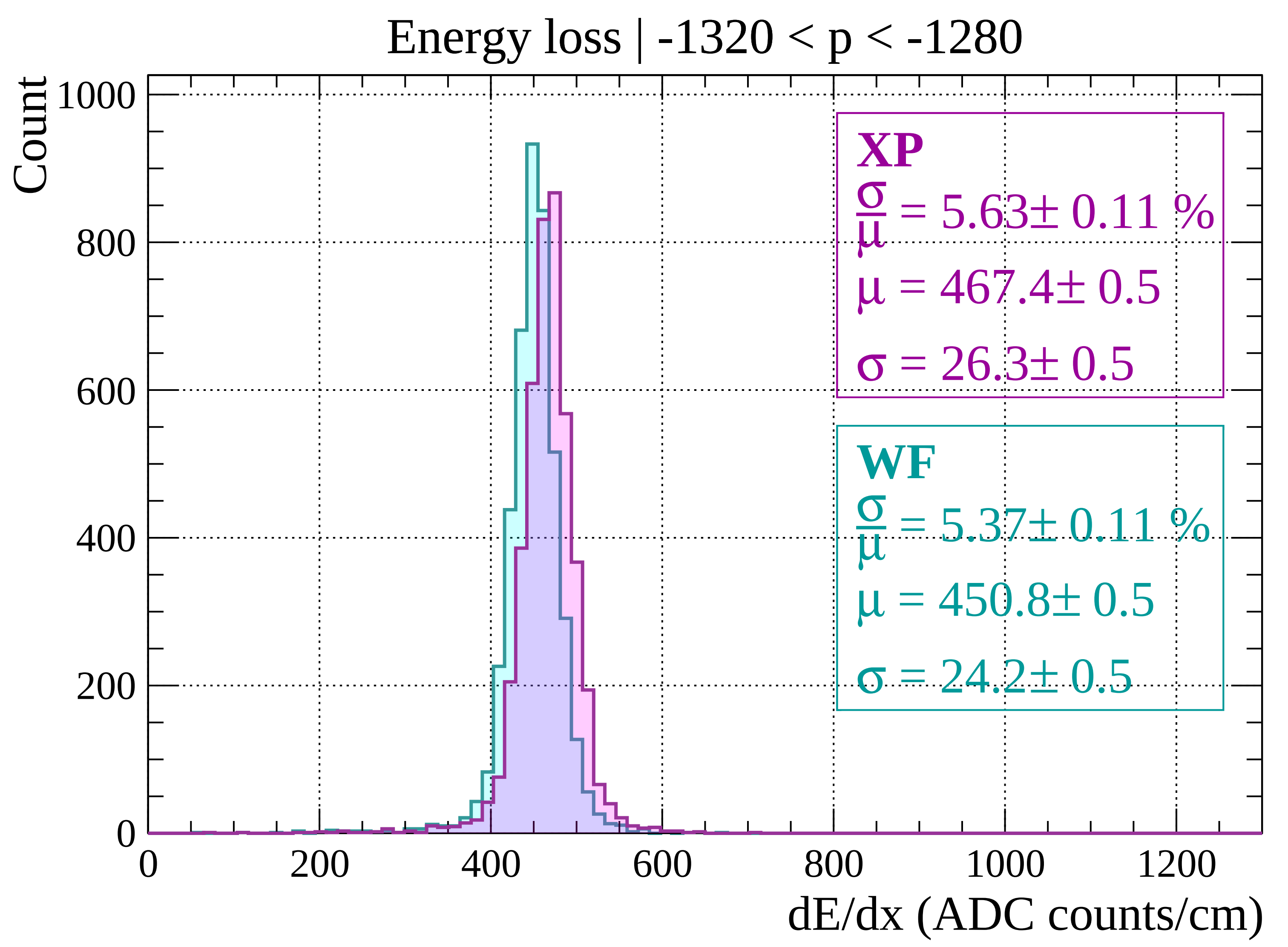


Energy loss | -1440 < p < -1400
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.45± 0.10 %
μ = 472.5± 0.5
σ = 25.7± 0.5
WF
σ/μ = 5.47± 0.10 %
μ = 456.4± 0.4
σ = 25.0± 0.4

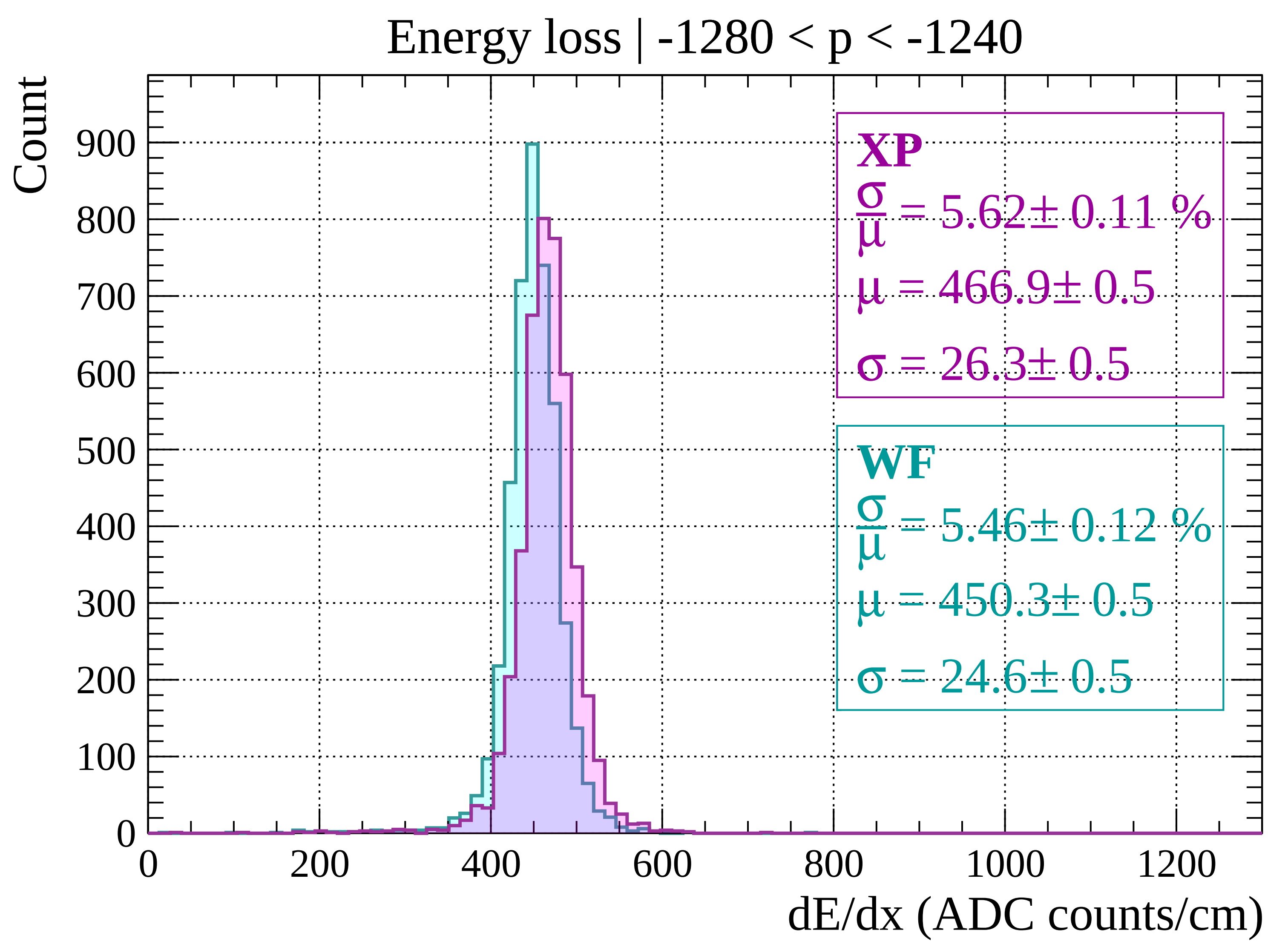

Energy loss | -1400 < p < -1360
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.19± 0.11 %
μ = 470.3± 0.5
σ = 24.4± 0.5
WF
σ/μ = 5.17± 0.11 %
μ = 454.1± 0.4
σ = 23.5± 0.5
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200

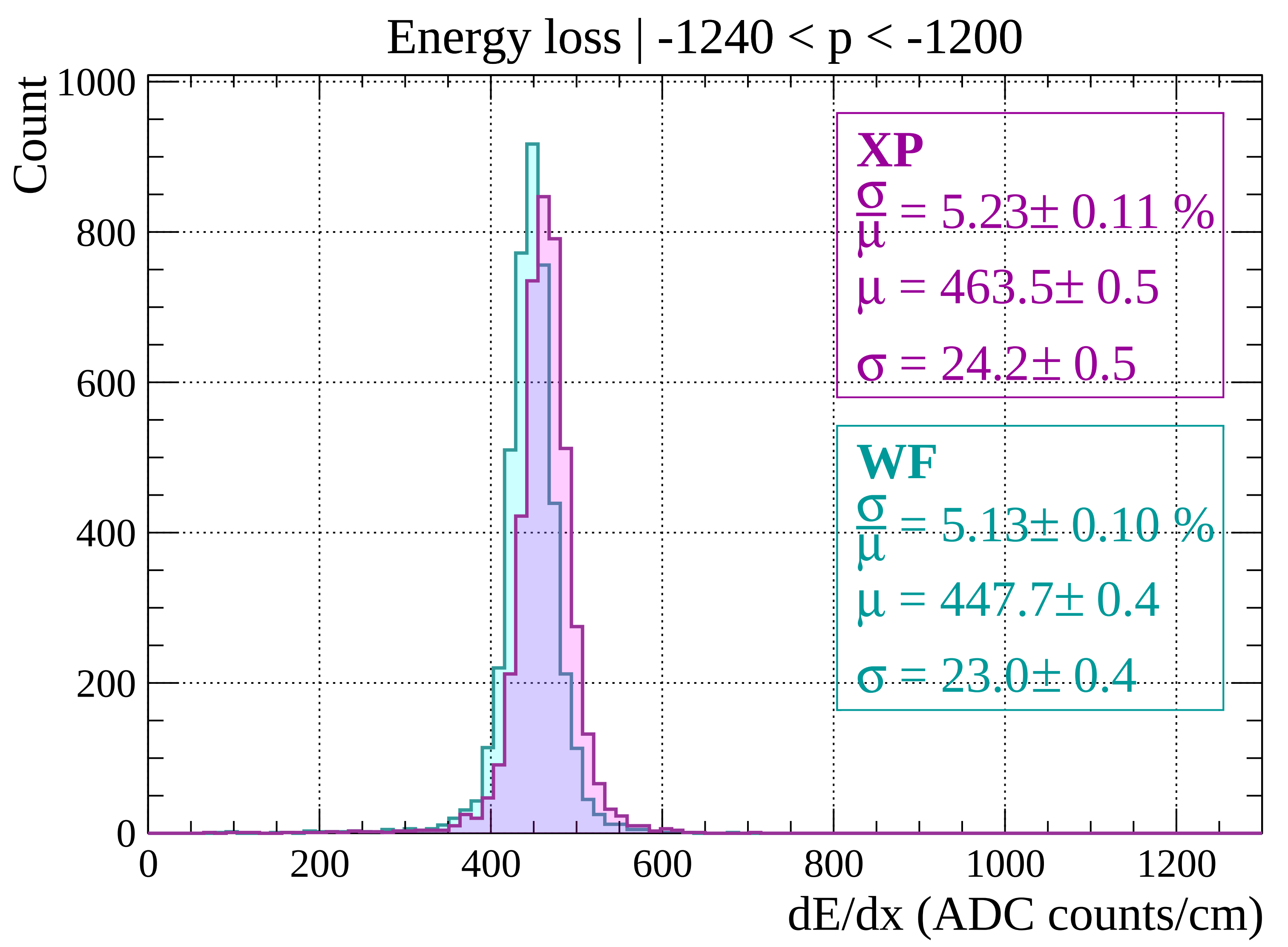


Energy loss | -1360 < p < -1320
Count
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 5.58 \pm 0.11$ %
$\mu = 469.1 \pm 0.5$
$\sigma = 26.2 \pm 0.5$
WF
$\frac{\sigma}{\mu} = 5.49 \pm 0.10$ %
$\mu = 453.0 \pm 0.4$
$\sigma = 24.9 \pm 0.4$

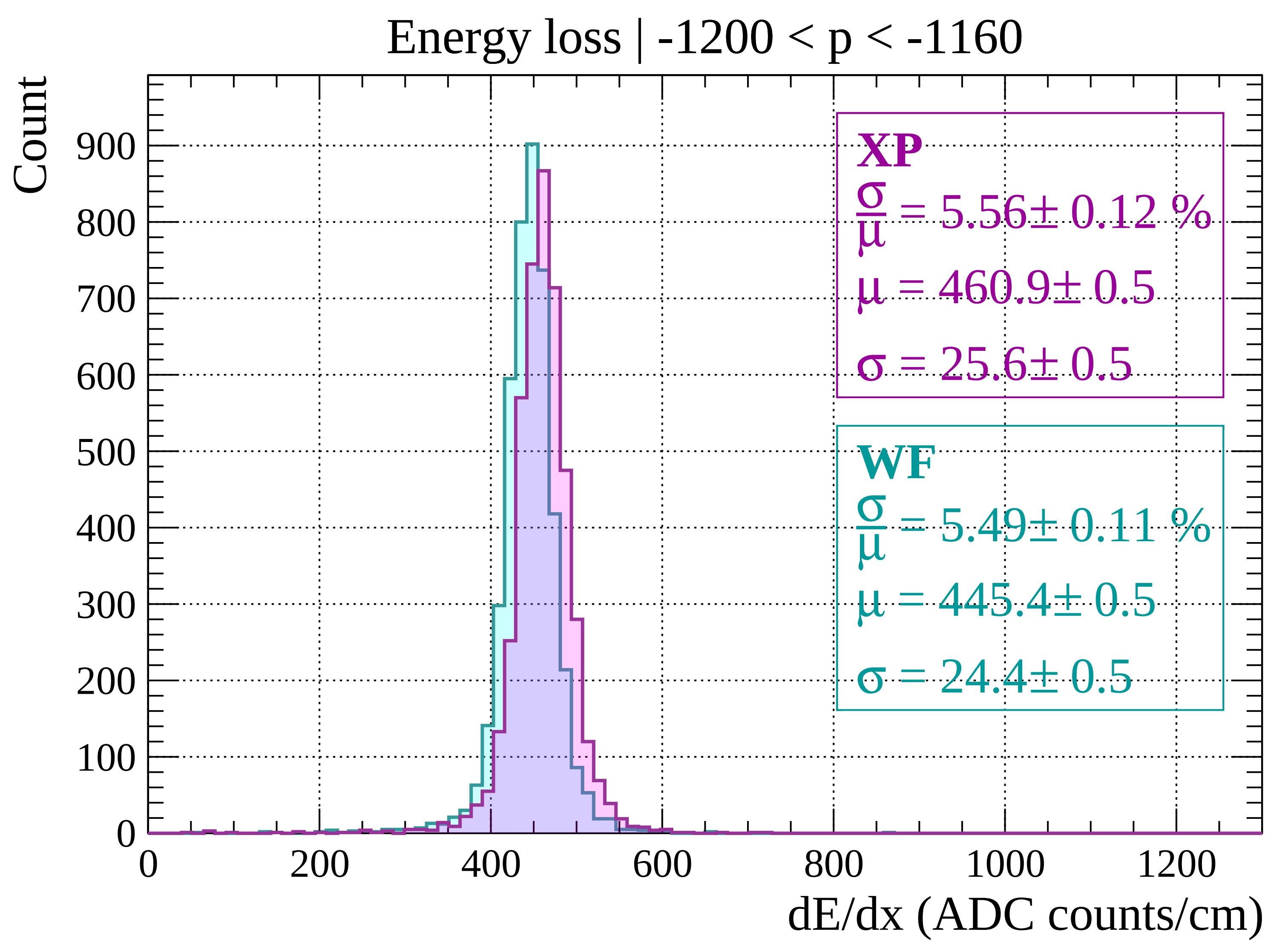


Energy loss | -1320 < p < -1280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 5.63± 0.11 %
μ = 467.4± 0.5
σ = 26.3± 0.5
WF
σ/μ = 5.37± 0.11 %
μ = 450.8± 0.5
σ = 24.2± 0.5

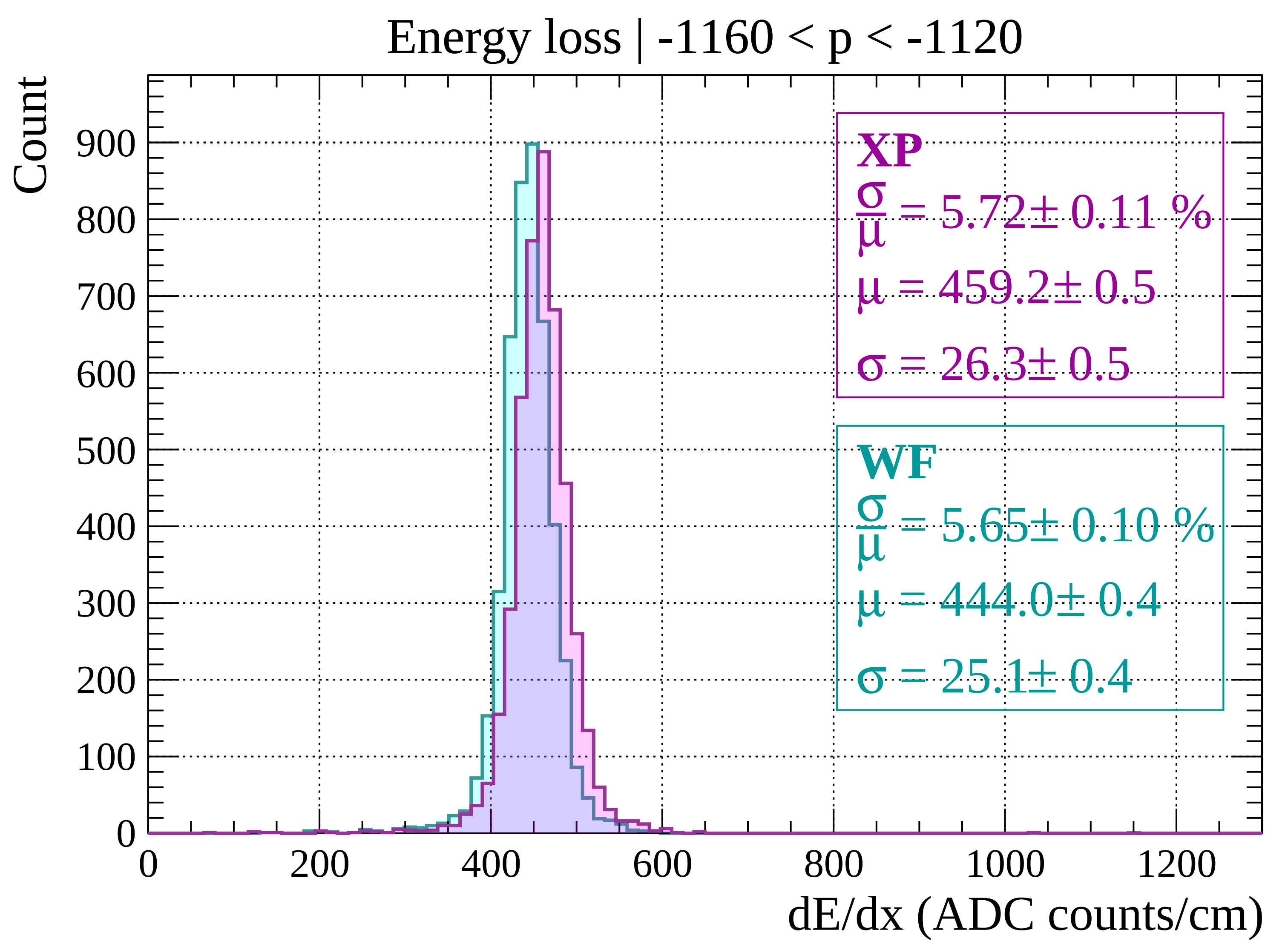

Energy loss | -1280 < p < -1240
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 5.62 \pm 0.11$ %
$\mu = 466.9 \pm 0.5$
$\sigma = 26.3 \pm 0.5$
WF
$\frac{\sigma}{\mu} = 5.46 \pm 0.12$ %
$\mu = 450.3 \pm 0.5$
$\sigma = 24.6 \pm 0.5$



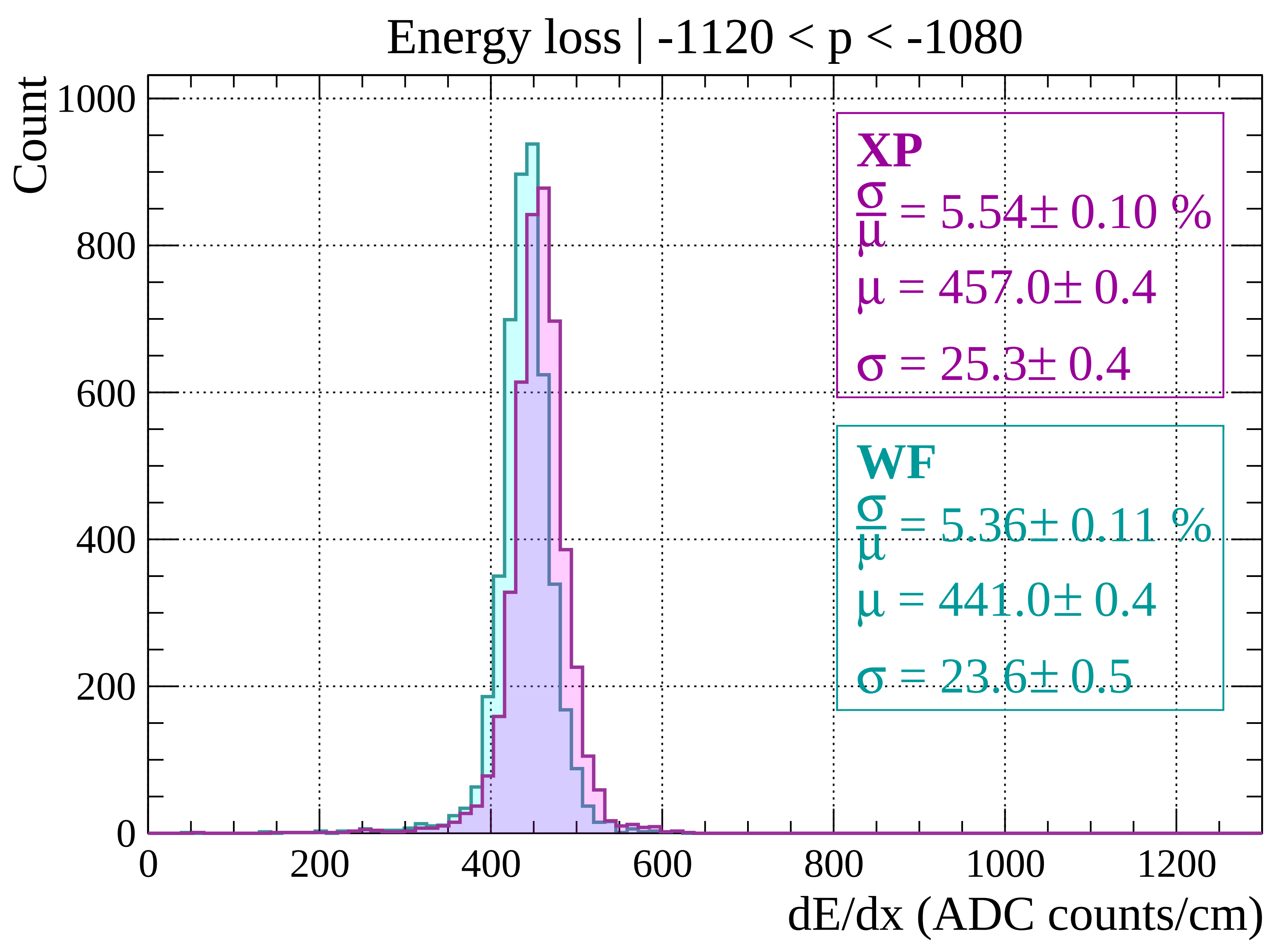
Energy loss | -1240 < p < -1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 5.23± 0.11 %
μ = 463.5± 0.5
σ = 24.2± 0.5
WF
σ/μ = 5.13± 0.10 %
μ = 447.7± 0.4
σ = 23.0± 0.4



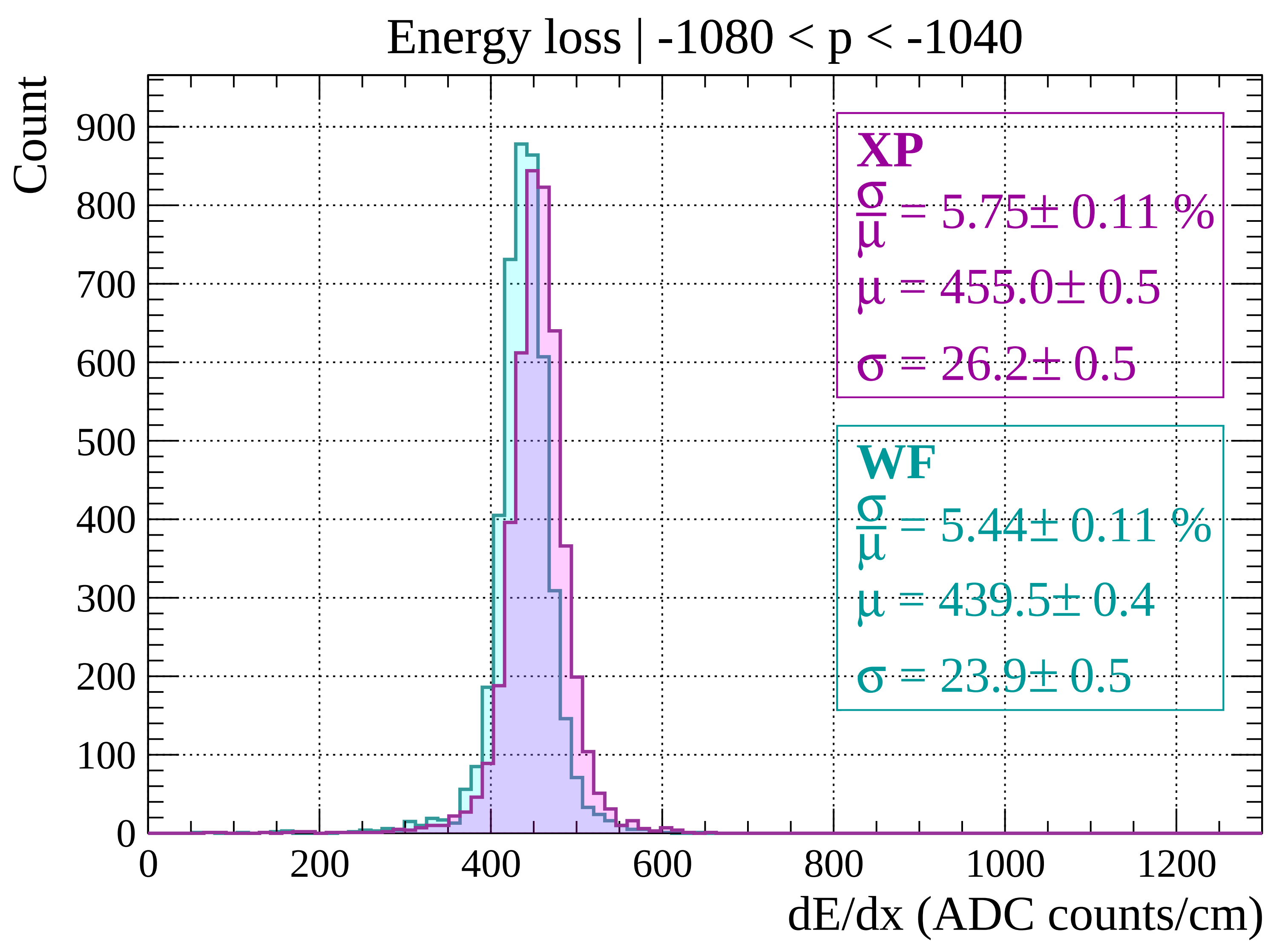
Energy loss | -1200 < p < -1160
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 5.56± 0.12 %
μ = 460.9± 0.5
σ = 25.6± 0.5
WF
σ/μ = 5.49± 0.11 %
μ = 445.4± 0.5
σ = 24.4± 0.5



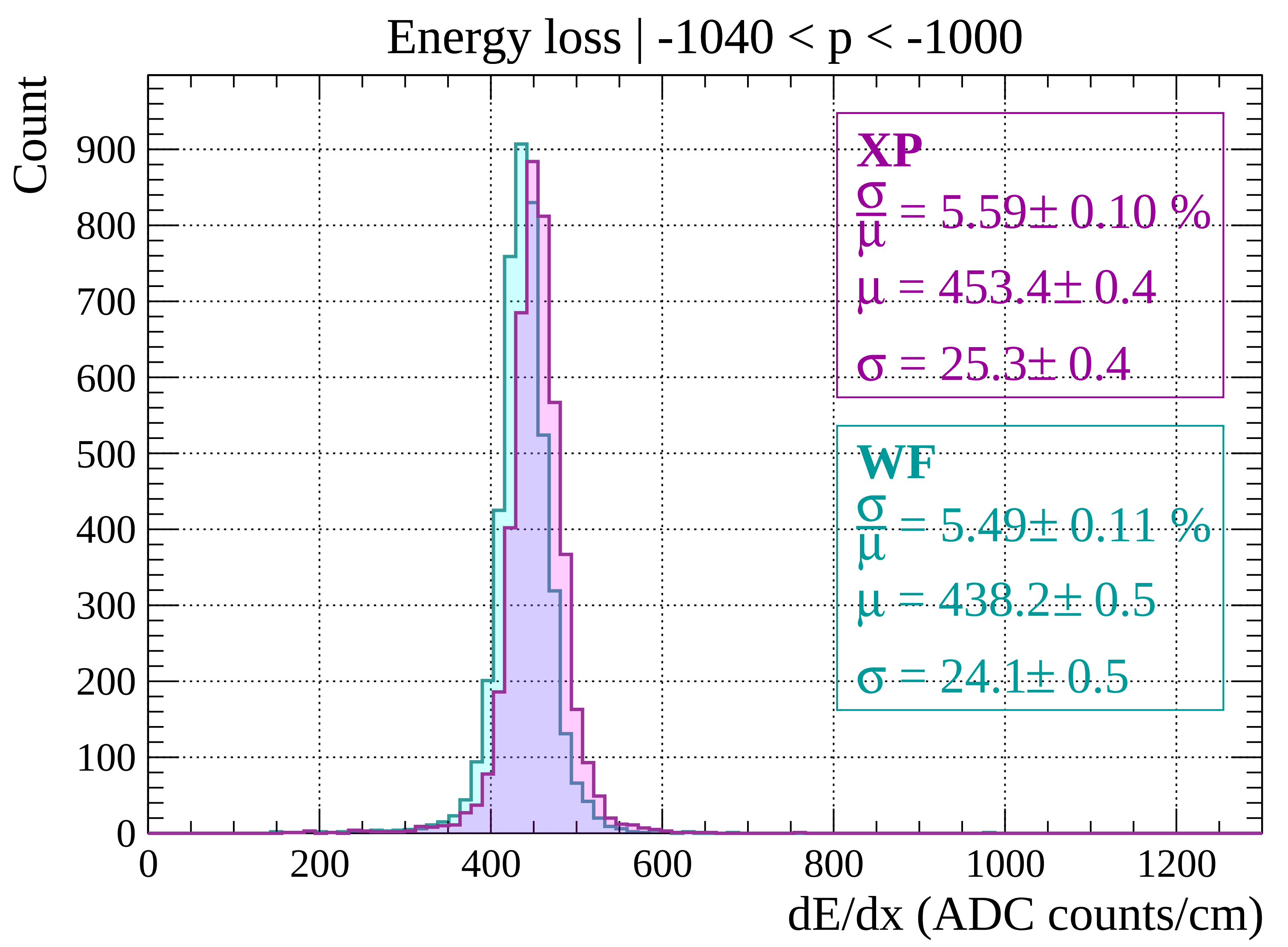
Energy loss | -1160 < p < -1120
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 5.72± 0.11 %
μ = 459.2± 0.5
σ = 26.3± 0.5
WF
σ/μ = 5.65± 0.10 %
μ = 444.0± 0.4
σ = 25.1± 0.4



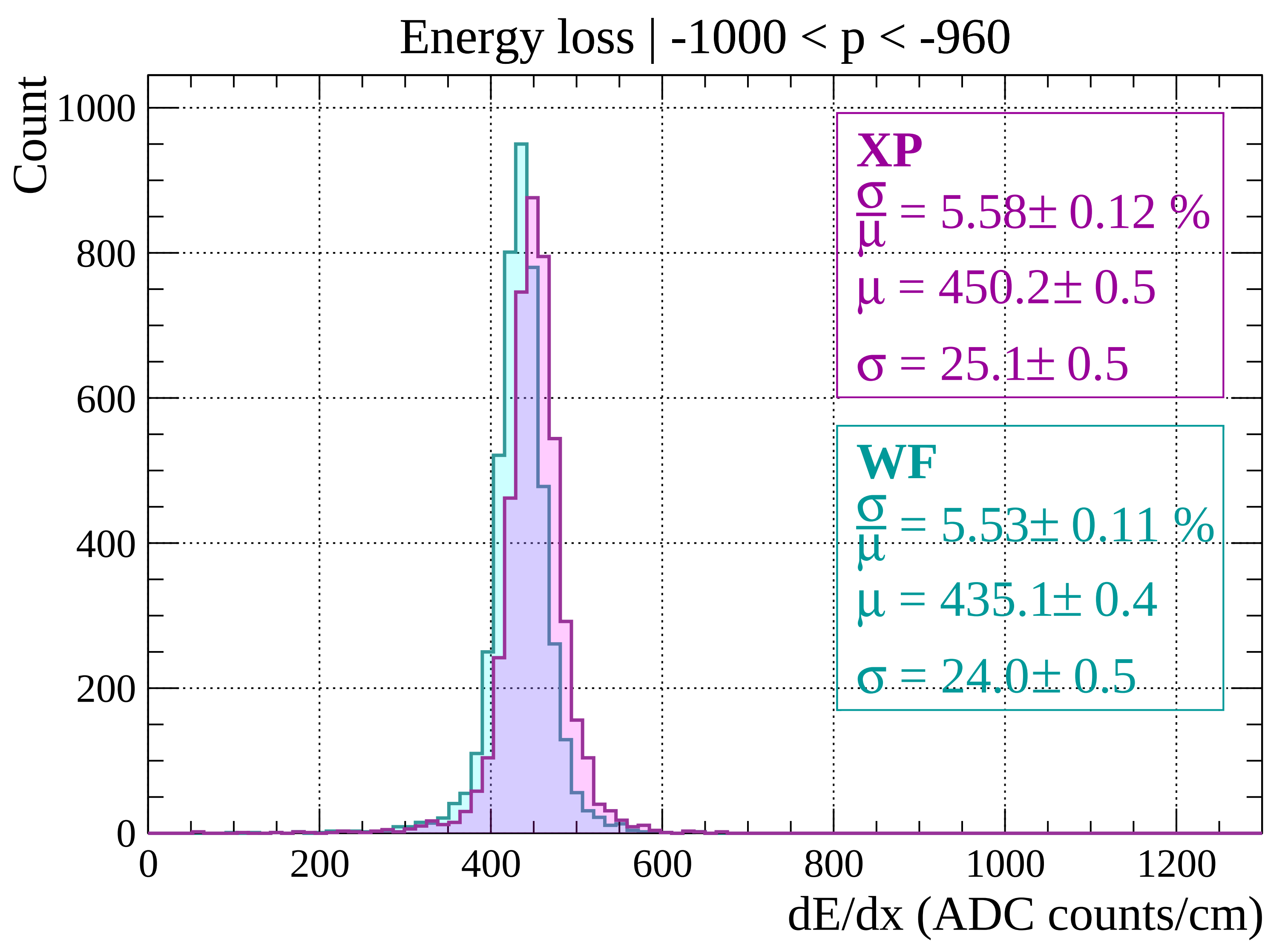
Energy loss | -1120 < p < -1080
Count
1000
800
600
400
200
0
XP
σ/μ = 5.54± 0.10 %
μ = 457.0± 0.4
σ = 25.3± 0.4
WF
σ/μ = 5.36± 0.11 %
μ = 441.0± 0.4
σ = 23.6± 0.5
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



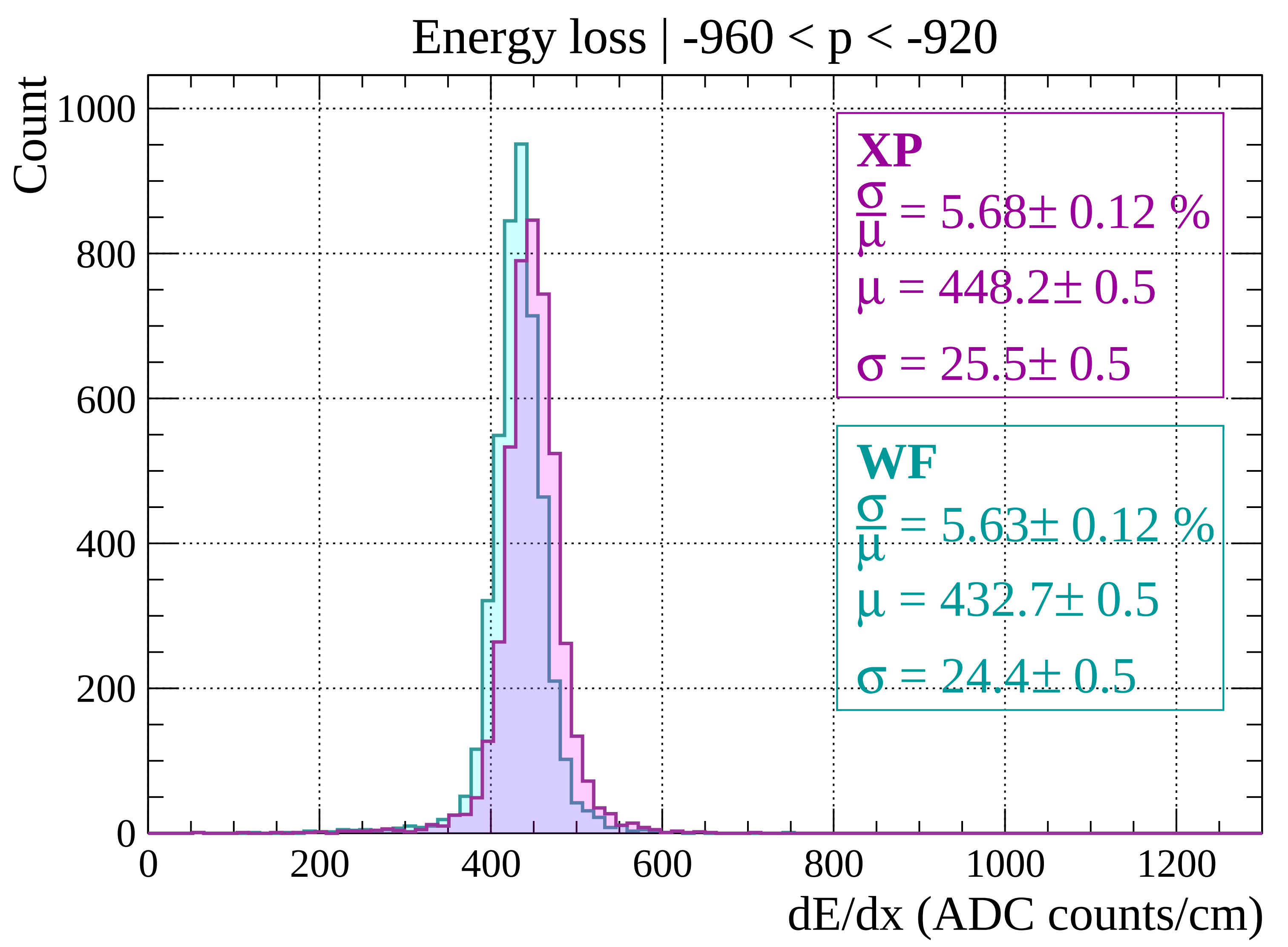
Energy loss | -1080 < p < -1040
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.75± 0.11 %
μ = 455.0± 0.5
σ = 26.2± 0.5
WF
$\frac{\sigma}{\mu}$ = 5.44± 0.11 %
μ = 439.5± 0.4
σ = 23.9± 0.5



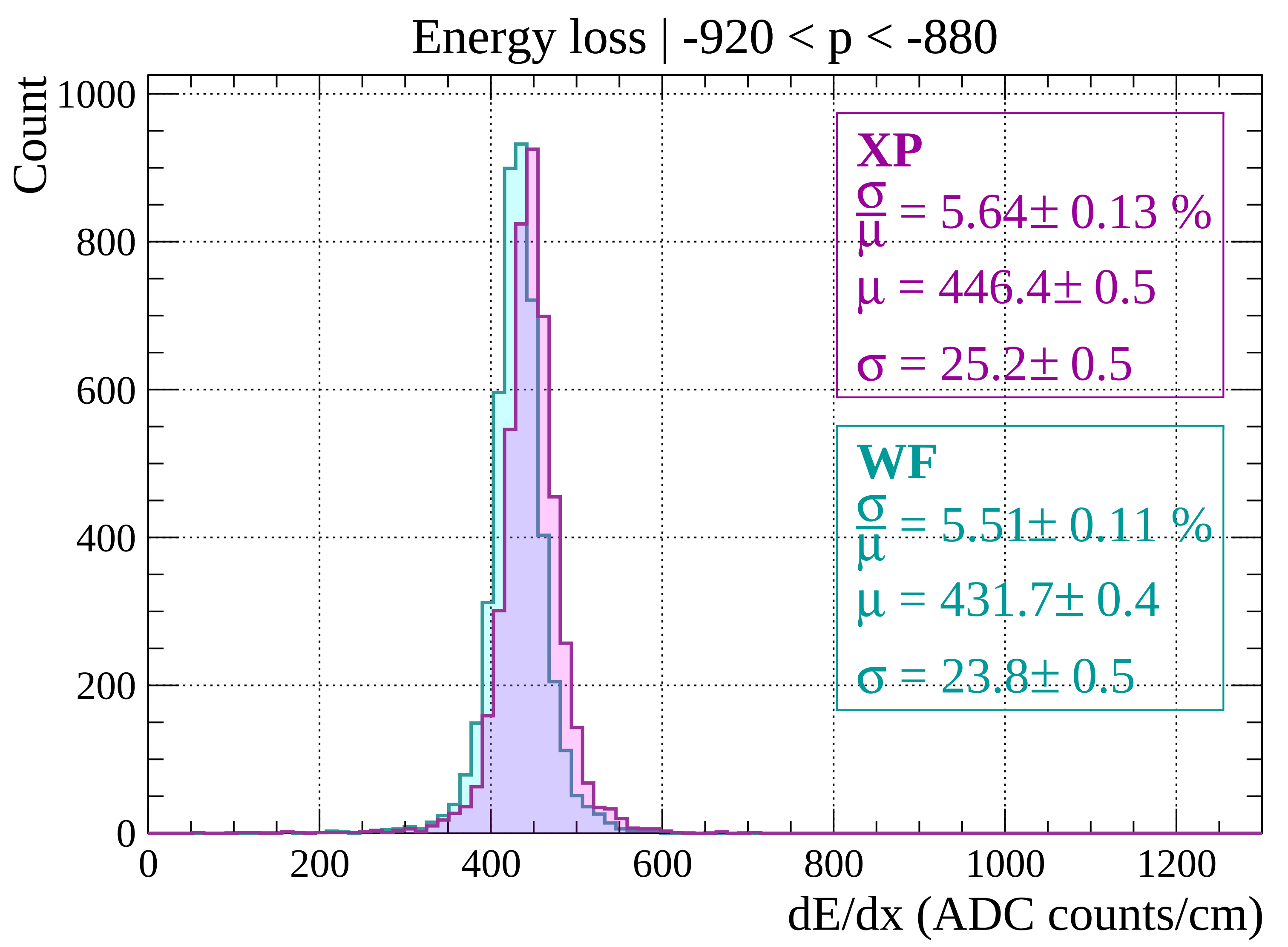
Energy loss | -1040 < p < -1000
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 5.59 \pm 0.10$ %
$\mu = 453.4 \pm 0.4$
$\sigma = 25.3 \pm 0.4$
WF
$\frac{\sigma}{\mu} = 5.49 \pm 0.11$ %
$\mu = 438.2 \pm 0.5$
$\sigma = 24.1 \pm 0.5$

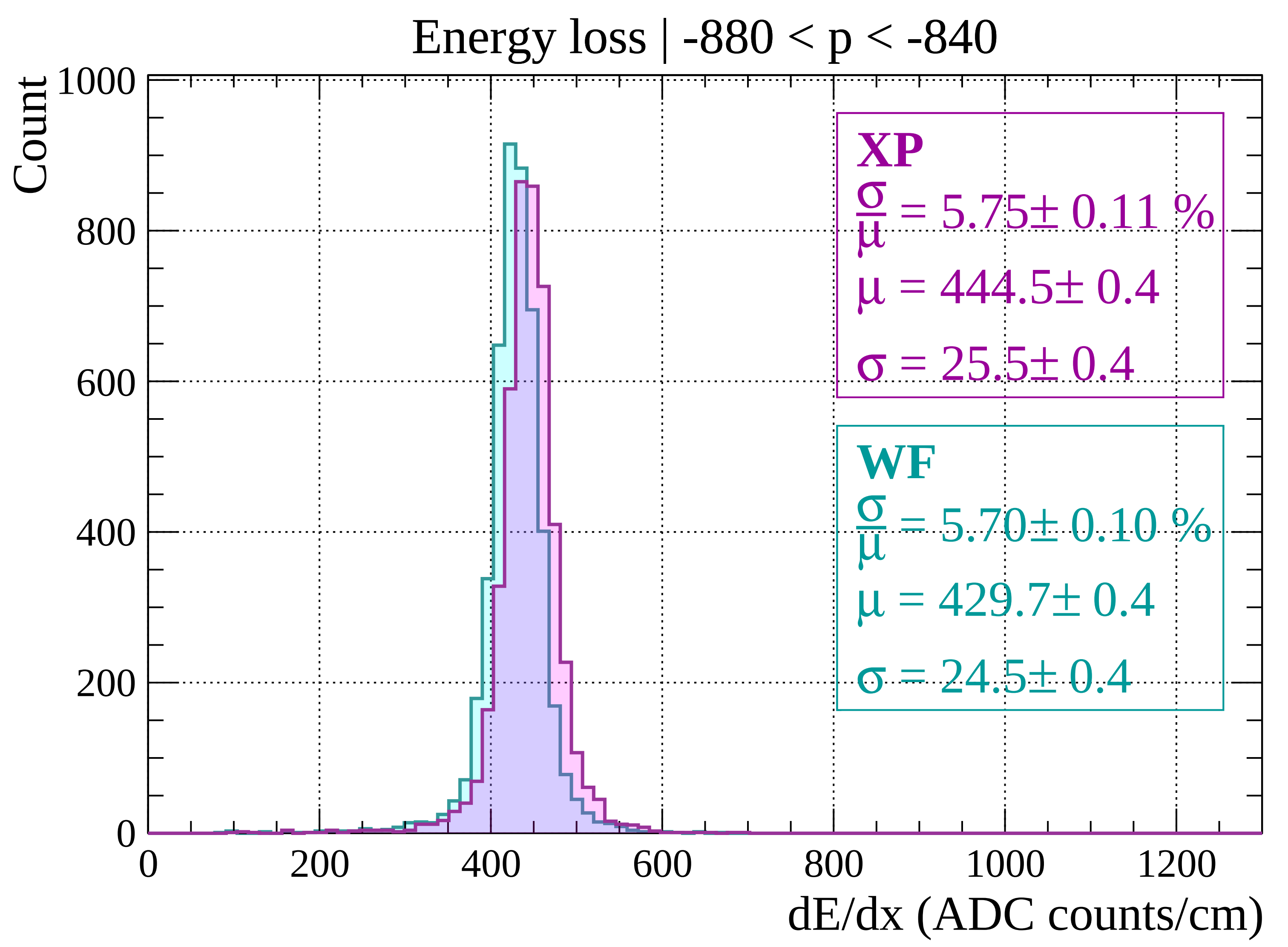


Energy loss | -1000 < p < -960
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.58± 0.12 %
μ = 450.2± 0.5
σ = 25.1± 0.5
WF
$\frac{\sigma}{\mu}$ = 5.53± 0.11 %
μ = 435.1± 0.4
σ = 24.0± 0.5

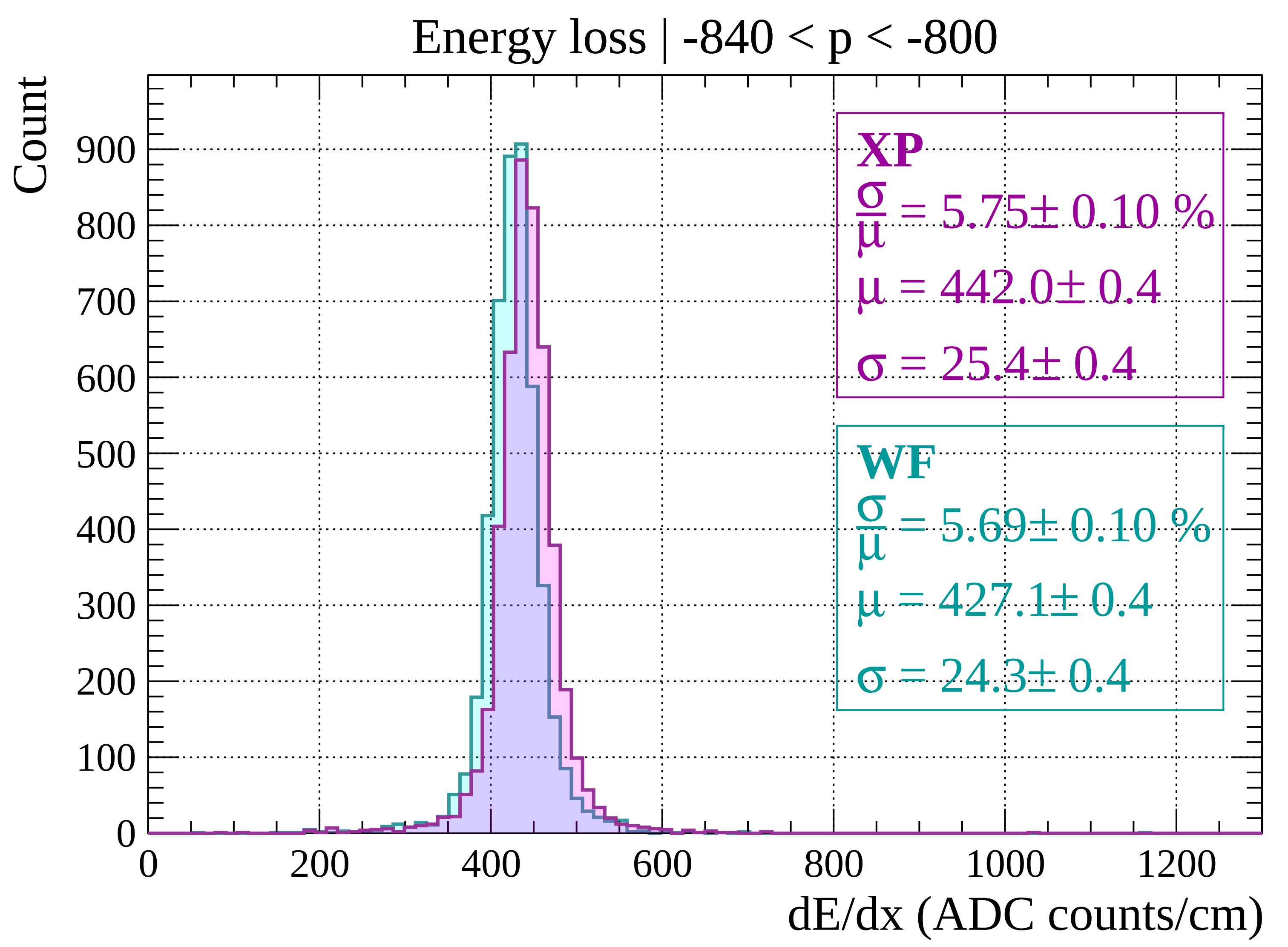


Energy loss | -960 < p < -920
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.68± 0.12 %
μ = 448.2± 0.5
σ = 25.5± 0.5
WF
$\frac{\sigma}{\mu}$ = 5.63± 0.12 %
μ = 432.7± 0.5
σ = 24.4± 0.5

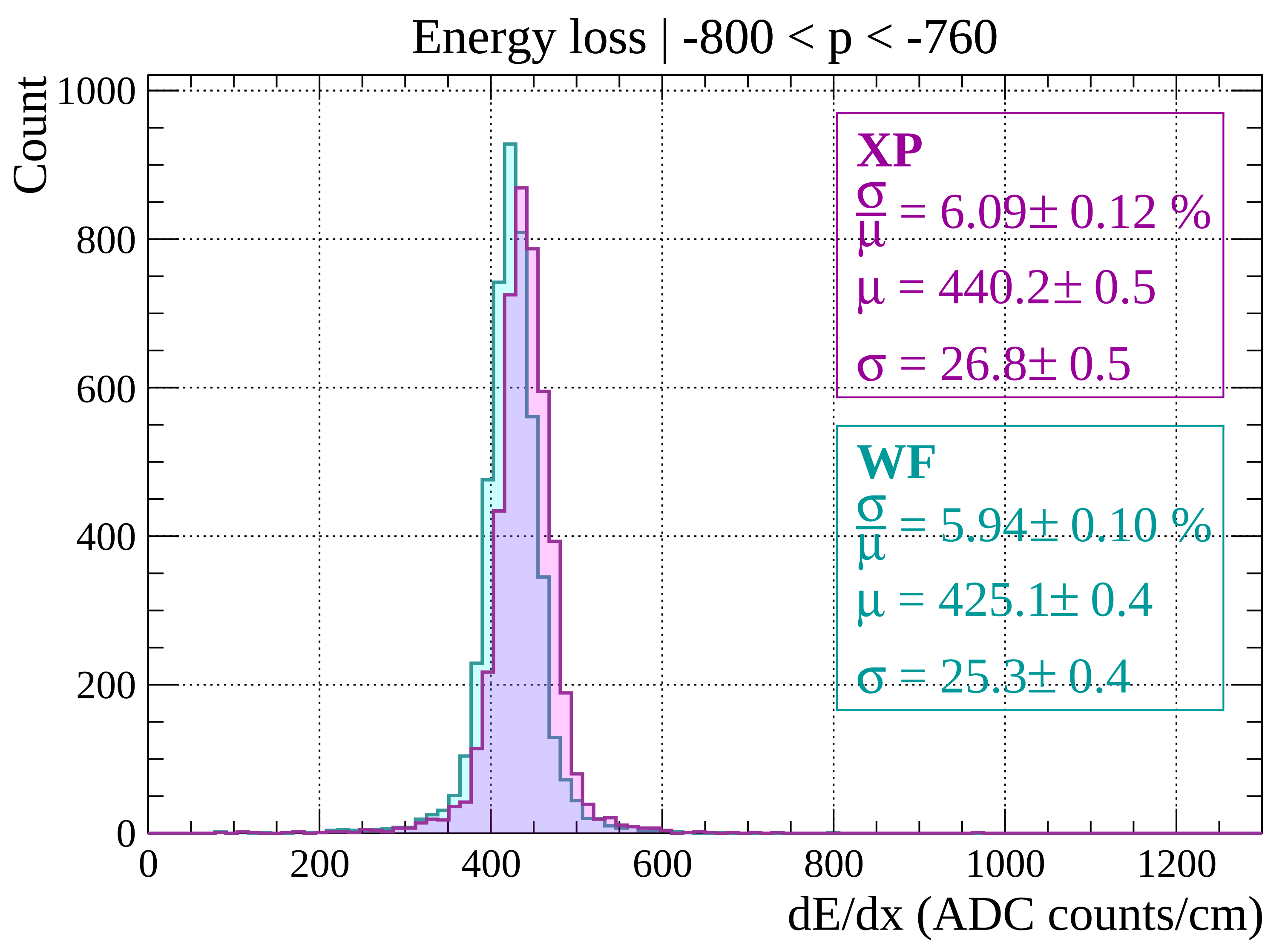

Energy loss | -920 < p < -880
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.64 ± 0.13 %
μ = 446.4 ± 0.5
σ = 25.2 ± 0.5
WF
σ/μ = 5.51 ± 0.11 %
μ = 431.7 ± 0.4
σ = 23.8 ± 0.5



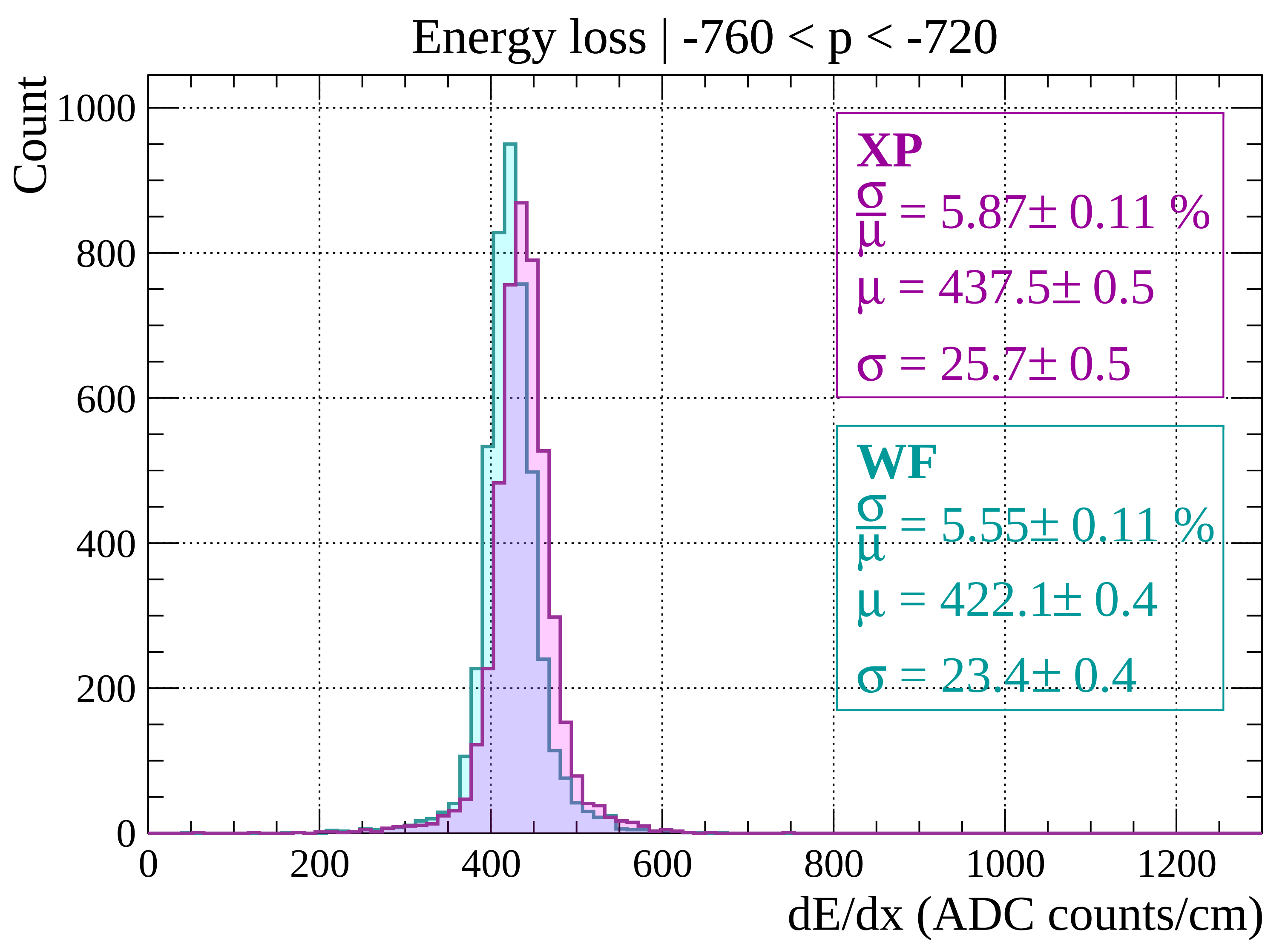
Energy loss | -880 < p < -840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.75± 0.11 %
μ = 444.5± 0.4
σ = 25.5± 0.4
WF
$\frac{\sigma}{\mu}$ = 5.70± 0.10 %
μ = 429.7± 0.4
σ = 24.5± 0.4



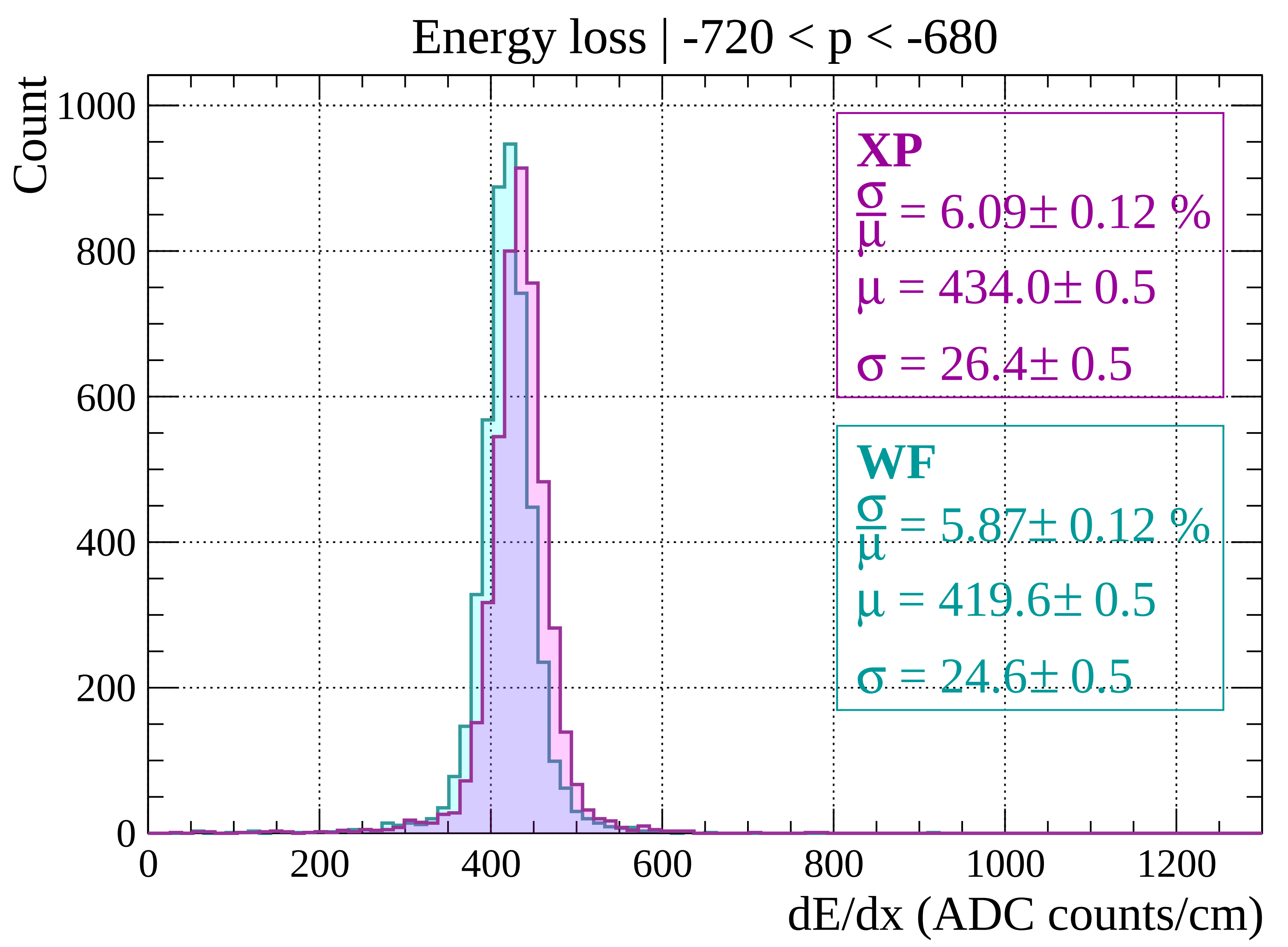
Energy loss | -840 < p < -800
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.75± 0.10 %
μ = 442.0± 0.4
σ = 25.4± 0.4
WF
σ/μ = 5.69± 0.10 %
μ = 427.1± 0.4
σ = 24.3± 0.4



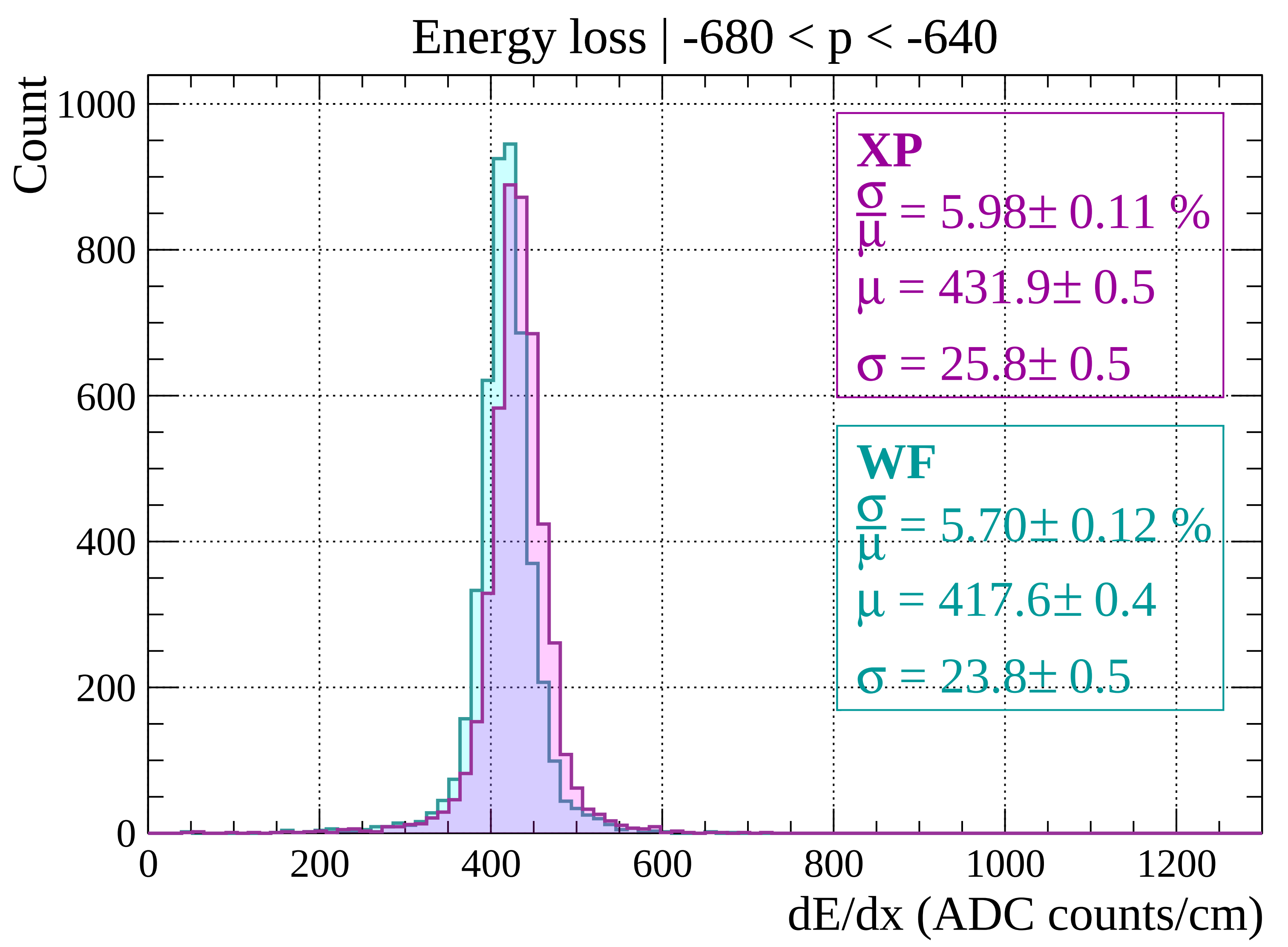
Energy loss | -800 < p < -760
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 6.09 \pm 0.12$ %
$\mu = 440.2 \pm 0.5$
$\sigma = 26.8 \pm 0.5$
WF
$\frac{\sigma}{\mu} = 5.94 \pm 0.10$ %
$\mu = 425.1 \pm 0.4$
$\sigma = 25.3 \pm 0.4$

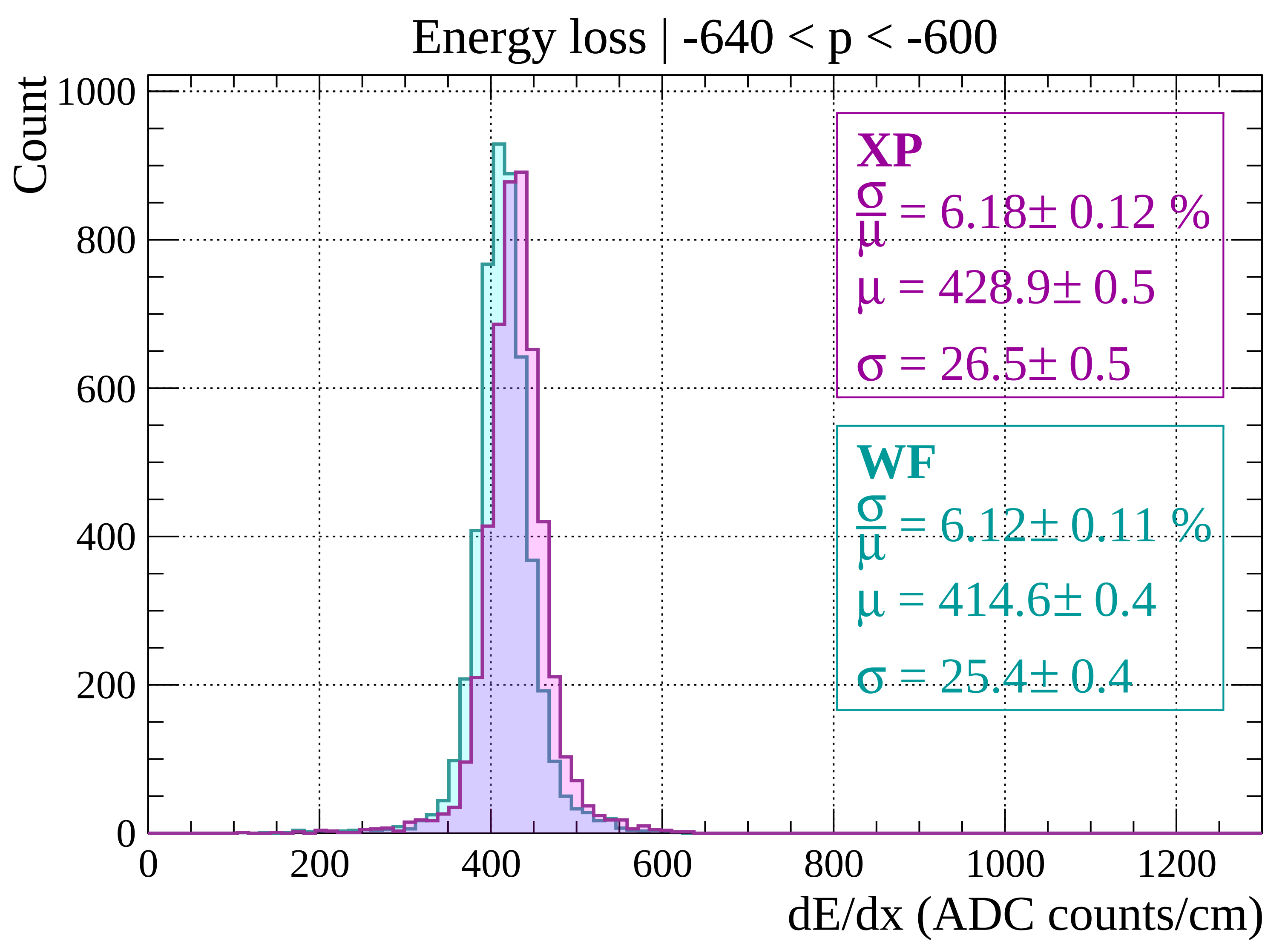


Energy loss | -760 < p < -720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 5.87± 0.11 %
μ = 437.5± 0.5
σ = 25.7± 0.5
WF
σ/μ = 5.55± 0.11 %
μ = 422.1± 0.4
σ = 23.4± 0.4

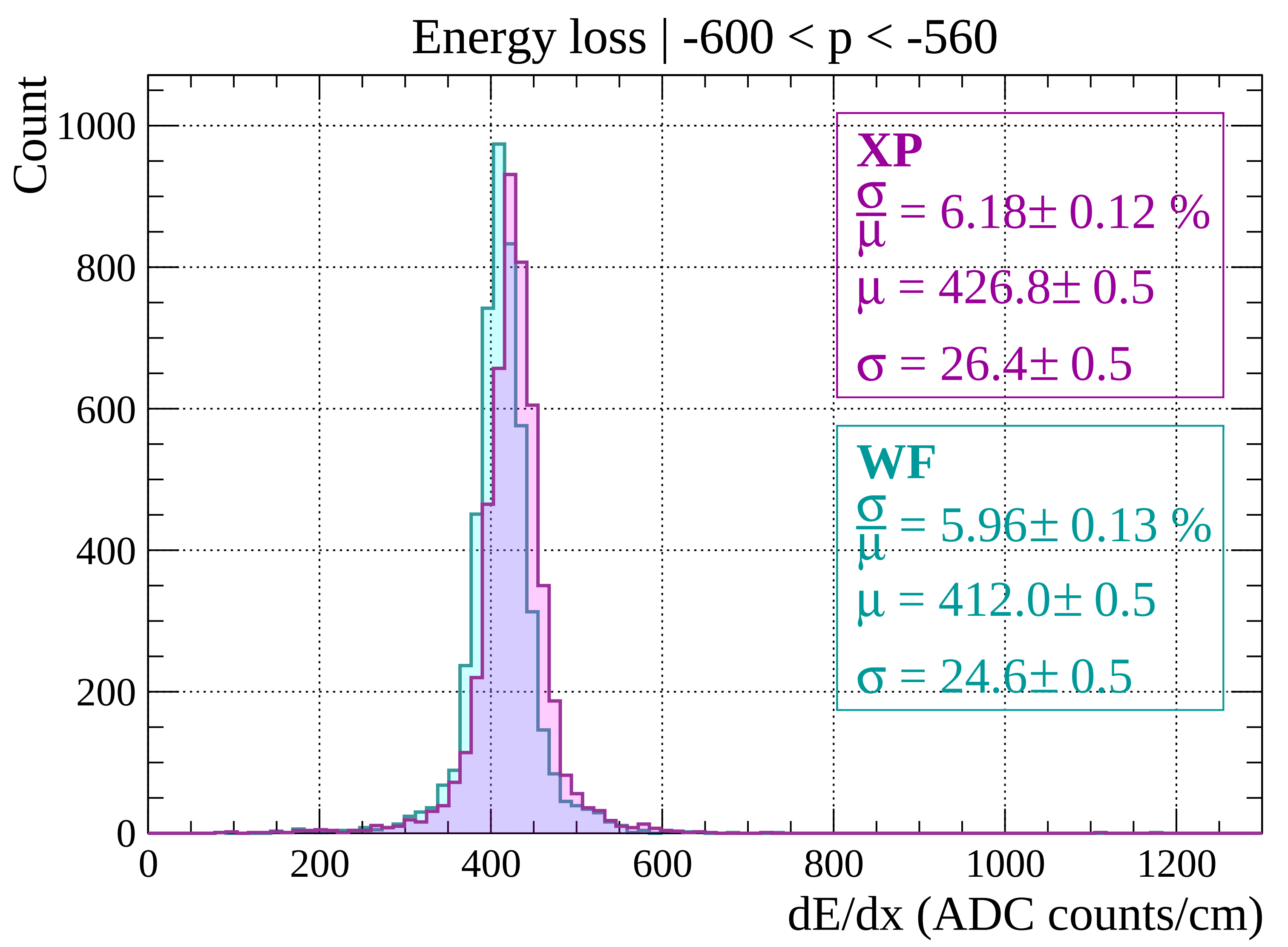


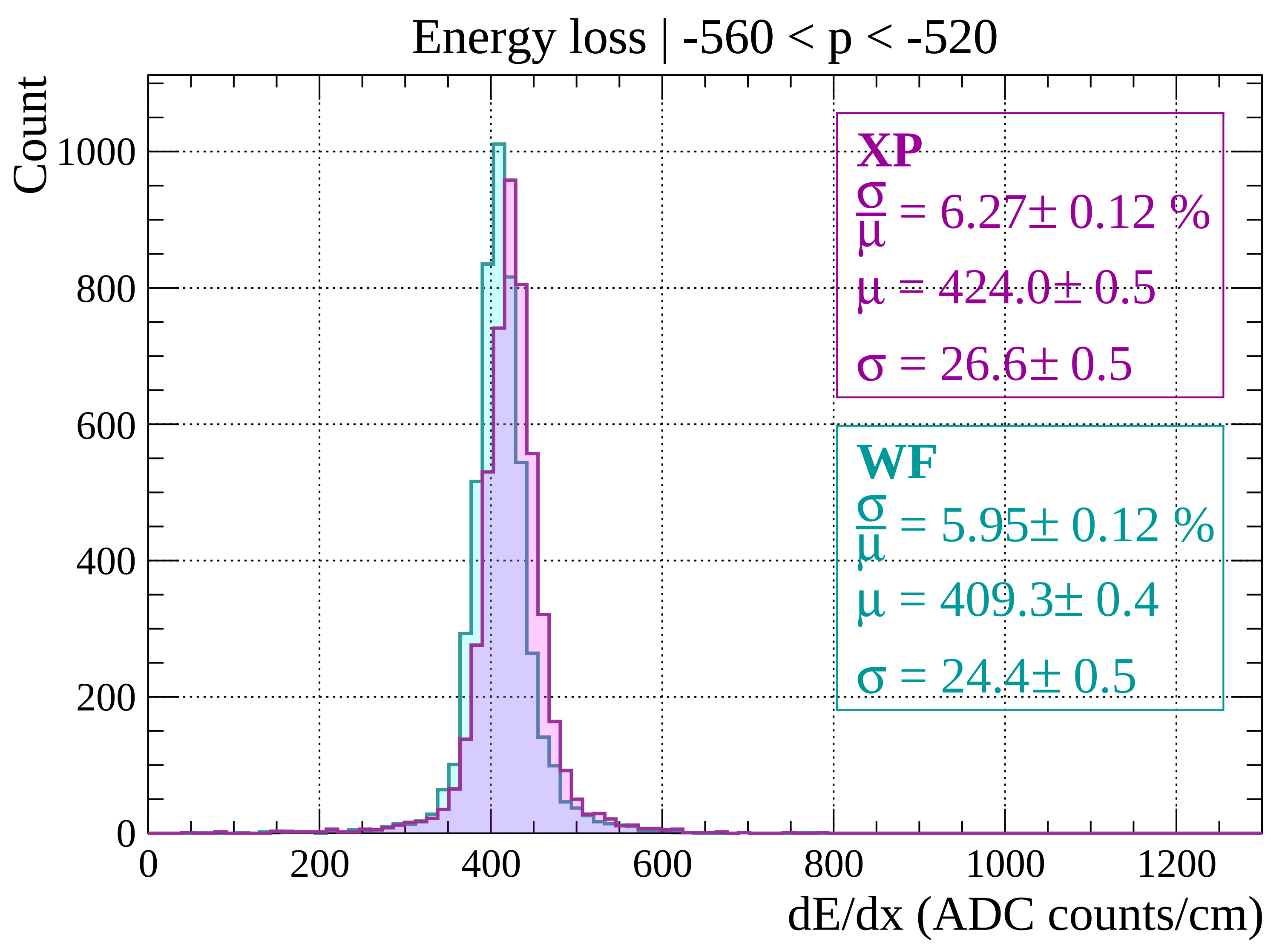
Energy loss | -720 < p < -680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 6.09 ± 0.12 %
μ = 434.0 ± 0.5
σ = 26.4 ± 0.5
WF
σ/μ = 5.87 ± 0.12 %
μ = 419.6 ± 0.5
σ = 24.6 ± 0.5

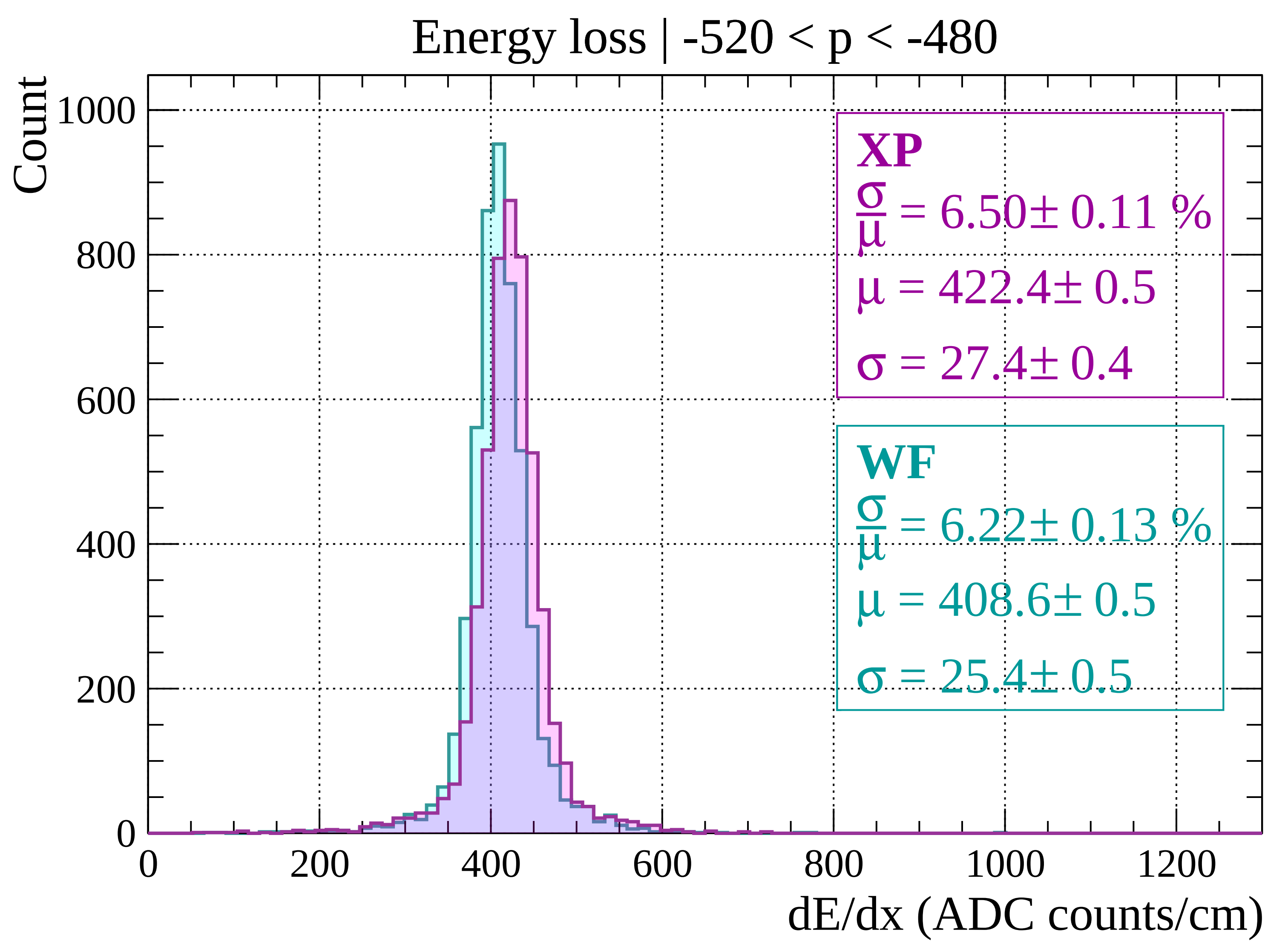

Energy loss | -680 < p < -640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 5.98± 0.11 %
μ = 431.9± 0.5
σ = 25.8± 0.5
WF
$\frac{\sigma}{\mu}$ = 5.70± 0.12 %
μ = 417.6± 0.4
σ = 23.8± 0.5



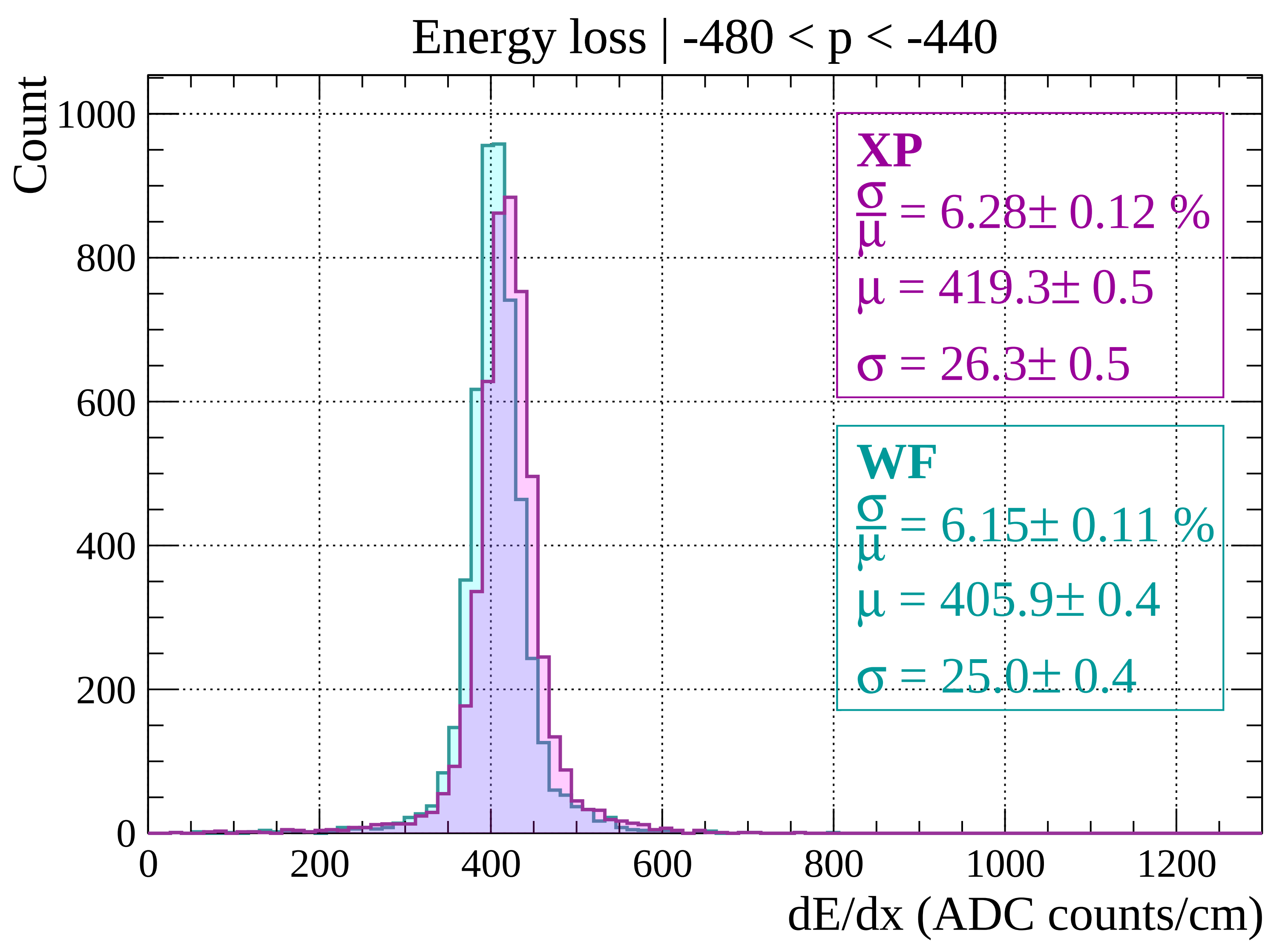
Energy loss | -640 < p < -600
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.18± 0.12 %
μ = 428.9± 0.5
σ = 26.5± 0.5
WF
$\frac{\sigma}{\mu}$ = 6.12± 0.11 %
μ = 414.6± 0.4
σ = 25.4± 0.4


Energy loss | -600 < p < -560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 6.18± 0.12 %
μ = 426.8± 0.5
σ = 26.4± 0.5
WF
σ/μ = 5.96± 0.13 %
μ = 412.0± 0.5
σ = 24.6± 0.5

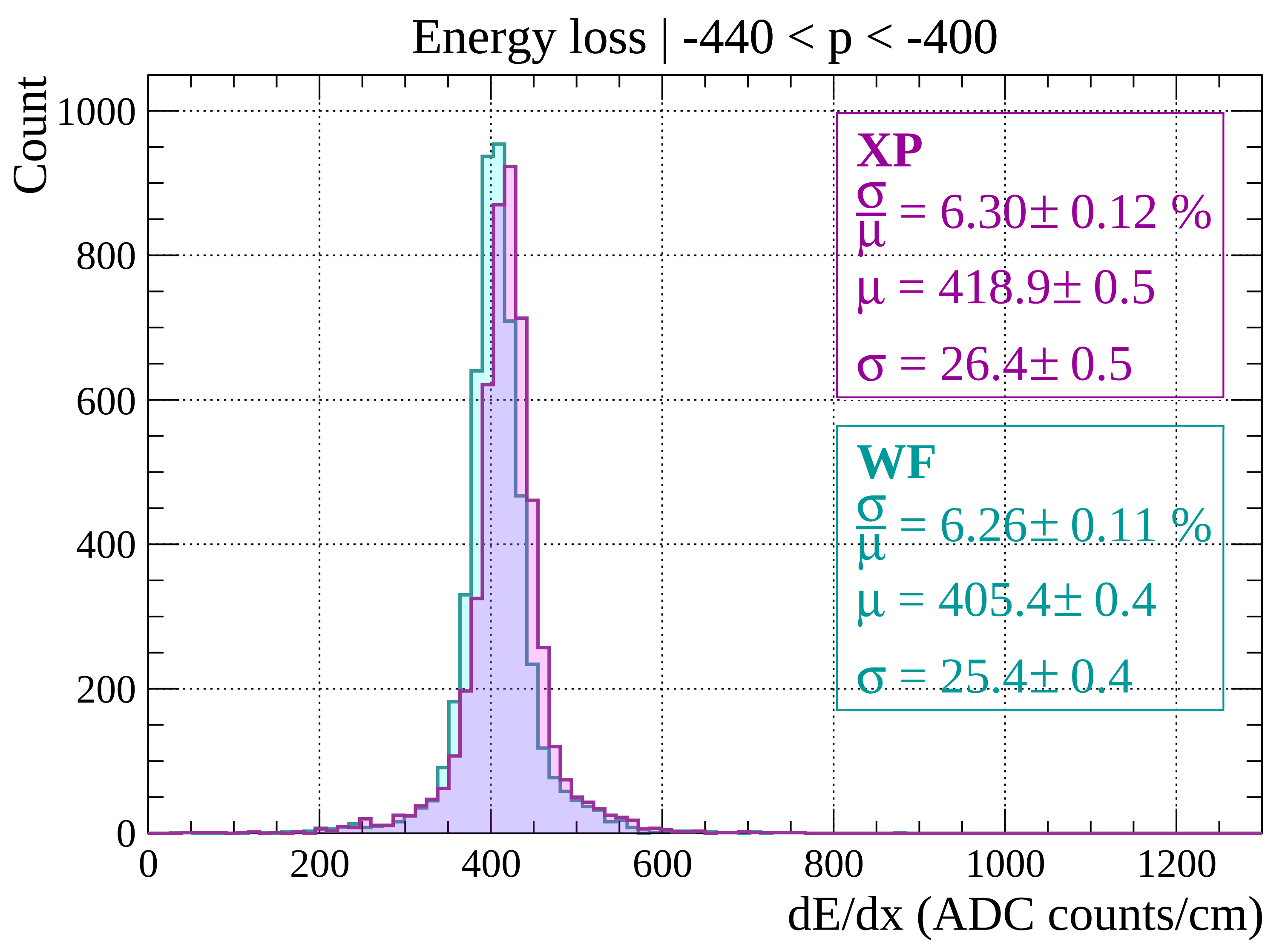

Energy loss | -560 < p < -520
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 6.27± 0.12 %
μ = 424.0± 0.5
σ = 26.6± 0.5
WF
σ/μ = 5.95± 0.12 %
μ = 409.3± 0.4
σ = 24.4± 0.5
0
200
400
600
800
1000
1200



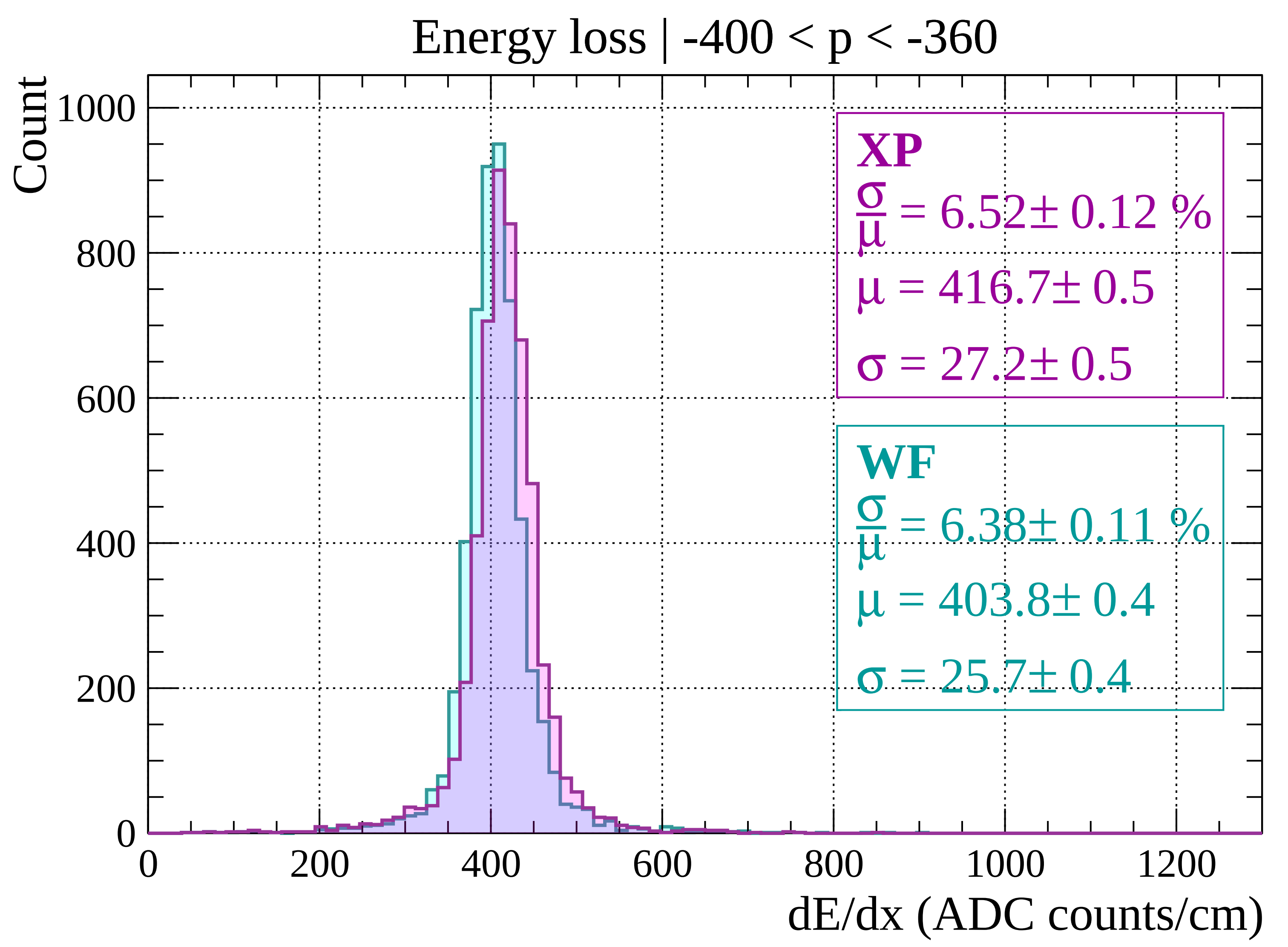
Energy loss | -520 < p < -480
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.50± 0.11 %
μ = 422.4± 0.5
σ = 27.4± 0.4
WF
$\frac{\sigma}{\mu}$ = 6.22± 0.13 %
μ = 408.6± 0.5
σ = 25.4± 0.5



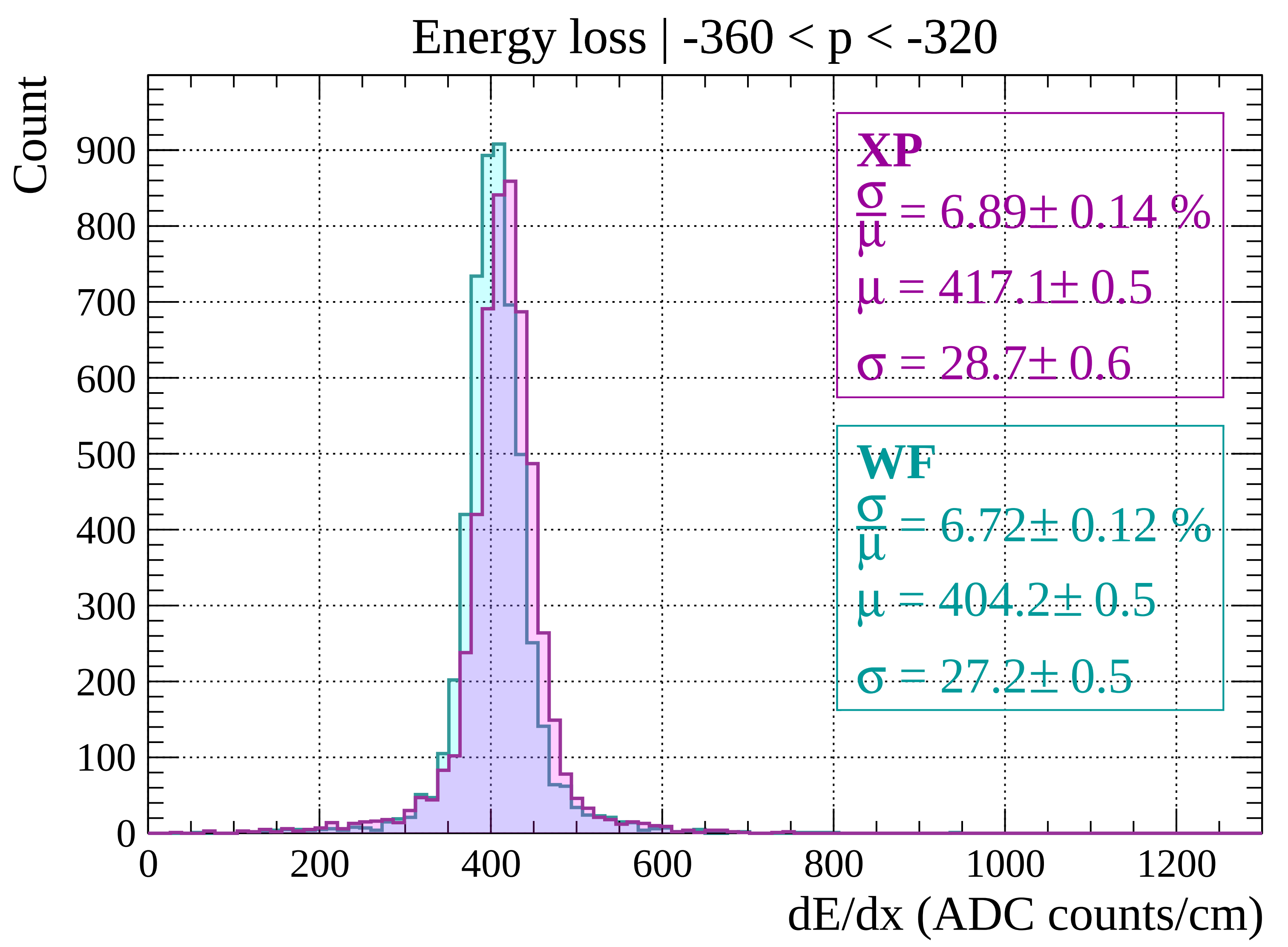
Energy loss | -480 < p < -440
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.28± 0.12 %
μ = 419.3± 0.5
σ = 26.3± 0.5
WF
σ/μ = 6.15± 0.11 %
μ = 405.9± 0.4
σ = 25.0± 0.4



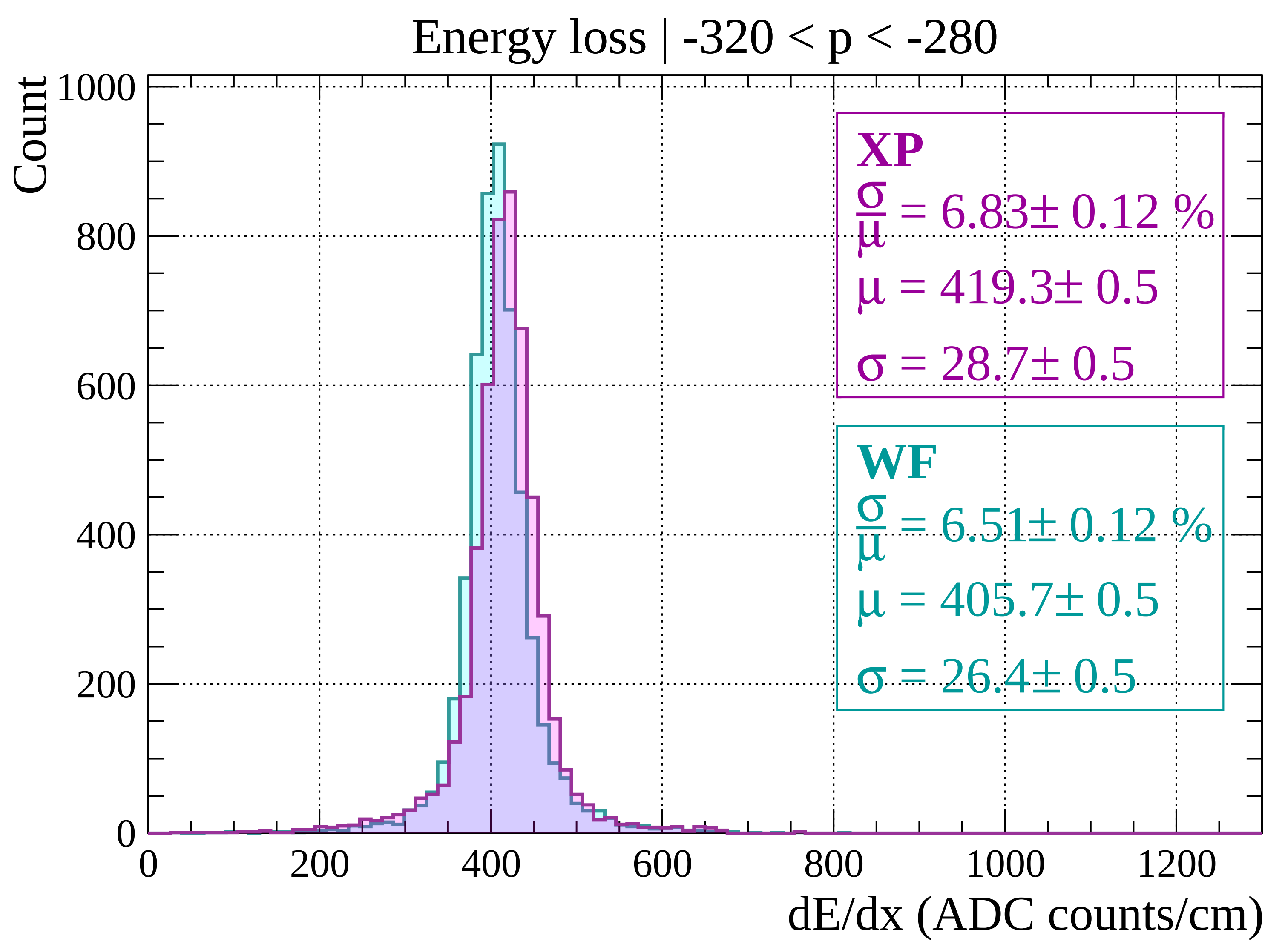
Energy loss | -440 < p < -400
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 6.30± 0.12 %
μ = 418.9± 0.5
σ = 26.4± 0.5
WF
σ/μ = 6.26± 0.11 %
μ = 405.4± 0.4
σ = 25.4± 0.4
0
200
400
600
800
1000
1200



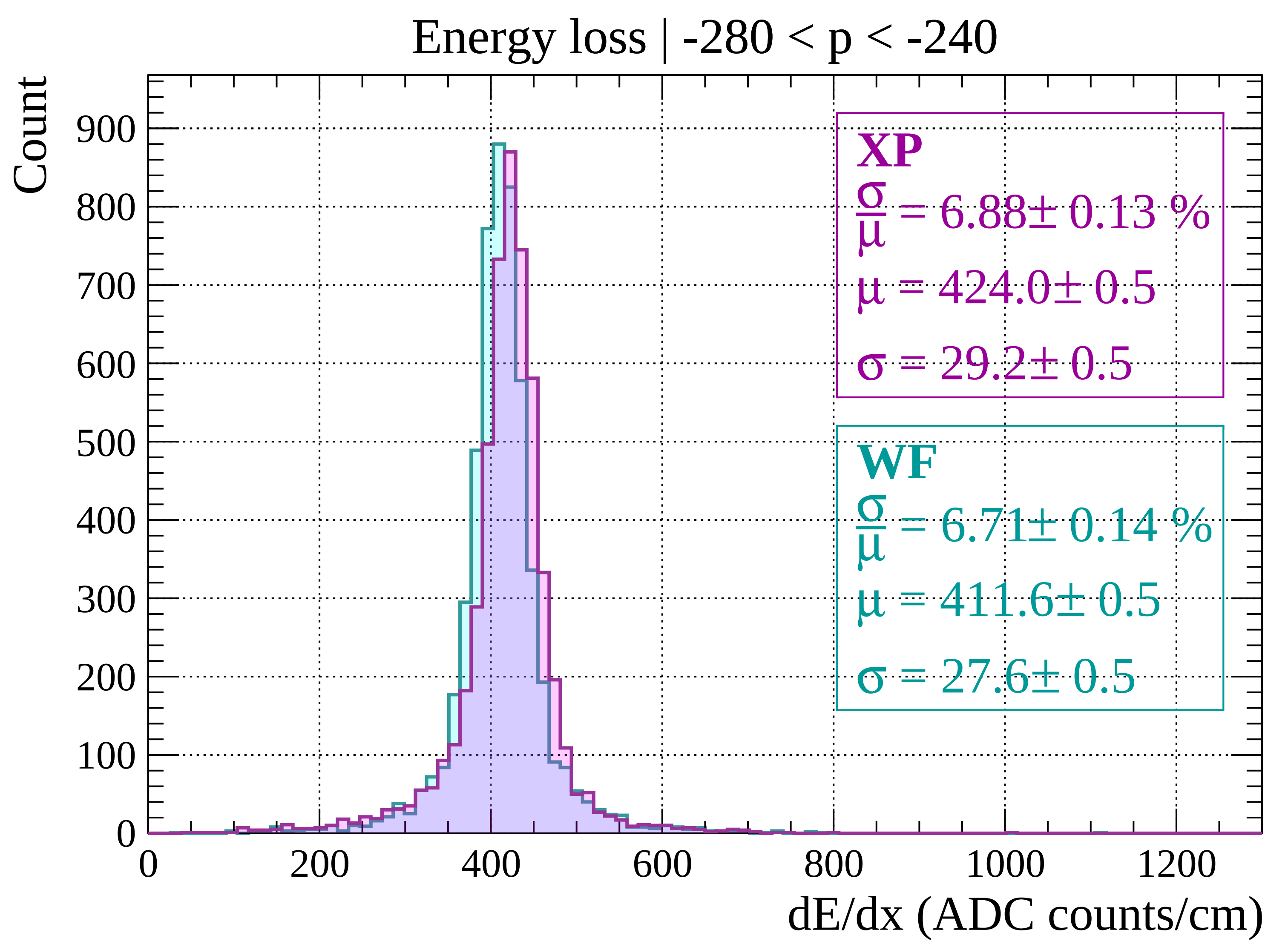
Energy loss | -400 < p < -360
Count
0
200
400
600
800
1000
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.52± 0.12 %
μ = 416.7± 0.5
σ = 27.2± 0.5
WF
σ/μ = 6.38± 0.11 %
μ = 403.8± 0.4
σ = 25.7± 0.4

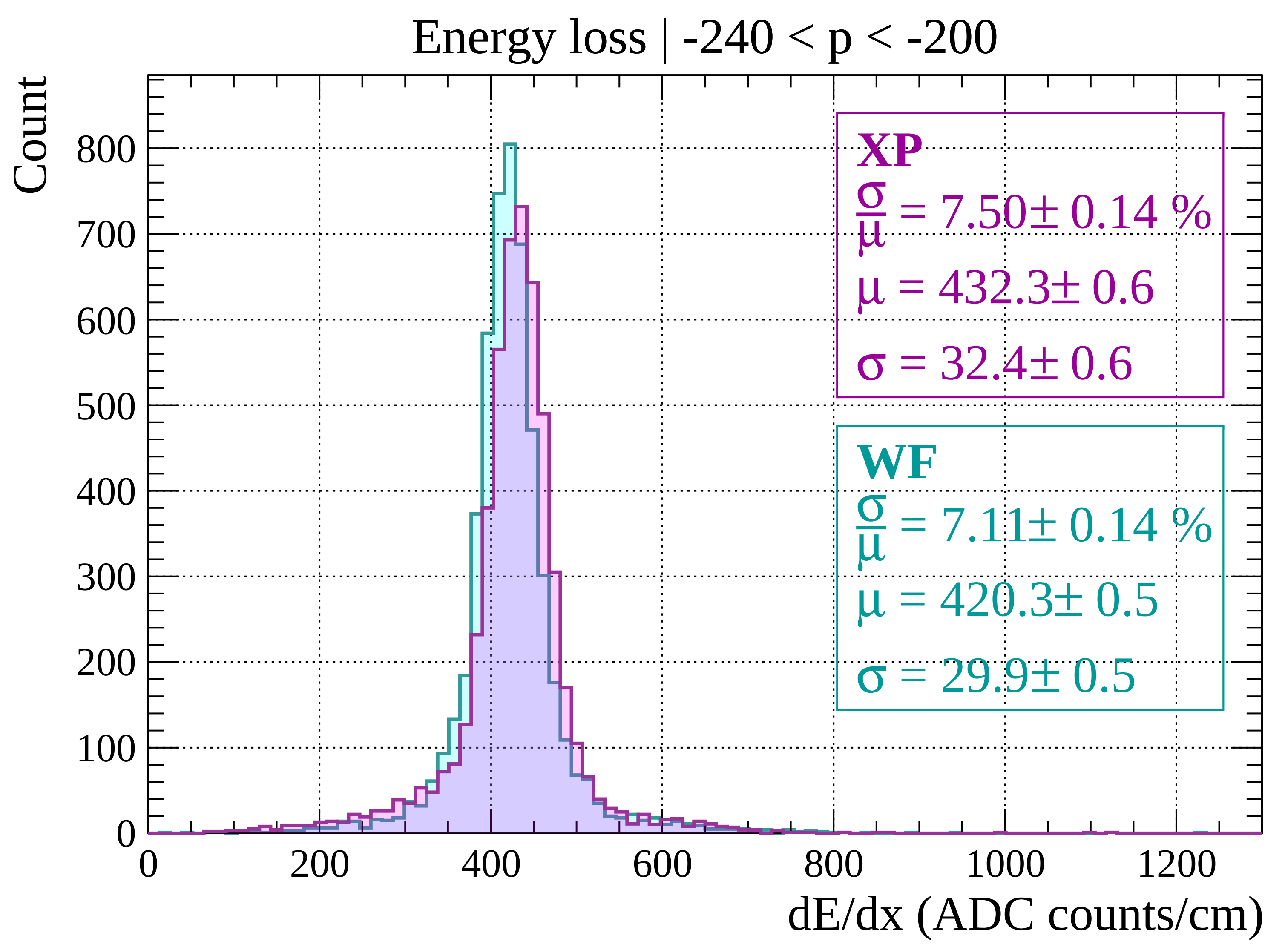


Energy loss | -360 < p < -320
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 6.89 \pm 0.14$ %
$\mu = 417.1 \pm 0.5$
$\sigma = 28.7 \pm 0.6$
WF
$\frac{\sigma}{\mu} = 6.72 \pm 0.12$ %
$\mu = 404.2 \pm 0.5$
$\sigma = 27.2 \pm 0.5$

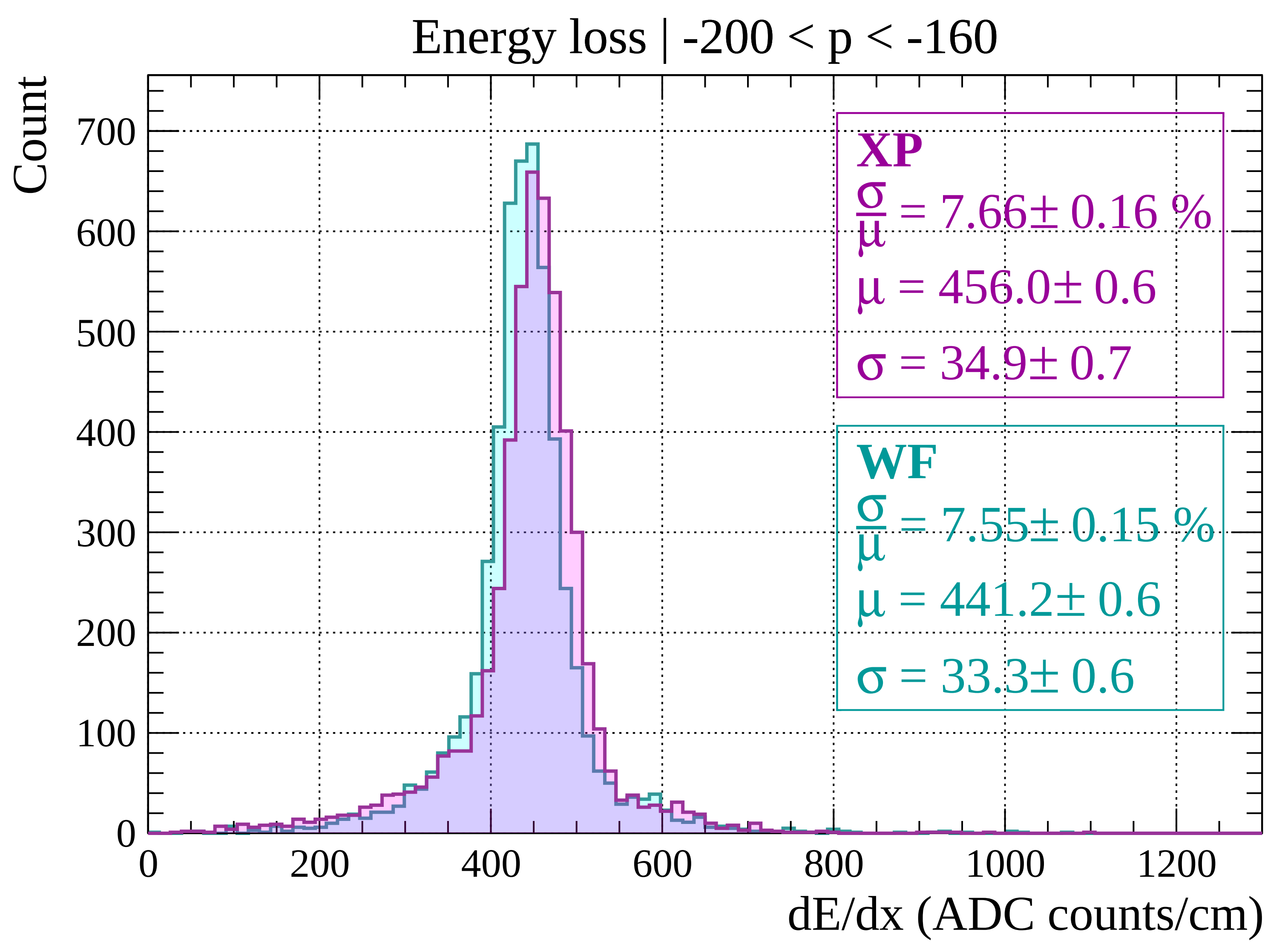


Energy loss | -320 < p < -280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 6.83± 0.12 %
μ = 419.3± 0.5
σ = 28.7± 0.5
WF
σ/μ = 6.51± 0.12 %
μ = 405.7± 0.5
σ = 26.4± 0.5

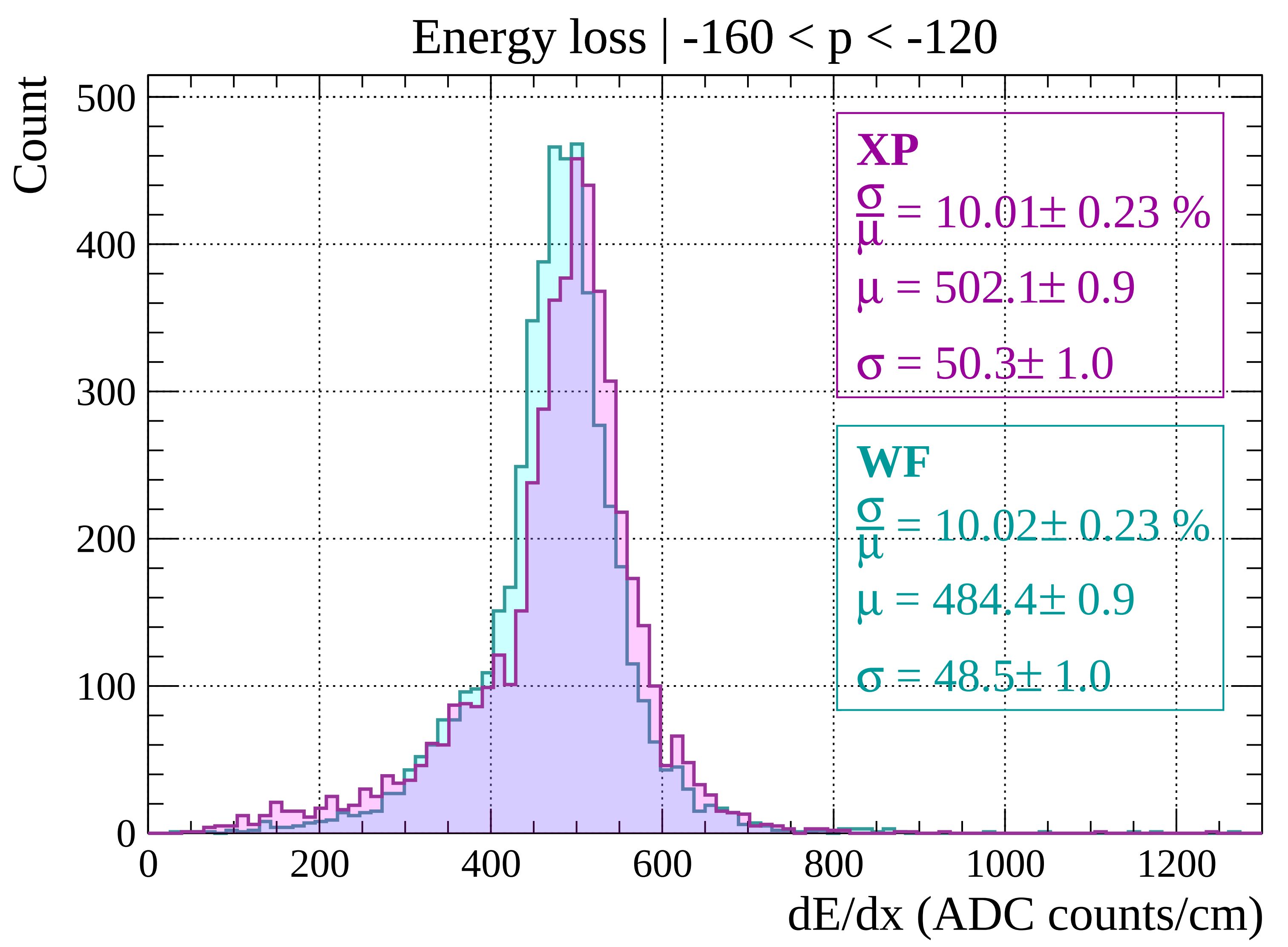

Energy loss | -280 < p < -240
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.88± 0.13 %
μ = 424.0± 0.5
σ = 29.2± 0.5
WF
$\frac{\sigma}{\mu}$ = 6.71± 0.14 %
μ = 411.6± 0.5
σ = 27.6± 0.5



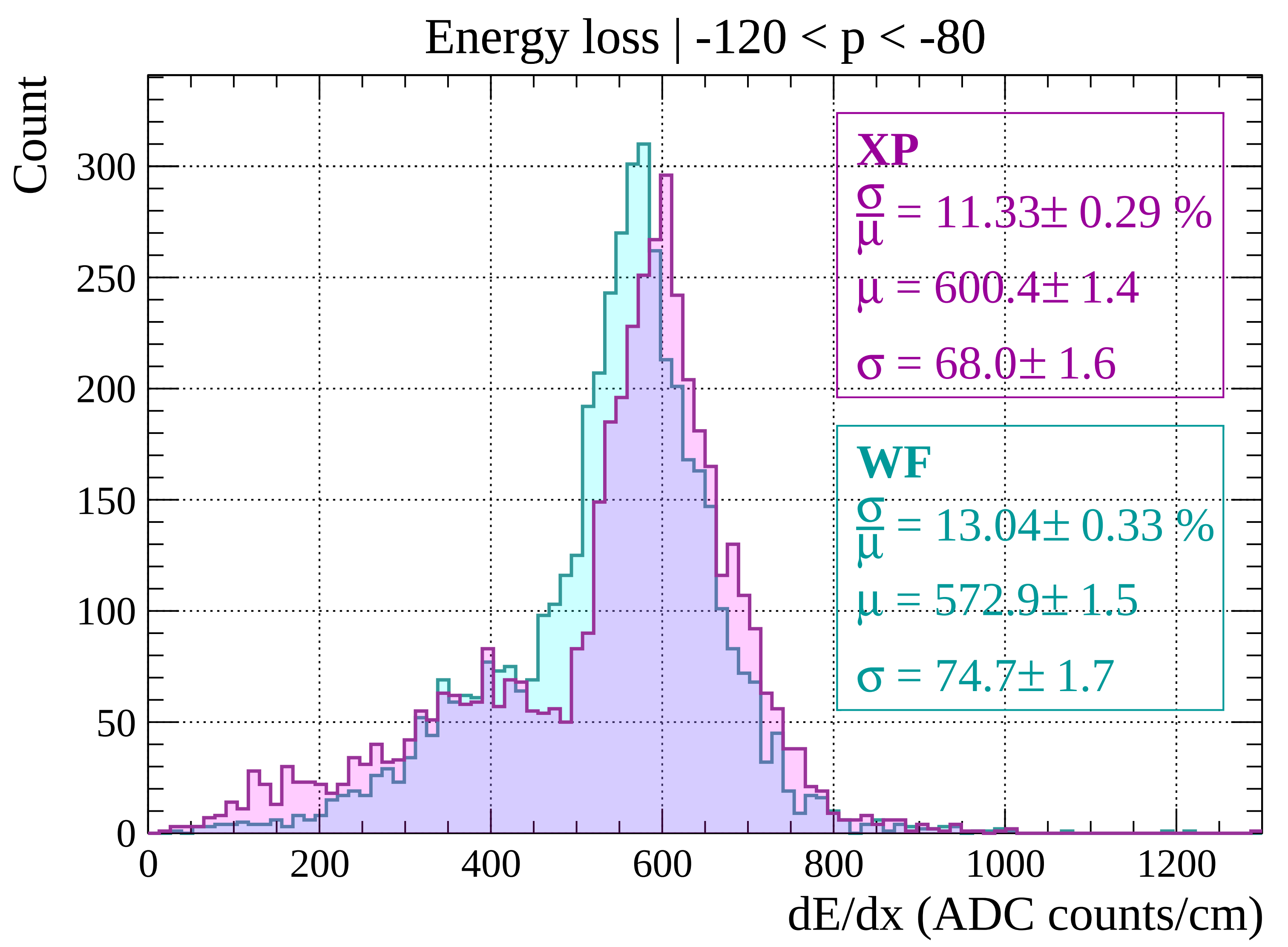
Energy loss | -240 < p < -200
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 7.50± 0.14 %
μ = 432.3± 0.6
σ = 32.4± 0.6
WF
σ/μ = 7.11± 0.14 %
μ = 420.3± 0.5
σ = 29.9± 0.5



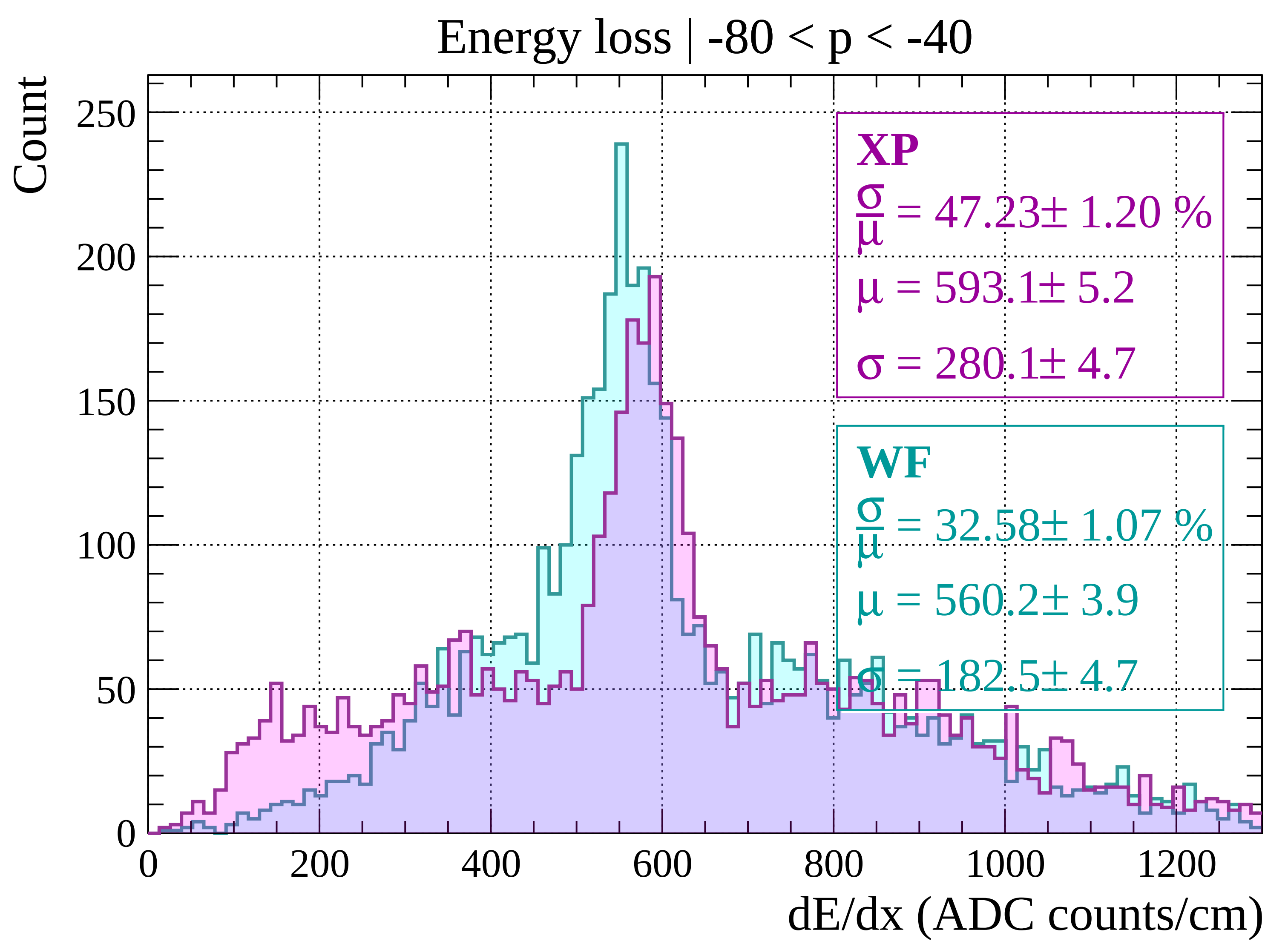
Energy loss | -200 < p < -160
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 7.66± 0.16 %
μ = 456.0± 0.6
σ = 34.9± 0.7
WF
σ/μ = 7.55± 0.15 %
μ = 441.2± 0.6
σ = 33.3± 0.6



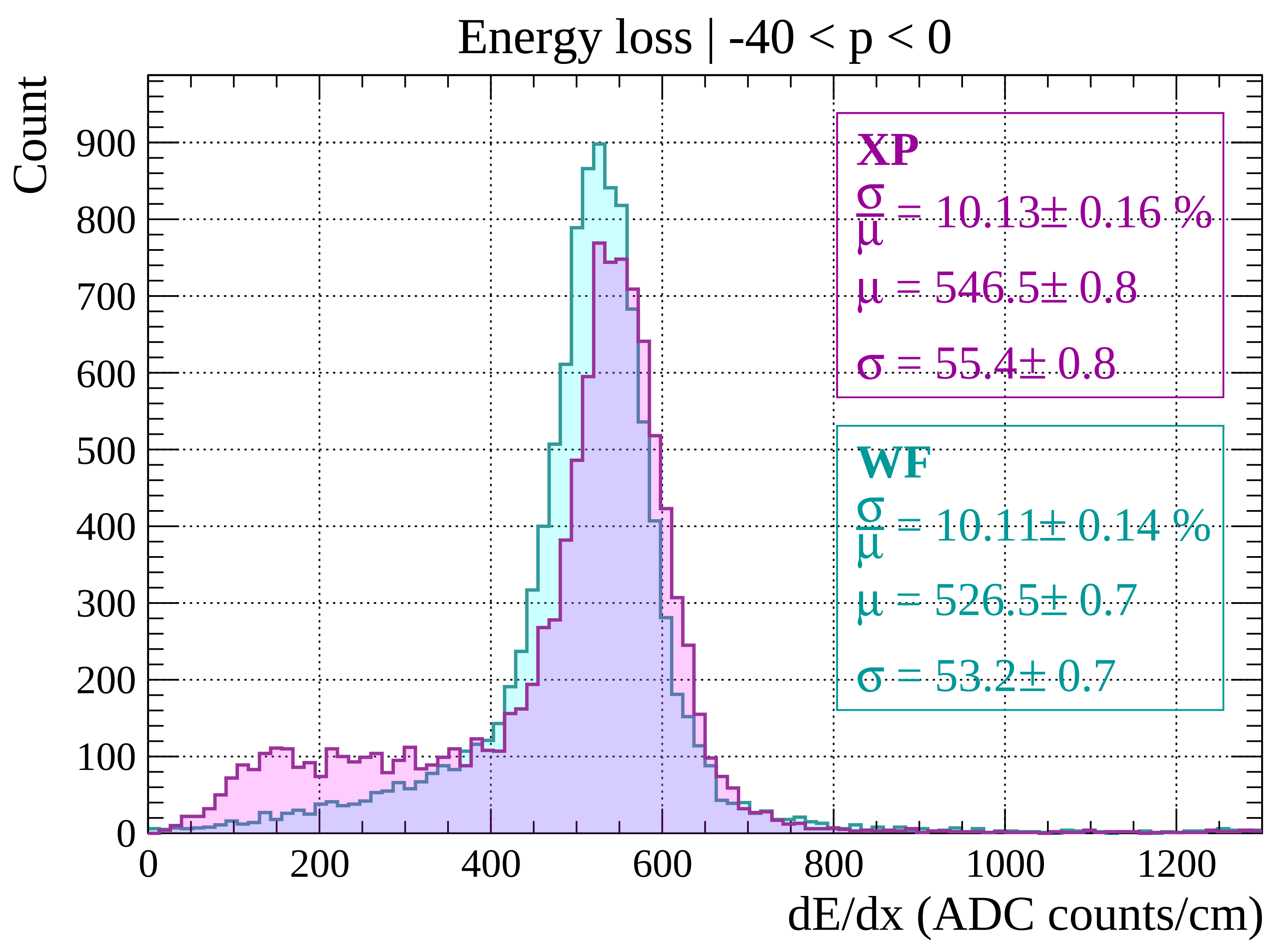
Energy loss | -160 < p < -120
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.01± 0.23 %
μ = 502.1± 0.9
σ = 50.3± 1.0
WF
σ/μ = 10.02± 0.23 %
μ = 484.4± 0.9
σ = 48.5± 1.0
0
100
200
300
400
500
0
200
400
600
800
1000
1200



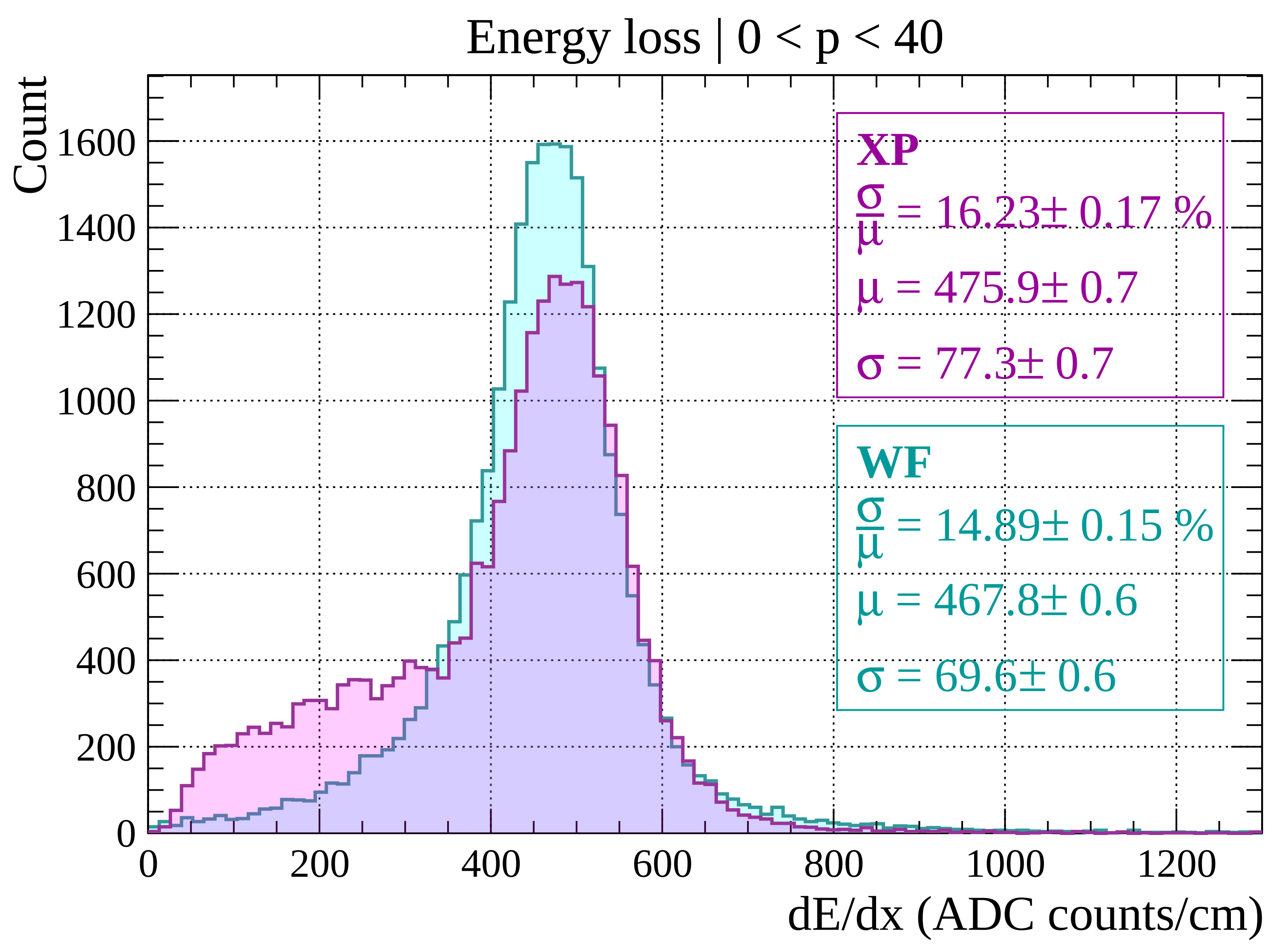
Energy loss | -120 < p < -80
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 11.33± 0.29 %
μ = 600.4± 1.4
σ = 68.0± 1.6
WF
$\frac{\sigma}{\mu}$ = 13.04± 0.33 %
μ = 572.9± 1.5
σ = 74.7± 1.7

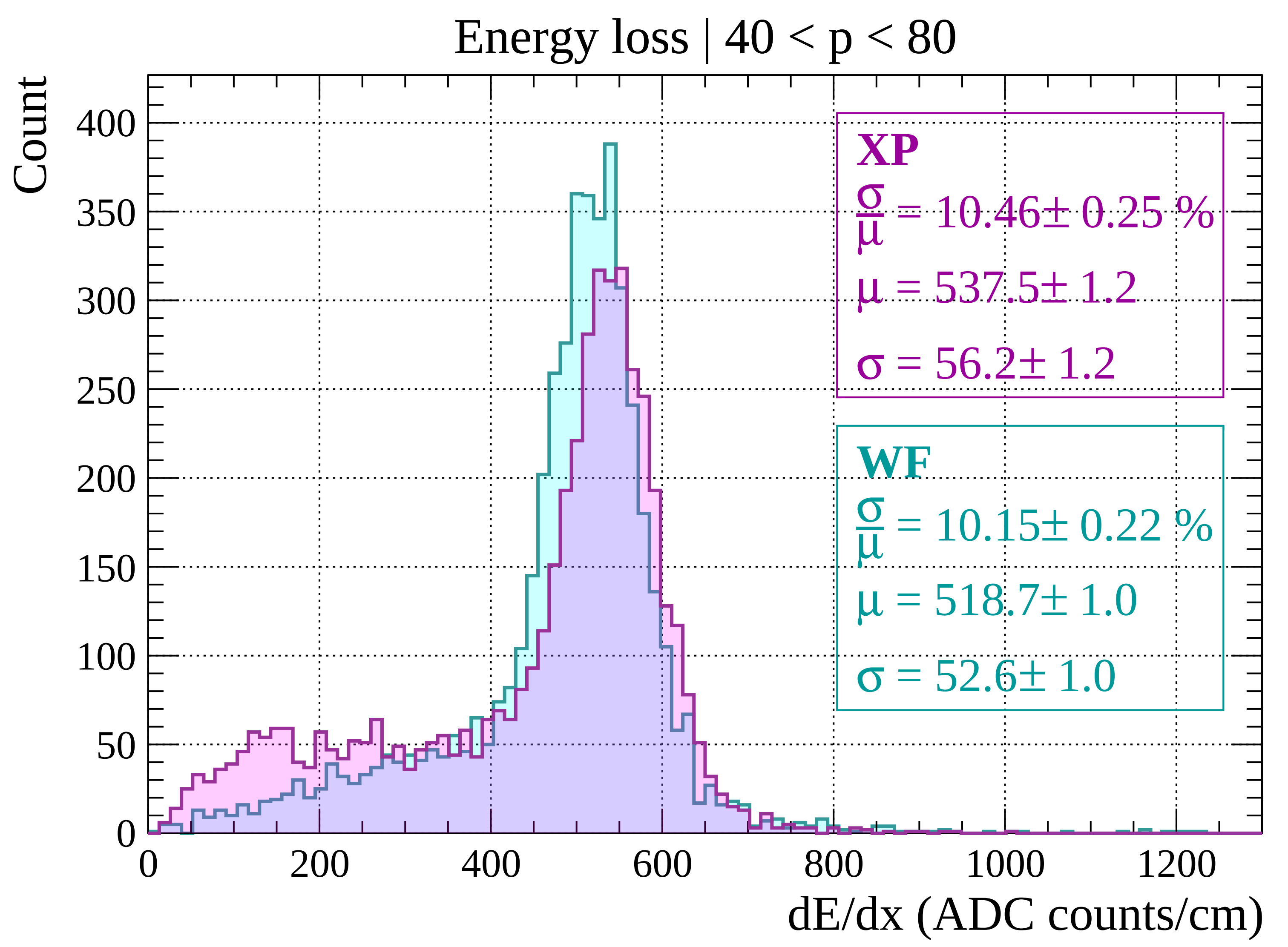


Energy loss | -80 < p < -40
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 47.23± 1.20 %
μ = 593.1± 5.2
σ = 280.1± 4.7
WF
σ/μ = 32.58± 1.07 %
μ = 560.2± 3.9
σ = 182.5± 4.7

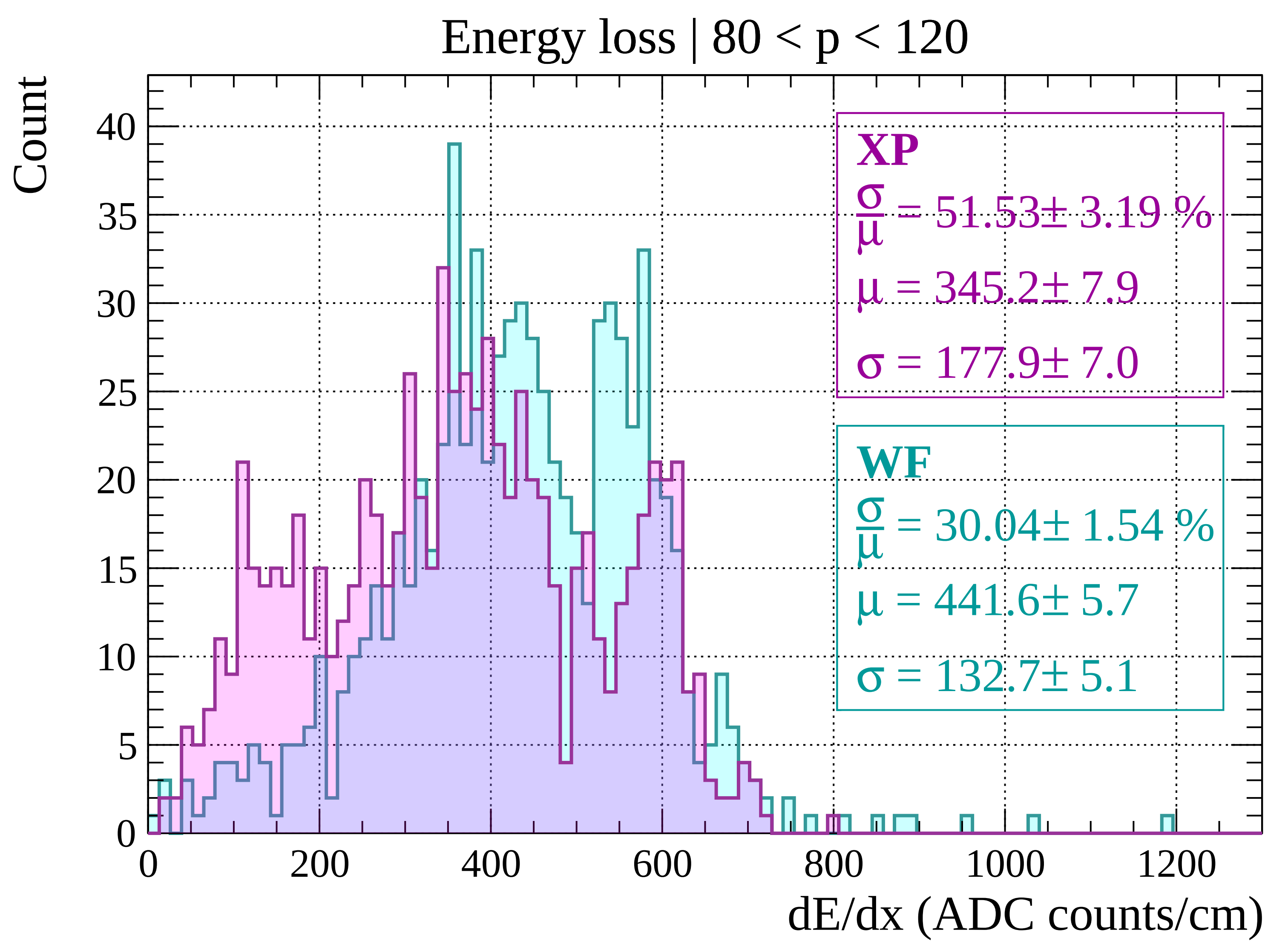


Energy loss | -40 < p < 0
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.13± 0.16 %
μ = 546.5± 0.8
σ = 55.4± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.11± 0.14 %
μ = 526.5± 0.7
σ = 53.2± 0.7
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200

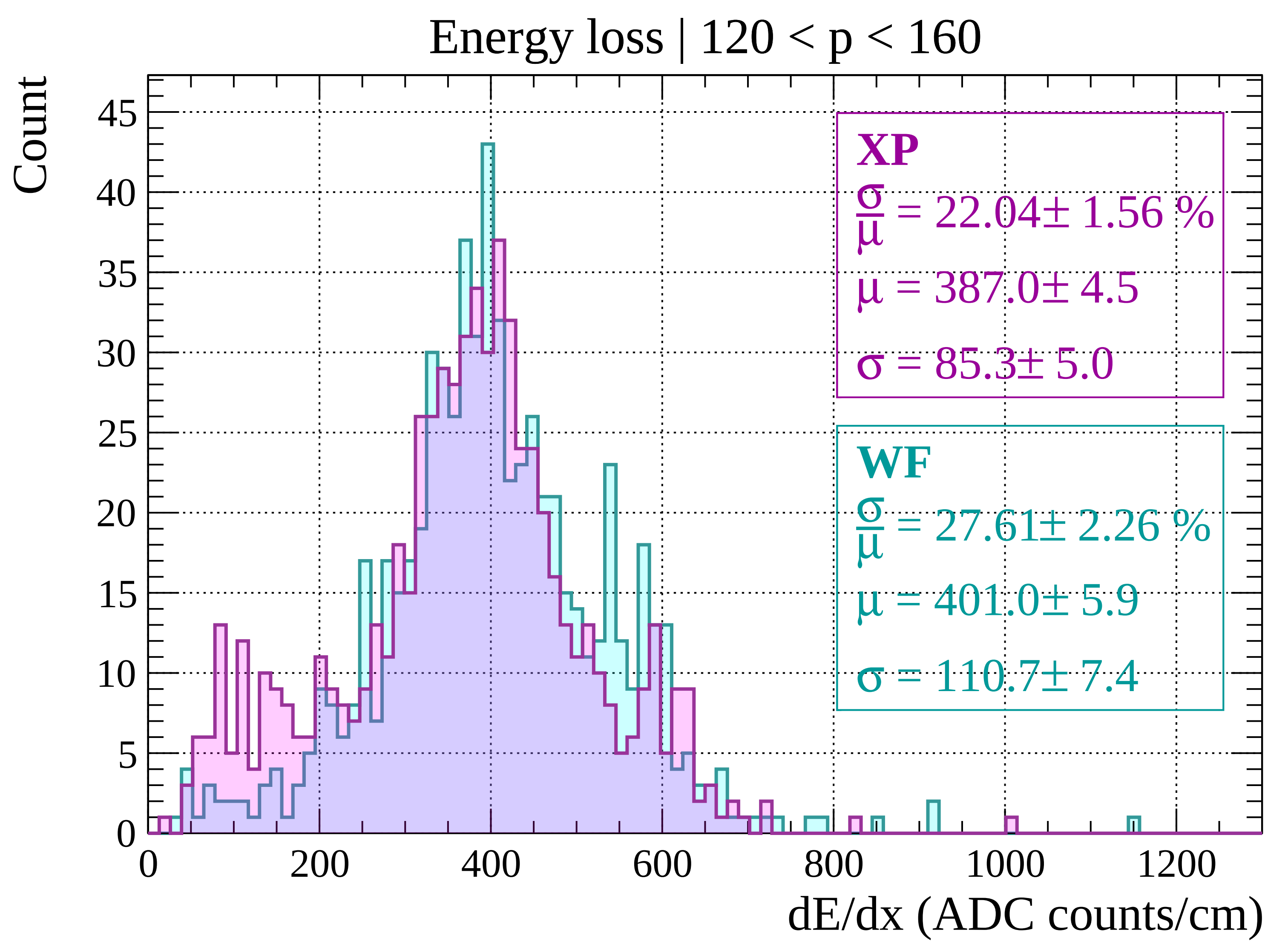

Energy loss | 0 < p < 40
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
XP
$\frac{\sigma}{\mu}$ = 16.23± 0.17 %
μ = 475.9± 0.7
σ = 77.3± 0.7
WF
$\frac{\sigma}{\mu}$ = 14.89± 0.15 %
μ = 467.8± 0.6
σ = 69.6± 0.6



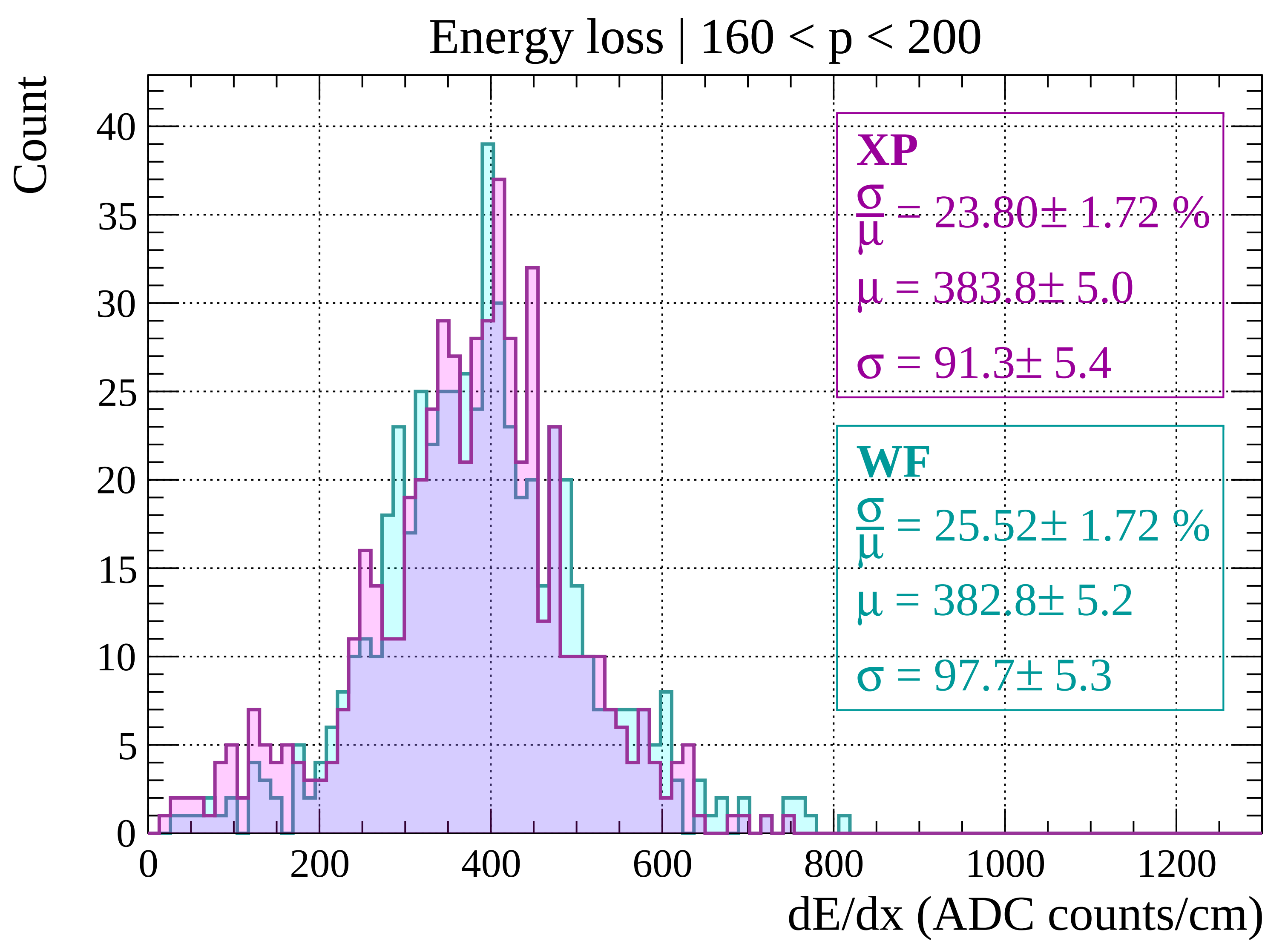
Energy loss | 40 < p < 80
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.46± 0.25 %
μ = 537.5± 1.2
σ = 56.2± 1.2
WF
σ/μ = 10.15± 0.22 %
μ = 518.7± 1.0
σ = 52.6± 1.0



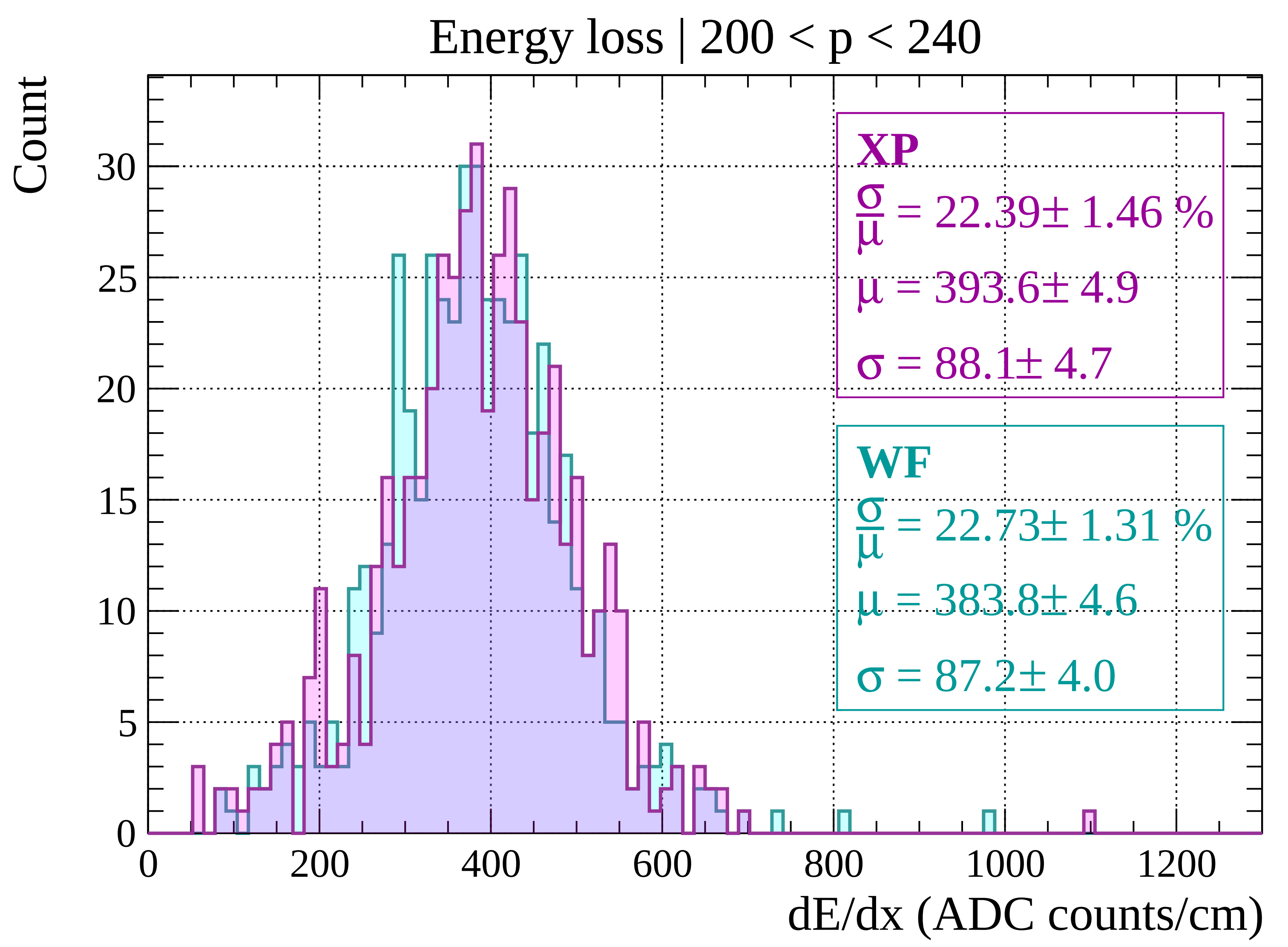
Energy loss | 80 < p < 120
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 51.53± 3.19 %
μ = 345.2± 7.9
σ = 177.9± 7.0
WF
$\frac{\sigma}{\mu}$ = 30.04± 1.54 %
μ = 441.6± 5.7
σ = 132.7± 5.1



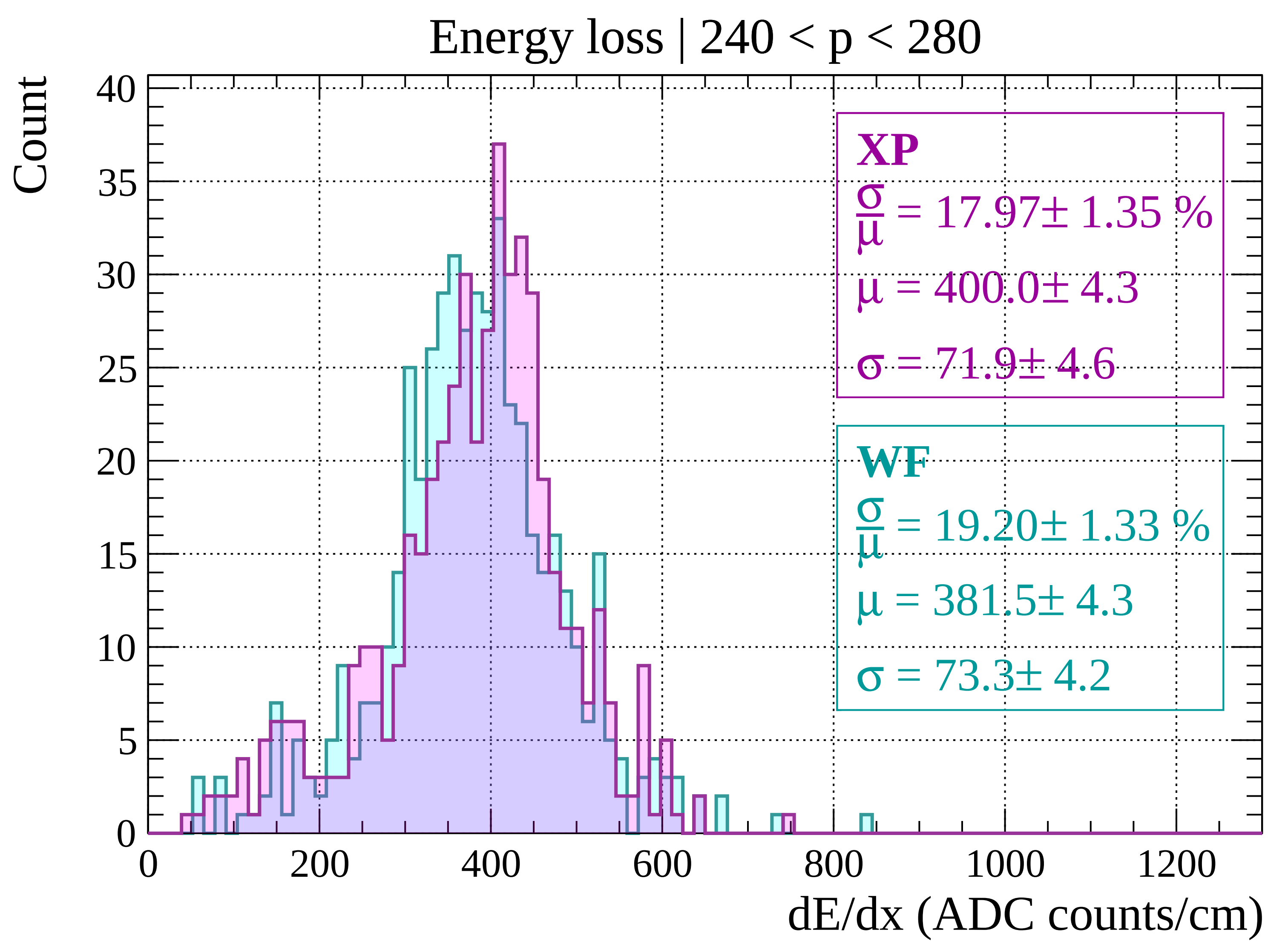
Energy loss | 120 < p < 160
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
XP
σ/μ = 22.04± 1.56 %
μ = 387.0± 4.5
σ = 85.3± 5.0
WF
σ/μ = 27.61± 2.26 %
μ = 401.0± 5.9
σ = 110.7± 7.4



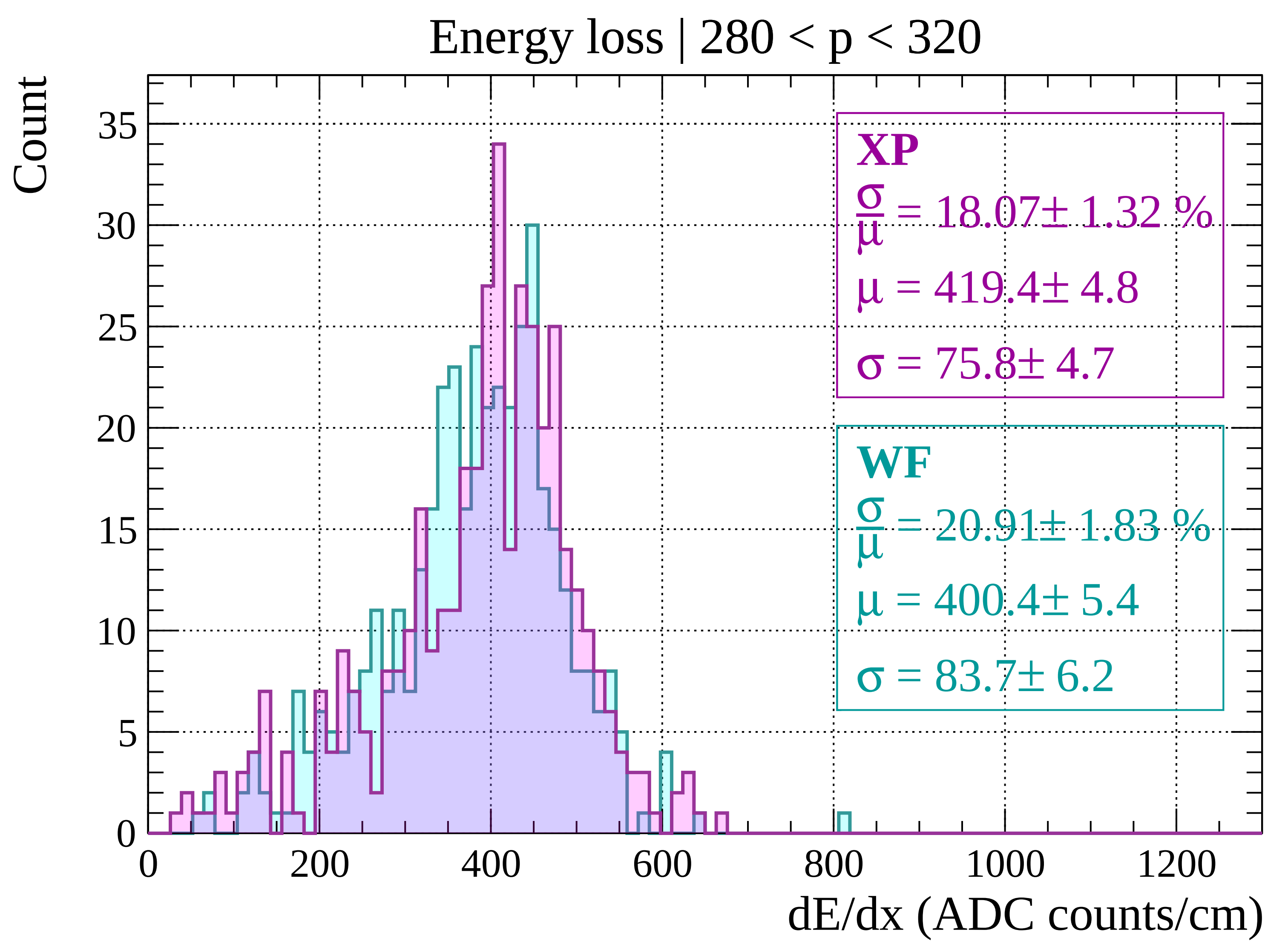
Energy loss | 160 < p < 200
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 23.80± 1.72 %
μ = 383.8± 5.0
σ = 91.3± 5.4
WF
$\frac{\sigma}{\mu}$ = 25.52± 1.72 %
μ = 382.8± 5.2
σ = 97.7± 5.3



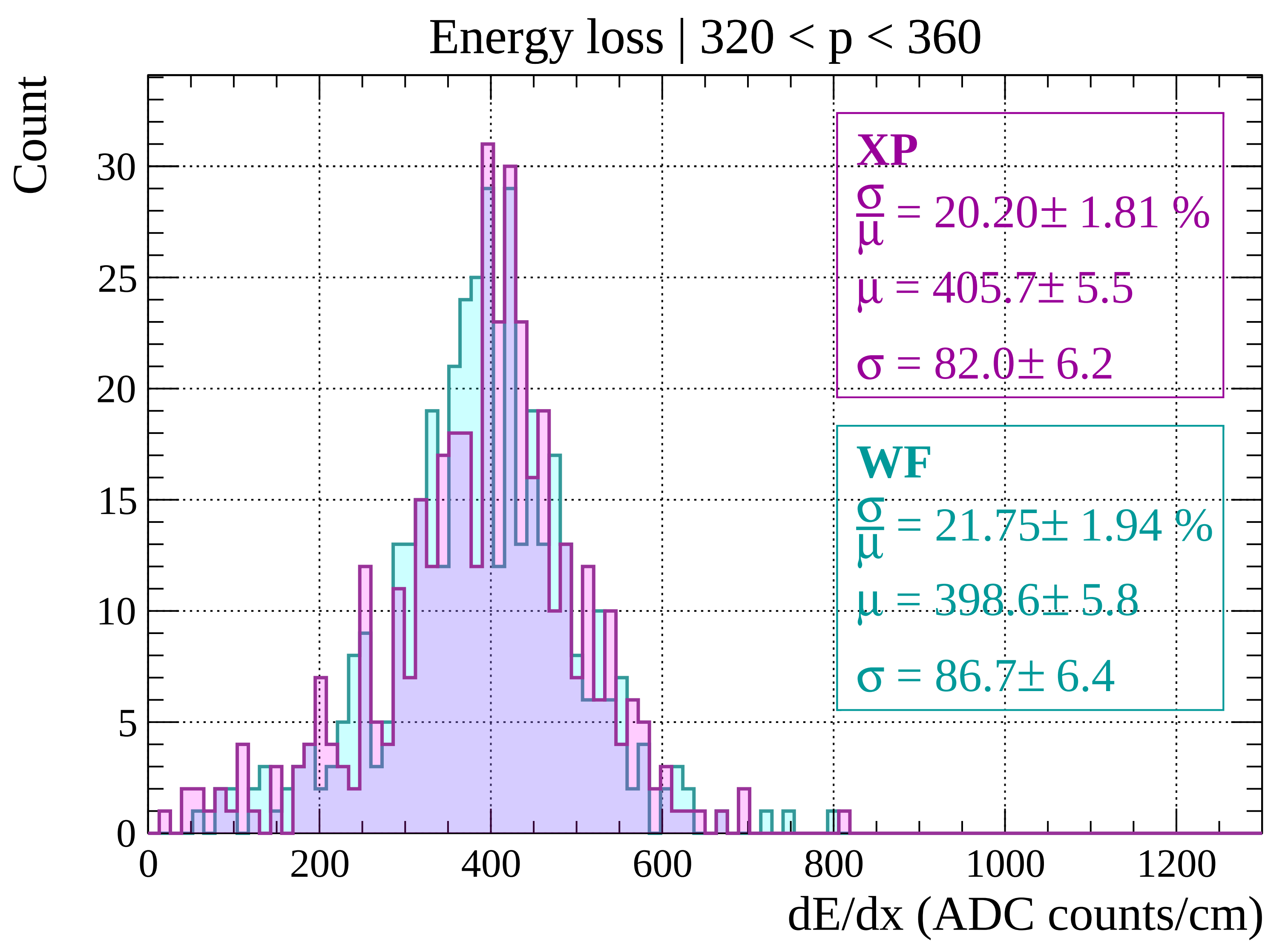
Energy loss | 200 < p < 240
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 22.39± 1.46 %
μ = 393.6± 4.9
σ = 88.1± 4.7
WF
σ/μ = 22.73± 1.31 %
μ = 383.8± 4.6
σ = 87.2± 4.0
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200



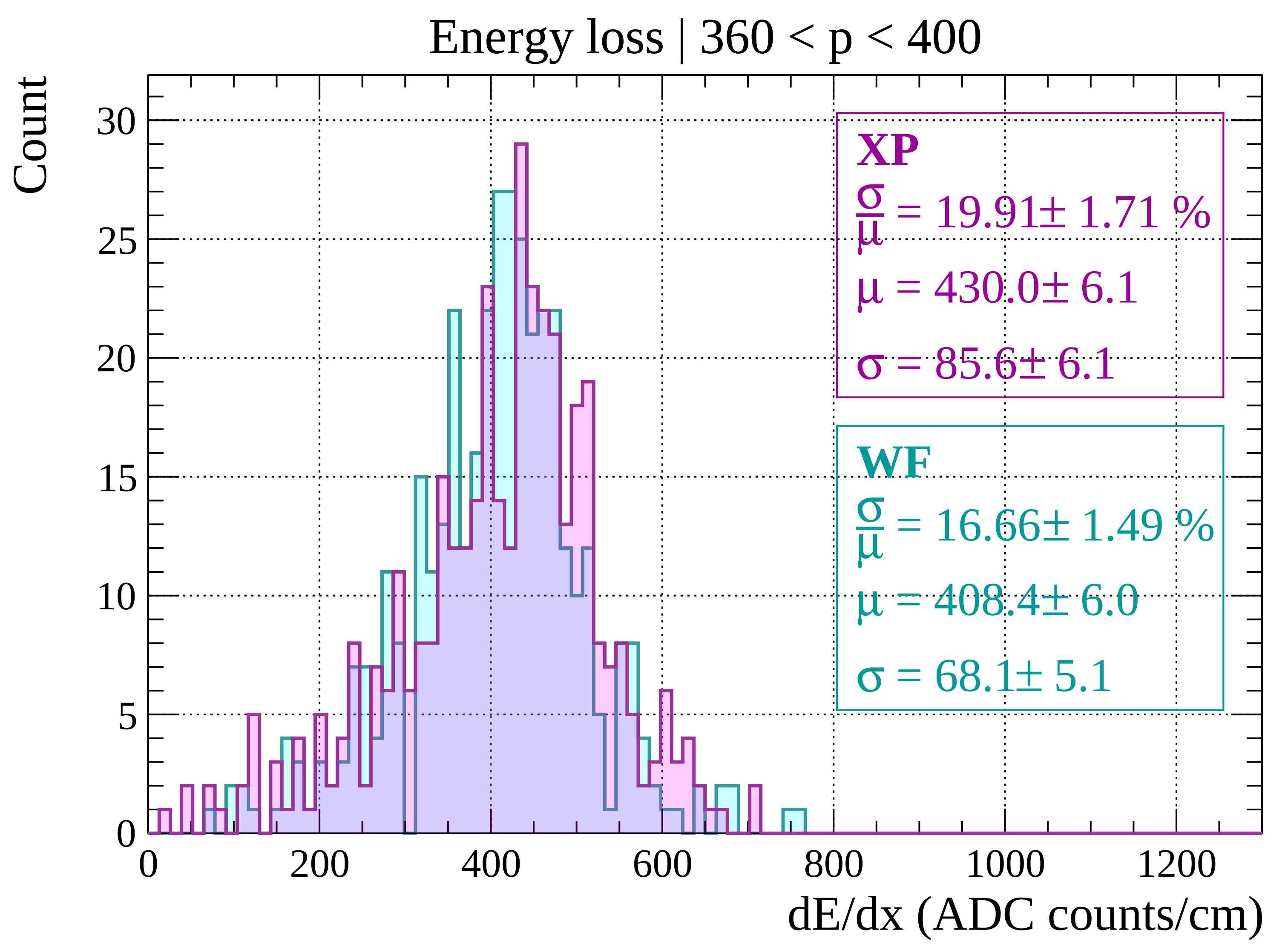
Energy loss | 240 < p < 280
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 17.97± 1.35 %
μ = 400.0± 4.3
σ = 71.9± 4.6
WF
$\frac{\sigma}{\mu}$ = 19.20± 1.33 %
μ = 381.5± 4.3
σ = 73.3± 4.2



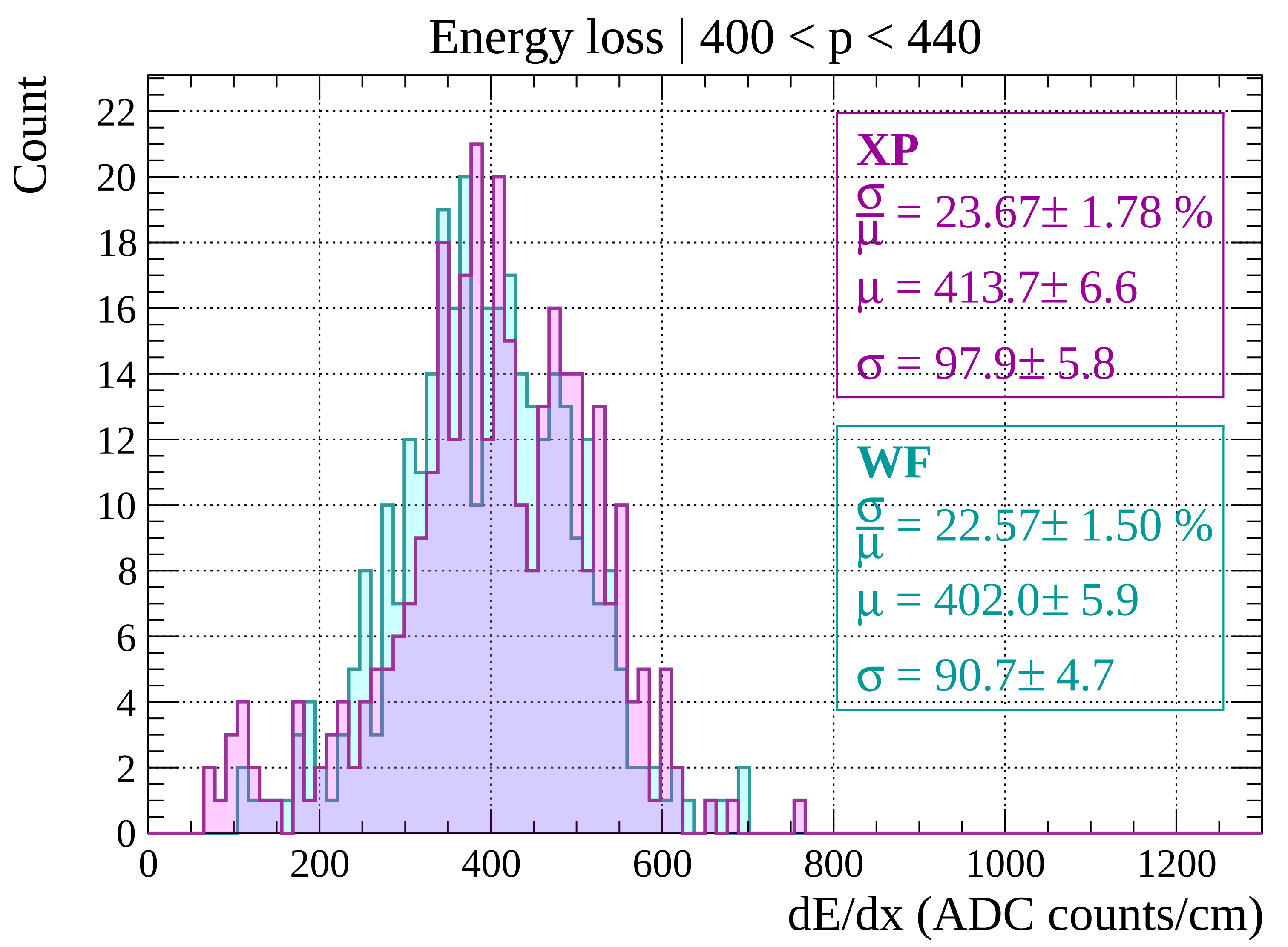
Energy loss | 280 < p < 320
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 18.07± 1.32 %
μ = 419.4± 4.8
σ = 75.8± 4.7
WF
$\frac{\sigma}{\mu}$ = 20.91± 1.83 %
μ = 400.4± 5.4
σ = 83.7± 6.2

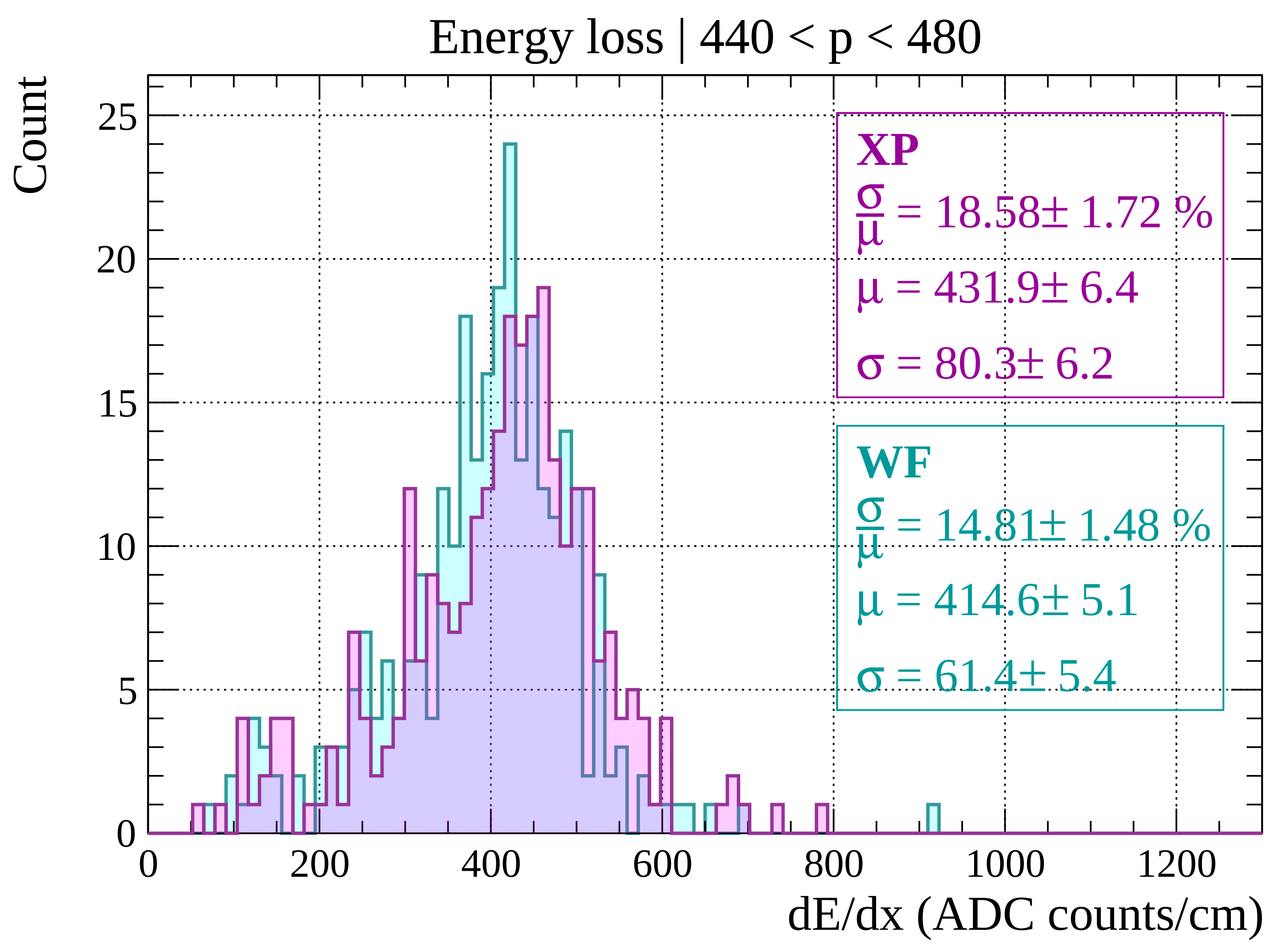


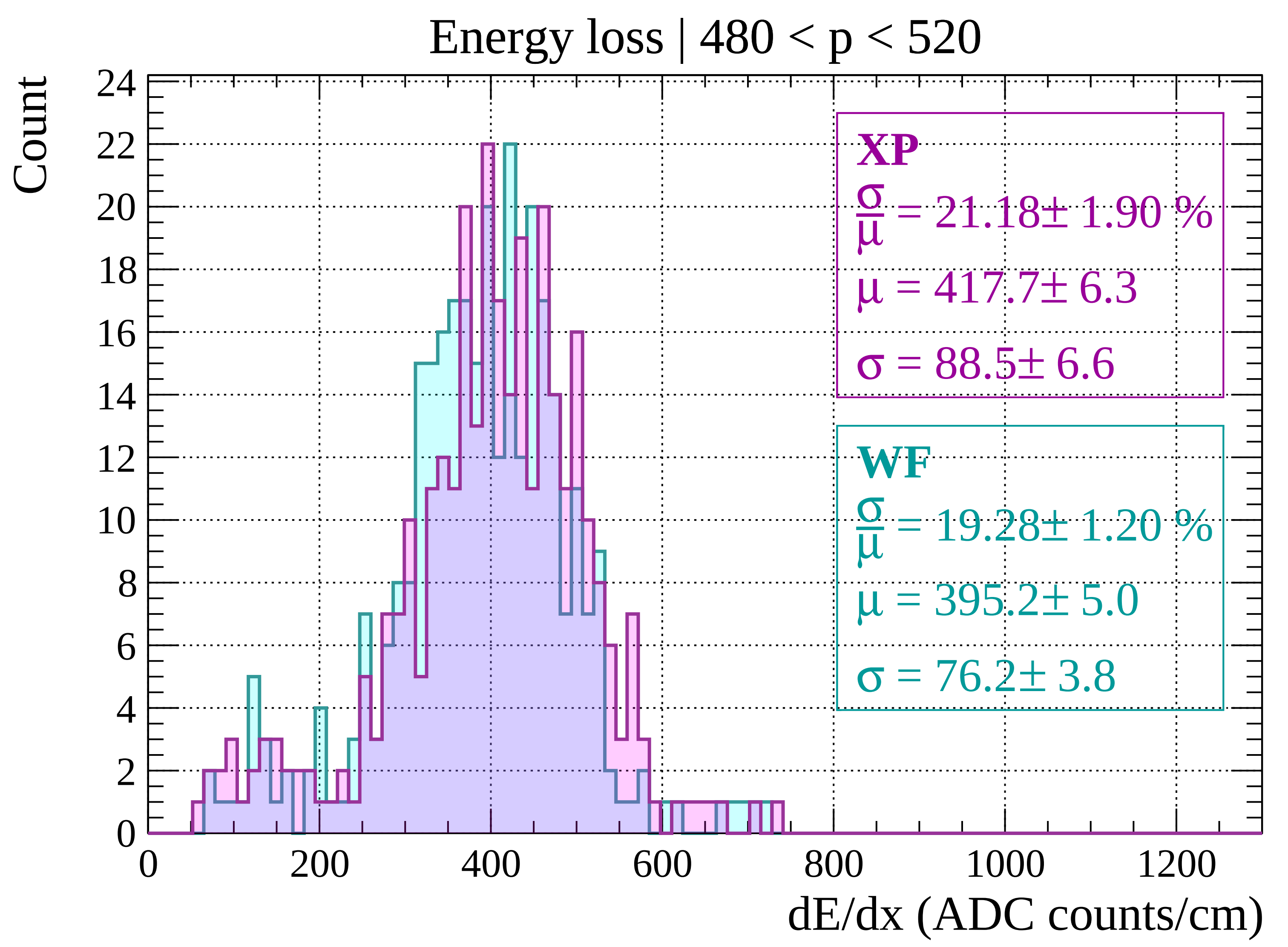
Energy loss | 320 < p < 360
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
XP
σ/μ = 20.20± 1.81 %
μ = 405.7± 5.5
σ = 82.0± 6.2
WF
σ/μ = 21.75± 1.94 %
μ = 398.6± 5.8
σ = 86.7± 6.4



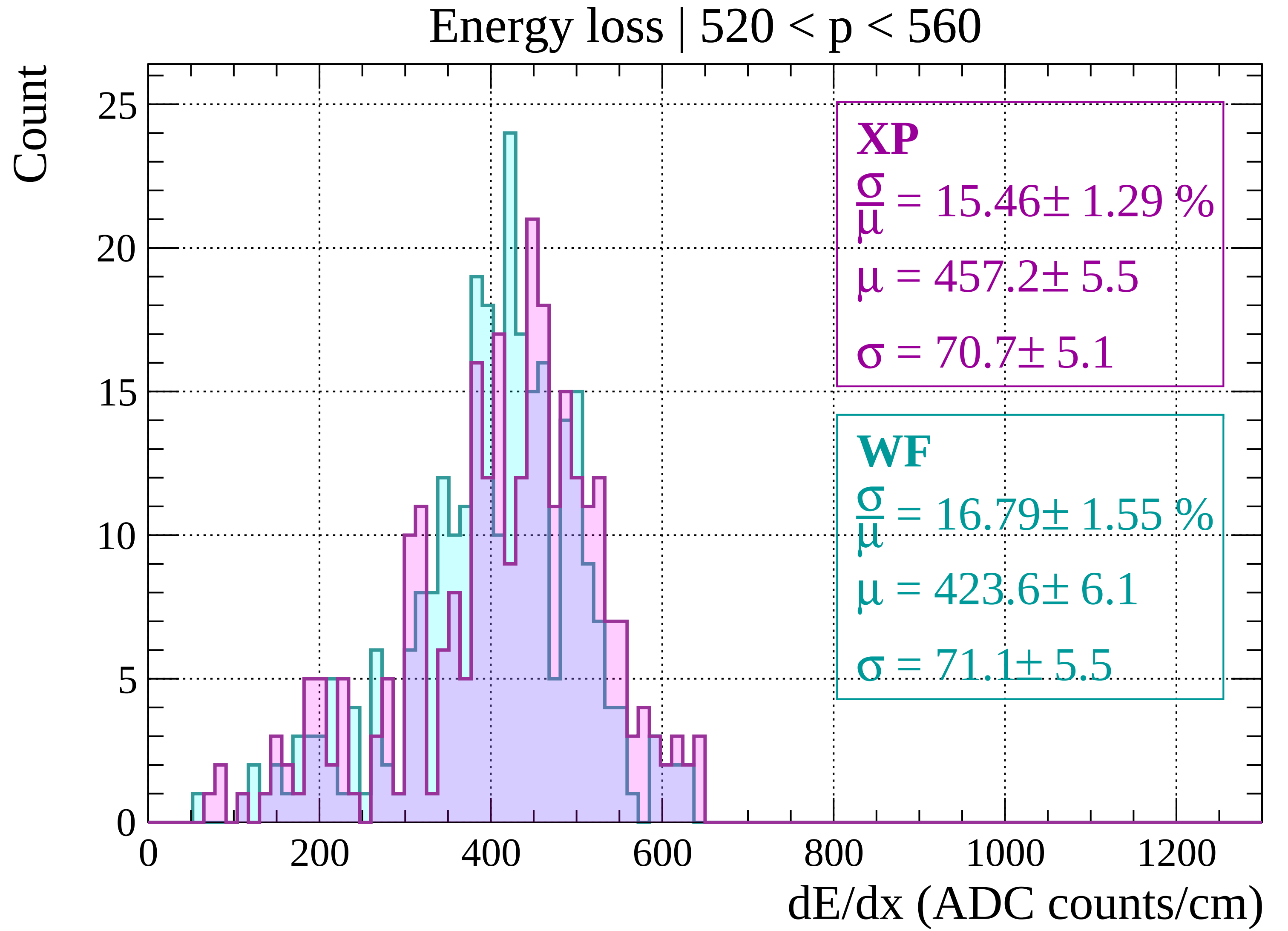
Energy loss | 360 < p < 400
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
XP
σ/μ = 19.91± 1.71 %
μ = 430.0± 6.1
σ = 85.6± 6.1
WF
σ/μ = 16.66± 1.49 %
μ = 408.4± 6.0
σ = 68.1± 5.1


Energy loss | 400 < p < 440
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 23.67± 1.78 %
μ = 413.7± 6.6
σ = 97.9± 5.8
WF
σ/μ = 22.57± 1.50 %
μ = 402.0± 5.9
σ = 90.7± 4.7
0
2
4
6
8
10
12
14
16
18
20
22
0
200
400
600
800
1000
1200

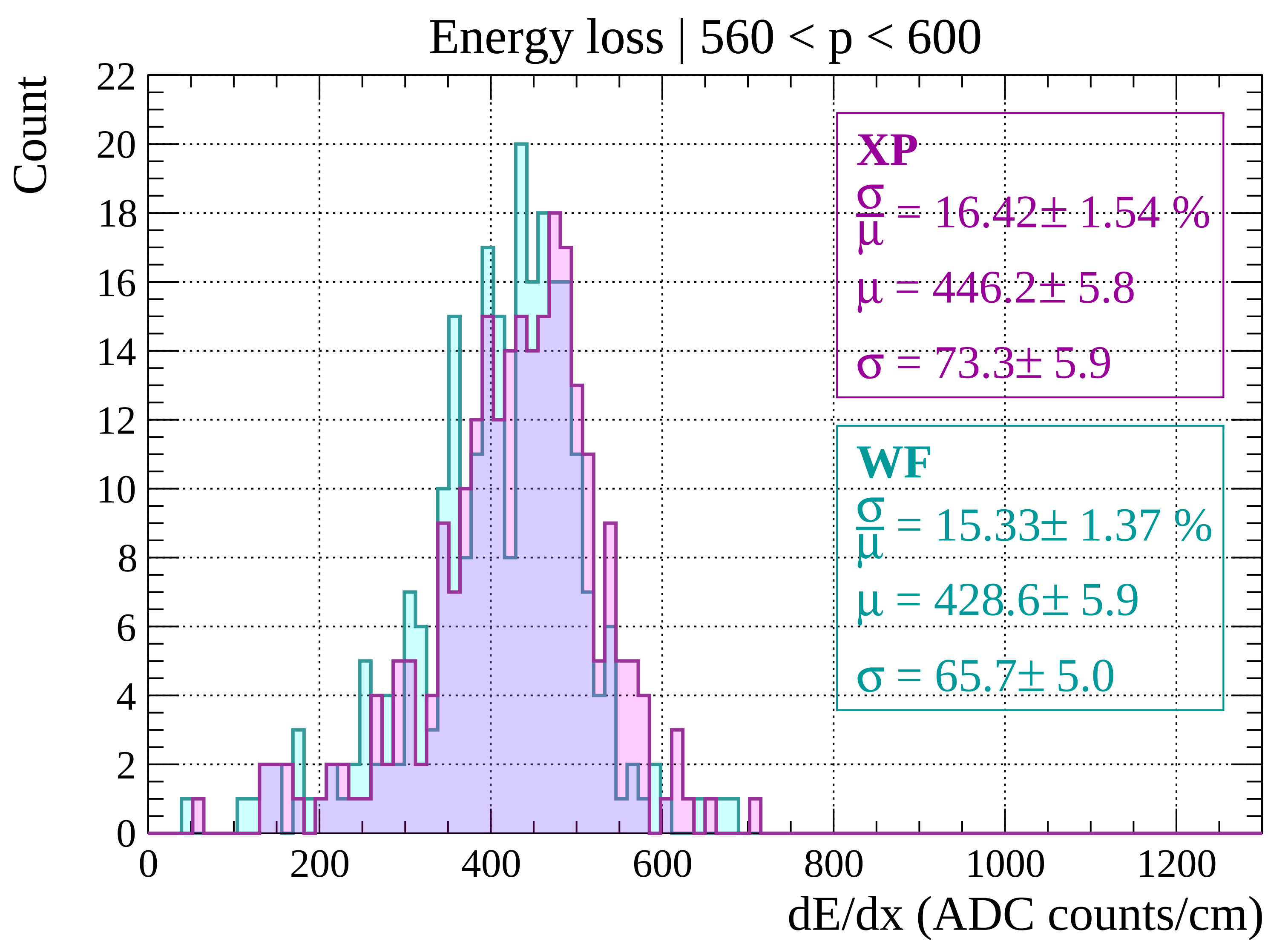

Energy loss | 440 < p < 480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP
$\frac{\sigma}{\mu}$ = 18.58± 1.72 %
μ = 431.9± 6.4
σ = 80.3± 6.2
WF
$\frac{\sigma}{\mu}$ = 14.81± 1.48 %
μ = 414.6± 5.1
σ = 61.4± 5.4



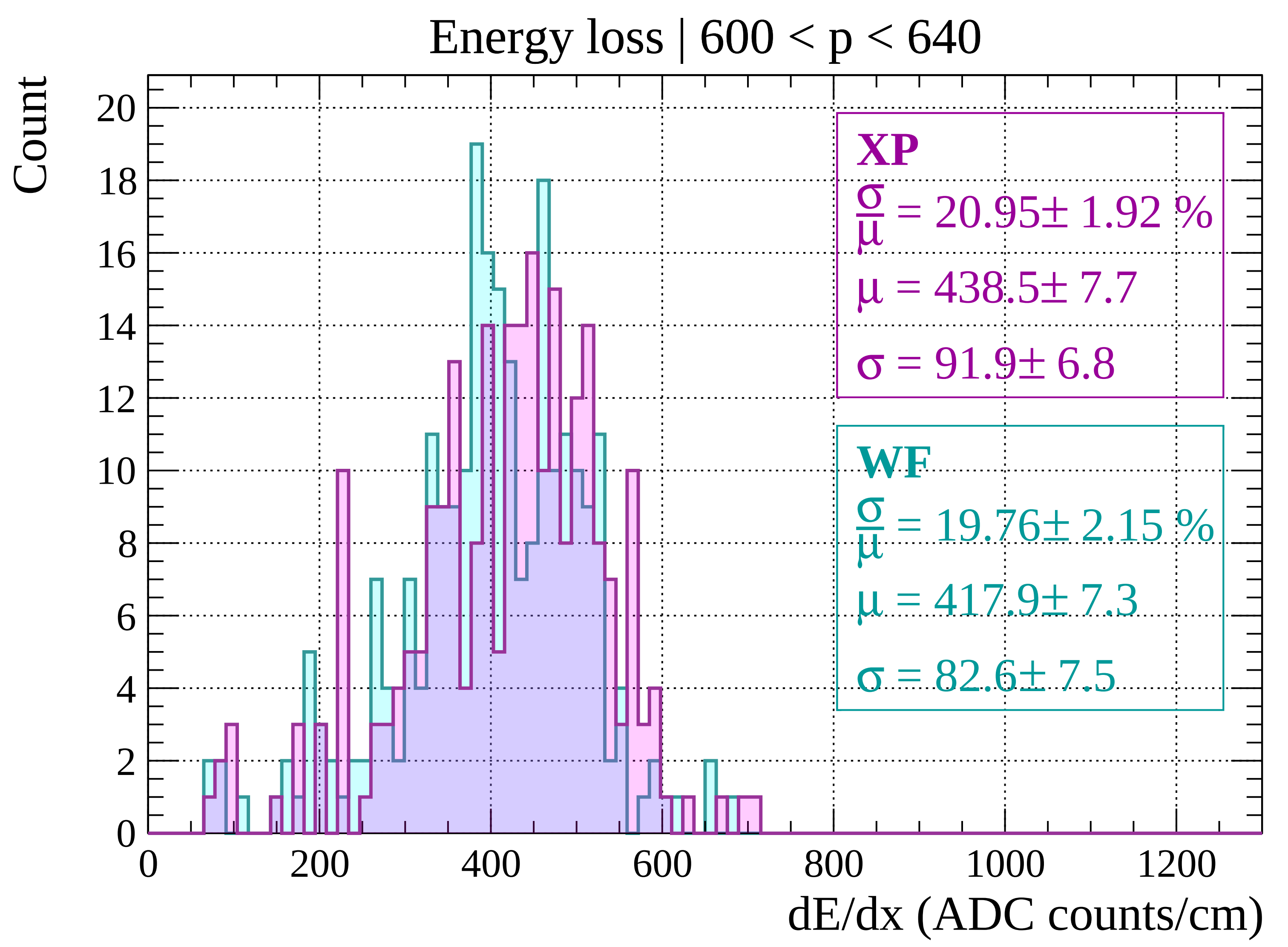
Energy loss | 480 < p < 520
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 21.18± 1.90 %
μ = 417.7± 6.3
σ = 88.5± 6.6
WF
σ/μ = 19.28± 1.20 %
μ = 395.2± 5.0
σ = 76.2± 3.8



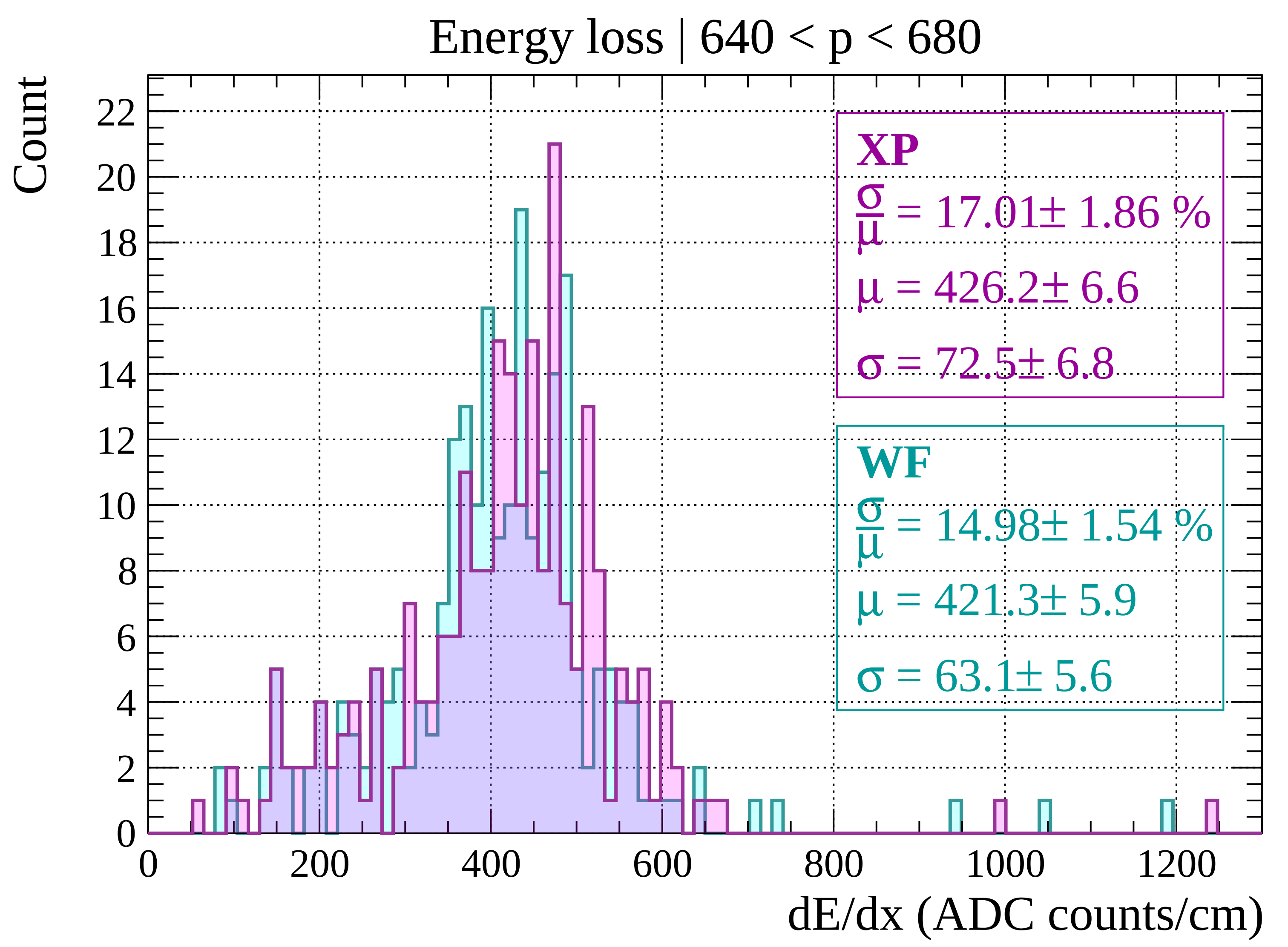
Energy loss | 520 < p < 560
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 15.46± 1.29 %
μ = 457.2± 5.5
σ = 70.7± 5.1
WF
σ/μ = 16.79± 1.55 %
μ = 423.6± 6.1
σ = 71.1± 5.5



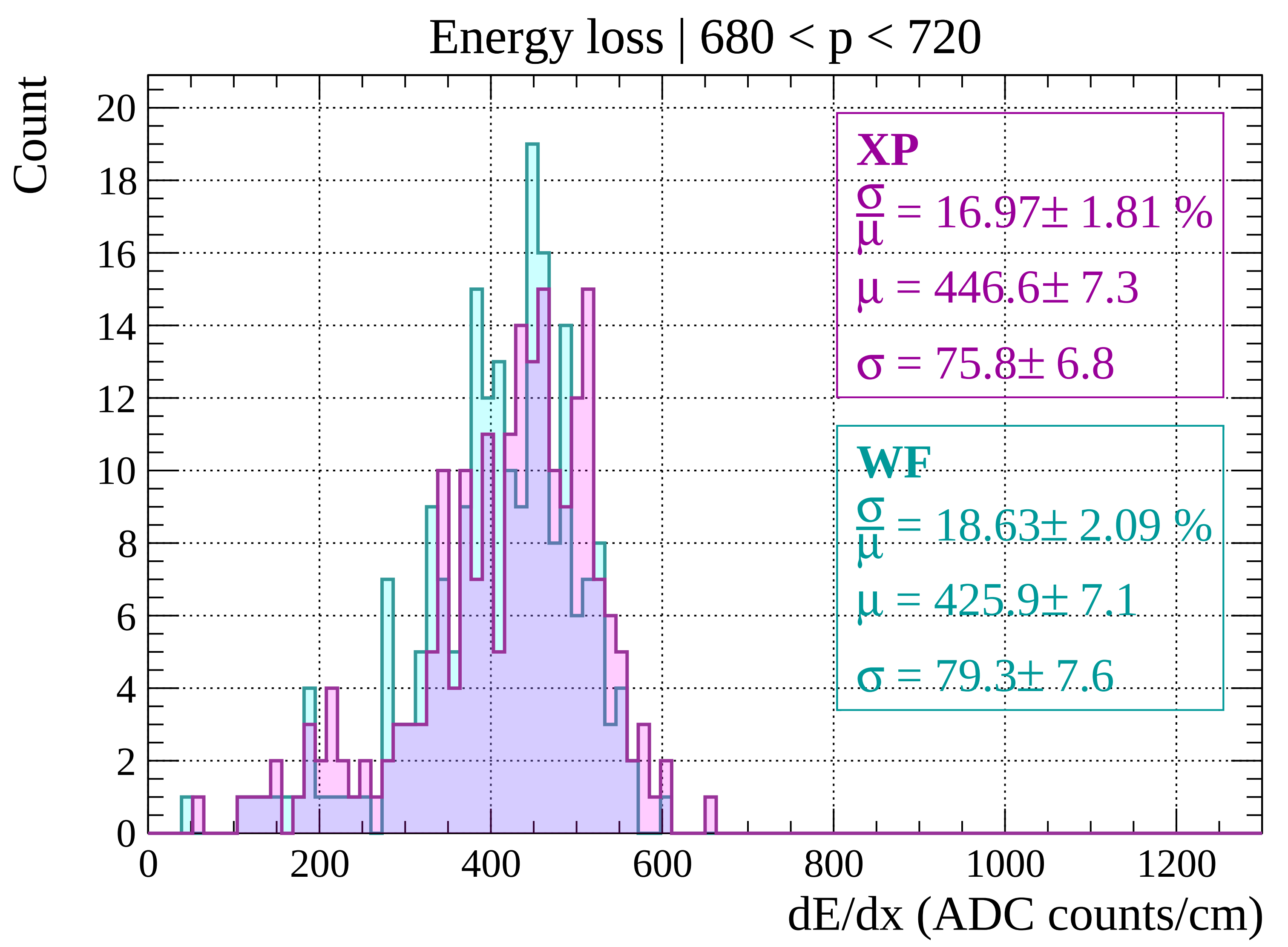
Energy loss | 560 < p < 600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
22
XP
$\frac{\sigma}{\mu}$ = 16.42± 1.54 %
μ = 446.2± 5.8
σ = 73.3± 5.9
WF
$\frac{\sigma}{\mu}$ = 15.33± 1.37 %
μ = 428.6± 5.9
σ = 65.7± 5.0



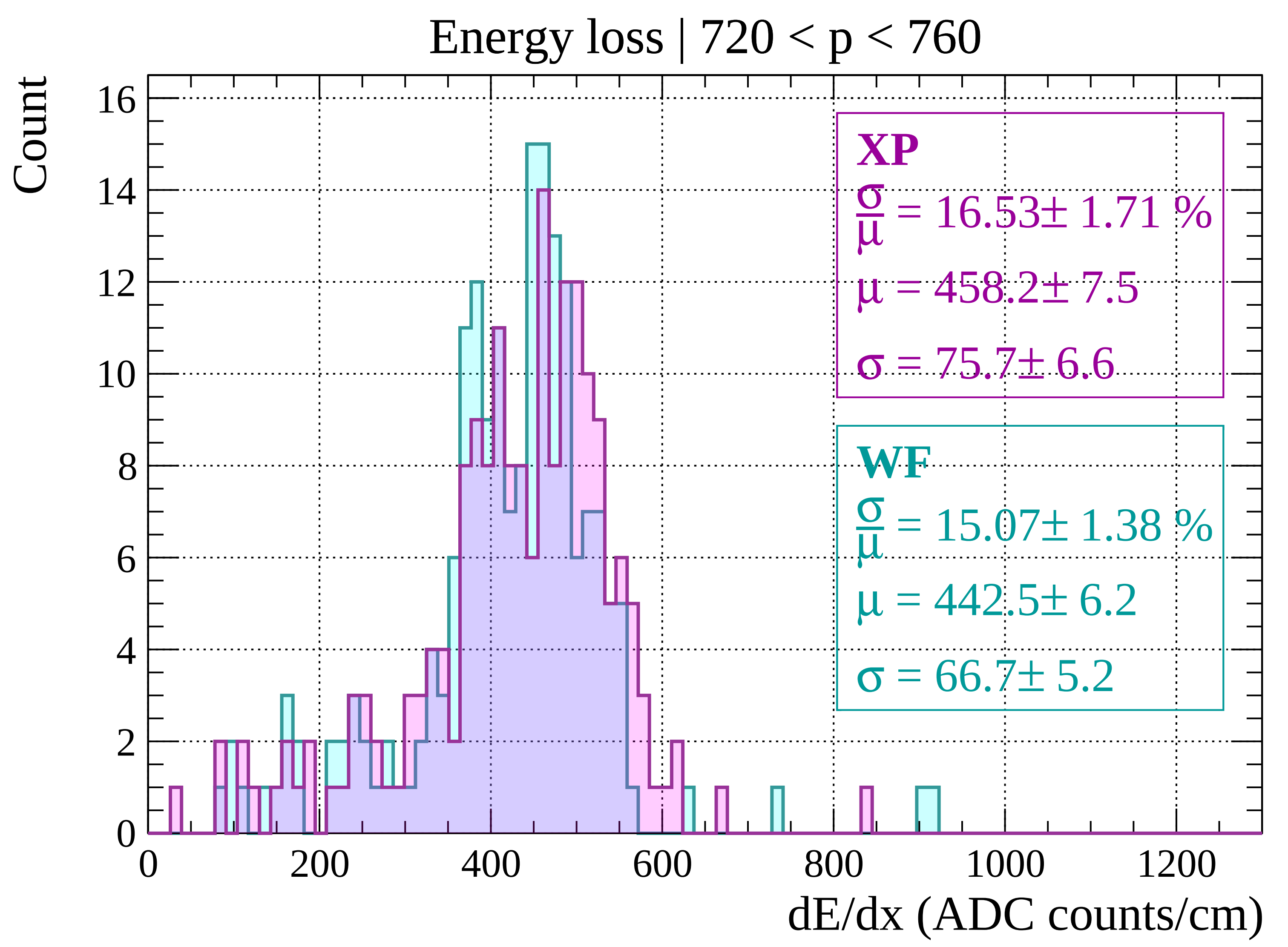
Energy loss | 600 < p < 640
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
20
0
200
400
600
800
1000
1200
XP
σ/μ = 20.95± 1.92 %
μ = 438.5± 7.7
σ = 91.9± 6.8
WF
σ/μ = 19.76± 2.15 %
μ = 417.9± 7.3
σ = 82.6± 7.5



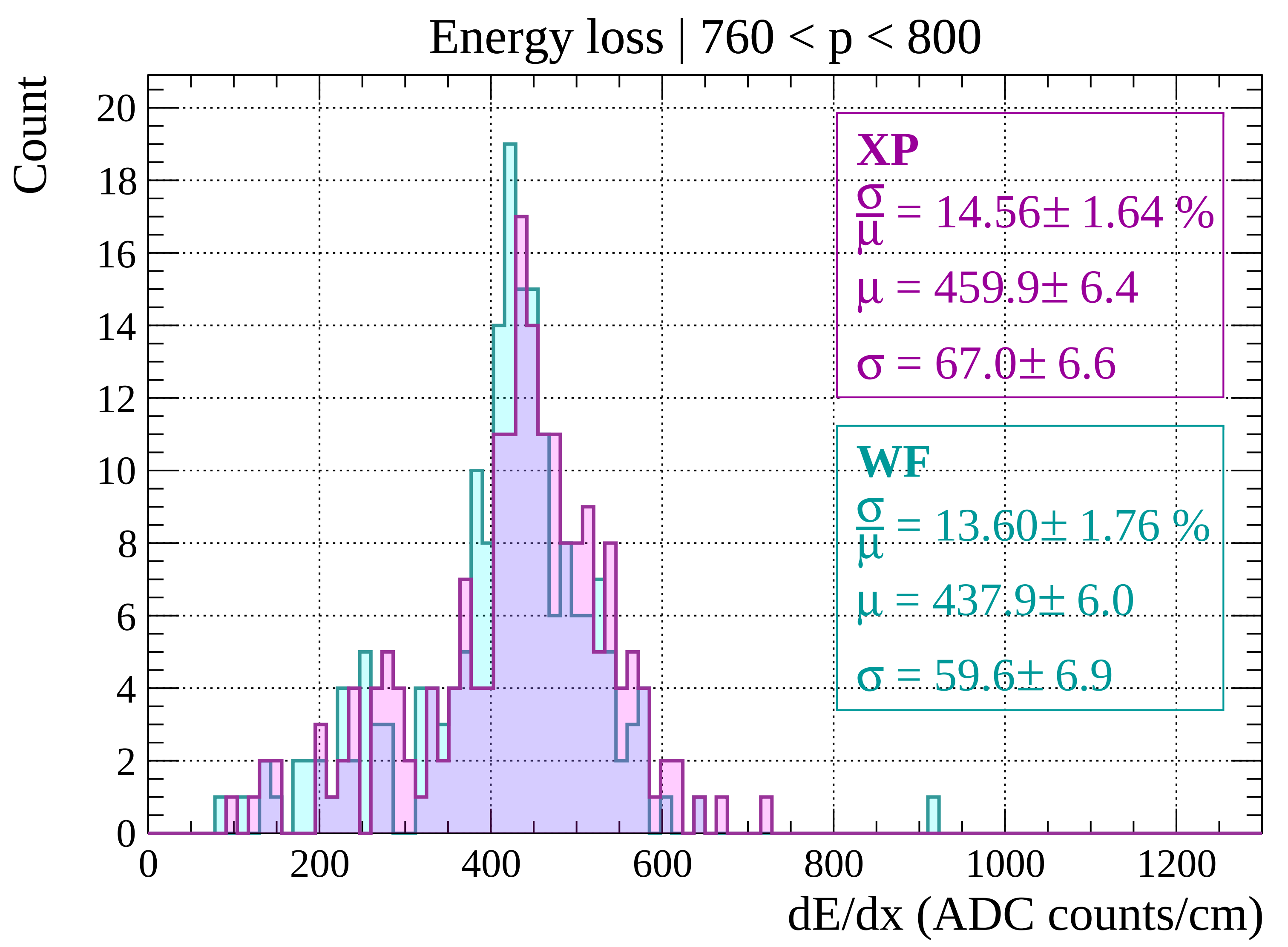
Energy loss | 640 < p < 680
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 17.01± 1.86 %
μ = 426.2± 6.6
σ = 72.5± 6.8
WF
σ/μ = 14.98± 1.54 %
μ = 421.3± 5.9
σ = 63.1± 5.6
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
22



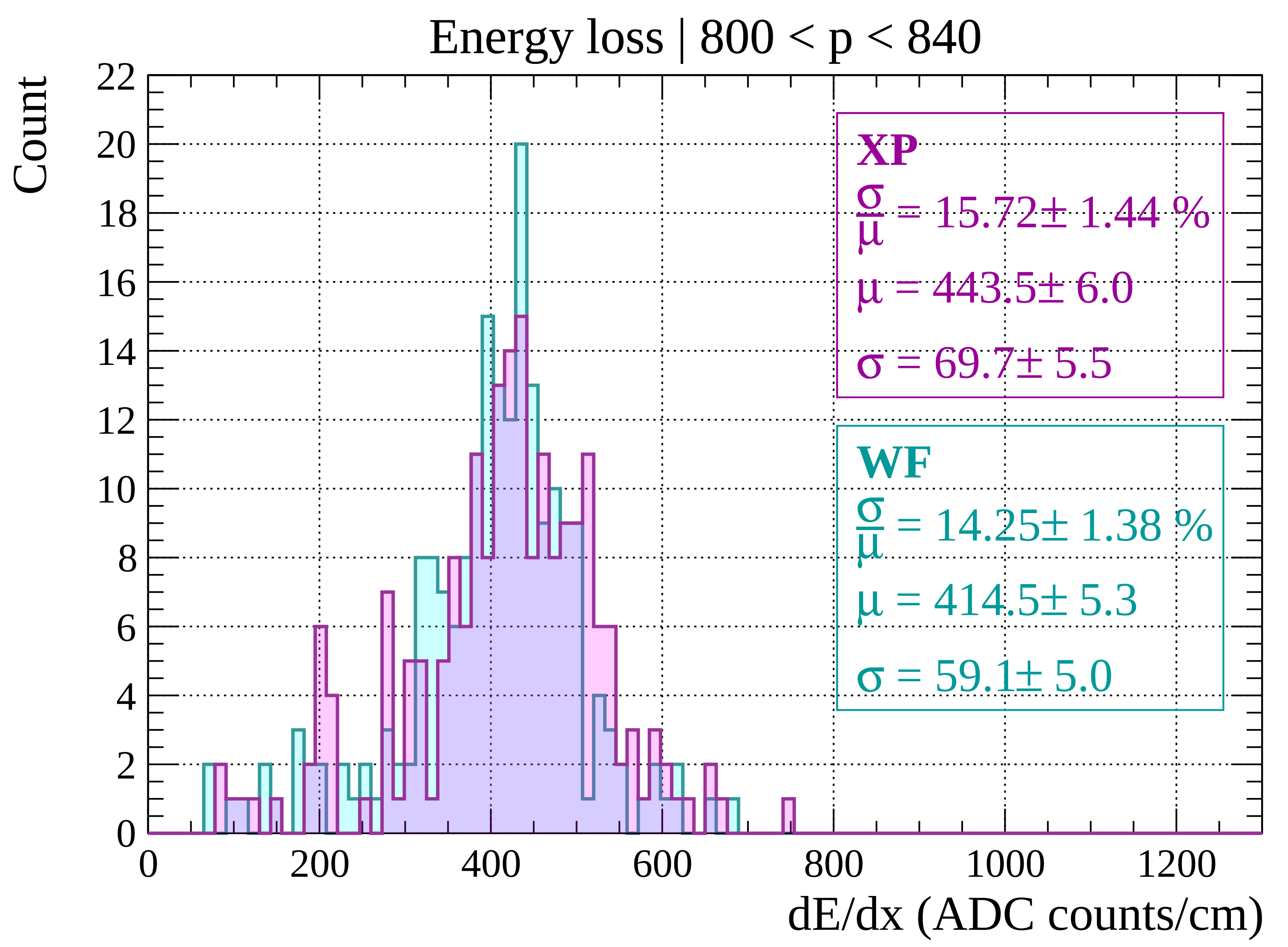
Energy loss | 680 < p < 720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
XP
σ/μ = 16.97± 1.81 %
μ = 446.6± 7.3
σ = 75.8± 6.8
WF
σ/μ = 18.63± 2.09 %
μ = 425.9± 7.1
σ = 79.3± 7.6



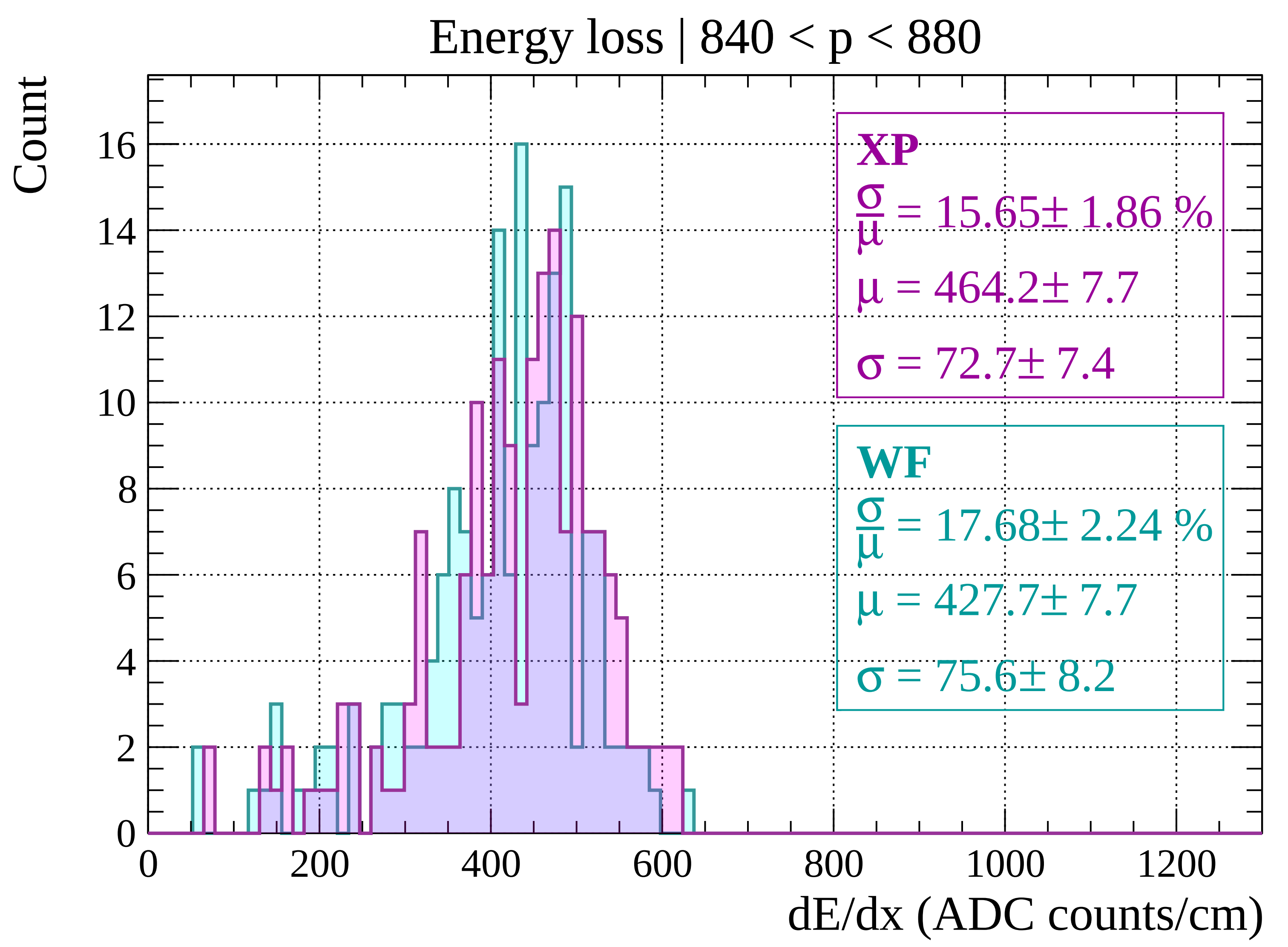
Energy loss | 720 < p < 760
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 16.53± 1.71 %
μ = 458.2± 7.5
σ = 75.7± 6.6
WF
$\frac{\sigma}{\mu}$ = 15.07± 1.38 %
μ = 442.5± 6.2
σ = 66.7± 5.2
0
2
4
6
8
10
12
14
16
0
200
400
600
800
1000
1200



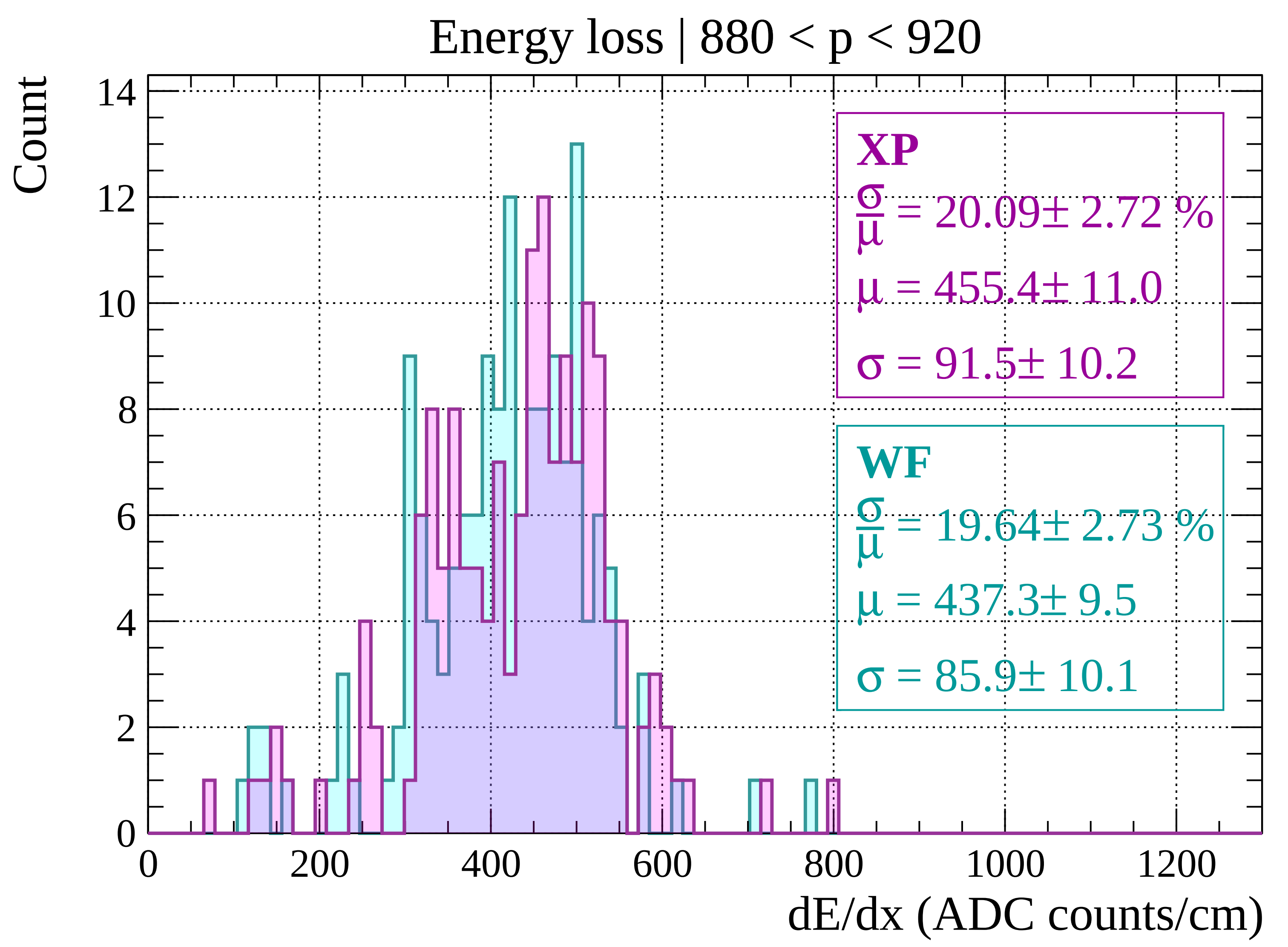
Energy loss | 760 < p < 800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
XP
σ/μ = 14.56± 1.64 %
μ = 459.9± 6.4
σ = 67.0± 6.6
WF
σ/μ = 13.60± 1.76 %
μ = 437.9± 6.0
σ = 59.6± 6.9

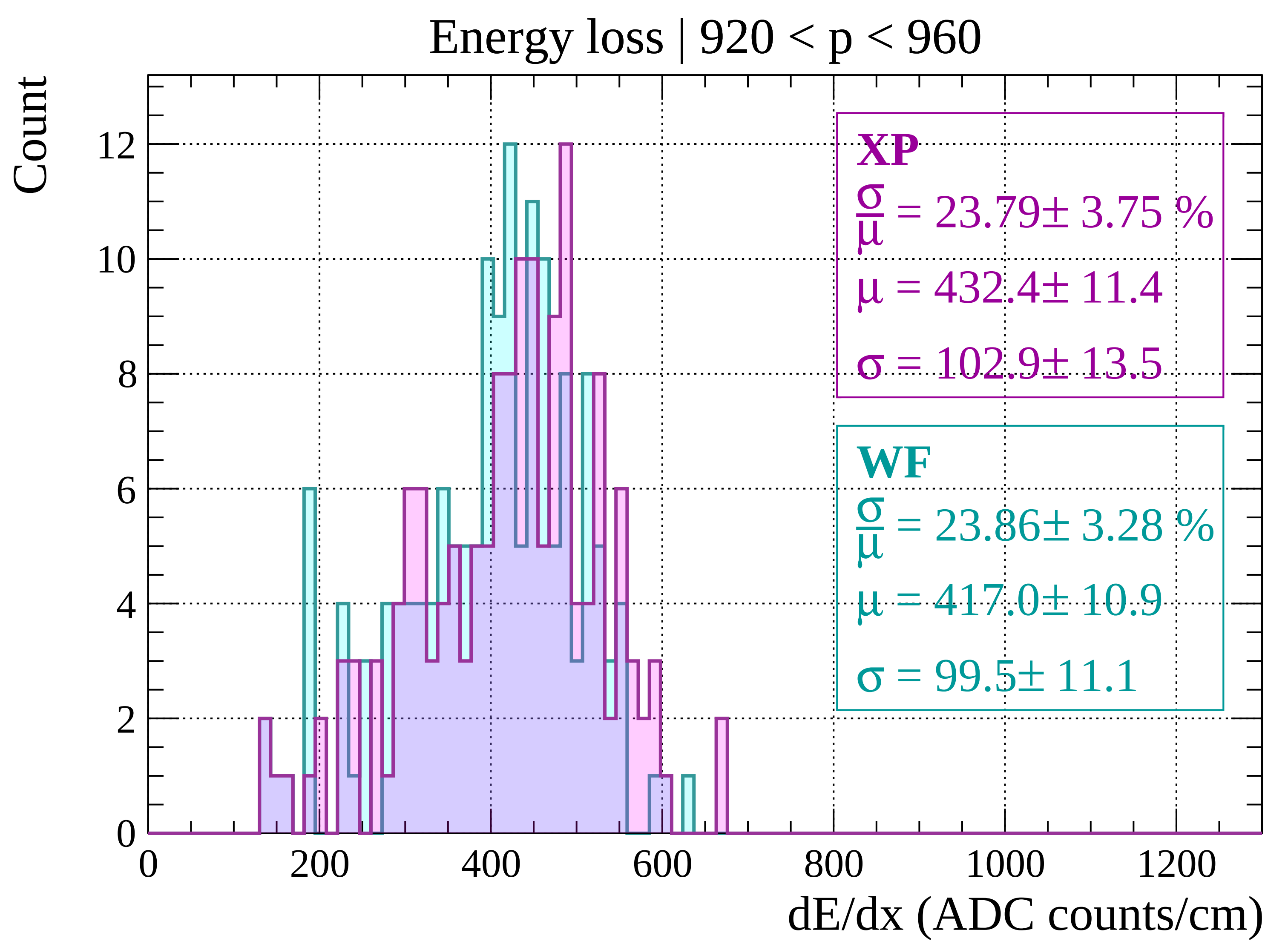


Energy loss | 800 < p < 840
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.72± 1.44 %
μ = 443.5± 6.0
σ = 69.7± 5.5
WF
σ/μ = 14.25± 1.38 %
μ = 414.5± 5.3
σ = 59.1± 5.0

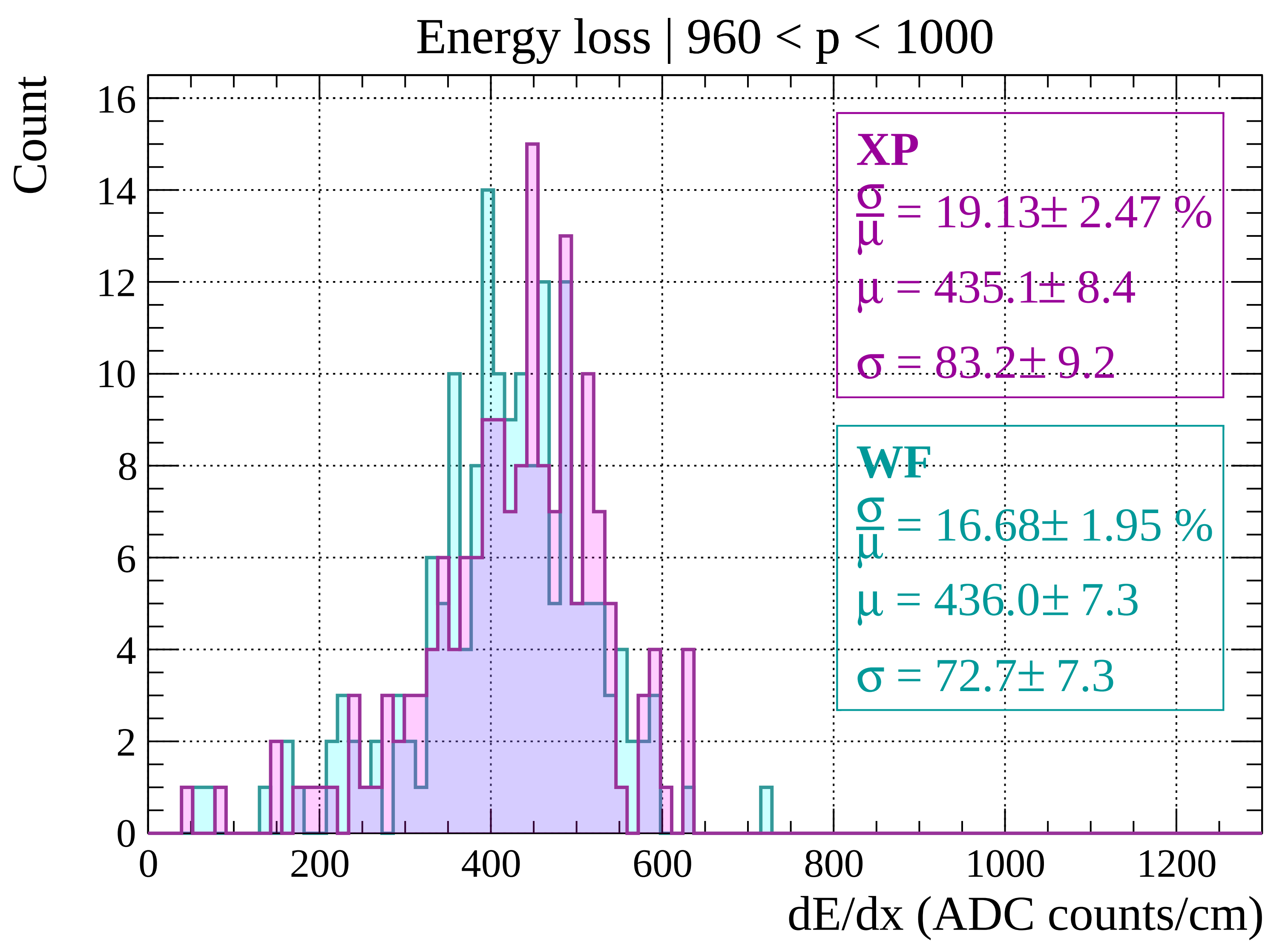


Energy loss | 840 < p < 880
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.65± 1.86 %
μ = 464.2± 7.7
σ = 72.7± 7.4
WF
$\frac{\sigma}{\mu}$ = 17.68± 2.24 %
μ = 427.7± 7.7
σ = 75.6± 8.2

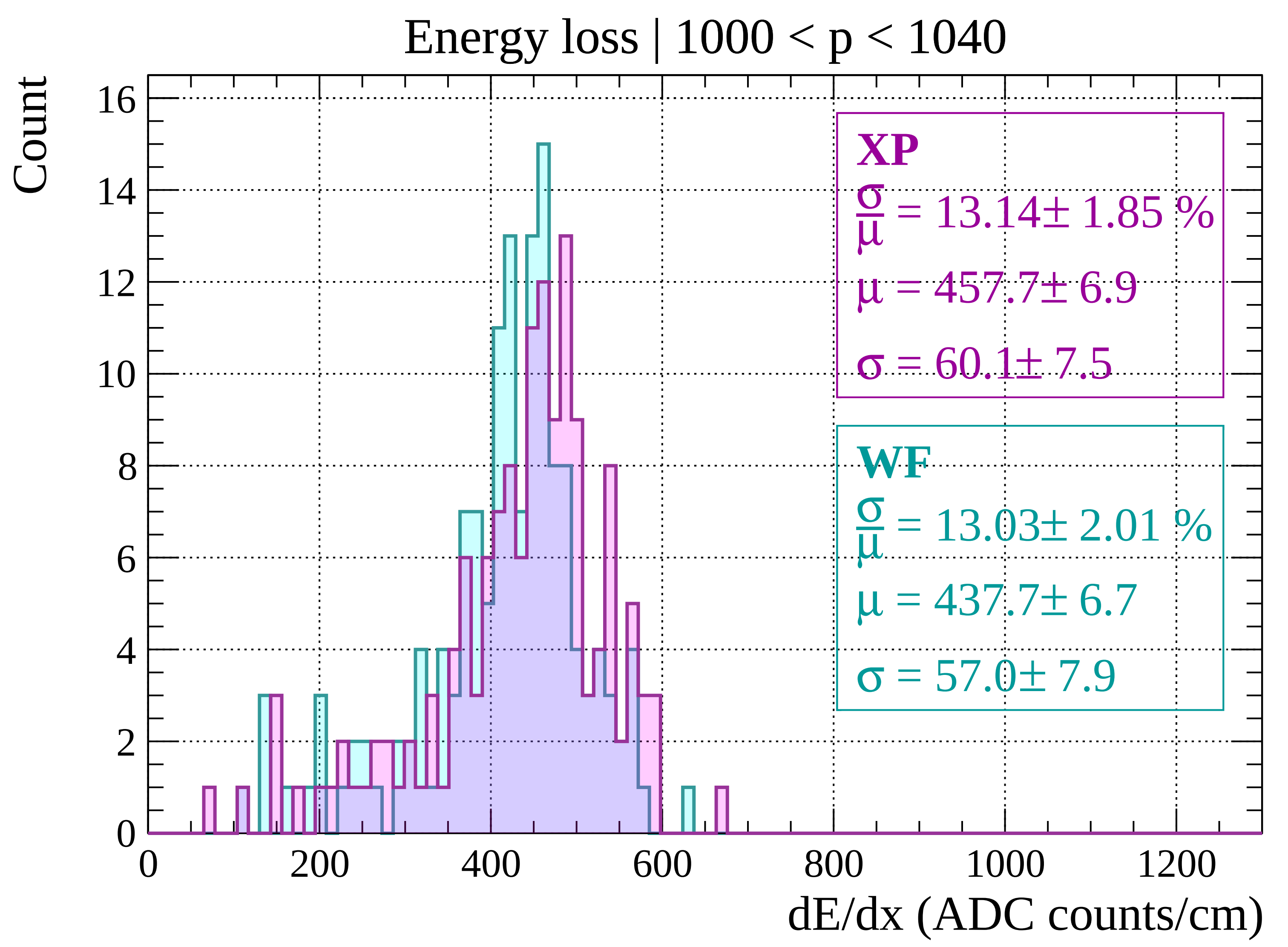

Energy loss | 880 < p < 920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
XP
σ/μ = 20.09± 2.72 %
μ = 455.4± 11.0
σ = 91.5± 10.2
WF
σ/μ = 19.64± 2.73 %
μ = 437.3± 9.5
σ = 85.9± 10.1



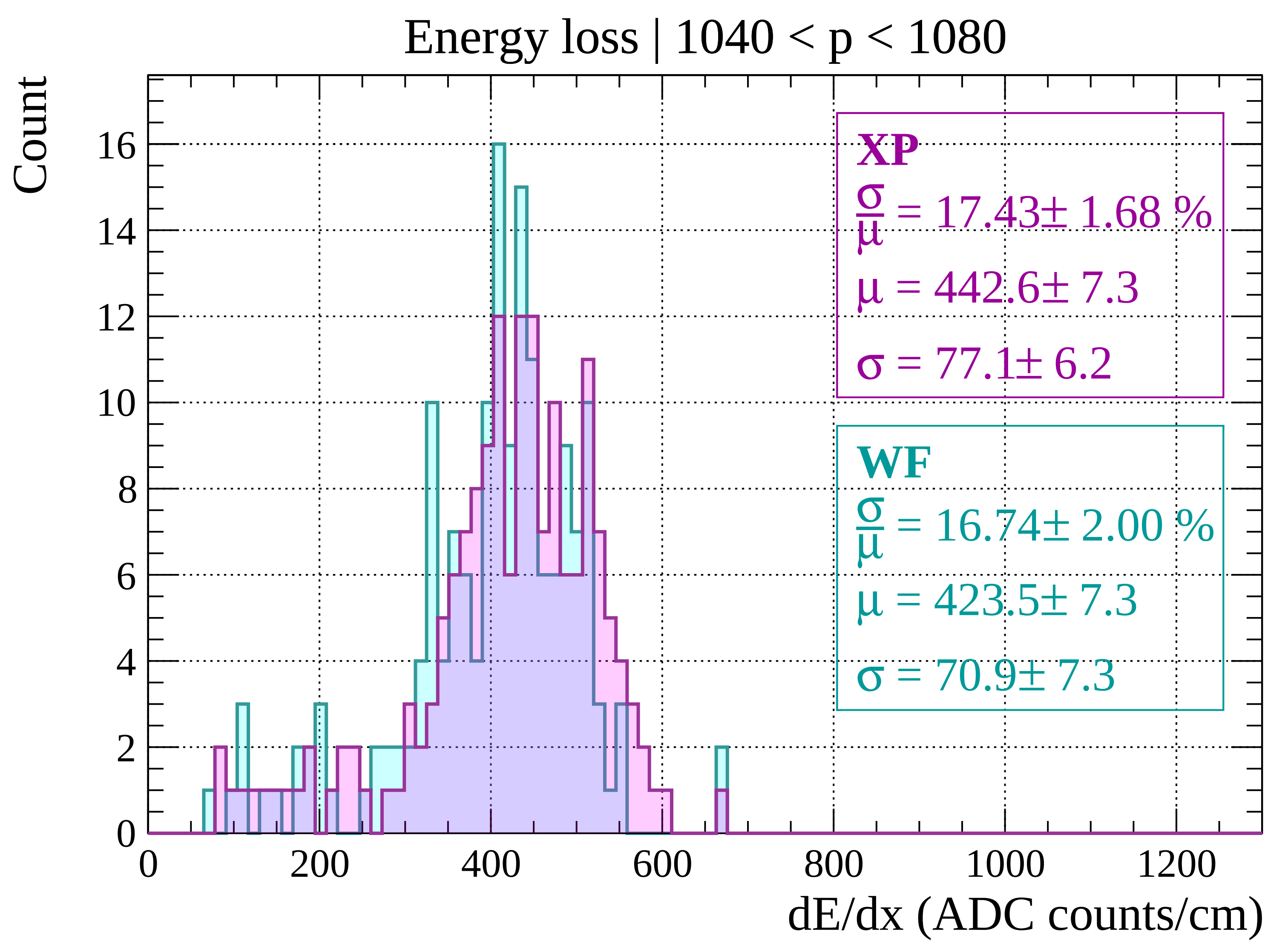
Energy loss | 920 < p < 960
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
σ/μ = 23.79± 3.75 %
μ = 432.4± 11.4
σ = 102.9± 13.5
WF
σ/μ = 23.86± 3.28 %
μ = 417.0± 10.9
σ = 99.5± 11.1

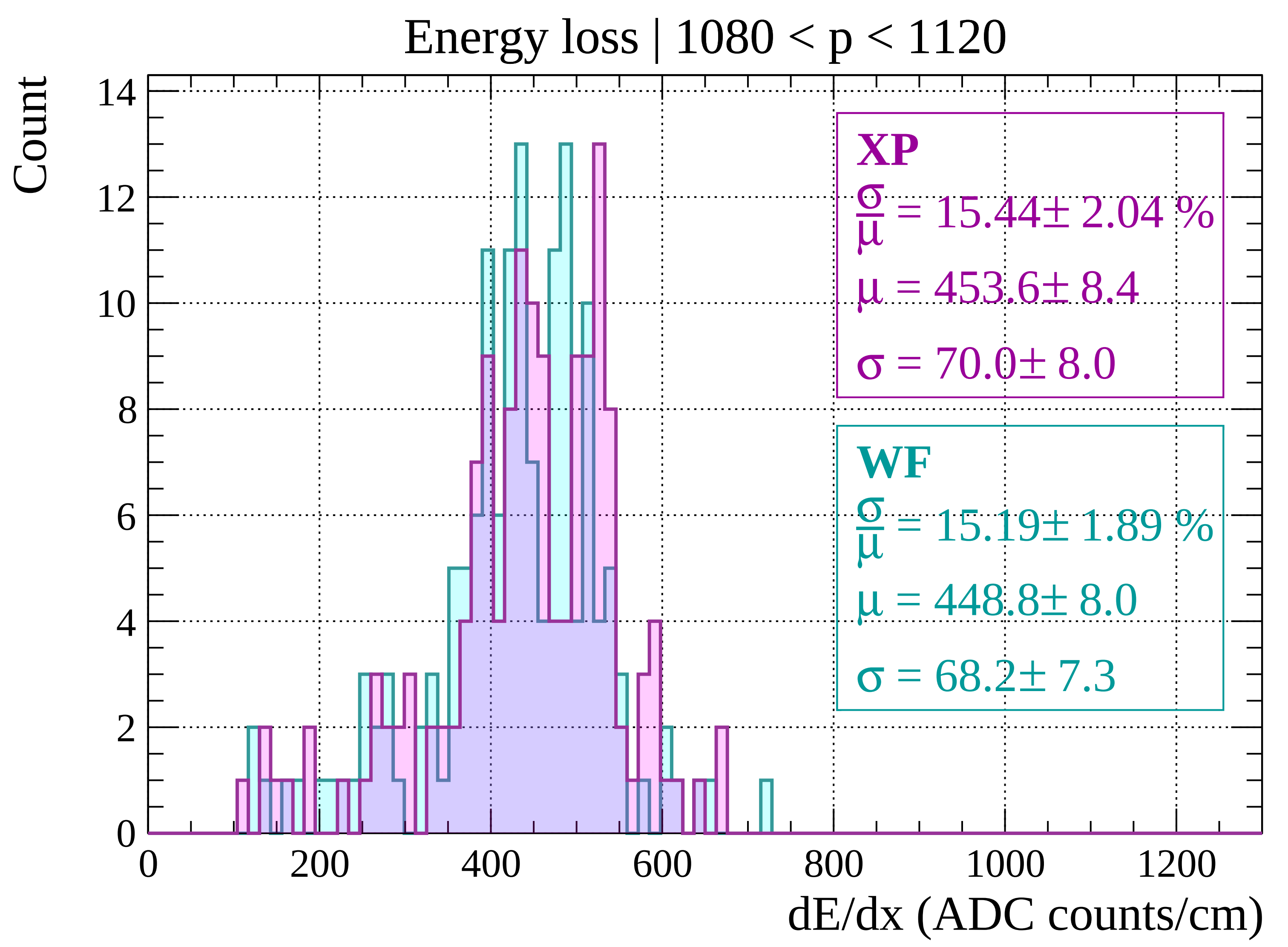


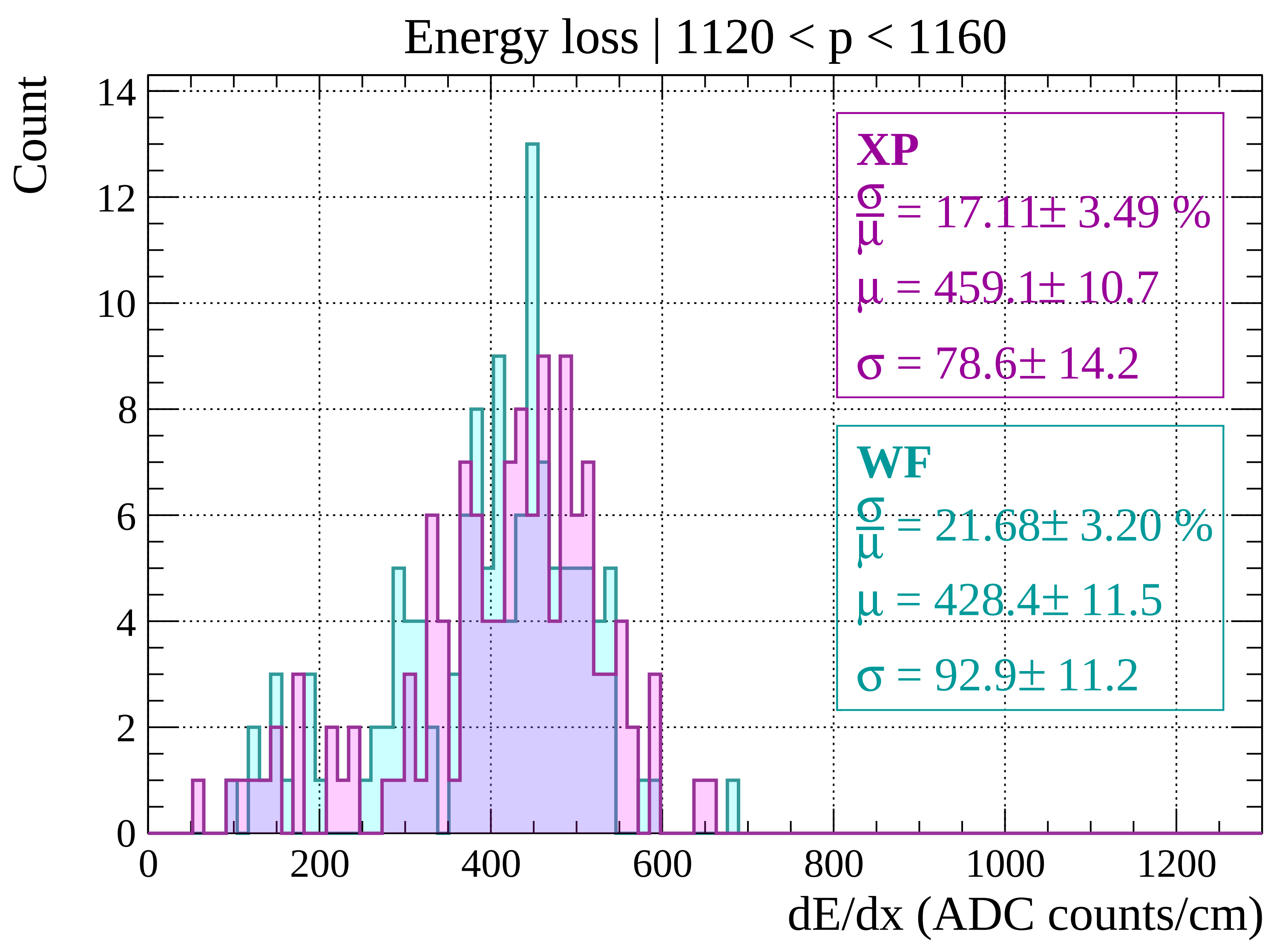
Energy loss | 960 < p < 1000
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
XP
$\frac{\sigma}{\mu}$ = 19.13± 2.47 %
μ = 435.1± 8.4
σ = 83.2± 9.2
WF
$\frac{\sigma}{\mu}$ = 16.68± 1.95 %
μ = 436.0± 7.3
σ = 72.7± 7.3



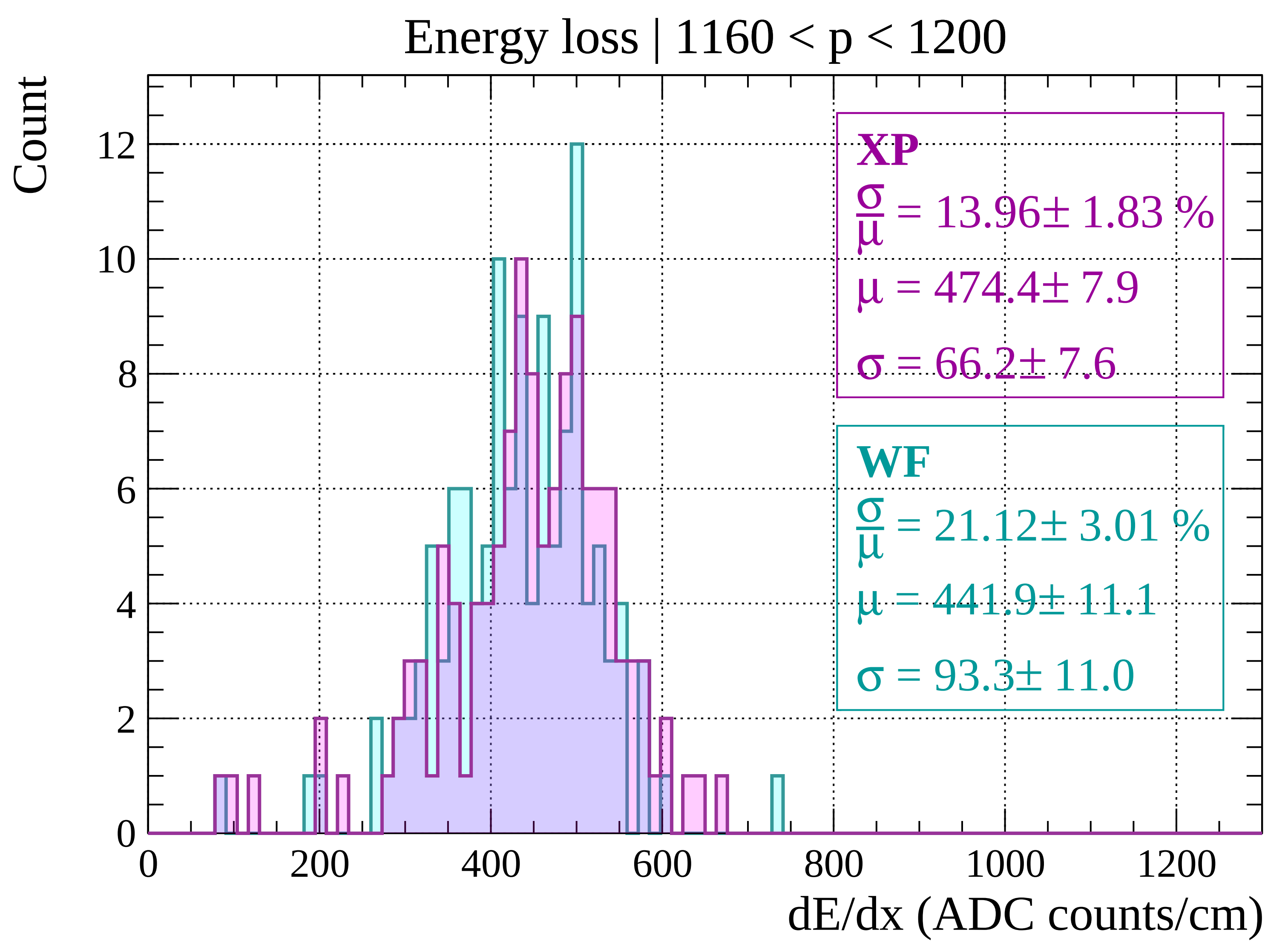
Energy loss | 1000 < p < 1040
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 13.14± 1.85 %
μ = 457.7± 6.9
σ = 60.1± 7.5
WF
$\frac{\sigma}{\mu}$ = 13.03± 2.01 %
μ = 437.7± 6.7
σ = 57.0± 7.9


Energy loss | 1040 < p < 1080
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
0
200
400
600
800
1000
1200
XP
σ/μ = 17.43± 1.68 %
μ = 442.6± 7.3
σ = 77.1± 6.2
WF
σ/μ = 16.74± 2.00 %
μ = 423.5± 7.3
σ = 70.9± 7.3

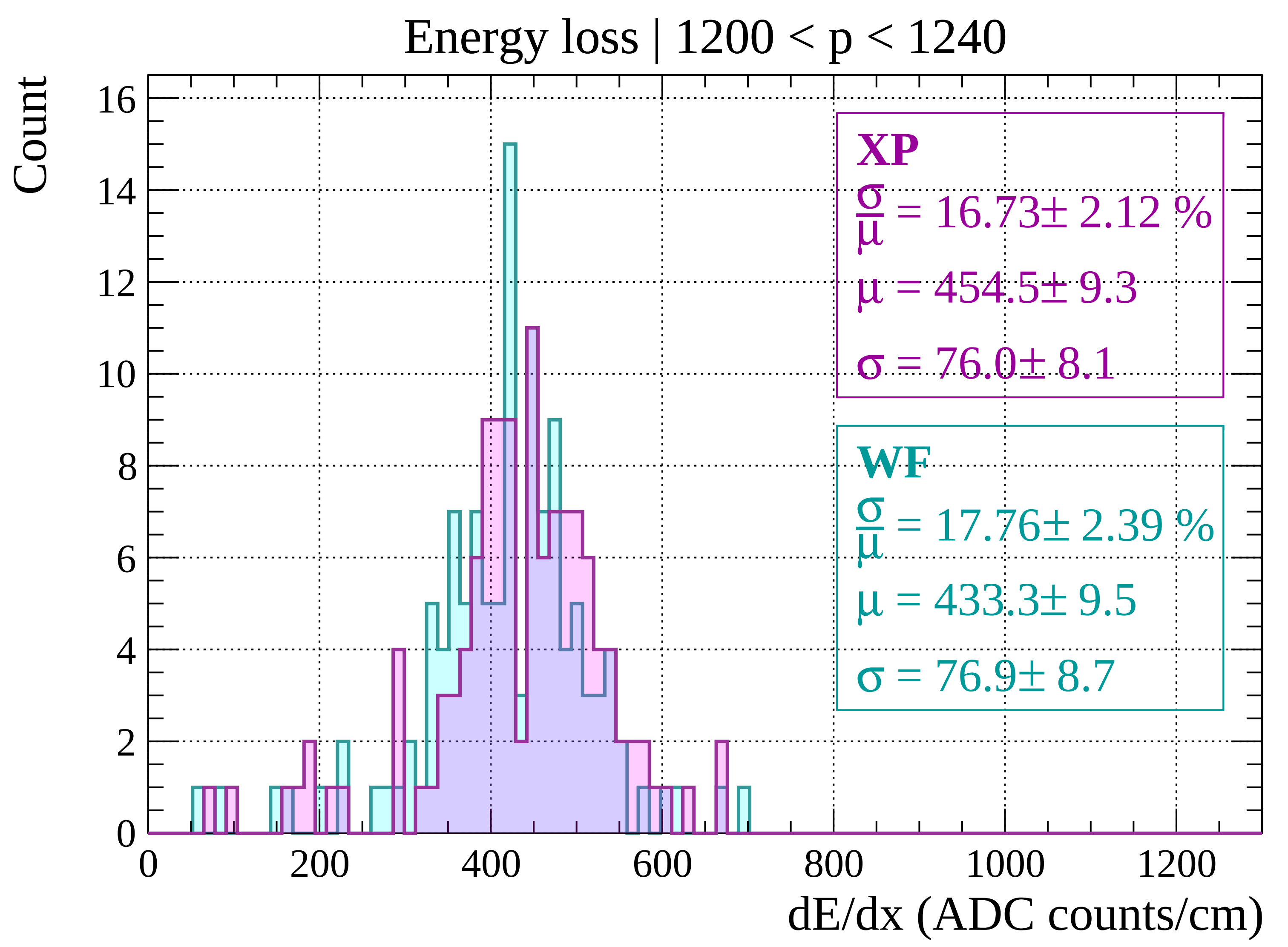

Energy loss | 1080 < p < 1120
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.44± 2.04 %
μ = 453.6± 8.4
σ = 70.0± 8.0
WF
σ/μ = 15.19± 1.89 %
μ = 448.8± 8.0
σ = 68.2± 7.3
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200

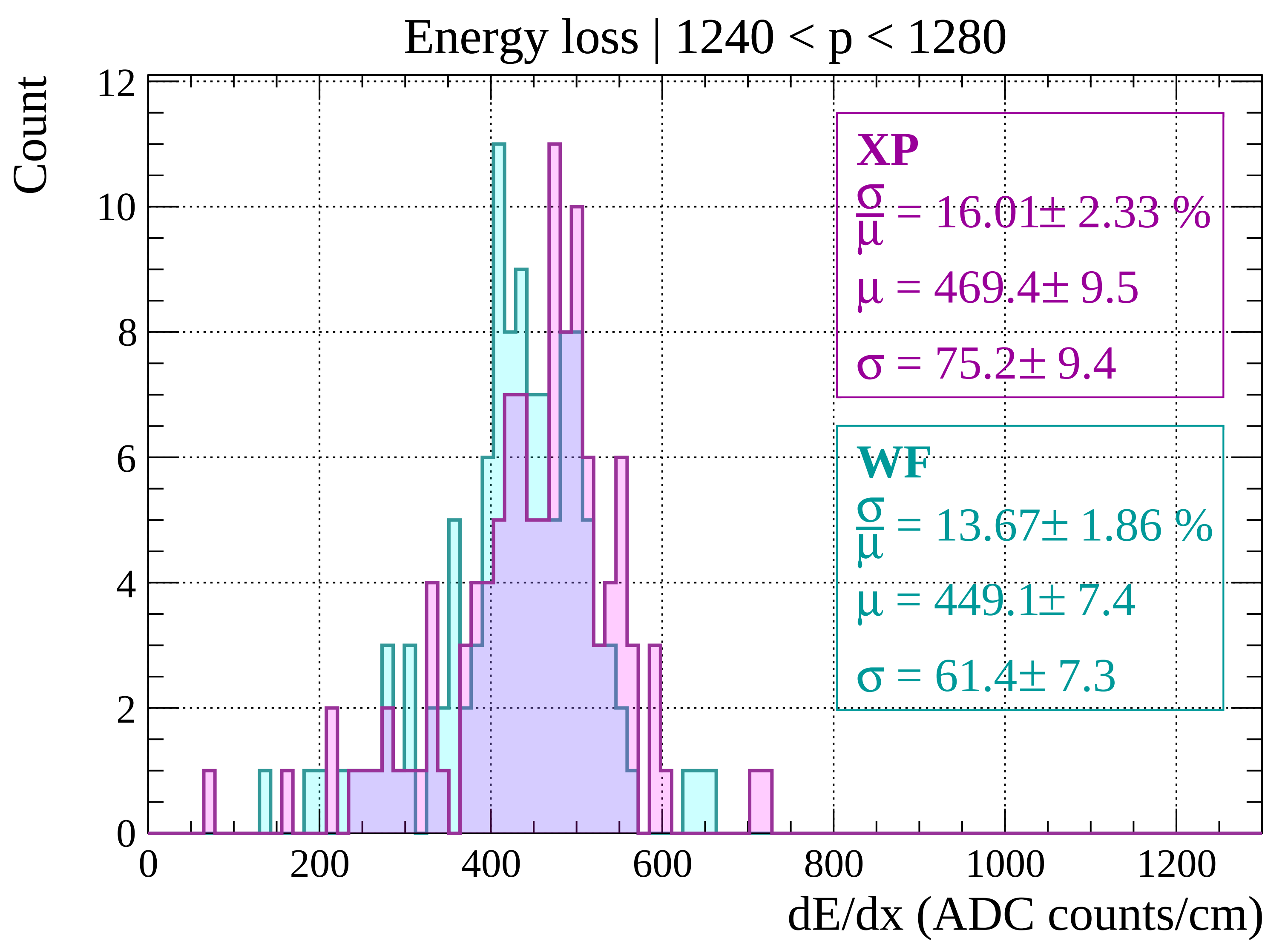


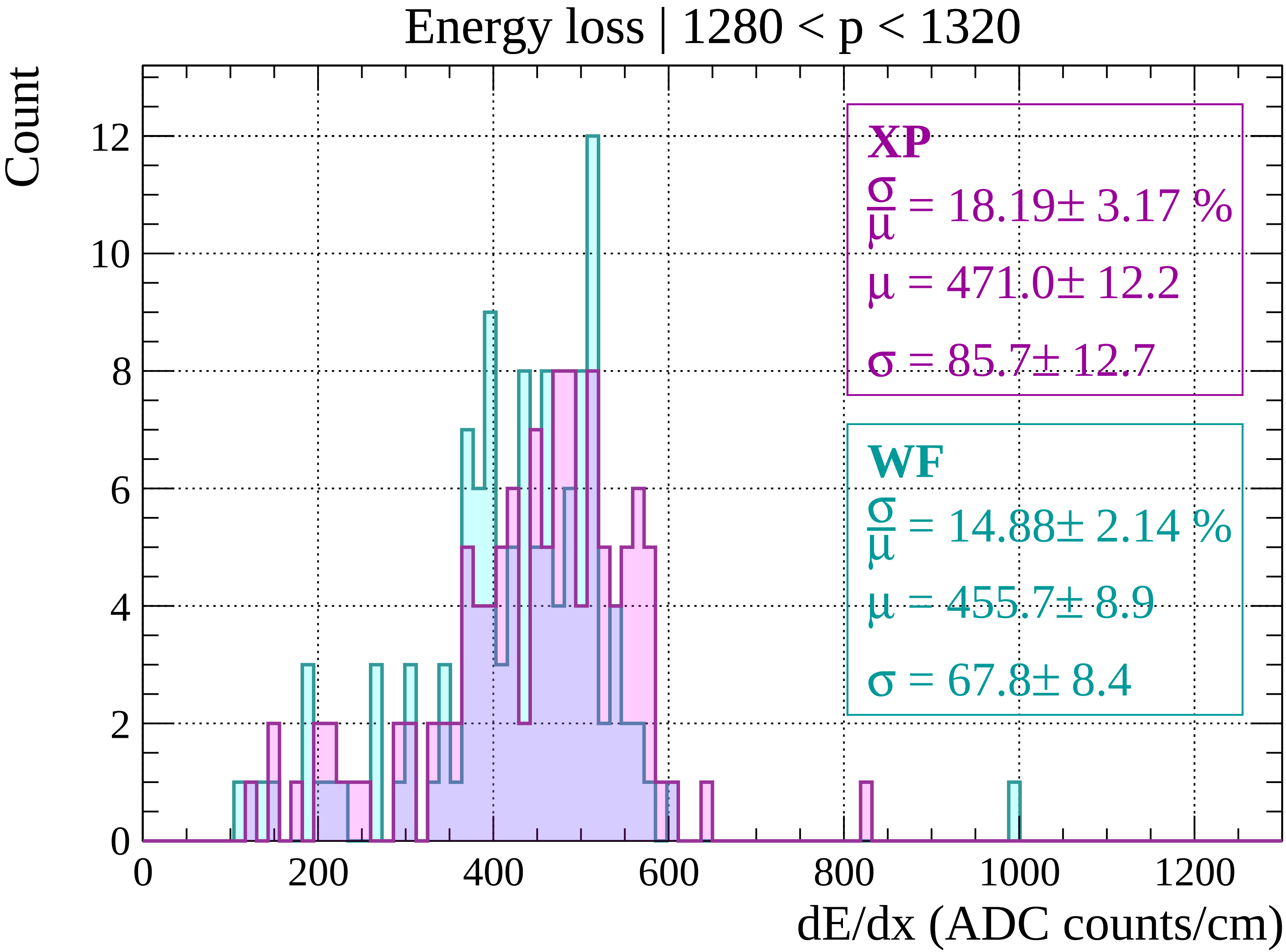
Energy loss | 1120 < p < 1160
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 17.11± 3.49 %
μ = 459.1± 10.7
σ = 78.6± 14.2
WF
$\frac{\sigma}{\mu}$ = 21.68± 3.20 %
μ = 428.4± 11.5
σ = 92.9± 11.2



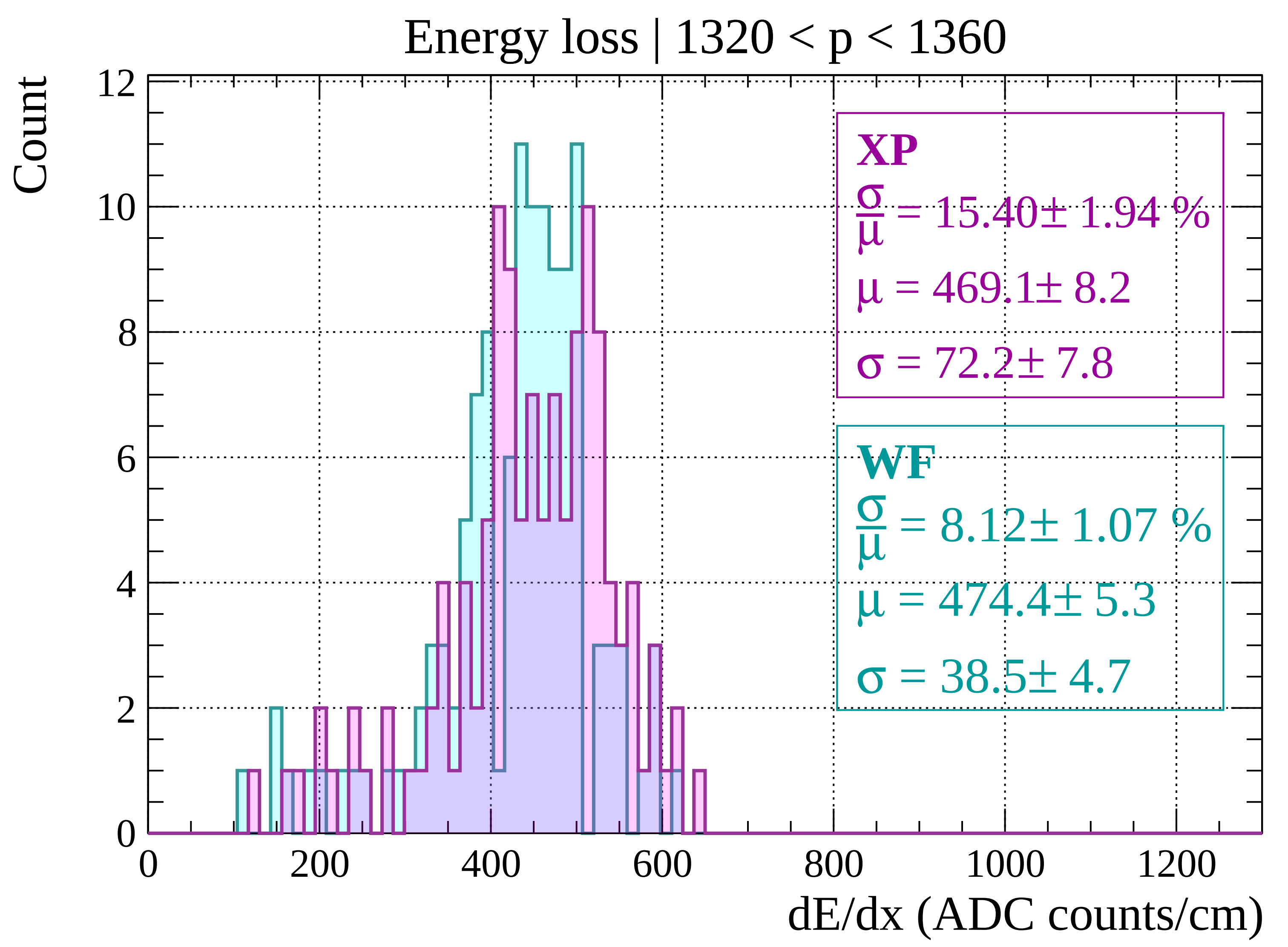
Energy loss | 1160 < p < 1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
XP
σ/μ = 13.96± 1.83 %
μ = 474.4± 7.9
σ = 66.2± 7.6
WF
σ/μ = 21.12± 3.01 %
μ = 441.9± 11.1
σ = 93.3± 11.0


Energy loss | 1200 < p < 1240
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.73± 2.12 %
μ = 454.5± 9.3
σ = 76.0± 8.1
WF
σ/μ = 17.76± 2.39 %
μ = 433.3± 9.5
σ = 76.9± 8.7
0
2
4
6
8
10
12
14
16
0
200
400
600
800
1000
1200

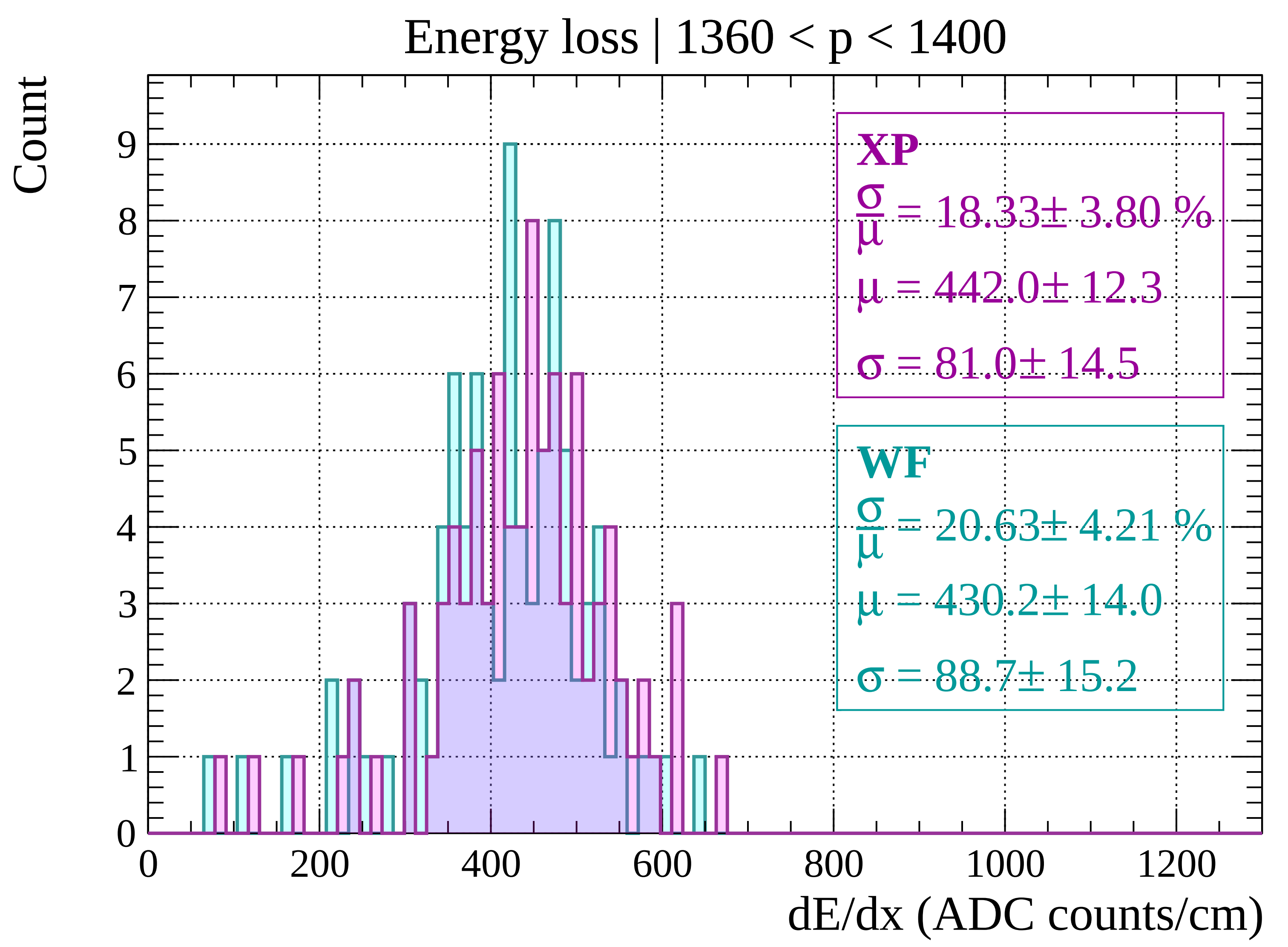

Energy loss | 1240 < p < 1280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
XP
$\frac{\sigma}{\mu}$ = 16.01± 2.33 %
μ = 469.4± 9.5
σ = 75.2± 9.4
WF
$\frac{\sigma}{\mu}$ = 13.67± 1.86 %
μ = 449.1± 7.4
σ = 61.4± 7.3



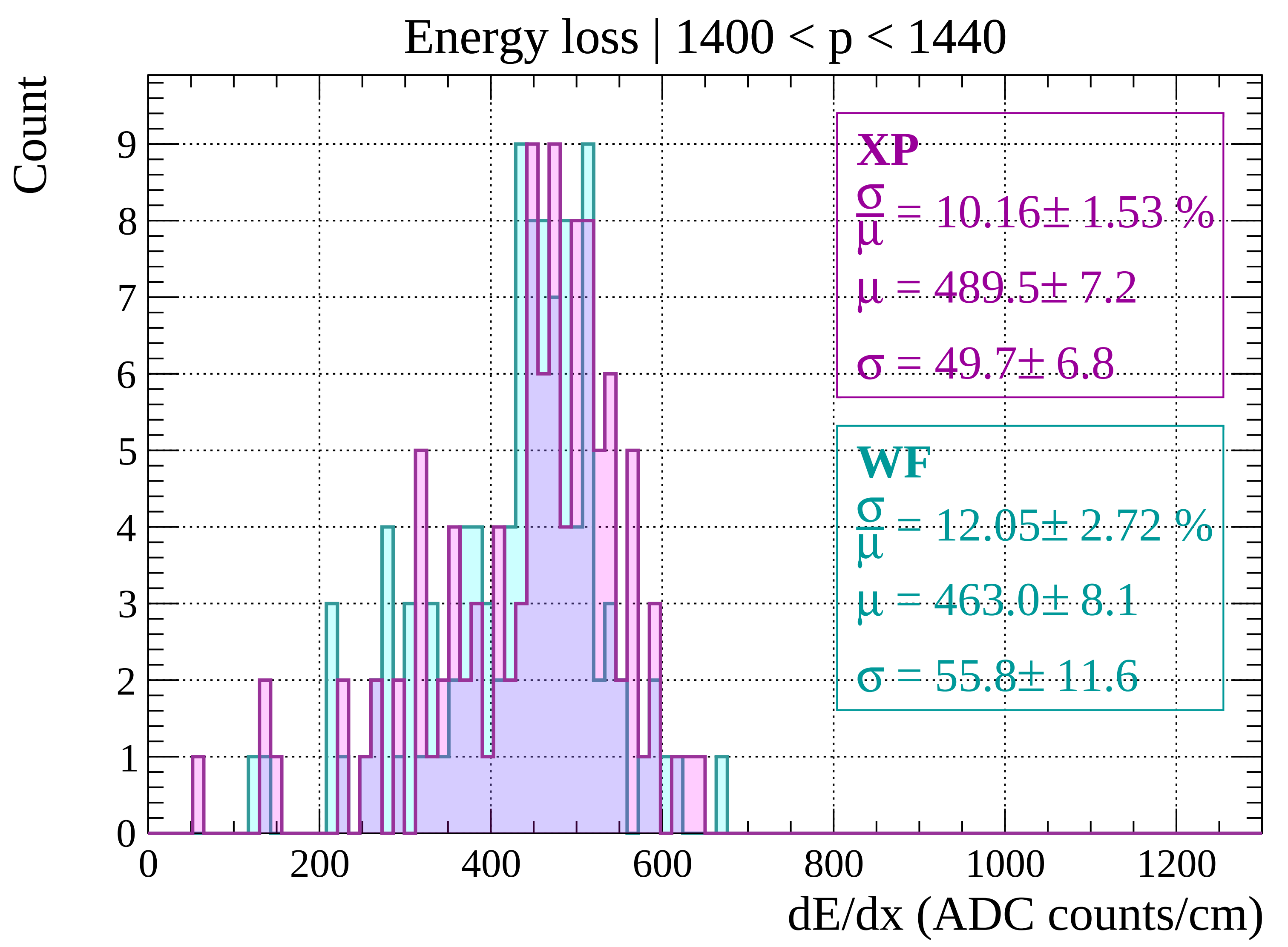
Energy loss | 1280 < p < 1320
Count
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 18.19± 3.17 %
μ = 471.0± 12.2
σ = 85.7± 12.7
WF
$\frac{\sigma}{\mu}$ = 14.88± 2.14 %
μ = 455.7± 8.9
σ = 67.8± 8.4



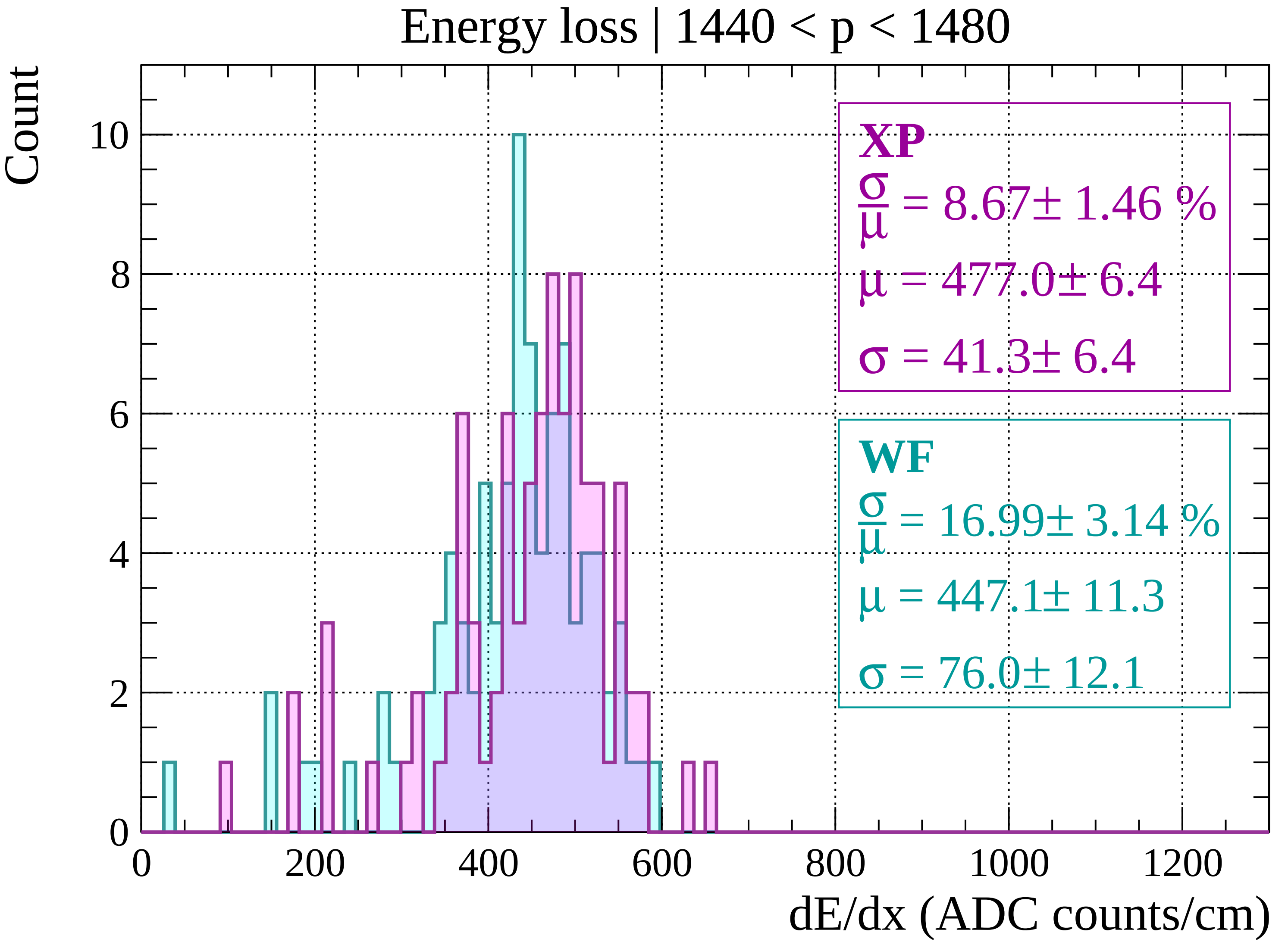
Energy loss | 1320 < p < 1360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
XP
σ/μ = 15.40± 1.94 %
μ = 469.1± 8.2
σ = 72.2± 7.8
WF
σ/μ = 8.12± 1.07 %
μ = 474.4± 5.3
σ = 38.5± 4.7



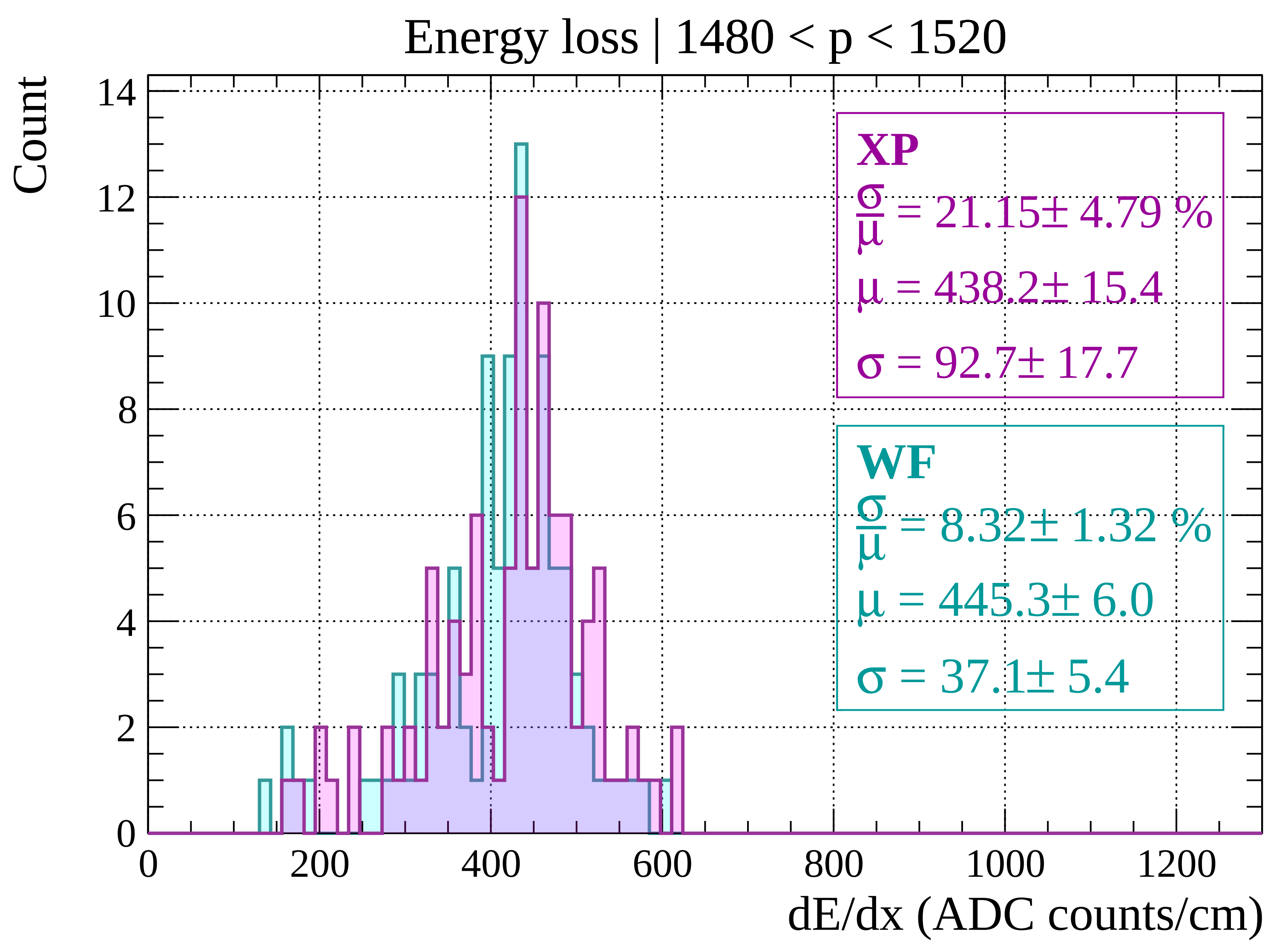
Energy loss | 1360 < p < 1400
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
8
9
0
200
400
600
800
1000
1200
XP
σ/μ = 18.33± 3.80 %
μ = 442.0± 12.3
σ = 81.0± 14.5
WF
σ/μ = 20.63± 4.21 %
μ = 430.2± 14.0
σ = 88.7± 15.2

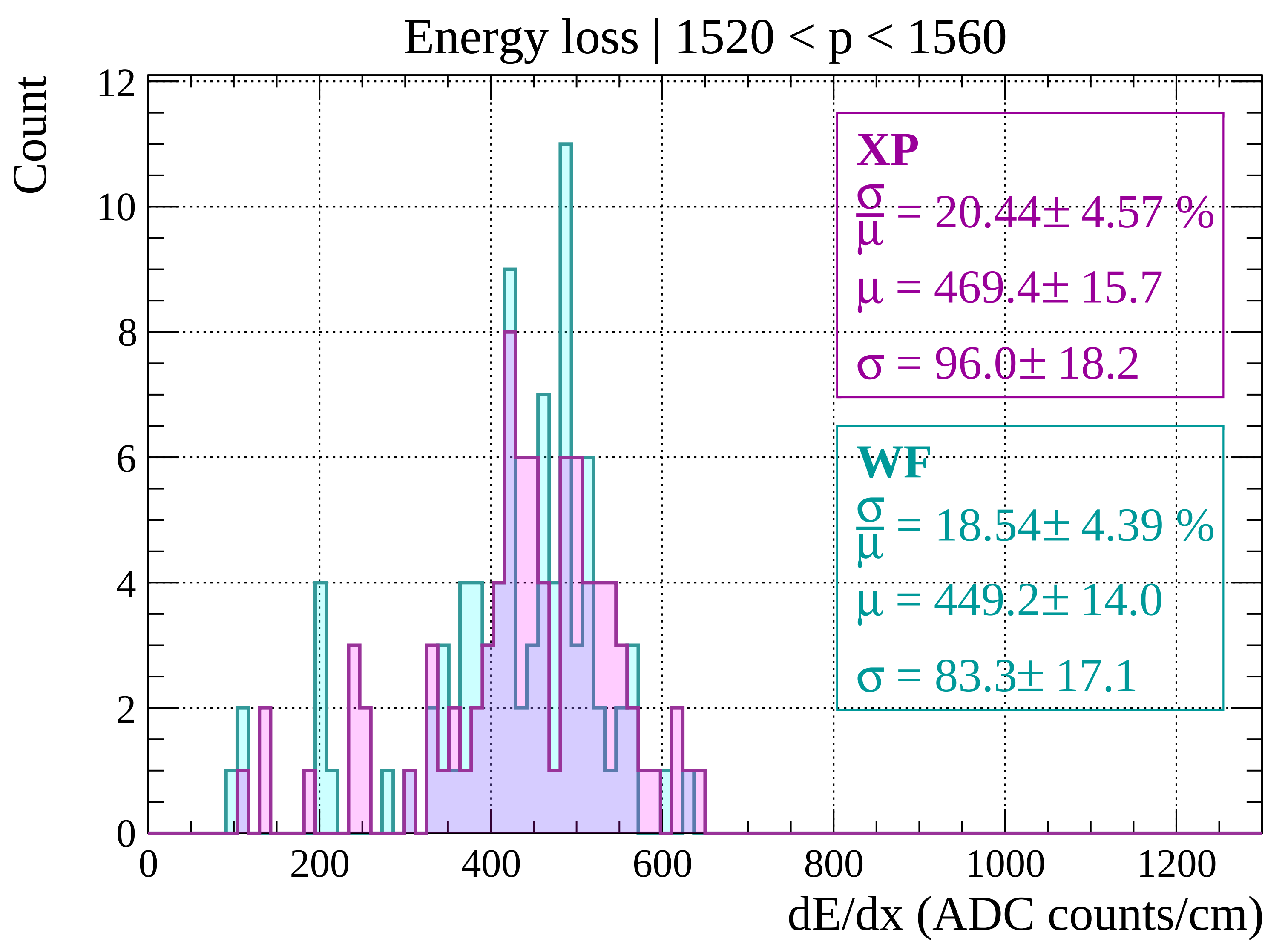


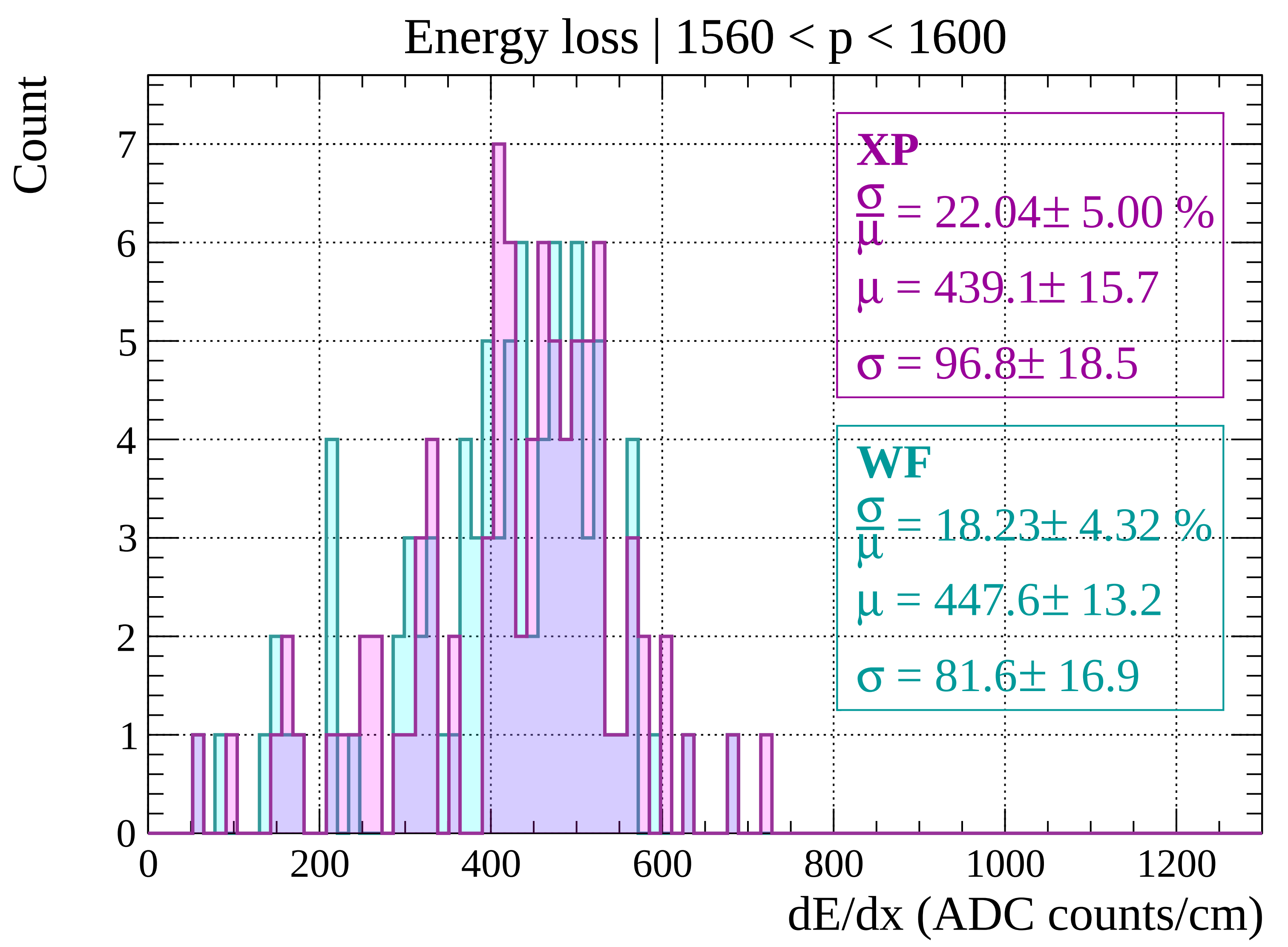
Energy loss | 1400 < p < 1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
8
9
XP
σ/μ = 10.16± 1.53 %
μ = 489.5± 7.2
σ = 49.7± 6.8
WF
σ/μ = 12.05± 2.72 %
μ = 463.0± 8.1
σ = 55.8± 11.6

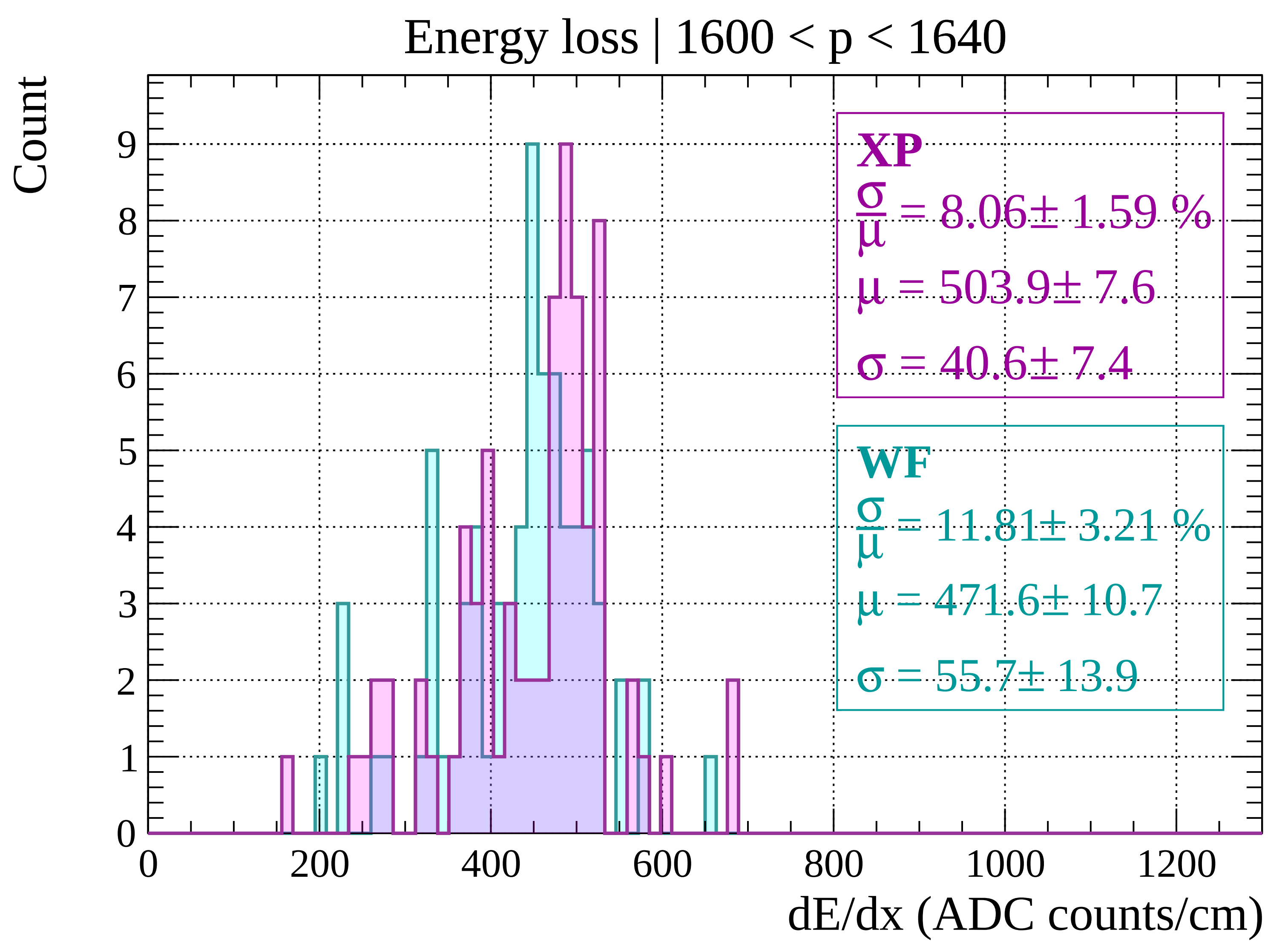


Energy loss | 1440 < p < 1480
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
0
200
400
600
800
1000
1200
XP
σ/μ = 8.67± 1.46 %
μ = 477.0± 6.4
σ = 41.3± 6.4
WF
σ/μ = 16.99± 3.14 %
μ = 447.1± 11.3
σ = 76.0± 12.1

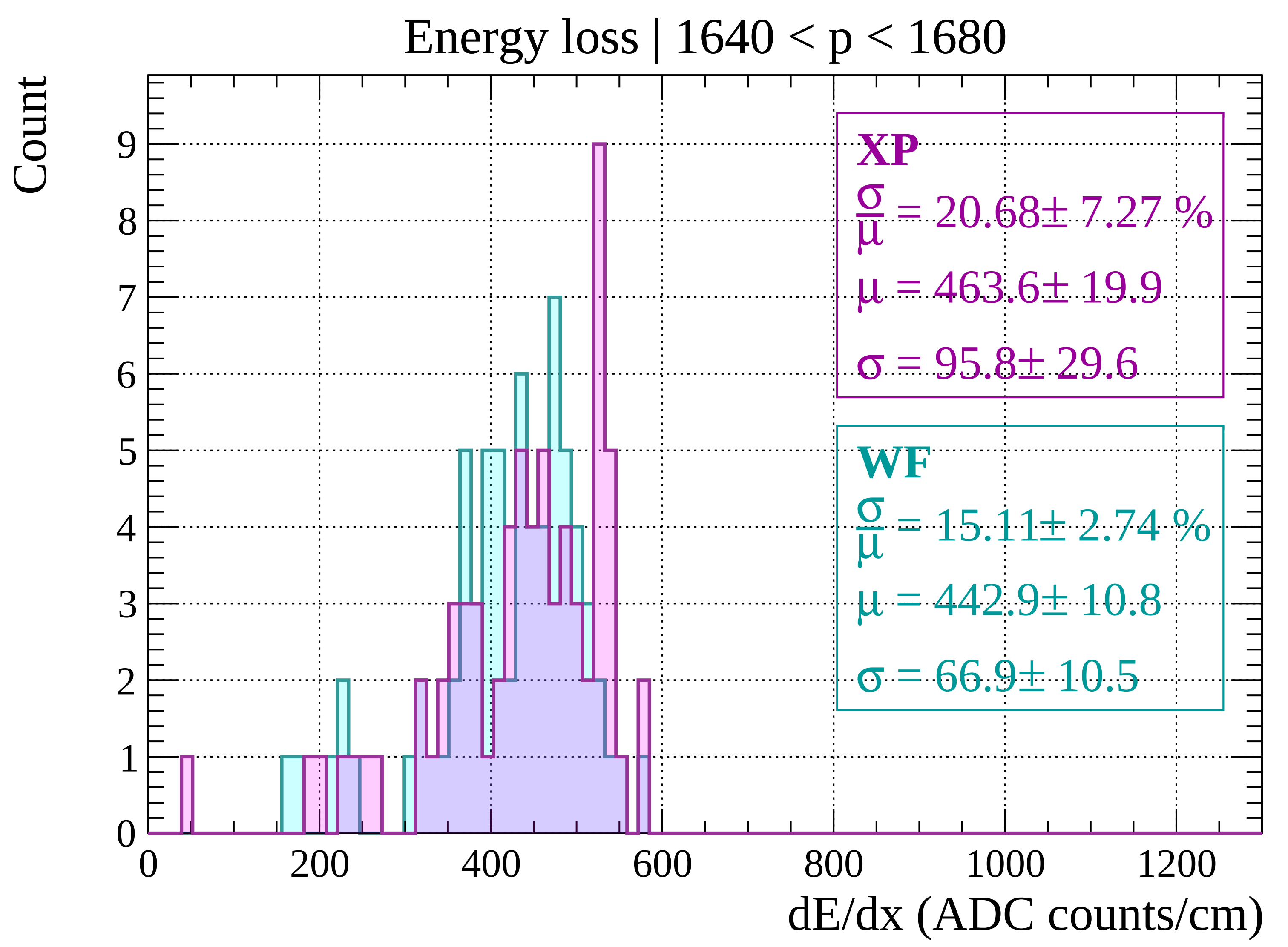

Energy loss | 1480 < p < 1520
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
XP
σ/μ = 21.15± 4.79 %
μ = 438.2± 15.4
σ = 92.7± 17.7
WF
σ/μ = 8.32± 1.32 %
μ = 445.3± 6.0
σ = 37.1± 5.4

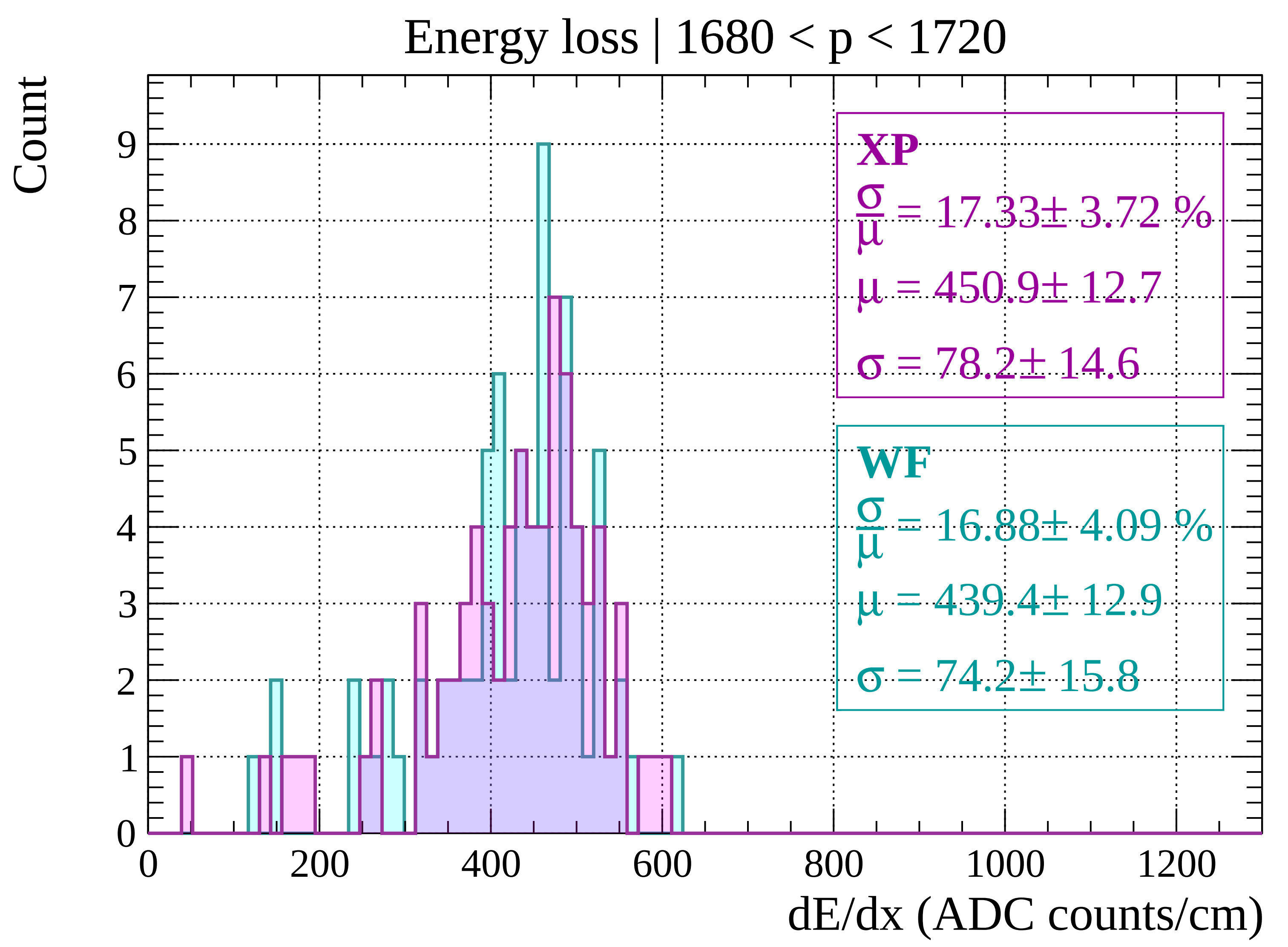

Energy loss | 1520 < p < 1560
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 20.44± 4.57 %
μ = 469.4± 15.7
σ = 96.0± 18.2
WF
$\frac{\sigma}{\mu}$ = 18.54± 4.39 %
μ = 449.2± 14.0
σ = 83.3± 17.1


Energy loss | 1560 < p < 1600
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 22.04± 5.00 %
μ = 439.1± 15.7
σ = 96.8± 18.5
WF
σ/μ = 18.23± 4.32 %
μ = 447.6± 13.2
σ = 81.6± 16.9
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7

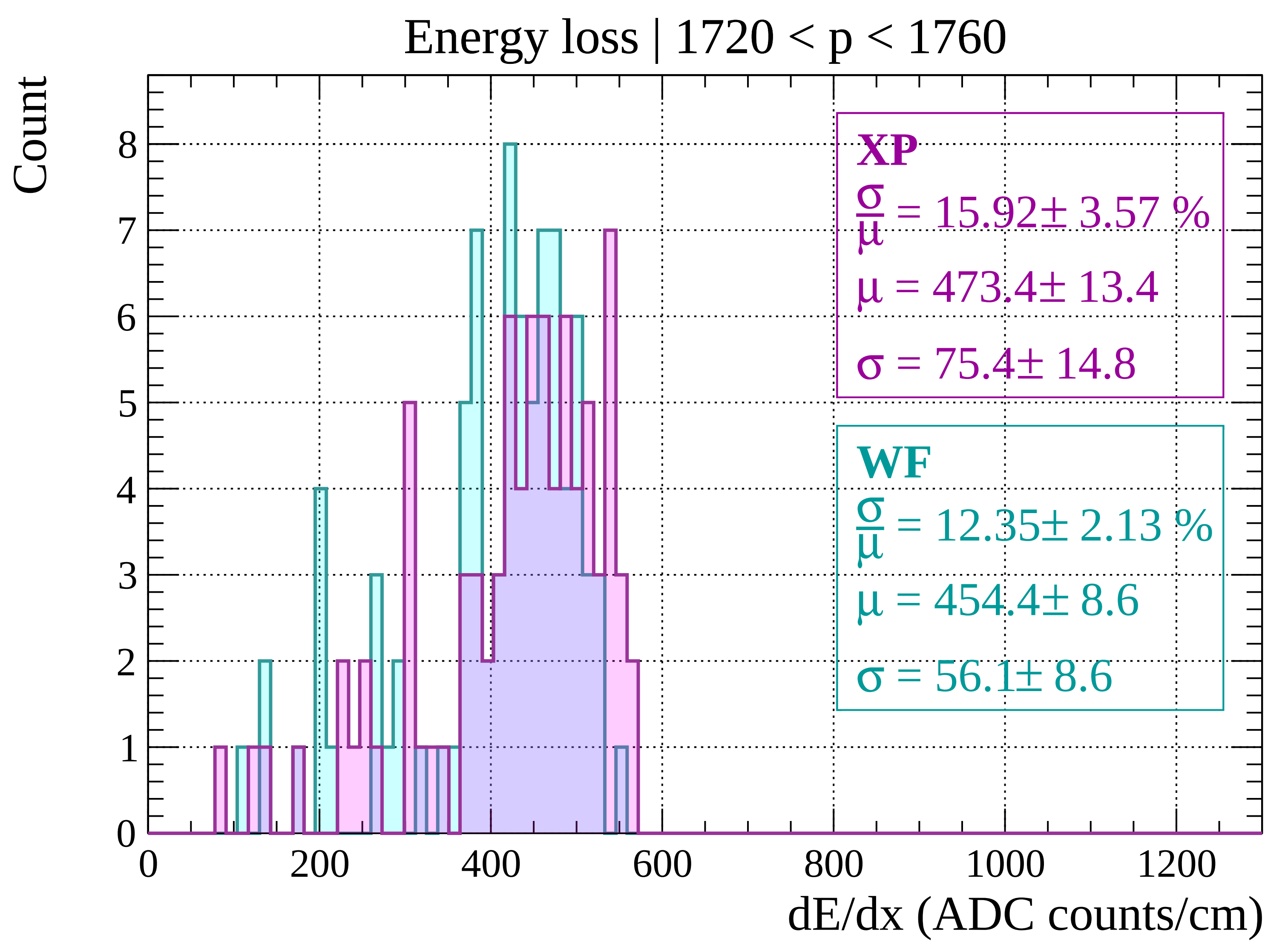


Energy loss | 1600 < p < 1640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
9
XP
$\frac{\sigma}{\mu}$ = 8.06± 1.59 %
μ = 503.9± 7.6
σ = 40.6± 7.4
WF
$\frac{\sigma}{\mu}$ = 11.81± 3.21 %
μ = 471.6± 10.7
σ = 55.7± 13.9

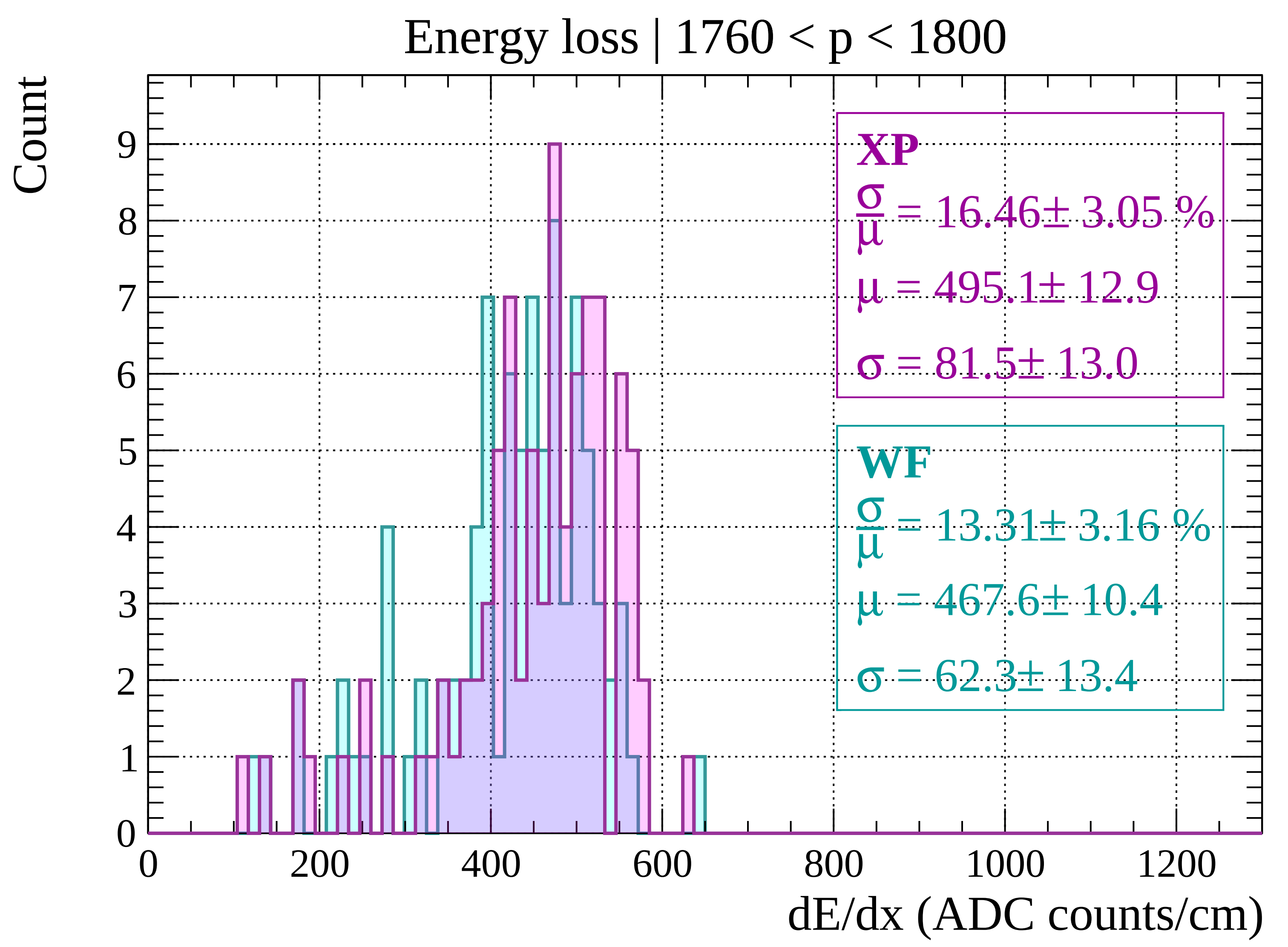

Energy loss | 1640 < p < 1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
8
9
XP
$\frac{\sigma}{\mu}$ = 20.68± 7.27 %
μ = 463.6± 19.9
σ = 95.8± 29.6
WF
$\frac{\sigma}{\mu}$ = 15.11± 2.74 %
μ = 442.9± 10.8
σ = 66.9± 10.5



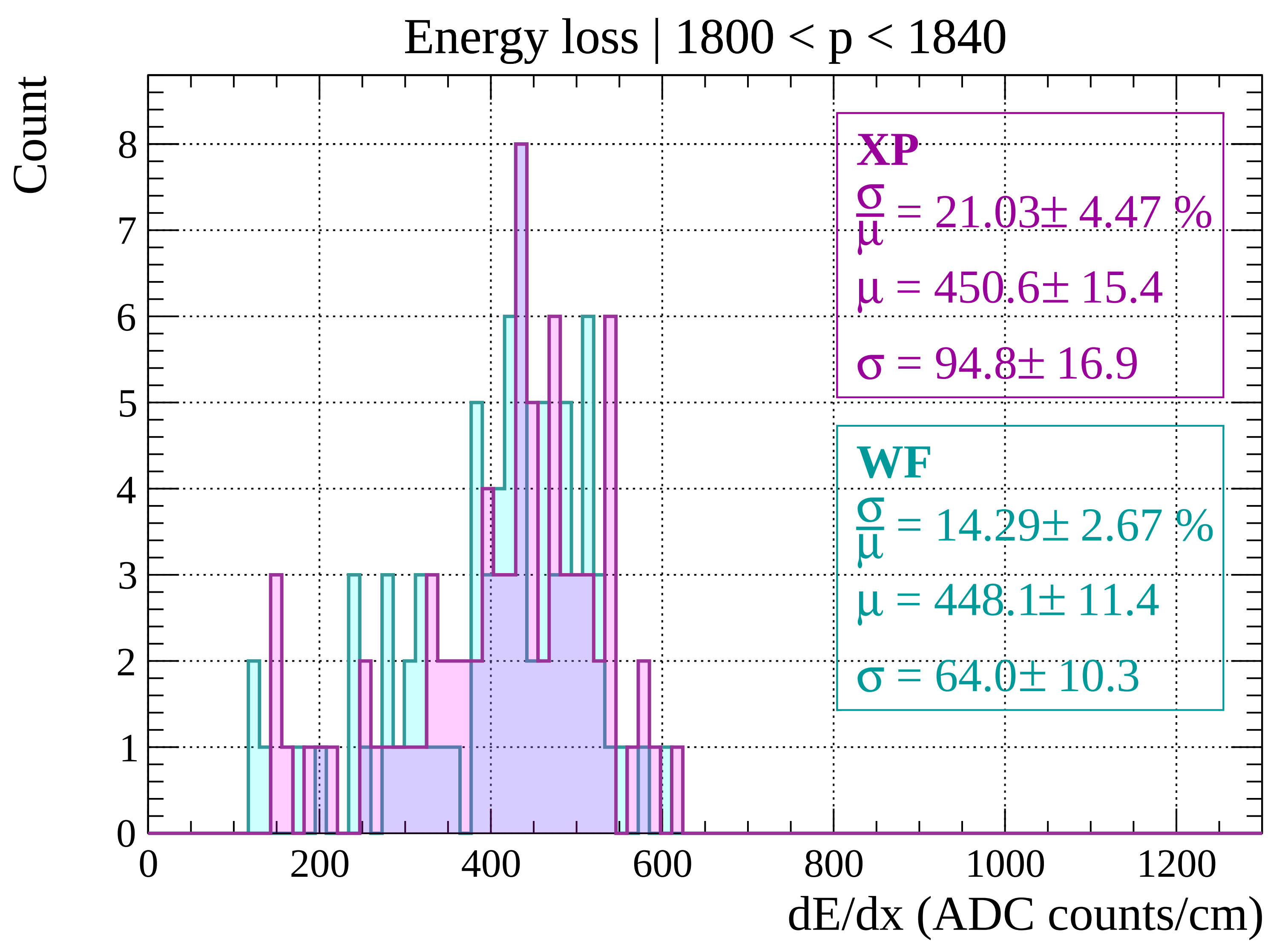
Energy loss | 1680 < p < 1720
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
8
9
0
200
400
600
800
1000
1200
XP
σ/μ = 17.33± 3.72 %
μ = 450.9± 12.7
σ = 78.2± 14.6
WF
σ/μ = 16.88± 4.09 %
μ = 439.4± 12.9
σ = 74.2± 15.8



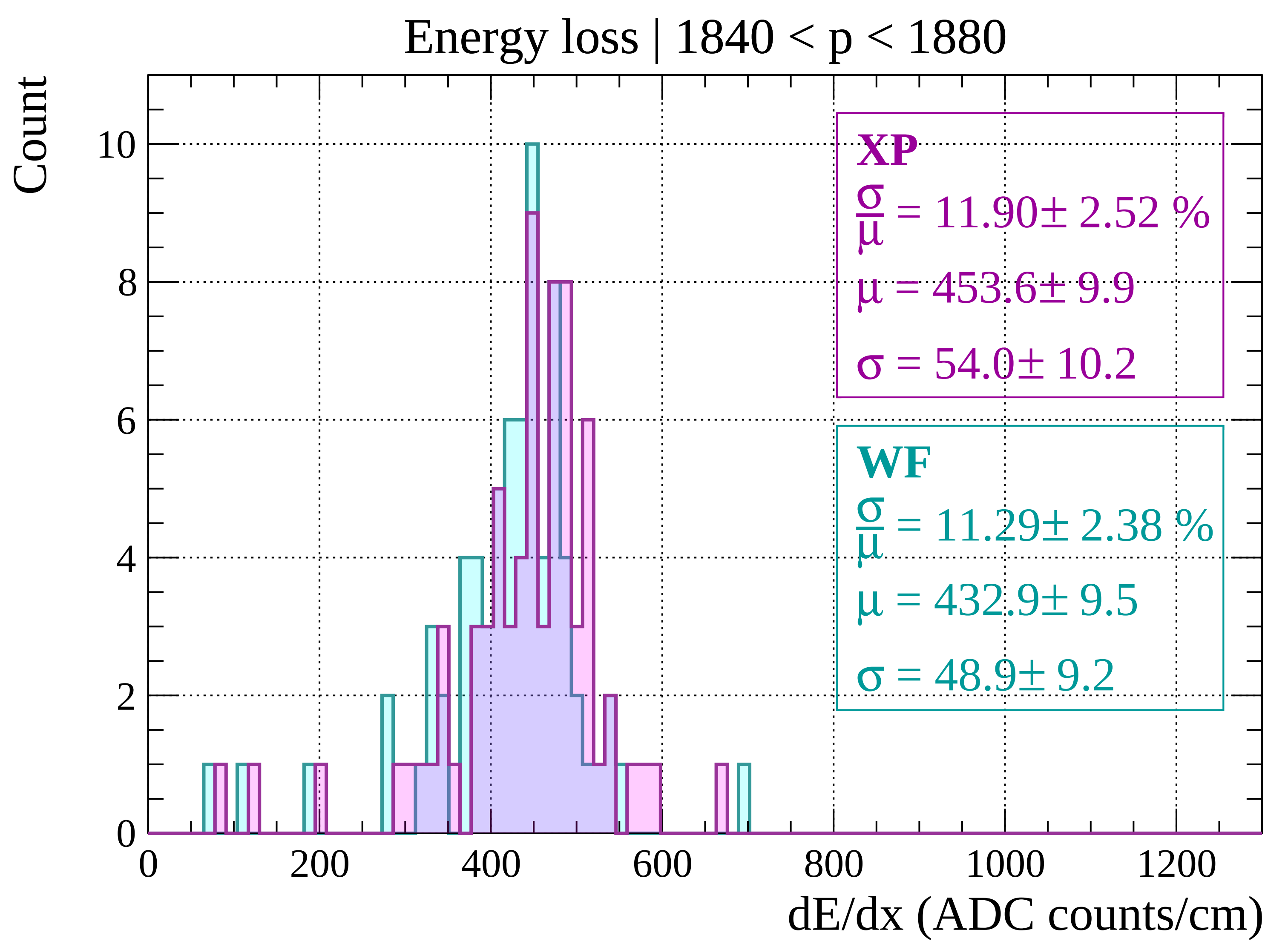
Energy loss | 1720 < p < 1760
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
8
XP
σ/μ = 15.92± 3.57 %
μ = 473.4± 13.4
σ = 75.4± 14.8
WF
σ/μ = 12.35± 2.13 %
μ = 454.4± 8.6
σ = 56.1± 8.6



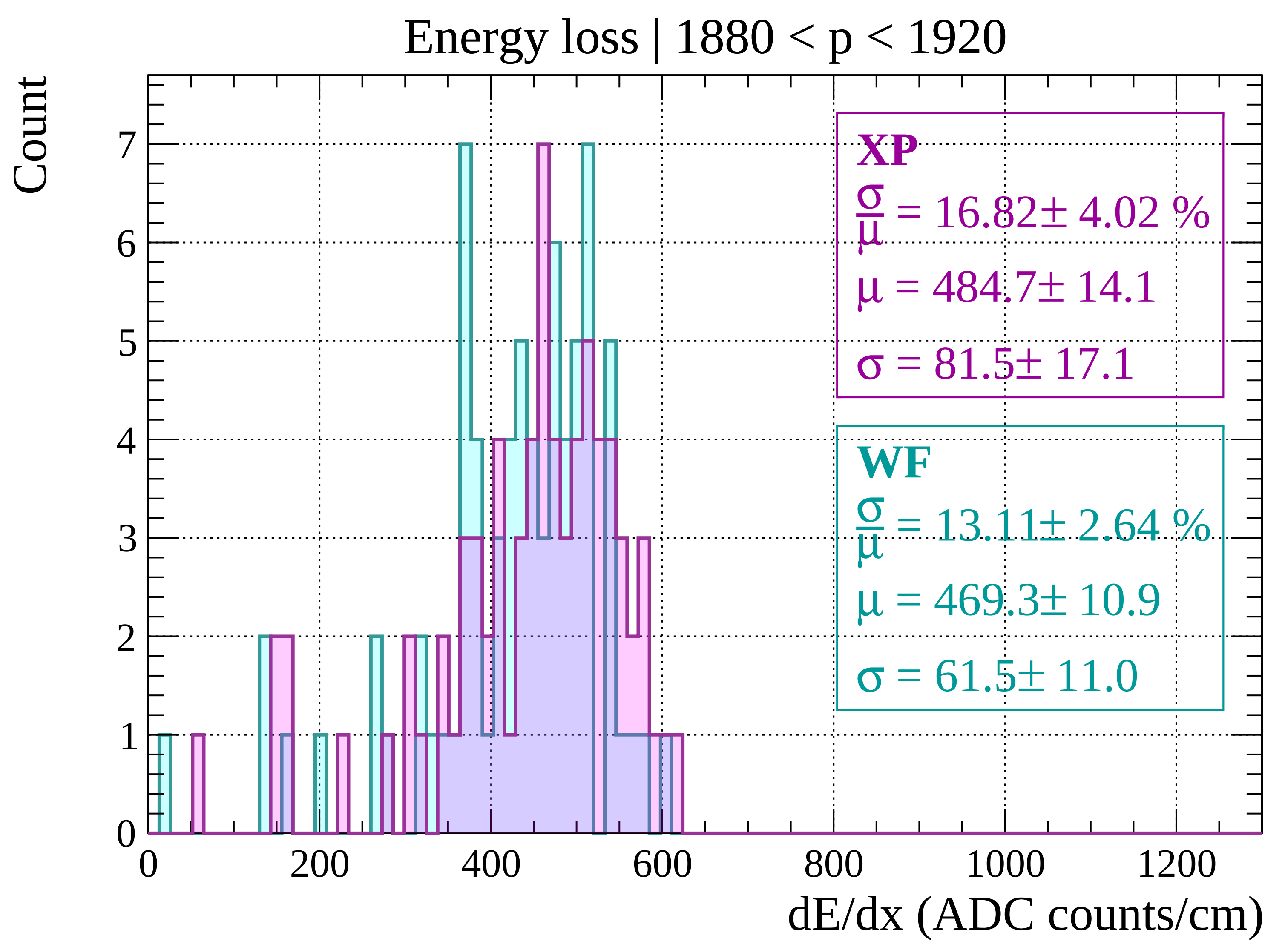
Energy loss | 1760 < p < 1800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
9
XP
$\frac{\sigma}{\mu}$ = 16.46± 3.05 %
μ = 495.1± 12.9
σ = 81.5± 13.0
WF
$\frac{\sigma}{\mu}$ = 13.31± 3.16 %
μ = 467.6± 10.4
σ = 62.3± 13.4



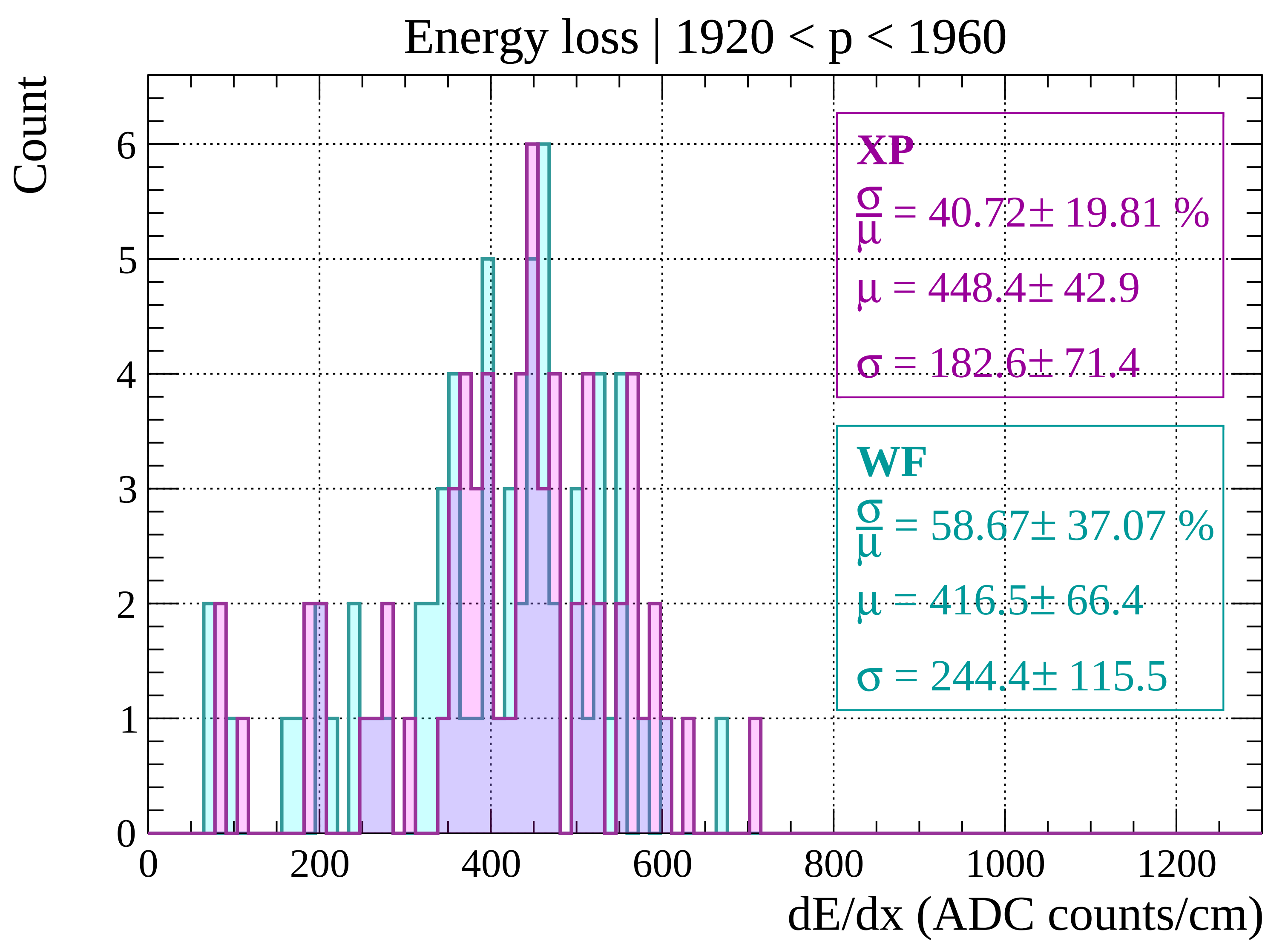
Energy loss | 1800 < p < 1840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
XP
σ/μ = 21.03± 4.47 %
μ = 450.6± 15.4
σ = 94.8± 16.9
WF
σ/μ = 14.29± 2.67 %
μ = 448.1± 11.4
σ = 64.0± 10.3

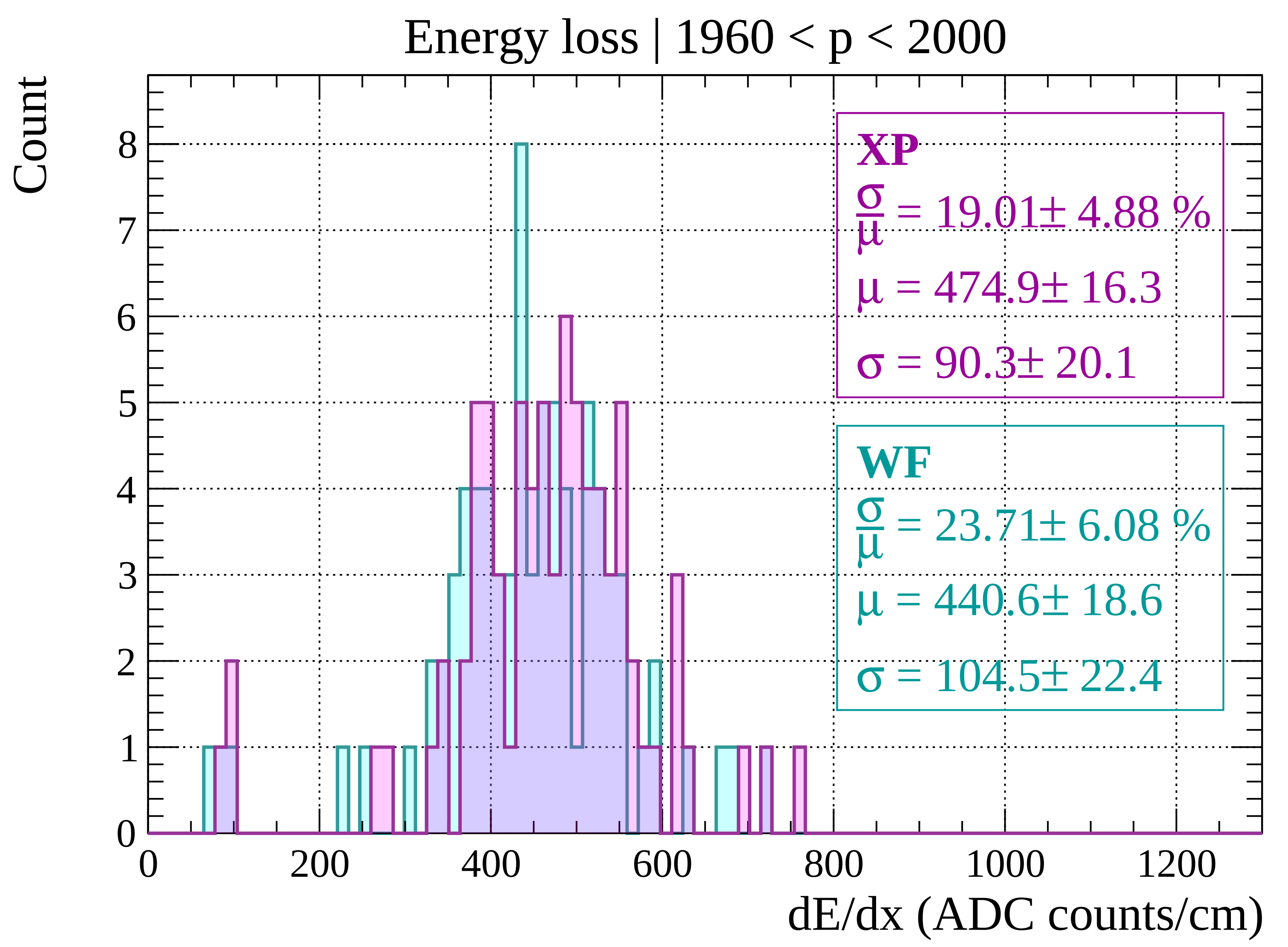


Energy loss | 1840 < p < 1880
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
0
200
400
600
800
1000
1200
XP
σ/μ = 11.90± 2.52 %
μ = 453.6± 9.9
σ = 54.0± 10.2
WF
σ/μ = 11.29± 2.38 %
μ = 432.9± 9.5
σ = 48.9± 9.2

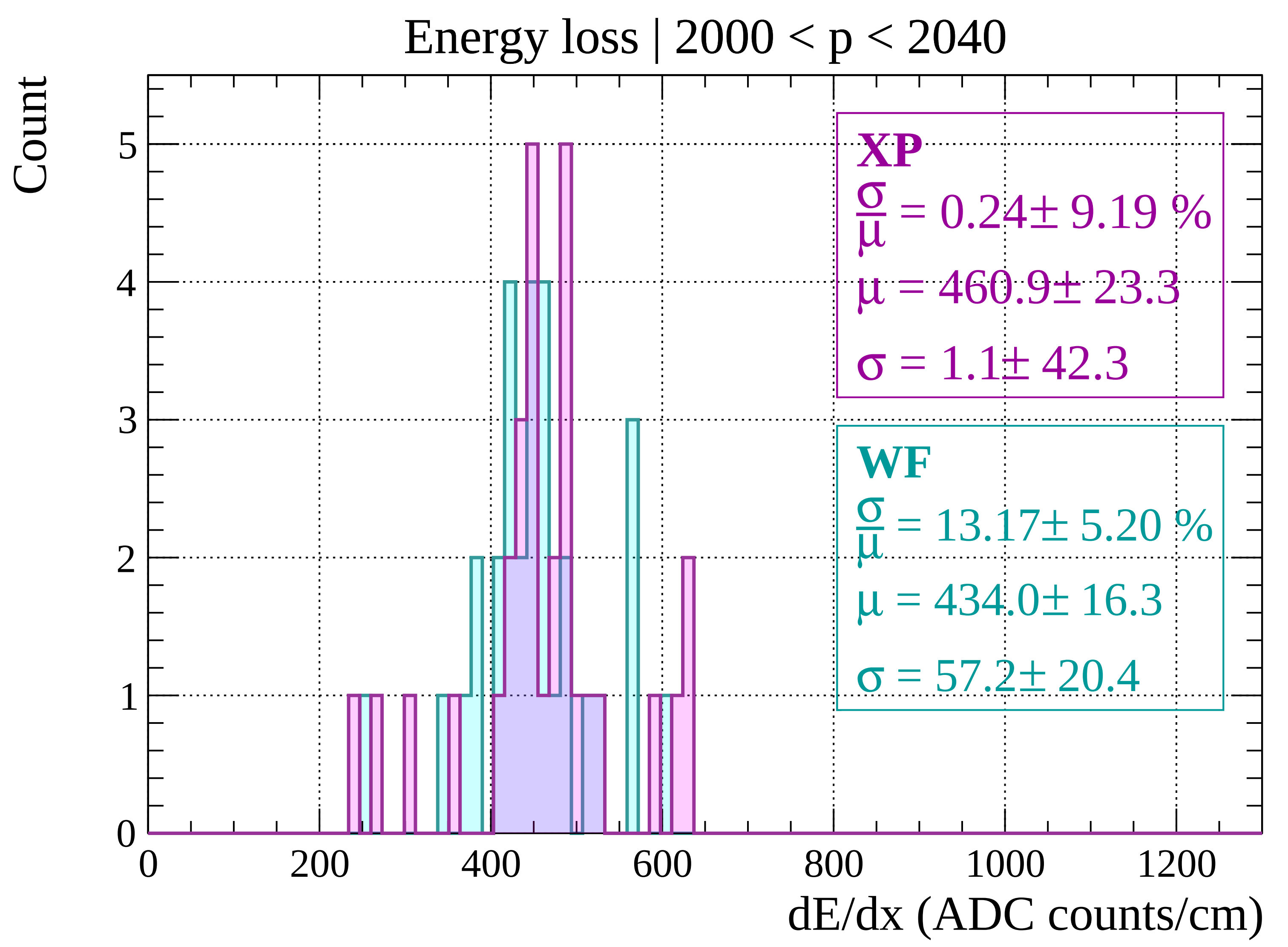


Energy loss | 1880 < p < 1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
XP
σ/μ = 16.82± 4.02 %
μ = 484.7± 14.1
σ = 81.5± 17.1
WF
σ/μ = 13.11± 2.64 %
μ = 469.3± 10.9
σ = 61.5± 11.0

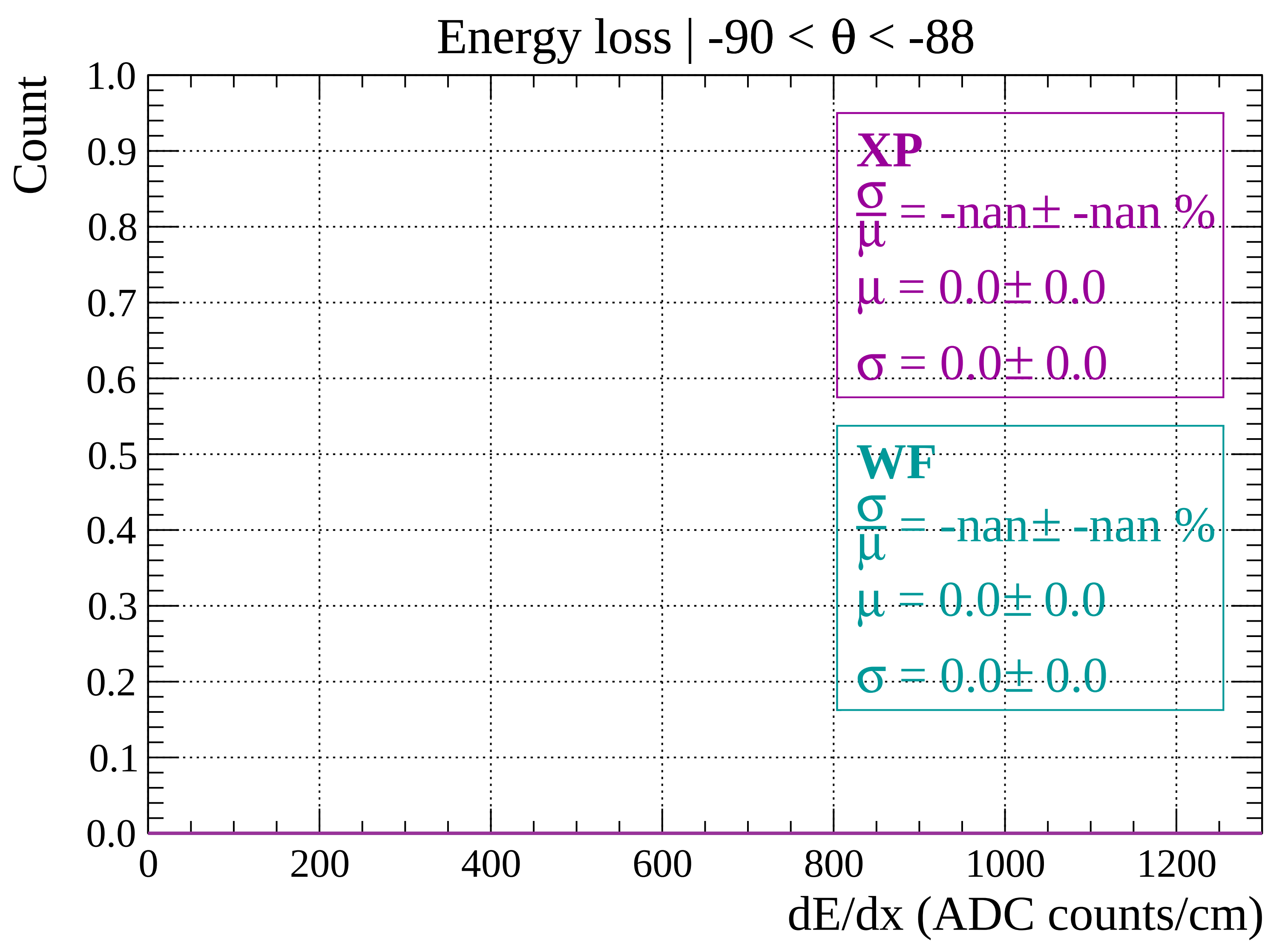

Energy loss | 1920 < p < 1960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
XP
σ/μ = 40.72± 19.81 %
μ = 448.4± 42.9
σ = 182.6± 71.4
WF
σ/μ = 58.67± 37.07 %
μ = 416.5± 66.4
σ = 244.4± 115.5



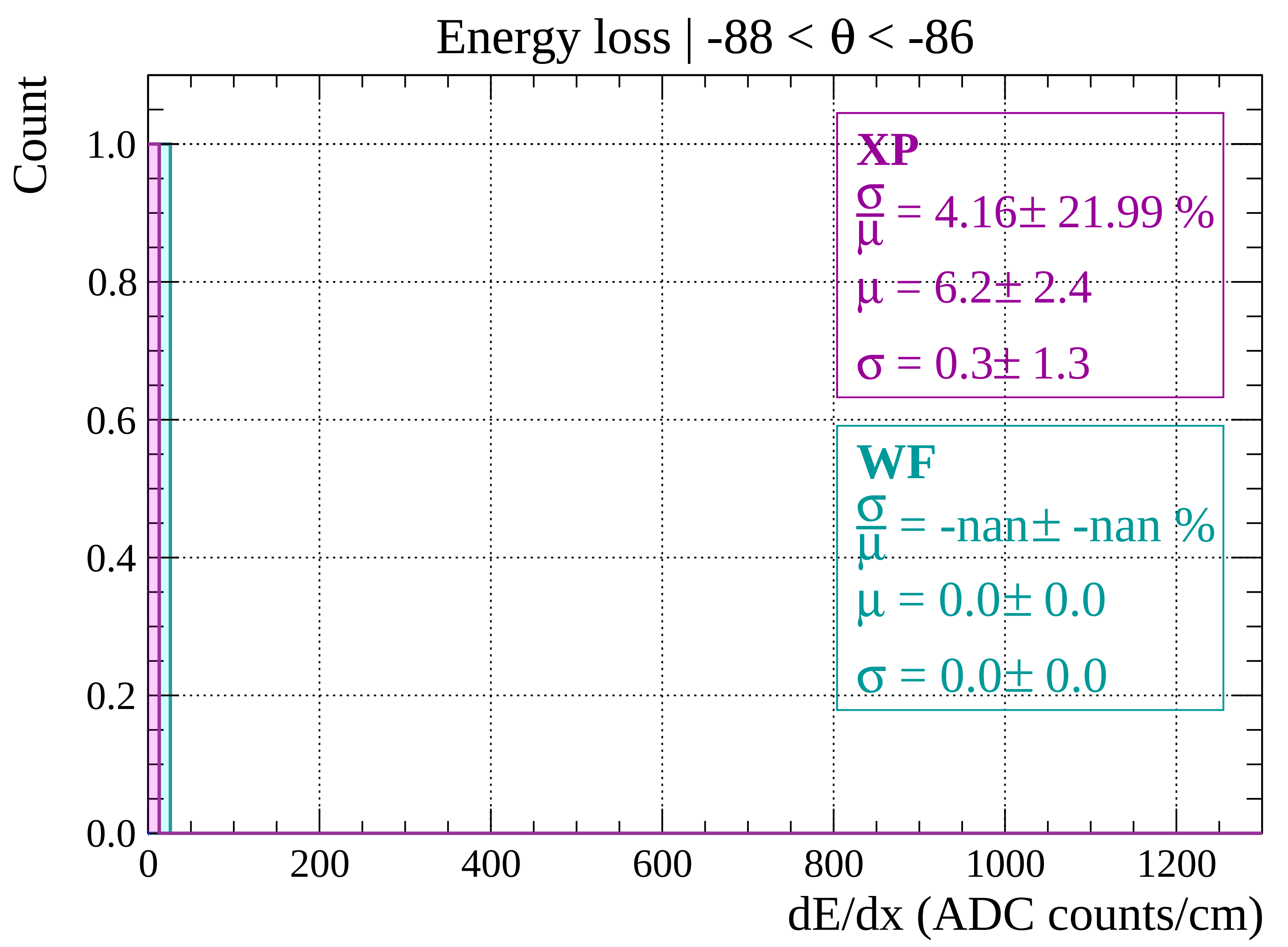
Energy loss | 1960 < p < 2000
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
8
XP
σ/μ = 19.01± 4.88 %
μ = 474.9± 16.3
σ = 90.3± 20.1
WF
σ/μ = 23.71± 6.08 %
μ = 440.6± 18.6
σ = 104.5± 22.4



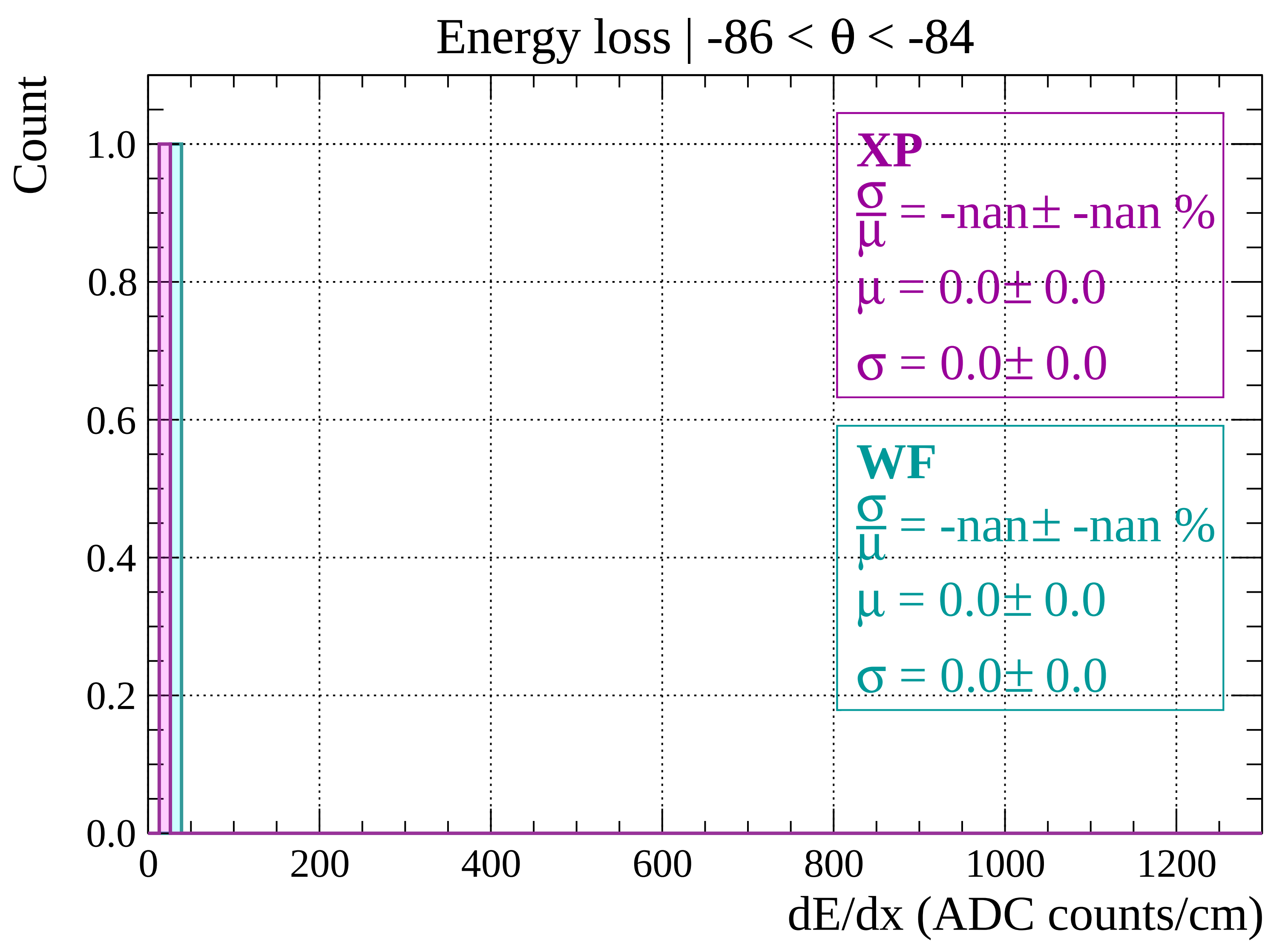
Energy loss | 2000 < p < 2040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
XP
σ/μ = 0.24± 9.19 %
μ = 460.9± 23.3
σ = 1.1± 42.3
WF
σ/μ = 13.17± 5.20 %
μ = 434.0± 16.3
σ = 57.2± 20.4



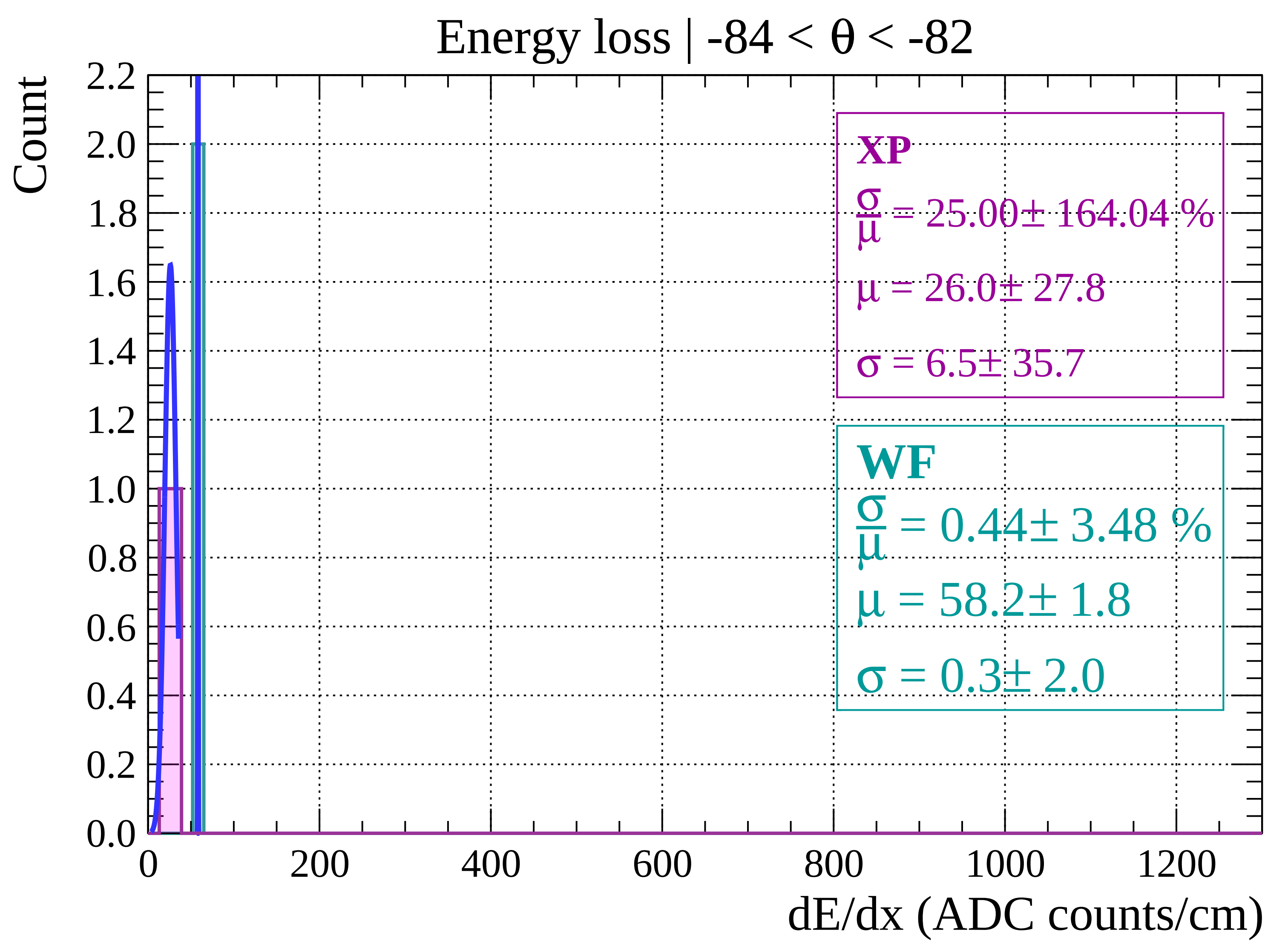
Energy loss | -90 < θ < -88
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



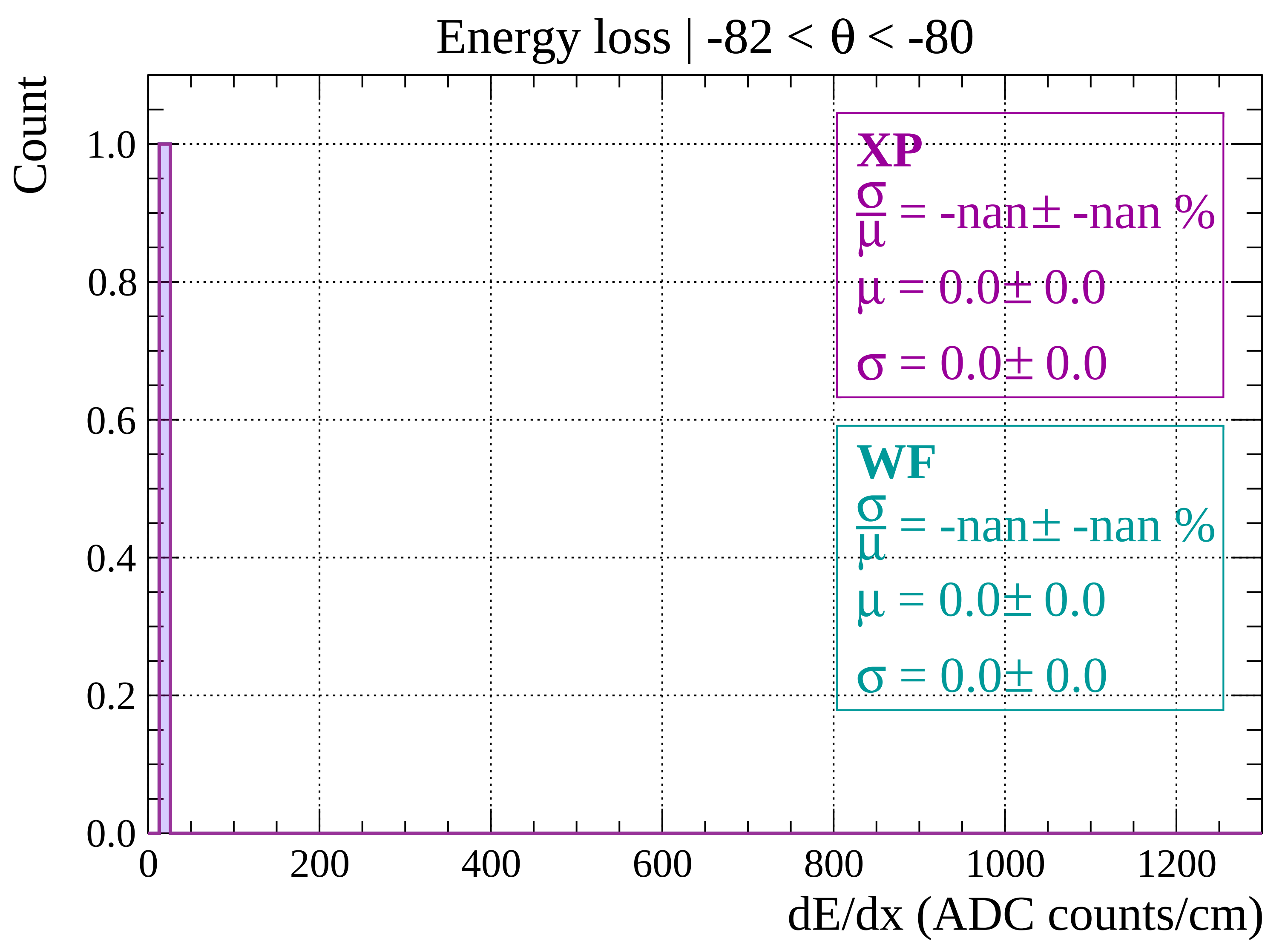
Energy loss | -88 < θ < -86
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.16± 21.99 %
μ = 6.2± 2.4
σ = 0.3± 1.3
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



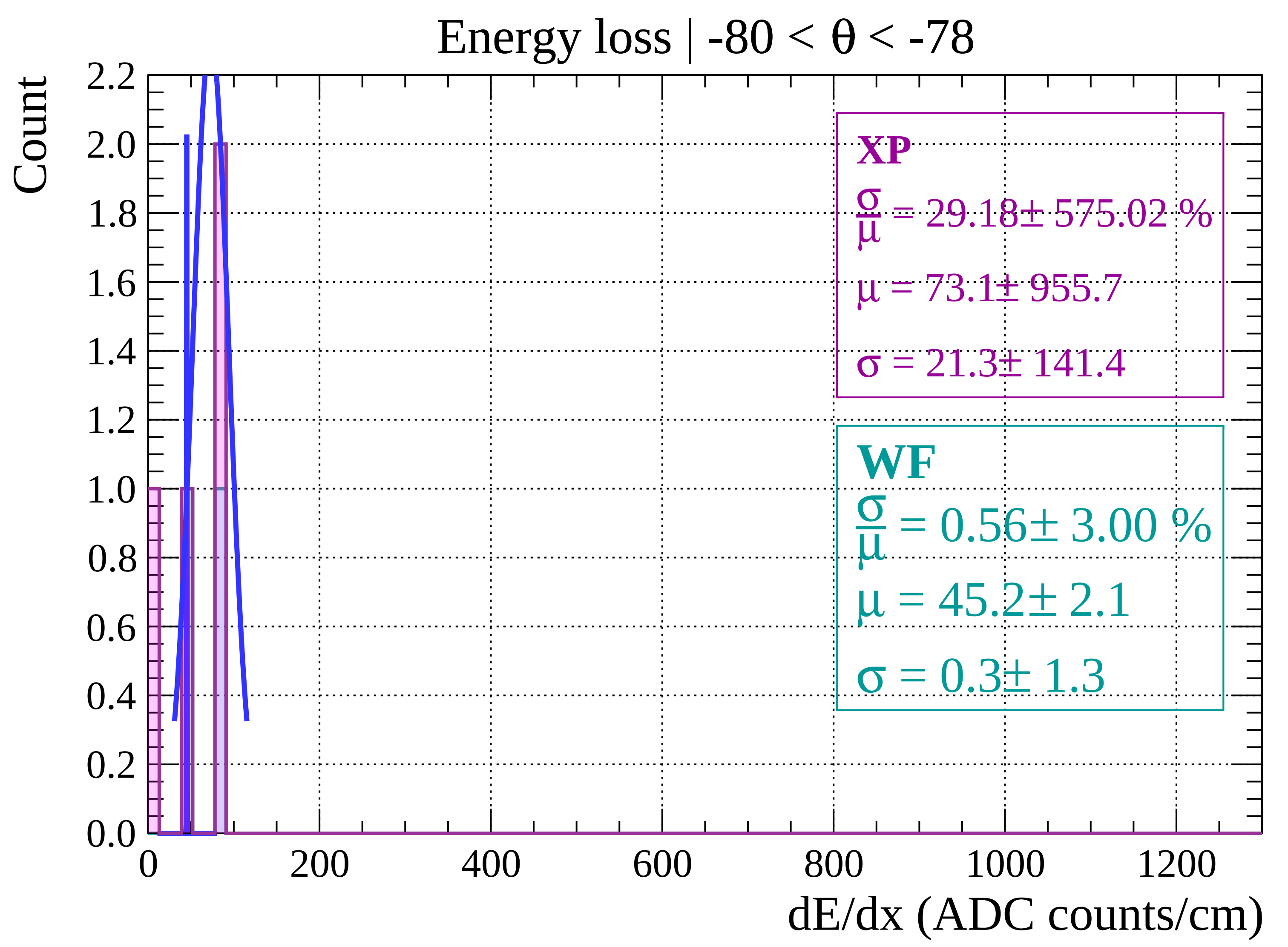
Energy loss | -86 < θ < -84
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



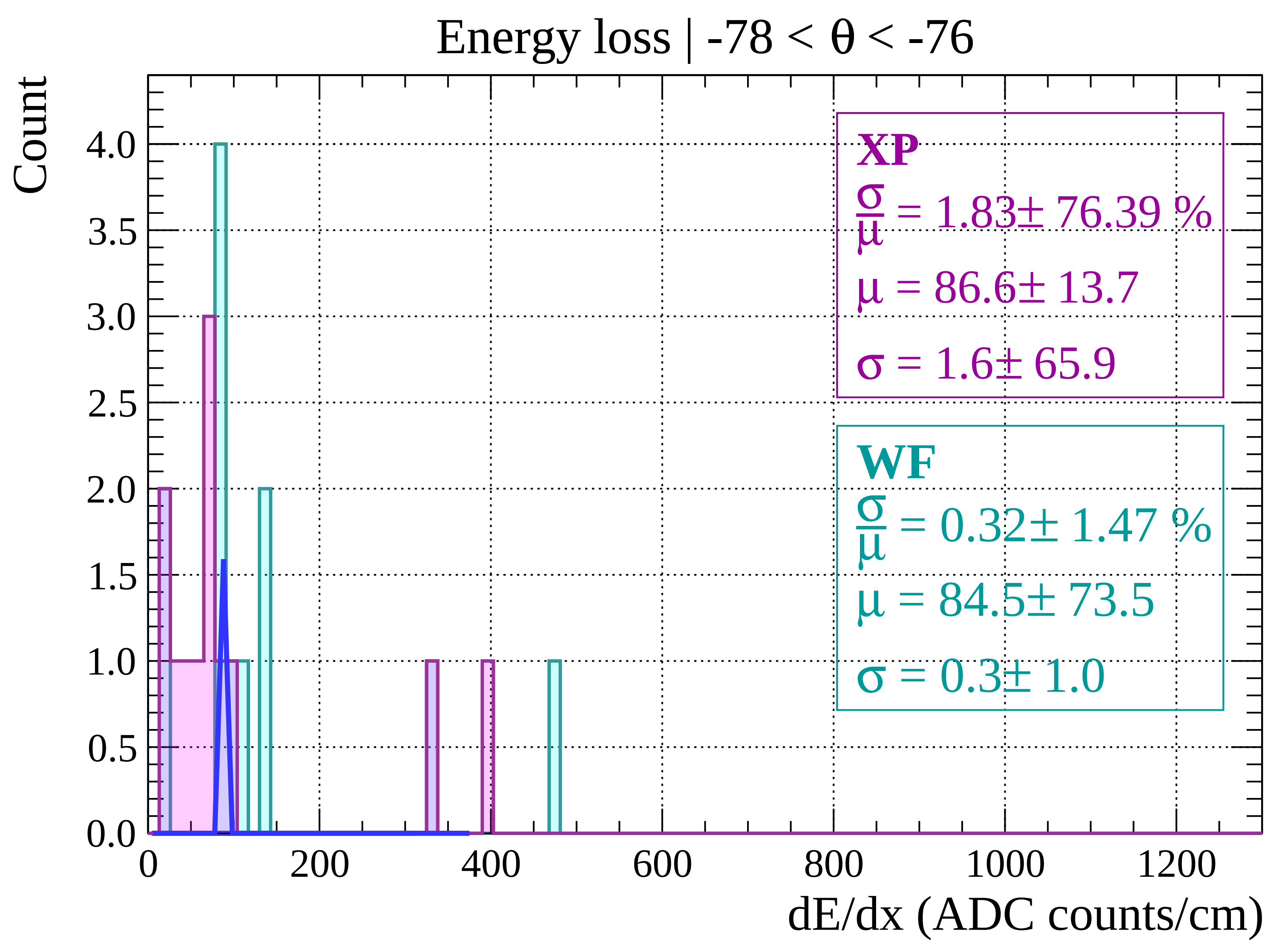
Energy loss | -84 < θ < -82
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 25.00± 164.04 %
μ = 26.0± 27.8
σ = 6.5± 35.7
WF
σ/μ = 0.44± 3.48 %
μ = 58.2± 1.8
σ = 0.3± 2.0

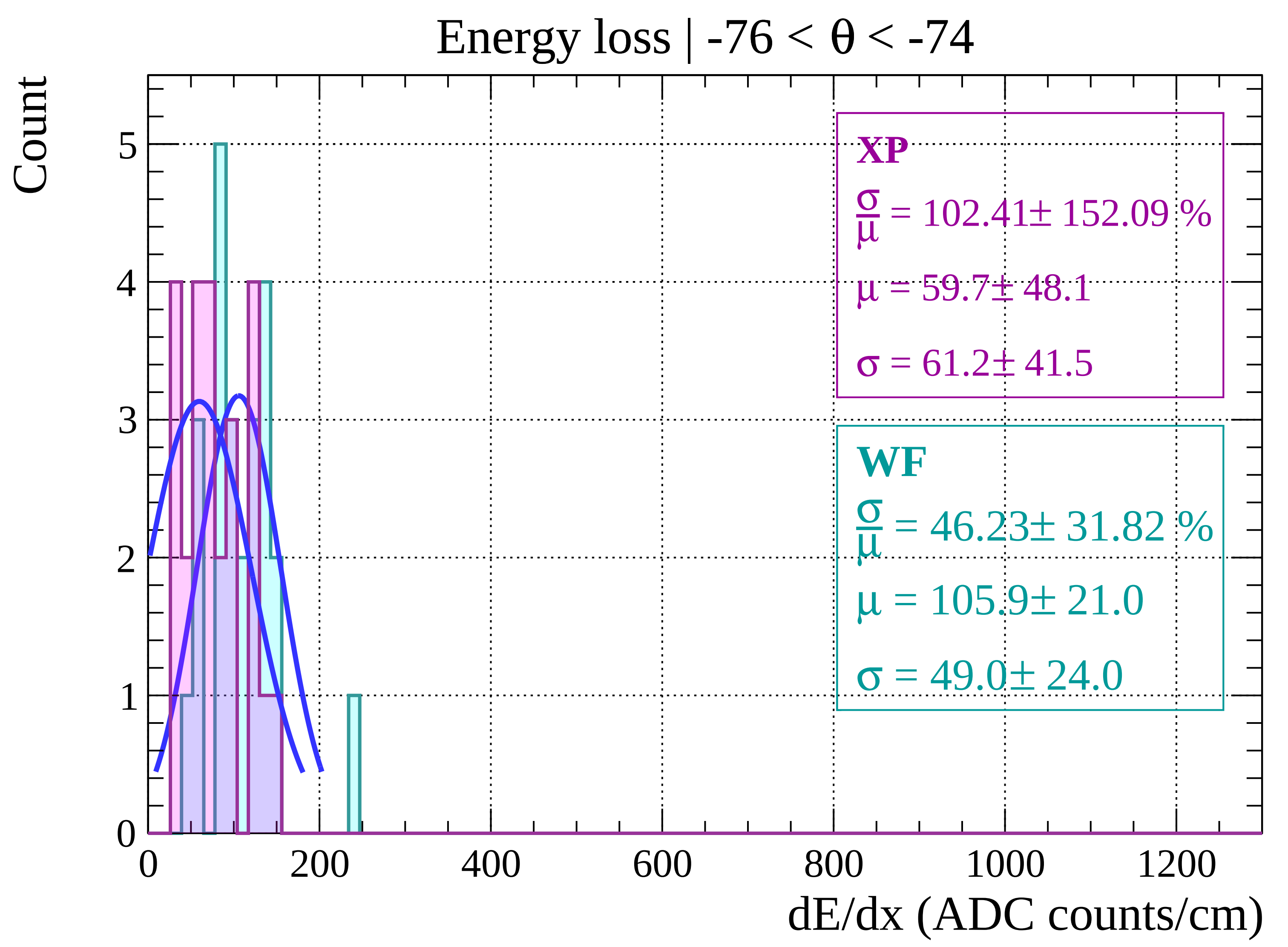


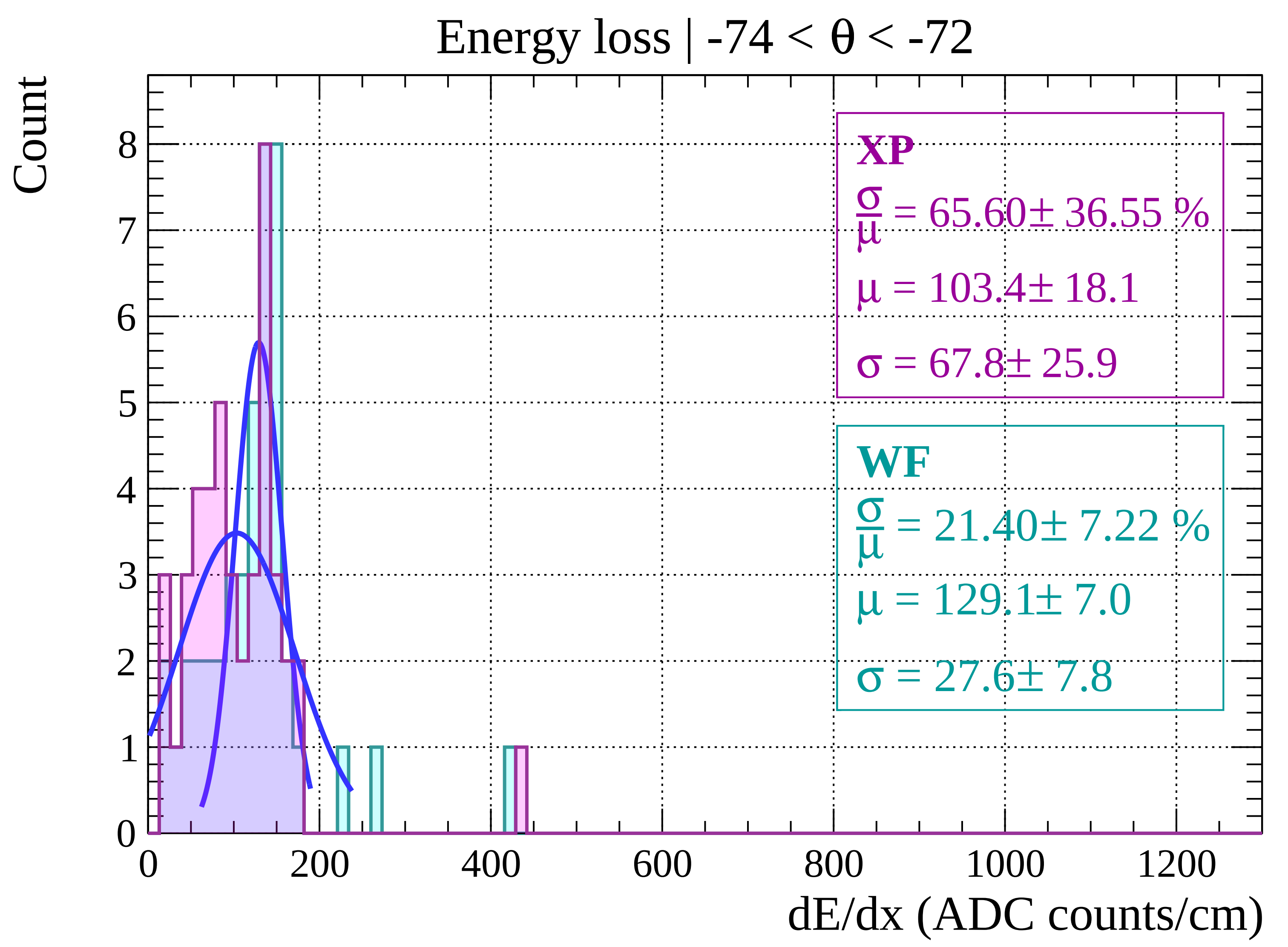
Energy loss | -82 < θ < -80
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



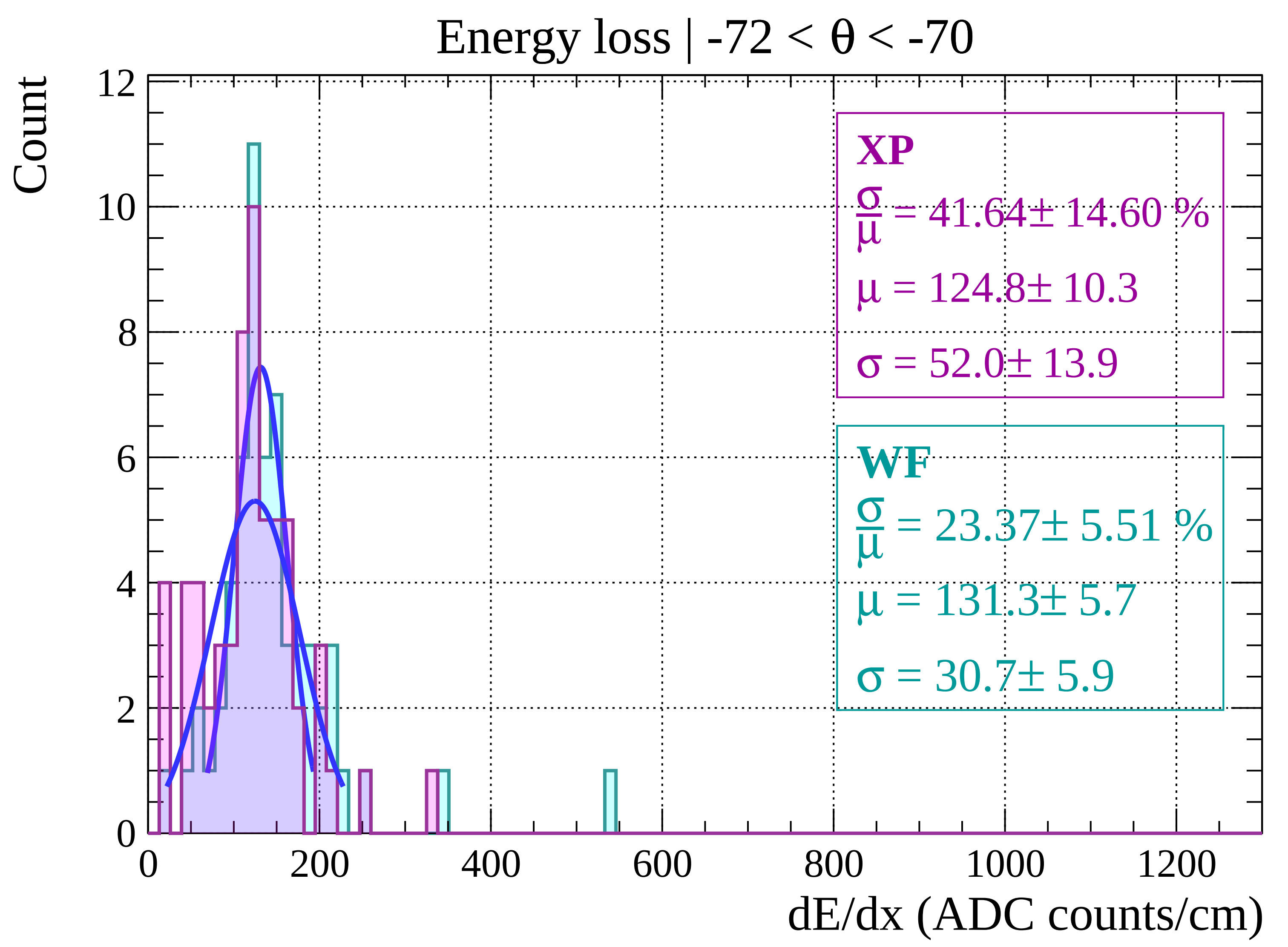
Energy loss | -80 < θ < -78
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 29.18± 575.02 %
μ = 73.1± 955.7
σ = 21.3± 141.4
WF
σ/μ = 0.56± 3.00 %
μ = 45.2± 2.1
σ = 0.3± 1.3


Energy loss | -78 < θ < -76
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
XP
σ/μ = 1.83± 76.39 %
μ = 86.6± 13.7
σ = 1.6± 65.9
WF
σ/μ = 0.32± 1.47 %
μ = 84.5± 73.5
σ = 0.3± 1.0

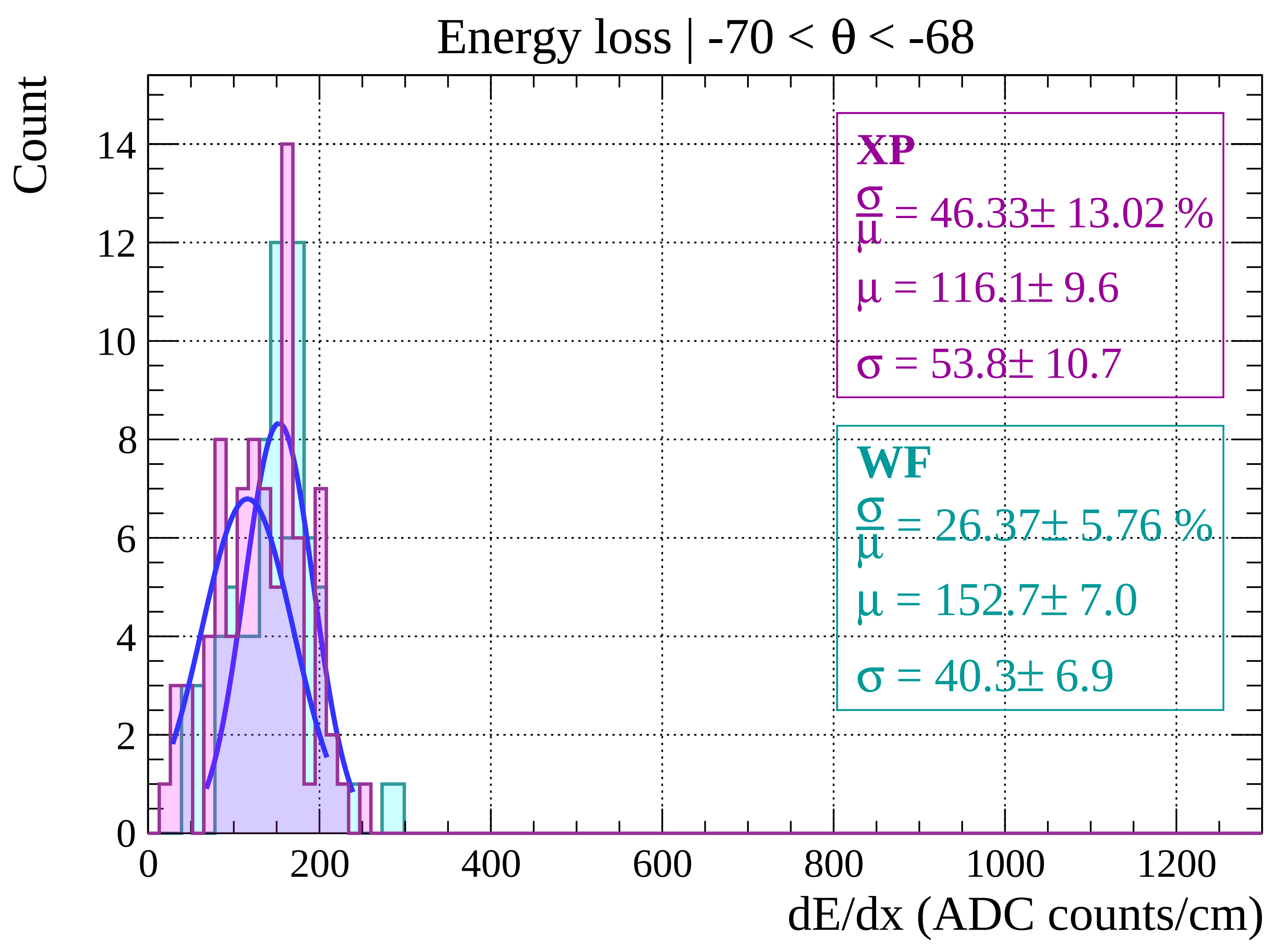

Energy loss | -76 < θ < -74
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
σ/μ = 102.41± 152.09 %
μ = 59.7± 48.1
σ = 61.2± 41.5
WF
σ/μ = 46.23± 31.82 %
μ = 105.9± 21.0
σ = 49.0± 24.0



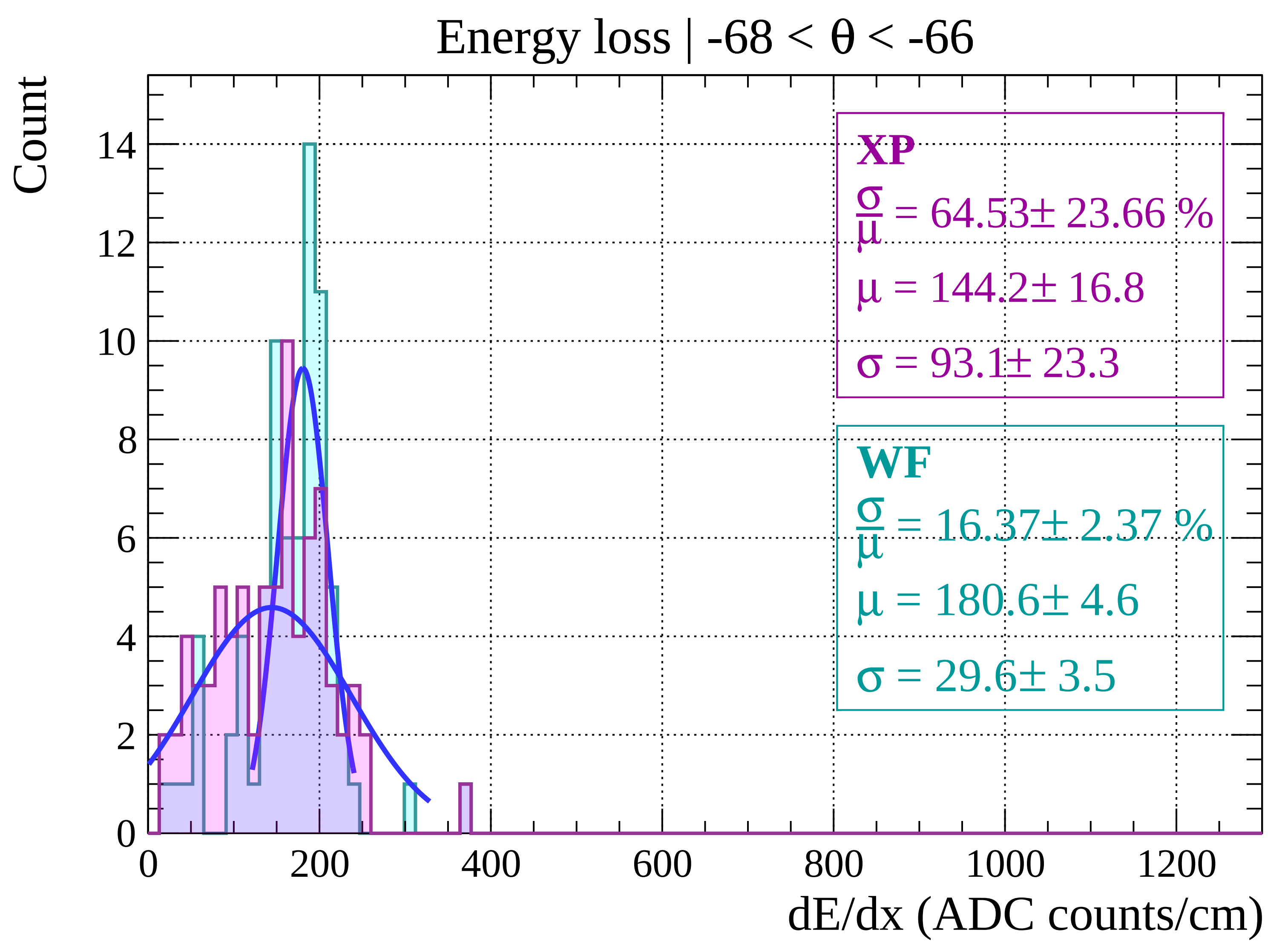
Energy loss | -74 < θ < -72
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 65.60± 36.55 %
μ = 103.4± 18.1
σ = 67.8± 25.9
WF
σ/μ = 21.40± 7.22 %
μ = 129.1± 7.0
σ = 27.6± 7.8
0
1
2
3
4
5
6
7
8
0
200
400
600
800
1000
1200



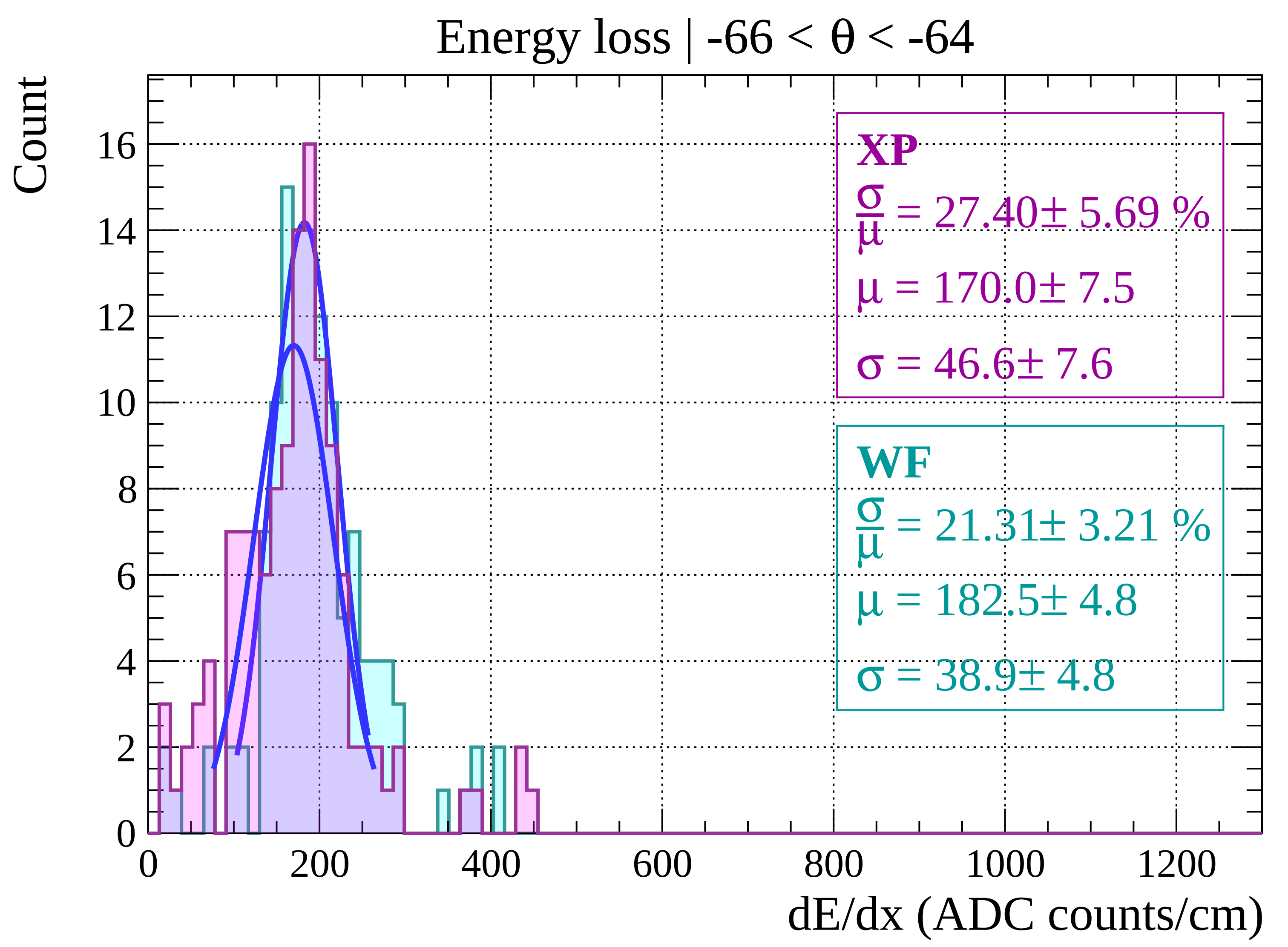
Energy loss | -72 < θ < -70
Count
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 41.64 ± 14.60 %
μ = 124.8 ± 10.3
σ = 52.0 ± 13.9
WF
σ/μ = 23.37 ± 5.51 %
μ = 131.3 ± 5.7
σ = 30.7 ± 5.9

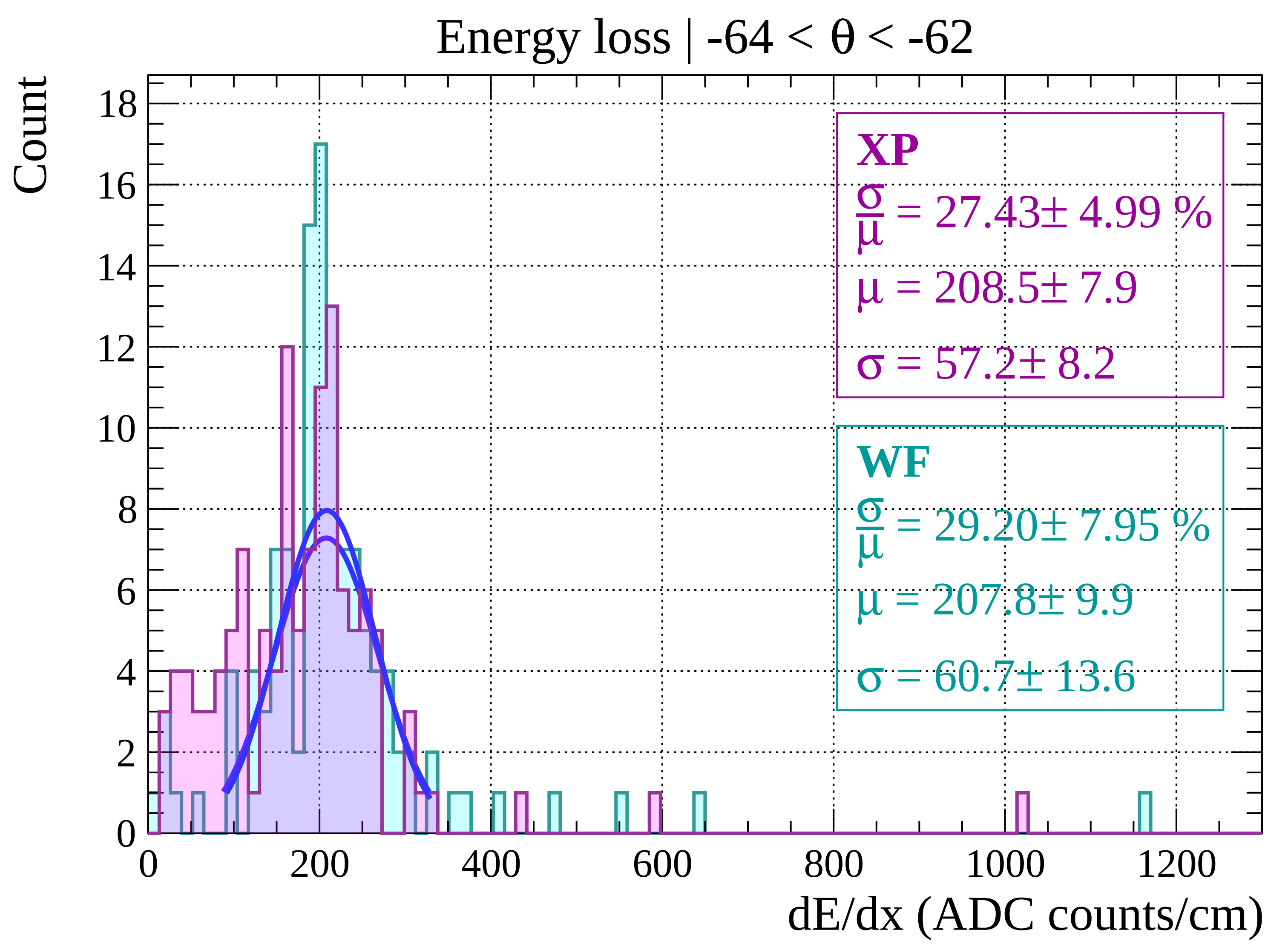


Energy loss | -70 < θ < -68
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 46.33± 13.02 %
μ = 116.1± 9.6
σ = 53.8± 10.7
WF
σ/μ = 26.37± 5.76 %
μ = 152.7± 7.0
σ = 40.3± 6.9

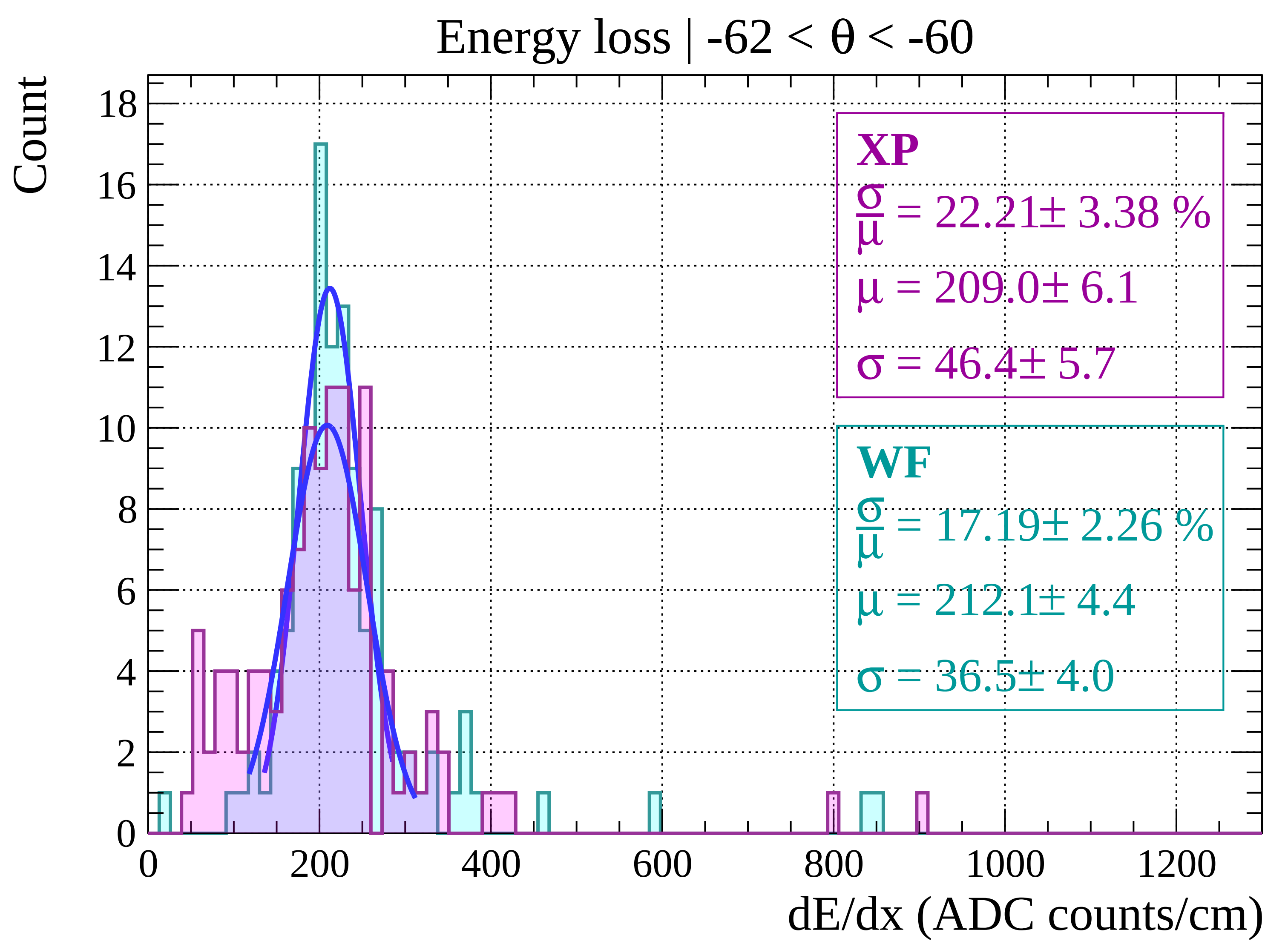


Energy loss | -68 < θ < -66
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
XP
σ/μ = 64.53± 23.66 %
μ = 144.2± 16.8
σ = 93.1± 23.3
WF
σ/μ = 16.37± 2.37 %
μ = 180.6± 4.6
σ = 29.6± 3.5

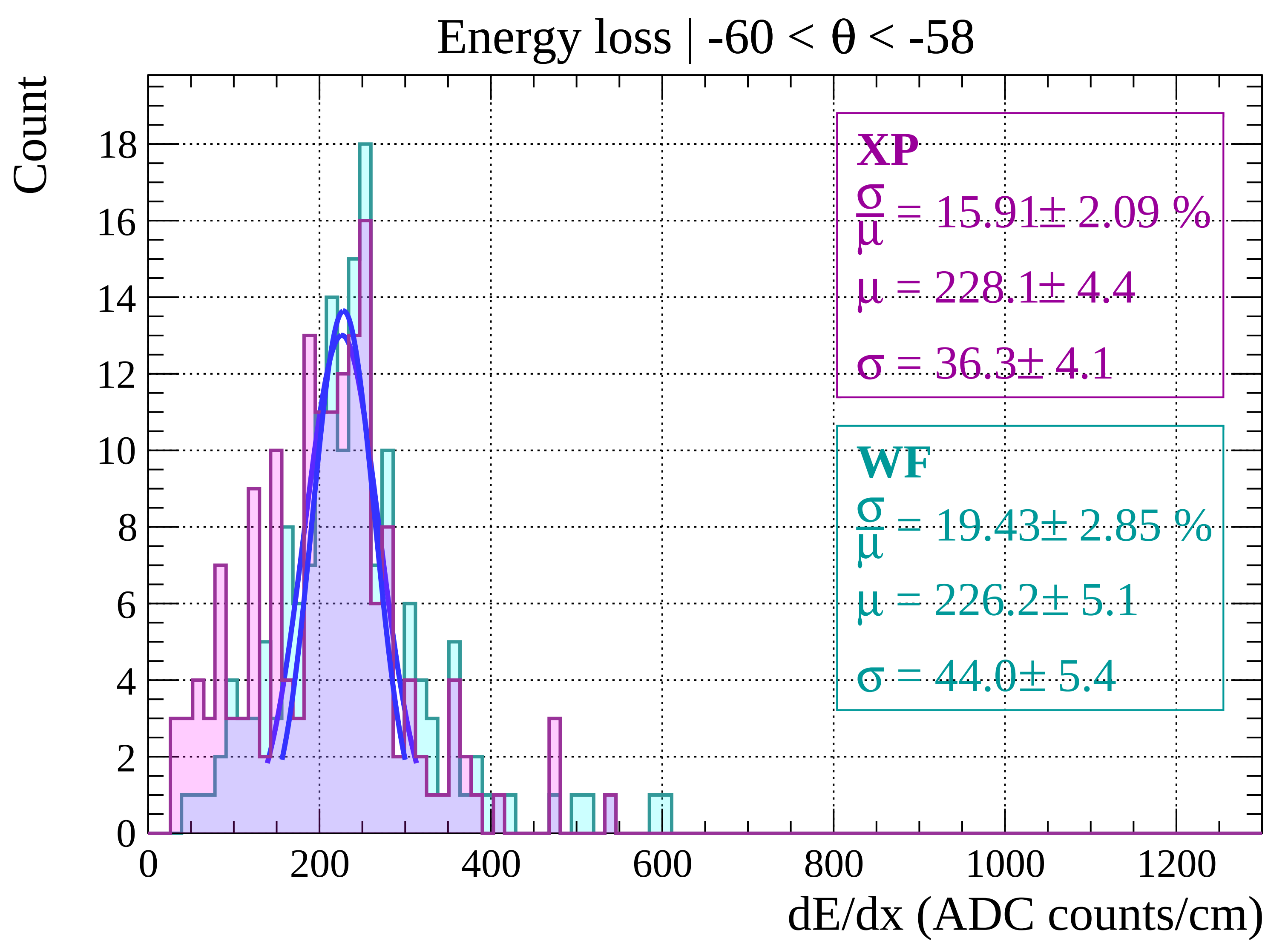

Energy loss | -66 < θ < -64
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
0
200
400
600
800
1000
1200
XP
σ/μ = 27.40± 5.69 %
μ = 170.0± 7.5
σ = 46.6± 7.6
WF
σ/μ = 21.31± 3.21 %
μ = 182.5± 4.8
σ = 38.9± 4.8



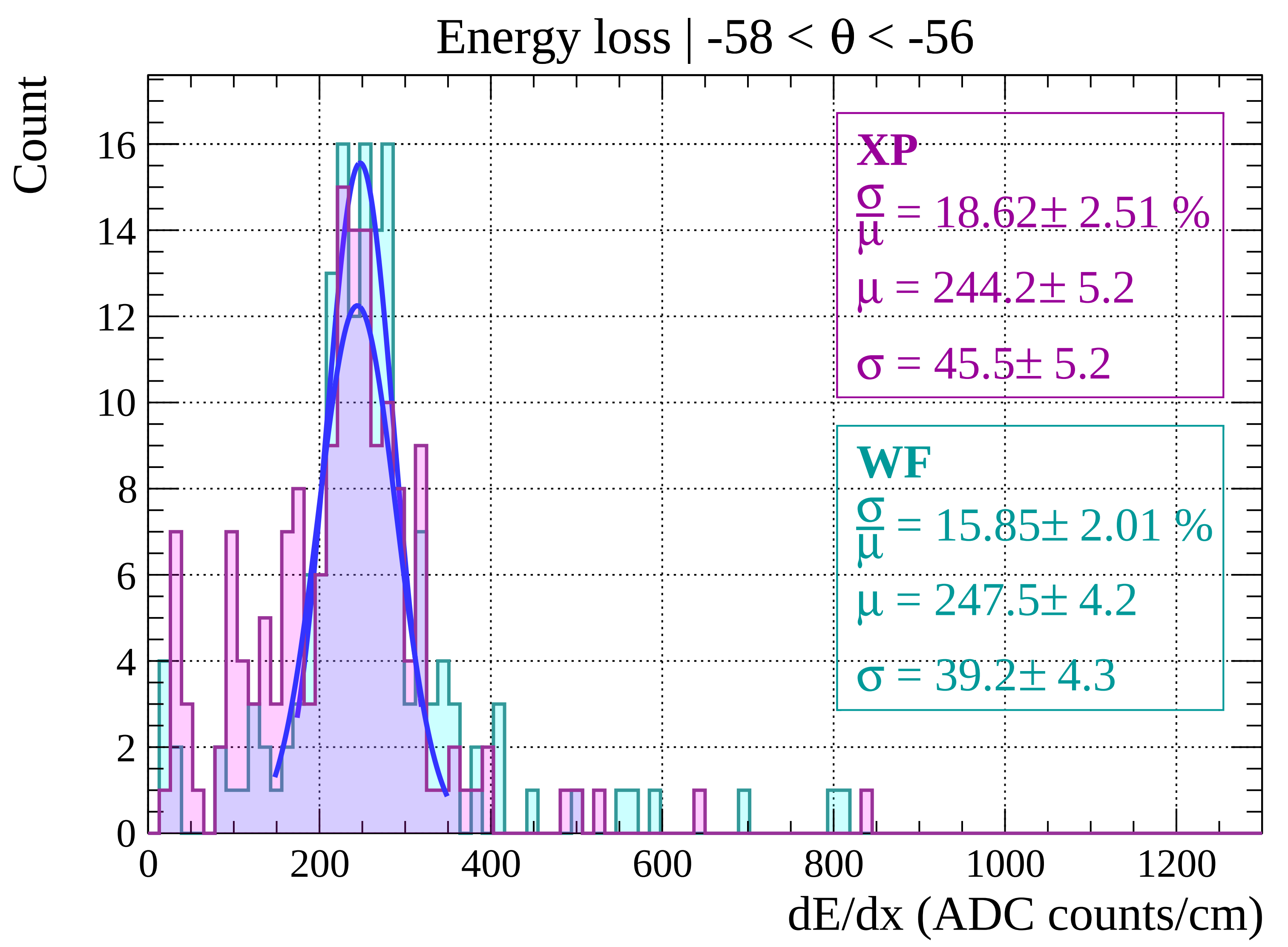
Energy loss | -64 < θ < -62
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
0
200
400
600
800
1000
1200
XP
σ/μ = 27.43± 4.99 %
μ = 208.5± 7.9
σ = 57.2± 8.2
WF
σ/μ = 29.20± 7.95 %
μ = 207.8± 9.9
σ = 60.7± 13.6



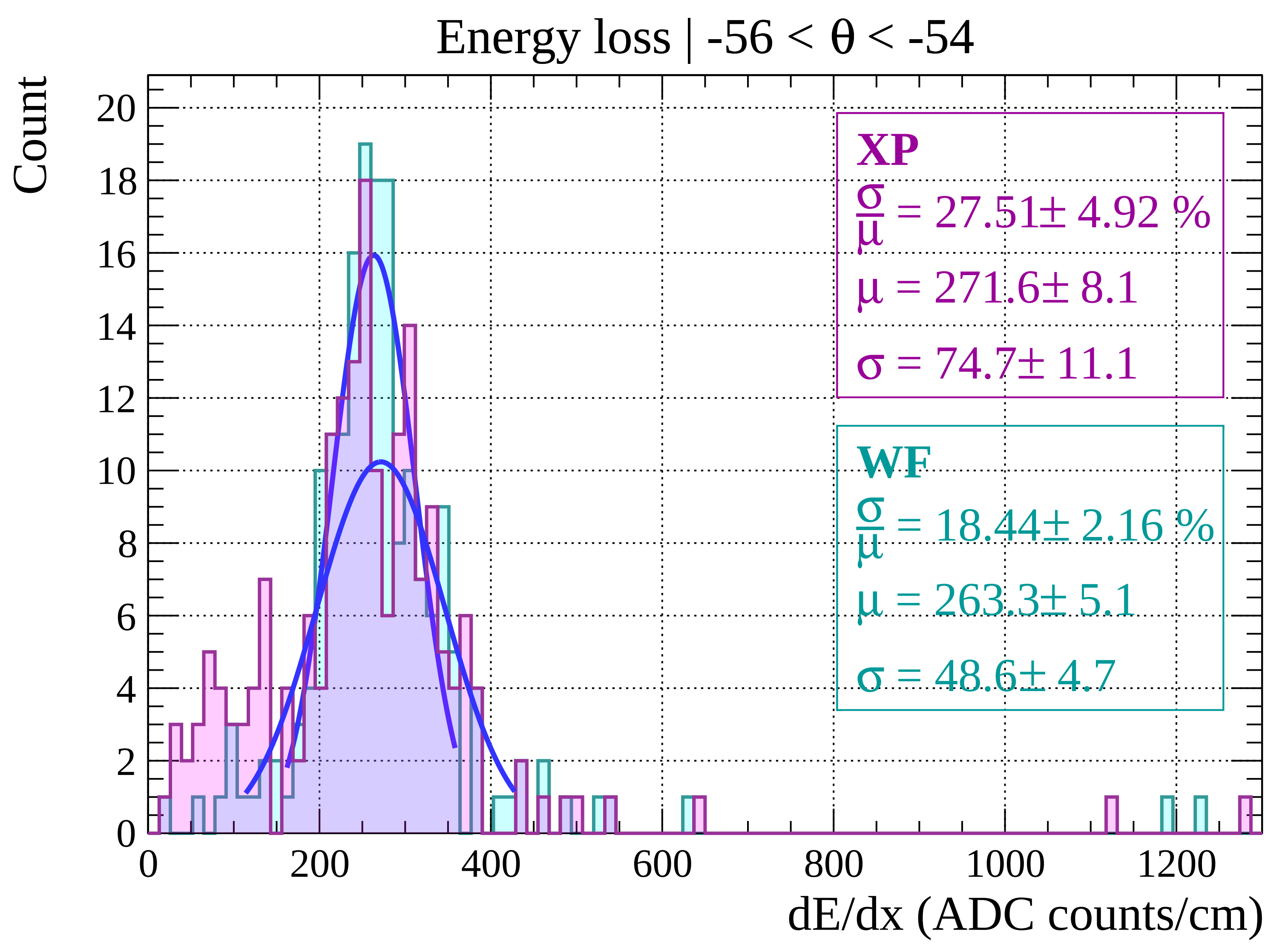
Energy loss | -62 < θ < -60
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
XP
σ/μ = 22.21± 3.38 %
μ = 209.0± 6.1
σ = 46.4± 5.7
WF
σ/μ = 17.19± 2.26 %
μ = 212.1± 4.4
σ = 36.5± 4.0



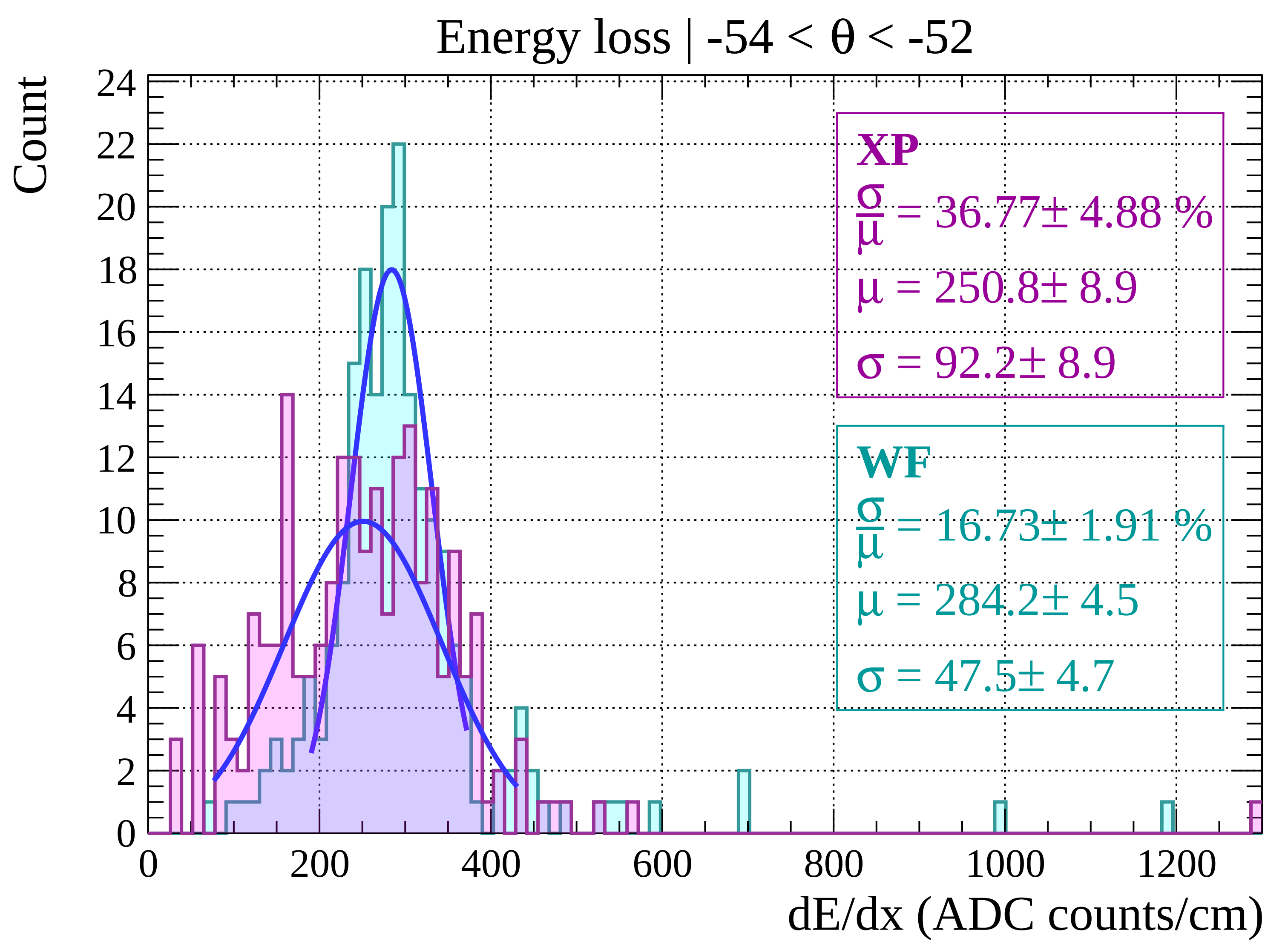
Energy loss | -60 < θ < -58
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
XP
σ/μ = 15.91± 2.09 %
μ = 228.1± 4.4
σ = 36.3± 4.1
WF
σ/μ = 19.43± 2.85 %
μ = 226.2± 5.1
σ = 44.0± 5.4



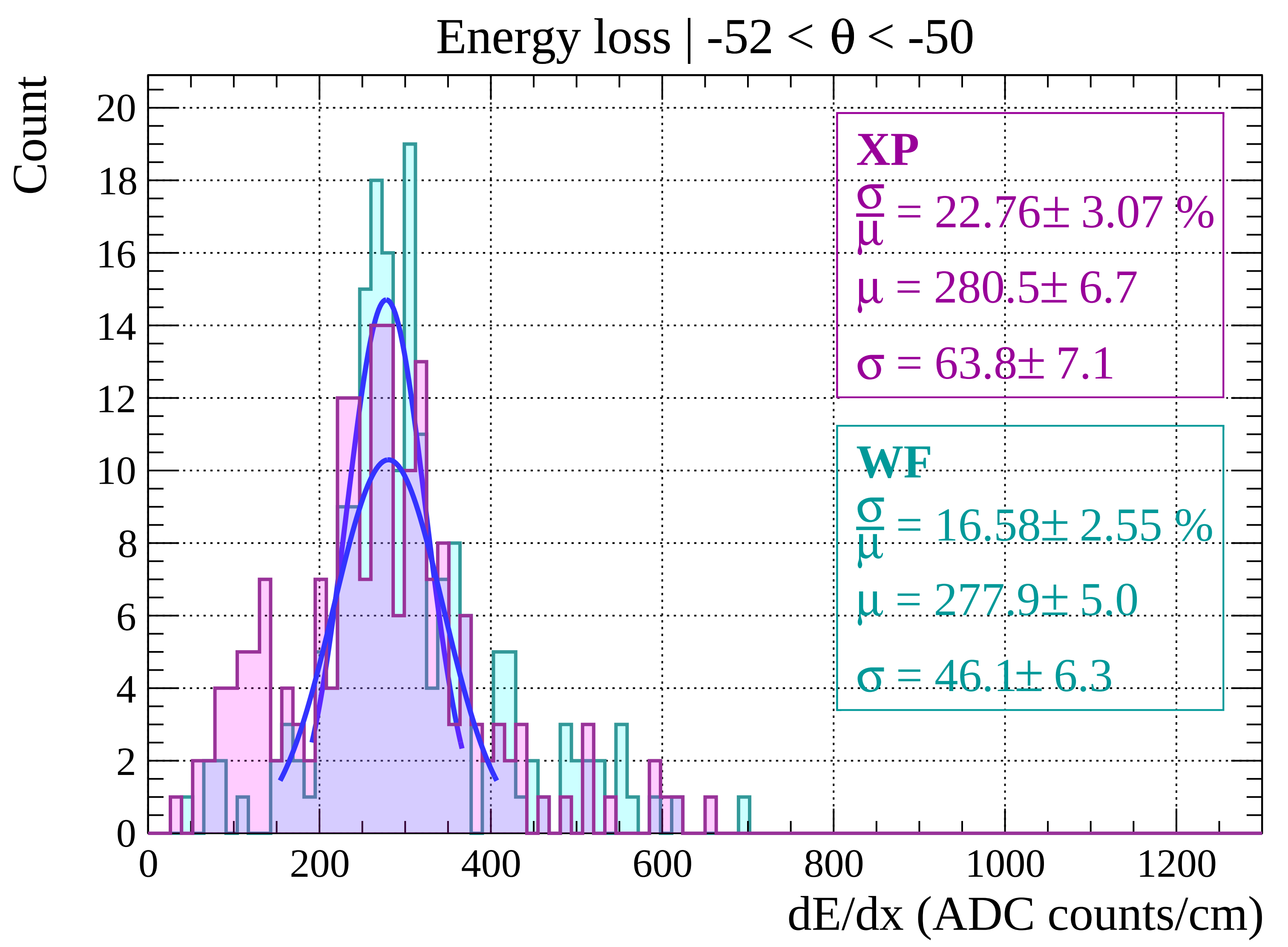
Energy loss | -58 < θ < -56
Count
0
2
4
6
8
10
12
14
16
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 18.62± 2.51 %
μ = 244.2± 5.2
σ = 45.5± 5.2
WF
σ/μ = 15.85± 2.01 %
μ = 247.5± 4.2
σ = 39.2± 4.3

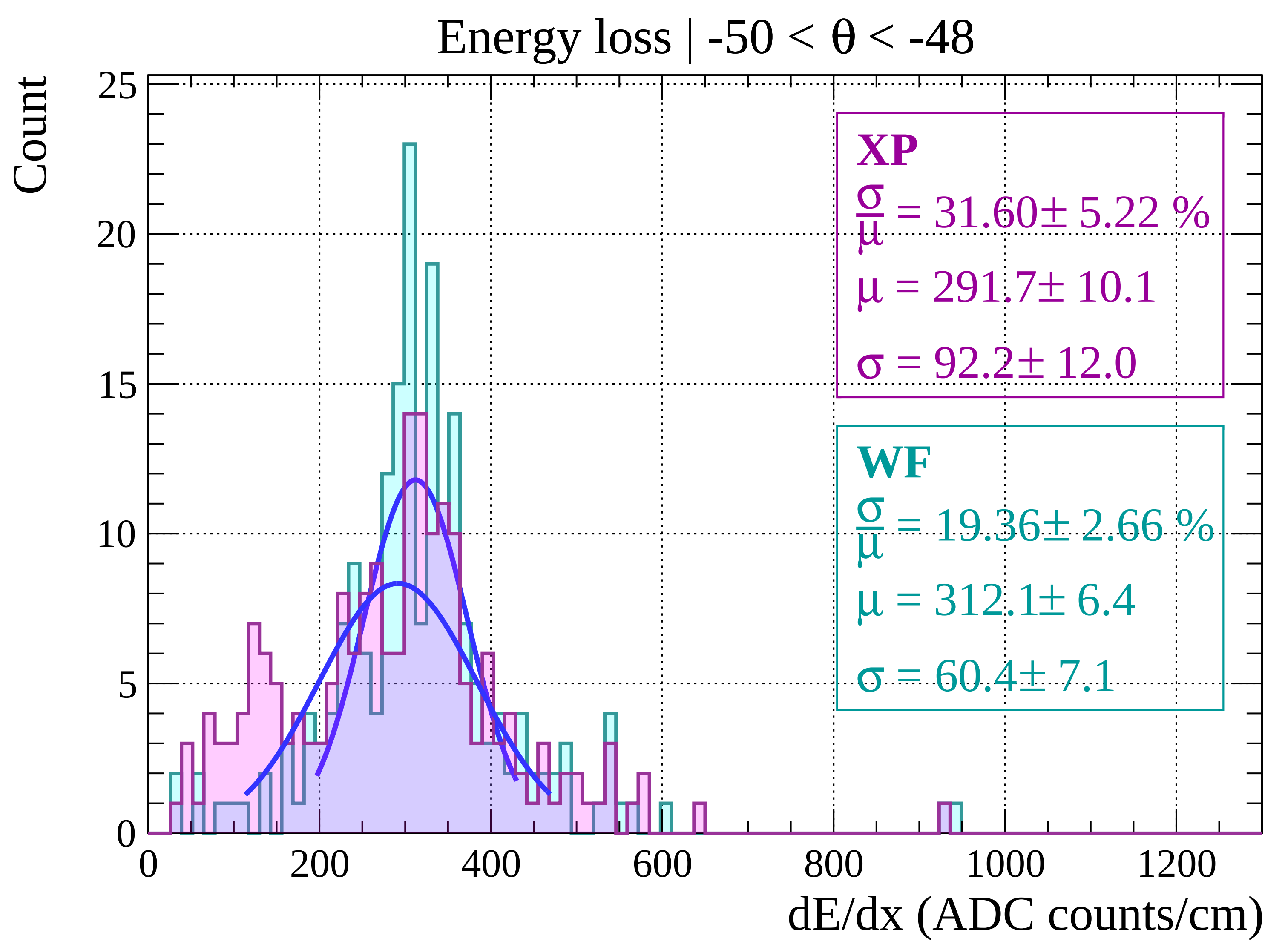


Energy loss | -56 < θ < -54
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
20
XP
$\frac{\sigma}{\mu}$ = 27.51± 4.92 %
μ = 271.6± 8.1
σ = 74.7± 11.1
WF
$\frac{\sigma}{\mu}$ = 18.44± 2.16 %
μ = 263.3± 5.1
σ = 48.6± 4.7

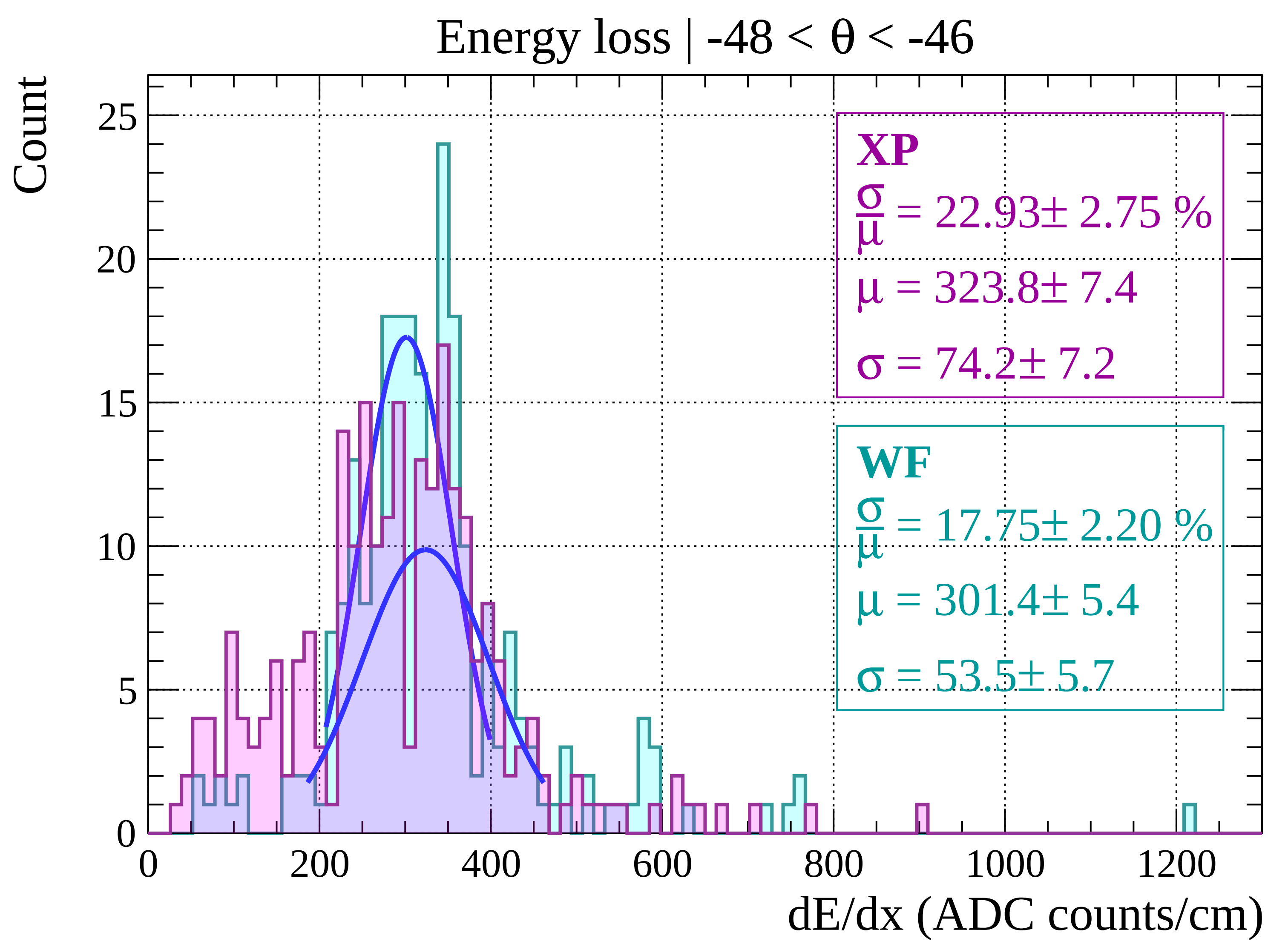


Energy loss | -54 < θ < -52
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
20
22
24
0
200
400
600
800
1000
1200
XP
σ/μ = 36.77± 4.88 %
μ = 250.8± 8.9
σ = 92.2± 8.9
WF
σ/μ = 16.73± 1.91 %
μ = 284.2± 4.5
σ = 47.5± 4.7

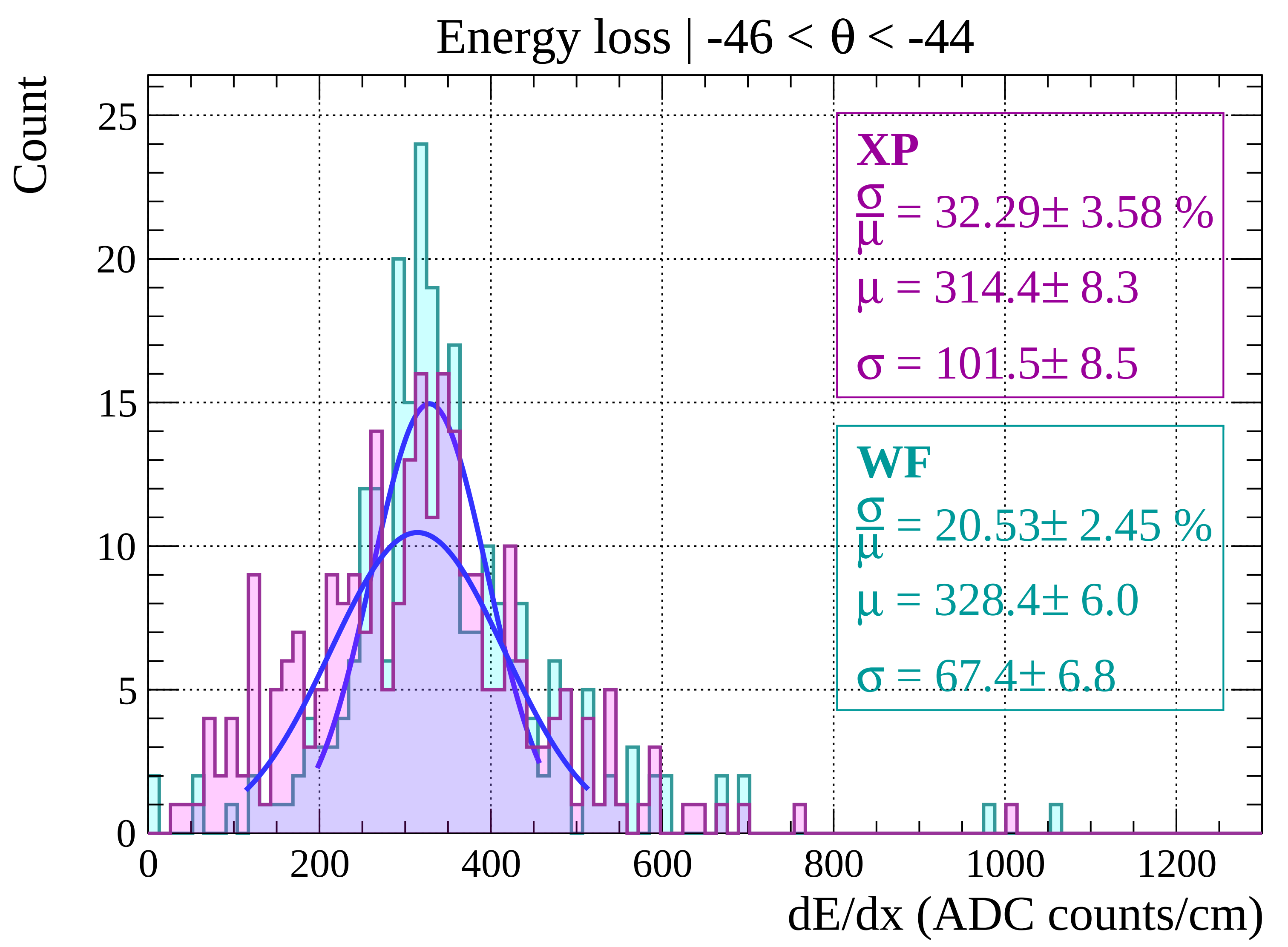

Energy loss | -52 < θ < -50
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 22.76± 3.07 %
μ = 280.5± 6.7
σ = 63.8± 7.1
WF
σ/μ = 16.58± 2.55 %
μ = 277.9± 5.0
σ = 46.1± 6.3
0
2
4
6
8
10
12
14
16
18
20
0
200
400
600
800
1000
1200

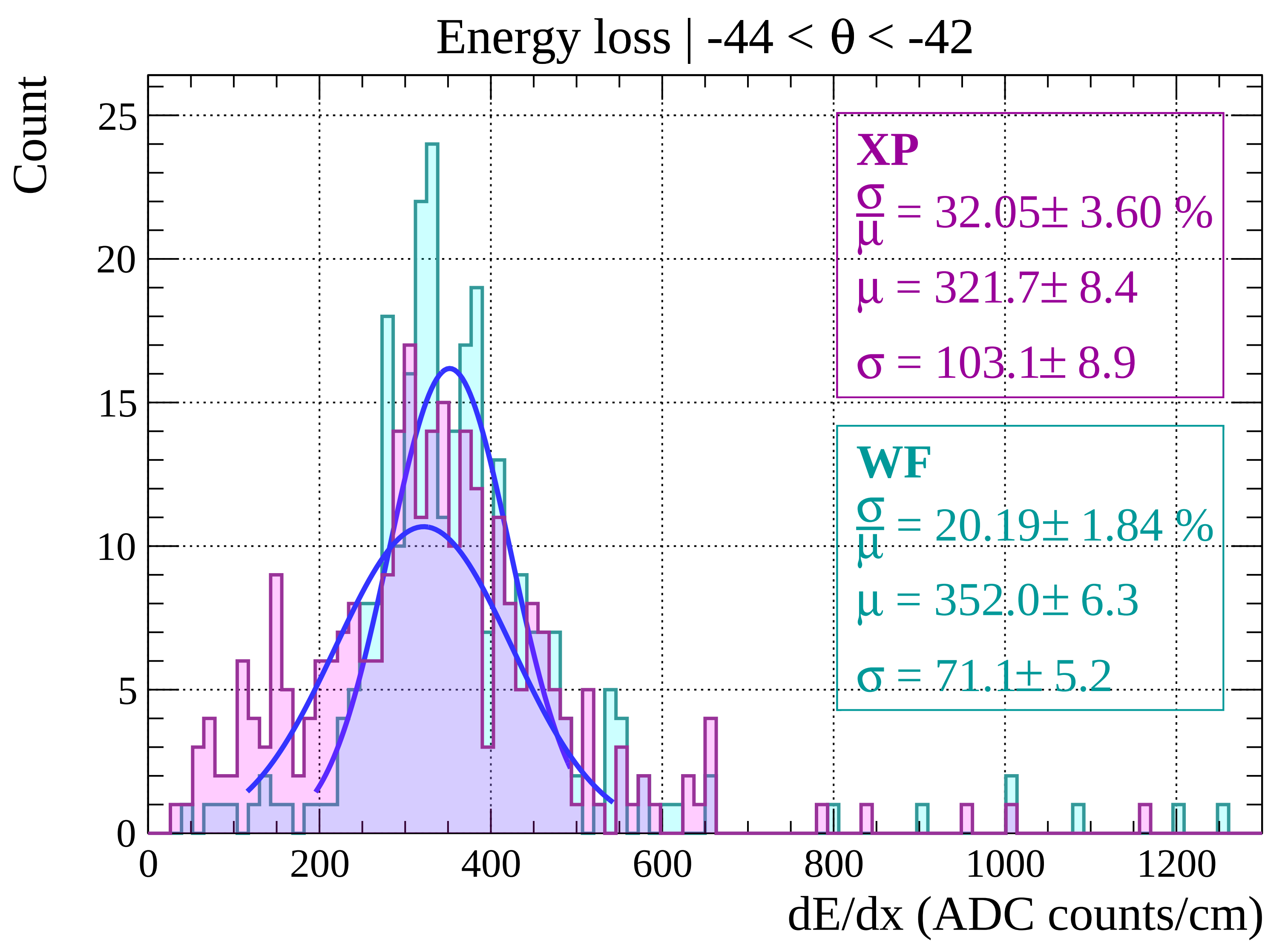


Energy loss | -50 < θ < -48
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 31.60± 5.22 %
μ = 291.7± 10.1
σ = 92.2± 12.0
WF
σ/μ = 19.36± 2.66 %
μ = 312.1± 6.4
σ = 60.4± 7.1
0
5
10
15
20
25
0
200
400
600
800
1000
1200

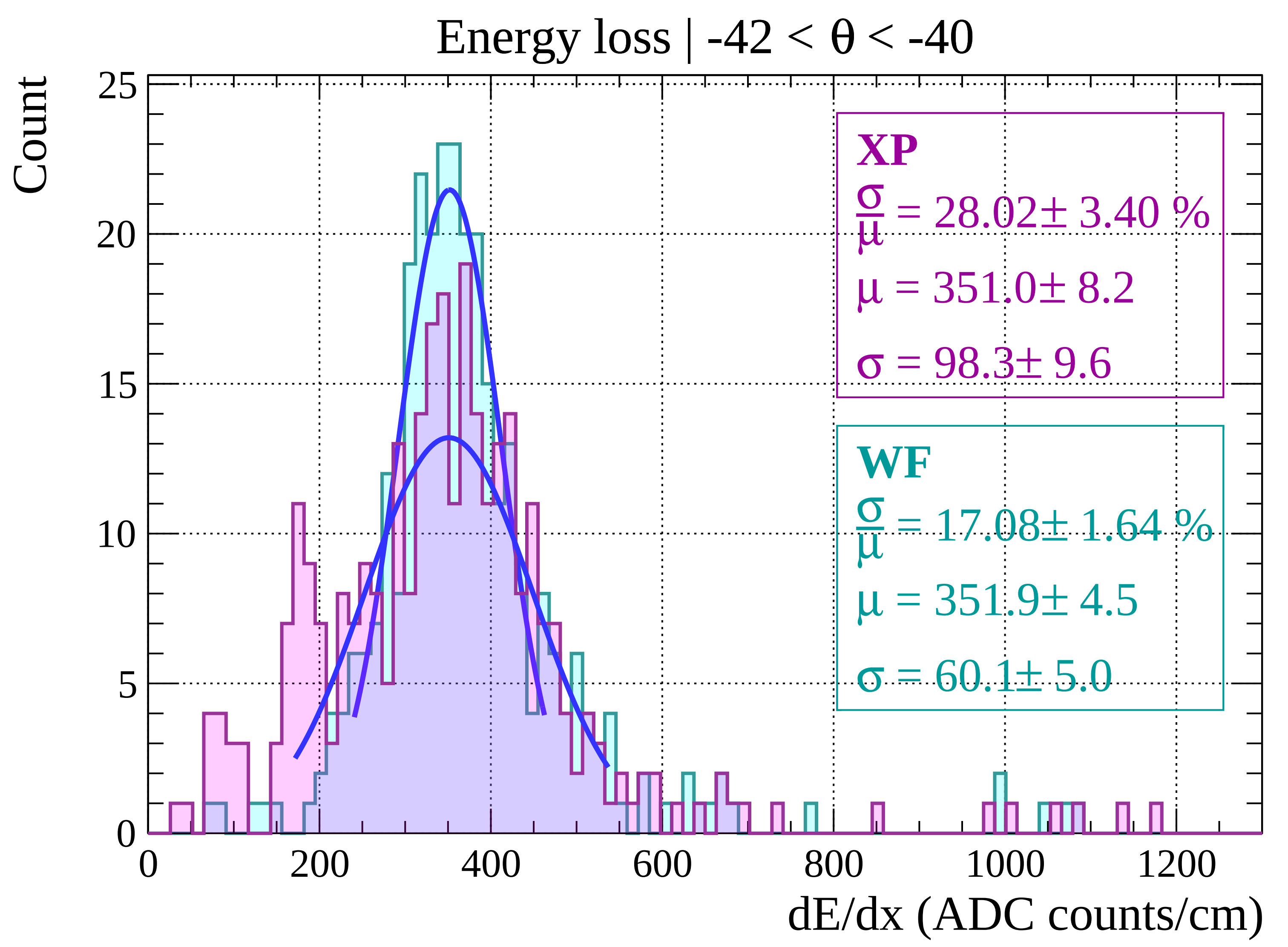


Energy loss | -48 < θ < -46
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 22.93± 2.75 %
μ = 323.8± 7.4
σ = 74.2± 7.2
WF
σ/μ = 17.75± 2.20 %
μ = 301.4± 5.4
σ = 53.5± 5.7

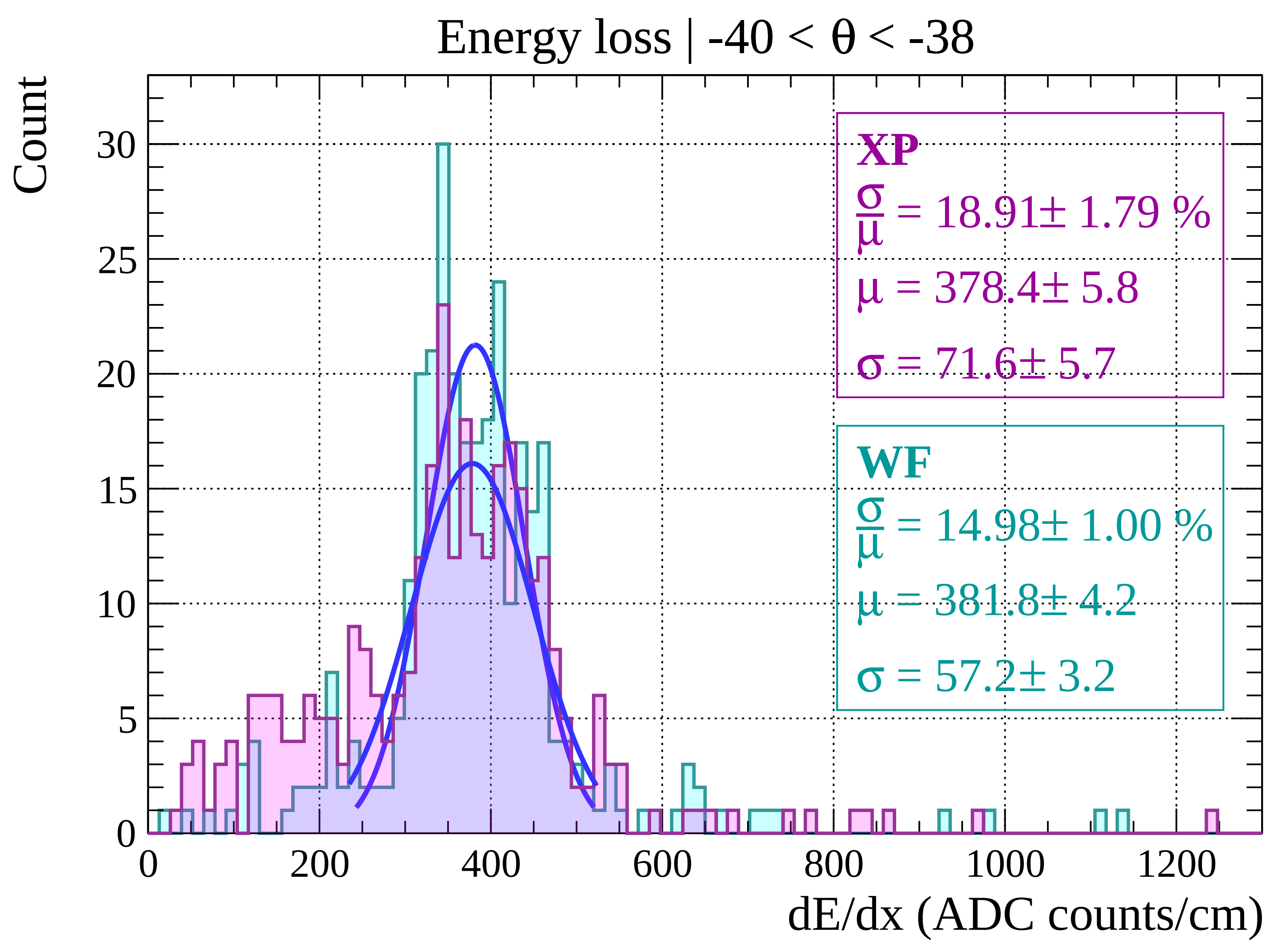

Energy loss | -46 < θ < -44
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 32.29± 3.58 %
μ = 314.4± 8.3
σ = 101.5± 8.5
WF
σ/μ = 20.53± 2.45 %
μ = 328.4± 6.0
σ = 67.4± 6.8
0
5
10
15
20
25
0
200
400
600
800
1000
1200



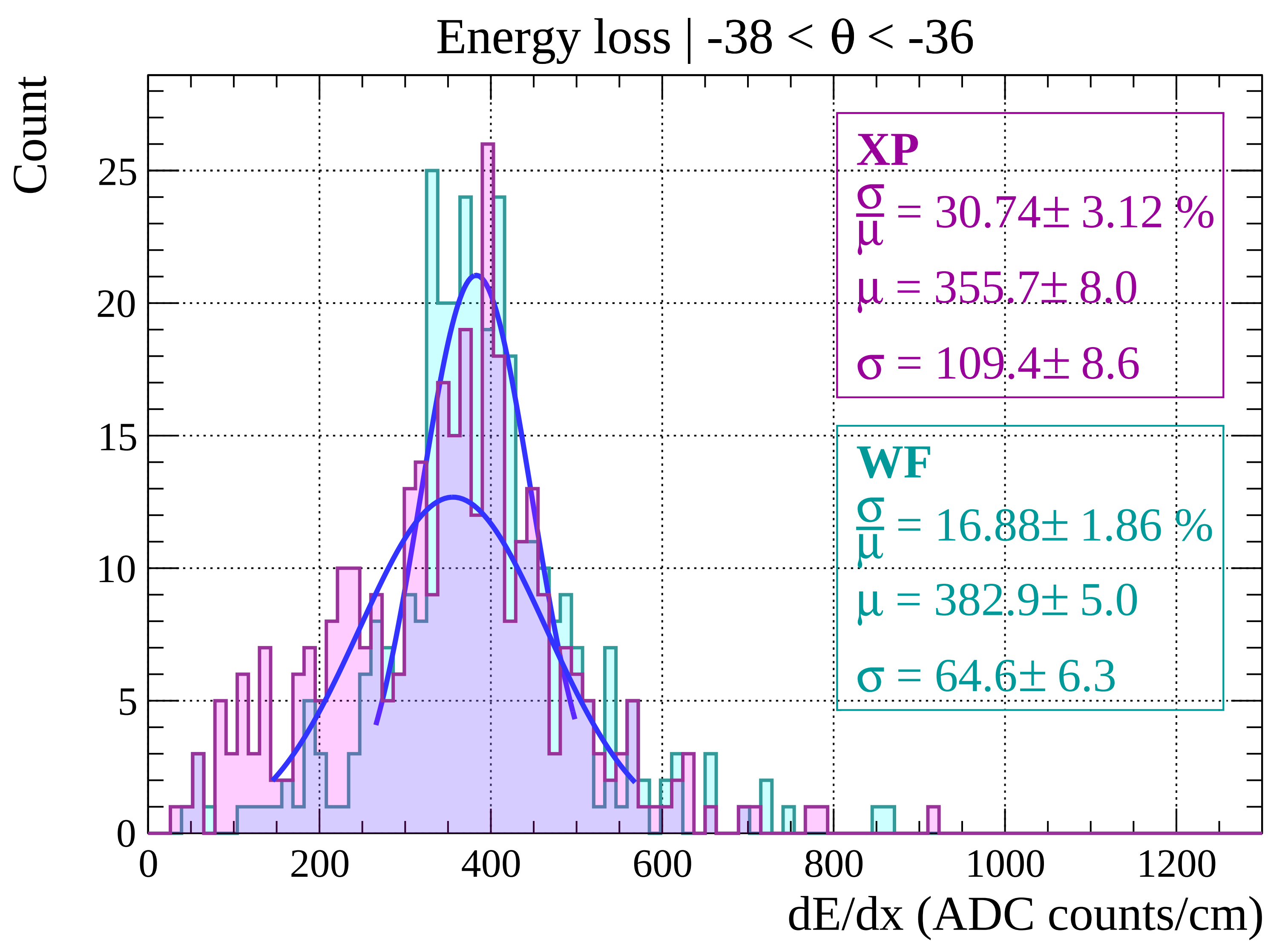
Energy loss | -44 < θ < -42
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 32.05± 3.60 %
μ = 321.7± 8.4
σ = 103.1± 8.9
WF
σ/μ = 20.19± 1.84 %
μ = 352.0± 6.3
σ = 71.1± 5.2



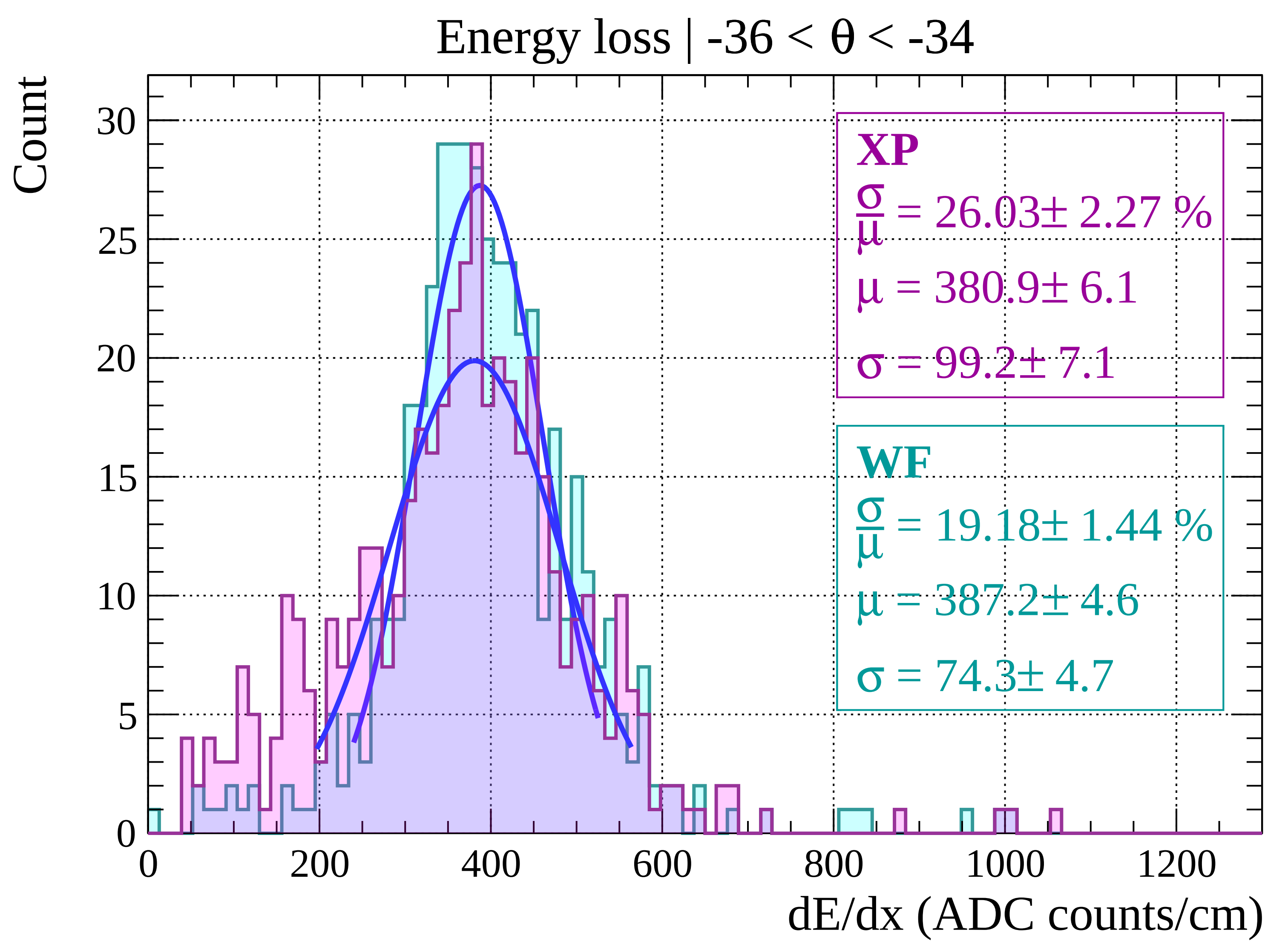
Energy loss | -42 < θ < -40
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 28.02± 3.40 %
μ = 351.0± 8.2
σ = 98.3± 9.6
WF
σ/μ = 17.08± 1.64 %
μ = 351.9± 4.5
σ = 60.1± 5.0



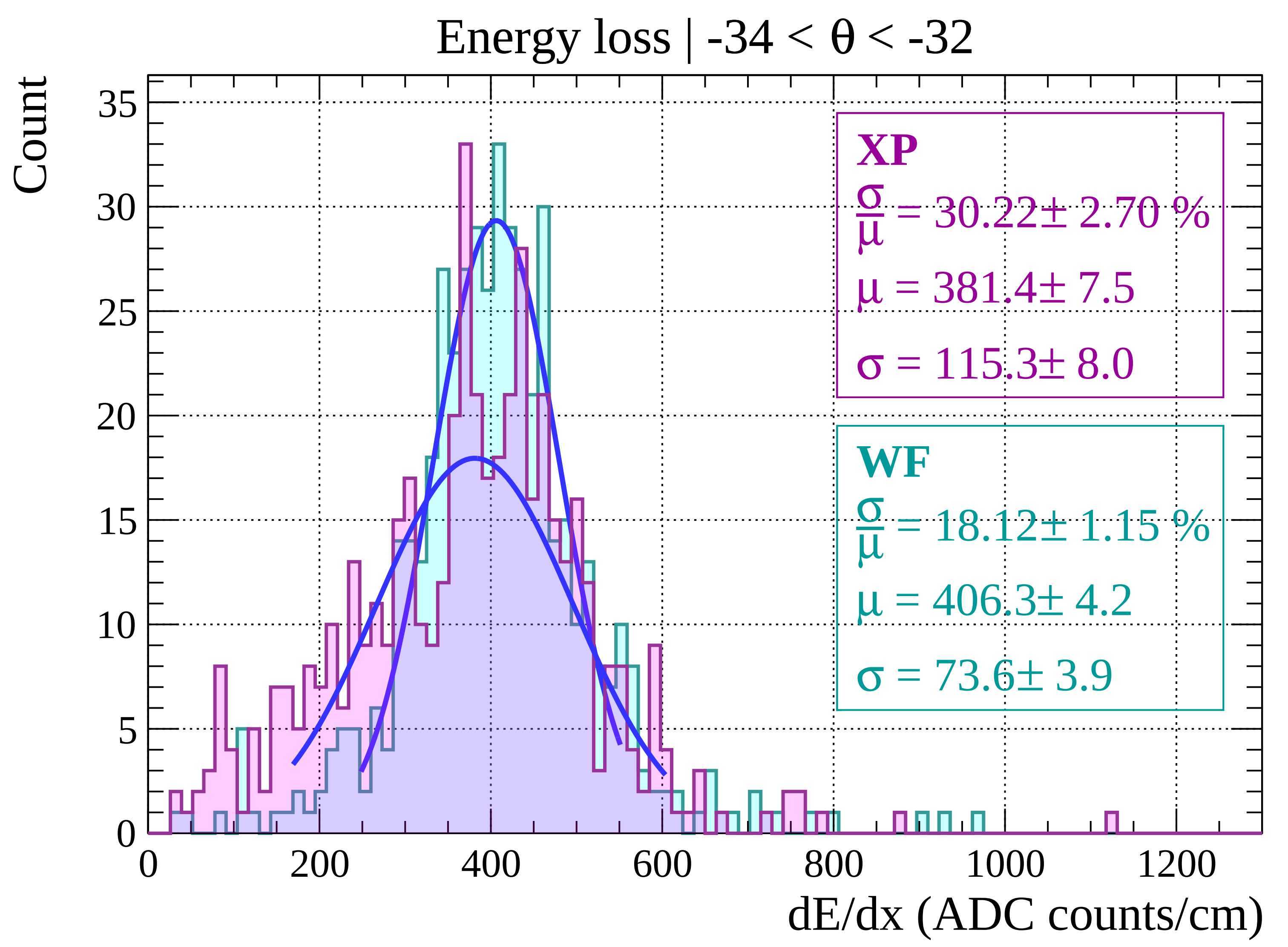
Energy loss | -40 < θ < -38
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
XP
σ/μ = 18.91± 1.79 %
μ = 378.4± 5.8
σ = 71.6± 5.7
WF
σ/μ = 14.98± 1.00 %
μ = 381.8± 4.2
σ = 57.2± 3.2



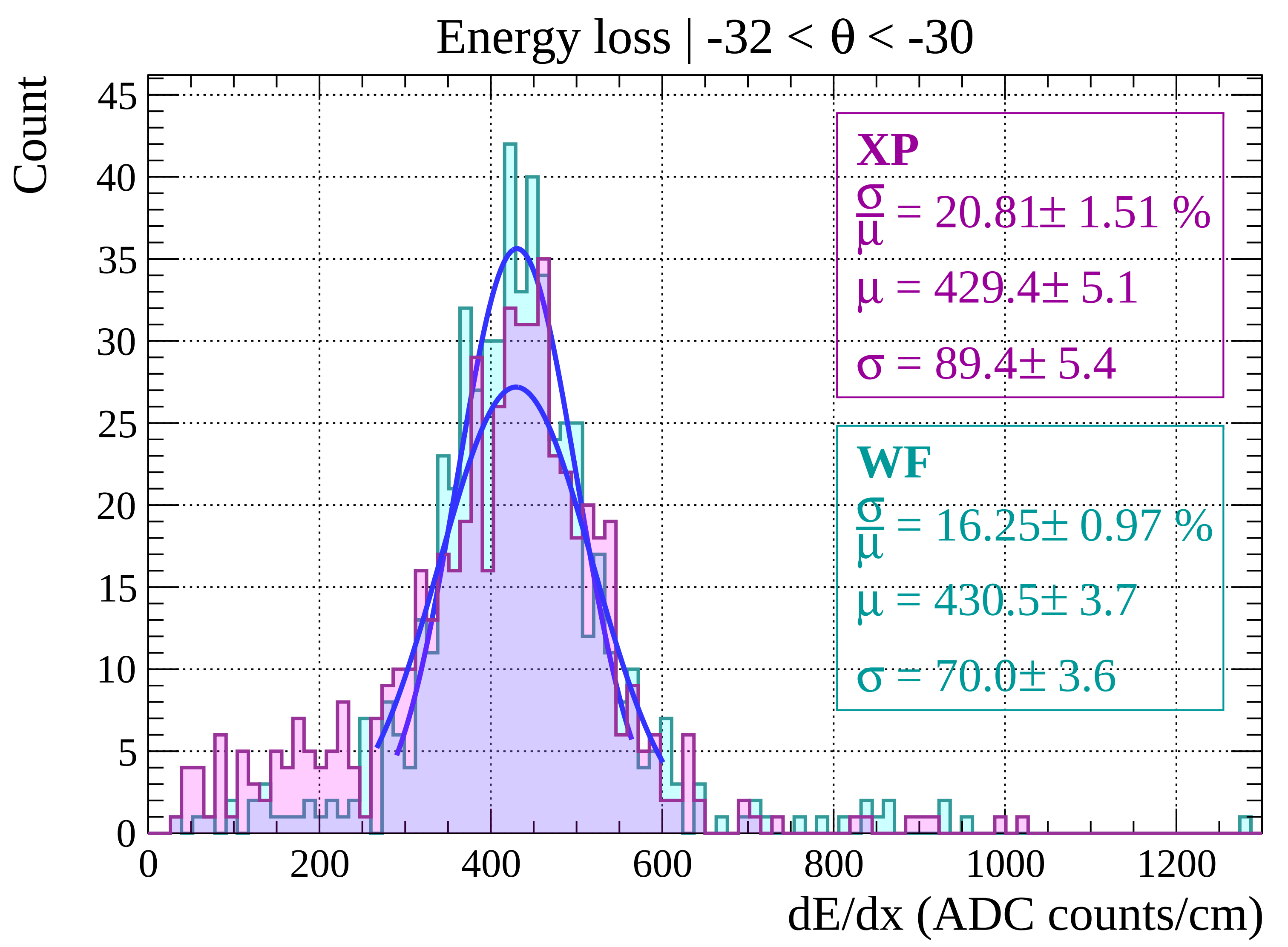
Energy loss | -38 < θ < -36
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 30.74± 3.12 %
μ = 355.7± 8.0
σ = 109.4± 8.6
WF
σ/μ = 16.88± 1.86 %
μ = 382.9± 5.0
σ = 64.6± 6.3



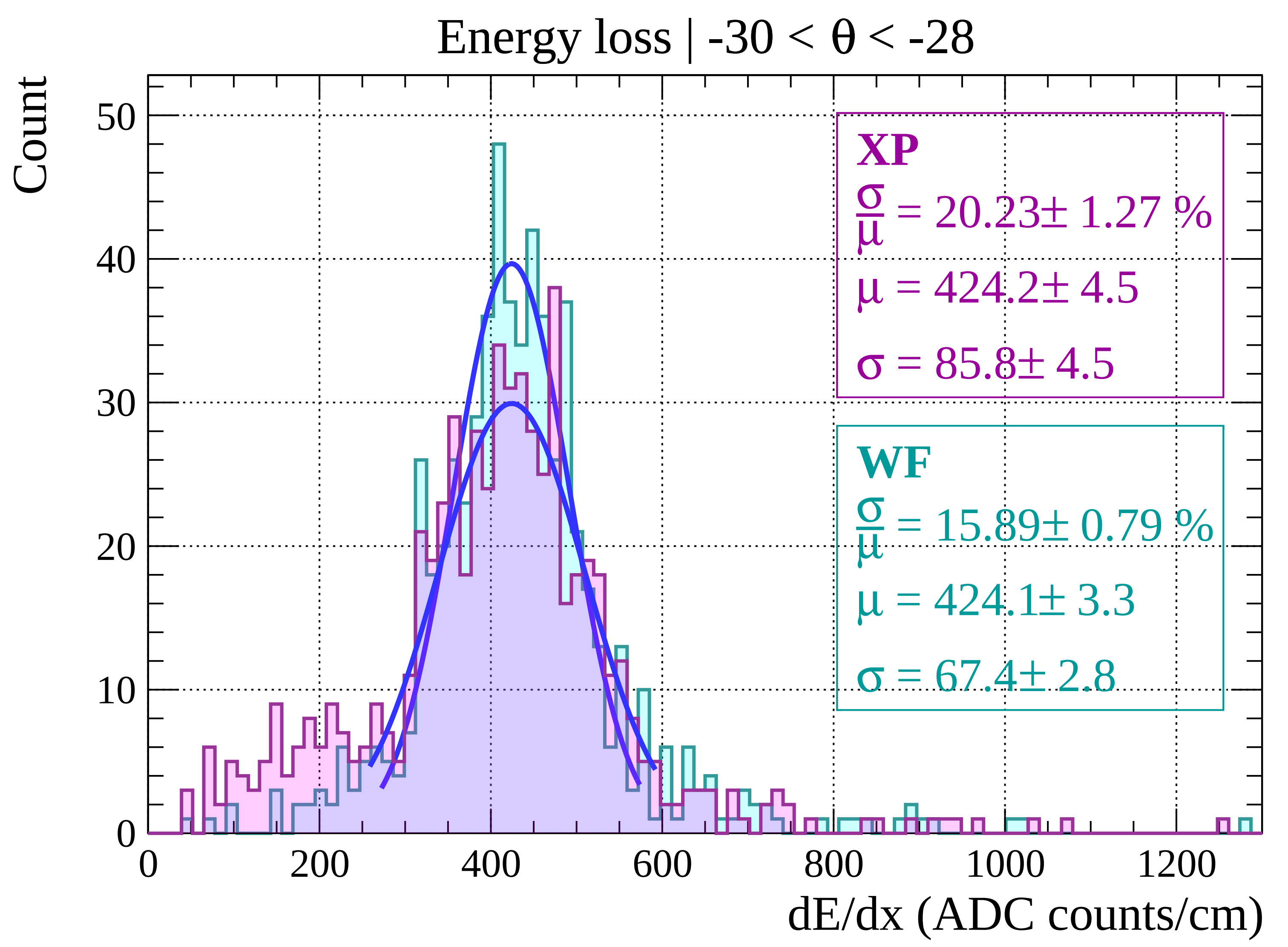
Energy loss | -36 < θ < -34
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 26.03± 2.27 %
μ = 380.9± 6.1
σ = 99.2± 7.1
WF
σ/μ = 19.18± 1.44 %
μ = 387.2± 4.6
σ = 74.3± 4.7



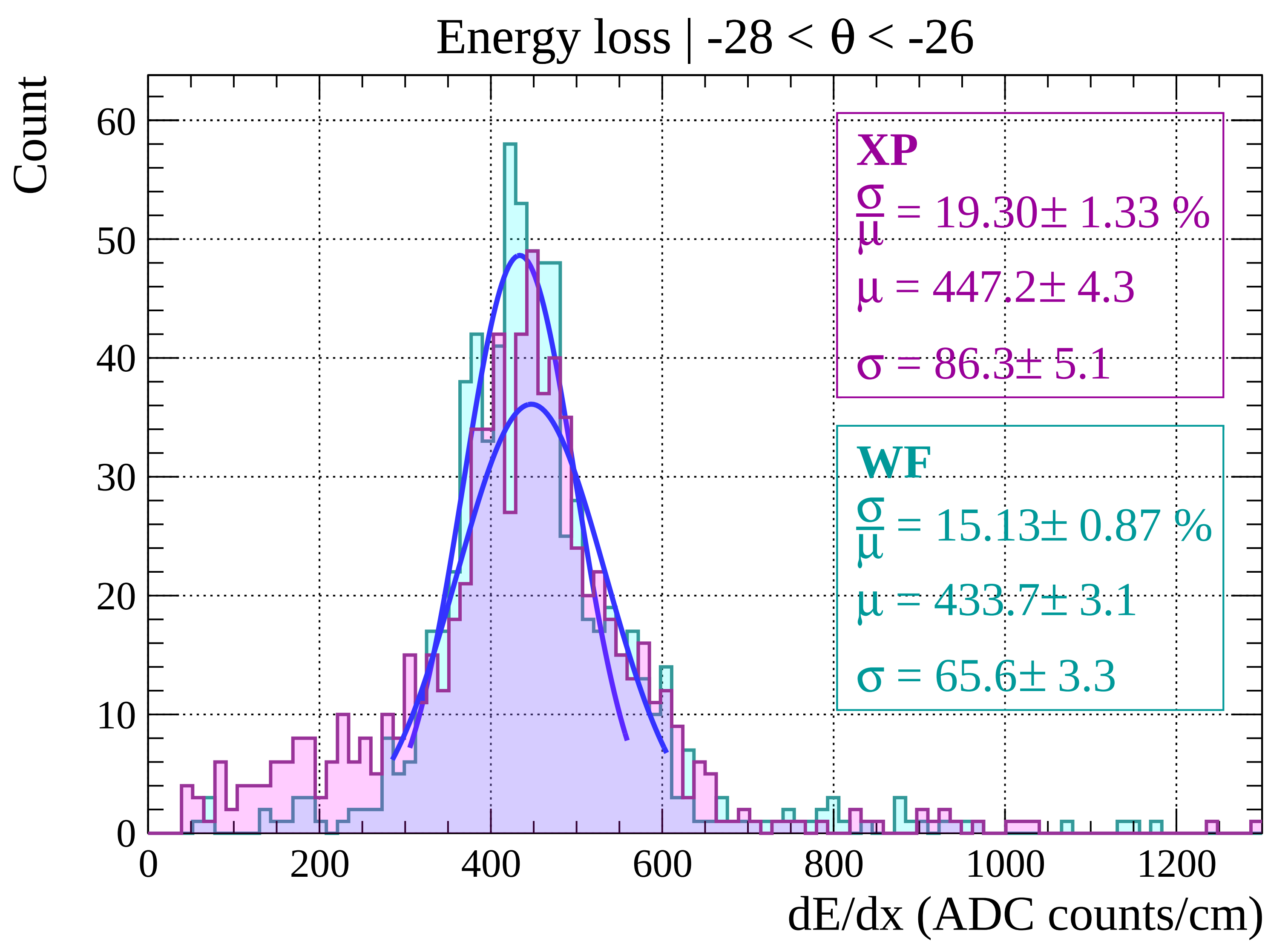
Energy loss | -34 < θ < -32
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP
σ/μ = 30.22± 2.70 %
μ = 381.4± 7.5
σ = 115.3± 8.0
WF
σ/μ = 18.12± 1.15 %
μ = 406.3± 4.2
σ = 73.6± 3.9



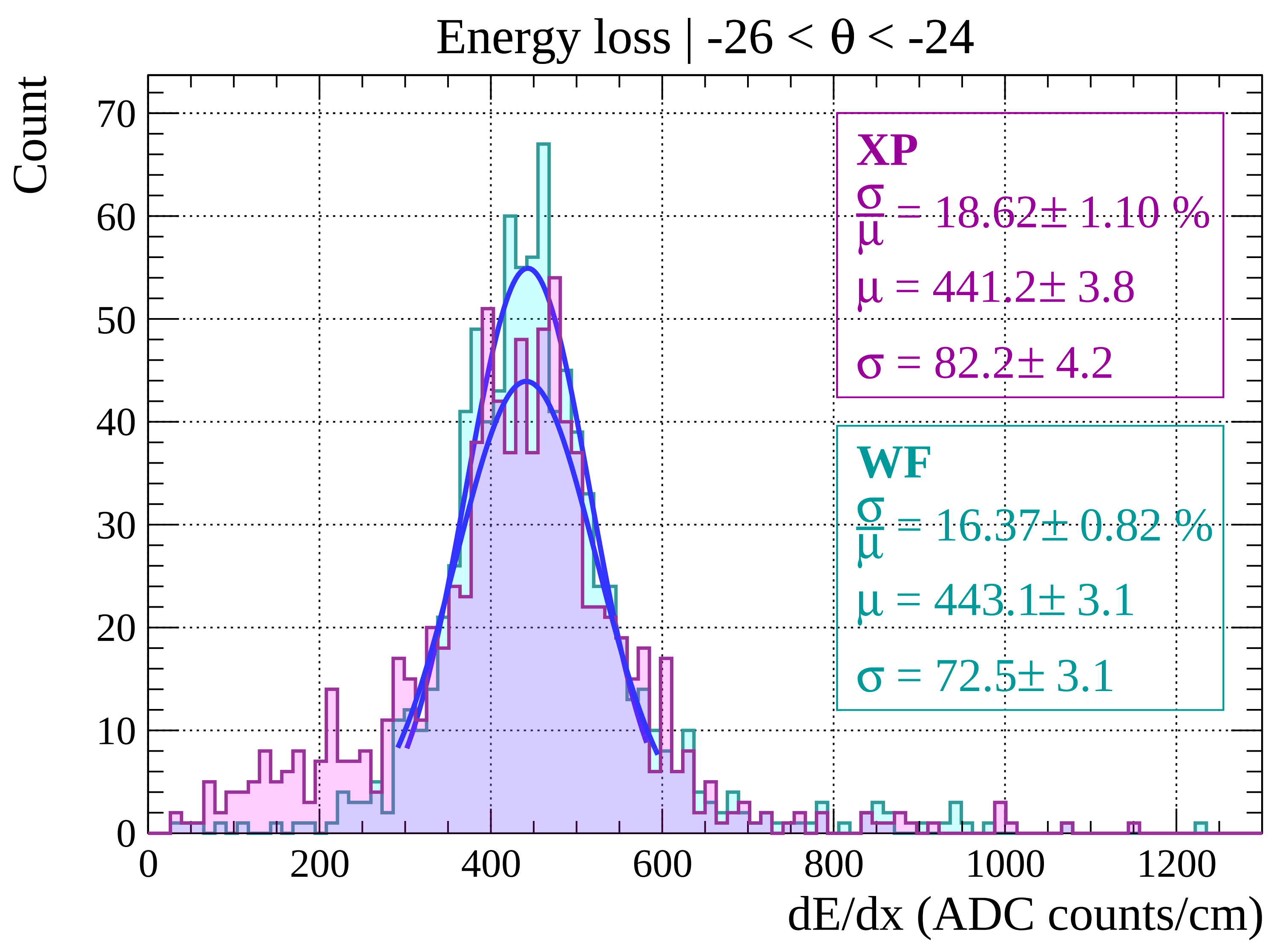
Energy loss | -32 < θ < -30
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 20.81± 1.51 %
μ = 429.4± 5.1
σ = 89.4± 5.4
WF
σ/μ = 16.25± 0.97 %
μ = 430.5± 3.7
σ = 70.0± 3.6
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
45



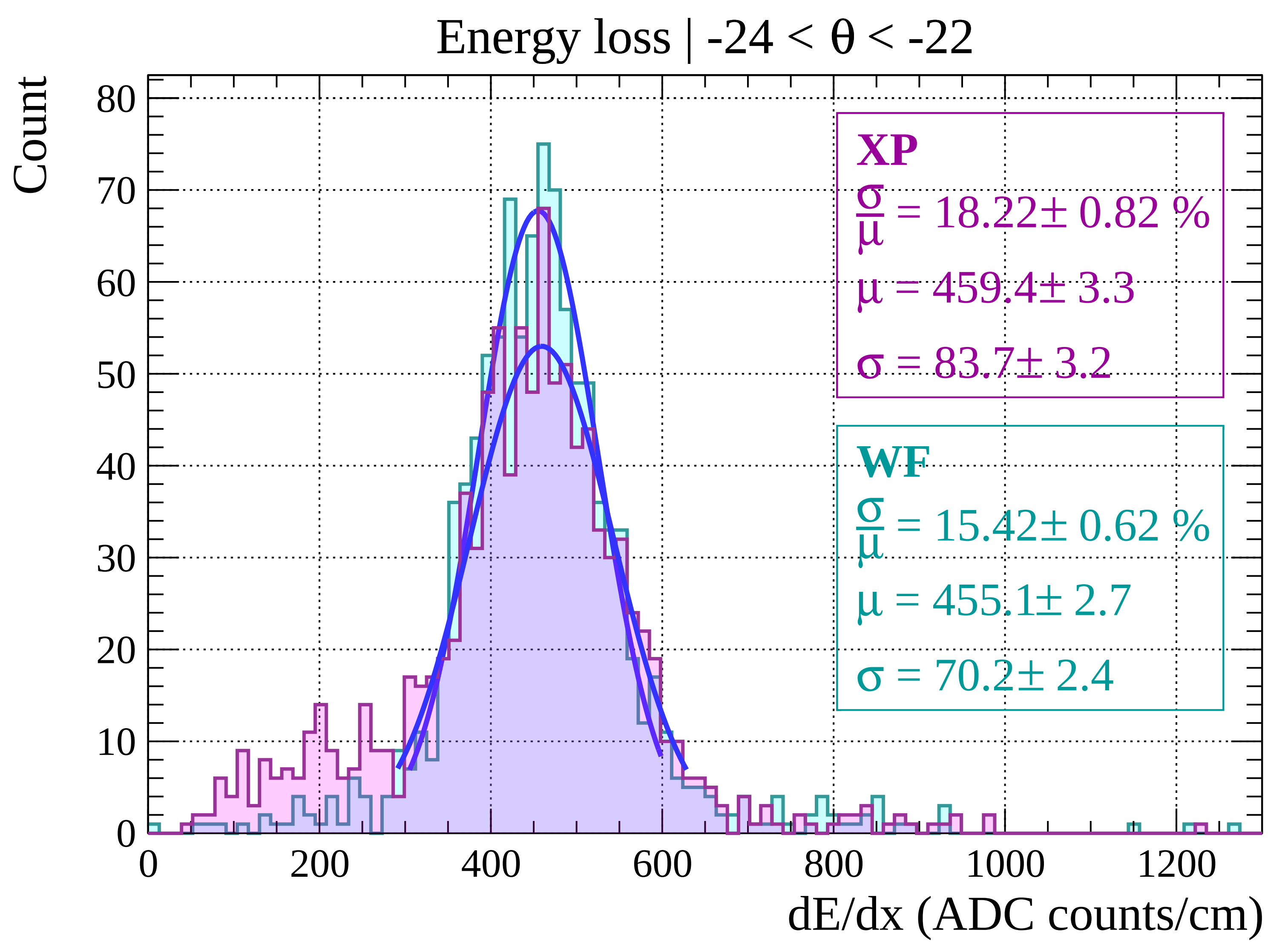
Energy loss | -30 < θ < -28
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
XP
σ/μ = 20.23± 1.27 %
μ = 424.2± 4.5
σ = 85.8± 4.5
WF
σ/μ = 15.89± 0.79 %
μ = 424.1± 3.3
σ = 67.4± 2.8



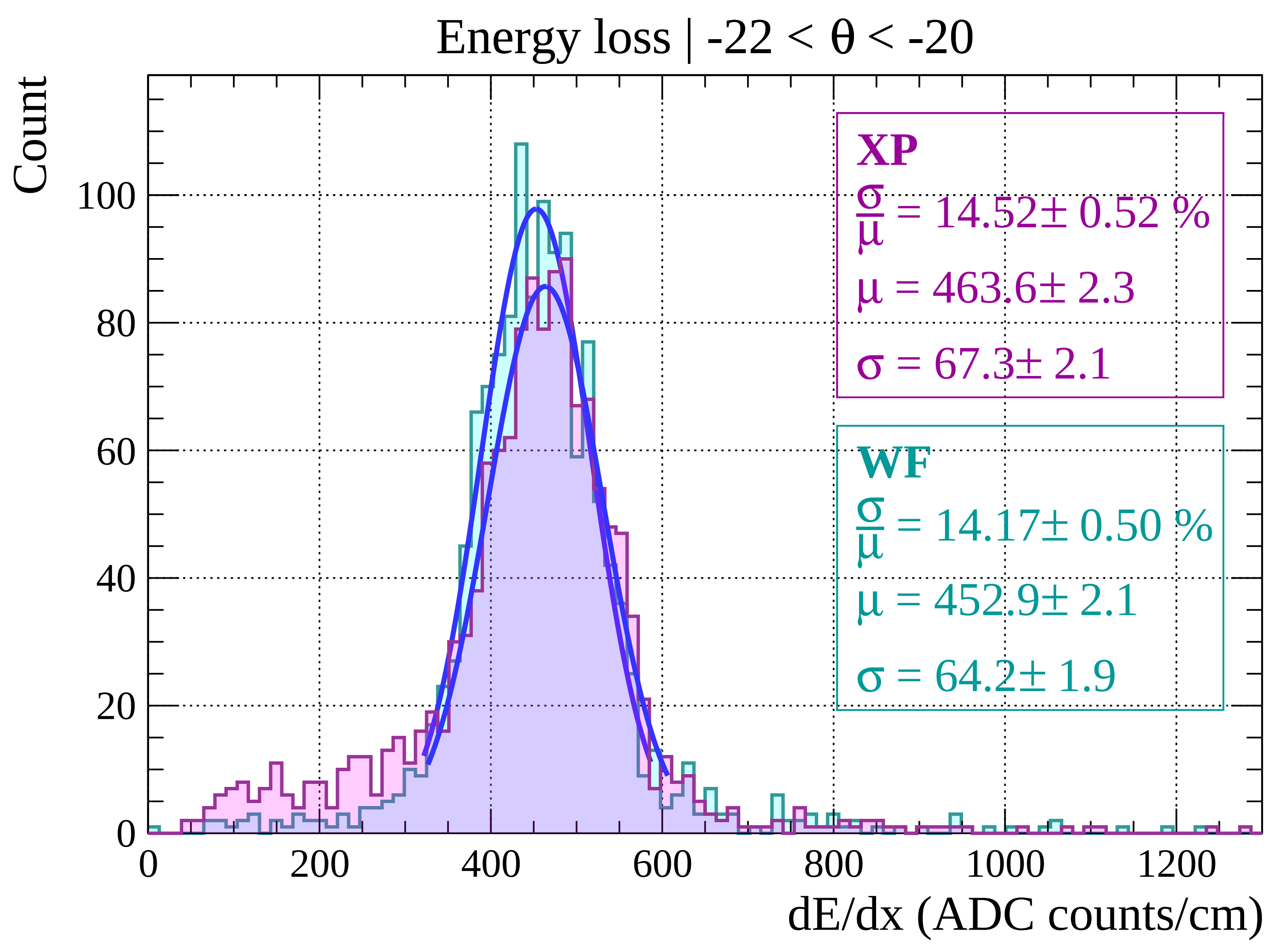
Energy loss | -28 < θ < -26
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 19.30± 1.33 %
μ = 447.2± 4.3
σ = 86.3± 5.1
WF
σ/μ = 15.13± 0.87 %
μ = 433.7± 3.1
σ = 65.6± 3.3



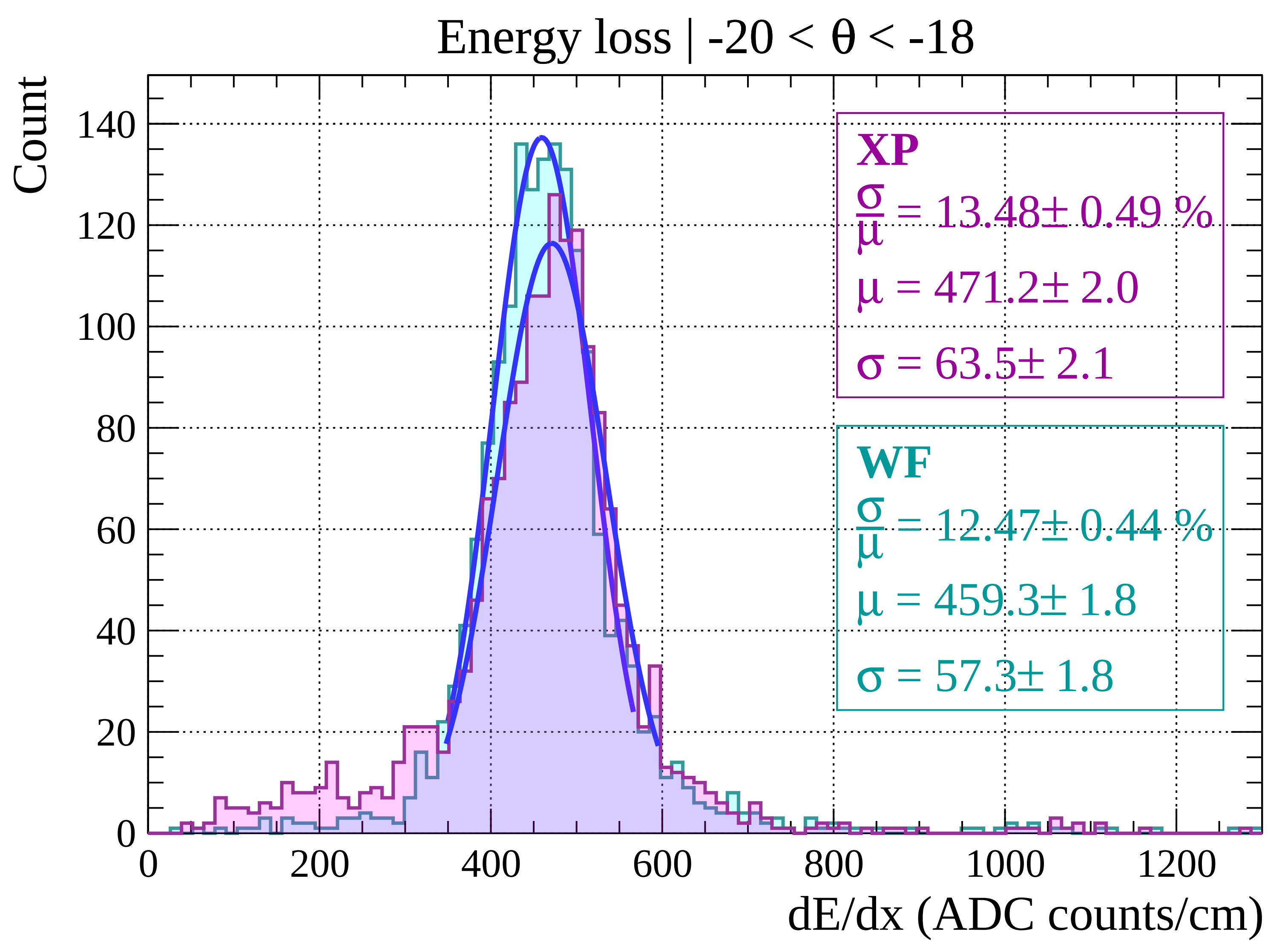
Energy loss | -26 < θ < -24
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 18.62± 1.10 %
μ = 441.2± 3.8
σ = 82.2± 4.2
WF
σ/μ = 16.37± 0.82 %
μ = 443.1± 3.1
σ = 72.5± 3.1



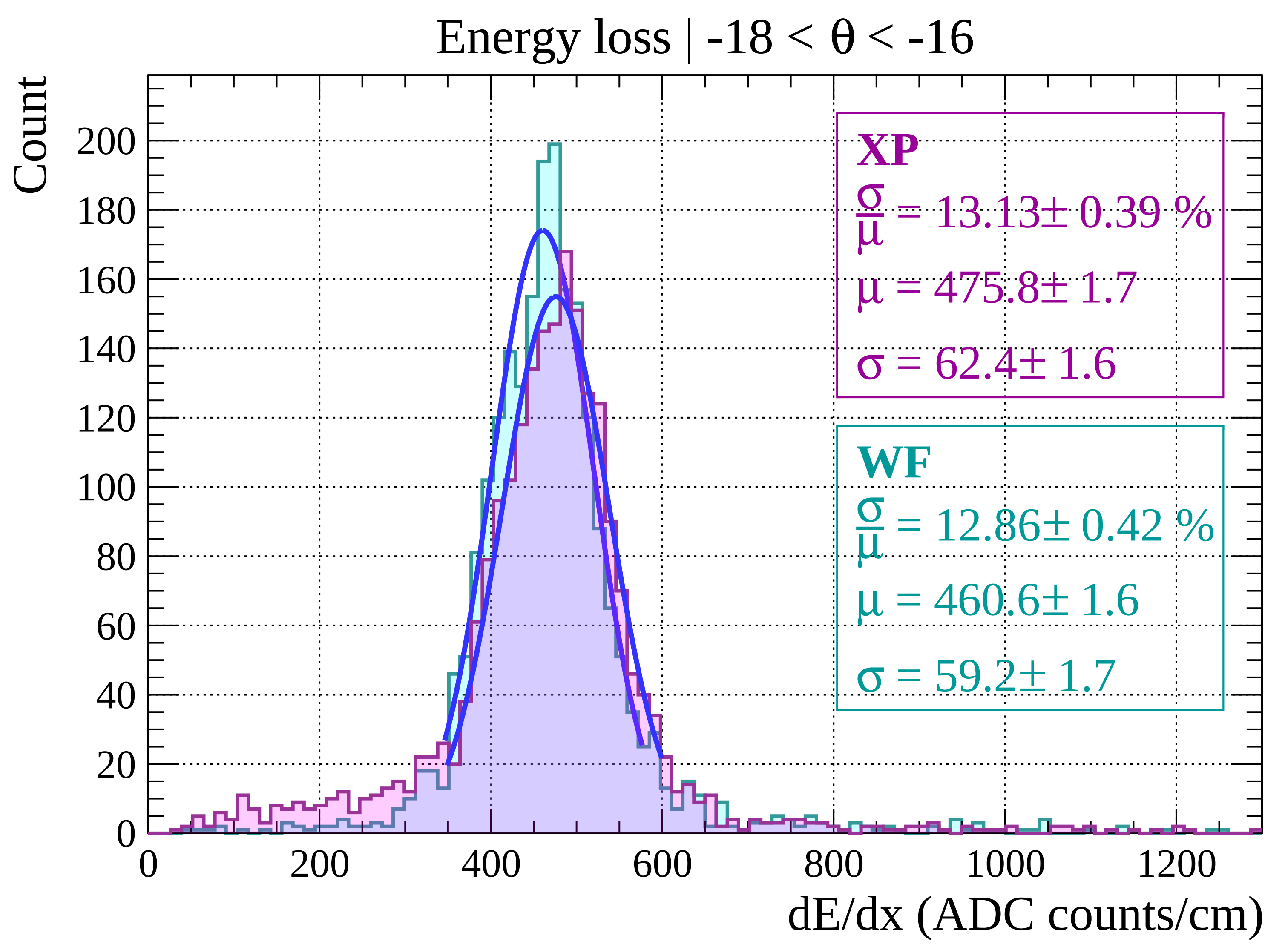
Energy loss | -24 < θ < -22
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP
σ/μ = 18.22± 0.82 %
μ = 459.4± 3.3
σ = 83.7± 3.2
WF
σ/μ = 15.42± 0.62 %
μ = 455.1± 2.7
σ = 70.2± 2.4



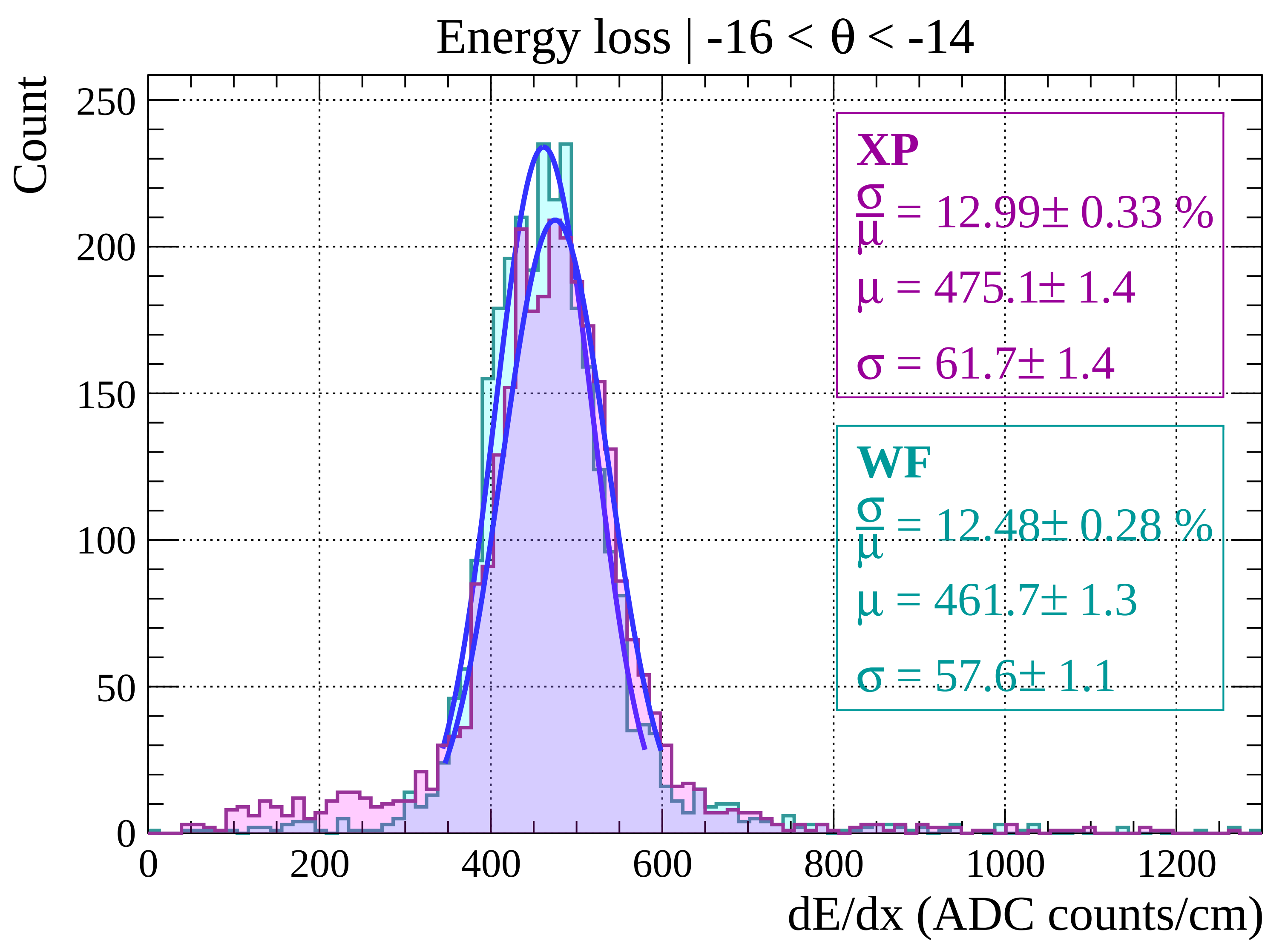
Energy loss | -22 < θ < -20
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 14.52± 0.52 %
μ = 463.6± 2.3
σ = 67.3± 2.1
WF
σ/μ = 14.17± 0.50 %
μ = 452.9± 2.1
σ = 64.2± 1.9



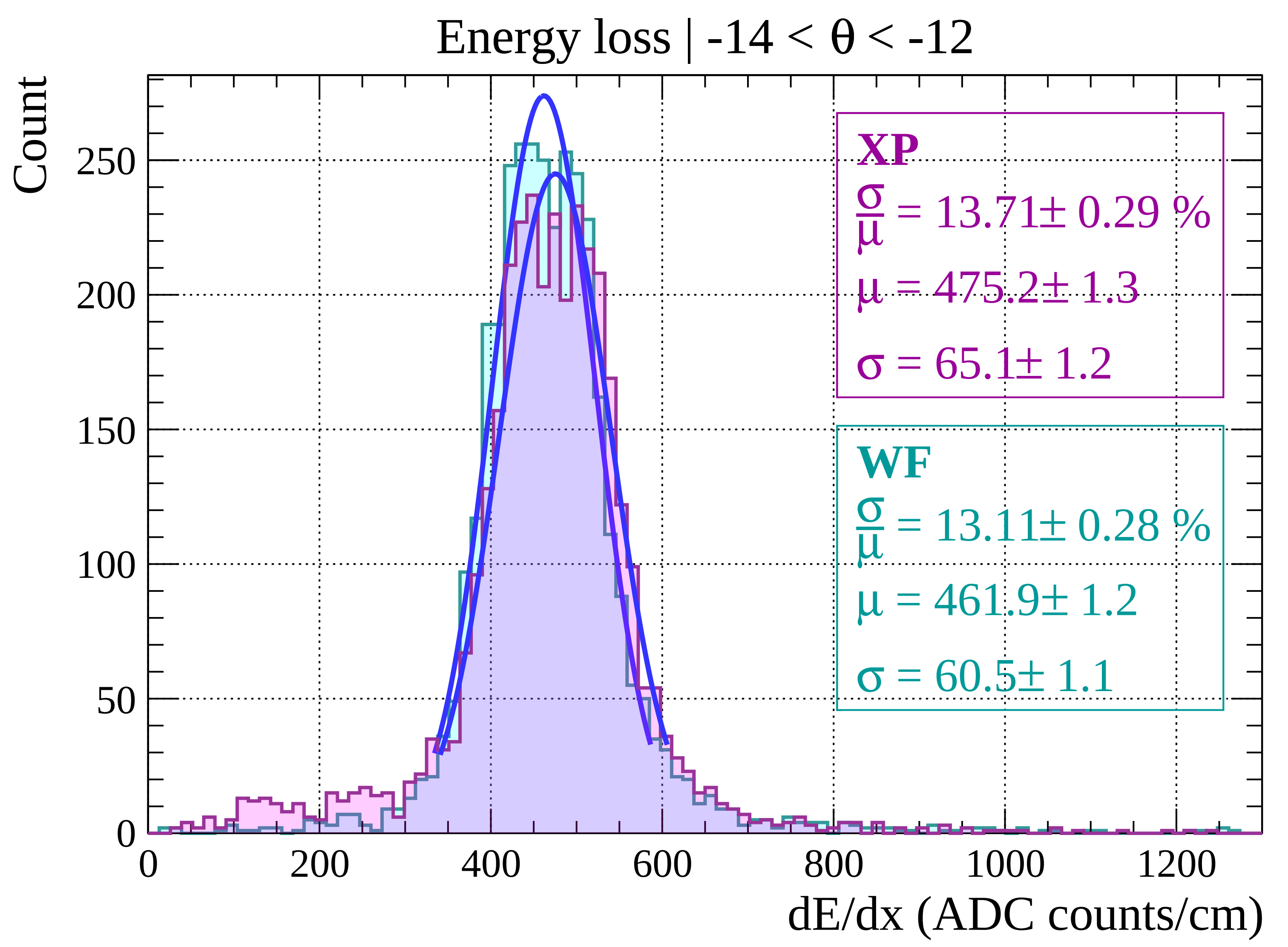
Energy loss | -20 < θ < -18
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 13.48 ± 0.49 %
μ = 471.2 ± 2.0
σ = 63.5 ± 2.1
WF
σ/μ = 12.47 ± 0.44 %
μ = 459.3 ± 1.8
σ = 57.3 ± 1.8



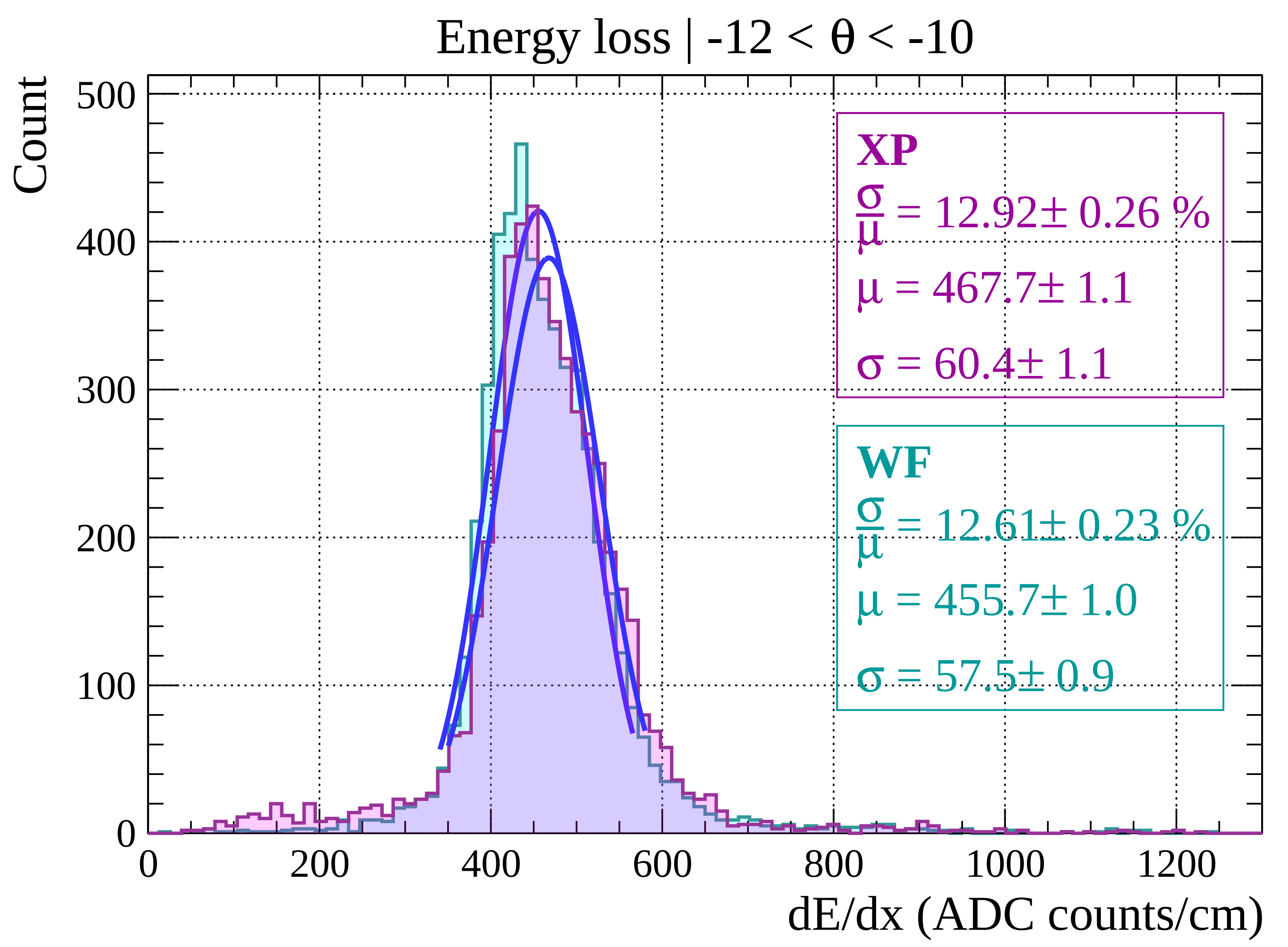
Energy loss | -18 < θ < -16
Count
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.13± 0.39 %
μ = 475.8± 1.7
σ = 62.4± 1.6
WF
σ/μ = 12.86± 0.42 %
μ = 460.6± 1.6
σ = 59.2± 1.7

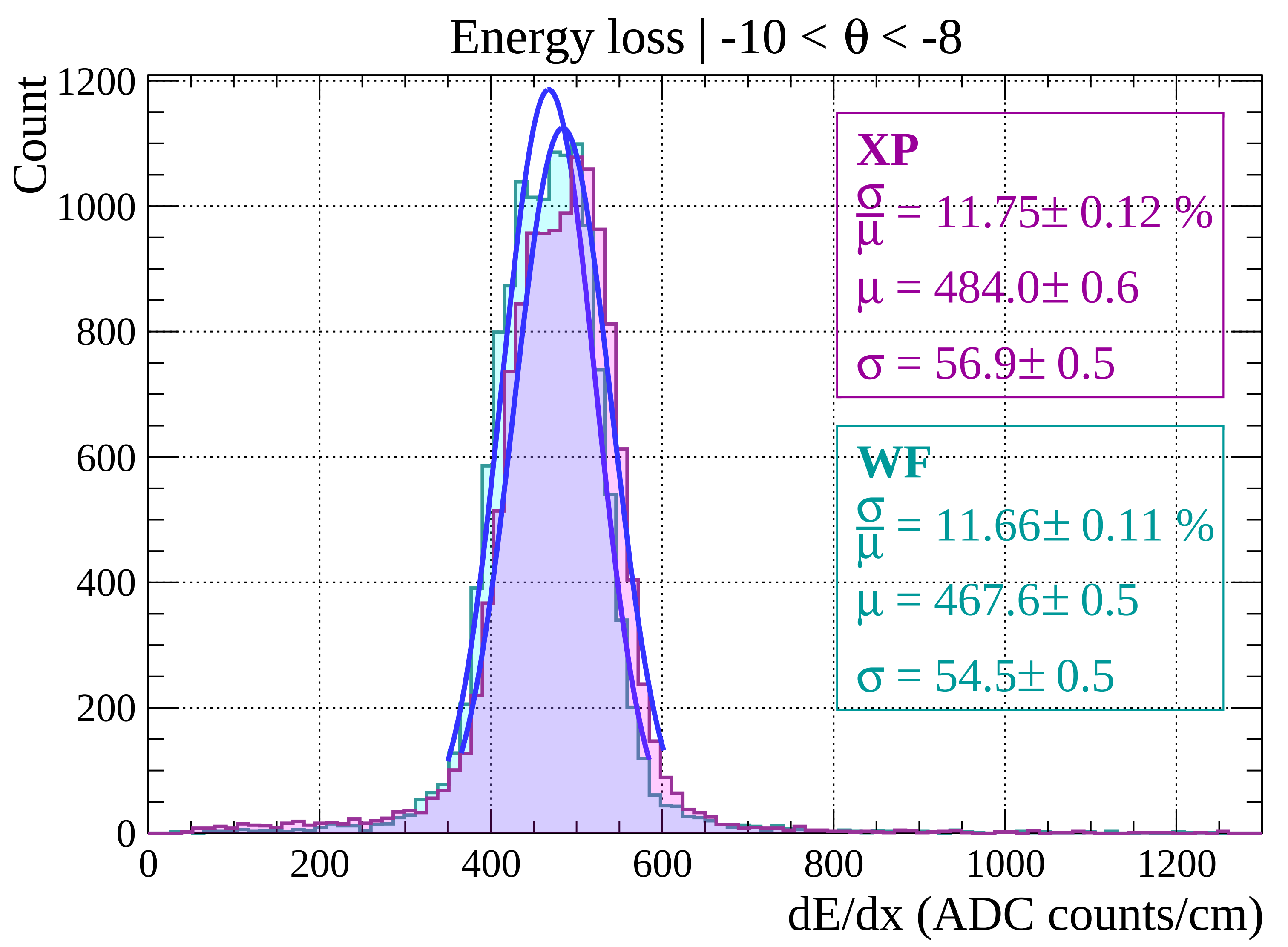


Energy loss | -16 < θ < -14
Count
XP
σ/μ = 12.99± 0.33 %
μ = 475.1± 1.4
σ = 61.7± 1.4
WF
σ/μ = 12.48± 0.28 %
μ = 461.7± 1.3
σ = 57.6± 1.1
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200

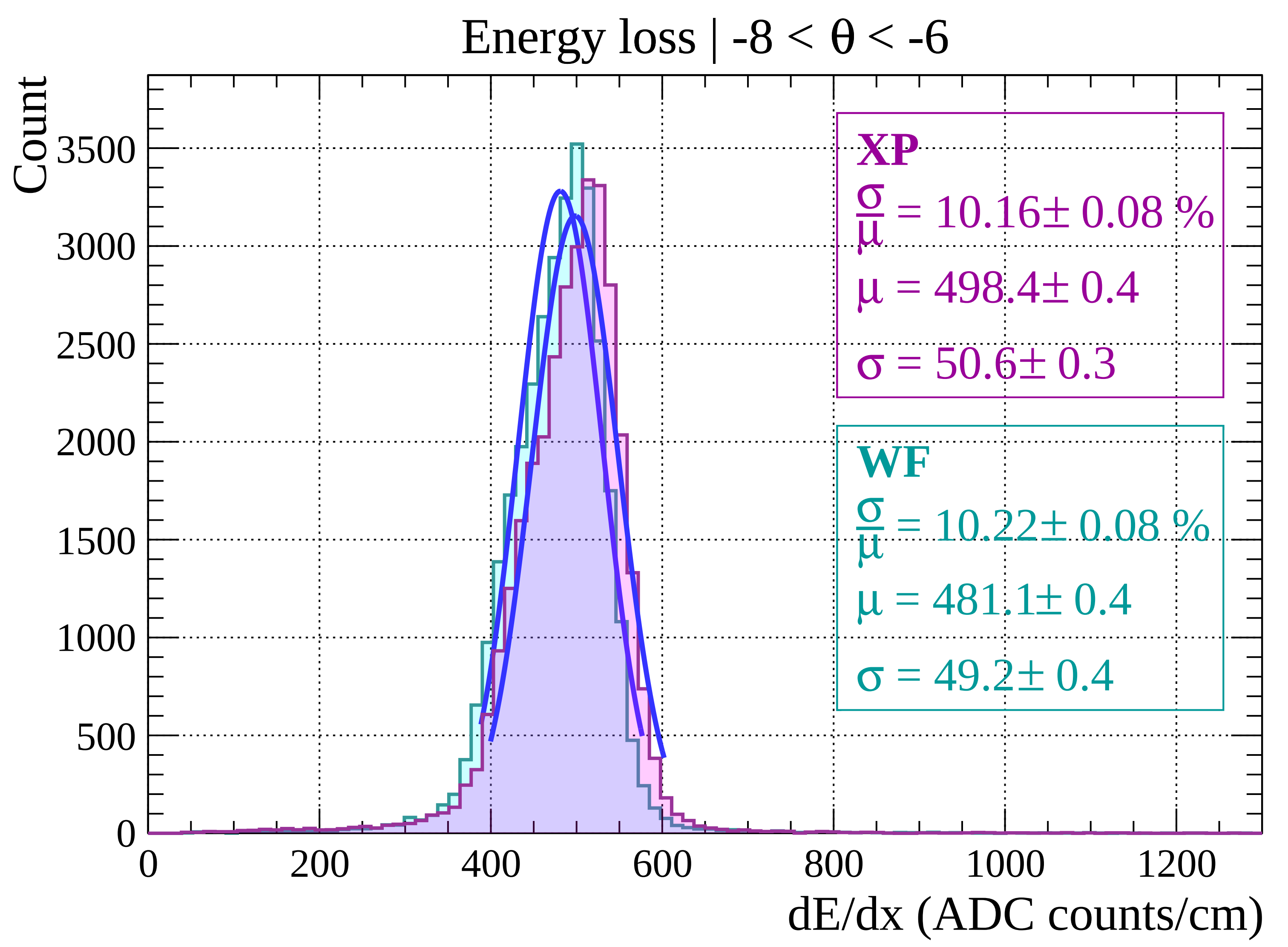


Energy loss | -14 < θ < -12
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.71± 0.29 %
μ = 475.2± 1.3
σ = 65.1± 1.2
WF
σ/μ = 13.11± 0.28 %
μ = 461.9± 1.2
σ = 60.5± 1.1

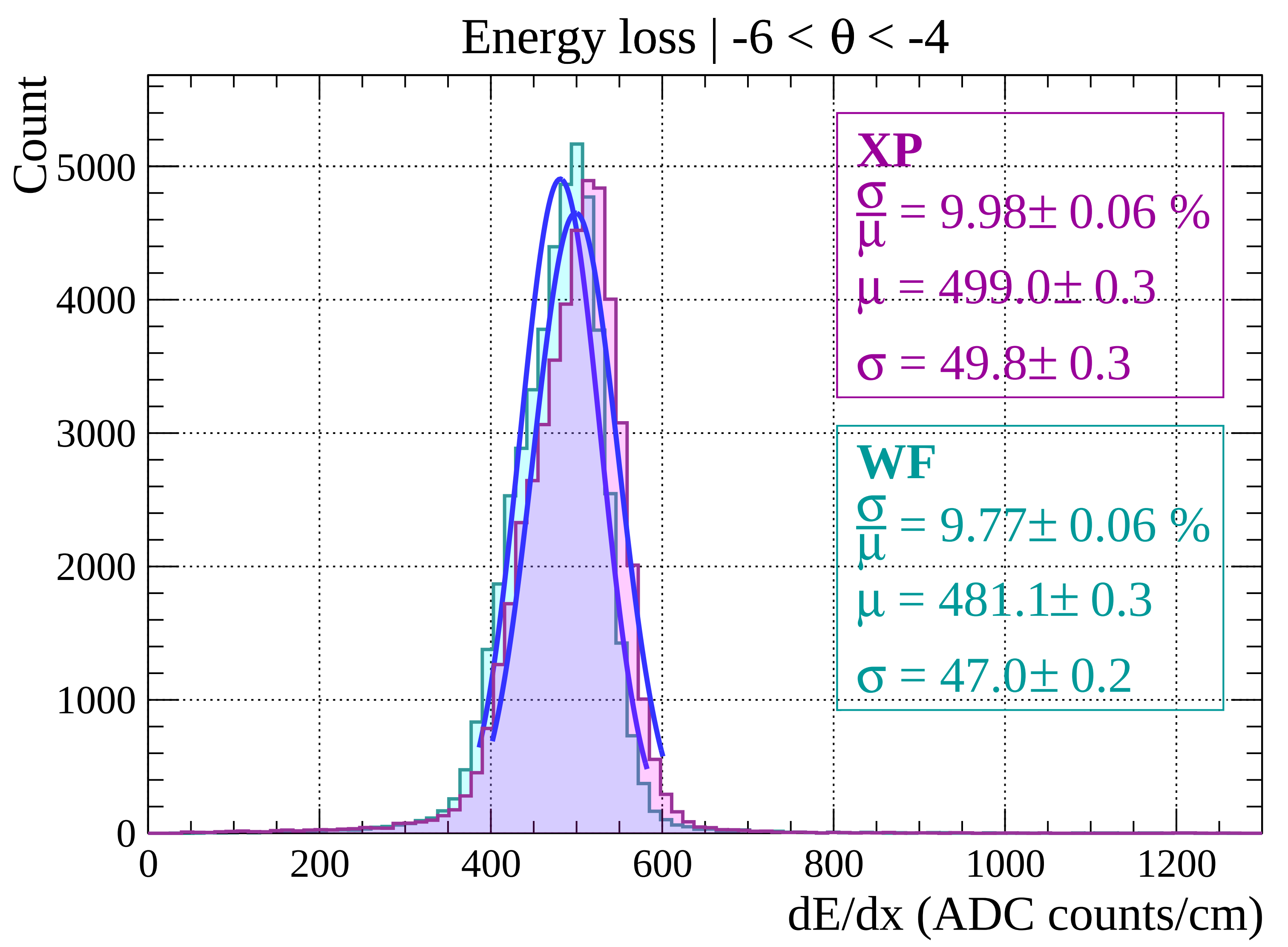

Energy loss | -12 < θ < -10
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 12.92± 0.26 %
μ = 467.7± 1.1
σ = 60.4± 1.1
WF
σ/μ = 12.61± 0.23 %
μ = 455.7± 1.0
σ = 57.5± 0.9



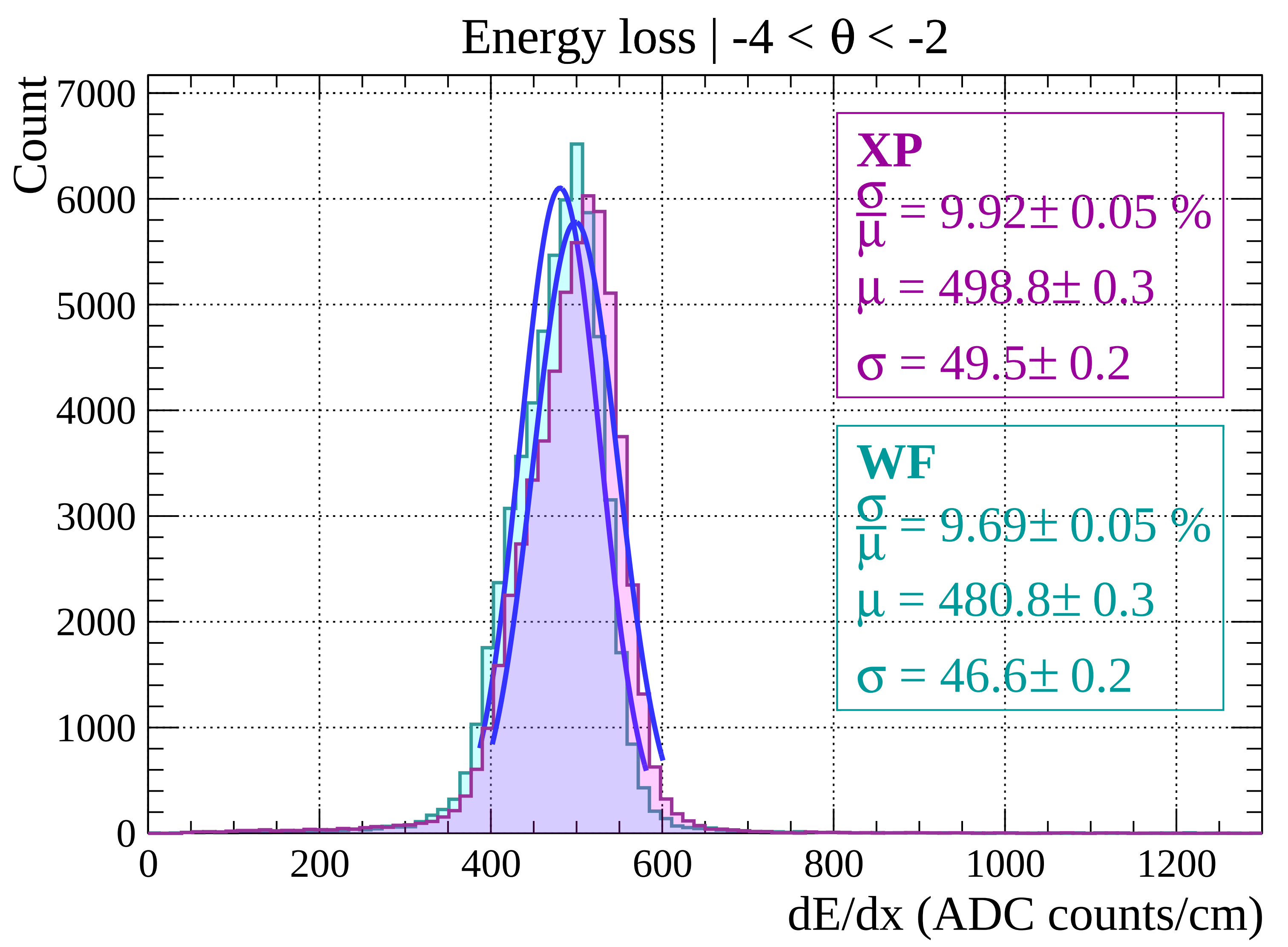
Energy loss | -10 < θ < -8
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 11.75± 0.12 %
μ = 484.0± 0.6
σ = 56.9± 0.5
WF
σ/μ = 11.66± 0.11 %
μ = 467.6± 0.5
σ = 54.5± 0.5



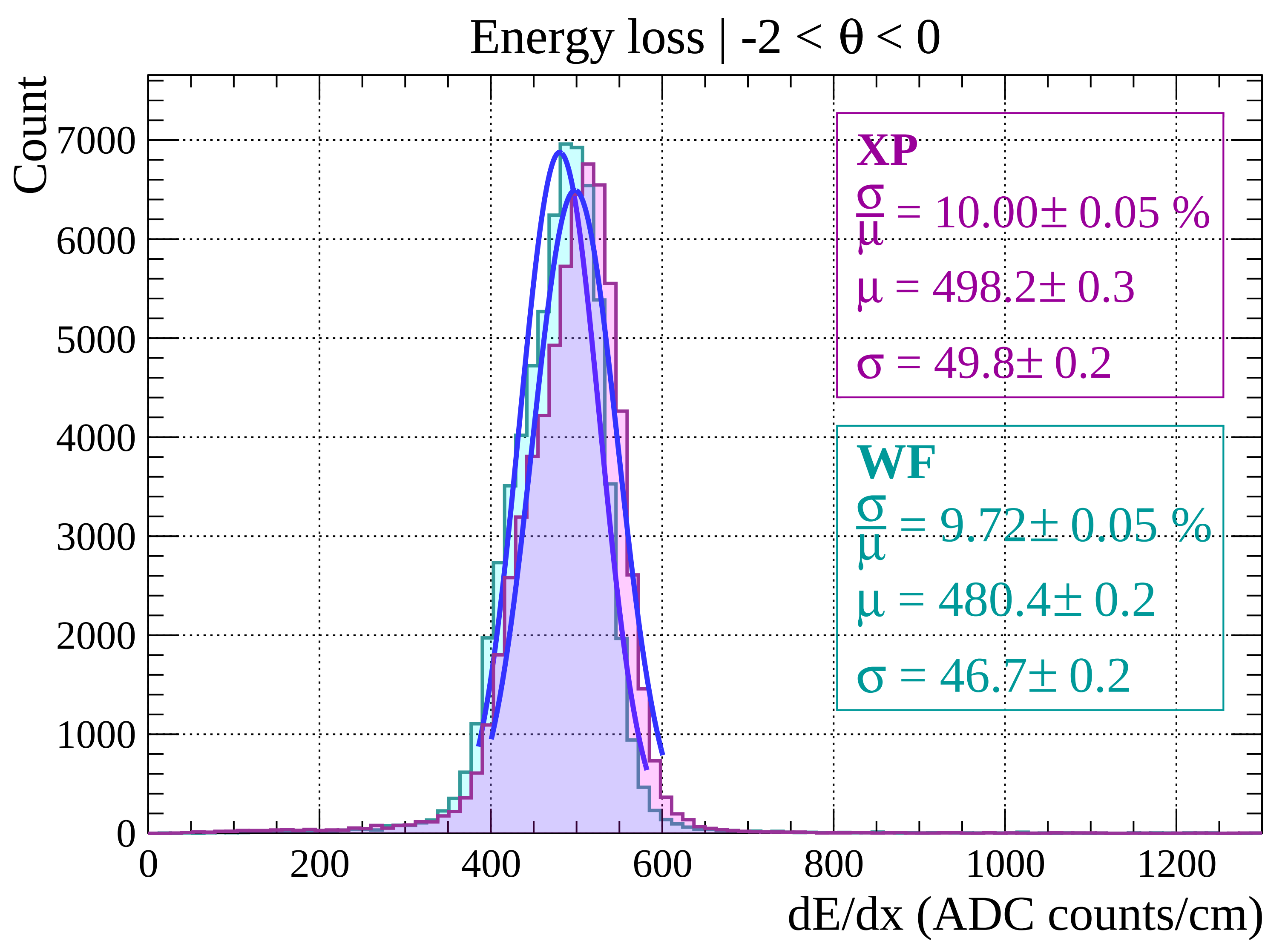
Energy loss | -8 < θ < -6
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.16± 0.08 %
μ = 498.4± 0.4
σ = 50.6± 0.3
WF
σ/μ = 10.22± 0.08 %
μ = 481.1± 0.4
σ = 49.2± 0.4
0
500
1000
1500
2000
2500
3000
3500
0
200
400
600
800
1000
1200



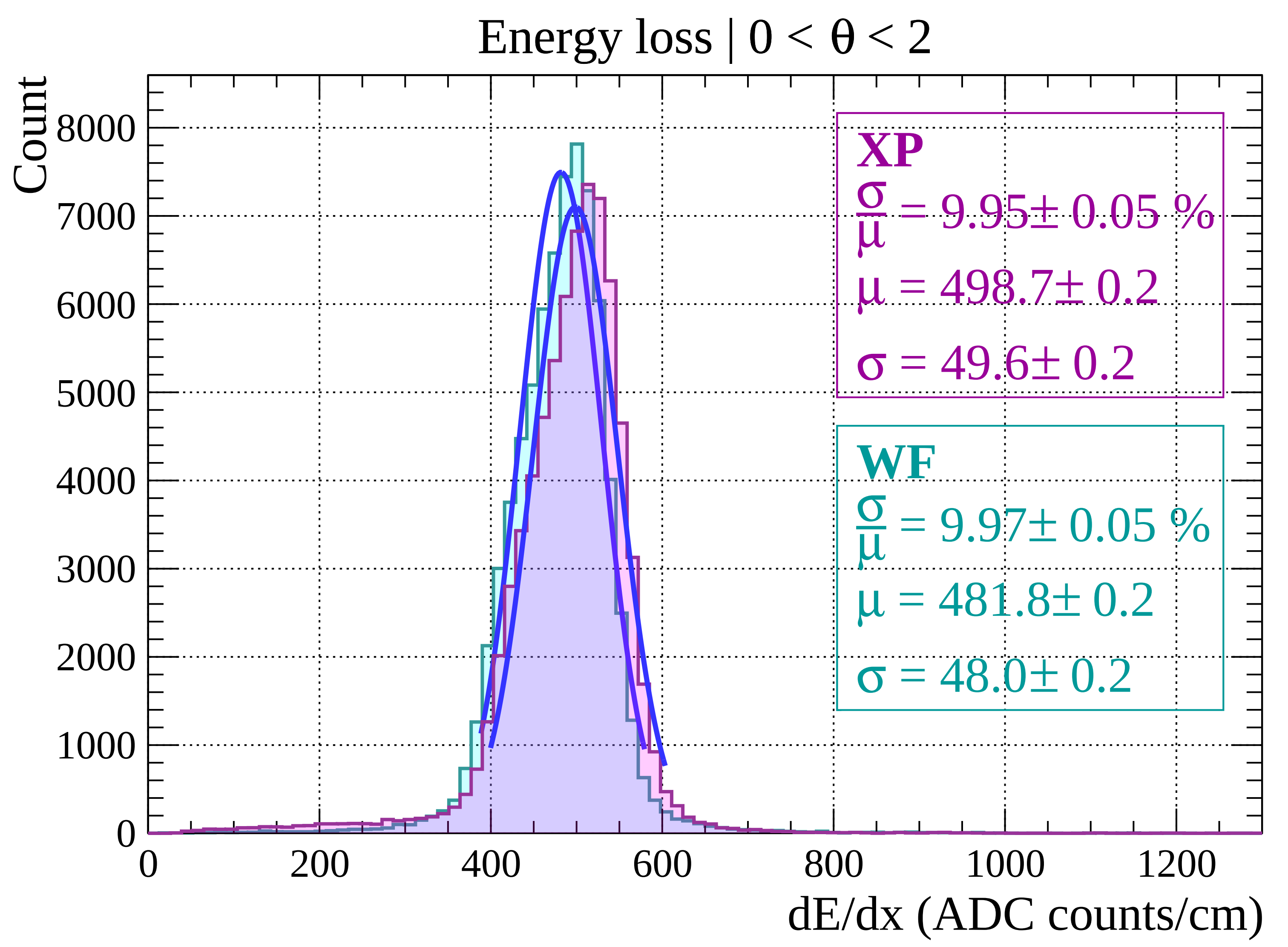
Energy loss | -6 < θ < -4
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.98± 0.06 %
μ = 499.0± 0.3
σ = 49.8± 0.3
WF
σ/μ = 9.77± 0.06 %
μ = 481.1± 0.3
σ = 47.0± 0.2
0
1000
2000
3000
4000
5000
0
200
400
600
800
1000
1200



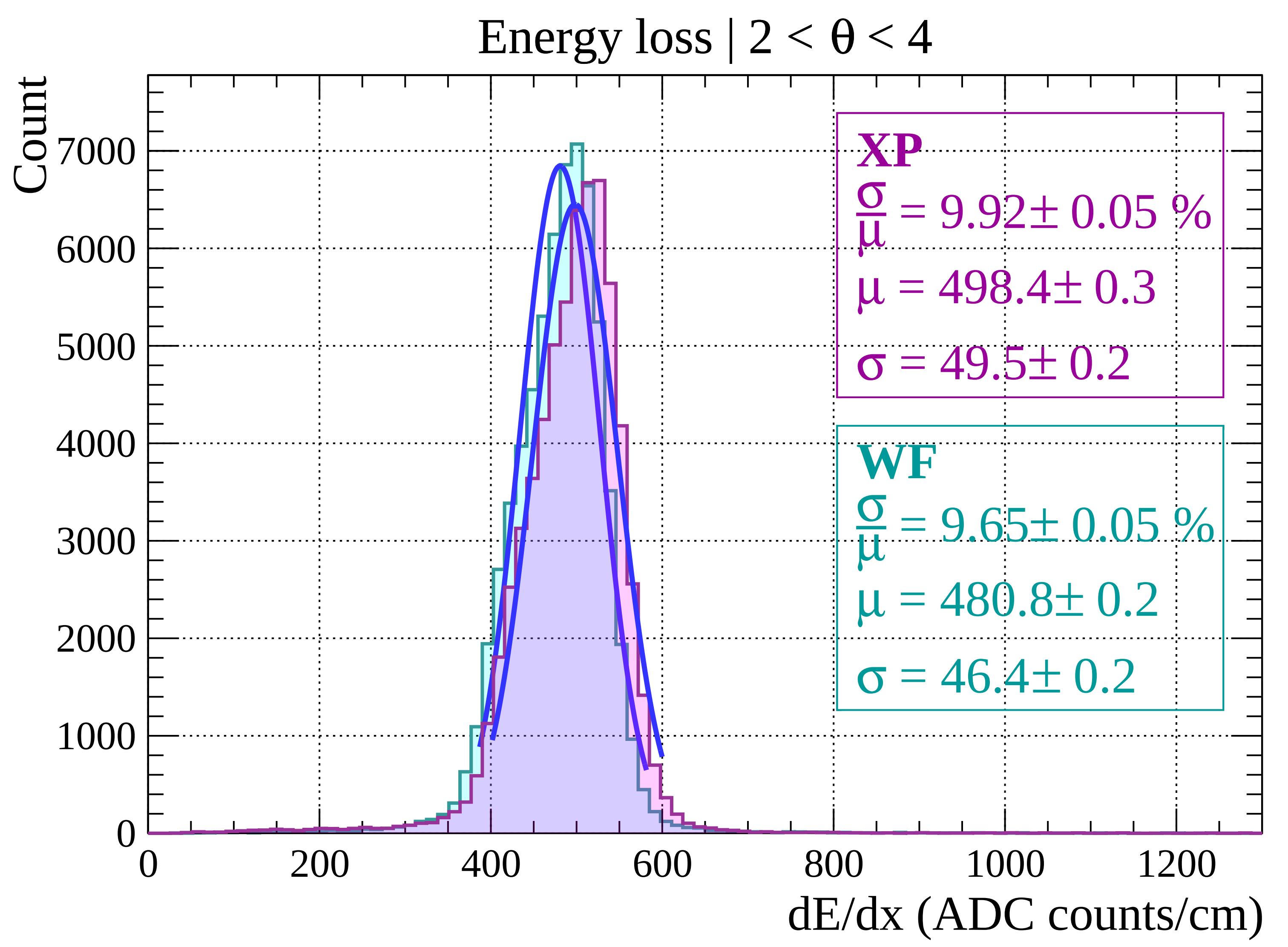
Energy loss | -4 < θ < -2
Count
dE/dx (ADC counts/cm)
0
1000
2000
3000
4000
5000
6000
7000
0
200
400
600
800
1000
1200
XP
σ/μ = 9.92 ± 0.05 %
μ = 498.8 ± 0.3
σ = 49.5 ± 0.2
WF
σ/μ = 9.69 ± 0.05 %
μ = 480.8 ± 0.3
σ = 46.6 ± 0.2



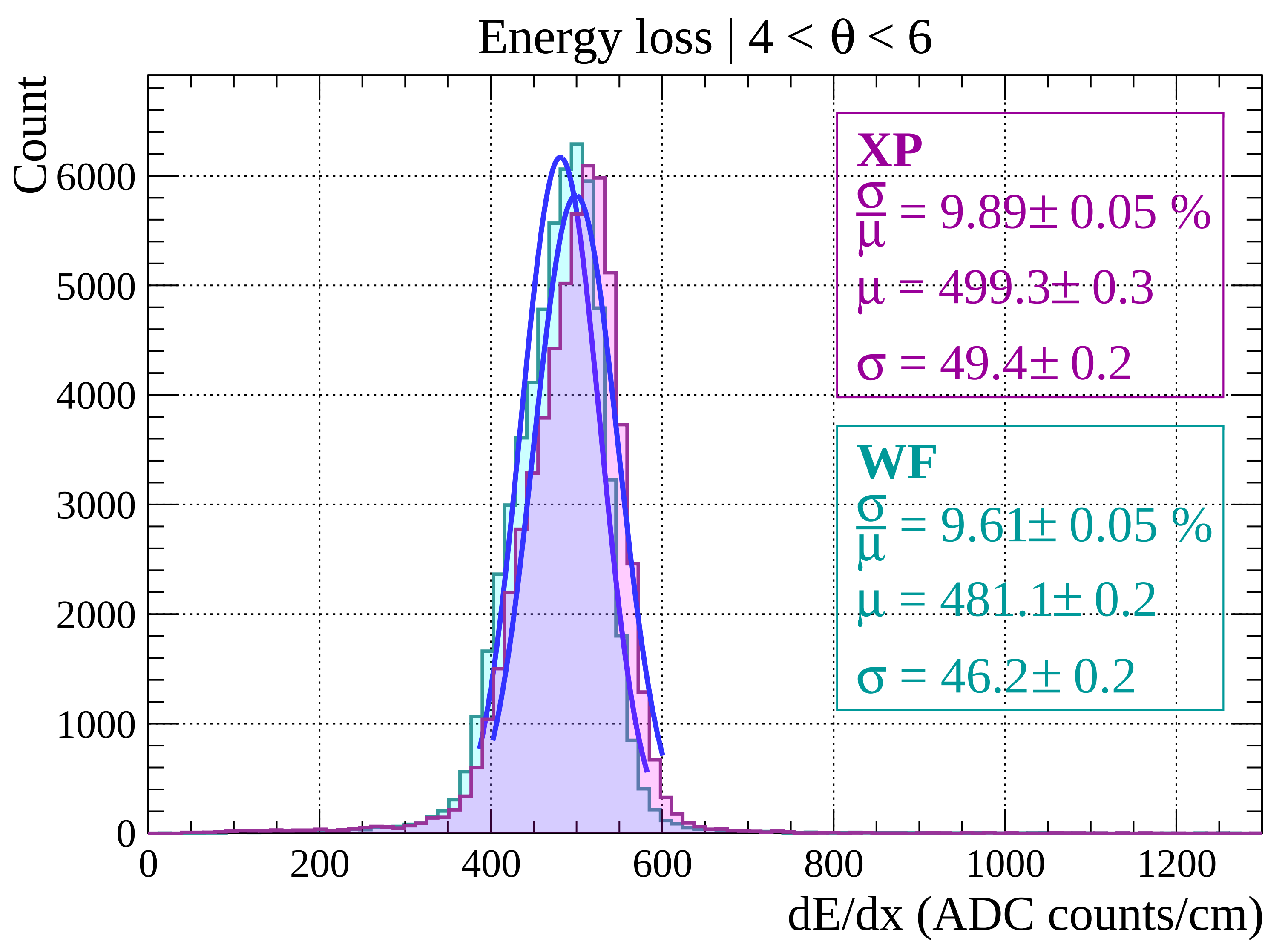
Energy loss | -2 < θ < 0
Count
7000
6000
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.00± 0.05 %
μ = 498.2± 0.3
σ = 49.8± 0.2
WF
σ/μ = 9.72± 0.05 %
μ = 480.4± 0.2
σ = 46.7± 0.2



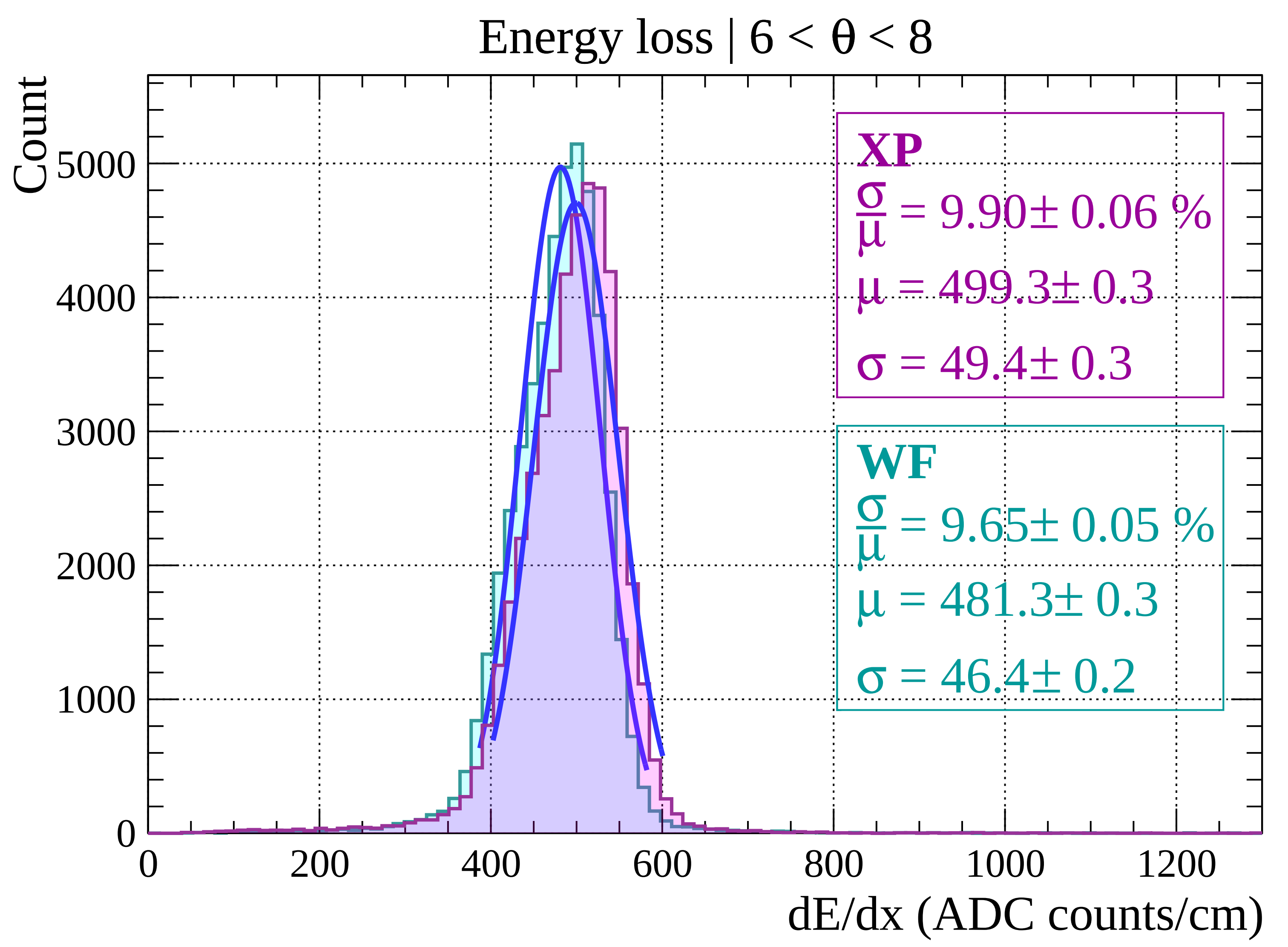
Energy loss | 0 < θ < 2
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.95± 0.05 %
μ = 498.7± 0.2
σ = 49.6± 0.2
WF
σ/μ = 9.97± 0.05 %
μ = 481.8± 0.2
σ = 48.0± 0.2
0
1000
2000
3000
4000
5000
6000
7000
8000
0
200
400
600
800
1000
1200

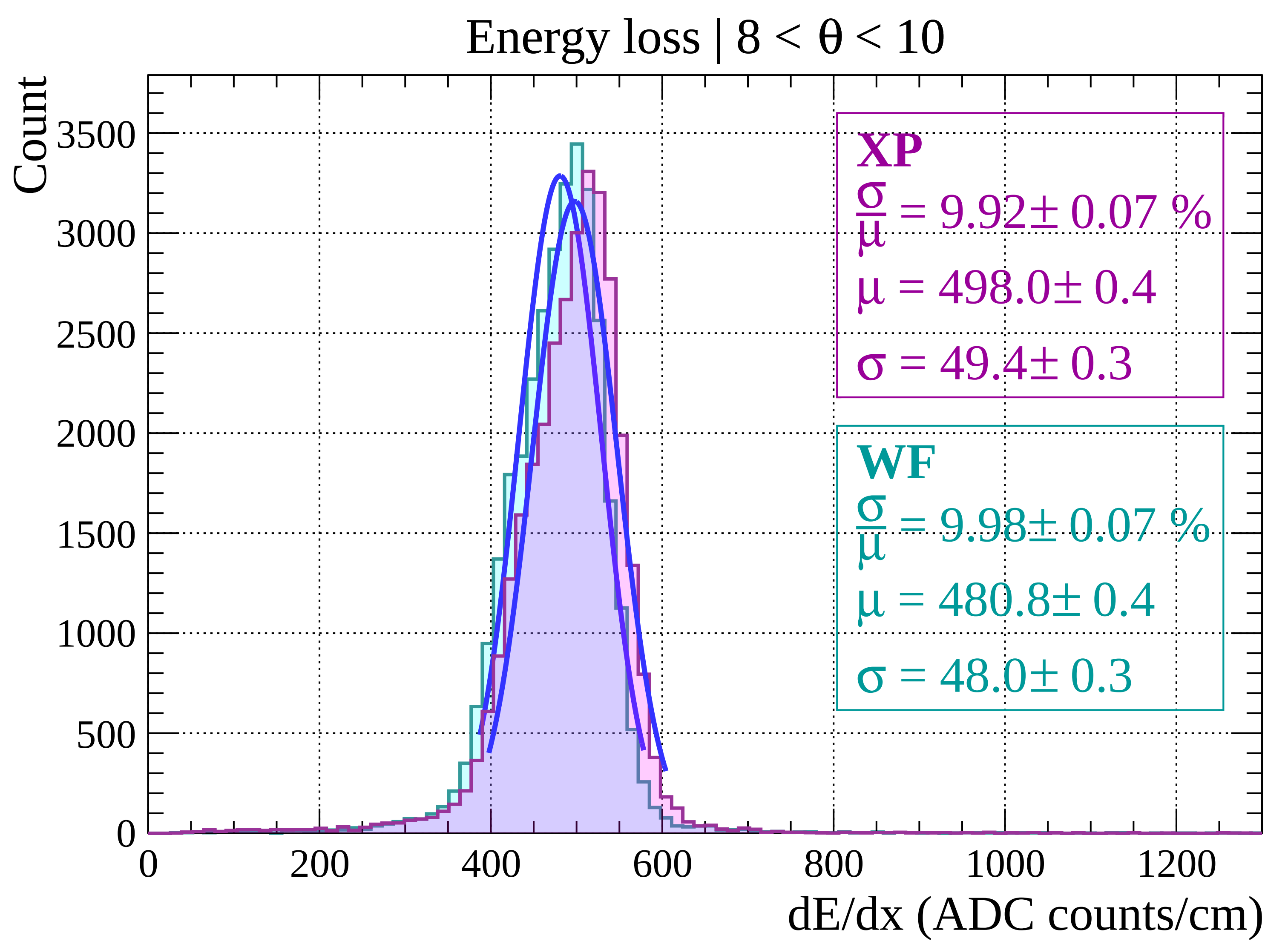


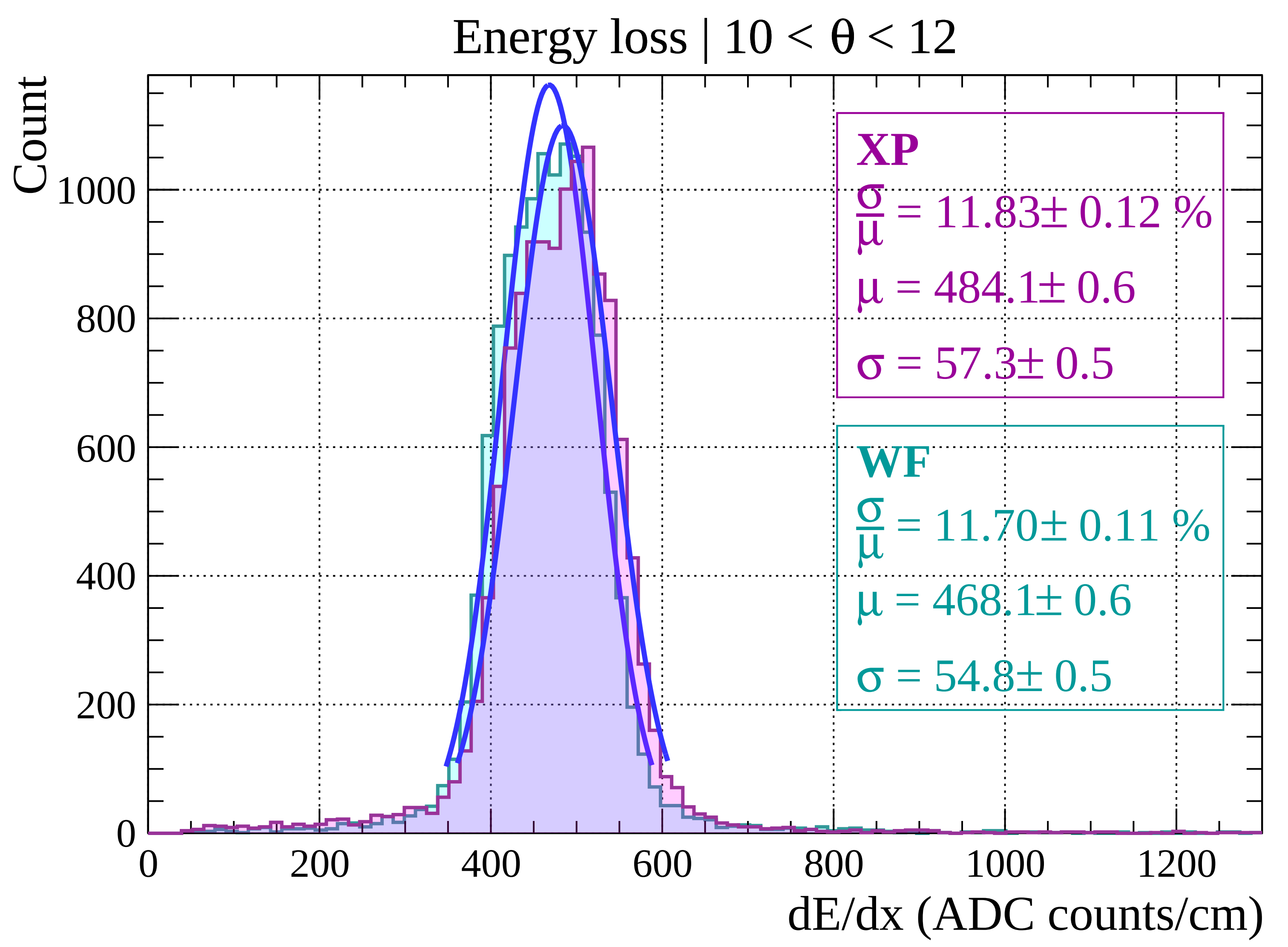
Energy loss | 2 < θ < 4
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1000
2000
3000
4000
5000
6000
7000
XP
σ/μ = 9.92± 0.05 %
μ = 498.4± 0.3
σ = 49.5± 0.2
WF
σ/μ = 9.65± 0.05 %
μ = 480.8± 0.2
σ = 46.4± 0.2



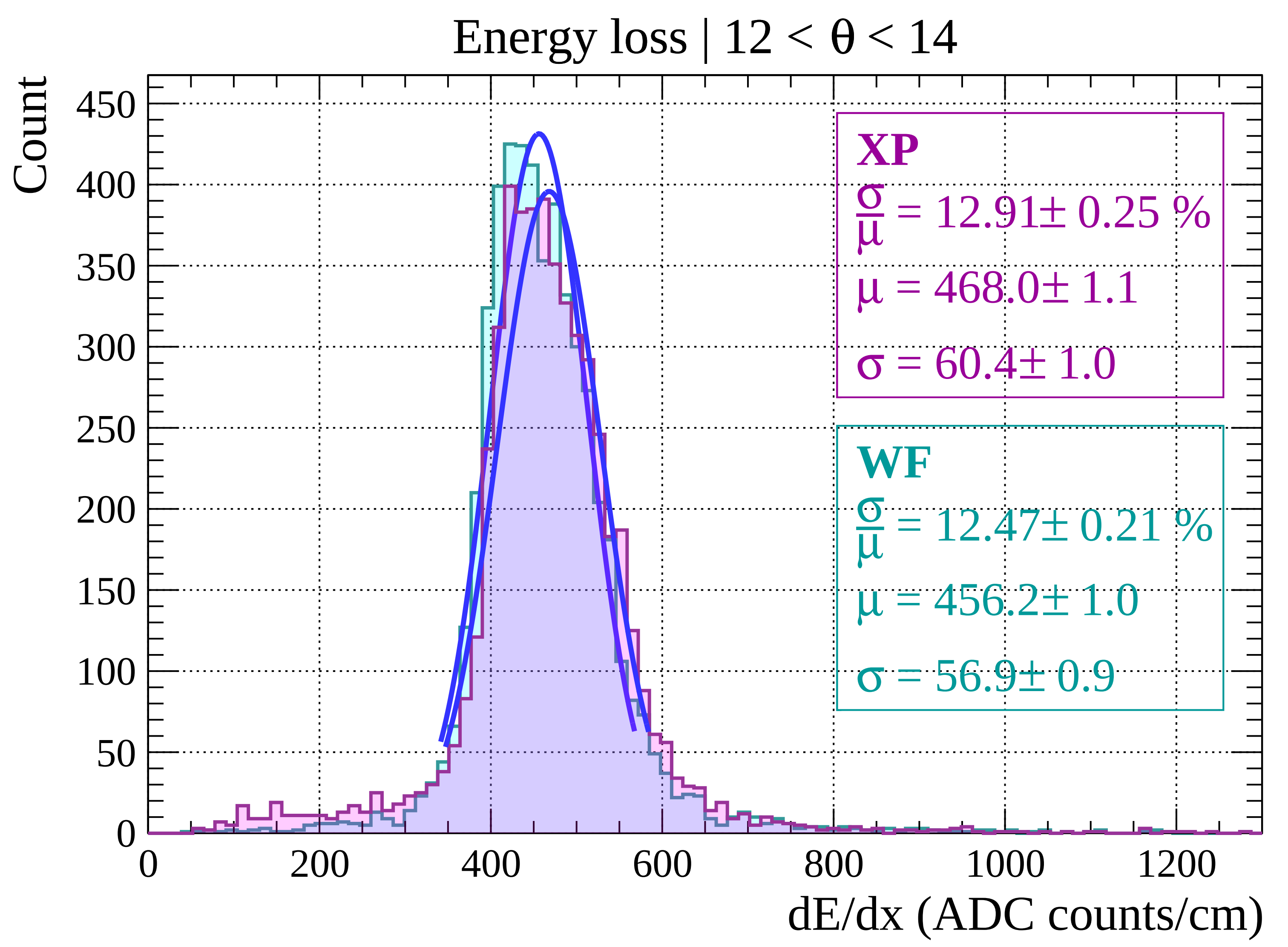
Energy loss | 4 < θ < 6
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.89± 0.05 %
μ = 499.3± 0.3
σ = 49.4± 0.2
WF
σ/μ = 9.61± 0.05 %
μ = 481.1± 0.2
σ = 46.2± 0.2
0
1000
2000
3000
4000
5000
6000
0
200
400
600
800
1000
1200


Energy loss | 6 < θ < 8
Count
dE/dx (ADC counts/cm)
0
1000
2000
3000
4000
5000
0
200
400
600
800
1000
1200
XP
σ/μ = 9.90± 0.06 %
μ = 499.3± 0.3
σ = 49.4± 0.3
WF
σ/μ = 9.65± 0.05 %
μ = 481.3± 0.3
σ = 46.4± 0.2

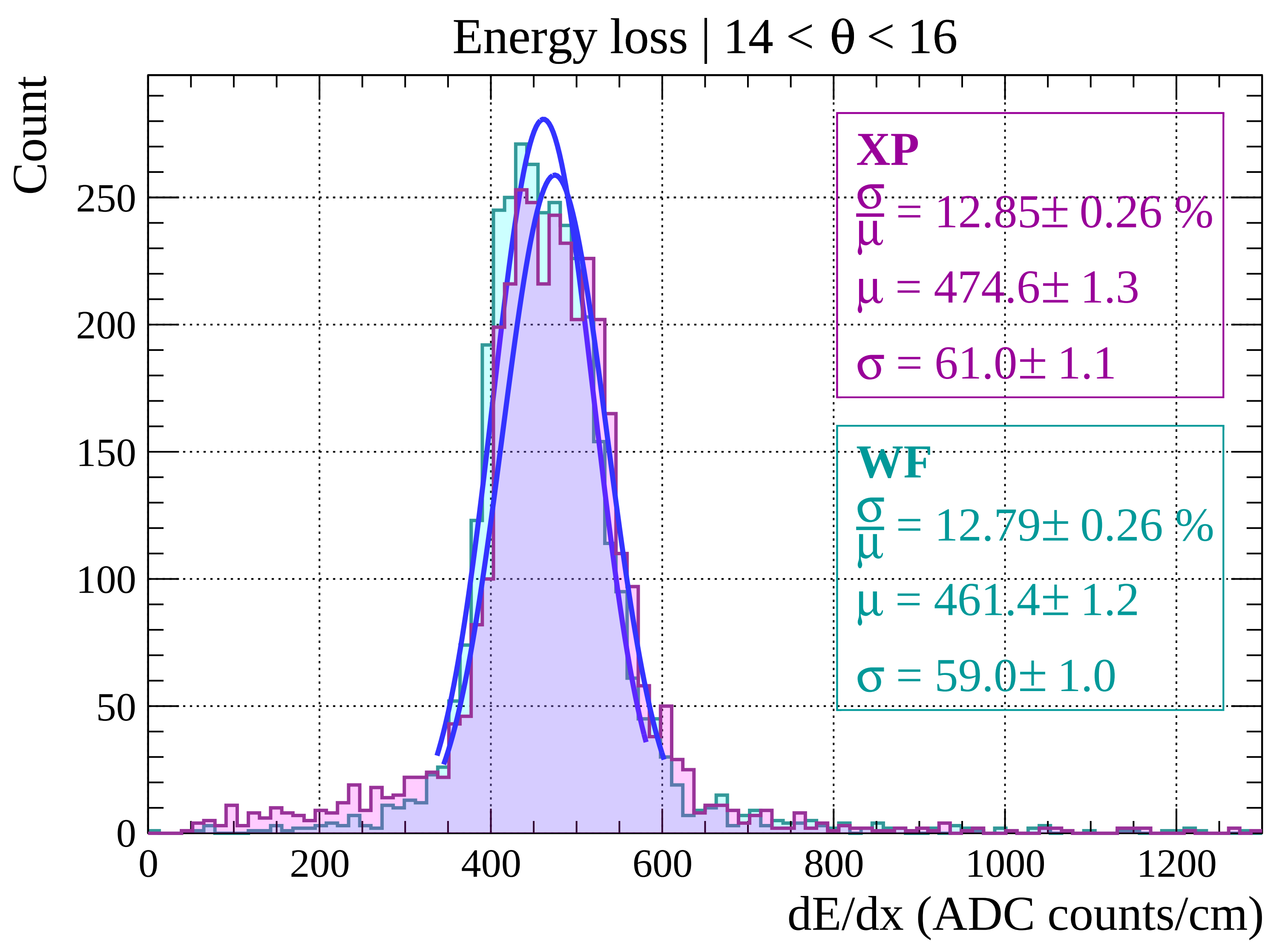

Energy loss | 8 < θ < 10
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
500
1000
1500
2000
2500
3000
3500
XP
σ/μ = 9.92± 0.07 %
μ = 498.0± 0.4
σ = 49.4± 0.3
WF
σ/μ = 9.98± 0.07 %
μ = 480.8± 0.4
σ = 48.0± 0.3



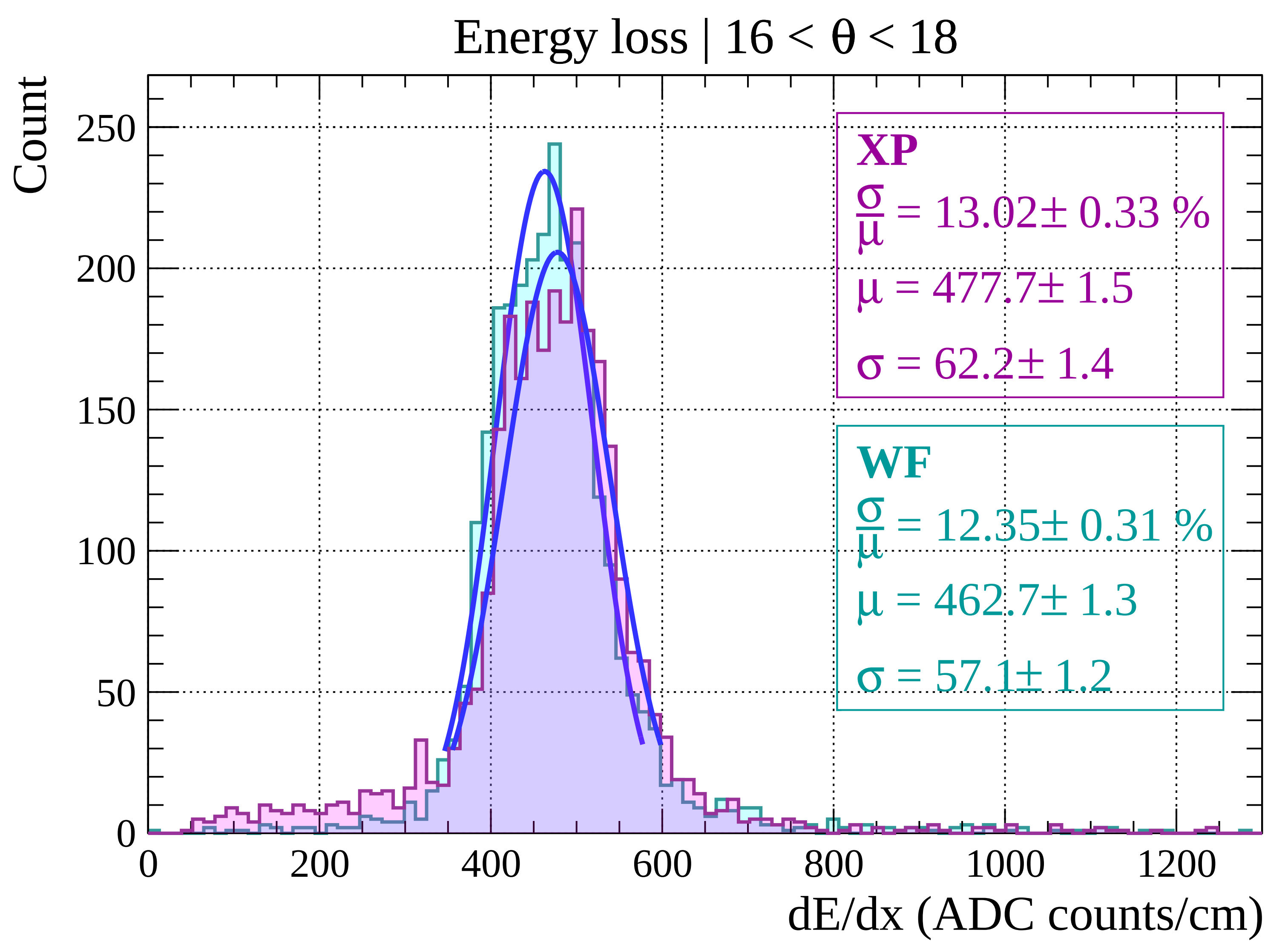
Energy loss | 10 < θ < 12
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 11.83± 0.12 %
μ = 484.1± 0.6
σ = 57.3± 0.5
WF
σ/μ = 11.70± 0.11 %
μ = 468.1± 0.6
σ = 54.8± 0.5



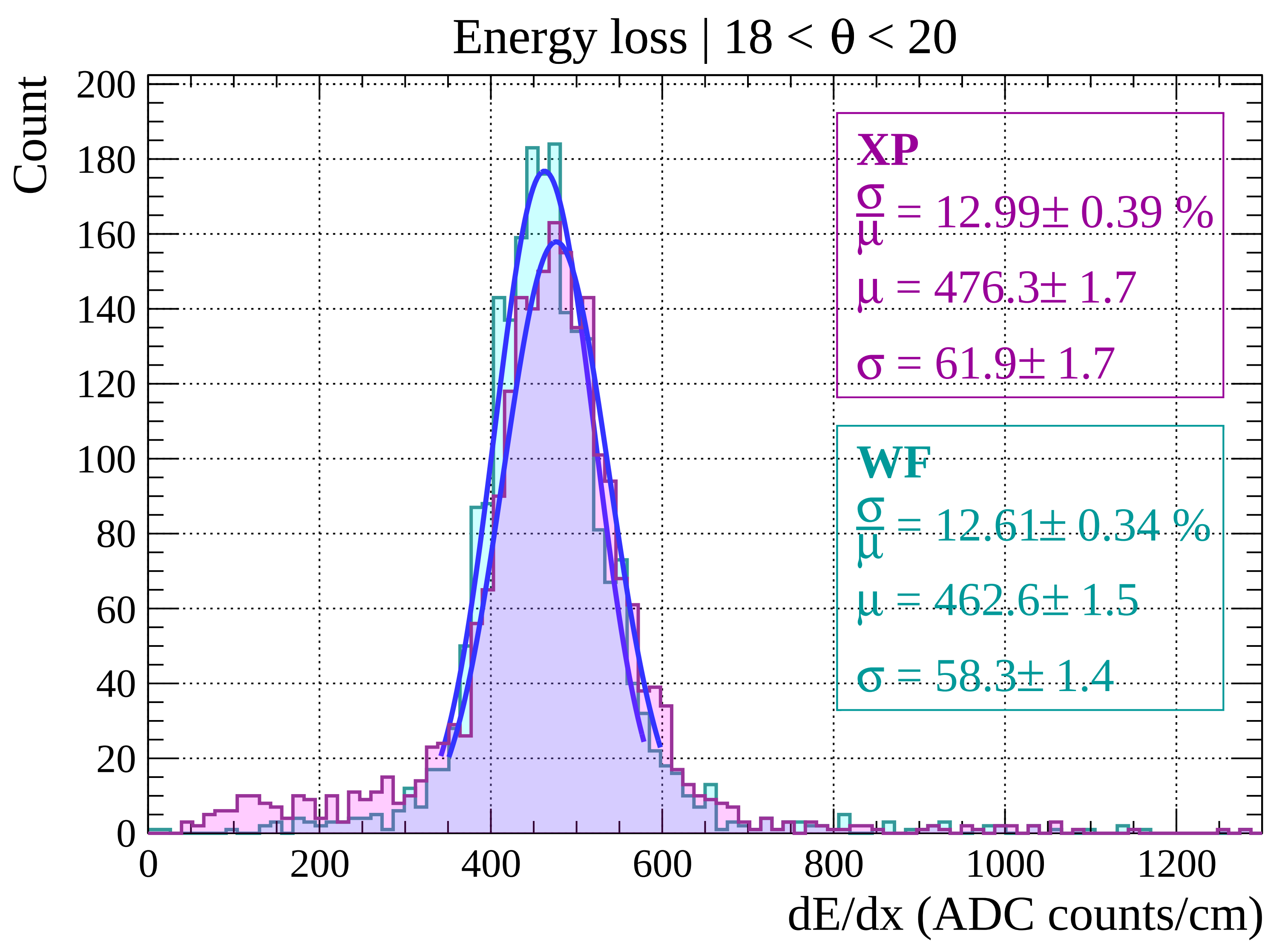
Energy loss | 12 < θ < 14
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
σ/μ = 12.91± 0.25 %
μ = 468.0± 1.1
σ = 60.4± 1.0
WF
σ/μ = 12.47± 0.21 %
μ = 456.2± 1.0
σ = 56.9± 0.9



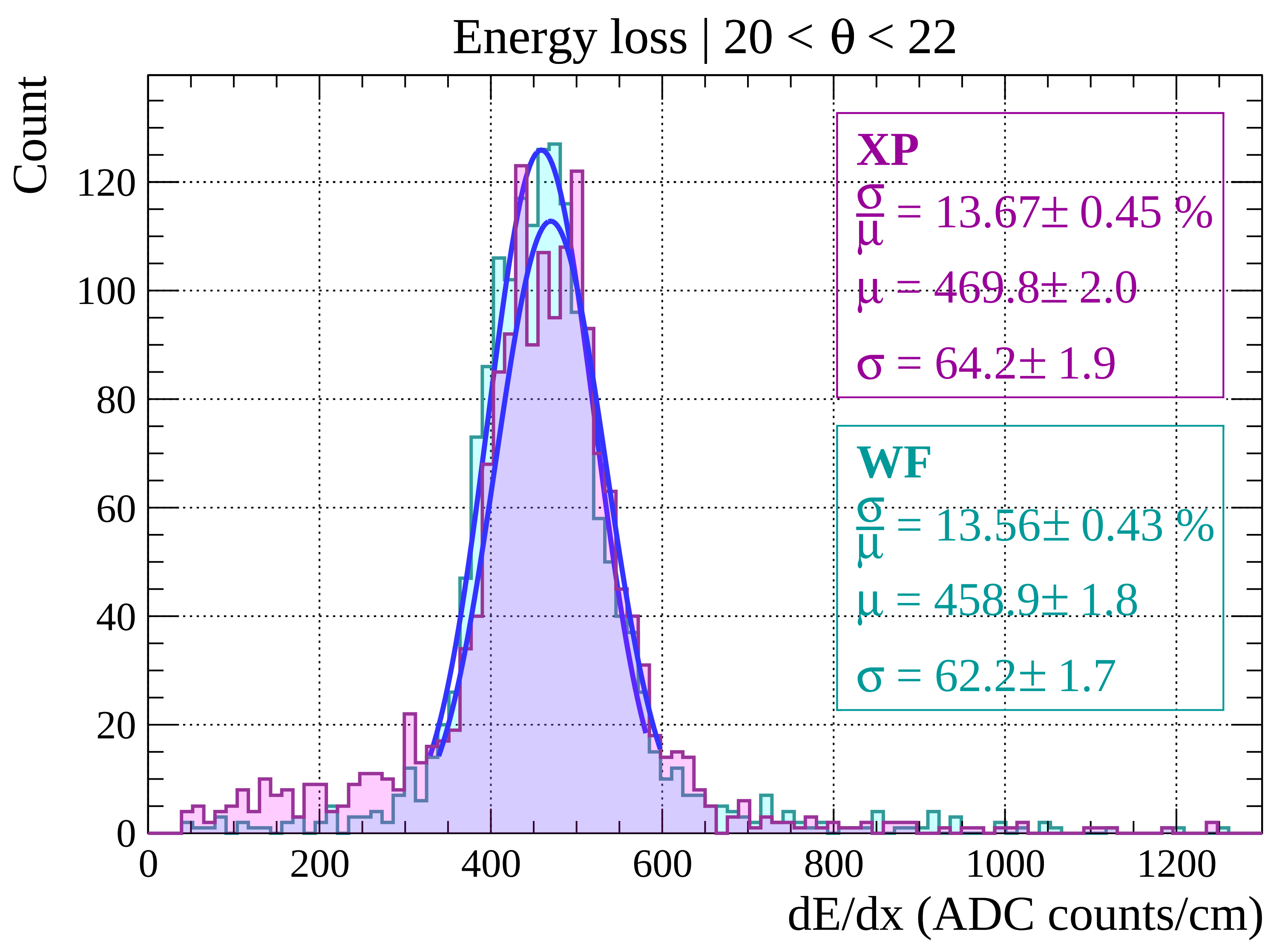
Energy loss | 14 < θ < 16
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 12.85± 0.26 %
μ = 474.6± 1.3
σ = 61.0± 1.1
WF
σ/μ = 12.79± 0.26 %
μ = 461.4± 1.2
σ = 59.0± 1.0



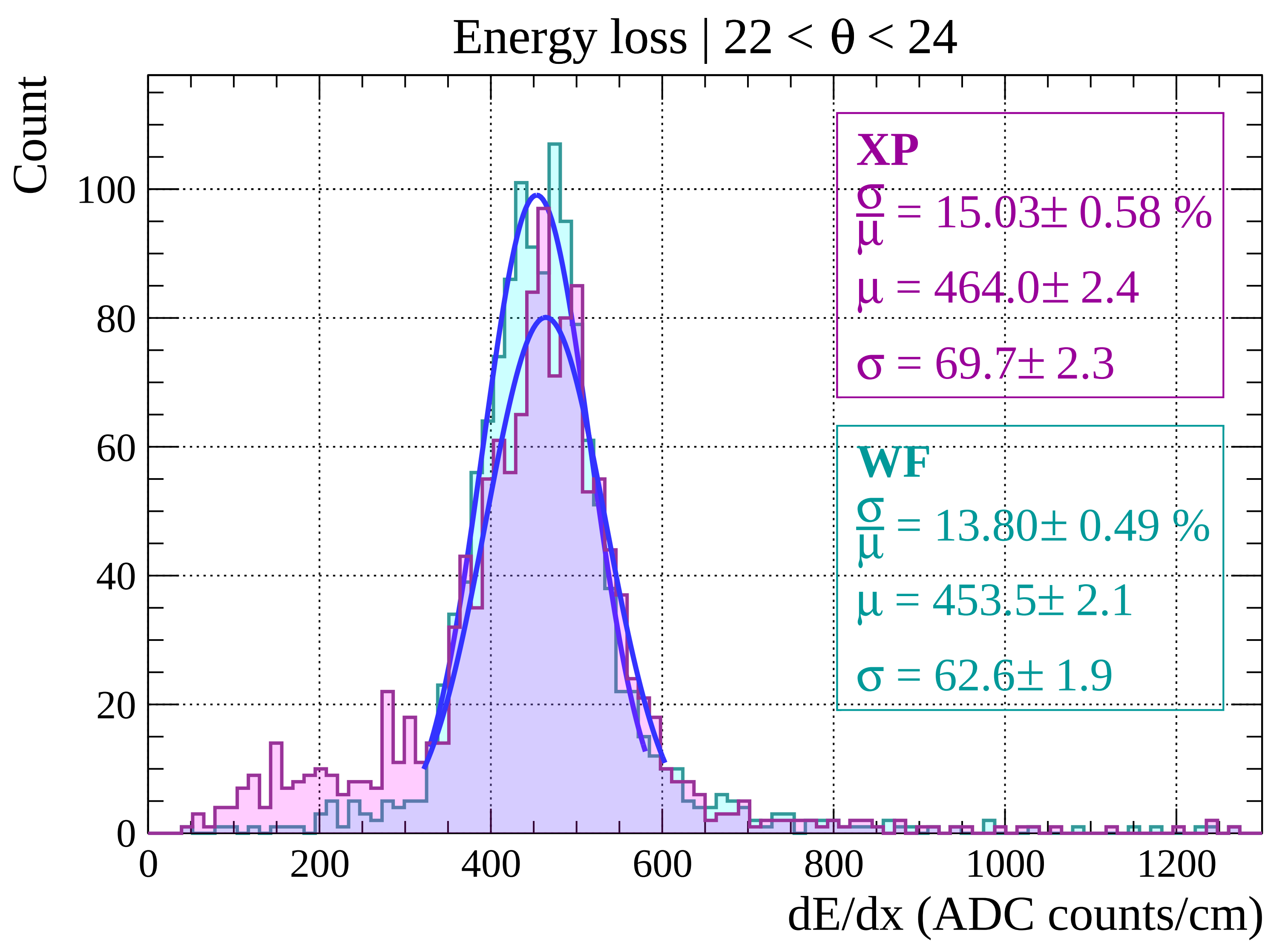
Energy loss | 16 < θ < 18
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 13.02± 0.33 %
μ = 477.7± 1.5
σ = 62.2± 1.4
WF
σ/μ = 12.35± 0.31 %
μ = 462.7± 1.3
σ = 57.1± 1.2



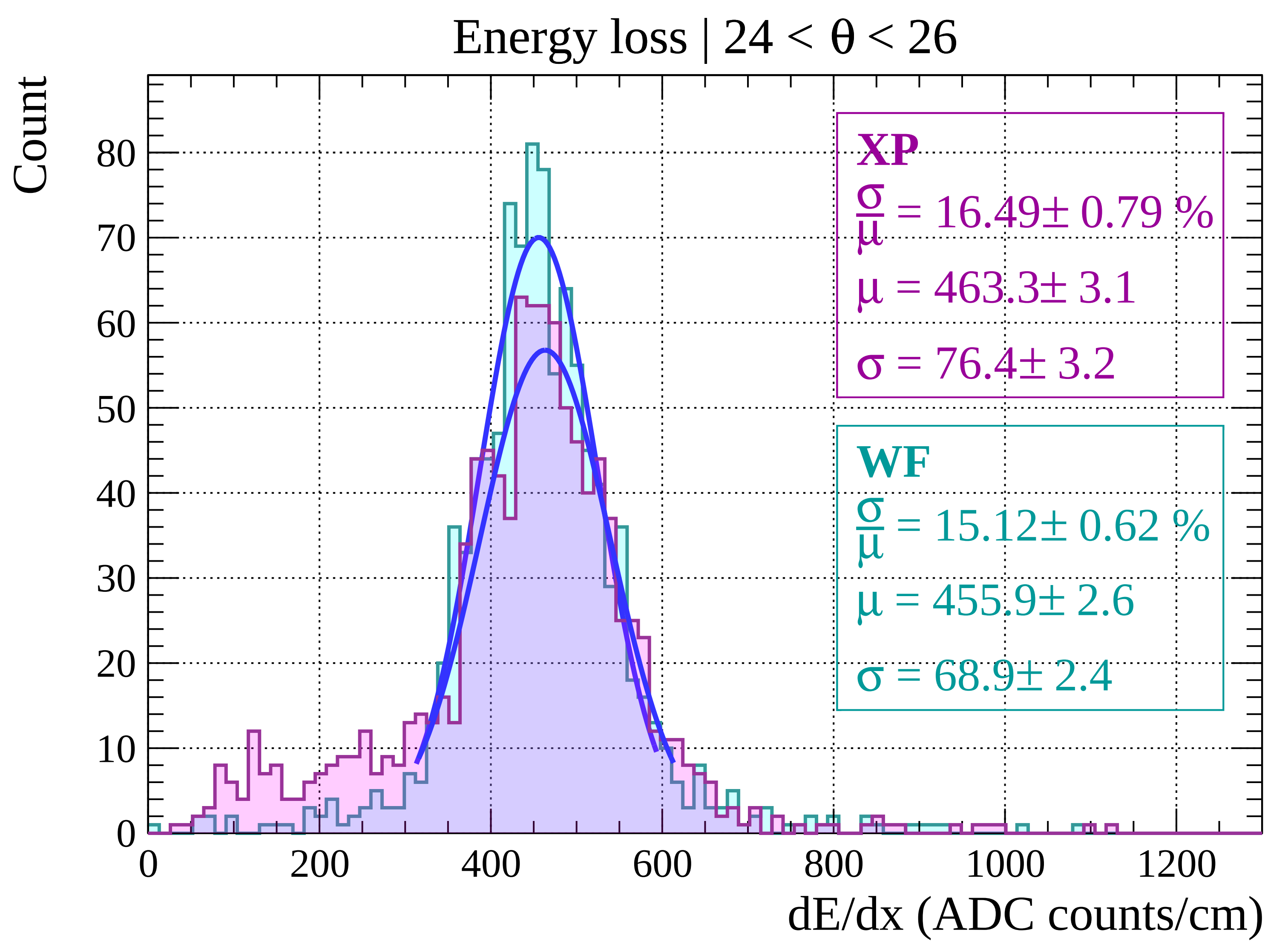
Energy loss | 18 < θ < 20
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP
σ/μ = 12.99± 0.39 %
μ = 476.3± 1.7
σ = 61.9± 1.7
WF
σ/μ = 12.61± 0.34 %
μ = 462.6± 1.5
σ = 58.3± 1.4



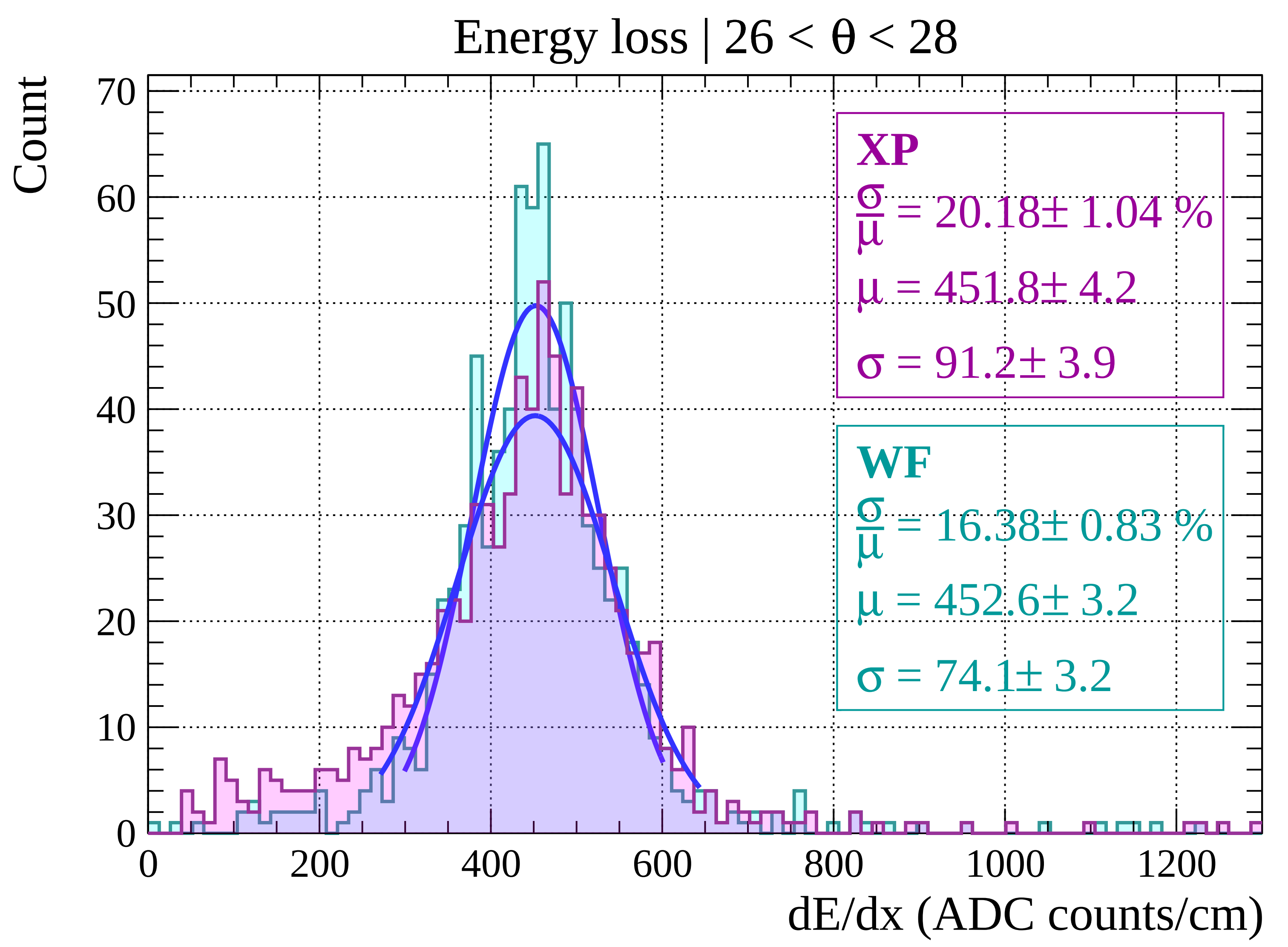
Energy loss | 20 < θ < 22
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
XP
σ/μ = 13.67± 0.45 %
μ = 469.8± 2.0
σ = 64.2± 1.9
WF
σ/μ = 13.56± 0.43 %
μ = 458.9± 1.8
σ = 62.2± 1.7



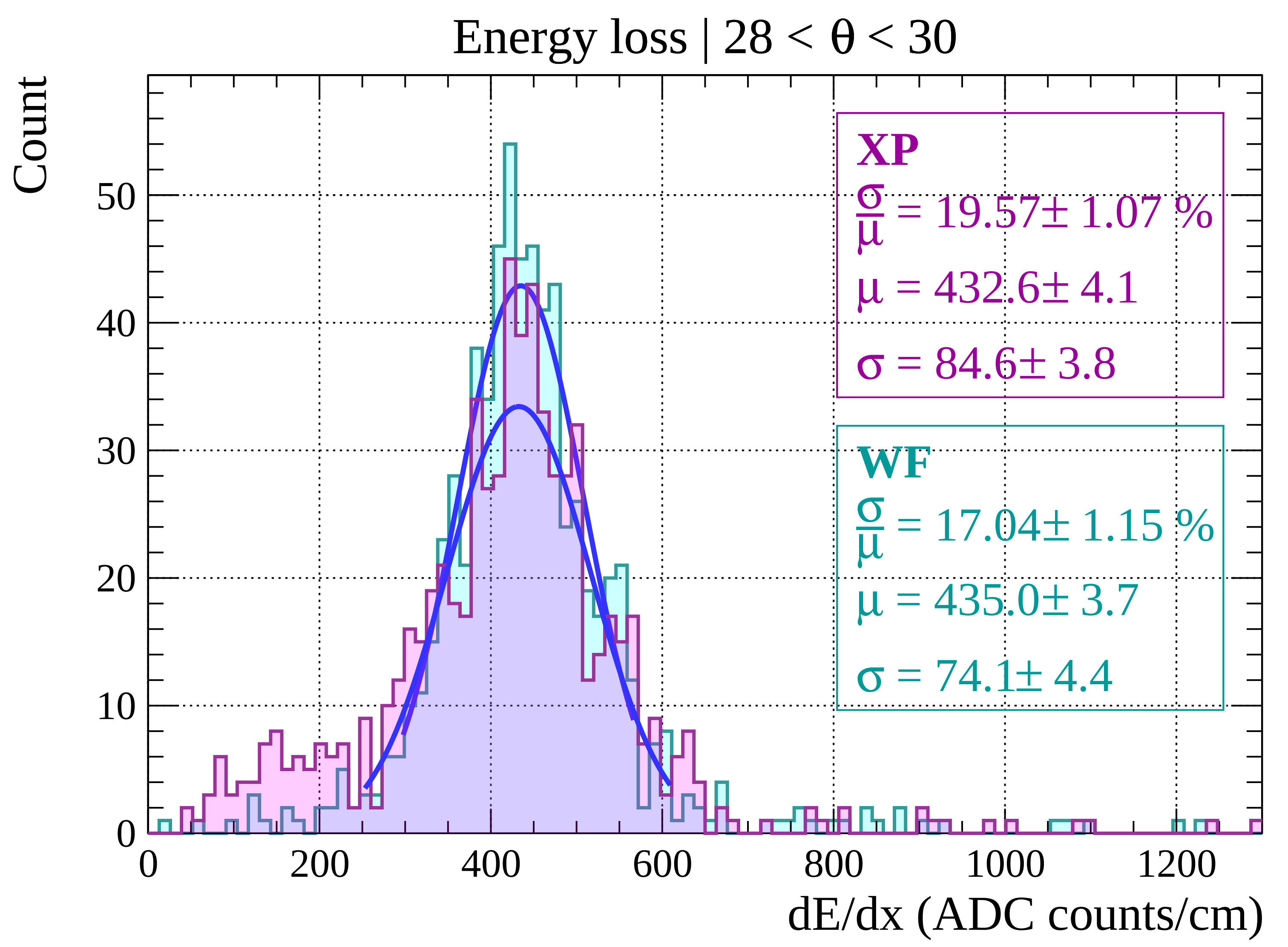
Energy loss | 22 < θ < 24
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.03± 0.58 %
μ = 464.0± 2.4
σ = 69.7± 2.3
WF
σ/μ = 13.80± 0.49 %
μ = 453.5± 2.1
σ = 62.6± 1.9



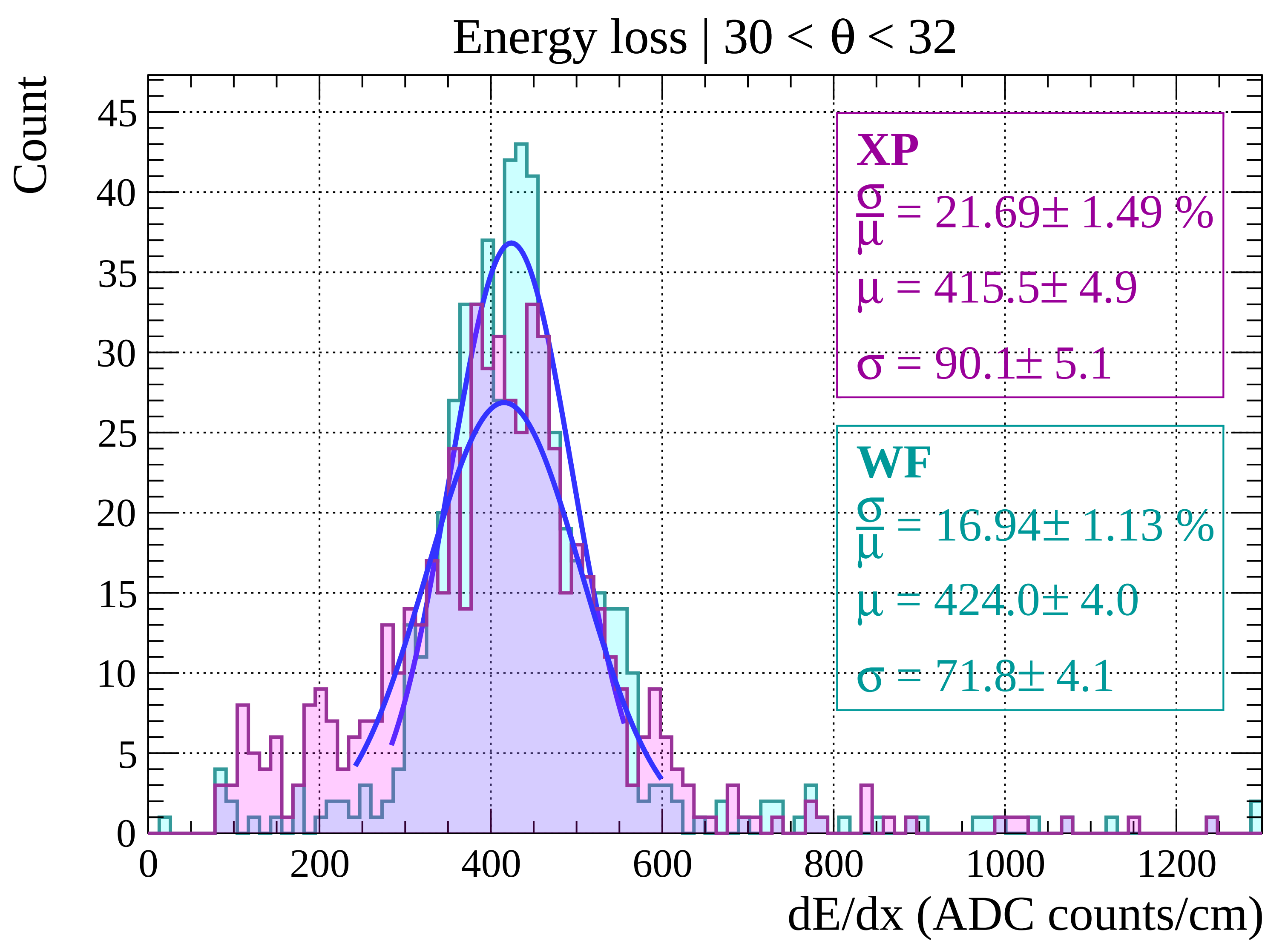
Energy loss | 24 < θ < 26
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
200
400
600
800
1000
1200
XP
σ/μ = 16.49± 0.79 %
μ = 463.3± 3.1
σ = 76.4± 3.2
WF
σ/μ = 15.12± 0.62 %
μ = 455.9± 2.6
σ = 68.9± 2.4



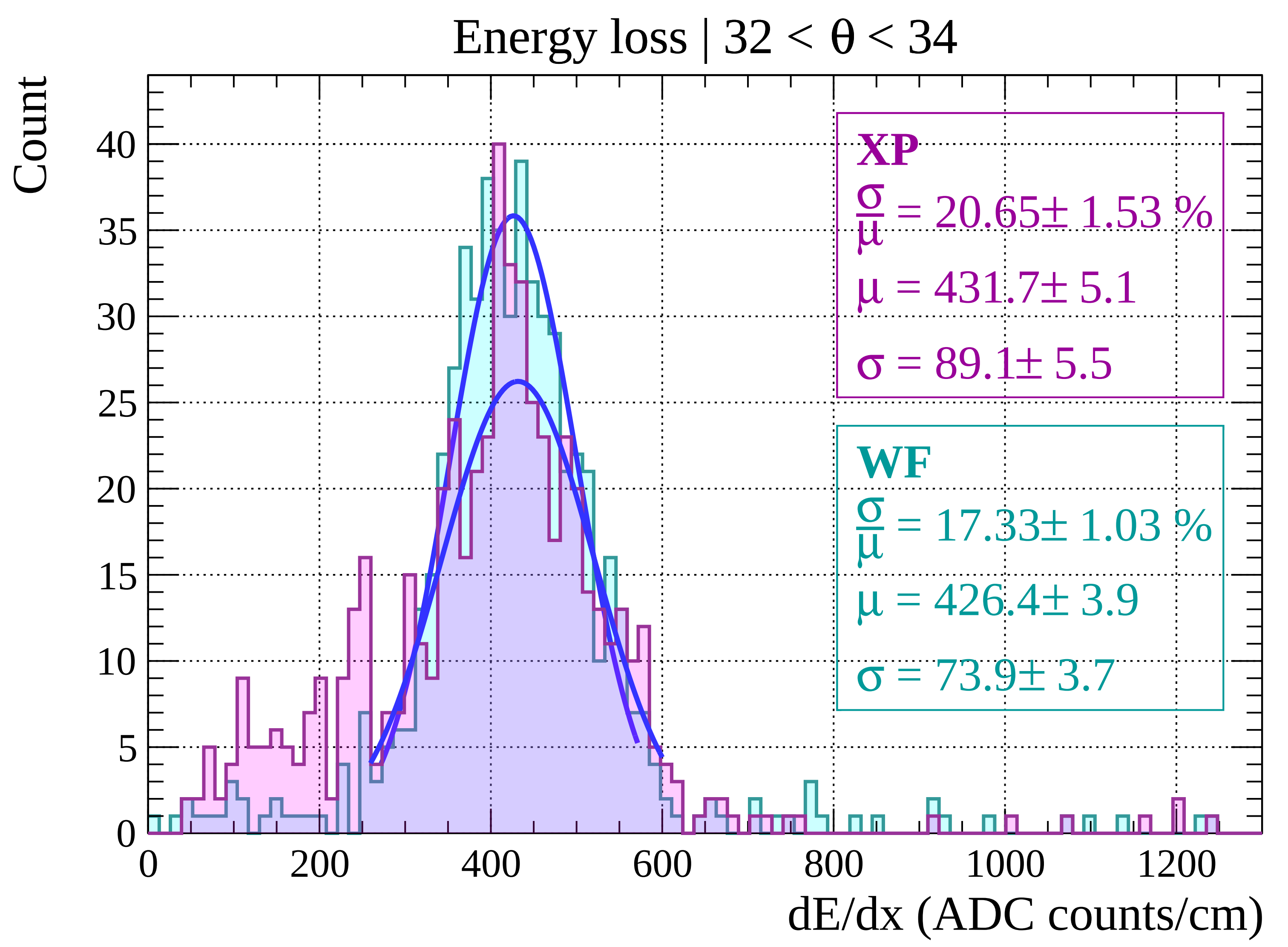
Energy loss | 26 < θ < 28
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP
σ/μ = 20.18± 1.04 %
μ = 451.8± 4.2
σ = 91.2± 3.9
WF
σ/μ = 16.38± 0.83 %
μ = 452.6± 3.2
σ = 74.1± 3.2



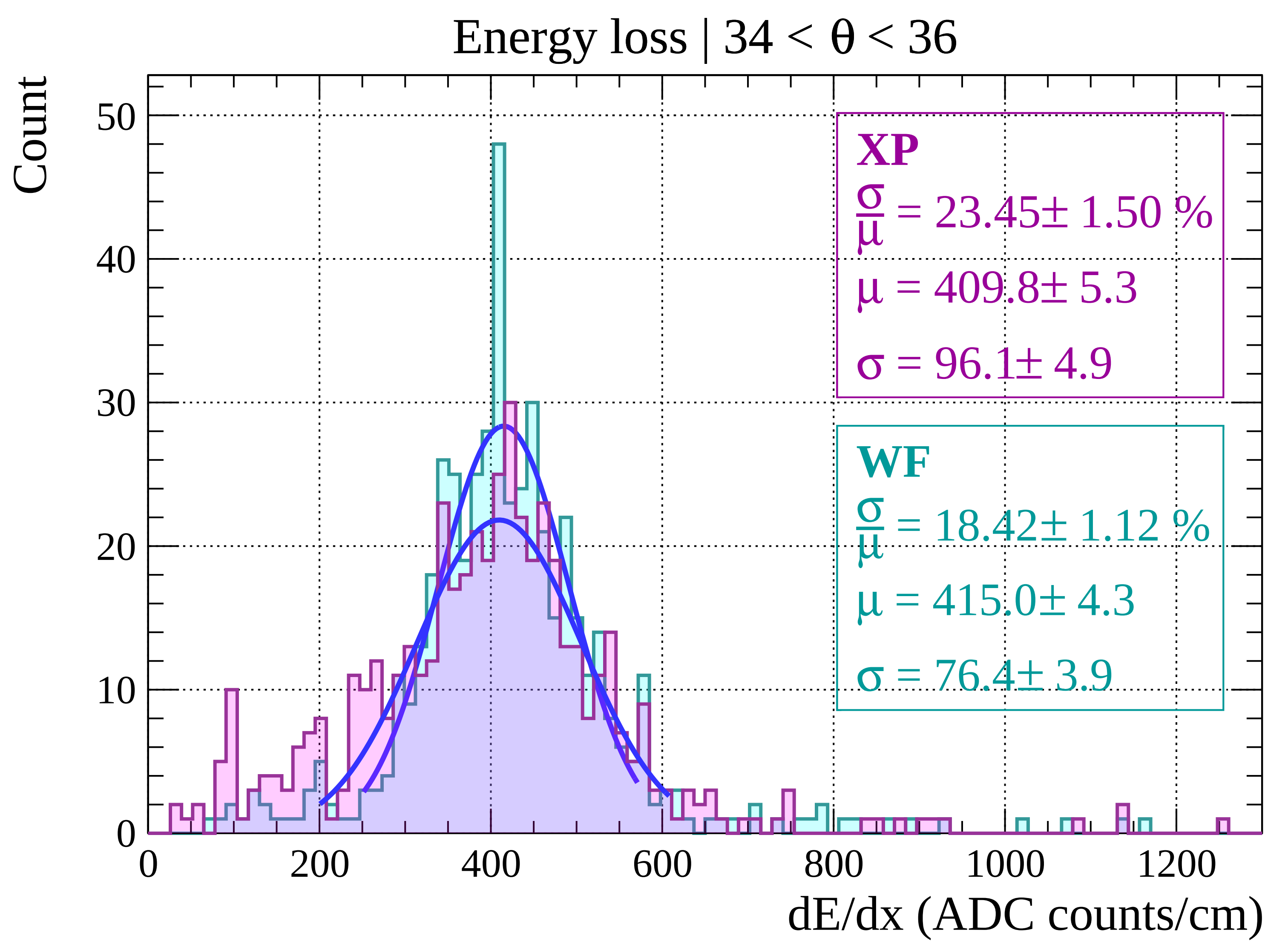
Energy loss | 28 < θ < 30
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 19.57± 1.07 %
μ = 432.6± 4.1
σ = 84.6± 3.8
WF
σ/μ = 17.04± 1.15 %
μ = 435.0± 3.7
σ = 74.1± 4.4



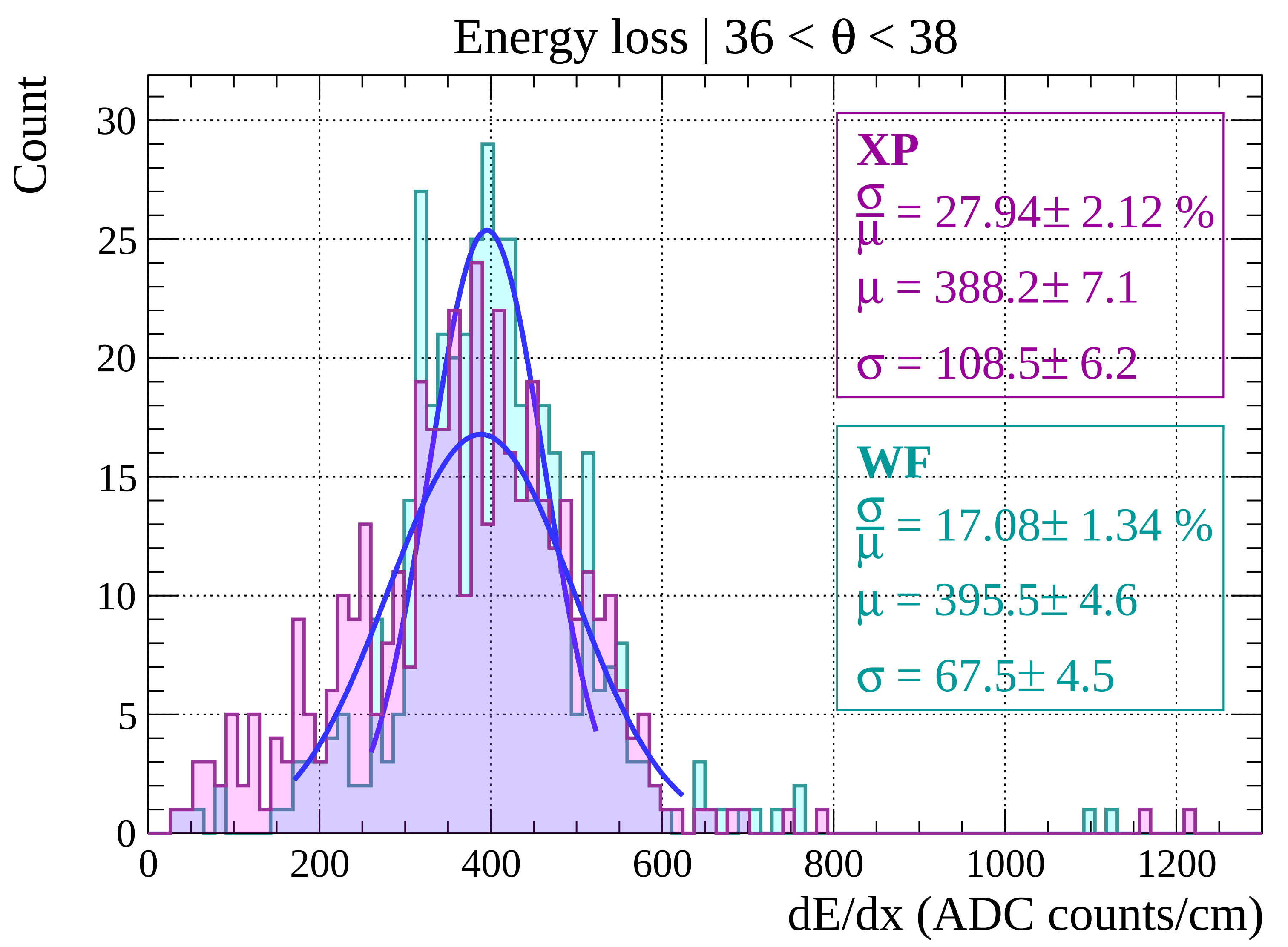
Energy loss | 30 < θ < 32
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
XP
σ/μ = 21.69± 1.49 %
μ = 415.5± 4.9
σ = 90.1± 5.1
WF
σ/μ = 16.94± 1.13 %
μ = 424.0± 4.0
σ = 71.8± 4.1



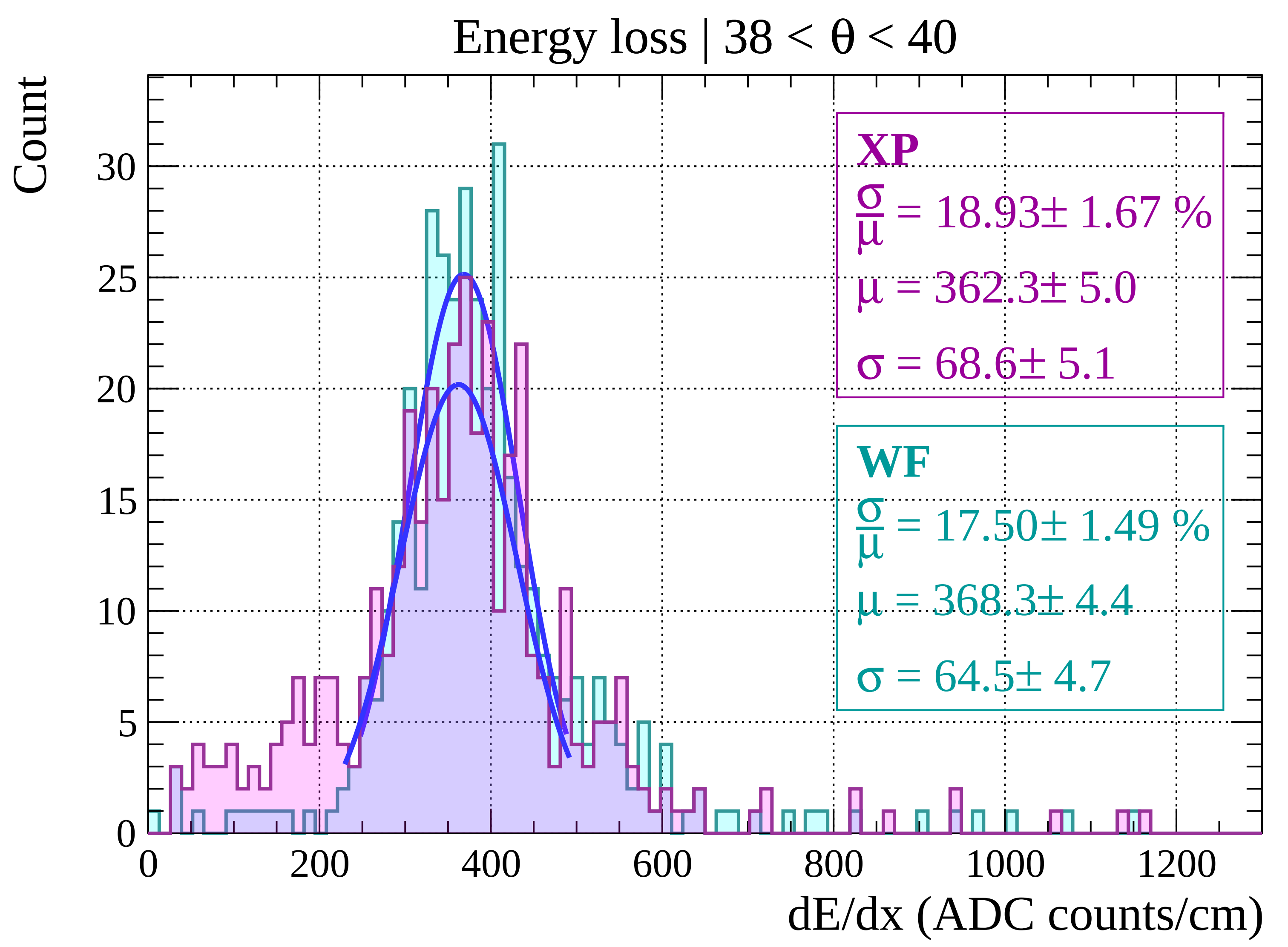
Energy loss | 32 < θ < 34
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
XP
σ/μ = 20.65± 1.53 %
μ = 431.7± 5.1
σ = 89.1± 5.5
WF
σ/μ = 17.33± 1.03 %
μ = 426.4± 3.9
σ = 73.9± 3.7



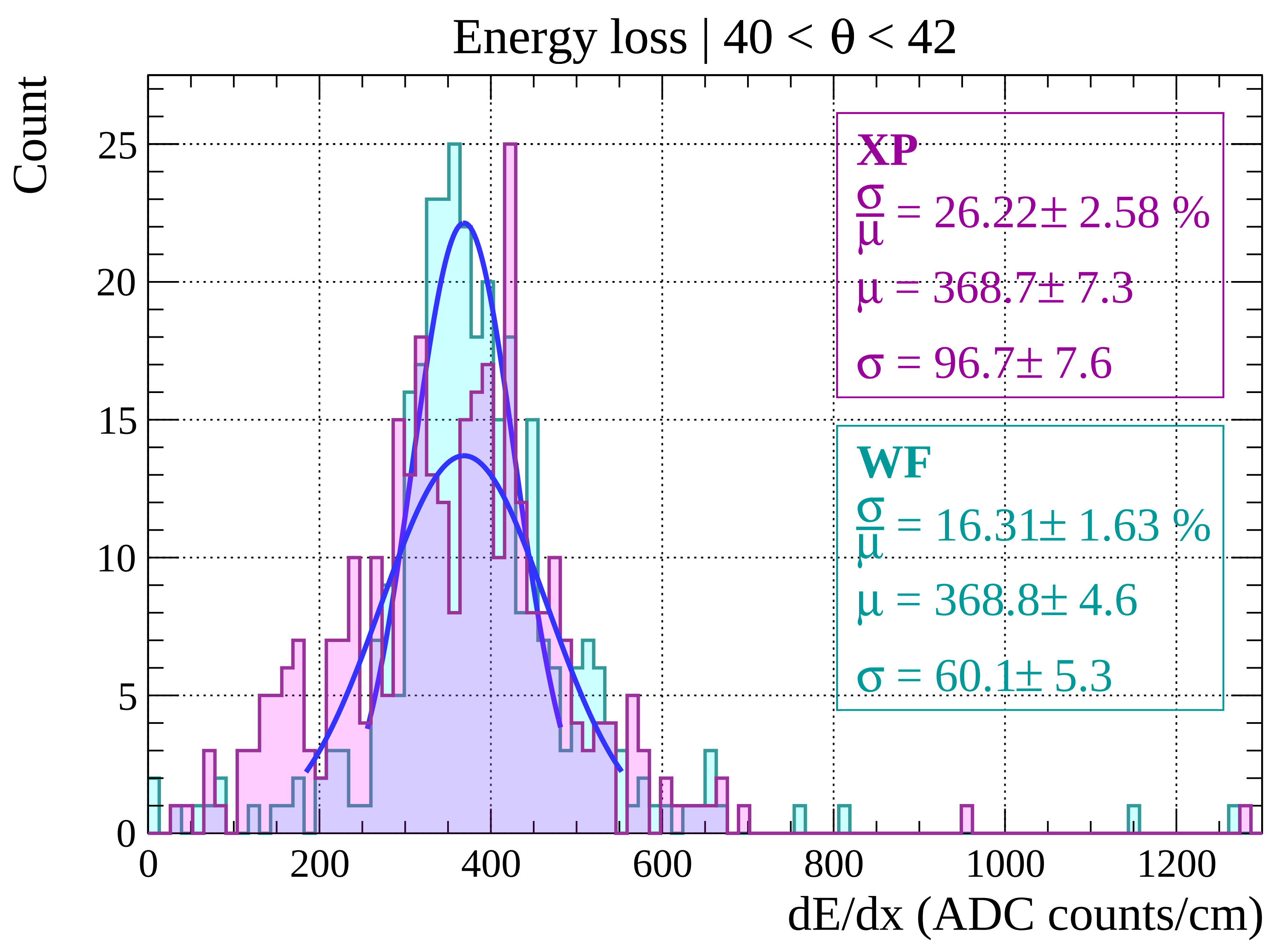
Energy loss | 34 < θ < 36
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 23.45± 1.50 %
μ = 409.8± 5.3
σ = 96.1± 4.9
WF
σ/μ = 18.42± 1.12 %
μ = 415.0± 4.3
σ = 76.4± 3.9



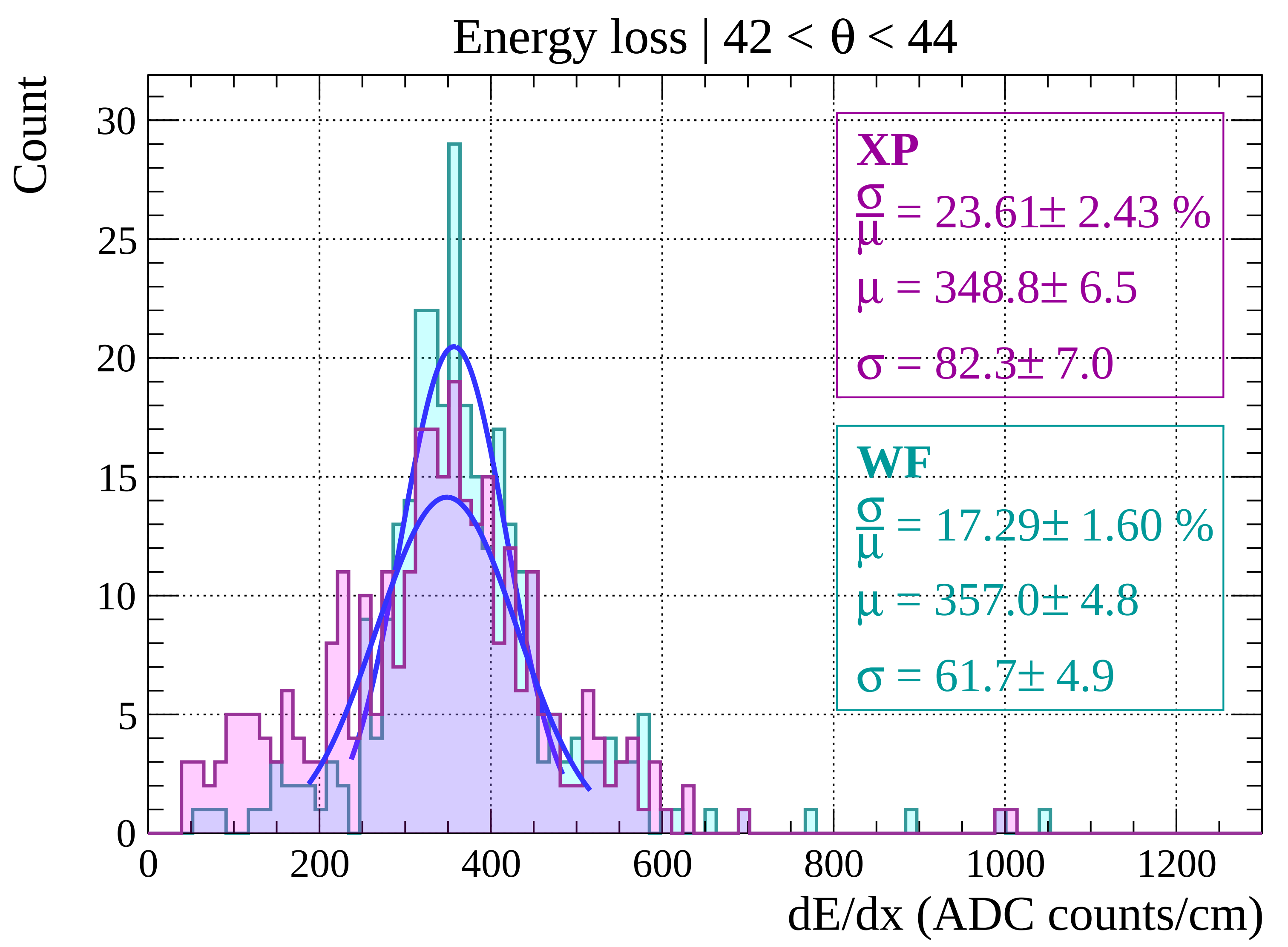
Energy loss | 36 < θ < 38
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
XP
σ/μ = 27.94± 2.12 %
μ = 388.2± 7.1
σ = 108.5± 6.2
WF
σ/μ = 17.08± 1.34 %
μ = 395.5± 4.6
σ = 67.5± 4.5



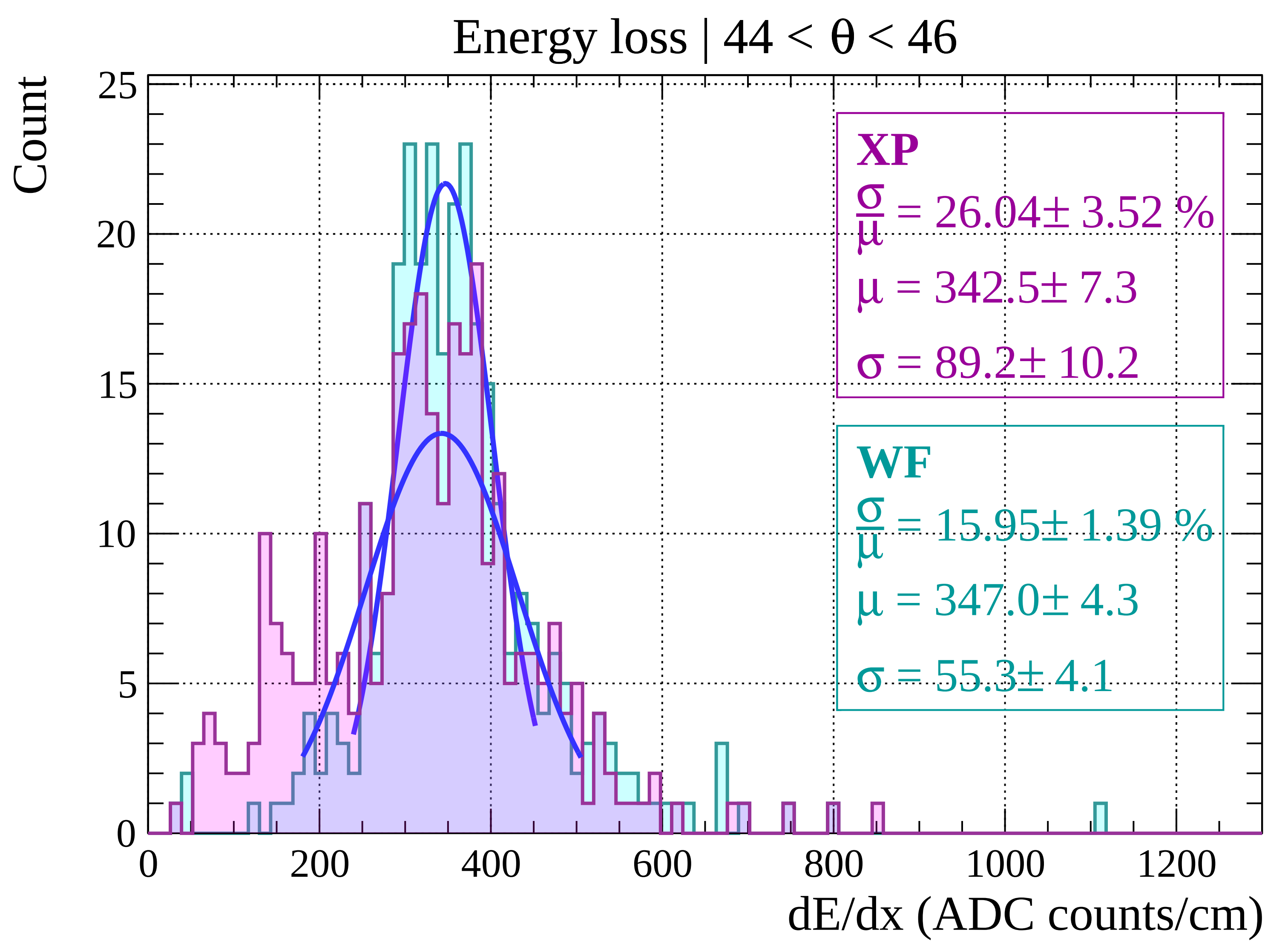
Energy loss | 38 < θ < 40
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 18.93± 1.67 %
μ = 362.3± 5.0
σ = 68.6± 5.1
WF
σ/μ = 17.50± 1.49 %
μ = 368.3± 4.4
σ = 64.5± 4.7

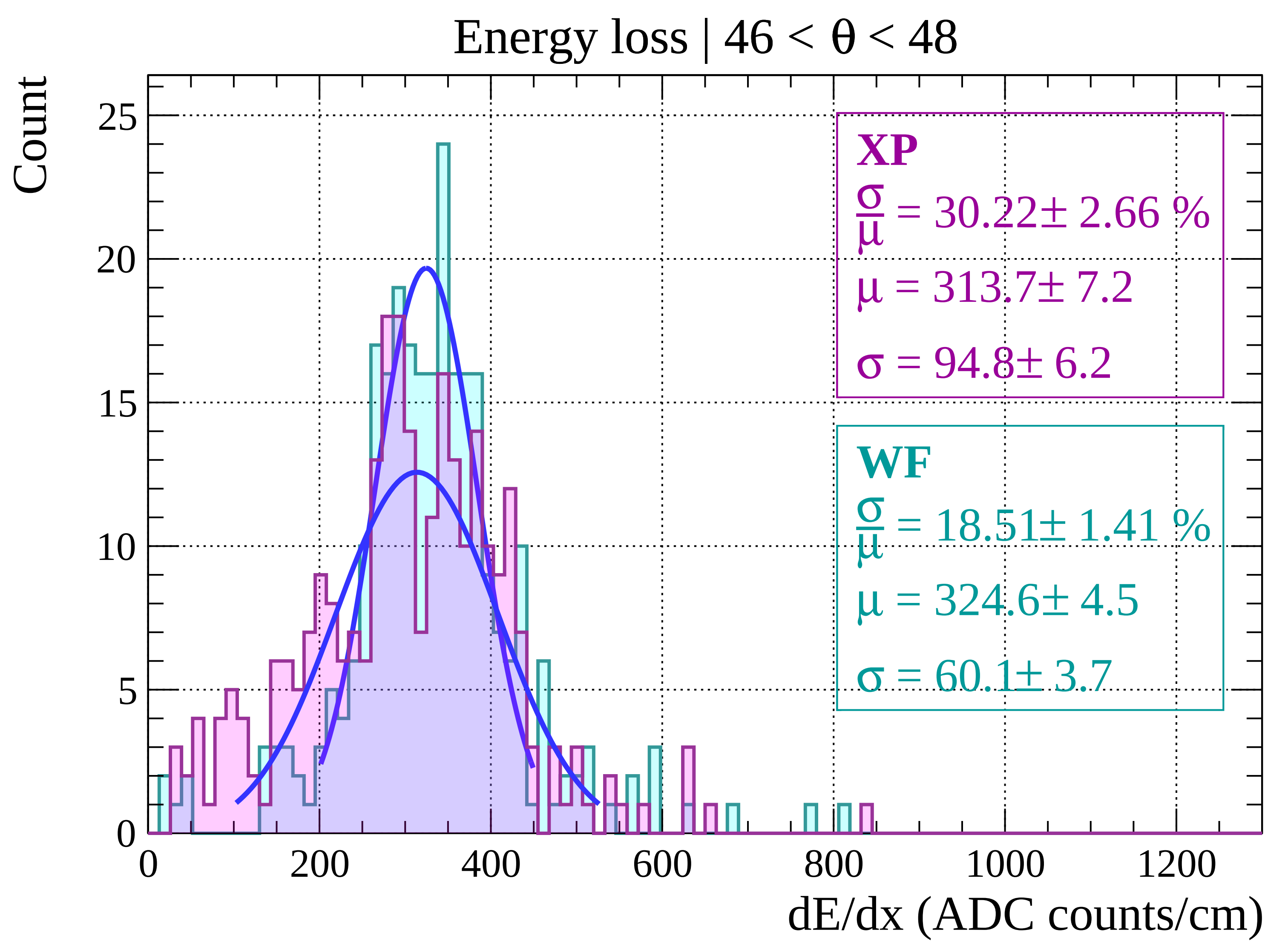


Energy loss | 40 < θ < 42
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 26.22± 2.58 %
μ = 368.7± 7.3
σ = 96.7± 7.6
WF
σ/μ = 16.31± 1.63 %
μ = 368.8± 4.6
σ = 60.1± 5.3

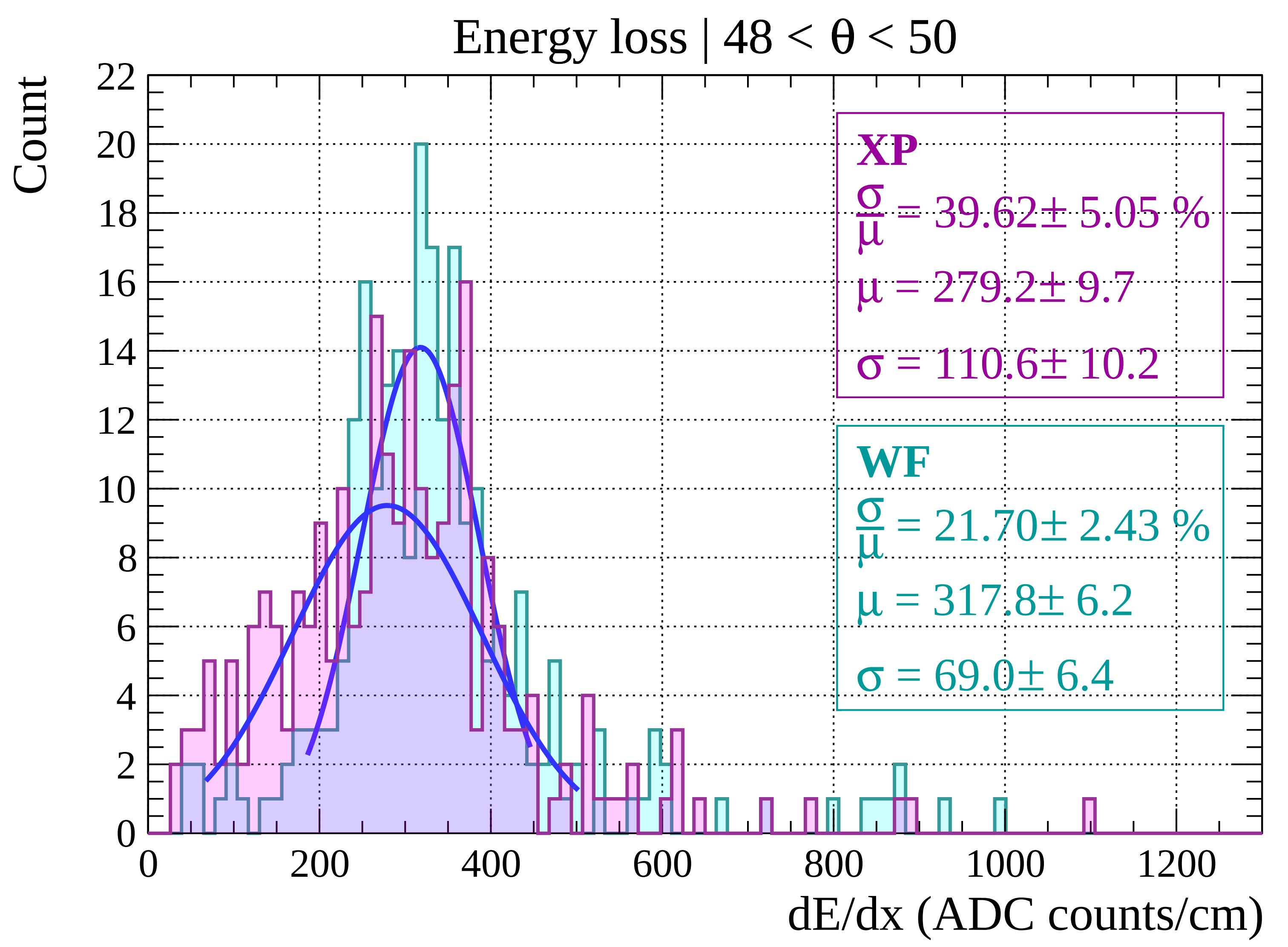


Energy loss | 42 < θ < 44
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
XP
σ/μ = 23.61± 2.43 %
μ = 348.8± 6.5
σ = 82.3± 7.0
WF
σ/μ = 17.29± 1.60 %
μ = 357.0± 4.8
σ = 61.7± 4.9

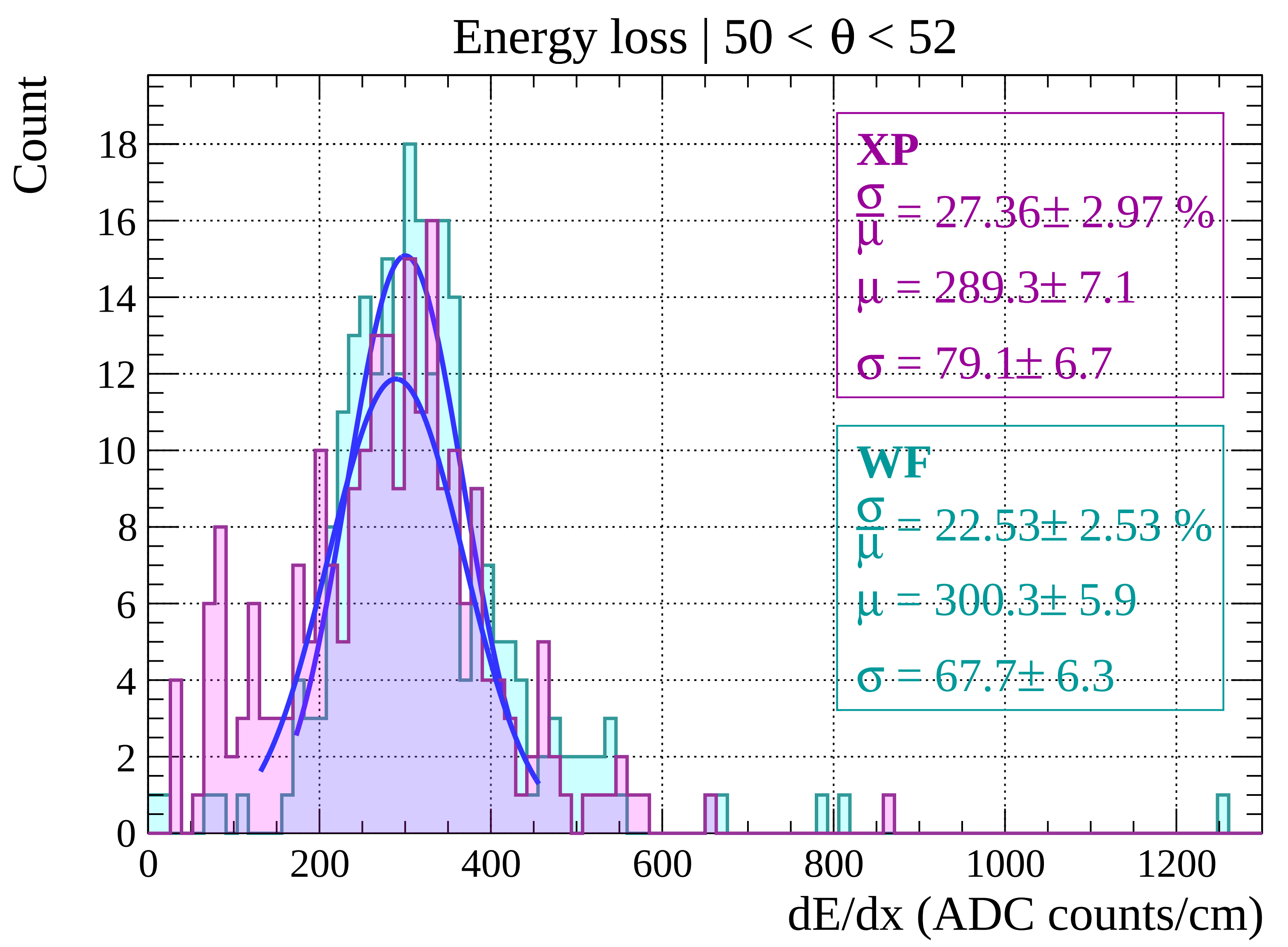

Energy loss | 44 < θ < 46
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP
σ/μ = 26.04± 3.52 %
μ = 342.5± 7.3
σ = 89.2± 10.2
WF
σ/μ = 15.95± 1.39 %
μ = 347.0± 4.3
σ = 55.3± 4.1



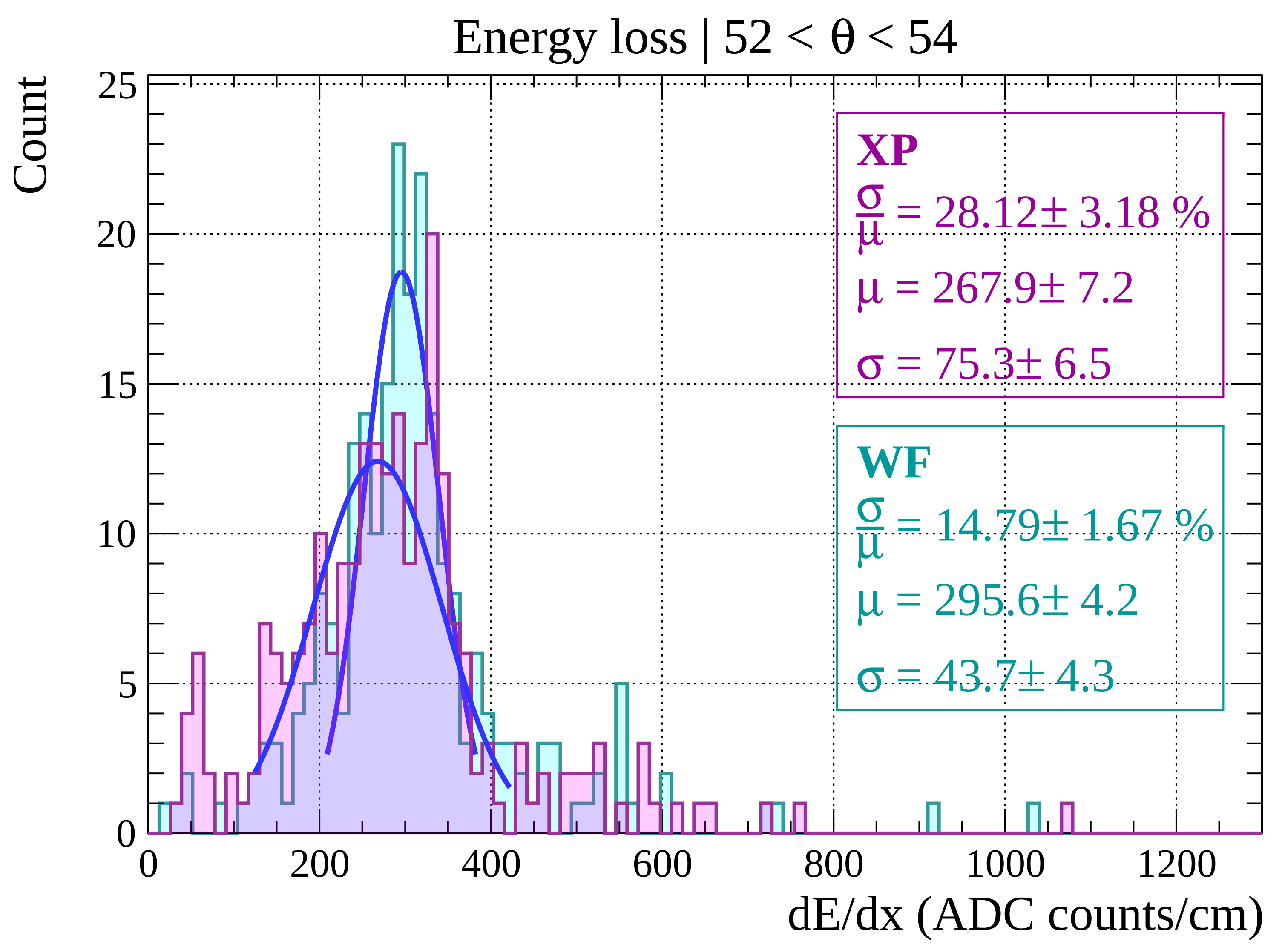
Energy loss | 46 < θ < 48
Count
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 30.22± 2.66 %
μ = 313.7± 7.2
σ = 94.8± 6.2
WF
σ/μ = 18.51± 1.41 %
μ = 324.6± 4.5
σ = 60.1± 3.7



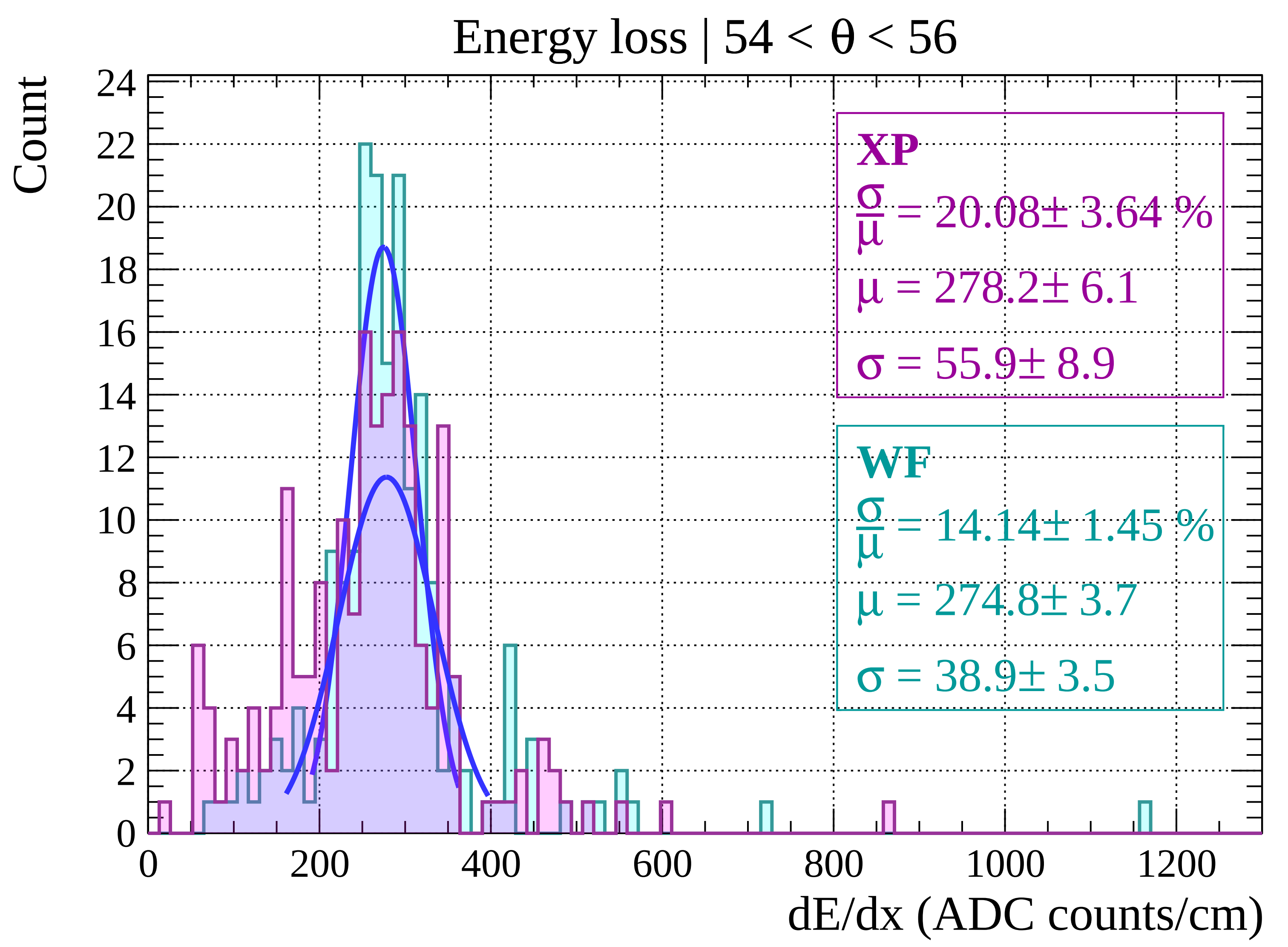
Energy loss | 48 < θ < 50
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
22
XP
σ/μ = 39.62± 5.05 %
μ = 279.2± 9.7
σ = 110.6± 10.2
WF
σ/μ = 21.70± 2.43 %
μ = 317.8± 6.2
σ = 69.0± 6.4



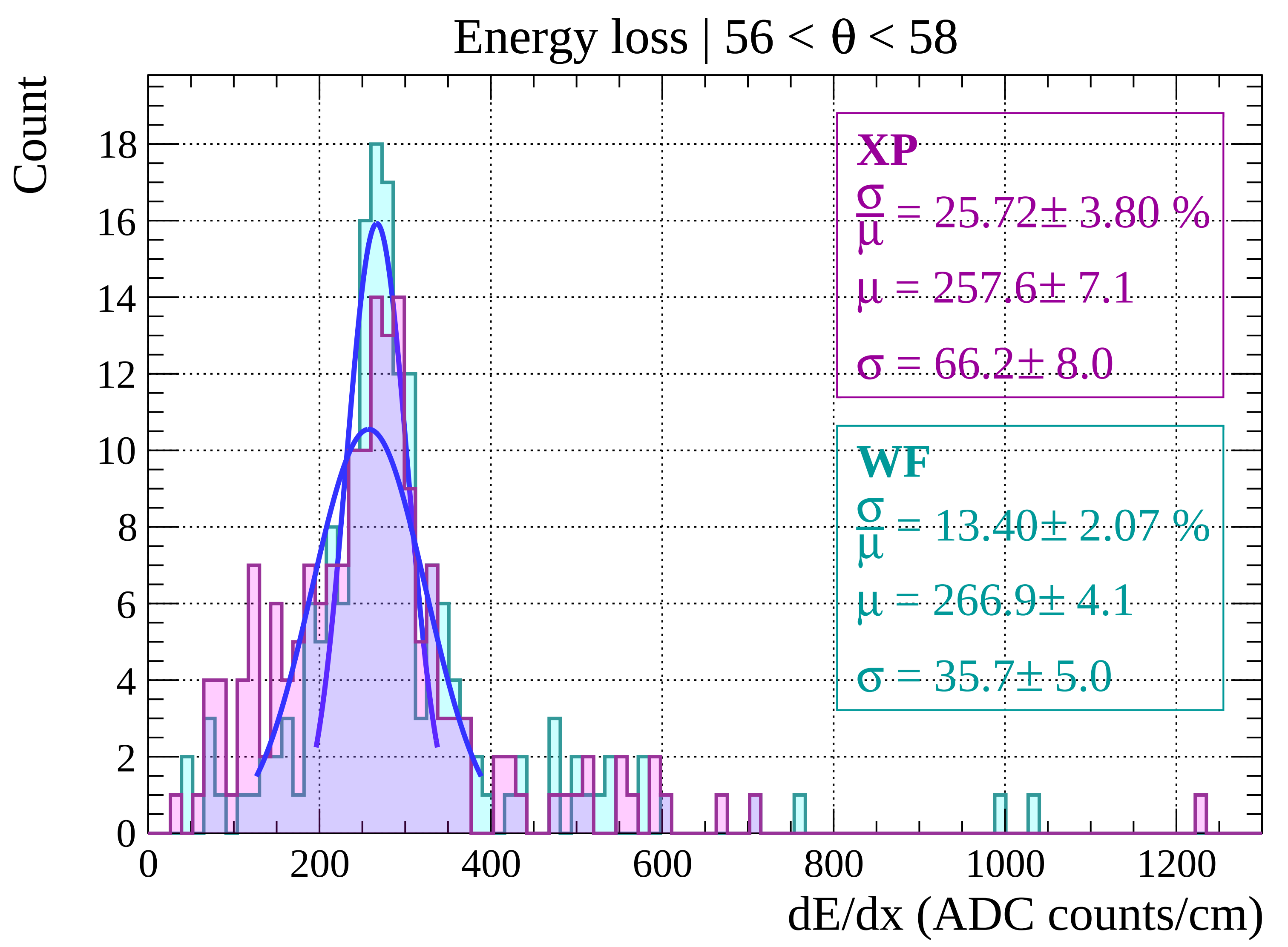
Energy loss | 50 < θ < 52
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
0
200
400
600
800
1000
1200
XP
σ/μ = 27.36± 2.97 %
μ = 289.3± 7.1
σ = 79.1± 6.7
WF
σ/μ = 22.53± 2.53 %
μ = 300.3± 5.9
σ = 67.7± 6.3



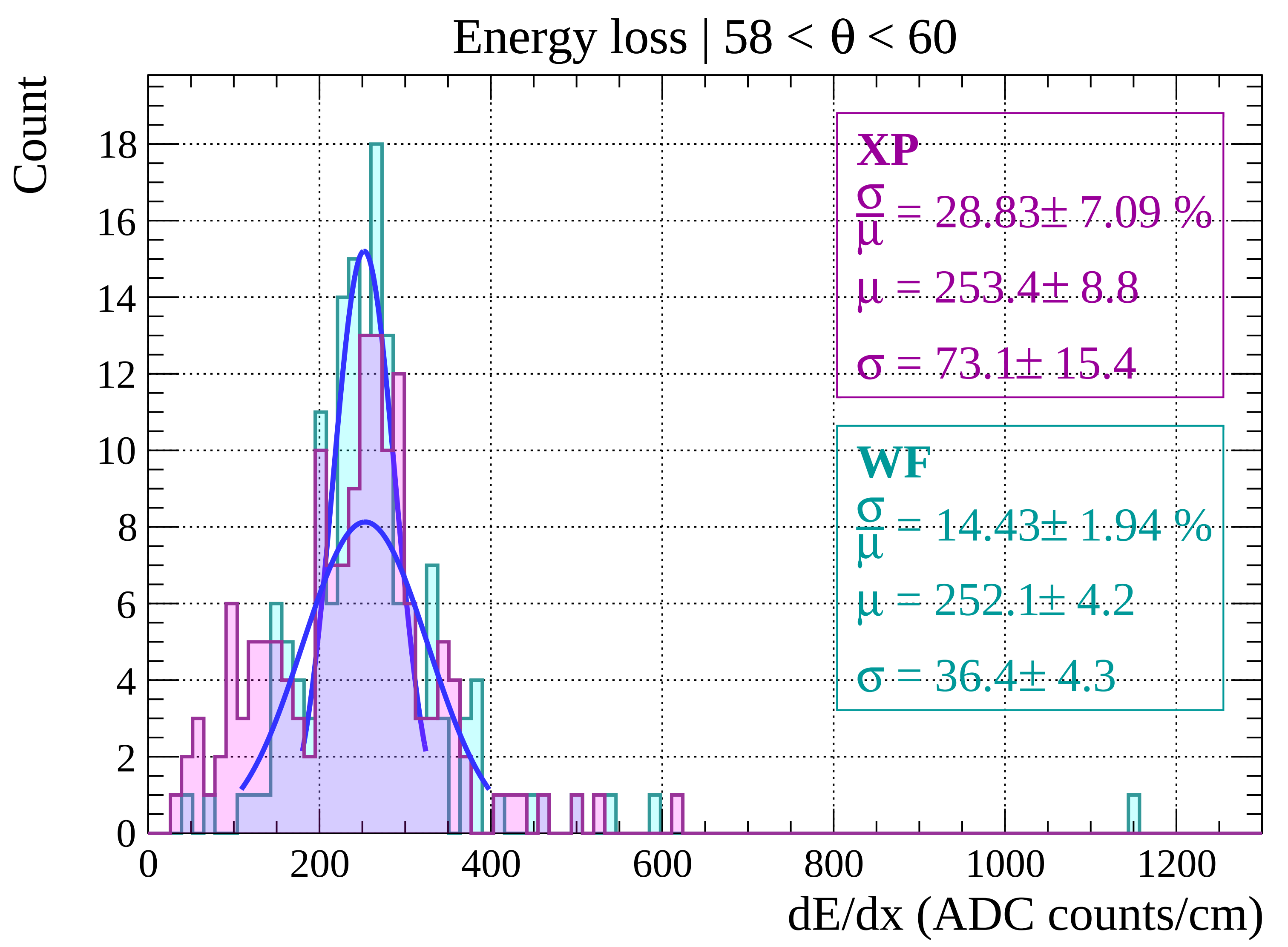
Energy loss | 52 < θ < 54
Count
0
5
10
15
20
25
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 28.12± 3.18 %
μ = 267.9± 7.2
σ = 75.3± 6.5
WF
σ/μ = 14.79± 1.67 %
μ = 295.6± 4.2
σ = 43.7± 4.3

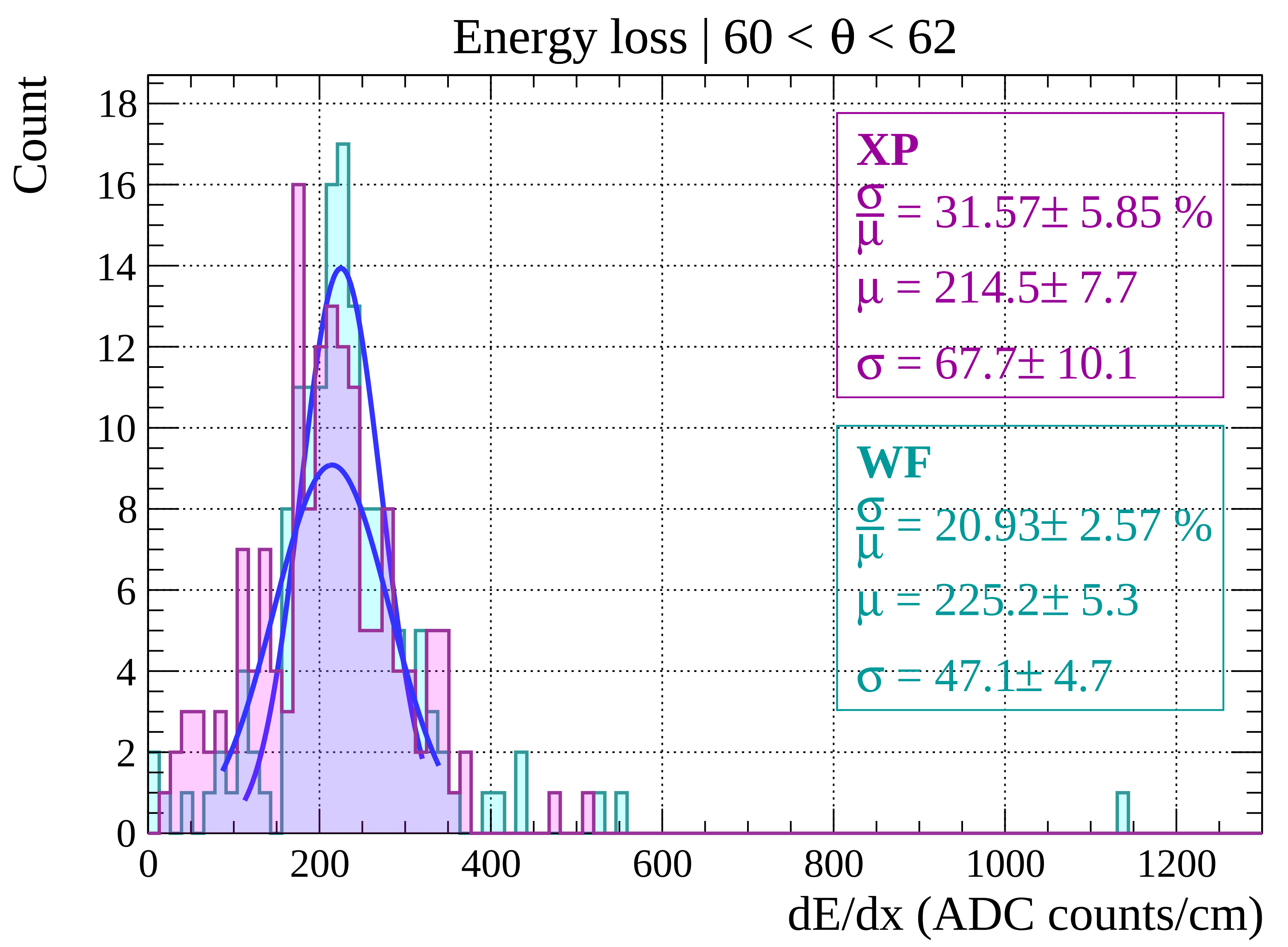


Energy loss | 54 < θ < 56
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 20.08± 3.64 %
μ = 278.2± 6.1
σ = 55.9± 8.9
WF
σ/μ = 14.14± 1.45 %
μ = 274.8± 3.7
σ = 38.9± 3.5

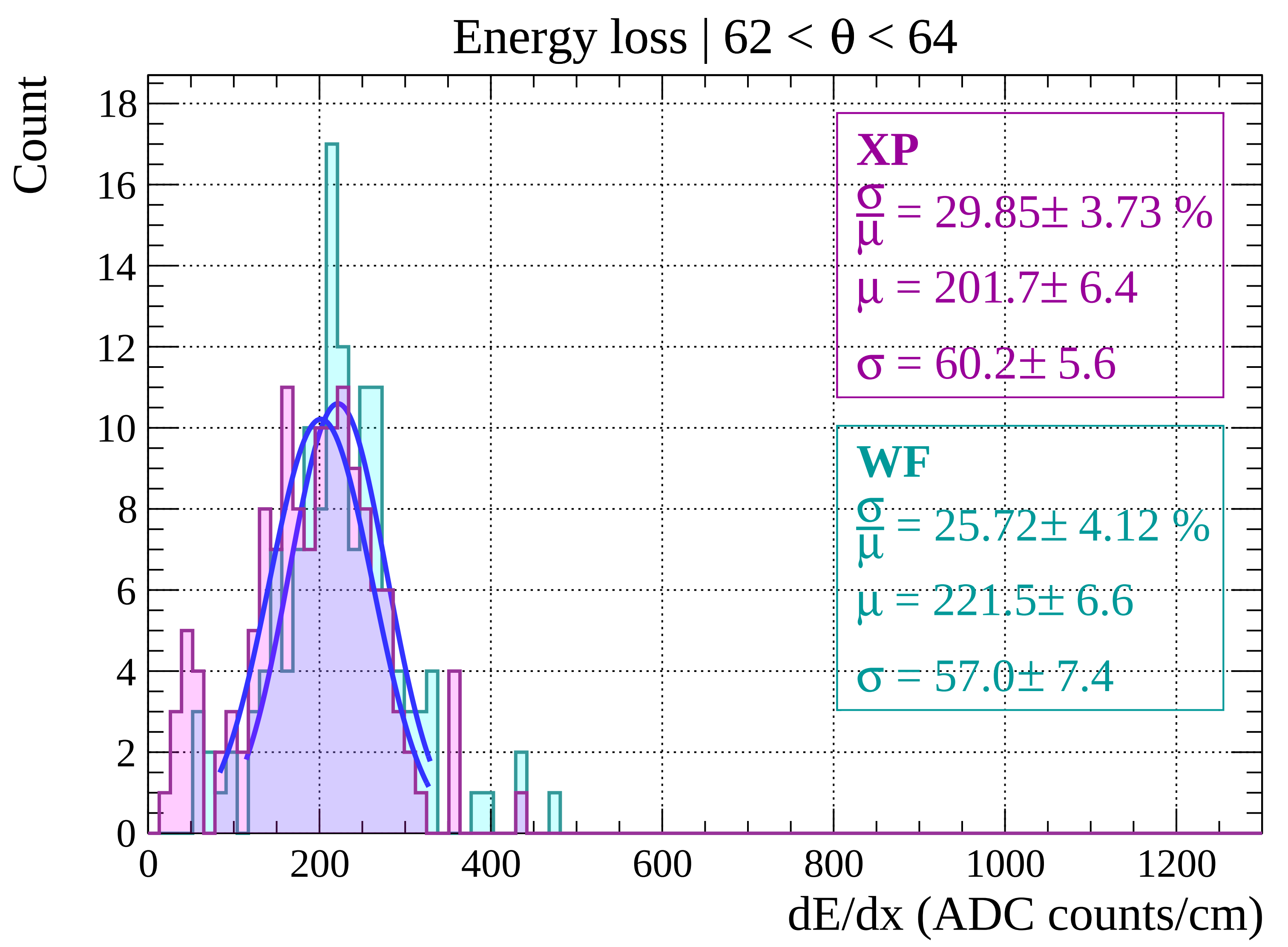


Energy loss | 56 < θ < 58
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 25.72 ± 3.80 %
μ = 257.6 ± 7.1
σ = 66.2 ± 8.0
WF
$\frac{\sigma}{\mu}$ = 13.40 ± 2.07 %
μ = 266.9 ± 4.1
σ = 35.7 ± 5.0

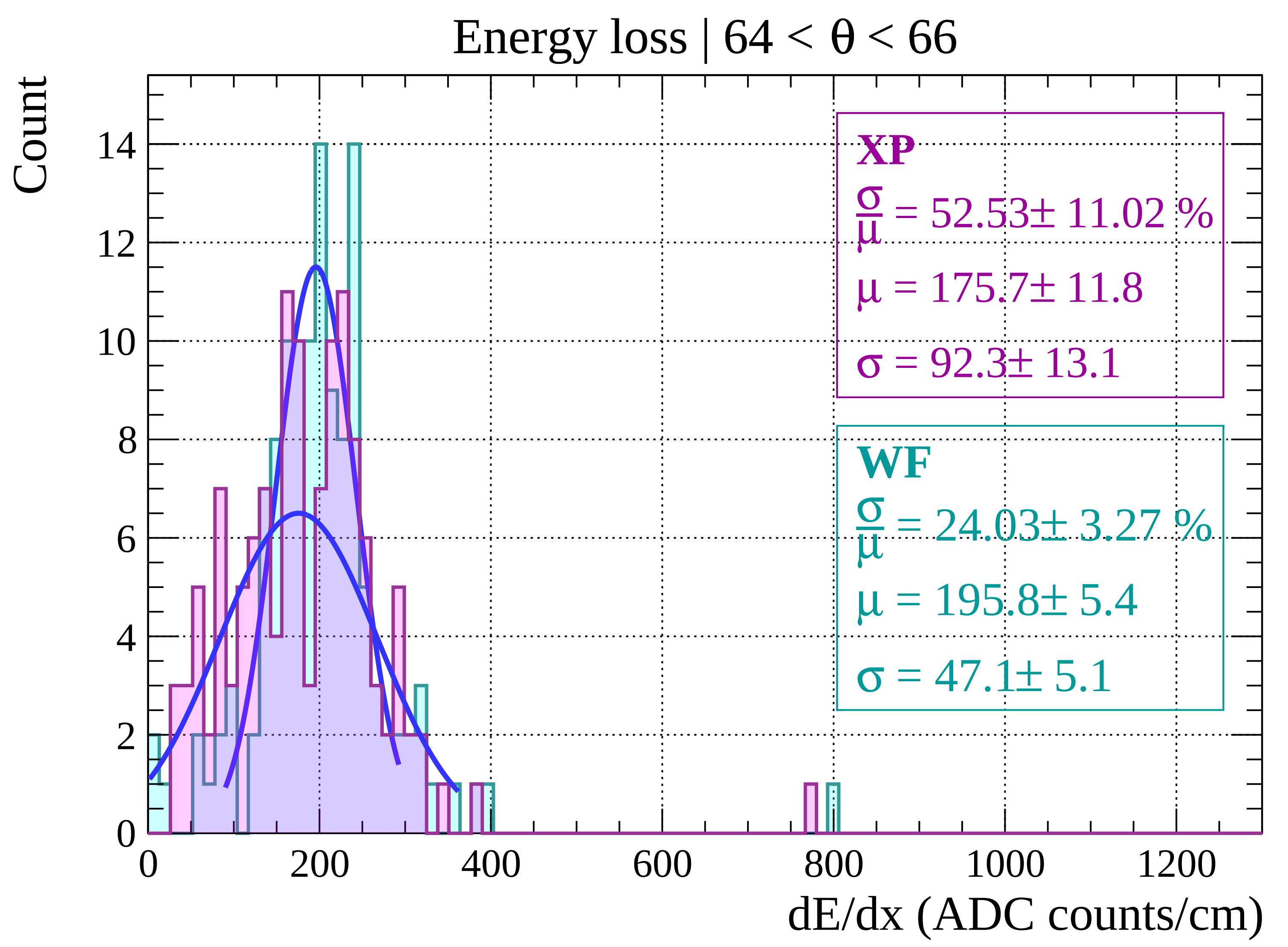

Energy loss | 58 < θ < 60
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 28.83± 7.09 %
μ = 253.4± 8.8
σ = 73.1± 15.4
WF
$\frac{\sigma}{\mu}$ = 14.43± 1.94 %
μ = 252.1± 4.2
σ = 36.4± 4.3



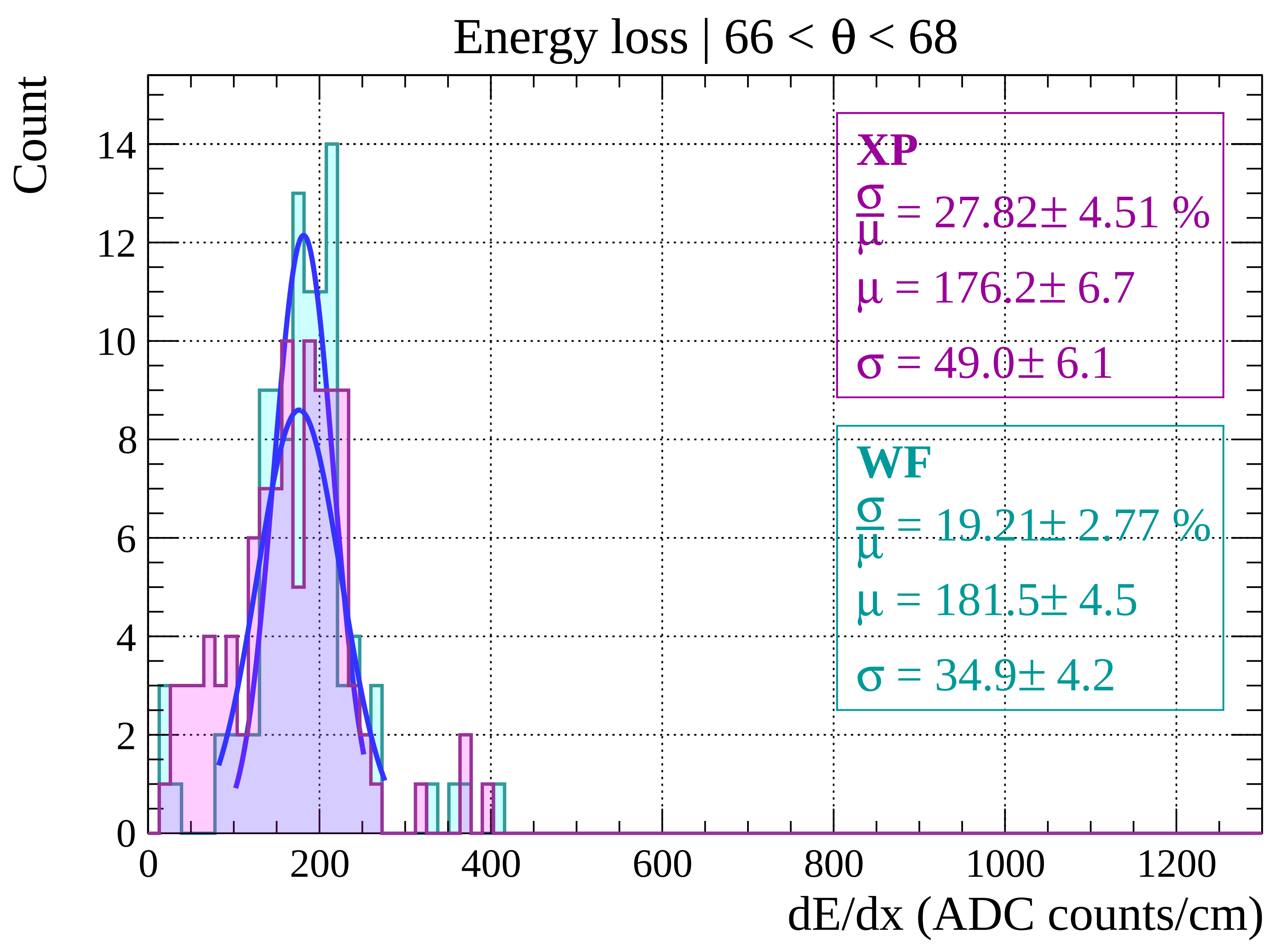
Energy loss | 60 < θ < 62
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
0
200
400
600
800
1000
1200
XP
σ/μ = 31.57± 5.85 %
μ = 214.5± 7.7
σ = 67.7± 10.1
WF
σ/μ = 20.93± 2.57 %
μ = 225.2± 5.3
σ = 47.1± 4.7



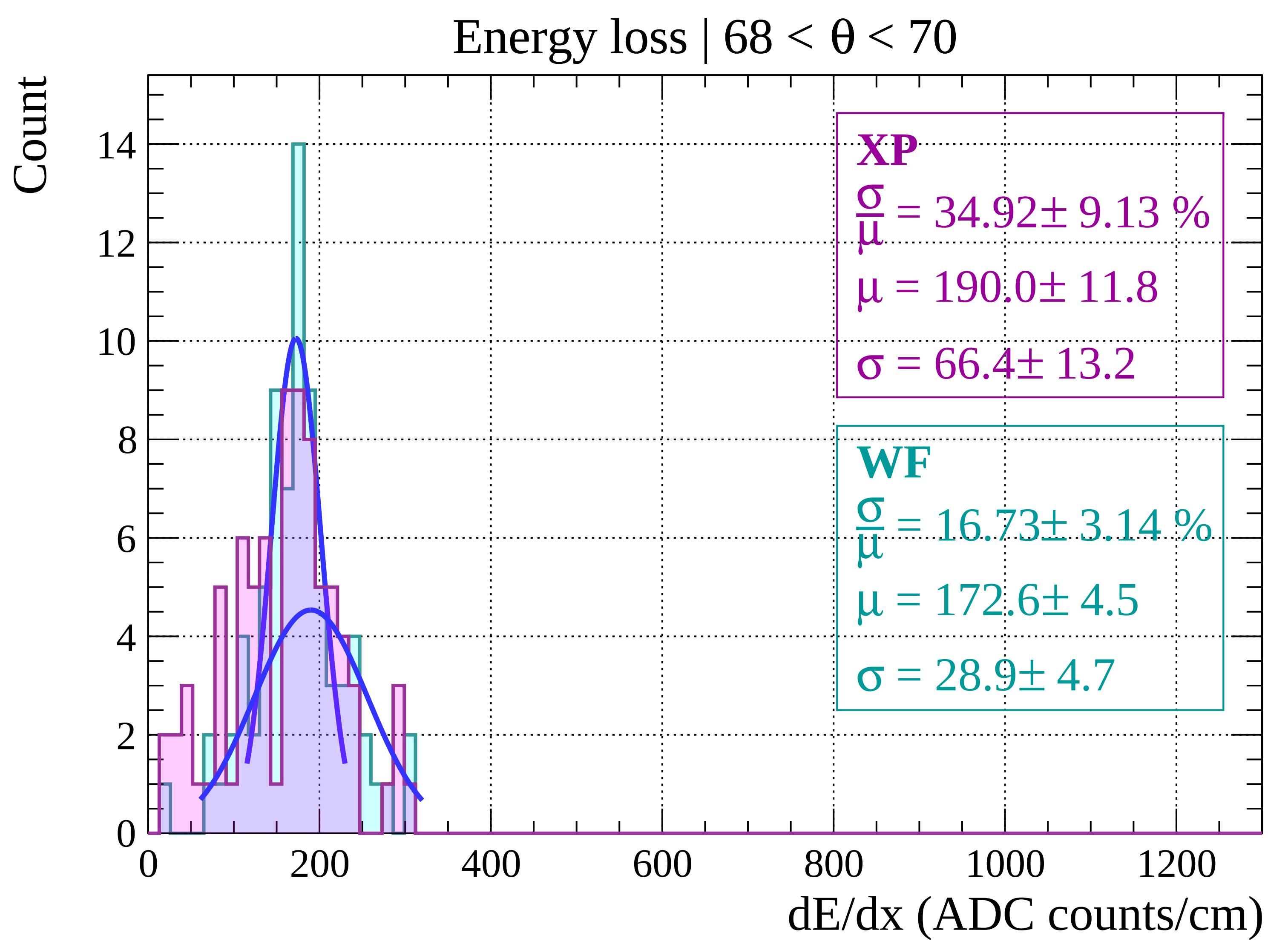
Energy loss | 62 < θ < 64
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
0
200
400
600
800
1000
1200
XP
σ/μ = 29.85± 3.73 %
μ = 201.7± 6.4
σ = 60.2± 5.6
WF
σ/μ = 25.72± 4.12 %
μ = 221.5± 6.6
σ = 57.0± 7.4

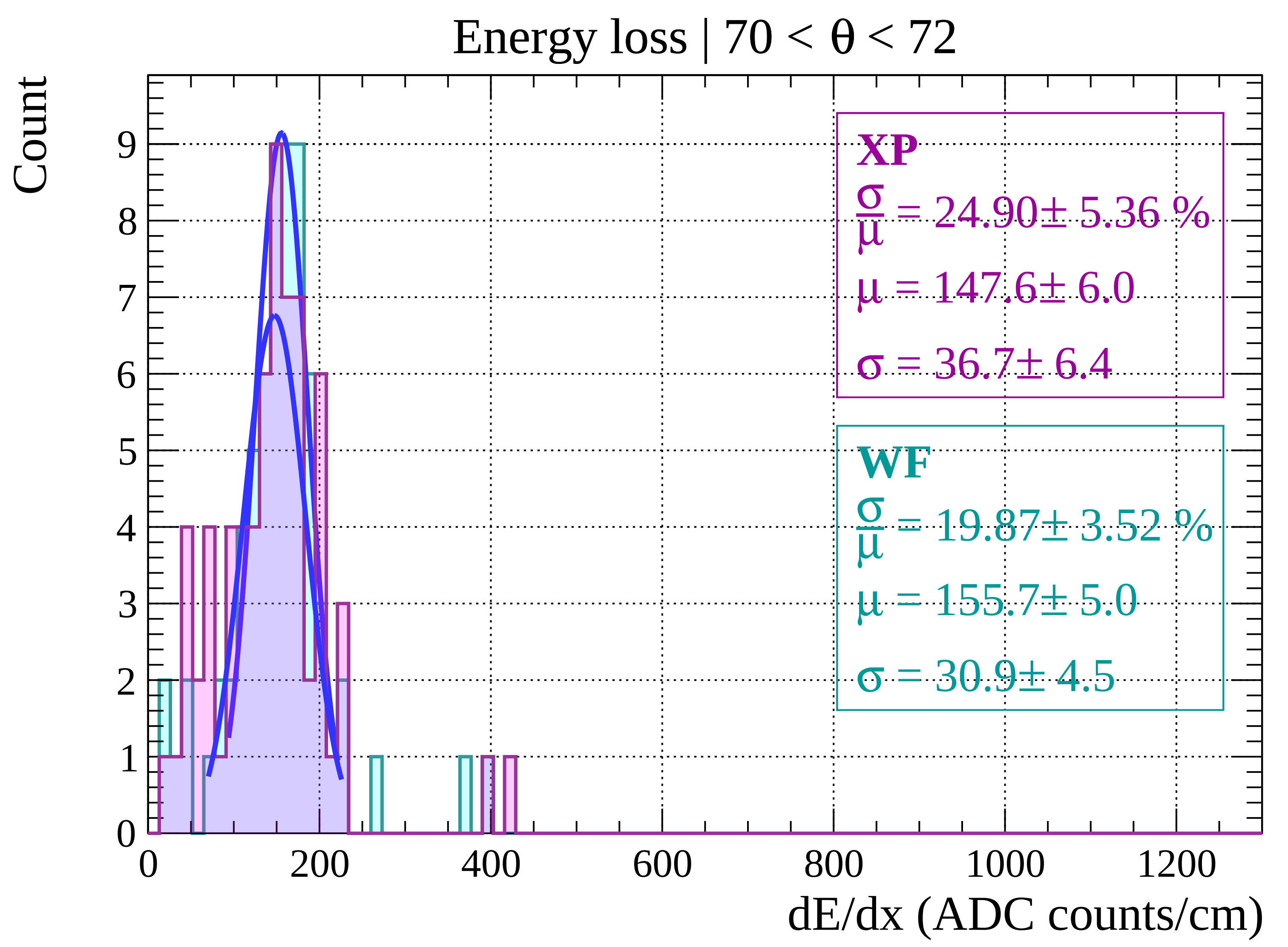


Energy loss | 64 < θ < 66
Count
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 52.53± 11.02 %
μ = 175.7± 11.8
σ = 92.3± 13.1
WF
σ/μ = 24.03± 3.27 %
μ = 195.8± 5.4
σ = 47.1± 5.1

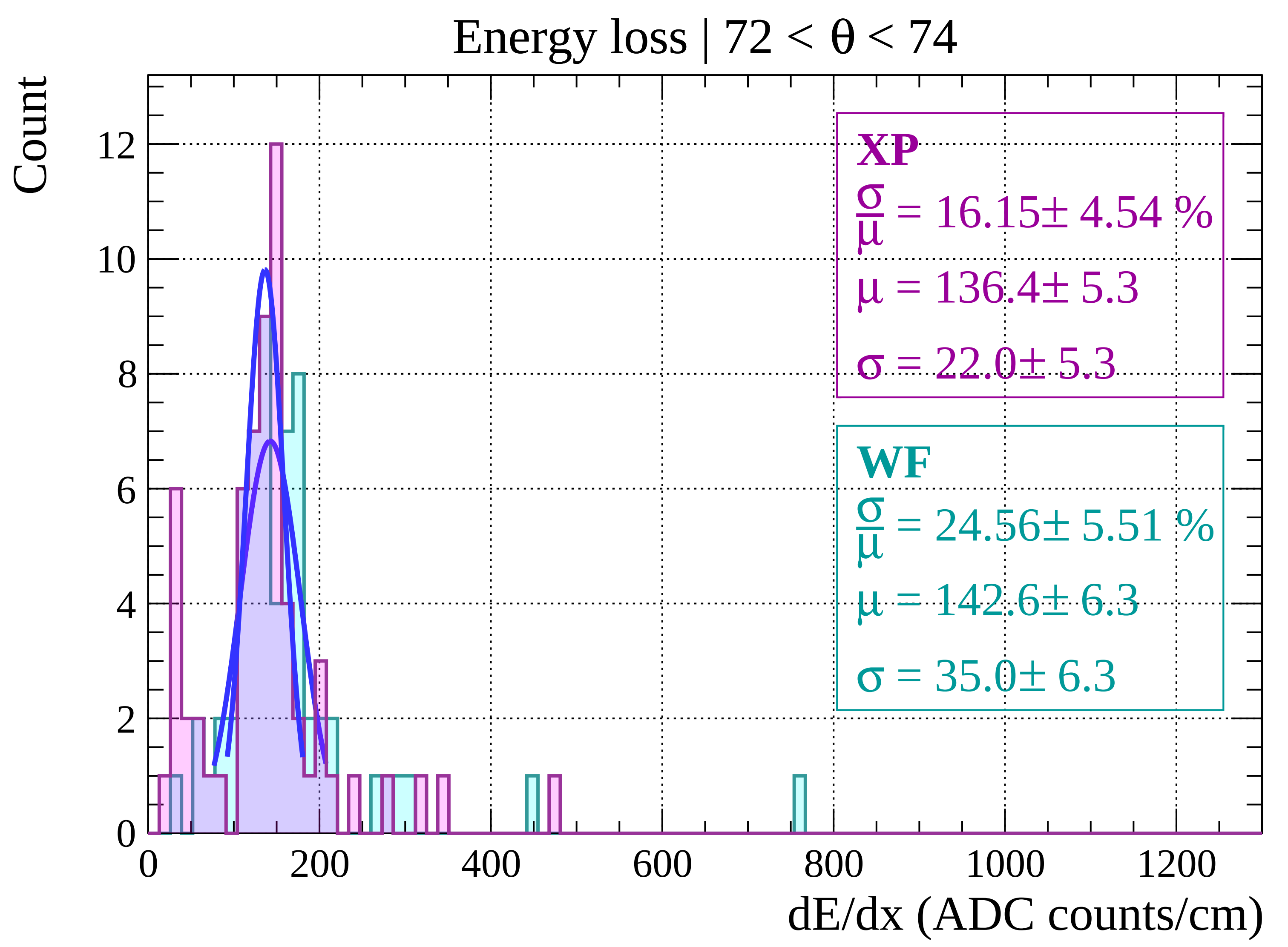


Energy loss | 66 < θ < 68
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
XP
σ/μ = 27.82± 4.51 %
μ = 176.2± 6.7
σ = 49.0± 6.1
WF
σ/μ = 19.21± 2.77 %
μ = 181.5± 4.5
σ = 34.9± 4.2

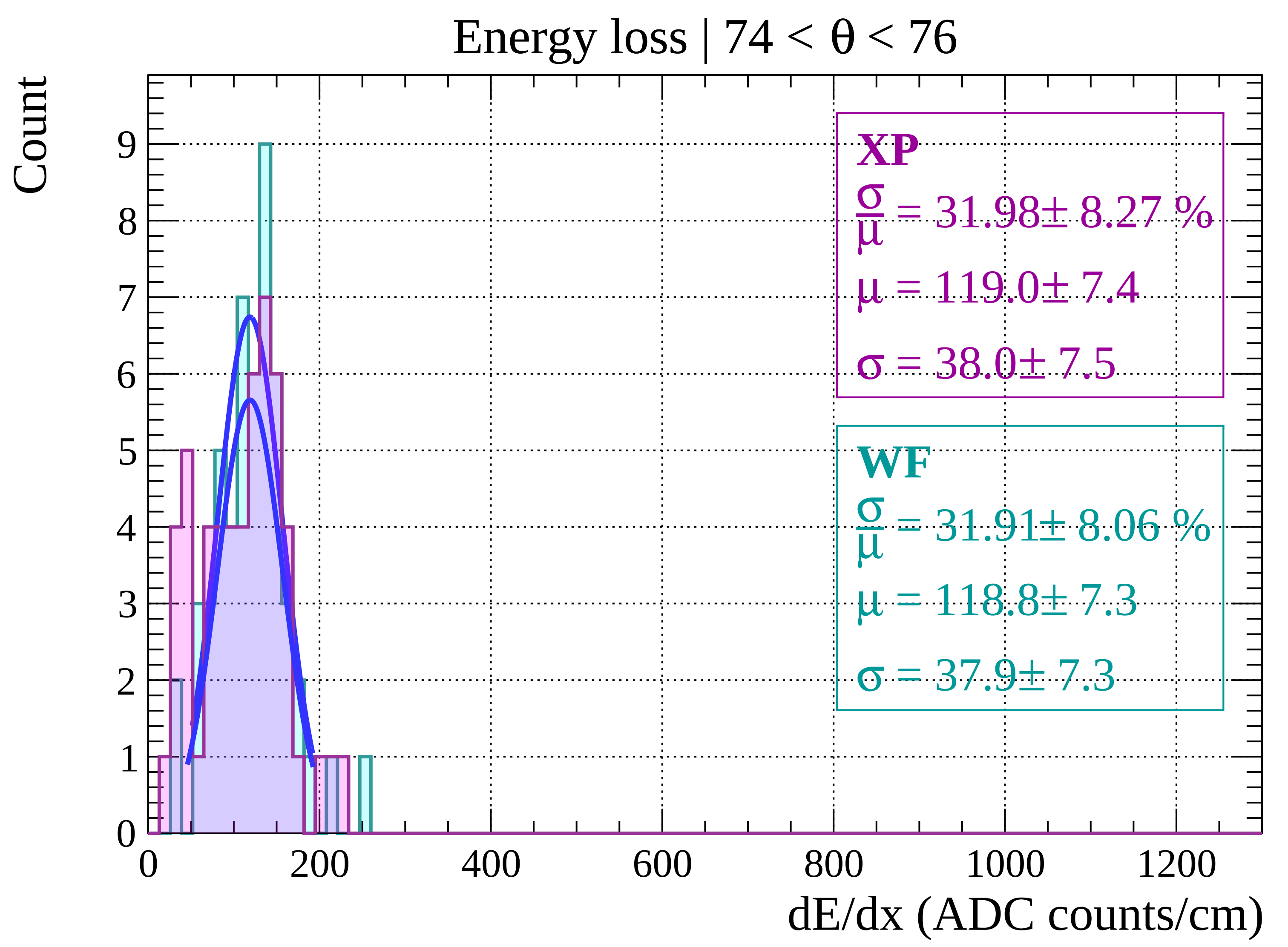

Energy loss | 68 < θ < 70
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
XP
σ/μ = 34.92± 9.13 %
μ = 190.0± 11.8
σ = 66.4± 13.2
WF
σ/μ = 16.73± 3.14 %
μ = 172.6± 4.5
σ = 28.9± 4.7



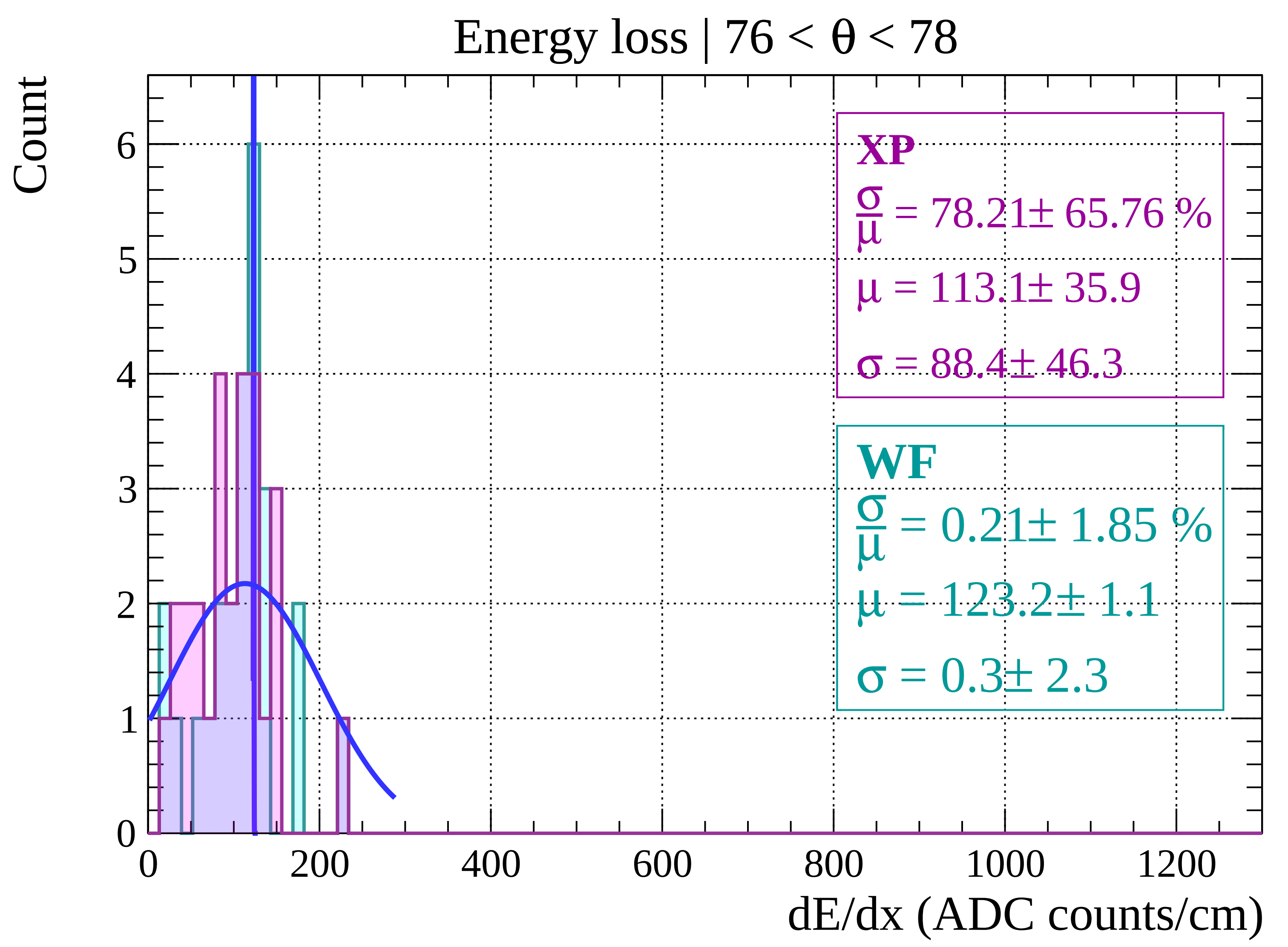
Energy loss | 70 < θ < 72
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
8
9
0
200
400
600
800
1000
1200
XP
σ/μ = 24.90± 5.36 %
μ = 147.6± 6.0
σ = 36.7± 6.4
WF
σ/μ = 19.87± 3.52 %
μ = 155.7± 5.0
σ = 30.9± 4.5

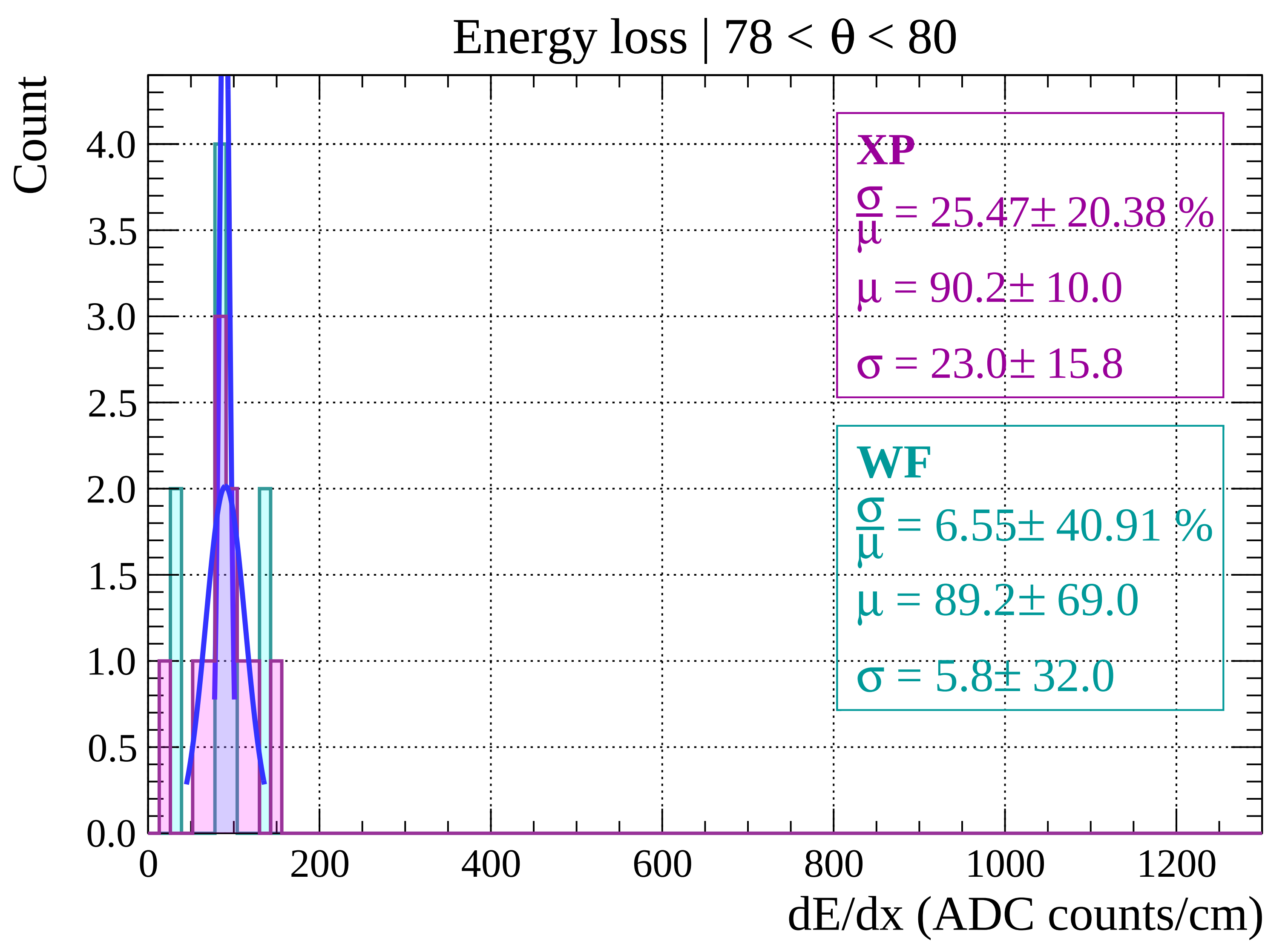


Energy loss | 72 < θ < 74
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
σ/μ = 16.15± 4.54 %
μ = 136.4± 5.3
σ = 22.0± 5.3
WF
σ/μ = 24.56± 5.51 %
μ = 142.6± 6.3
σ = 35.0± 6.3

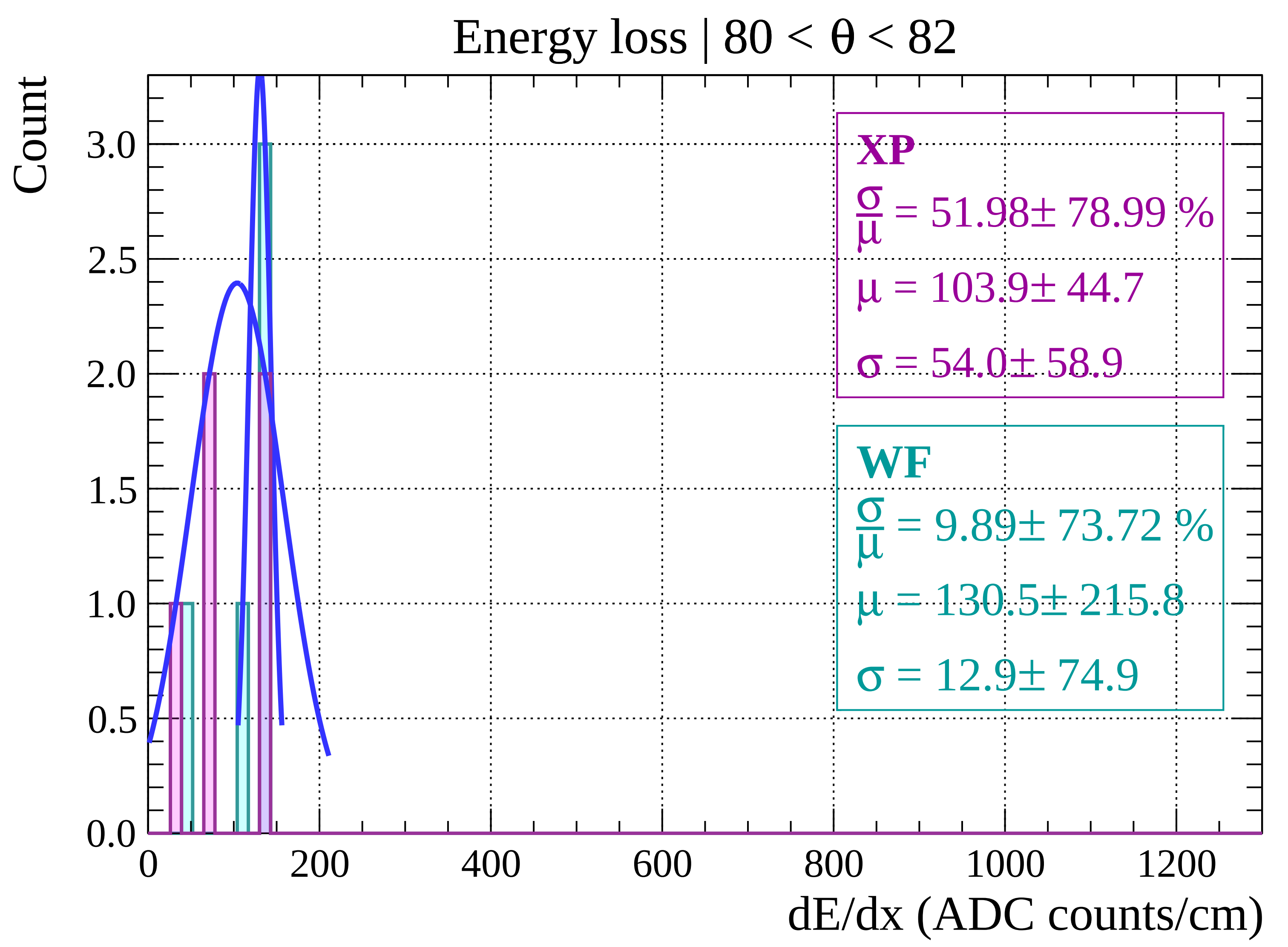


Energy loss | 74 < θ < 76
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
8
9
0
200
400
600
800
1000
1200
XP
σ/μ = 31.98± 8.27 %
μ = 119.0± 7.4
σ = 38.0± 7.5
WF
σ/μ = 31.91± 8.06 %
μ = 118.8± 7.3
σ = 37.9± 7.3

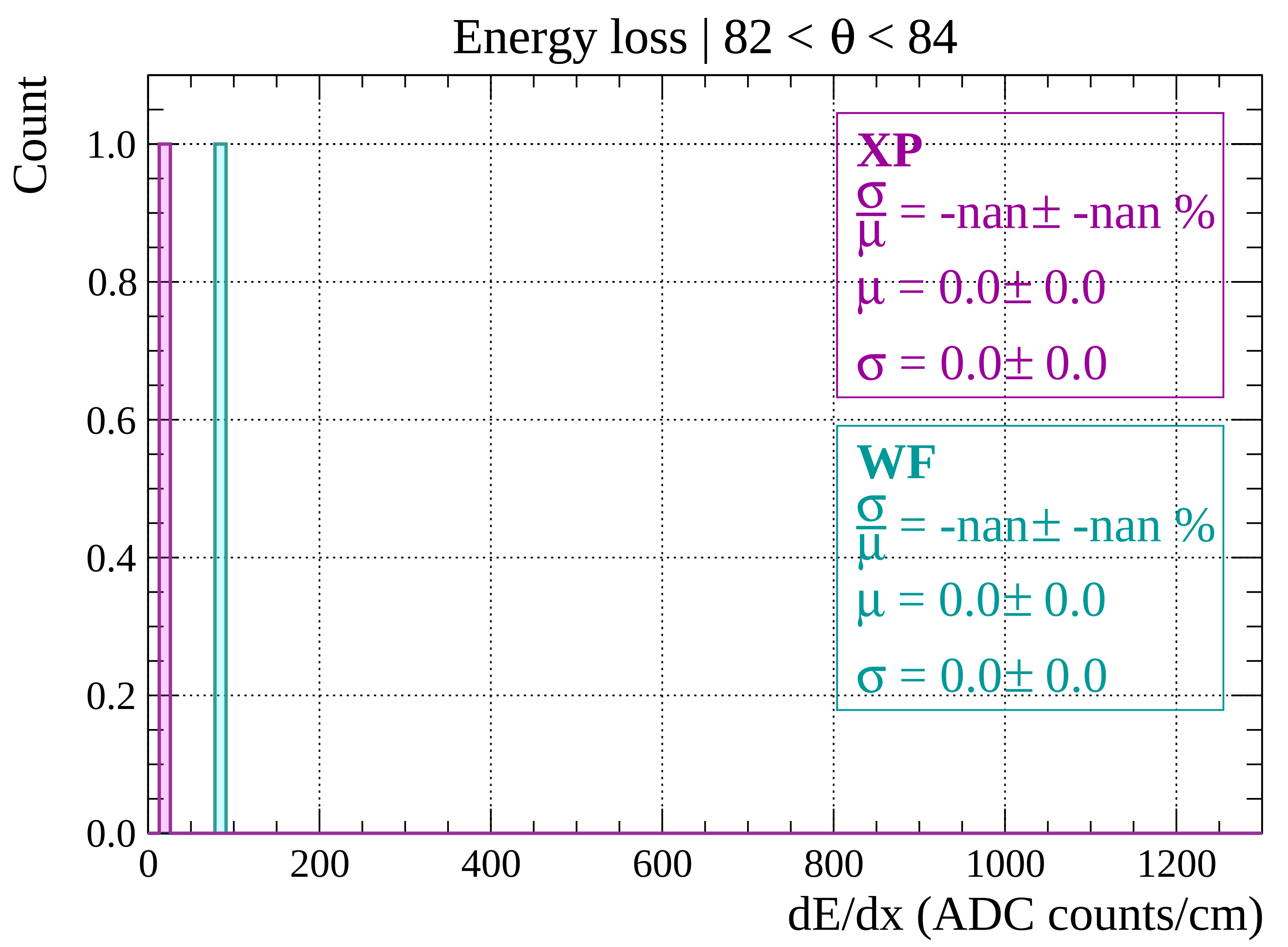

Energy loss | 76 < θ < 78
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
XP
σ/μ = 78.21± 65.76 %
μ = 113.1± 35.9
σ = 88.4± 46.3
WF
σ/μ = 0.21± 1.85 %
μ = 123.2± 1.1
σ = 0.3± 2.3



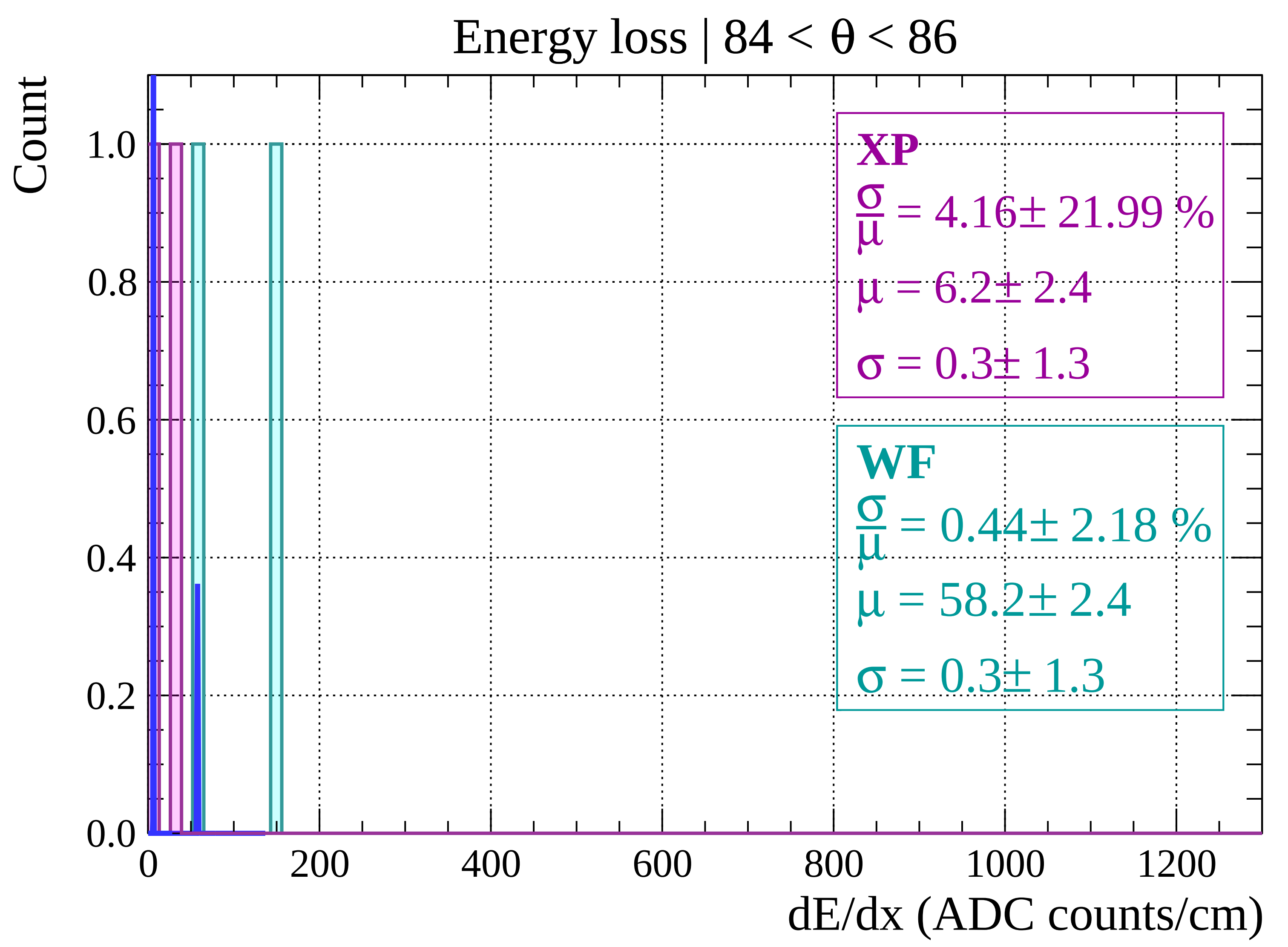
Energy loss | 78 < θ < 80
Count
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 25.47± 20.38 %
μ = 90.2± 10.0
σ = 23.0± 15.8
WF
σ/μ = 6.55± 40.91 %
μ = 89.2± 69.0
σ = 5.8± 32.0

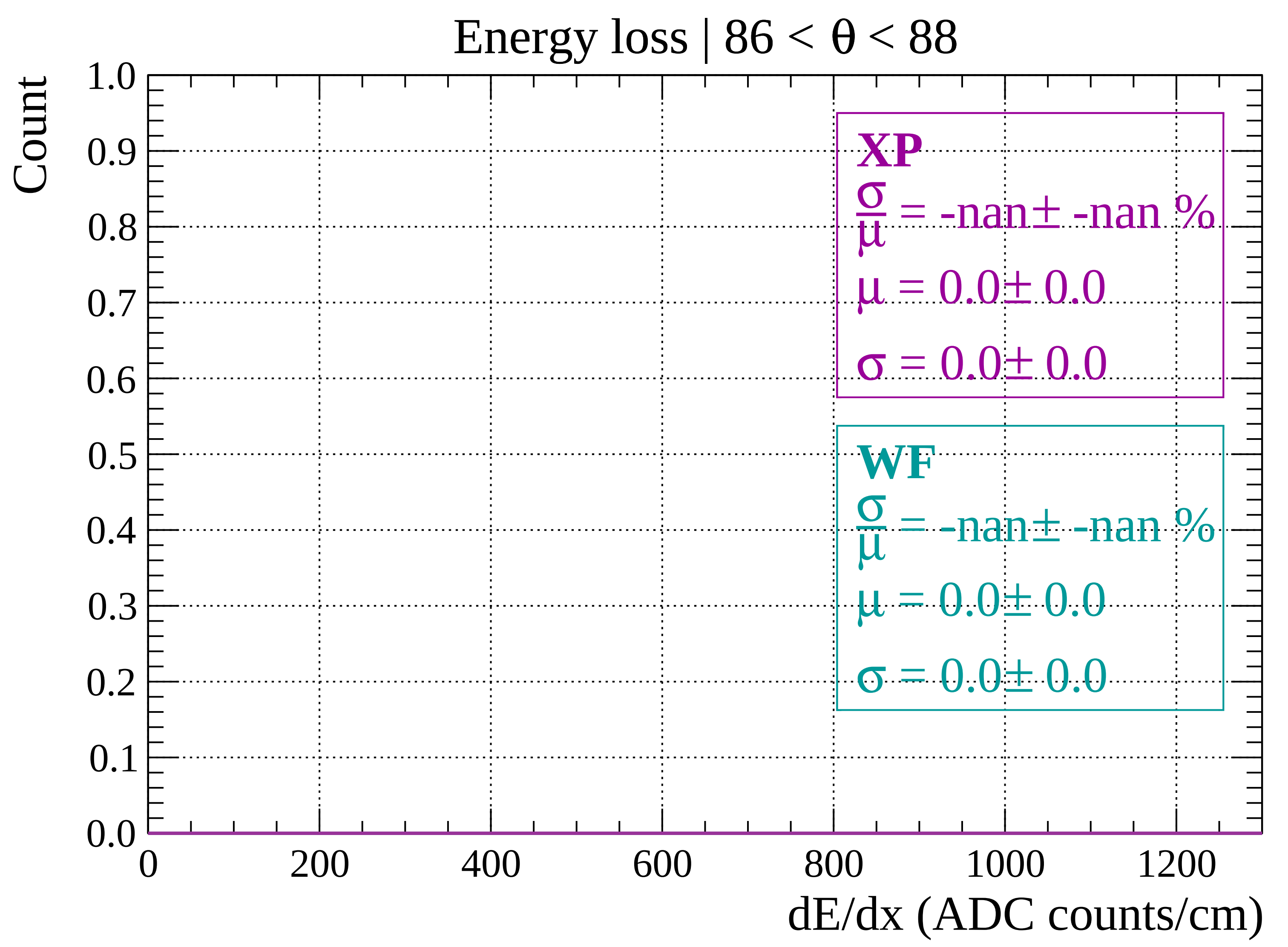


Energy loss | 80 < θ < 82
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 51.98± 78.99 %
μ = 103.9± 44.7
σ = 54.0± 58.9
WF
σ/μ = 9.89± 73.72 %
μ = 130.5± 215.8
σ = 12.9± 74.9

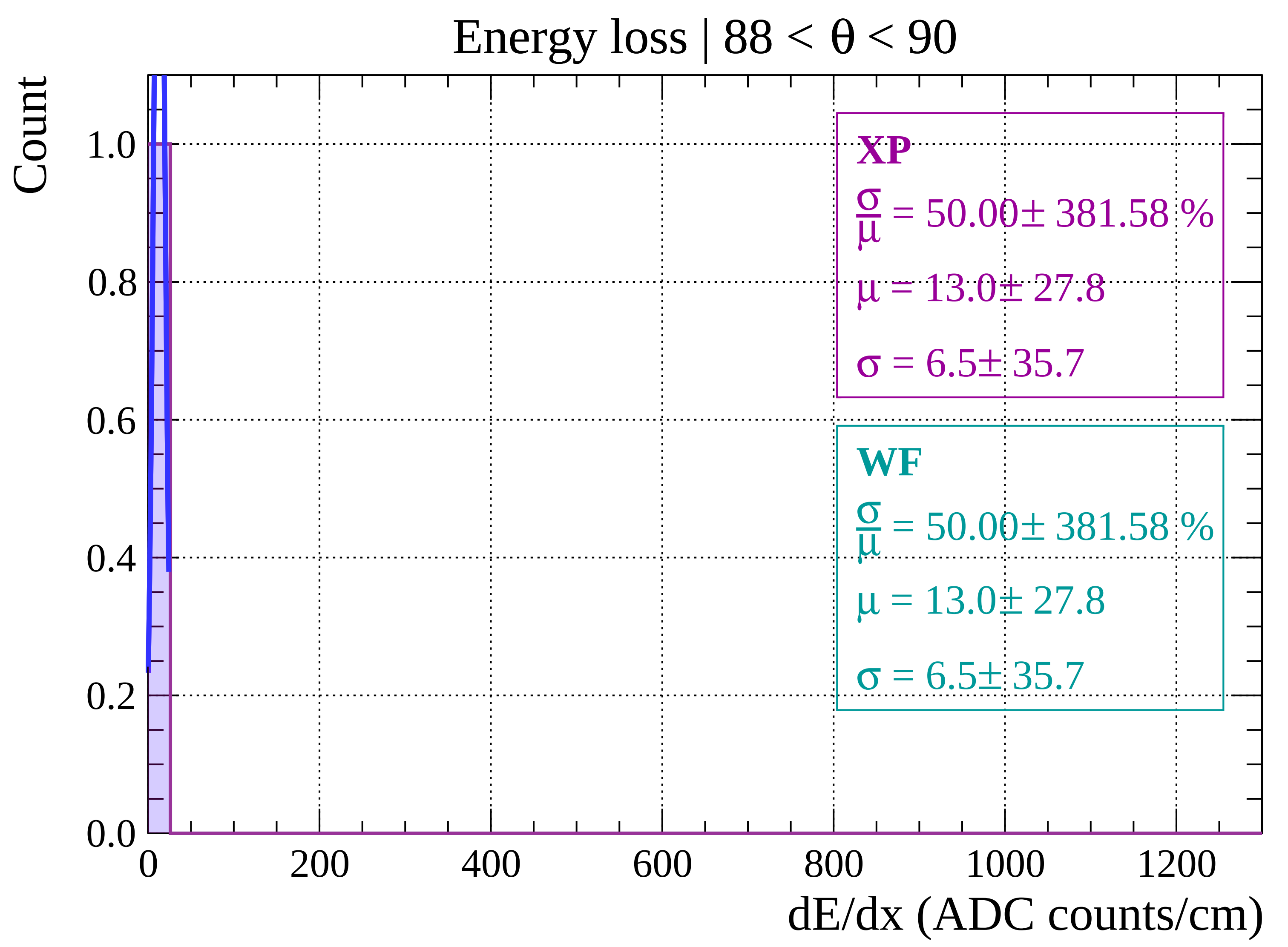


Energy loss | 82 < θ < 84
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

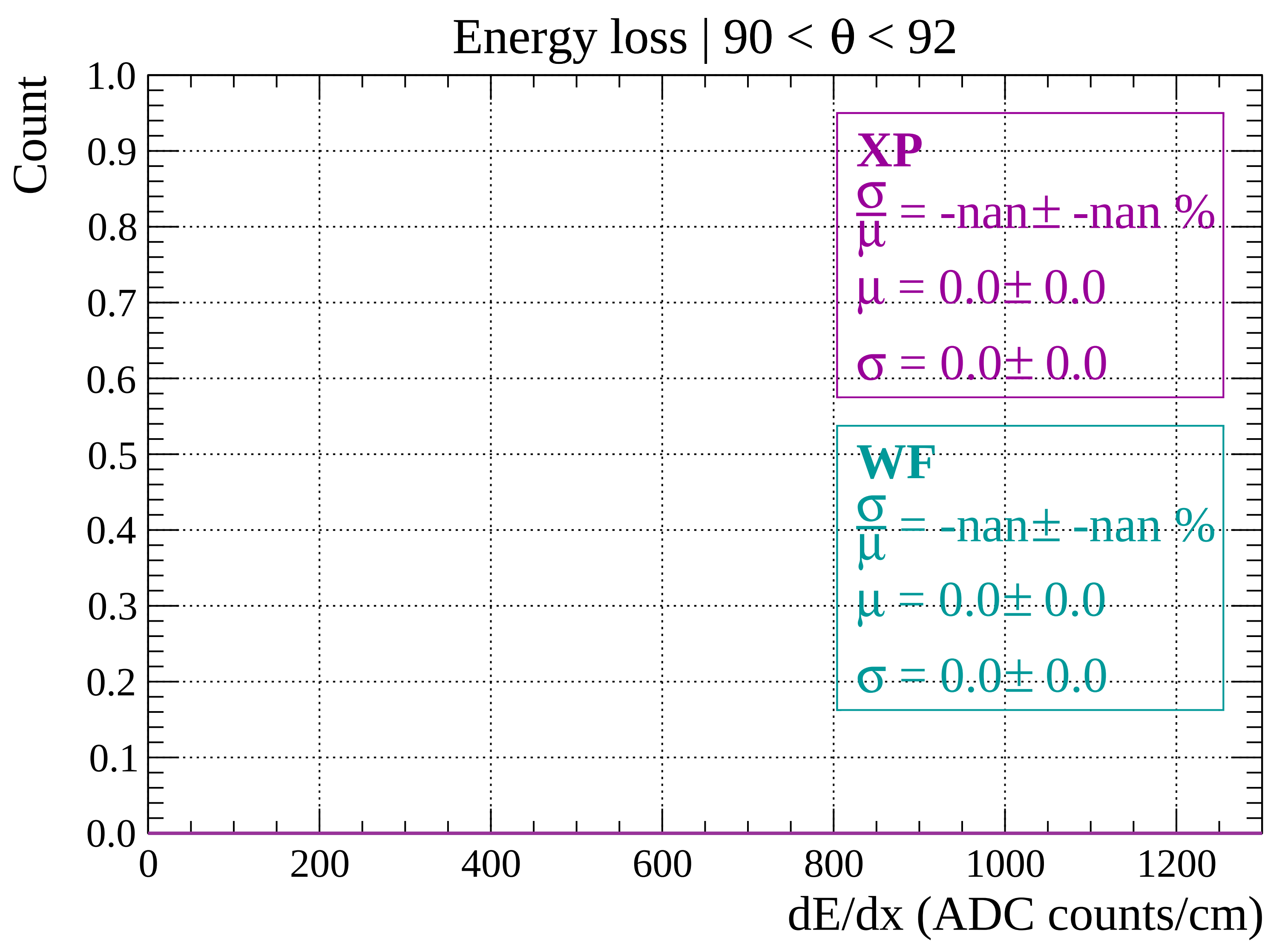

Energy loss | 84 < θ < 86
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.16± 21.99 %
μ = 6.2± 2.4
σ = 0.3± 1.3
WF
σ/μ = 0.44± 2.18 %
μ = 58.2± 2.4
σ = 0.3± 1.3



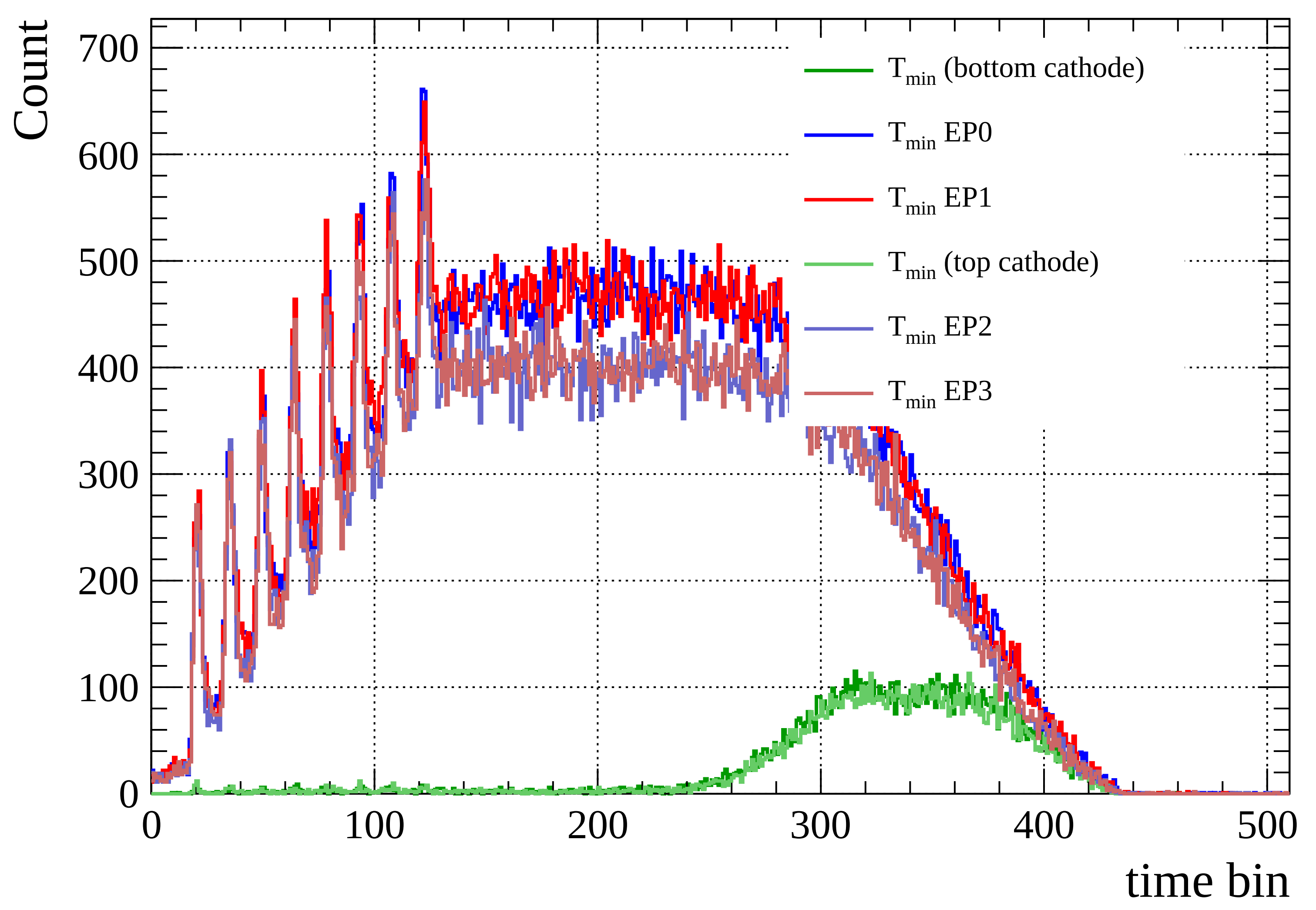
Energy loss | 86 < θ < 88
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



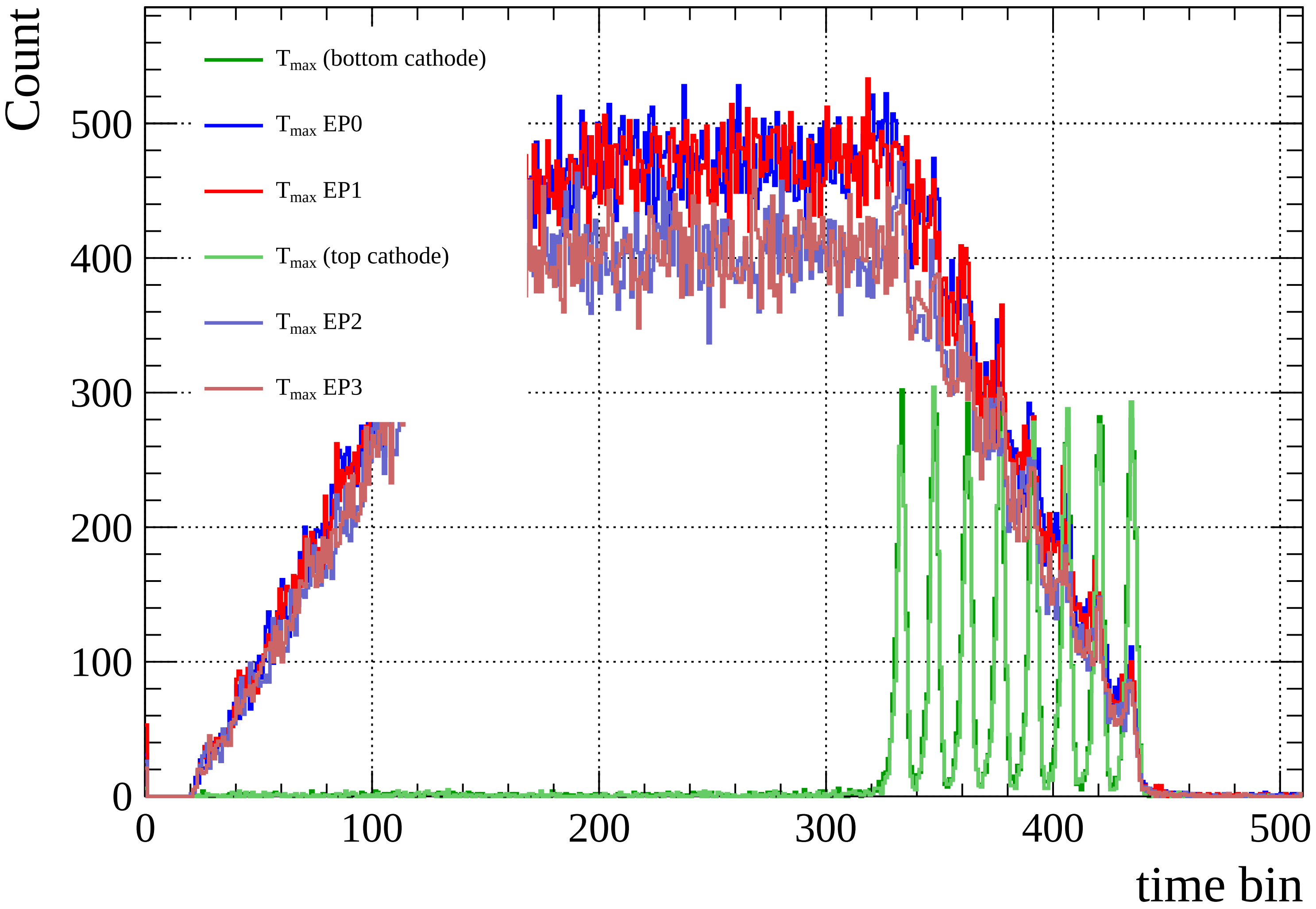
Energy loss | 88 < θ < 90
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 50.00± 381.58 %
μ = 13.0± 27.8
σ = 6.5± 35.7
WF
σ/μ = 50.00± 381.58 %
μ = 13.0± 27.8
σ = 6.5± 35.7



Energy loss | 90 < θ < 92
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
700
600
500
400
300
200
100
0
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
time bin
0
100
200
300
400
500
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3